NAVIGATION SYSTEM DATA BASE

ARINC SPECIFICATION 424-19

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A description of the changes introduced by each supplement is included on Goldenrod paper at the end of this document.

FOREWORD

Aeronautical Radio, Inc., the AEEC, and ARINC Standards

ARINC organizes aviation industry committees and participates in related industry activities that benefit aviation at large by providing technical leadership and guidance. These activities directly support aviation industry goals: promote safety, efficiency, regularity, and cost-effectiveness in aircraft operations.

ARINC Industry Activities organizes and provides the secretariat for international aviation organizations (AEEC, AMC, FSEMC) which coordinate the work of aviation industry technical professionals and lead the development of technical standards for airborne electronic equipment, aircraft maintenance equipment and practices and flight simulator equipment and used in commercial, military, and business aviation. The AEEC, AMC, and FSEMC develop consensus-based, voluntary standards that are published by ARINC and are known as ARINC Standards. The use of ARINC Standards results in substantial benefits to the aviation industry by allowing avionics interchangeability and commonality and reducing avionics cost by promoting competition.

There are three classes of ARINC Standards:

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In order to facilitate the continuous product improvement of this ARINC Standard, two items are included in the back of this volume:

An Errata Report solicits any corrections to the text or diagrams in this ARINC Standard.

An ARINC IA Project Initiation/Modification (APIM) form solicits any recommendations for addition of substantive material to this volume which would be the subject of a new supplement.

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1.0 INTRODUCTION

1.1 Purpose of this Document

This document sets forth the air transport industry's recommended standards for the preparation of airborne navigation system reference data files. The data on these files are intended for merging with airborne navigation computer operational software to produce media for use by such computers on board aircraft. Since the industry does not desire to standardize the operational software of these computers, this merging process is not described in this document, nor do the standards set forth necessarily apply to the aircraft-employed data.

The data bases prescribed by this document are also used by computer flight planning systems, flight simulators, and other applications.

The purpose of the ARINC 424 Navigation System Data Base Standard is to be an enabling document. It enables data base suppliers, avionics systems, and other users of the data bases to fly and flight plan procedures as prescribed by procedure designers. The document is not meant to be a prescriptive document for procedure designers.

This document is also not a requirements document for airborne navigation systems.

Procedures that are not compatible with this data base standard in some cases cannot be coded for inclusion in some airborne data bases.

1.1.1 Coverage of Flight Simulator Needs

Supplement 4 to this document added material related to the special navigation data base needs of flight simulators. The approach taken, i.e., the definition of three new subsections to the master file and the exploitation of previously unused continuation record capability, was designed to ensure that users who wish to continue using the document solely as the basis for supporting airborne navigation system operation can do so without simulator-related records nor be concerned that the software used to merge Specification 424 data with airborne equipment operational software will need modification as the result of the changes. Users who wish to support both airborne navigation system and flight simulator operations can also do so without having to modify this merging software. Only the simulator navigation data base compilers need take into account the presence of the simulator-related components in the input (Specification 424) data.

1.1.2 Coverage of Flight Planning Needs

Supplement 5 of this document added material related to the special navigation data base needs to flight planning computer systems. The approach taken, i.e., the definition of the new material and the exploitation of previously unused continuation record capability, was designed to ensure that users who wish to continue using the document solely as the basis for supporting airborne navigation system operations can do so without penalty. Such users need not obtain the flight planning related records nor be concerned that the software used to merge Specification 424 data with airborne equipment operational software will need modification as the result of the changes. Users who wish to support both airborne navigation system and flight planning system operations can also do so without having to modify this merging software. Only the flight planning navigation data base systems need to take into account the presence of the flight planning related components in the input (Specification 424) data.

1.2 Data Format Standardization Philosophy

The production of navigation data for use with onboard navigation computers may be viewed as a four-step process (Attachment 1). The first step is the assembly of a data bank. The second is the production of data files organized such that individual airlines' operational needs can be met. The third step is the merging of these data with the operational software of those airlines' navigation computers. The final step is the production of final storage media containing these merged data for use on individual aircraft.

Data banks will contain world-wide navigation reference information obtained both from "public" sources (ICAO, governments, etc.) and from navigation system users. They could be assembled and maintained by public bodies (e.g., government agencies or international organizations), by commercial institutions, or both. The information needed by an airline to make use of a navigation system over its own routes will consist of a section from the "public" part of the bank and the data it requires from the "user" part of the bank. It will occupy one of the "airline nav. data files" shown at the step 2 level in Attachment 1.

To facilitate the sorting process necessary to produce individual airline files, every record in the data bank is encoded as to type. Those in the "public" part of the bank are termed "standard" records, and may appear in any airline's file. The "Master Airline User File" shown at the step 1 level in the diagram of Attachment 1 is made up of such records. They contain the data specified in Chapter 3 of this document, and are formatted according to the rules set forth in Chapters 4 and 5. Records in the "user" part of the bank are termed "tailored" records, and each one is entered into the bank to support the operations of the particular user (airline) that requires it. Chapter 4 of this document sets forth a standard format for encoding tailored route information, while Chapter 5 includes definitions of certain fields used exclusively for this purpose.

Individual airline files are used in step 3 of the airborne navigation system media production process. This may be performed either by the airline itself (as may step 4), or by an agency contracted to support the airline's navigation system operations, such as the airborne equipment manufacturer.

It can readily be seen that in the absence of air transport industry guidance, individual navigation system manufacturers could follow equipment design approaches that impose different requirements on the format of the navigation reference data. Although, as implied in paragraph 1.1 above, the airlines do not wish unnecessarily to constrain equipment design, the cost to them as an industry of supporting the production of files in several different formats would be prohibitively high. For this reason they have produced in this document data format and encoding standards to be applied in the

1.0 INTRODUCTION

production of these files. These standards are not intended to be used in the final two steps of file production, nor are any obligation imposed on anyone to make use of every data element defined. In this way manufacturers are free to optimize their hardware and software designs as they see fit, and reference data acquisition costs are minimized.

COMMENTARY

> In some cases in this document data fields are defined offering greater resolution than is usually available for the data in question from the source data bases. This is intended to reflect the airlines' desire for the use of the best available data. It is not, however, intended to suggest a need for special surveys in order to provide the data to the resolutions shown. Also, consideration of the application of the data base described in this document, with the aim of determining whether or not a standard earth model reference should be defined, produced the conclusion that such action was not necessary.

Readers should note that ARINC 424 Specification is not a data base specification per se. It is a standard for the preparation and transmission of data for assembly of airborne navigation system data bases.

1.3 Organization of this Document

A glossary of data processing and special navigation terms precedes the chapters of the document in which the recommended standards are defined. In the first of these chapters, the organization and content of the "master airlines user file" (see Attachment 1) is defined. The next chapter describes records in terms of their field structures. Following that, individual fields are defined in terms of the data elements from which they are constructed. Figure 1-1 pictorially relates these methods of information presentation to the layout of data on one of the individual airline files shown at the step 2 level in the diagram of Attachment 1. With the file structure definition complete, attention is turned to the encoding of data for computer processing.

1.3.1 Coverage of Helicopter Operation Needs

Supplement 14 of this document added material related to the special navigation database needs of rotorwing flight operations. The approach taken was to define as "dual use" as much of the database as possible, specifically the ground base navigation and landing aids. These records are defined as the Master Airline User File. Where dual use was not possible, new content was defined as the Master Helicopter User File. The new content was all related directly to the heliport and flight operations into and out of heliports. It included helicopter SIDs, STARs and Approach Procedures and Heliport Terminal Waypoint. Minor adjustments to the content of records that are dual usage were required; an example would be a new route type code for Enroute Airways dedicated to helicopter airways. All changes were made so as to have no impact on any other application of ARINC Specification 424, provided database suppliers avail themselves of the data selection capabilities built into that revision. **Supplement 19 of this document added material required to provide for Helicopter Company Routes.**

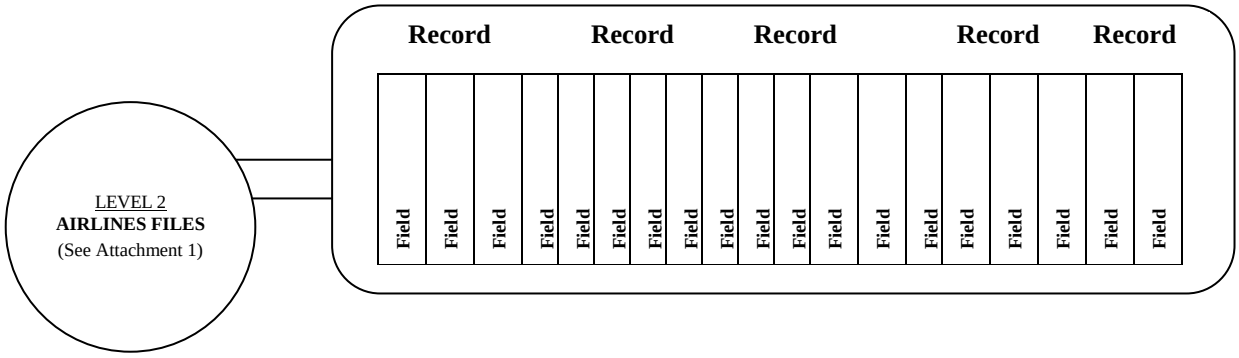
1.4 Reference Documentation

ARINC Characteristic 702: Flight Management Computer System

ARINC Characteristic 702A: Advanced Flight Management Computer System

ARINC Characteristic 756: GNSS Navigation and Landing Unit (GNLU)

1.0 INTRODUCTION



CHAPTER 3 **defines content and organization of the Master Airline User File (Attachment 1)**

CHAPTER 4 **defines locations for fields in RECORDS**

CHAPTER 5 **describes FIELDS**

CHAPTER 6 **defines data ENCODING STANDARDS**

Figure 1-1 - ARINC Specification 424 Information Presentation

2.0 GLOSSARY OF TERMS

2.1 Data Processing Terms

This Section contains definitions for the data processing terms used in this document. They are listed alphabetically.

Alpha	The terms employed to describe any letter of the alphabet (A through Z); any punctuation; or any printable character, other than a numeric, including space**.**
Character	The basic human-oriented data element, e.g., a single letter of the alphabet or a single number (0 through 9). The entry RW26L is said to consist of five characters.
Column	The spaces for data entry on each record. One column can accommodate one character.
Field	The collection of characters needed to define one item of information. The entry RW26L identifies "runway 26 left" and is described as a five-character field.
Numeric	The term employed to describe any single number in the range 0 through 9.
Record	A single line of computer data made up of the fields necessary to define fully a single useful piece of data. A VORTAC station record, for example, contains fields for station name, coordinates, frequency, elevation, variation, ICAO code, ident code, plus certain administrative data pertaining to the record itself.
Subsection	A collection of records of functional data items. The records for Approach routes form a subsection of the Airport data base.
Section	The first division of the database. Each section is made up of subsections as defined above.

2.2 Special Navigation Terms

This section contains definitions of certain special navigation-related terms used in this Specification. **This section is divided into four subsections: Special Data Terms, Procedure and Route Terms, Support Terms, and Precision RNAV Terms.**

2.2.1 Special Data Terms

ATC Compulsory Reporting Point	Essential or nonessential waypoints may be classified as ATC compulsory points. ATC requires the pilot to make a communications report at these waypoints. All other waypoints may be classified as non-compulsory reporting points and are reported only when specifically requested by ATC.
Essential Waypoints	An Essential Waypoint is defined as any waypoint at which a change in course is required or as the intersection of two or more airways.
Gateway Fix	A Gateway Fix is a waypoint associated with organized track systems across large areas which no ATS Routes have been established such as the Atlantic Ocean. It is coded into the database to indicate the point at which a change is made from ATS Route flying to random track flying.
Nonessential Waypoints	Nonessential Waypoints include all other waypoints of an airway not included under Essential Waypoints.
Off-Route Floating Waypoint	Waypoints which are not part of any route system but are designated by the ATC authority to be charted are considered to be Off-Route Floating Waypoints.
Phantom Waypoint	A database waypoint established during procedure coding to facilitate more accurate navigation by the Flight Management Computer than would be allowed using air-mass related Path Terminators to replicate source data. The waypoint finds use when such considerations as increased environmental restrictions and the congestion of the available airspace come into play. Used to permit route construction with "track to a fix" (TF) legs.
Transition Essential Waypoints	A waypoint which normally would be classified as non-essential might be required to transition from the enroute structure to the terminal structure. Waypoints falling into this category are classified as Transition Essential Waypoints.
Uncharted Airway Intersection	**A database waypoint established during airway coding that is not designated as part of that airway by government source. Used generally to establish intersections at route crossings and transition points for airways to terminal procedures that have not been provided by government source, but are necessary to provide the route or procedure in accordance with the rules of this Specification .**

2.0 GLOSSARY OF TERMS

2.2.2 Procedure and Route Terms

Approach Transition	**An Approach Transition is that series of sequences of a coded procedure that represent the government source data for Arrival Routes or Feeder Routes, generally used to define a path from a fix in the enroute environment to an Initial Approach Fix (IAF), as well as the series of procedure sequences from the IAF to the Intermediate Fix (IF or to the Final Approach Fix (FAF) when no Intermediate Segment is published.**
Final Approach Course Fix (FACF)	**The FACF is a waypoint located at the beginning of the Final Approach Coding. For Localizer-based Approach Procedures, the location of the FACF is on the localizer beam center, usually at a distance of 2 to 8 NM from the coded FAF. If the government source provides a named fix or an unnamed fix that is designated as the IF within this distance range or beyond this range, but within the published reception range of the localizer, it will be designated as the FACF. For non-localizer based procedures, if an FACF is coded, the FACF will be positioned 8 NM or less from the coded FAF. A government source provided named fix or unnamed fix designated, as the IF will be used. For localizer based procedures, if the government source does not include a named fix or unnamed fix designated as IF at a suitable location, the FACF will be a database waypoint created to meet the criteria established above.**
Final Approach Fix (FAF)	**The FAF is a waypoint located at a position that represents the Final Approach Fix or Final Approach Point defined in the government procedure source and or procedure design documentation. For non-precision approach procedures, including non-precision localizer based procedures, the waypoint will be located at the fix designated by the government source as the procedure FAF, when one is published. When none is published, the FAF waypoint will be established according to the rules defined in Attachment Five of this Specification. For precision localizer-based procedures, the coded FAF will be at the OM position, when an OM is present and located on the localizer beam center, or at the glide slope intercept point.**
Final End Point (FEP)	**The FEP is a waypoint located in the coded Final Approach Course (FAC). It is located at a point defined by the intersection of the FAC and a line perpendicular to that course through the runway threshold for procedures designed to straight-in criteria or threshold the first usable landing surface for circling only procedures. Rules governing when a FEP waypoint is coded are contained in Attachment Five of this Specification.**
Final Approach Coding	**The Final Approach Coding is that series of sequences of a coded procedure that represent the government source data for the Intermediate Approach Segment (IF to Final Approach Fix (FAF)), when appropriate source is available, as well as the Final Approach Segment (FAF to Missed Approach Point (MAP)).**
Final Approach Course (FAC)	**A straight line extension of a localizer course, a final approach radial/bearing or an extended runway centerline, all without regard to distance.**
Initial Approach Fix (IAF)	**An Initial Approach Fix is that fix designated by the government source documentations as the fix at which the Initial Approach segment begins. An Approach Procedure may have no IAF or multiple IAFs waypoints.**
Intermediate Fix (IF)	**An Intermediate Fix is that fix designated by government source documentation as the fix at which the Intermediate Approach Segment of the approach begins. An approach procedure can have no IF or one IF.**
Missed Approach Point (MAP)	**The MAP is the coded waypoint at which the Missed Approach Procedure begins. This waypoint can be a fix designated in government source documentation as the MAP or the Landing Threshold Point. The location of this point is determined by the rules in Attachment Five.**
Missed Approach Procedure	**The Missed Approach Procedure is that series of sequences of a coded procedure that represent the government source data for the Missed Approach Segment (MAP to Missed Approach Holding Fix).**
Precision Approach Procedures	**A Precision Approach Procedure is any procedure for which specific altitude and angle information with reference to an electronic glide slope is included in the coding of that procedure.**
Precision Final Approach Fix (PFAF)	**Used in MLS Procedure Coding. The PFAF is located at a point where the glide path intercepts the intermediate altitude. This point is the beginning of the MLS precision final approach segment.**

2.0 GLOSSARY OF TERMS

Terminal Procedure	**Collectively, all Standard Instrument Departures (SIDs), Standard Terminal Arrival Routes (STARs) and Instrument Approach Procedures (IAPs) coded to the standards in this Specification are referred to as Terminal Procedures.**
VNAV Path	**The Vertical Navigation or VNAV Path is the term used to identify the angular data provided on non-precision Final Approach Coding. Refer to Attachment Five, Section 8.9, of this Specification for information on how the VNAV Path data is provided.**

2.2.3 Support Terms

Enroute Airway (ER) to Restrictive Airspace (UR) Link	**The "ER" to "UR" Link indicates the physical affect of Enroute Airway to a Restrictive Airspace through which that Airway is designated, defined by the airway segment centerline. The Link is reflected in one or more dedicated Enroute Airway Continuation Records.**
Localizer	**Except where this term is specifically related to a particular type of approach procedure, the term Localizer is used as a general reference to all types of approach facilities that provide an electronic course guidance signal, including ILS, LOC, BC, IGS, LDA and SDF type signals.**
Mandatory Hold	**Any Flight maneuver of a holding nature defined in a terminal procedure where execution is part of the source defined flight path.**
Precision ARC	**A circular arc flight path between two known points, whose construction is tangent to the inbound and outbound paths to and from the known paths.**

2.2. **4** Precision RNAV Terms

COMMENTARY

The term RNAV-GPS/GLS is used to reference RNAV procedure formerly referenced as RNP.

Final Approach Flight Path	**This is the path that is defined laterally and vertically by the GPA and three precision approach path points that lie in a vertical plane coincident with the center of the geodetic reference ellipsoid. These three points are the LTP/FTP, FPAP and the FPCP.**
Final Approach Segment (FAS) Data Block	The FAS Data Block defines the lateral and vertical paths and associated criteria for the final approach segment of a SBAS (FAS) Data Block or GBAS Approach Procedure. Within an ARINC 424 output file, FAS Data Block data is carried in the Path Point Record and the two terms have been used interchangeably
Flight Path Alignment Point (FPAP)	**The FPAP is a point used to define the lateral alignment of the vertical plane containing the precision Final Approach Segment. For approach procedures that are aligned with the runway centerline, it is located at the designated center of the opposite runway threshold or an extension of a geodesic line calculated between the LTP and the designated center of the opposite runway landing threshold. It is positioned at a distance from the LTP to support a prescribed angular splay of lateral deviations. The FPAP is defined by latitude and longitude. The FPAP may be located beyond the opposite end of the landing runway, particularly on short runways.**
Flight Path Control Point (FPCP)	**The FPCP is a point above the LTP used to define the vertical component of the precision Final Approach Segment. It is in the vertical plane containing the LTP and the FPAP. Horizontally, the FPCP has the same latitude/longitude as the LTP. Vertically, the elevation of the FPCP is the LTP ellipsoidal elevation plus the threshold crossing height (TCH).**
Glide Path Angle (GPA)	**The GPA defines the descent angle of the precision final approach segment. It is defined relative to the horizontal plane, tangent to the WGS-84 ellipsoid at the LTP.**
Landing Threshold Point/Fictitious Threshold Point (LTP/FTP)	**The LTP is a point at the designated center of the landing runway threshold, defined by latitude, longitude, and height above the WGS-84 reference ellipsoid. The LTP is used in conjunction with the Flight Path Alignment Point (FPAP) to determine the lateral alignment of the vertical plane containing the precision Final Approach Segment. The FTP is the name applied to the LTP when that point is offset from the actual runway surface. When used in calculation and databases, it will generally be shown/referred as LTP/FTP.**

2.0 GLOSSARY OF TERMS

Level of Service (LPV, LNAV/VNAV, and LNAV and RNP)	The terms LPV, LNAV/VNAV, and LNAV as used in this document starting with Supplement 18, the terminology developed was by the FAA to denote operating criteria for RNAV procedures. Within this document, these terms are used strictly with regard to operations based on SBAS, although LNAV/VNAV and LNAV operations are often authorized both with and without SBAS for the same procedure. Other government authorities may use other terms to define these criteria. The use of this terminology starting with Supplement 18 of this Specification does not rule out using these terms for that source, as long as the intention of the government source identical to that of the FAA. Should other terms be developed that do not have the identical intent, they will be added to the appropriate portions of this Specification.

3.0 NAVIGATION DATA

3.1 User File Organization

The records defined in Chapter 4 of this document are sorted such that they appear on the master file in alphabetical/numerical order by column. The sorting necessary to achieve this process is as follows. Records are first divided into "standard" and "tailored" groups by the content of the first column. "Standard" or "S" records are located on the file ahead of the "tailored" or "T" records. The next columns order the "standard" records alphabetically by AREA Code and "tailored" records by Airline Code. After that the column content orders both "standard" and "tailored" records by sections. This process is illustrated in Figure 3-1. Sorting continues this way, column by column, until each record is uniquely defined.

The column number at which this occurs for each record type may be determined by inspecting the record layout forms of Figure 4-1 of this document. The master file may then be assembled with records located in the positions thus defined.

While the sorting process is basically alphabetical, it has to accommodate columns that are permitted to contain blanks or numeric characters. When this occurs, blank characters will be sorted before numerics and numeric characters will be sorted before alphabetic characters.

3.2 Master Airline User File Content

3.2.1 General

This section of this document defines the content of each section of the Master Airline User File. As indicated in Section 1.2 of this document, this file can be composed of the "standard" records or "standard" and "tailored" records, sorted according to the procedure set forth in Section 3.1 above.

The Master Airline User File includes all records listed in Section 3.2.

3.2.2 Navaid Section (D)

3.2.2.1 VHF Navaid Section (D), Subsection (Blank)

The VHF NAVAID Subsection should contain all the VORs, VORDMEs, VORTACs, DMEs, ILS DMEs and MLS DMEs as well as all TACANs paired with civil-use VHF NAVAID frequencies. It may also contain TACANs paired with military-use VHF frequencies for specific applications. As a minimum, all VHF NAVAIDs referenced by records in Sections 3.2.3.3, 3.2.3.4, 3.2.4.1, 3.2.4.4, 3.2.4.5, 3.2.4.6, 3.2.4.11, 3.2.4.12, 3.2.5, 3.2.10, 3.3.5, 3.3.6, 3.3.7 and 3.3.8 should be available in the VHF NAVAID Subsection. ILS DMEs and MLS DMEs included can be for either Airports or Heliports.

3.2.2.2 NDB Navaid Section (D), Subsection (B)

The NDB NAVAID Subsection file should contain all LF and MF NDBs and selected Marine Beacons defined in the enroute structure. As a minimum, all Enroute NDB NAVAIDs referenced by records in Sections 3.2.3.3, 3.2.3.4, 3,2,3.6, 3.2.4.4, 3.2.4.5, 3.2.4.6, 3.2.4.11, 3.2.4.12, 3.2.5, 3.2.10, 3.3.5, 3.3.6, 3.3.7 and 3.3.8 should be available in the NDB NAVAID Subsection.

3.2.3 Enroute Section

3.2.3.1 Enroute Waypoint Section (E), Subsection (A)

The Enroute Waypoint Subsection file should contain all named intersections defined in the enroute structure. The file will also contain those VFR waypoints not associated with Airports or Heliports. As a minimum, all enroute waypoints referenced in Sections 3.2.3.3, 3.2.3.4, 3.2.4.4, 3.2.4.5, 3.2.4.6, 3.2.5, 3.2.10, 3.3.5, 3.3.6, and 3.3.7 should be available in the Enroute Waypoint Subsection.

3.2.3.2 Enroute Airway Marker Section (E), Subsection (M)

The Enroute Airway Markers Subsection file should contain all government-published airways marker facilities.

3.2.3.3 Holding Patterns (E), Subsection (P)

The Holding Pattern Subsection file should contain all holding patterns shown on aeronautical charts.

3.2.3.4 Enroute Airways Section (E), Subsection (R)

The Enroute Airways Subsection file should contain all government-designated airways.

3.2.3.5 Enroute Airways Restrictions Section (E), Subsection (U)

The Enroute Airways Restrictions Subsection file contains the official altitude, time and usage restrictions for Enroute Airways referenced in Section 3.2.3.4.

3.2.3.6 Enroute Communications Section (E), Subsection (V)

The Enroute Communications Subsection file should contain all government-published enroute communications facilities.

3.2.4 Airport Section (P)

3.2.4.1 Airport Reference Points Section (P), Subsection (A)

The Airport Reference Points subsection file should contain reference points for all airports having at least one hard surfaced runway. As a minimum, all airport reference points referenced in Sections 3.2.3.4, 3.2.3.5 and 3.2.5 should be included**.**

3.2.4.2 Airport Gates Section (P), Subsection (B)

The Airport Gates Subsection should contain all gates published in official government documents to support the Airport referenced in Section 3.2.4. If the Airport is provided as "standard" data, the Gates may be provided as "standard" or "tailored" data, depending on whether the gate owner and operator is the public sector or a specific airline. If the Airport is provided as "tailored" data, the Gates must also be provided as "tailored" data.

3.0 NAVIGATION DATA

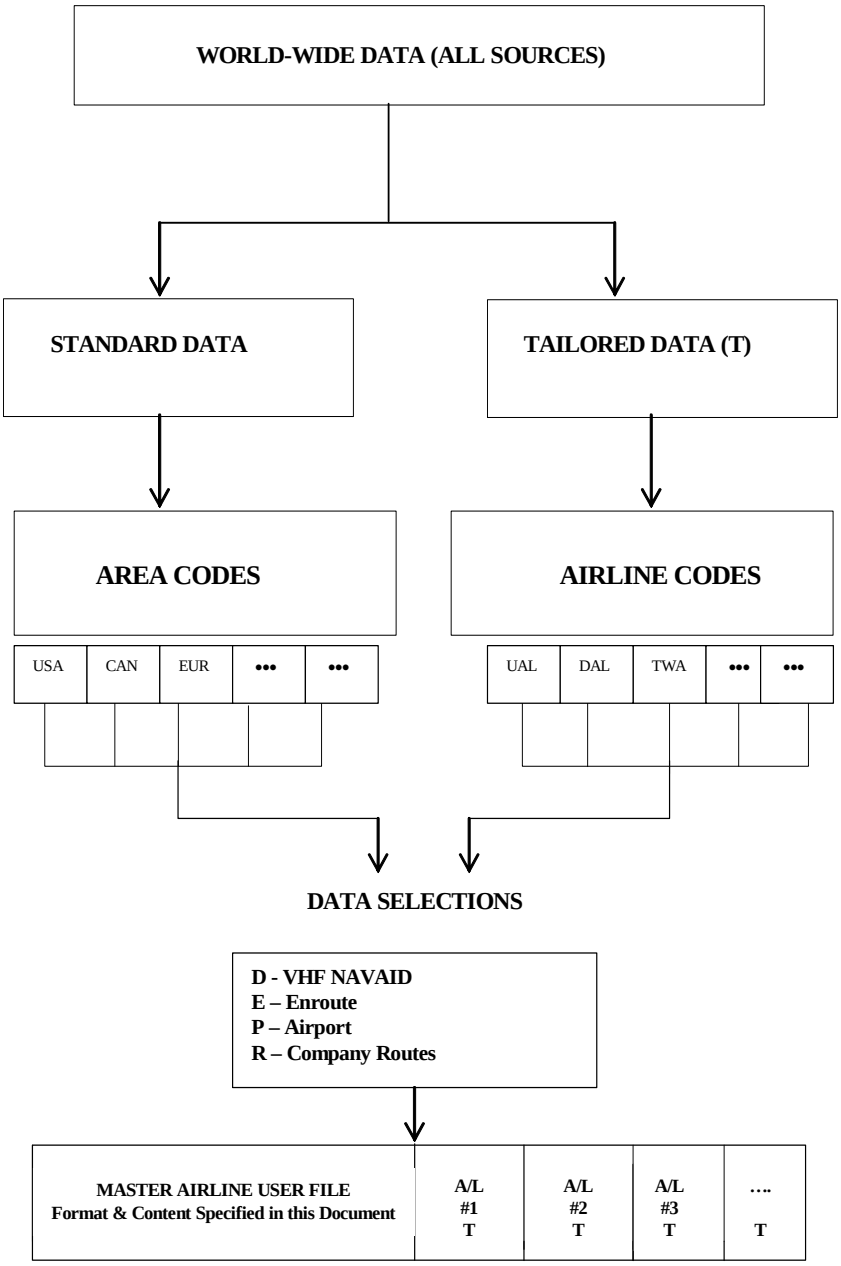


Figure 3-1 – Data Sorting Necessary to Achieve Step 1 of FDSU File Production Process

3.2.4.3 Airport Terminal Waypoints Section (P), Subsection (C)

The Terminal Waypoints Subsection file should contain those waypoints necessary to support Standard Instrument Departures (SIDs), Standard Terminal Arrival Routes (STARs) and Approaches specified in Sections 3.2.4.4, 3.2.4.5 and 3.2.4.6, excluding the landing threshold as a fix. The file will also contain those VFR waypoints associated with Airports. If a waypoint is used in both the terminal and enroute areas, it should appear in the Enroute (EA) file.

3.2.4.4 Airport Standard Instrument Departures (SIDs) Section (P), Subsection (D)

The SIDs Subsection file should contain all government published SIDs from airports referenced in Section 3.2.4.1.

3.2.4.5 Airport Standard Terminal Arrival Routes (STARs) Section (P), Subsection (E)

The STARs Subsection file should contain all government published STARs to the airports referenced in Section 3.2.4.1.

3.2.4.6 Airport Approaches Section (P), Subsection (F)

The Approach Route Subsection file should contain at least one instrument approach, if published, for each runway to the airports referenced in Section 3.2.4.1 except Radar Approaches. Approach Procedures types have been identified and are covered by coding rules elsewhere in this specification.

This specification originally subscribed to an approach procedures coding system known as the Multiple Approach Coding Concept. The concept is defined as one approach procedure for a given reference facility to a given single runway. For example, an ILS based and a VOR based procedure to the same runway may be included but not an ILS and an ILS Localizer only or a VORDME and a VOR only to the same runway. Through several Supplements to this specification, modifications to this concept have been incorporated and it is now possible to have multiples of the same reference facility or to address reference facilities in a more specific manner. For details see Chapter Five, Sections 5.7 and 5.10. Data Suppliers are requested to supply to either the original or to the expanded concept.

3.2.4.7 Airport Runway Section (P), Subsection (G)

The Runway Subsection file should contain all runways referenced in Sections 3.2.4.4, 3.2.4.5, and 3.2.4.6.

3.2.4.8 Airport and Heliport Localizer/Glide Slope Section (P), Subsection (I)

The Localizer/Glide Slope Subsection file should contain all government published localizer facilities to airport runways and/or helipad referenced in Section 3.2.4.7 or 3.3.3. As a minimum, the section should contain all localizer facilities referenced in Sections 3.2.4.6 and 3.3.7.

3.2.4.9 Airport and Heliport MLS Section (P), Subsection (L)

The MLS Subsection file should contain all government published MLS facilities for airport runways and/or helipads referenced in Section 3.2.4.7 or 3.3.3.

3.2.4.10 Airport and Heliport Marker/Localizer Section (P), Subsection (M)

The Airport and Heliport Localizer Marker Subsection file should contain all government published Markers and locators associated with the localizers referenced in Section 3.2.4.8. As a minimum, this Subsection should contain all markers referenced in Sections 3.2.4.6 and 3.3.7.

3.2.4.11 MSA Section (P), Subsection (S)

The MSA (Minimum Sector Altitude) Subsection should contain the Sector Altitude for all government published SIDs referenced in Section 3.2.4.4, published STARs referenced in Section 3.2.4.5 and approach procedures referenced in Section 3.2.4.6.

3.2.4.12 Airport Communications Section (P), Subsection (V)

The Airport Communications Subsection file should contain all government-published airport communications facilities for airports referenced in Section 3.2.4.1.

3.2.4.13 Airport and Heliport Terminal NDB Section (P), Subsection (N)

The Terminal NDB Subsection file should contain those Terminal NDB NAVAIDS referenced by records in Sections 3.2.3.3, 3.2.4.4, 3.2.4.5, 3.2.4.6, 3.2.4.11, 3.2.5, 3.2.9, 3.3.5, 3.3.6, 3.3.7, and 3.3.8. If a NDB is used in both the terminal and enroute environments, it should appear in the Enroute NDB NAVAID (DB) file.

3.2.4.14 Airport and Heliport Path Point Section (P), Subsection (P)

The Path Point Subsection file should contain the Path Point records required to support all RNAV-GPS/GLS Approach Procedures referenced in Sections 3.2.4.6 and 3.3.7.

Note: The pathpoint concept is currently being developed by the FAA in cooperation with industry and other governments. Most of the items in the pathpoint record have been finalized; however, many of the fields have not had final resolution and therefore care should be taken before creating software for implementation. The pathpoint record continues to be included in ARINC Specification 424 as a concept that will have changes as the concept is matured.

3.2.4.15 Flight Planning Arrival/Departure Data Record Section (P), Subsection (R)

The Flight Planning Arrival/Departure Data Subsection should contain a set of data that meet the needs of computerized flight planning for Arrival and Departure designations, transitions and distances for airports referenced in Section 3.2.4.1.

3.2.4.16 GNSS Landing System (GLS) Section (P), Subsection (T)

The GLS Subsection file should contain all those government-published GNSS Landing System approaches for airport runways and/or helipads referenced in Sections 3.2.4.7 and 3.3.3. As a minimum, the section should contain all GLS approaches referenced in Sections 3.2.4.6 and 3.3.7.

3.2.4.17 Airport Terminal Arrival Altitude Section (P), Subsection (K)

The Airport TAA (Terminal Arrival Altitude) Subsection should contain the Sector Bearings, Sector Radii, and Sector Altitudes for all government published Approach Procedures referenced in Section 3.2.4.6.

3.2.5 Company Route and Alternate Destination Section (R)

This section supports Company Route information for the Master Airline User File. The Company Route information is available only as tailored data records.

3.0 NAVIGATION DATA

3.2.5.1 Company Route Section (R), Subsection (Blank)

This section supports Company Route information for the Master Airline User File. The Company Route information is available only as tailored data records.

3.2.5.2 The Alternate Record Section (R), Subsection (A)

The Alternate Record Section defines departure, destination or enroute alternate airports or alternate company routes. The data is only available as tailored data records. **This section supports Company Route information for the Master Airline User File.**

3.2.6 Special Use Airspace Section (U)

3.2.6.1 Restrictive Airspace Section (U), Subsection (R)

The Restrictive Airspace Subsection should contain all government published restrictive airspace areas containing their lateral and vertical limits.

3.2.6.2 FIR/UIR Section (U), Subsection (F)

The FIR/UIR Subsection file should contain all government-published FIR and UIR boundaries, including both lateral and vertical limits.

3.2.6.3 Controlled Airspace Section (U), Subsection (C)

The Controlled Airspace Subsection file should contain those government-published airspaces required to support the specific needs of this Specification, see Chapter Five, Section 5.217, as they relate to Airports and Heliports, including their lateral and vertical limits.

3.2.7 Cruising Tables Section (T)

3.2.7.1 Cruising Tables Section (T), Subsection (C)

The Cruising Table Subsection file should contain the standard ICAO Cruising Level Table and all modified Cruising Level Tables required to support Sections 3.2.3.4 and 3.2.6.2.

3.2.7.2 Geographical Reference Table Section (T), Subsection (G)

The Geographical Reference Table Subsection file should contain all geographical cross reference entries required to create linkage to Preferred Route Identifiers, Section 3.2.9, for wide area origin or destination entries.

3.2.8 MORA Section (A), Subsection (S)

The MORA Subsection should contain all grid MORA values for each degree of latitude and longitude.

3.2.9 Preferred Routes Section (E), Subsection (T)

The Preferred Route Subsection file will contain frequently used routes (i.e., North American Preferred Routes, North American Routes to the North Atlantic Traffic, and Europe Preferential Route System). These routes will, in effect, combine existing Subsection files [SID (PD), STAR (PE), Enroute Airway (ER), Enroute Waypoint (EA), Terminal Waypoint (PC, VHF NAVAID (D), NDB NAVAID (DB), Airport (P)] to form a continuous route structure. This route structure may be referenced by the Company Route records.

3.3 Master Helicopter User File Content

3.3.1 General

The Master Helicopter User File will incorporate the use of records from Section 3.2, Master Airline User File as well as sections unique to helicopter operations.

3.3.2 Jointly and Specifically Used Sections/Subsections

Section 3.3, Master Helicopter User File will jointly use the following sections from Section 3.2, Master Airline User File:

3.2.2 VHF Navaid Section
3.2.3 Enroute Section
3.2.4 Airport Section, but limited to Airports with Helipads **and Airports with published helicopter procedures from/to runways.**
3.2.4.7 Airport Runways Section, but limited to Airports with published helicopter procedures from runways.
3.2.4.8 Airport and Heliport Localizer/Glide Slope Section
3.2.4.9 Airport and Heliport MLS Section
3.2.4.10 Airport and Heliport Localizer Marker Section
3.2.4.13 Airport and Heliport Terminal NDB Section
3.2.5 Company Route Section
3.2.6 Special Use Airspace Section
3.2.7 Tables Section
3.2.8 MORA Section
3.2.9 Preferred Route Section
3.2.11 GLS

Section 3.3, Master Helicopter User File will include the following specifically used paragraphs:

3.3.3 Heliport Section (H), Subsection (A)
3.3.4 Heliport Terminal Waypoint Section (H), Subsection (C)

Heliport Terminal Procedure Section (H)

3.3.5 SID Subsection (D)
3.3.6 STAR Subsection (E)
3.3.7 Approach Subsection (F)
3.3.8 Heliport MSA Section (H), Subsection (S)
3.3.9 Heliport Communications Section (H), Subsection (V)

3.3.3 Heliport Section (H), Subsection (A)

The Heliport Subsection file should contain reference points for all government-published helipads at heliport and airport facilities.

3.3.4 Heliport Terminal Waypoints Section (H), Subsection (C)

The Heliport Terminal Waypoint Subsection should contain those waypoints necessary to support Standard Terminal Departures (SIDs), Standard Terminal Arrival Routes (STARs) and Approaches specified in Sections

3.3.5, 3.3.6, and 3.3.7, excluding Helipads as a fix. The file will also contain those VFR waypoints associated with Heliports. If a waypoint is used in both the terminal area and the enroute areas, is should appear in the Enroute (EA) file.

3.3.5 Heliport Standard Instrument Departures (SIDs) Section (H), Subsection (D)

The SIDs Subsection file should contain all government published SIDs from Heliports referenced in Section 3.3.3.

3.3.6 Heliport Standard Terminal Arrival Routes (STARs) Section (H), Subsection (E)

The STARs Subsection file should contain all government published STARs to Heliports referenced in Section 3.3.3.

3.3.7 Heliport Approaches Section (H), Subsection (F)

The Approach Route Subsection file should contain all government published approaches to Heliports referenced in Section 3.3.3. Approach procedure types have been identified and are covered by coding rules elsewhere in this specification.

3.3.8 Heliport MSA Section (H), Subsection (S)

The MSA (Minimum Sector Altitude) Subsection should contain the Sector Altitude for all government published SIDS referenced in Section 3.3.5, published STARs referenced in Section 3.3.6 and approach procedures referenced in Section 3.3.7.

3.3.9 Heliport Communications Section (H), Subsection (V)

The Heliport Communications Subsection file should contain all government published heliport communications facilities for heliports referenced in Section 3.3.3.

3.3.10 Helicopter Terminal Arrival Altitude Section (H), Subsection (K)

The Heliport TAA (Terminal Arrival Altitude) Subsection should contain the Sector Bearings, Sector Radii, and Sector Altitudes for all government published Approach Procedures referenced in Section 3.3.7.

3.3.11 Helicopter Operations Company Route Section (R), Subsection (H)

This section supports Company Route information for the Master Helicopter User File. The Helicopter Operations Company Route Subsection should contain any helicopter operations specific Company Route including operations from Heliport, Helipads at Airports and operations from Runways at Airports. Helicopter Company Route information is available only as tailored data records.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.0.1 General

In an effort to describe the Master Airline and Master Helicopter sections, Section 4 is divided into Section 4.1 Navigation Data - Record Layout, Master Airline User Content and Section 4.2 Navigation Data - Record Layout, Master Helicopter User Content.

Each record is made up of combinations of the fields described in Chapter 5 of this document. This chapter sets forth the standard layout of each type of record found in the data base. These layouts are also presented diagrammatically at the end of this section. Paragraphs and Tables in the 4.1 series are the record types, which have been identified as being a part of Master Airline User Content. Paragraphs and Tables in the 4.2 series are the record types, which have been identified as being part of the Master Helicopter User Content. This paragraph and table numbering system does not prevent any given data base from including any of the records defined in this document. The separation is for editorial and reference purposes only.

Each record contains 132 character positions or columns. Not all of these are used in every record. Some are left blank to permit like information to appear in the same columns of different records and others are reserved for the possible future expansion of the record's content. In the tables that follow, the former are identified by the term "Blank (Spacing)" under the "Field" heading. The latter are identified by the term "Reserved," followed by the function for which the reservation is made (where it can specifically be stated).

The tables show the record columns occupied by each field. For convenience, the number of characters in each field is shown in brackets following the field name. Also, the paragraph numbers in Chapter 5 of this document wherein individual fields are defined are referenced. Each table appears under a paragraph heading that is followed by the data base section and subsection codes employed in the record described.

4.1 Master Airline User File

4.1.2 VHF NAVAID Record (D)

The VHF NAVAID file contains details of all VOR, VOR/DME, VORTAC, DME and TACAN stations within the geographical area of interest. For VOR and TACAN stations having the same identifier but different operating frequencies, the TACAN is available and the VOR is suppressed unless the VOR is required to support Sections 3.2.3.3, 3.2.3.4, 3.2.4.4, 3.2.4.5, 3.2.4.6, 3.2.4.11, 3.2.5, 3.2.9, 3.3.5, 3.3.6, 3.3.7 or 3.3.8. In such cases the VOR is available and the TACAN is suppressed.

4.1.2.1 VHF NAVAID Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 10	Airport ICAO Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Blank (Spacing) (1)	
14 thru 17	VOR Identifier (4)	5.33
18 thru 19	Blank (Spacing) (2)	
20 thru 21	ICAO Code (2)	5.14
22	Continuation Record No. (1)	5.16
23 thru 27	VOR Frequency (5)	5.34
28 thru 32	NAVAID Class (5)	5.35
33 thru 41	VOR Latitude (9)	5.36
42 thru 51	VOR Longitude (10)	5.37
52 thru 55	DME Ident (4)	5.38
56 thru 64	DME Latitude (9)	5.36
65 thru 74	DME Longitude (10)	5.37
75 thru 79	Station Declination (5)	5.66
80 thru 84	DME Elevation (5)	5.40
85	Figure of Merit (1)	5.149
86 thru 87	ILS/DME Bias (2)	5.90
88 thru 90	Frequency Protection (3)	5.150
91 thru 93	Datum Code (3)	5.197
94 thru **118**	VOR Name (*25*)	5.71
119 thru 121	**Blank (Spacing) (3)**	
122	**Route Inappropriate DME (1)**	**5.297**
123	**DME Operational Service Volume (1)**	**5.277**
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.2.2 VHF NAVAID Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 92	Notes (69)	5.61
93 thru 123	Reserved (Expansion) (31)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.2.3 VHF NAVAID Simulation Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 27	Blank (Spacing) (4)	
28 thru 32	Facility Characteristics (5)	5.93
33 thru 74	Reserved (Spacing) (42)	
75 thru 79	Magnetic Variation (5)	5.39
80 thru 84	Facility Elevation (5)	5.92
85 thru 123	Reserved (Expansion) (39)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.2.4 VHF NAVAID Flight Planning Continuation Records

This Continuation Record is used to indicate the FIR and UIR within which the VHF NAVAID defined in the Primary Record is located and the Start/End validity dates/times of the Primary Record.

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 27	FIR Identifier (4)	5.116
28 thru 31	UIR Identifier (4)	5.116
32	Start/End Indicator (1)	5.152
33 thru 43	**Blank (Spacing) (11)**	
44 thru 123	Reserved (Expansion) (80)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.2.5 VHF NAVAID Flight Planning Continuation Records

Deleted by Supplement 19.

4.1.2.6 VHF NAVAID Limitation Continuation Record

This Continuation Record is used to provide details on signal limitations of the VHF Navaid contained in the Primary Record Section 4.1.2.1. Note that multiple records formatted as in Section 4.1.2.6 may be included for a single Primary Record. As “Service Volume” or “Designated Operational Coverage” may also be considered limitations, this information is also provided for each navaid listed in the Primary Records, where such information is available.

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24	Navaid Limitation Code (1)	5.205
25	Component Affected Indicator (1)	5.206
26 thru 27	Sequence Number (2)	5.12
28 thru 29	Sector From/Sector To (2)	5.207
30	Distance Description (1)	5.187
31 thru 36	Distance Limitation (6)	5.208
37	Altitude Description (1)	5.29
38 thru 43	Altitude Limitation (6)	5.209
44 thru 45	Sector From/Sector To (2)	5.207
46	Distance Description (1)	5.187
47 thru 52	Distance Limitation (6)	5.208
53	Altitude Description (1)	5.29
54 thru 59	Altitude Limitation (6)	5.209
60 thru 61	Sector From/Sector To (2)	5.207
62	Distance Description (1)	5.187
63 thru 68	Distance Limitation (6)	5.208
69	Altitude Description (1)	5.29
70 thru 75	Altitude Limitation (6)	5.209
76 thru 77	Sector From/Sector To (2)	5.207
78	Distance Description (1)	5.187
79 thru 84	Distance Limitation (6)	5.208
85	Altitude Description (1)	5.29
86 thru 91	Altitude Limitation (6)	5.209
92 thru 93	Sector From/Sector To (2)	5.207
94	Distance Description (1)	5.187
95 thru 100	Distance Limitation (6)	5.208
101	Altitude Description (1)	5.29
102 thru 107	Altitude Limitation (6)	5.209
108	Sequence End Indicator (1)	5.210
109 thru 123	Blank (Spacing) (15)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.3 NDB NAVAID Record (DB or PN)

The Enroute NDB NAVAID file (DB) contains all enroute on-airway and off-airway NDBs within the geographical area of interest. The Terminal NDB NAVAID file (PN) contains NDBs associated with the Airports contained in Subsection 3.2.4.1 and Heliport contained in Section 3.3.3. Terminal NDBs referenced to two or more Airports or Heliports will be available in the Enroute NDB Subsection unless that handling would create duplicate NDB identifiers within that Subsection. Marine Beacons shown on aeronautical charts may also be included in this record type.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.3.1 NDB NAVAID Primary Records

Columns	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 10	Airport ICAO Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Blank (Spacing) (1)	
14 thru 17	NDB Identifier (4)	5.33
18 thru 19	Blank (Spacing) (2)	
20 thru 21	ICAO Code (2)	5.14
22	Continuation Record No. (1)	5.16
23 thru 27	NDB Frequency (5)	5.34
28 thru 32	NDB Class (5)	5.35
33 thru 41	NDB Latitude (9)	5.36
42 thru 51	NDB Longitude (10)	5.37
52 thru 74	Blank (Spacing) (23)	
75 thru 79	Magnetic Variation (5)	5.39
80 thru 85	Blank (Spacing) (6)	
86 thru 90	Reserved (Expansion) (5)	
91 thru 93	Datum Code (3)	5.197
94 thru 123	NDB Name (30)	5.71
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Data (4)	5.32

4.1.3.2 NDB NAVAID Continuation Records

Columns	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 92	Notes (69)	5.61
93 thru 123	Reserved (Expansion) (31)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Data (4)	5.32

4.1.3.3 NDB NAVAID Simulation Continuation Record

Columns	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 27	Blank (Spacing) (4)	
28 thru 32	Facility Characteristics (5)	5.93
33 thru 79	Reserved (Spacing) (47)	
80 thru 84	Facility Elevation (5)	5.92
85 thru 123	Reserved (Expansion) (39)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Data (4)	5.32

4.1.3.4 NDB NAVAID Flight Planning Continuation Records

This Continuation Record is used to indicate the FIR and UIR within which the NDB NAVAID defined in the Primary Record is located and the Start/End validity dates/times of the Primary Record.

Columns	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 27	FIR Identifier (4)	5.116
28 thru 31	UIR Identifier (4)	5.116
32	Start/End Indicator (1)	5.152
33 thru 43	**Blank (Spacing) (11)**	
44 thru 123	Reserved (Expansion) (80)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.3.5 NDB NAVAID Flight Planning Continuation Records

Deleted by Supplement 19.

4.1.4 Waypoint Record (EA) or (PC)

The Enroute Waypoint file (EA) contains all enroute on-airway and off-airway waypoints within a desired geographical area. The Airport Terminal Waypoint file (PC) contains all terminal waypoints and VFR waypoints within the geographical area of each airport. Airport Terminal Waypoints utilized by two or more airports will be stored in the Enroute Waypoint Subsection (EA) to eliminate duplication. Terminal Waypoints used jointly by an airport and a heliport are also stored in the Enroute Waypoint file. The Enroute Waypoint File will contain waypoints established for Helicopter Airways. For Heliport Terminal Waypoints (HC) see Section 4.2.2.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.4.1 Waypoint Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5 Note 1
7 thru 10	Region Code (4)	5.41 Note 2
11 thru 12	ICAO Code (2)	5.14
13	Subsection (1)	5.5 Note 1
14 thru 18	Waypoint Identifier (5)	5.13
19	Blank (Spacing) (1)	
20 thru 21	ICAO Code (2)	5.14
22	Continuation Record No. (1)	5.16
23 thru 26	Blank (Spacing) (4)	
27 thru 29	Waypoint Type (3)	5.42
30 thru 31	Waypoint Usage (2)	5.82
32	Blank (Spacing) (1)	
33 thru 41	Waypoint Latitude (9)	5.36
42 thru 51	Waypoint Longitude (10)	5.37
52 thru 74	Blank (Spacing) (23)	
75 thru 79	Dynamic Mag. Variation (5)	5.39
80 thru 84	Reserved (Expansion) (5)	
85 thru 87	Datum Code (3)	5.197
88 thru 95	Reserved (Expansion) (8)	
96 thru 98	Name Format Indicator (3)	5.196
99 thru 123	Waypoint Name/ Description (25)	5.43
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: In Enroute Waypoint Records, the Subsection Code occupies column 6, with column 13 blank. In Airport or Heliport Terminal Waypoint Records, the Subsection Code occupies column 13, with column 6 blank.

Note 2: In Enroute Waypoint Records the code “ENRT” is used. In Terminal Waypoint records, the region code field contains the Airport ICAO Identification code.

4.1.4.2 Waypoint Continuation Records

Columns	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 92	Notes (69)	5.61
93 thru 123	Reserved (Expansion) (31)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Data (4)	5.32

4.1.4.3 Waypoint Flight Planning Continuation Records

This Continuation Record is used to indicate the FIR and UIR within which the Waypoint defined in the Primary Record is located and the Start/End validity dates/times of the Primary Record.

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 27	FIR Identifier (4)	5.116
28 thru 31	UIR Identifier (4)	5.116
32	Start/End Indicator (1)	5.152
33 thru 43	**Blank (Spacing) (11)**	
44 thru 123	Reserved (Expansion) (80)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.4.4 Waypoint Flight Planning Continuation Records

Deleted by Supplement 19.

4.1.5 Holding Pattern Records (EP)

The Enroute Holding Patterns contained in this file are holding patterns recommended by the official government authority for inclusion on enroute aeronautical charts. The Terminal Holding Patterns included in this file are holding patterns recommended for aeronautical charts for the geographical area of an airport or heliport. The type, Enroute or Terminal, will be determined by the Subsection of the fix upon which the holding is predicated.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.5.1 Holding Pattern Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 10	Region Code (4)	5.41 Note 1
11 thru 12	ICAO Code (2)	5.14 Note 1
13 thru 27	Blank (Spacing) (15)	
28 thru 29	Duplicate Identifier (2)	5.114
30 thru 34	Fix Identifier (5)	5.13
35 thru 36	ICAO Code (2)	5.14
37	Section Code (1)	5.4
38	Subsection Code (1)	5.5
39	Continuation Record No. (1)	5.16
40 thru 43	Inbound Holding Course (4)	5.62
44	Turn Direction (1)	5.63
45 thru 47	Leg Length (3)	5.64
48 thru 49	Leg Time (2)	5.65
50 thru 54	Minimum Altitude (5)	5.30
55 thru 59	Maximum Altitude (5)	5.127
60 thru 62	Holding Speed (3)	5.175
63 thru 65	RNP (3)	5.211
66 thru 71	Arc Radius (6)	5.204
72 thru 74	**Vertical Scale Factor (3)**	**5.293**
75 thru 77	**RVSM Minimum Level (3)**	**5.294**
78 thru 80	**RVSM Maximum Level (3)**	**5.295**
81 thru 98	Reserved (Expansion) (**18**)	
99 thru 123	Name (25)	5.60
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note1: In Enroute Fix Holding Pattern records, the code of “ENRT” is used in the Region Code field and the ICAO Code field is blank. In Terminal Fix Holding Records, the Region Code field contains the identifier of the Airport or Heliport with which the holding is associated. The ICAO Code field will not be blank. This information will uniquely identify the Terminal NDB, Airport Terminal Waypoint or Heliport Terminal Waypoint.

4.1.5.2 Holding Pattern Continuation Records

Column	Field Name (Length)	Reference
1 thru 38	Fields as on Primary Records	
39	Continuation Record No. (1)	5.16
40	Application Type (1)	5.91
41 thru 109	Notes (69)	5.61
110 thru 123	Reserved (Expansion) (14)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.6 Enroute Airways Records (ER)

The Enroute Airways file will contain the sequential listing of officially published airways and other established ATS Routes by geographical areas. The file also contains published airways specific to helicopter operations.

4.1.6.1 Enroute Airways Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 13	Blank (Spacing) (7)	
14 thru 18	Route Identifier (5)	5.8
19	Reserved (1)	Note 1
20 thru 25	Blank (Spacing) (6)	
26 thru 29	Sequence Number (4)	5.12
30 thru 34	Fix Identifier (5)	5.13
35 thru 36	ICAO Code (2)	5.14
37	Section Code (1)	5.4
38	Subsection (1)	5.5
39	Continuation Record No. (1)	5.16
40 thru 43	Waypoint Description Code (4)	5.17
44	Boundary Code (1)	5.18
45	Route Type (1)	5.7
46	Level (1)	5.19
47	Direction Restriction (1)	5.115
48 thru 49	Cruise Table Indicator (2)	5.134
50	EU Indicator (1)	5.164
51 thru 54	Recommended NAVAID (4)	5.23
55 thru 56	ICAO Code (2)	5.14
57 thru 59	RNP (3)	5.211
60 thru 62	Blank (Spacing) (3)	
63 thru 66	Theta (4)	5.24
67 thru 70	Rho (4)	5.25
71 thru 74	Outbound Magnetic Course (4)	5.26
75 thru 78	Route Distance From (4)	5.27
79 thru 82	Inbound Magnetic Course (4)	5.28
83	Blank (Spacing) (1)	
84 thru 88	Minimum Altitude (5)	5.30
89 thru 93	Minimum Altitude (5)	5.30
94 thru 98	Maximum Altitude (5)	5.127
99 thru 101	Fix Radius Transition Indicator (3)	5.254
102 thru 104	**Vertical Scale Factor (3)**	**5.293**
105 thru 107	**RVSM Minimum Level (3)**	**5.294**
108 thru 110	**VSF RVSM Maximum Level (3)**	**5.295**
111 thru 114	**Reserved (4)**	
115 thru 116	**ICAO Code (2)**	**5.14**
117 thru 120	**Reserved (4)**	
121 thru 123	**Blank (Spacing) (3)**	
124 thru 128	File Record No (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: The standard length for the Route Identifier is five characters. Some users envisage the need for a six-character field. This reserved column will permit this usage. Some data suppliers may use this position for the ATS Service suffix associated with some Route Identifiers.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.6.2 Enroute Airways Continuation Records

Column	Field Name (Length)	Reference
1 thru 38	Fields as on Primary Records	
39	Continuation Record No. (1)	5.16
40	Application Type (1)	5.91
41 thru 109	Notes (69)	5.61
110 thru 123	Reserved (Expansion) (14)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.6.3 Enroute Airways Flight Planning Continuation Records

This Continuation Record is used to indicate the Start/End validity times of the Primary Record, and to indicate restrictive airspace that affects the Primary Record according to the definition given in Section 2.0, Glossary of Terms.

Column	Field Name (Length)	Reference
1 thru 38	Fields as on Primary Records	
39	Continuation Record No. (1)	5.16
40	Application Type (1)	5.91
41	Start/End Indicator (1)	5.152
42 thru **66**	**Blank (Spacing) (25)**	
67 thru 68	Restr. Airspace ICAO Code (2)	5.14
69	Restr. Airspace Type (1)	5.128
70 thru 79	Restr. Airspace Designation (10)	5.129
80	Restr. Airspace Multiple Code (1)	5.130
81 thru 82	Restr. Airspace ICAO Code (2)	5.14
83	Restr. Airspace Type (1)	5.12
84 thru 93	Restr. Airspace Designation (10)	5.129
94	Restr. Airspace Multiple Code (1)	5.130
95 thru 96	Restr. Airspace ICAO Code (2)	5.14
97	Restr. Airspace Type (1)	5.12
98 thru 107	Restr. Airspace Designation (10)	5.129
108	Restr. Airspace Multiple Code (1)	5.130
109 thru 110	Restr. Airspace ICAO Code (2)	5.14
111	Restr. Airspace Type (1)	5.128
112 thru 121	Restr. Airspace Designation (10)	5.129
122	Restr. Airspace Multiple Code (1)	5.130
123	Restr. Airspace Link Continuation (1)	5.174
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.6.4 Enroute Airways Flight Planning Continuation Records

Deleted by Supplement 19.

4.1.7 Airport Records (PA)

This file contains airport information for all airports within the desired geographical reference area and meeting other criteria on available runways. Additionally, the file contains all airports required to support Enroute Airway structure coding for those areas where Airport reference points are used as enroute airway fixes.

4.1.7.1 Airport Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport ICAO Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 16	ATA/IATA Designator (3)	5.107
17 thru 18	Reserved (Expansion) (2)	
19 thru 21	Blank (Spacing) (3)	
22	Continuation Record Number (1)	5.16
23 thru 27	Speed Limit Altitude (5)	5.73
28 thru 30	Longest Runway (3)	5.54
31	IFR Capability (1)	5.108
32	Longest Runway Surface Code (1)	5.249
33 thru 41	Airport Reference Pt. Latitude (9)	5.36
42 thru 51	Airport Reference Pt. Longitude (10)	5.37
52 thru 56	Magnetic Variation (5)	5.39
57 thru 61	Airport Elevation (5)	5.55
62 thru 64	Speed Limit (3)	5.72
65 thru 68	Recommended Navaid (4)	5.23
69 thru 70	ICAO Code (2)	5.14
71 thru 75	Transitions Altitude (5)	5.53
76 thru 80	Transition Level (5)	5.53
81	Public/Military Indicator (1)	5.177
82 thru 84	Time Zone (3)	5.178
85	Daylight Indicator (1)	5.179
86	Magnetic/True Indicator (1)	5.165
87 thru 89	Datum Code (3)	5.197
90 thru 93	Reserved (Expansion) (4)	
94 thru 123	Airport Name (30)	5.71
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.7.2 Airport Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Field as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 92	Notes (69)	5.61
93 thru 123	Reserved (Expansion) (31)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.7.3 Airport Flight Planning Continuation Records

This Continuation Record is used to indicate the FIR and UIR within which the Airport define in the Primary Record resides in and the Start/End validity dates/times of the Primary Record and provide an indication if the Airport defined in the Primary Record is associated with Controlled Airspace.

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 27	FIR Identifier (4)	5.116
28 thru 31	UIR Identifier (4)	5.116
32	Start/End Indicator (1)	5.152
33 thru 66	**Blank (Spacing) (34)**	
67	Controlled A/S Indicator (1)	5.217
68 thru 71	Controlled A/S Arpt Ident (4)	5.6
72 thru 73	Controlled A/S Arpt ICAO (2)	5.14
74 thru 123	Blank (Spacing) (50)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Data (4)	5.32

4.1.7.4 Airport Flight Planning Continuation Records

Deleted by Supplement 19.

4.1.8 Airport Gate Records (PB)

This file contains passenger gate information.

4.1.8.1 Airport Gate Primary Record

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport ICAO Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 18	Gate Identifier (5)	5.56
19 thru 21	Blank (Spacing) (3)	
22	Continuation Record No. (1)	5.16
23 thru 32	Blank (Spacing) (10)	
33 thru 41	Gate Latitude (9)	5.36
42 thru 51	Gate Longitude (10)	5.37
52 thru 98	Reserved (Expansion) (47)	
99 thru 123	Name (25)	5.60
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.8.2 Airport Gate Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Field as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 92	Notes (69)	5.61
93 thru 123	Reserved (Expansion) (31)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.9 Airport SID/STAR/Approach (PD, PE and PF)

Airport SIDs, STARs and Approach Procedures are contained in three separate Section/Subsection groupings, using this single record format. Section/Subsection PD contains a sequential listing of those published Airport Standard Instrument Departures that can be encoded according to this Specification. Section/Subsection PE contains a sequential list of those published Airport Standard Terminal Arrival Routes that can be encoded according to this Specification. Section/Subsection PF contains a sequential listing of those published Airport Standard Instrument Approach Procedures that can be encoded according to this specification.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.9.1 Airport SID/STAR/Approach Primary Records

Column	Field Name (Length)		Reference
1	Record Type (1)		5.2
2 thru 4	Customer/Area Code (3)		5.3
5	Section Code (1)		5.4
6	Blank (Spacing) (1)		
7 thru 10	Airport Identifier (4)		5.6
11 thru 12	ICAO Code (2)		5.14
13	Subsection Code (1)		5.5
14 thru 19	SID/STAR/Approach Identifier (6)	Note 1	5.9, 5.10
20	Route Type (1)		5.7
21 thru 25	Transition Identifier (5)		5.11
26	Blank (Spacing) (1)		
27 thru 29	Sequence Number (3)		5.12
30 thru 34	Fix Identifier (5)		5.13
35 thru 36	ICAO Code (2)		5.14
37	Section Code (1)		5.4
38	Subsection Code (1)		5.5
39	Continuation Record Number (1)		5.16
40 thru 43	Waypoint Description Code (4)		5.17
44	Turn Direction (1)		5.20
45 thru 47	RNP (3)	Note 4	5.211
48 thru 49	Path and Termination (2)		5.21
50	Turn Direction Valid (1)		5.22
51 thru 54	Recommended Navaid (4)		5.23
55 thru 56	ICAO Code (2)		5.14
57 thru 62	ARC Radius (6)		5.204
63 thru 66	Theta (4)		5.24
67 thru 70	Rho (4)		5.25
71 thru 74	Magnetic Course (4)		5.26
75 thru 78	Route Distance/Holding Distance or Time (4)		5.27
79	RECD NAV Section (1)		5.4
80	RECD NAV Subsection (1)		5.5
81 thru 82	Reserved (expansion) (2)		
83	Altitude Description (1)		5.29
84	ATC Indicator (1)		5.81
85 thru 89	Altitude (5)		5.30
90 thru 94	Altitude (5)		5.30
95 thru 99	Transition Altitude (5)		5.53
100 thru 102	Speed Limit (3)		5.72
103 thru 106	Vertical Angle (4)		5.70
107 thru 111	Center Fix or TAA Procedure Turn Indicator (5)		5.144 or 5.271
112	Multiple Code or TAA Sector Identifier (1)		5.130 or 5.272
113 thru 114	ICAO Code (2)	Note 3	5.14
115	Section Code (1)	Note 3	5.4
116	Subsection Code (1)	Note 3	5.5
117	GNSS/FMS Indication (1)		5.222
118	Speed Limit Description (1)		5.261
119	Apch Route Qualifier 1 (1)	Note 2	5.7
120	Apch Route Qualifier 2 (1)	Note 2	5.7
121 thru 123	**Vertical Scale Factor (3)**		**5.293**
124 thru 128	File Record Number (5)		5.31
129 thru 132	Cycle Date (4)		5.32

Note 1: For approach route idents including "Multiple Indicator," see Section 5.10.

Note 2: Columns 119 and 120 (Approach Route Qualifier 1 and 2) are required to match the Primary Record to the Continuation Record(s). This non-standard column sorting sequence was selected to preserve the Primary Record for SID/STAR/Approach Records as much as possible as these new fields were introduced in Supplement 14.

Note 3: When columns 107 thru 116 are providing a reference to a MSA or the center fix for an RF leg, all of the columns are used. When they are providing a reference to a TAA, only columns 107 thru 112 are used and 113 thru 116 are blank.

Note 4: If there is only one set of RNP criteria for the RNAV procedure, that criteria is provided in the RNP value field for Primary Record. Otherwise, the Primary Record contains one consistent set of RNP values for the least restrictive RNAV operating criteria and not a mix of RNP values for different RNP operating criteria.

4.1.9.2 Airport SID/STAR/ Primary Extension Approach Continuation Records

Column	Field Name (Length)		Reference
1 thru 38	Fields as on Primary Records		
39	Continuation Record Number (1)		5.16
40	Application Type (1)		5.91
41 thru 44	CAT A Decision Height (4)		5.170
45 thru 48	CAT B Decision Height (4)		5.170
49 thru 52	CAT C Decision Height (4)		5.170
53 thru 56	CAT D Decision Height (4)		5.170
57 thru 60	CAT A Minimum Descent Altitude (4)		5.171
61 thru 64	CAT B Minimum Descent Altitude (4)		5.171
65 thru 68	CAT C Minimum Descent Altitude (4)		5.171
69 thru 72	CAT D Minimum Descent Altitude (4)		5.171
73 thru 75	Procedure TCH (3)		5.67
76	Localizer Only Altitude Description (1)		5.29
77 thru 81	Localizer Only Altitude (5)		5.30
82 thru 85	Localizer Only Vertical Angle (4)		5.70
86 thru 89	**Procedure Category (4)**		**5.242**
90 thru 92	RNP (3)	Note 2	5.211
93 thru 97	**Procedure Design Mag Var (5)**	**Note 3**	**5.290**
98	**Procedure Design Mag Var Ind (1)**	**Note 3**	**5.291**
99 thru 100	**CAT A Radii (2)**		**5.292**
101 thru 102	**CAT B Radii (2)**		**5.292**
103 thru 104	**CAT C Radii (2)**		**5.292**
105 thru 106	**CAT D Radii (2)**		**5.292**
107 thru 110	**Reserved**		
111 thru 112	**ICAO Code (2)**		**5.14**
113 thru 116	**Reserved**		
117 thru 118	**ICAO Code (2)**		**5.14**
119	Apch Route Qualifier 1 (1)	Note 1	5.7
120	Apch Route Qualifier 2 (1)	Note 1	5.7
121 thru 123	Blank (Spacing) (3)		
124 thru 128	File Record Number (5)		5.31
129 thru 132	Cycle Date (4)		5.32

Note 1: Columns 119 and 120, (Approach Route Qualifier 1 and 2) are required to match the Primary Record to the Continuation Record(s). This non-standard column sorting sequence was

4.0 NAVIGATION DATA - RECORD LAYOUT

selected to preserve the Primary Record for SID/STAR/Approach Records as much as possible as these new fields were introduced in Supplement 14.

Note 2: For records used to define RNAV Procedures with multiple RNP values and for RNAV Procedures published with multiple RNP operating criteria, the least restrictive values are carried in columns 45 thru 47 of the Primary record for each leg. Columns 90 thru 92 of this Primary Extension Continuation Record will carry the additional government source RNP values. There may be more than one Primary Extension Record provided when the procedure has more than two sets of distinct RNP values.

Note 3: When government source provides Procedure Design Mag Var at the procedure level, a single Primary Extension Continuation Record will be provided, associated to the first sequence in each transition and the Procedure Design Mag Var Indicator will be set to "P". This is consistent with the intent of this continuation record. When government source provides Procedure Design Mag Var the leg level, a Primary Extension Continuation Record will be provided associates with each sequence of each transitions and the Procedure Design Mag Var Indicator will be set to "L".

4.1.9.3 Airport SID/STAR/Approach Flight Planning Continuation Records

This Continuation Record is used to indicate the Leg Distance for each segment of the Route and the Start/End validity times of the Primary Record.

Column	Field Name (Length)	Reference
1 thru 38	Field as on Primary	
39	Continuation Record No. (1)	5.16
40	Application Type (1)	5.91
41	Start/End Indicator (1)	5.152
42 thru **74**	**Blank (Spacing) (33)**	
75 thru 78	Leg Distance (4)	5.260
79 thru 123	Reserved (Expansion) (45)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.9.4 Airport SID/STAR Flight Planning Continuation Records

Deleted by Supplement 19.

4.1.9.5 Airport Procedure Data Continuation Record

The Airport Procedure Data Continuation Record is used to provide information on SBAS authorization levels. This Continuation Record is provided once per procedure as a Continuation to Primary Approach Procedure Record that contains the encoding for Final Approach Fix (FAF) of the procedure.

Column	Field Name (Length)	Reference
1 thru 38	Fields as on Primary Record	
39	Continuation Record Number (1)	5.16
40	Application Type (1)	5.91
41	FAS Block Provided	5.276
42 thru 51	FAS Block Provided Lev of Serv Name (10)	5.275
52	LNAV/VNAV Authorized for SBAS (1)	5.276
53 thru 62	LNAV/VNAV Level of Service Name (10)	5.275
63	LNAV Authorized for SBAS(1)	5.276
64 thru 73	LNAV Level of Service Name (10)	5.275
74 thru 88	**Blank (Spacing) (11)**	
89	**RNP Authorized (1)**	**5.276**
90 thru 92	**RNP Level of Service value (3)**	**5.29**6
93	**RNP Authorized (1)**	**5.276**
94 thru 96	**RNP Level of Service value (3)**	**5.29**6
97	**RNP Authorized (1)**	**5.276**
98 thru 100	**RNP Level of Service value (3)**	**5.29**6
101	**RNP Authorized (1)**	**5.276**
102 thru 104	**RNP Level of Service value (3)**	**5.29**6
105 thru 118	**Blank (Spacing) (14)**	
119	Approach Route Type Qualifier 1 (1) Note 1	5.7
120	Approach Route Type Qualifier 2 (1) Note 1	5.7
121 thru 123	Blank (3)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: Columns 119 and 120 (Approach Route Type Qualifiers 1 and 2) are required to match the Primary Record to the Continuation Record(s). This non-standard column sorting sequence was selected to preserve the Primary Record for SID/STAR/Approach Records as much as possible as these new fields were introduced in Supplement 14.

Note 2: For records used to define RNAV procedures with route segment RNP values, the least restrictive value will be carried in columns 45-47 of the primary record for each leg. Columns 90-92 of this Primary Extension Continuation record will carry the additional government source RNP value. There may be more than one Primary Extension Record provided when the procedure route segment has more than two distinct RNP values. RNP operational minima criteria for procedures will be identified in the Airport Procedure Data Continuation record 4.1.9.5.

4.1.10 Runway Records (PG)

This file contains runway information.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.10.1 Runway Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport ICAO Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 18	Runway Identifier (5)	5.46
19 thru 21	Blank (Spacing) (3)	
22	Continuation Record No. (1)	5.16
23 thru 27	Runway Length (5)	5.57
28 thru 31	Runway Magnetic Bearing (4)	5.58
32	Blank (Spacing) (1)	
33 thru 41	Runway Latitude (9)	5.36
42 thru 51	Runway Longitude (10)	5.37
52 thru 56	Runway Gradient (5)	5.212
57 thru 60	Blank (Spacing) (4)	
61 thru 66	(LTP) Ellipsoid Height (6)	5.225
67 thru 71	Landing Threshold Elevation (5)	5.68
72 thru 75	Displaced Threshold Distance (4)	5.69
76 thru 77	Threshold Crossing Height (2)	5.67
78 thru 80	Runway Width (3)	5.109
81	TCH Value Indicator (1)	5.270
82 thru 85	Localizer/MLS/ GLS Ref Path Identifier (4)	5.44
86	Localizer/ MLS/ GLS Category/ Class (1)	5.80
87 thru 90	Stopway (4)	5.79
91 thru 94	Second Localizer/ MLS/ GLS Ref Path Ident (4)	5.44
95	Second Localizer/ MLS/ GLS Category/ Class (1)	5.80
96 thru 101	Reserved (Expansion) (6)	
102 thru 123	Runway Description (22)	5.59
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.10.2 Runway Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 92	Notes (69)	5.61
93 thru 123	Reserved (Expansion) (31)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.10.3 Runway Simulation Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 51	Reserved (Spacing) (28)	
52 thru 56	Runway True Bearing (5)	5.94
57	True Bearing Source (1)	5.95
58 thru 65	Reserved (Spacing) (8)	
66	TDZE Location (1)	5.98
67 thru 71	Touchdown Zone Elevation (5)	5.97
72 thru 123	Reserved (Expansion) (52)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.11 Airport and Heliport Localizer and Glide Slope Records (PI)

This file will contain a sequential listing of all localizers and glide slopes associated with those localizers. The glide slope portion of the record may contain blanks if no glide slope is associated with the localizer (Category 0 localizer, see Section 5.80). When a glide slope is installed, a glide slope angle will be provided. The latitude and longitude fields for the glide slope may be set to blanks when such information is not available to the data supplier for a particular glide slope installation due to insufficient government source.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.11.1 Airport and Heliport Localizer and Glide Slope Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 17	Localizer Identifier (4)	5.44
18	ILS Category (1)	5.80
19 thru 21	Blank (Spacing) (3)	
22	Continuation Record No. (1)	5.16
23 thru 27	Localizer Frequency (5)	5.45
28 thru 32	Runway Identifier (5)	5.46
33 thru 41	Localizer Latitude (9)	5.36
42 thru 51	Localizer Longitude (10)	5.37
52 thru 55	Localizer Bearing (4)	5.47
56 thru 64	Glide Slope Latitude (9)	5.36
65 thru 74	Glide Slope Longitude (10)	5.37
75 thru 78	Localizer Position (4)	5.48
79	Localizer Position Reference (1)	5.49
80 thru 83	Glide Slope Position (4)	5.50
84 thru 87	Localizer Width (4)	5.51
88 thru 90	Glide Slope Angle (3)	5.52
91 thru 95	Station Declination (5)	5.66
96 thru 97	Glide Slope Height at Landing Threshold (2)	5.67
98 thru 102	Glide Slope Elevation (5)	5.74
103 thru 106	Supporting Facility ID (4) Note 1	5.33
107 thru 108	Supporting Facility ICAO Code (2) Note 1	5.14
109	Supporting Facility Section Code (1) Note 1	5.4
110	Supporting Facility Subsection Code (1) Note 1	5.5
111 thru 123	Reserved (Expansion) (13)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: Terminal Navaids used as Supporting Facilities must be located at the same airport as the ILS.

4.1.11.2 Airport and Heliport Localizer and Glide Slope Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 92	Notes (69)	5.61
93 thru 123	Reserved (Expansion) (31)	
124 thru 128	File Record (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.11.3 Airport and Heliport Localizer and Glide Slope Simulation Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Record	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 27	Blank (Spacing) (4)	
28 thru 32	Facility Characteristics (5)	5.93
33 thru 51	Blank (Spacing) (19)	
52 thru 56	Localizer True Bearing (5)	5.94
57	Localizer Bearing Source (1)	5.95
58 thru 87	Reserved (Spacing) (30)	
88 thru 90	Glide Slope Beam Width (3)	5.96
91 thru 96	Approach Route Ident (6)	5.10
97 thru 102	Approach Route Ident (6)	5.10
103 thru 108	Approach Route Ident (6)	5.10
109 thru 114	Approach Route Ident (6)	5.10
115 thru 120	Approach Route Ident (6)	5.10
121 thru 123	Blank (Spacing) (3)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.12 Company Route Records (R)

This file contains company tailored route information.

4.1.12.1 Company Route Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 11	From Airport/Fix (5)	5.75
12	Blank (Spacing) (1)	
13 thru 14	ICAO Code (2)	5.14
15	Section Code (1)	5.4
16	Subsection Code (1)	5.5
17 thru 21	To Airport/Fix (5)	5.75
22	Blank (Spacing) (1)	
23 thru 24	ICAO Code (2)	5.14
25	Section Code (1)	5.4
26	Subsection Code (1)	5.5
27 thru 36	Company Route ID (10)	5.76
37 thru 39	Sequence No. (3)	5.12
40 thru 42	VIA (3)	5.77
43 thru 48	SID/STAR/App/Awy (6)	5.78
49 thru 51	Area Code (3)	5.3
52 thru 57	To Fix (6)	5.83
58 thru 59	ICAO Code (2)	5.14
60	Section Code (1)	5.4
61	Subsection Code (1)	5.5
62 thru 66	Runway Trans (5)	5.84
67 thru 71	ENRT Trans (5)	5.85
72	Reserved (1)	
73 thru 77	Cruise Altitude (5)	5.86
78 thru 81	Terminal/Alternate Airport (4)	5.87
82 thru 83	ICAO Code (2)	5.14
84 thru 87	Alternate Distance (4)	5.88
88 thru 90	Cost Index (3)	5.89
91 thru 94	Enroute Alternate Airport (4)	5.148
95	**SID/STAR/App/Awy Route Type (1)**	**5.7**
96	**S/S/A Route Type Qualifier 1 (1)**	**5.7**
97	**S/S/A Route Type Qualifier 2 (1)**	**5.7**
98 thru 123	**Reserved (Expansion) (26)**	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: This Company Route Record is defined for use with fixed wing aircraft and rotor wing aircraft operating from airports. Airports referenced will be in Section/SubSection PA, Runway in Section/SubSection PG, and Terminal Procedure referenced will be in Section/SubSections PD/PE/PF. For Helicopter Operations Company Routes see Section 4.2.7.

4.1.13 Airport and Heliport Localizer Marker Records (PM)

The Airport and Heliport Localizer Marker File (PM) contains details of all markers and locators associated with all types of localizers. It does not contain airway markers, see Section 4.1.15.

4.1.13.1 Airport and Heliport Localizer Marker Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 17	Localizer Identifier (4)	5.44
18 thru 20	Marker Type (3)	5.99
21	Blank (Spacing) (1)	
22	Continuation Record No. (1)	5.16
23 thru 27	Locator Frequency (5)	5.34
28 thru 32	Runway Identifier (5)	5.46
33 thru 41	Marker Latitude (9)	5.36
42 thru 51	Marker Longitude (10)	5.37
52 thru 55	Minor Axis Bearing (4)	5.100
56 thru 64	Locator Latitude (9)	5.36
65 thru 74	Locator Longitude (10)	5.37
75 thru 79	Locator Class (5)	5.35
80 thru 84	Locator Facility Characteristics (5)	5.93
85 thru 88	Locator Identifier (4)	5.33
89 thru 90	Blank (Spacing) (2)	
91 thru 95	Magnetic Variation (5)	5.39
96 thru 97	Blank (Spacing) (2)	
98 thru 102	Facility Elevation (5)	5.92
103 thru 123	Reserved (Expansion) (21)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.13.2 Airport and Heliport Localizer Marker Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 123	Reserved (Expansion) (100)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.14 Airport Communications Records (PV)

4.1.14.1 Airport Communications Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru **15**	**Blank (Spacing) (2)**	
16 thru 19	**Communication Class (4)**	**5.283**
20 thru 21	**Sequence Number (2)**	**5.12**
22	**Continuation Number (1)**	**5.16**
23 thru 25	**Communication Types (3)**	**5.101**
26 thru 32	**Transmit Frequency (7)**	**5.103**
33 thru 39	**Receive Frequency (7)**	**5.103**
40	**Frequency Units (1)**	**5.104**
41	**Radar Units (1)**	**5.102**
42	**H24 Indicator (1)**	**5.181**
43 thru 67	**Call Signs (25)**	**5.105**
68	**Multi-Sector Indicator (1)**	**5.286**
69 thru 74	**Sectorization (6)**	**5.183**
75 thru 78	**Sector Facility (4)**	**5.185**
79 thru 80	**ICAO (2)**	**5.14**
81	**Section Code (1)**	**5.4**
82	**Subsection Code (1)**	**5.5**
83	**Altitude Description Code (1)**	**5.29**
84 thru 86	**Communication Altitude 1 (3)**	**5.184**
87 thru 89	**Communication Altitude 2 (3)**	**5.184**
90	**Distance Description Code (1)**	**5.187**
91 thru 92	**Communication Distance (2)**	**5.188**
93 thru 101	**Transmitter Latitude (9)**	**5.36**
102 thru 111	**Transmitter Longitude (10)**	**5.37**
112 thru 114	**Service Indicator (3)**	**5.106**
115	**Modulation (1)**	**5.198**
116	**Signal Emission (2)**	**5.199**
117 thru 123	**Blank (Spacing) (7)**	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.14.2 Airport Communications Continuation Records

Column	Field Name (Length)	Reference
1 thru **21**	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 27	**Remote Facility (4)**	**5.200**
28 thru 29	**ICAO (2)**	**5.14**
30	**Section (1)**	**5.4**
31	**Subsection (1)**	**5.5**
32 thru 36	**Transmitter Site Mag Var (5)**	**5.39**
37 thru 41	**Transmitter Site Elevation (5)**	**5.92**
42 thru 47	**Additional Sectorization 1 (6)**	**5.183**
48	**Additional Sectorization 1 Altitude Description (1)**	**5.29**
49 thru 51	**Additional Sectorization 1 Altitude 1 (3)**	**5.184**
52 thru 53	**Additional Sectorization 1 Altitude 2 (3)**	**5.184**
54 thru 59	**Additional Sectorization 2 (6)**	**5.183**
60	**Additional Sectorization 2 Altitude Description (1)**	**5.29**
61 thru 63	**Additional Sectorization 2 Altitude 1 (3)**	**5.184**
64 thru 66	**Additional Sectorization 2 Altitude 2 (3)**	**5.184**
67	**Time Code (1)**	**5.131**
68	**NOTAM (1)**	**5.132**
69	**Time Indicator (1)**	**5.138**
70 thru 79	**Time of Operation (10)**	**5.195**
80 thru 89	**Time of Operation (10)**	**5.195**
90 thru 99	**Time of Operation (10)**	**5.195**
100 thru 109	**Time of Operation (10)**	**5.195**
110 thru 119	**Time of Operation (10)**	**5.195**
120 thru 123	**Blank (Spacing) (4)**	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.14.3 Airport **Communications** Continuation Records

Column	Field Name (Length)	Reference
1 thru **21**	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	**Application Type (1)**	**5.91**
24thru 83	**Sectorization Narrative (60)**	**5.186**
84 thru 123	**Reserved (Spacing) (40)**	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.14.4 Airport Communications Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	**Field as on Primary Record**	
22	**Continuation Record Number (1)**	**5.16**
23	**Application Type (1)**	**5.91**
24 thru 69	**Blank (Spacing) (46)**	
70 thru 79	**Time of Operation (10)**	**5.195**
80 thru 89	**Time of Operation (10)**	**5.195**
90 thru 99	**Time of Operation (10)**	**5.195**
100 thru 109	**Time of Operation (10)**	**5.195**
110 thru 119	**Time of Operation (10)**	**5.195**
120 thru 123	**Blank (Spacing) (4)**	
124 thru 128	**File Record numbers (5)**	**5.31**
129 thru 132	**Cycle Date (4)**	**5.32**

4.1.14.5 Airport Communications Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	**Field as on Primary Record**	
22	**Continuation Record Number (1)**	**5.16**
23	**Application Type (1)**	**5.91**
24 thru 123	**Time Narrative (100)**	**5.285**
124 thru 128	**File Record numbers (5)**	**5.31**
129 thru 132	**Cycle Date (4)**	**5.32**

4.1.15 Airways Marker Records (EM)

The Airways Marker file contains details of all airways markers.

4.1.15.1 Airways Marker Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 13	Blank (Spacing) (7)	
14 thru 17	Marker Identifier (4)	5.110
18 thru 19	Blank (Spacing) (2)	
20 thru 21	ICAO Code (2)	5.14
22	Continuation Record No. (1)	5.16
23 thru 26	Marker Code (4)	5.111
27	Reserved (Expansion) (1)	
28	Marker Shape (1)	5.112
29	Marker Power (1)	5.113
30 thru 32	Blank (Spacing) (3)	
33 thru 41	Marker Latitude (9)	5.36
42 thru 51	Marker Longitude (10)	5.37
52 thru 55	Minor Axis (4)	5.100
56 thru 74	Blank (Spacing) (19)	
75 thru 79	Magnetic Variation (5)	5.39
80 thru 84	Facility Elevation (5)	5.92
85 thru 87	Datum Code (3)	5.197
88 thru 93	Blank (Spacing) (6)	
94 thru 123	Marker Name (30)	5.71
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.15.2 Airways Marker Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 123	Reserved (Expansion) (100)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.16 Cruising Tables Records (TC)

The Cruising Tables file contains details relating to available Cruising Levels for IFR flights.

4.1.16.1 Cruising Table Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Blank (Spacing) (3)	
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 8	Cruise Table Identifier (2)	5.134
9	Sequence Number (1)	5.12
10 thru 28	Blank (Spacing) (19)	
29 thru 32	Course From (4)	5.135
33 thru 36	Course To (4)	5.135
37	Mag/True (1)	5.165
38 thru 39	Blank (Spacing) (2)	
40 thru 44	Cruise Level From (5)	5.136
45 thru 49	Vertical Separation (5)	5.137
50 thru 54	Cruise Level To (5)	5.136
55 thru 59	Cruise Level From (5)	5.136
60 thru 64	Vertical Separation (5)	5.137
65 thru 69	Cruise Level To (5)	5.136
70 thru 74	Cruise Level From (5)	5.136
75 thru 79	Vertical Separation (5)	5.137
80 thru 84	Cruise Level To (5)	5.136
85 thru 89	Cruise Level From (5)	5.136
90 thru 94	Vertical Separation (5)	5.137
95 thru 99	Cruise Level To (5)	5.136
100 thru 123	Reserved (Expansion) (24)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.17 FIR/UIR Records (UF)

The FIR/UIR file contains the lateral boundary description of the FIR/UIR in a sequence of records and the vertical boundary description of the FIR/UIR in the first of the sequence.

4.1.17.1 FIR/UIR Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 10	FIR/UIR Identifier (4)	5.116
11 thru 14	FIR/UIR Address (4)	5.151
15	FIR/UIR Indicator (1)	5.117
16 thru 19	Sequence Number (4)	5.12
20	Continuation Record No. (1)	5.16
21 thru 24	Adjacent FIR Identifier (4)	5.116
25 thru 28	Adjacent UIR Identifier (4)	5.116
29	Reporting Units Speed (1)	5.122
30	Reporting Units Altitude (1)	5.123
31	Entry Report (1)	5.124
32	Blank (Spacing) (1)	
33 thru 34	Boundary Via (2)	5.118
35 thru 43	FIR/UIR Latitude (9)	5.36
44 thru 53	FIR/UIR Longitude (10)	5.37
54 thru 62	Arc Origin Latitude (9)	5.36
63 thru 72	Arc Origin Longitude (10)	5.37
73 thru 76	Arc Distance (4)	5.119
77 thru 80	Arc Bearing (4)	5.120
81 thru 85	FIR Upper Limit (5)	5.121
86 thru 90	UIR Lower Limit (5)	5.121
91 thru 95	UIR Upper Limit (5)	5.121
96 thru 97	Cruise Table Ind (2)	5.134
98	Reserved (Expansion) (1)	
99 thru 123	FIR/UIR Name (25)	5.125
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.17.2 FIR/UIR Continuation Records

Column	Field Name (Length)	Reference
1 thru 19	Fields as on Primary Records	
20	Continuation Record No. (1)	5.16
21	Application Type (1)	5.91
22 thru 123	Reserved (Expansion) (102)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.18 Restrictive Airspace Records (UR)

The Restrictive Airspace Record File contains a sequential listing of vertical and lateral limits of restrictive areas.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.18.1 Restrictive Airspace Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 8	ICAO Code (2)	5.14
9	Restrictive Type (1)	5.128
10 thru 19	Restrictive Airspace Designation (10)	5.129
20	Multiple Code (1)	5.130
21 thru 24	Sequence Number (4)	5.12
25	Continuation Record No. (1)	5.16
26	Level (1)	5.19
27	Time Code (1)	5.131
28	NOTAM (1)	5.132
29 thru 30	Blank (Spacing) (2)	
31 thru 32	Boundary Via (2)	5.118
33 thru 41	Latitude (9)	5.36
42 thru 51	Longitude (10)	5.37
52 thru 60	Arc Origin Latitude (9)	5.36
61 thru 70	Arc Origin Longitude (10)	5.37
71 thru 74	Arc Distance (4)	5.119
75 thru 78	Arc Bearing (4)	5.120
79 thru 81	Blank (Spacing) (3)	
82 thru 86	Lower Limit (5)	5.121
87	Unit Indicator (1)	5.133
88 thru 92	Upper Limit (5)	5.121
93	Unit Indicator (1)	5.133
94 thru 123	Restrictive Airspace Name (30)	5.126
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.18.2 Restrictive Airspace Continuation Records

Column	Filed Name (Length)	Reference
1 thru 24	Fields as on Primary Records	
25	Continuation Record Number (1)	5.16
26	Application Type (1)	5.91
27	Time Code (1)	5.131
28	NOTAM (1)	5.132
29	Time Indicator (1)	5.138
30 thru 39	Time of Operations (10)	5.195
40 thru 49	Time of Operations (10)	5.195
50 thru 59	Time of Operations (10)	5.195
60 thru 69	Time of Operations (10)	5.195
70 thru 79	Time of Operations (10)	5.195
80 thru 89	Time of Operations (10)	5.195
90 thru 99	Time of Operations (10)	5.195
100 thru 123	Controlling Agency (24)	5.140
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.18.3 Restrictive Airspace Flight Planning Continuation Record

Column	Field Name (Length)	Reference
1 thru 24	Fields as on Primary Records	
25	Continuation Record No. (1)	5.16
26	Application Type (1)	5.91
27 thru 29	Blank (Spacing) (3)	
30	Start/End Indicator (1)	5.152
31 thru 41	Start/End Date (11)	5.153
42 thru 123	Reserved (Expansion) (82)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.19 Grid MORA Records (AS)

The Grid MORA (Minimum Off Route Altitude) file contains a table of Minimum Off Route Altitudes.

4.1.19.1 Grid MORA Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Blank (Spacing) (3)	
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 13	Blank (Spacing) (7)	
14 thru 16	Starting Latitude (3)	5.141
17 thru 20	Starting Longitude (4)	5.142
21 thru 30	Blank (Spacing) (10)	
31 thru 33	MORA (3)	5.143
34 thru 36	MORA (3)	5.143
37 thru 39	MORA (3)	5.143
40 thru 42	MORA (3)	5.143
43 thru 45	MORA (3)	5.143
46 thru 48	MORA (3)	5.143
49 thru 51	MORA (3)	5.143
52 thru 54	MORA (3)	5.143
55 thru 57	MORA (3)	5.143
58 thru 60	MORA (3)	5.143
61 thru 63	MORA (3)	5.143
64 thru 66	MORA (3)	5.143
67 thru 69	MORA (3)	5.143
70 thru 72	MORA (3)	5.143
73 thru 75	MORA (3)	5.143
76 thru 78	MORA (3)	5.143
79 thru 81	MORA (3)	5.143
82 thru 84	MORA (3)	5.143
85 thru 87	MORA (3)	5.143
88 thru 90	MORA (3)	5.143
91 thru 93	MORA (3)	5.143
94 thru 96	MORA (3)	5.143
97 thru 99	MORA (3)	5.143
100 thru 102	MORA (3)	5.143
103 thru 105	MORA (3)	5.143
106 thru 108	MORA (3)	5.143
109 thru 111	MORA (3)	5.143
112 thru 114	MORA (3)	5.143
115 thru 117	MORA (3)	5.143
118 thru 120	MORA (3)	5.143
121 thru 123	Reserved (Expansion) (3)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.20 Airport MSA (Minimum Sector Altitude) Records (PS)

The Minimum Sector Altitude (MSA) file contains details relating to available sector altitudes.

4.1.20.1 Airport MSA Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 18	MSA Center (5)	5.144
19 thru 20	ICAO Code (2)	5.14
21	Section Code (1)	5.4
22	Subsection Code (1)	5.5
23	Multiple Code (1)	5.130
24 thru 38	Reserved (Expansion) (15)	
39	Continuation Record No. (1)	5.16
40 thru 42	Reserved (Spacing) (3)	
43 thru 48	Sector Bearing (6)	5.146
49 thru 51	Sector Altitude (3)	5.147
52 thru 53	Sector Radius (2)	5.145
54 thru 59	Sector Bearing (6)	5.146
60 thru 62	Sector Altitude (3)	5.147
63 thru 64	Sector Radius (2)	5.145
65 thru 70	Sector Bearing (6)	5.146
71 thru 73	Sector Altitude (3)	5.147
74 thru 75	Sector Radius (2)	5.145
76 thru 81	Sector Bearing (6)	5.146
82 thru 84	Sector Altitude (3)	5.147
85 thru 86	Sector Radius (2)	5.145
87 thru 92	Sector Bearing (6)	5.146
93 thru 95	Sector Altitude (3)	5.147
96 thru 97	Sector Radius (2)	5.145
98 thru103	Sector Bearing (6)	5.146
104 thru 106	Sector Altitude (3)	5.147
107 thru 108	Sector Radius (2)	5.145
109 thru 114	Sector Bearing (6)	5.146
115 thru 117	Sector Altitude (3)	5.147
118 thru 119	Sector Radius (2)	5.145
120	Magnetic/True Indicator (1)	5.165
121 thru 123	Reserved (Expansion) (3)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.20.2 Airport MSA Primary Record Extension

Column	Field Name (Length)	Reference
1 thru 38	Field as on Primary Records	
39	Continuation Record No. (1)	5.16
40	Application Type (1)	5.91
41 thru 42	**Reserved (Spacing) (2)**	
43 thru 48	**Sector Bearing (6)**	**5.146**
49 thru 51	**Sector Altitude (3)**	**5.147**
52 thru 53	**Sector Radius (2)**	**5.145**
54 thru 59	**Sector Bearing (6)**	**5.146**
60 thru 62	**Sector Altitude (3)**	**5.147**
63 thru 64	**Sector Radius (2)**	**5.145**
65 thru 70	**Sector Bearing (6)**	**5.146**
71 thru 73	**Sector Altitude (3)**	**5.147**
74 thru 75	**Sector Radius (2)**	**5.145**
76 thru 81	**Sector Bearing (6)**	**5.146**
82 thru 84	**Sector Altitude (3)**	**5.147**
85 thru 86	**Sector Radius (2)**	**5.145**
87 thru 92	**Sector Bearing (6)**	**5.146**
93 thru 95	**Sector Altitude (3)**	**5.147**
96 thru 97	**Sector Radius (2)**	**5.145**
98 thru 103	**Sector Bearing (6)**	**5.146**
104 thru 106	**Sector Altitude (3)**	**5.147**
107 thru 108	**Sector Radius (2)**	**5.145**
109 thru 114	**Sector Bearing (6)**	**5.146**
115 thru 117	**Sector Altitude (3)**	**5.147**
118 thru 119	**Sector Radius (2)**	**5.145**
120 thru 123	**Field as on Primary Records**	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.20.3 Airport MSA Continuation Records

Column	Field Name (Length)	Reference
1 thru 38	Field as on Primary Records	
39	Continuation Record No. (1)	5.16
40	Application Type (1)	5.91
41 thru 109	Notes (69)	5.61
110 thru 123	Reserved (Expansion) (14)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.21 Enroute Airways Restriction Records (EU)

The Enroute Airway Restriction file will contain altitude and time restrictions for an airway, airway segment or sequence of airway segments. The Enroute Airway Restriction file may contain four different types of primary records, dependent on the type of restriction. A Restriction Code will identify the type of restriction contained in the record. Continuation Records may be used if a single record does not provide sufficient space for coding a single, complete restriction.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.21.1 Enroute Airways Restriction Altitude Exclusion Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 11	Route Identifier (5)	5.8
12	Reserved (1)	Note 1
13 thru 15	Restriction Identifier (3)	5.154
16 thru 17	Restriction Type (2)	5.201
18	Continuation Record No. (1)	5.16
19 thru 23	Start Fix Identifier (5)	5.13
24 thru 25	Start Fix ICAO Code (2)	5.14
26	Start Fix Section Code (1)	5.4
27	Start Fix Subsection Code(1)	5.5
28 thru 32	End Fix Identifier (5)	5.13
33 thru 34	End Fix ICAO Code (2)	5.14
35	End Fix Section Code (1)	5.4
36	End Fix Subsection Code (1)	5.5
37	Blank (Spacing) (1)	
38 thru 44	Start Date (7)	5.157
45 thru 51	End Date (7)	5.157
52	Time Code (1)	5.131
53	Time Indicator (1)	5.138
54 thru 63	Time of Operation (10)	5.195
64 thru 73	Time of Operation (10)	5.195
74 thru 83	Time of Operation (10)	5.195
84 thru 93	Time of Operation (10)	5.195
94	Exclusion Indicator (1)	5.202
95	Units of Altitude (1)	5.160
96 thru 98	Restriction Altitude (3)	5.161
99	Block Indicator (1)	5.203
100 thru 102	Restriction Altitude (3)	5.161
103	Block Indicator (1)	5.203
104 thru 106	Restriction Altitude (3)	5.161
107	Block Indicator (1)	5.203
108 thru 110	Restriction Altitude (3)	5.161
111	Block Indicator (1)	5.203
112 thru 114	Restriction Altitude (3)	5.161
115	Block Indicator (1)	5.203
116 thru 118	Restriction Altitude (3)	5.161
119	Block Indicator (1)	5.203
120 thru 122	Restriction Altitude (3)	5.161
123	Block Indicator (1)	5.203
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: The standard length for the Route Identifier is five characters. Some users envisage the need for a six-character field. This reserved column will permit this usage.

4.1.21A.1 Enroute Airways Restriction Note Restriction Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 11	Route Identifier (5)	5.8
12	Reserved (1)	Note 1
13 thru 15	Restriction Identifier (3)	5.154
16 thru 17	Restriction Type (2)	5.201
18	Continuation Record No. (1)	5.16
19 thru 23	Start Fix Identifier (5)	5.13
24 thru 25	Start Fix ICAO Code (2)	5.14
26	Start Fix Section Code (1)	5.4
27	Start Fix Subsection Code (1)	5.5
28 thru 32	End Fix Identifier (5)	5.13
33 thru 34	End Fix ICAO Code (2)	5.14
35	End Fix Section Code (1)	5.4
36	End Fix Subsection Code (1)	5.5
37	Blank (Spacing) (1)	
38 thru 44	Start Date (7)	5.157
45 thru 51	End Date (7)	5.157
52 thru 120	Restriction Notes (69)	5.163
121 thru 123	Blank (Spacing) (3)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.21

Note 1: The standard length for the Route Identifier is five characters. Some users envisage the need for a six-character field. This reserved column will permit this usage.

4.1.21A.2 Enroute Airways Restriction Note Restriction Continuation Records

Column	Field Name (Length)	Reference
1 thru 17	Fields as on Primary Records	
18	Continuation Record No. (1)	5.16
19	Application Type (1)	5.91
20 thru 51	Reserved (Expansion) (32)	
52 thru 120	Restriction Notes (69)	5.163
121 thru 123	Blank (Spacing) (3)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.21B.1 Enroute Airways Restriction Seasonal Closure Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 11	Route Identifier (5)	5.8
12	Reserved (1)	Note 1
13 thru 15	Restriction Identifier (3)	5.154
16 thru 17	Restriction Type (2)	5.201
18	Continuation Record No. (1)	5.16
19 thru 23	Start Fix Identifier (5)	5.13
24 thru 25	Start Fix ICAO Code (2)	5.14
26	Start Fix Section Code (1)	5.4
27	Start Fix Subsection Code (1)	5.5
28 thru 32	End Fix Identifier (5)	5.13
33 thru 34	End Fix ICAO Code (2)	5.14
35	End Fix Section Code (1)	5.4
36	End Fix Subsection Code (1)	5.5
37	Blank (Spacing) (1)	
38 thru 44	Start Date (7)	5.157
45 thru 51	End Date (7)	5.157
52	Time Code (1)	5.131
53	Time Indicator (1)	5.138
54 thru 63	Time of Operation (10)	5.195
64 thru 73	Time of Operation (10)	5.195
74 thru 83	Time of Operation (10)	5.195
84 thru 93	Time of Operation (10)	5.195
94 thru 95	Cruise Table Ident (2)	5.134
96 thru 123	Blank (Spacing) (28)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: The standard length for the Route Identifier is five characters. Some users envisage the need for a six-character field. This reserved column will permit this usage.

4.1.21C.1 Enroute Airways Restriction Cruising Table Replacement Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 11	Route Identifier (5)	5.8
12	Reserved (1)	Note 1
13 thru 15	Restriction Identifier (3)	5.154
16 thru 17	Restriction Type (2)	5.201
18	Continuation Record No. (1)	5.16
19 thru 23	Start Fix Identifier (5)	5.13
24 thru 25	Start Fix ICAO Code (2)	5.14
26	Start Fix Section Code (1)	5.4
27	Start Fix Subsection Code (1)	5.5
28 thru 32	End Fix Identifier (5)	5.13
33 thru 34	End Fix ICAO Code (2)	5.14
35	End Fix Section Code (1)	5.4
36	End Fix Subsection Code (1)	5.5
37	Blank (Spacing) (1)	
38 thru 44	Start Date (7)	5.157
45 thru 51	End Date (7)	5.157
52	Time Code (1)	5.131
53	Time Indicator (1)	5.138
54 thru 63	Time of Operation (10)	5.195
64 thru 73	Time of Operation (10)	5.195
74 thru 83	Time of Operation (10)	5.195
84 thru 93	Time of Operation (10)	5.195
94 thru 95	Cruise Table Ident (2)	5.134
96 thru 123	Blank (Spacing) (28)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: The standard length for the Route Identifier is five characters. Some users envisage the need for a six-character field. This reserved column will permit this usage.

4.1.21C.2 Enroute Airways Restriction Cruising Table Replacement Continuation Records

Column	Field Name (Length)	Reference
1 thru 17	Fields as on Primary Records	
18	Continuation Record No. (1)	5.16
19	Application Type (1)	5.91
20 thru 51	Reserved (Expansion) (32)	
52	Time Code (1)	5.131
53	Time Indicator (1)	5.138
54 thru 63	Time of Operation (10)	5.195
64 thru 73	Time of Operation (10)	5.195
74 thru 83	Time of Operation (10)	5.195
84 thru 93	Time of Operation (10)	5.195
94 thru 95	Cruise Table Ident (2)	5.134
96 thru 123	Blank (Spacing) (28)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.21.2 Enroute Airways Restriction Altitude Exclusion Continuation Records

Column	Field Name (Length)	Reference
1 thru 17	Fields as on Primary Records	
18	Continuation Record No. (1)	5.16
19	Application Type (1)	5.91
20 thru 51	Reserved (Expansion) (32)	
52	Time Code (1)	5.131
53	Time Indicator (1)	5.138
54 thru 63	Time of Operation (10)	5.195
64 thru 73	Time of Operation (10)	5.195
74 thru 83	Time of Operation (10)	5.195
84 thru 93	Time of Operation (10)	5.195
94	Exclusion Indicator (1)	5.202
95	Units of Altitude (1)	5.160
96 thru 98	Restriction Altitude (3)	5.161
99	Block Indicator (1)	5.203
100 thru 102	Restriction Altitude (3)	5.161
103	Block Indicator (1)	5.203
104 thru 106	Restriction Altitude (3)	5.161
107	Block Indicator (1)	5.203
108 thru 110	Restriction Altitude (3)	5.161
111	Block Indicator (1)	5.203
112 thru 114	Restriction Altitude (3)	5.161
115	Block Indicator (1)	5.203
116 thru 118	Restriction Altitude (3)	5.161
119	Block Indicator (1)	5.203
120 thru 122	Restriction Altitude (3)	5.161
123	Block Indicator (1)	5.203
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.22 Airport and Heliport MLS (Azimuth, Elevation and Back Azimuth) Records (PL)

This file will contain a listing of all Microwave Landing Systems, including the Azimuth station, the Elevation station and the Back Azimuth station if installed.

4.1.22.1 Airport and Heliport MLS Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing (1)	
7 thru 10	Airport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 17	MLS Identifier (4)	5.44
18	MLS Category (1)	5.80
19 thru 21	Blank (Spacing) (3)	
22	Continuation Record No. (1)	5.16
23 thru 25	Channel (3)	5.166
26 thru 27	Blank (Spacing) (2)	
28 thru 32	Runway Identifier (5)	5.46
33 thru 41	Azimuth Latitude (9)	5.36
42 thru 51	Azimuth Longitude (10)	5.37
52 thru 55	Azimuth Bearing (4)	5.167
56 thru 64	Elevation Latitude (9)	5.36
65 thru 74	Elevation Longitude (10)	5.37
75 thru 78	Azimuth Position (4)	5.48
79	Azimuth Position Reference (1)	5.49
80 thru 83	Elevation Position (4)	5.50
84 thru 86	Azimuth Proportional Angle Right (3)	5.168
87 thru 89	Azimuth Proportional Angle Left (3)	5.168
90 thru 92	Azimuth Coverage Right (3)	5.172
93 thru 95	Azimuth Coverage Left (3)	5.172
96 thru 98	Elevation Angle Span (3)	5.169
99 thru 103	Magnetic Variation (5)	5.39
104 thru 108	EL Elevation (5)	5.74
109 thru 112	Nominal Elevation Angle (4)	5.173
113 thru 115	Minimum Glide Path Angle (3)	5.52
116 thru 119	Supporting Facility Identifier (4)	5.33 Note 1
120 thru 121	Supporting Facility ICAO Code (2)	5.14 Note 1
122	Supporting Facility Section Code (1)	5.4 Note 1
123	Supporting Facility Subsection Code (1)	5.5 Note 1
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: Terminal Navaids used as Supporting Facilities must be located at the same airport as the MLS.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.22.2 Airport and Heliport MLS Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Field as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 27	Blank (Spacing) (4)	
28 thru 32	Facility Characteristics (5)	5.93
33 thru 41	Back Azimuth Latitude (9)	5.36
42 thru 51	Back Azimuth Longitude (10)	5.37
52 thru 55	Back Azimuth Bearing (4)	5.167
56 thru 64	MLS Datum Point Latitude (9)	5.36
65 thru 74	MLS Datum Point Longitude (10)	5.37
75 thru 78	Back Azimuth Position (4)	5.48
79	Back Azimuth Position Reference (1)	5.49
80 thru 83	Blank (Spacing) (4)	
84 thru 86	Back Azimuth Proportional Sector Right (3)	5.168
87 thru 89	Back Azimuth Proportional Sector Left (3)	5.168
90 thru 92	Back Azimuth Coverage Right (3)	5.172
93 thru 95	Back Azimuth Coverage Left (3)	5.172
96 thru 100	Back Azimuth True Bearing (5)	5.94
101	Back Azimuth Bearing Source (1)	5.95
102 thru 106	Azimuth True Bearing (5)	5.94
107	Azimuth Bearing Source (1)	5.95
108 thru 109	Glide Path Height at Landing Threshold (2)	5.67
110 thru 123	Reserved (Expansion) (14)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.23 Enroute Communications Records (EV)

4.1.23.1 Enroute Communications Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 10	FIR/RDO Ident (4)	5.190
11 thru 14	FIR/UIR Address (4)	5.151
15	Indicator (1)	5.117
16 thru 19	**Communication Class (4)**	**5.283**
20 thru 21	**Sequence Number (2)**	**5.12**
22	Continuation Record No. (1)	5.16
23 thru 25	Communications Type (3)	5.101
26 thru 32	**Transmit Frequency (7)**	**5.103**
33 thru 39	**Receive Frequency (7)**	**5.103**
40	Frequency Units (1)	5.104
41	Radar Service (1)	5.102
42	H24 Indicator (1)	5.181
43 thru 67	**Call Sign (25)**	**5.105**
68 thru 92	**Position Narrative** (25)	5.189
93 thru 101	Latitude (9)	5.36
102 thru 111	Longitude (10)	5.37
112 thru 114	Service Indicator (3)	5.106
115	Modulation (1)	5.198
116	Signal Emission (1)	5.199
117	Altitude Descript. (1)	5.29
118 thru 120	Communication Altitude 1 (3)	5.184
121 thru 123	Communication Altitude 2 (3)	5.184
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.23.2 Enroute Communications Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 27	**Remote Facility (4)**	**5.200**
28 thru 29	**ICAO (2)**	**5.14**
30	**Section (1)**	**5.4**
31	**Subsection (1)**	**5.5**
32 thru 36	**Transmitter Site Mag Var (5)**	**5.39**
37 thru 41	**Transmitter Site Elevation (5)**	**5.92**
42 thru 66	**Assigned Sector Name (25)**	**5.284**
67	Time Code (1)	5.131
68	NOTAM (1)	5.132
69	Time Indicator (1)	5.138
70 thru 79	**Time of Operation (10)**	**5.195**
80 thru 89	**Time of Operation (10)**	**5.195**
90 thru 99	**Time of Operation (10)**	**5.195**
100 thru 109	**Time of Operation (10)**	**5.195**
110 thru 119	**Time of Operation (10)**	**5.195**
120 thru 123	**Blank (Spacing) 4**	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.23.3 Enroute Communications Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 69	Blank (Spacing) (46)	
70 thru 79	Time of Operation (10)	5.195
80 thru 89	Time of Operation (10)	5.195
90 thru 99	Time of Operation (10)	5.195
100 thru 109	Time of Operation (10)	5.195
110 thru 119	Time of Operation (10)	5.195
120 thru 123	Reserved (Expansion) (4)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.23.4 Enroute Communications Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	**Fields as on Primary Record**	
22	**Continuation Record No. (1)**	**5.16**
23	**Application Type (1)**	**5.91**
24 thru 123	**Time Narrative (100)**	**5.285**
124 thru 128	**File Record No. (5)**	**5.31**
129 thru 132	**Cycle Date (4)**	**5.32**

4.1.24 Preferred Routes Records (ET)

The Preferred Routes file contains details defining the Preferred Routes, North America Routes for North Atlantic Traffic, the Traffic Orientation System, and the similar predefined routings that do not meet the requirements of the Enroute Airway Record.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.24.1 Preferred Route Primary Records

Columns	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 13	Blank (Spacing) (7)	
14 thru 23	Route Identifier (10)	5.8
24 thru 25	Preferred Route Use Ind (2)	5.220
26 thru 29	Sequence Number (4)	5.12
30 thru 38	Blank (Spacing) (9)	
39	Continuation Record No. (1)	5.16
40 thru 44	To Fix Identifier (5)	5.83
45 thru 46	ICAO Code (2)	5.14
47	Section Code (1)	5.4
48	Subsection Code (1)	5.5
49 thru 51	VIA Code (3)	5.77
52 thru 57	SID/STAR/AWY Ident (6) (Note 1)	5.78
58 thru 60	AREA Code (3)	5.3
61	Level (1)	5.19
62	Route Type (1)	5.7
63 thru 67	Initial Airport/Fix (5)	5.194
68 thru 69	ICAO Code (2)	5.14
70	Section Code (1)	5.4
71	Subsection Code (1)	5.5
72 thru 76	Terminus Airport/Fix (5)	5.194
77 thru 78	ICAO Code (2)	5.14
79	Section Code (1)	5.4
80	Subsection Code (1)	5.5
81 thru 85	Minimum Altitude (5)	5.30
86 thru 90	Maximum Altitude (5)	5.127
91	Time Code (1)	5.131
92 thru 93	Aircraft Use Group (2)	5.221
94	Direction Restriction (1)	5.115
95	Altitude Description (1)	5.29
96 thru 100	Altitude One (5)	5.30
101 thru 105	Altitude Two (5)	5.30
106	**SID/STAR/App/Awy Route Type (1)**	**5.7**
107	**S/S/A Route Type Qualifier 1 (1)**	**5.7**
108	**S/S/A Route Type Qualifier 2 (1)**	**5.7**
109 thru 123	**Reserved (Expansion) (15)**	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: The Standard Enroute Airway Identifier is five characters. Some users envision the need for a sixth character. This field length will permit such coding, see Section 5.8.

4.1.24.2 Preferred Route Continuation Records

Column	Field Name (Length)	Reference
1 thru 38	Fields as on Primary Records	
39	Continuation Record No. (1)	5.16
40	Application Type (1)	5.91
41	Time Code (1)	5.131
42	Time Indicator (1)	5.138
43 thru 52	Time of Operation (10)	5.195
53 thru 62	Time of Operation (10)	5.195
63 thru 72	Time of Operation (10)	5.195
73 thru 82	Time of Operation (10)	5.195
83 thru 92	Time of Operation (10)	5.195
93 thru 102	Time of Operation (10)	5.195
103 thru 112	Time of Operation (10)	5.195
113 thru 123	Reserved (Expansion) (11)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.24.3 Preferred Route Continuation Record (ET)

Columns	Field Name (Length)	Reference
1 thru 38	Fields as on Primary Records	
39	Continuation Record No. (1)	5.16
40	Application Type (1)	5.91
41 thru 109	Notes (69)	5.61
110 thru 123	Reserved (Expansion) (14)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note: Section 5.221 describes the use of this record for Aircraft Use Groups.

4.1.25 Controlled Airspace Records (UC)

The Controlled Airspace Record file contains a sequential listing of vertical and lateral limits of all types and classifications of Controlled Airspace. It includes Controlled Airspace associated with Airports and Heliports.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.25.1 Controlled Airspace Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 8	ICAO Code (2)	5.14
9	Airspace Type (1)	5.213
10 thru 14	Airspace Center (5)	5.214
15	Section Code (1)	5.4
16	Subsection Code (1)	5.5
17	Airspace Classification (1)	5.215
18 thru 19	Reserved (Spacing) (2)	
20	Multiple Code (1)	5.130
21 thru 24	Sequence Number (4)	5.12
25	Continuation Record Number (1)	5.16
26	Level (1)	5.19
27	Time Code (1)	5.131
28	NOTAM (1)	5.132
29 thru 30	Blank (Spacing) (2)	
31 thru 32	Boundary Via (2)	5.118
33 thru 41	Latitude (9)	5.36
42 thru 51	Longitude (10)	5.37
52 thru 60	Arc Origin Latitude (9)	5.36
61 thru 70	Arc Origin Longitude (10)	5.37
71 thru 74	Arc Distance (4)	5.119
75 thru 78	Arc Bearing (4)	5.120
79 thru 81	RNP (3)	5.211
82 thru 86	Lower Limit (5)	5.121
87	Unit Indicator (1)	5.133
88 thru 92	Upper Limit (5)	5.121
93	Unit Indicator (1)	5.133
94 thru 123	Controlled Airspace Name (30)	5.216
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.25.2 Controlled Airspace Continuation Records

Column	Filed Name (Length)	Reference
1 thru 24	Fields as on Primary Records	
25	Continuation Record Number (1)	5.16
26	Application Type (1)	5.91
27	Time Code (1)	5.131
28	NOTAM (1)	5.132
29	Time Indicator (1)	5.138
30 thru 39	Time of Operations (10)	5.195
40 thru 49	Time of Operations (10)	5.195
50 thru 59	Time of Operations (10)	5.195
60 thru 69	Time of Operations (10)	5.195
70 thru 79	Time of Operations (10)	5.195
80 thru 89	Time of Operations (10)	5.195
90 thru 99	Time of Operations (10)	5.195
100 thru 123	Controlling Agency (24)	5.140
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.26 Geographical Reference Table Records (TG)

The Geographical Reference Table file contains information that permits the cross referencing of otherwise undefined geographical entities and Route Identifiers in the Preferred Route file. The contents are not standardized and may vary from data supplier to data supplier. The contents of such a file can only be used in conjunction with the Preferred Route file of the same database in which the file is presented.

4.1.26.1 Geographical Reference Table Primary Records (TG)

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 8	Geographical Ref Table ID (2)	5.218
9	Sequence Number (1)	5.12
10 thru 38	Geographical Entity (29)	5.219
39	Continuation Record No (1)	5.16
40	Reserved (1)	
41 thru 50	Preferred Route Ident (10)	5.8
51 thru 52	Preferred Route Use Ind (2)	5.220
53 thru 62	Preferred Route Ident (10)	5.8
63 thru 64	Preferred Route Use Ind (2)	5.220
65 thru 74	Preferred Route Ident (10)	5.8
75 thru 76	Preferred Route Use Indi (2)	5.220
77 thru 86	Preferred Route Ident (10)	5.8
87 thru 88	Preferred Route Use Indi (2)	5.220
89 thru 98	Preferred Route Ident (10)	5.8
99 thru 100	Preferred Route Use Indi (2)	5.220
101 thru 110	Preferred Route Ident (10)	5.8
111 thru 112	Preferred Route Use Ind (2)	5.220
113 thru 123	Blank (Spacing) (11)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.26.2 Geographical Reference Table Continuation Records

Column	Field Name (Length)	Reference
1 thru 38	Fields as on Primary Records	
39	Continuation Record No. (1)	5.16
40	Application Type (1)	5.91
41 thru 123	Reserved (Expansion) (83)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.27 Flight Planning Arrival/Departure Data Records (PR)

The Flight Planning Arrival/Departure Data Record is used to provide the sub-set of data defining SIDs (PD), STARs (PE) and Approach Procedures (PF) from Section 4.1.9 required for the computer generation of Flight Plans which include Terminal Procedures. The file contains a sequential listing of published Arrival Procedures, Approach Procedures and Departure Procedures, the available Enroute and Runway Transitions for those procedures, the Transition waypoints, the appropriate along track distance fields and the intermediate fixes along those routes.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.27.1 Flight Planning Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 19	SID/STAR/Approach Identifier (6)	5.9, 5.10
20	Procedure Type (1)	5.230
21 thru 25	Runway Transition Identifier (5)	5.11
26 thru 30	Runway Transition Fix (5)	5.13
31 thru 32	ICAO Code (2)	5.14
33	Section Code (1)	5.4
34	Subsection Code (1)	5.5
35 thru 37	Runway Transition Along Track Distance (3)	5.231
38 thru 42	Common Segment Transition Fix (5)	5.13
43 thru 44	ICAO Code (2)	5.14
45	Section Code (1)	5.4
46	Subsection Code (1)	5.5
47 thru 49	Common Segment Along Track Distance (3)	5.231
50 thru 54	Enroute Transition Identifier (5)	5.11
55 thru 59	Enroute Transition Fix (5)	5.13
60 thru 61	ICAO Code (2)	5.14
62	Section Code (1)	5.4
63	Subsection Code (1)	5.5
64 thru 66	Enroute Transition Along Track Distance (3)	5.231
67 thru 69	Sequence Number (3)	5.12
70	Continuation Number (1)	5.16
71 thru 74	Number of Engines (4)	5.232
75	Turboprop/Jet Indicator (1)	5.233
76	RNAV Flag (1)	5.234
77	ATC Weight Category (1)	5.235
78 thru 84	ATC Identifier (7)	5.236
85	Time Code (1)	5.131
86 thru 100	Procedure Description (15)	5.237
101 thru 102	Leg Type Code (2)	5.238
103	Reporting Code (1)	5.239
104 thru 107	Initial Departure Magnetic Course (4)	5.26
108	Altitude Description (1)	5.29
109 thru 111	Altitude (3)	5.240
112 thru 114	Altitude (3)	5.240
115 thru 117	Speed Limit (3)	5.72
118 thru 119	Initial Cruise Table (2)	5.134
120	Speed Limit Description (1)	5.261
121 thru 123	Blank (Spacing) (3)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.27.2 Flight Planning Continuation Records

This Flight Planning Arrival/Departure Data Continuation Record is provided when Intermediate Fix information is required for the procedure coded in the Primary Record.

Column	Field Name (Length)	Reference
1 thru 69	Fields as on Primary Records	
70	Continuation Number (1)	5.16
71	Application Type (1)	5.91
72 thru 76	Intermediate Fix Identifier (5)	5.13
77 thru 78	ICAO Code (2)	5.14
79	Section Code (1)	5.4
80	Subsection Code (1)	5.5
81 thru 83	Intermediate Distance (3)	5.231
84	Fix Related Transition Code (1)	5.241
85 thru 89	Intermediate Fix Identifier (5)	5.13
90 thru 91	ICAO Code (2)	5.14
92	Section Code (1)	5.4
93	Subsection Code (1)	5.5
94 thru 96	Intermediate Distance (3)	5.231
97	Fix Related Transition Code (1)	5.241
98 thru 102	Intermediate Fix Identifier (5)	5.13
103 thru 104	ICAO Code (2)	5.14
105	Section Code (1)	5.4
106	Subsection Code (1)	5.5
107 thru 109	Intermediate Distance (3)	5.231
110	Fix Related Transition Code (1)	5.241
111 thru 115	Intermediate Fix Identifier (5)	5.13
116 thru 117	ICAO Code (2)	5.14
118	Section Code (1)	5.4
119	Subsection Code (1)	5.5
120 thru 122	Intermediate Distance (3)	5.231
123	Fix Related Transition Code (1)	5.241
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.27.3 Flight Planning Continuation Records

Column	Field Name (Length)	Reference
1 thru 69	Fields as on Primary Records	
70	Continuation Number (1)	5.16
71	Application Type (1)	5.91
72	Time Code (1)	5.131
73	Time Indicator (1)	5.138
74 thru 83	Time of Operation (10)	5.195
84 thru 93	Time of Operation (10)	5.195
94 thru 103	Time of Operation (10)	5.195
104 thru 113	Time of Operation (10)	5.195
114 thru 123	Time of Operation (10)	5.195
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note: Record 4.1.27.4 would carry Time of Operation in “note” form starting with column 74 and ending in column 123, where required.

4.1.28 Path Point Records (PP)

This file will contain Path Point records for RNAV GPS/GLS Approach Procedures.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.28.1 Path Point Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (1)	
7 thru 10	*Airport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 19	Approach Procedure Ident (6)	5.10
20 thru 24	*Runway or Helipad Identifier (5) Note 3	5.46 or 5.180
25 thru 26	*Operation Type (2)	5.223
27	Continuation Record Number (1)	5.16
28	*Route Indicator (1)	5.224
29 thru 30	*SBAS Service Provider Identifier (2)	5.255
31 thru 32	*Reference Path Data Selector (2)	5.256
33 thru 36	*Reference Path Identifier (4)	5.257
37	*Approach Performance Designator (1)	5.258
38 thru 48	*Landing Threshold Point Latitude (11)	5.267
49 thru 60	*Landing Threshold Point Longitude (12)	5.268
61 thru 66	*(LTP) Ellipsoid Height (6)	5.225
67 thru 70	*Glide Path Angle (4)	5.226
71 thru 81	*Flight Path Alignment Point Latitude (11)	5.267
82 thru 93	*Flight Path Alignment Point Longitude (12)	5.268
94 thru 98	*Course Width at Threshold (5) Note 4	5.228
99 thru 102	*Length Offset (4)	5.259
103 thru 108	*Path Point TCH (6)	5.265
109	*TCH Units Indicator (1)	5.266
110 thru 112	*HAL (3)	5.263
113 thru 115	*VAL (3)	5.264
116 thru 123	SBAS FAS Data CRC Remainder (8)	5.229
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: In the Path Point Record description, the field prefixed with * in the Field Name are those columns that have been determined as required for the data wrap for CRC calculations

Note 2: In order to properly convert values and binary pack these fields for the CRC data wrap, refer to RTCA/DO-229 "Minimum Operational Performance Standards for Global Positioning System/Wide Area Augmentation System Airborne Equipment for Final Approach Segment (FAS) Data Block CRC standards.

Note 3: For procedures designed for helicopter options to a pad or point in space, the runway number is set to "00", the runway letter is set to blank and the pad identifier and/or final approach course are carried in the Continuation Record.

Note 4: When the runway number is set to 00, then the Course Width field is ignored and the course width is 38 meters.

4.1.28.2 Path Point Continuation Records

Column	Field Name (Length)	Reference
1 thru 26	Fields as on Primary Record Type	
27	Continuation Record Number (1)	5.16
28	Application Type (1)	5.91
29 thru 34	(FPAP) Ellipsoid Height (6)	5.225
35 thru 40	(FPAP) Orthometric Height (6)	5.227
41 thru 46	(LTP) Orthometric Height (6)	5.227
47 thru 56	Approach Type Identifier (10)	5.262
57 thru 61	GNSS Channel Number (5)	5.244
62 thru 71	Blank (Spacing) (10)	
72 thru 74	Helicopter Procedure Course (3)	5.269
75 thru 123	Blank (Spacing) (49)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.29 GLS Record (PT)

This record contains a sequential listing of all GNSS Landing Systems (GLS) approaches, including the slope, course and reference path idents of the GLS approach. A GLS approach is identified by its ident and channel. Note that several GLS approaches can be supported by a single differential GLS ground station.

4.1.29.1 GLS Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (1)	
7 thru 10	Airport or Heliport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection code (1)	5.5
14 thru 17	GLS Ref Path Identifier (4)	5.44
18	GLS Category (1)	5.80
19 thru 21	Blank (3)	
22	Continuation Number (1)	5.16
23 thru 27	GLS Channel (5)	5.244
28 thru 32	Runway Identifier (5)	5.46
33 thru 51	Blank (19)	
52 thru 55	GLS Approach Bearing (4) (Note 1)	5.47
56 thru 64	Station Latitude (9)	5.36
65 thru 74	Station Longitude (10)	5.37
75 thru 78	GLS Station ident (4)	5.243
79 thru 83	Blank (5)	
84 thru 85	Service Volume Radius (2)	5.245
86 thru 87	TDMA Slots (2)	5.246
88 thru 90	GLS Approach Slope (3)	5.52
91 thru 95	Magnetic Variation (5)	5.39
96 thru 97	Reserved (2)	
98 thru 102	Station Elevation (5)	5.74
103 thru 105	Datum Code (3)	5.197
106 thru 108	Station Type (3)	5.247
109 thru 110	Blank (2)	
111 thru 115	Station Elevation WGS 84 (5)	5.248
116 thru 123	Blank (8)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: GLS reference point should be equal to IFR landing threshold position, as it is the trajectory reference point.

Note 2: All the latitudes/longitudes of the record refer to the same datum code.

4.1.29.2 GLS Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 123	Reserved (Expansion) (100)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.30 Alternate Record (RA)

The Alternate Record file contains a listing of up to six Alternate Airport Identifiers or, up to six Alternate Company Route Identifiers or any combination of Alternate Airport or Alternate Route Identifiers for a given departure airport, destination airport or enroute fix. The data content of the record is customer defined.

4.1.30.1 Alternate Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 11	Alternate Related Airport or Fix (5)	5.75
12 thru 13	Alternate Related ICAO Code (2)	5.14
14	Alternate Related Section Code (1)	5.4
15	Alternate Related Subsection Code (1)	5.5
16 thru 17	Alternate Record Type (2)	5.250
18 thru 19	Blank (Spacing) (2)	
20 thru 22	Distance to Alternate (3)	5.251
23	Alternate Type (1)	5.252
24 thru 33	Primary Alternate Identifier (10)	5.253
34 thru 35	Blank (Spacing) (2)	
36 thru 38	Distance to Alternate (3)	5.251
39	Alternate Type (1)	5.252
40 thru 49	Additional Alternate Identifier One (10)	5.253
50 thru 51	Blank (Spacing) (2)	
52 thru 54	Distance to Alternate (3)	5.251
55	Alternate Type (1)	5.252
56 thru 65	Additional Alternate Identifier Two (10)	5.253
66 thru 67	Blank (Spacing) (2)	
68 thru 70	Distance to Alternate (3)	5.251
71	Alternate Type (1)	5.252
72 thru 81	Additional Alternate Identifier Three (10)	5.253
82 thru 83	Blank (Spacing) (2)	
84 thru 86	Distance to Alternate (3)	5.251
87	Alternate Type (1)	5.252
88 thru 97	Additional Alternate Identifier Four (10)	5.253
98 thru 99	Blank (Spacing) (2)	
100 thru 102	Distance to Alternate (3)	5.251
103	Alternate Type (1)	5.252
104 thru 113	Additional Alternate Identifier Five (10)	5.253
114 thru 123	Reserved (expansion) (10)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.31 Airport TAA (PK)

The Airport Terminal Arrival Altitude (TAA) file contains details relating to TAA sectorization and sector altitudes.

4.1.31.1 Airport TAA Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 19	Approach Identifier (6)	5.10
20 thru 24	**TAA IAF Waypoint (5)**	**5.273**
25 thru 26	**ICAO Code (2)**	**5.14**
27	**Section Code (1)**	**5.4**
28	**Subsection Code (1)**	**5.5**
29	**TAA Fix Position Indicator (1)**	**5.272**
30	**Continuation Record No. (1)**	**5.16**
31 thru 32	**Blank (2)**	
33 thru 38	**Sector Bearing (6)**	**5.146**
39 thru 41	**Sector Minimum Altitude (3)**	**5.147**
42 thru 45	**Sector Radius 1 (4)**	**5.274**
46	**Procedure Turn Indicator (1)**	**5.271**
47 thru 52	**Sector Bearing (6)**	**5.146**
53 thru 55	**Sector Minimum Altitude (3)**	**5.147**
56 thru 59	**Sector Radius 1 (4)**	**5.274**
60	**Procedure Turn Indicator (1)**	**5.271**
61 thru 66	**Sector Bearing (6)**	**5.146**
67 thru 69	**Sector Minimum Altitude (3)**	**5.147**
70 thru 73	**Sector Radius 1 (4)**	**5.274**
74	**Procedure Turn Indicator (1)**	**5.271**
75 thru 80	**Sector Bearing (6)**	**5.146**
81 thru 83	**Sector Minimum Altitude (3)**	**5.147**
84 thru 87	**Sector Radius 1 (4)**	**5.274**
88	**Procedure Turn Indicator (1)**	**5.271**
89 thru 94	**Sector Bearing (6)**	**5.146**
95 thru 101	**Sector Radius 1 (4)**	**5.274**
102	**Procedure Turn Indicator (1)**	**5.271**
103 thru 108	**Sector Bearing (6)**	**5.146**
109 thru 111	**Sector Minimum Altitude (3)**	**5.147**
112 thru 115	**Sector Radius 1 (4)**	**5.274**
116	**Procedure Turn Indicator (1)**	**5.271**
117 thru 119	**Blank (Spacing) (3)**	
120	**Mag/True Indicator (1)**	**5.165**
121 thru 123	**Blank (Spacing) (3)**	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.0 NAVIGATION DATA - RECORD LAYOUT

4.1.31.2 Airport Terminal Arrival Altitude Continuation Records

Column	Field Name (Length)	Reference
1 thru 30	Fields as on Primary Records	
31	Application Type (1)	5.91
32	Blank (Spacing) (1)	
33 thru 38	Sector Bearing (6)	5.146
39 thru 41	Sector Minimum Altitude (3)	5.147
42 thru 45	Sector Radius 1 (4)	5.274
46	Procedure Turn Indicator (1)	5.271
47 thru 52	Sector Bearing (6)	5.146
53 thru 55	Sector Minimum Altitude (3)	5.147
56 thru 59	Sector Radius 1 (4)	5.274
60	Procedure Turn Indicator (1)	5.271
61 thru 66	Sector Bearing (6)	5.146
67 thru 69	Sector Minimum Altitude (3)	5.147
70 thru 73	Sector Radius 1 (4)	5.274
74	Procedure Turn Indicator (1)	5.271
75 thru 80	Sector Bearing (6)	5.146
81 thru 83	Sector Minimum Altitude (3)	5.147
84 thru 87	Sector Radius 1 (4)	5.274
88	Procedure Turn Indicator (1)	5.271
89 thru 109	Notes (21)	5.61
110 thru 123	Reserved (Expansion) (14)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.32 TACAN-Only NAVAID Record (DT)

The TACAN-only NAVAID file contains certain details of TACAN stations with a duplicate identifier (same ident and ICAO Code) navaid in the VHF Navaid (D) file.

4.1.32.1 TACAN-Only NAVAID Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 10	Airport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Blank (Spacing) (1)	
14 thru 17	VOR Identifier (4)	5.33
18 thru 21	Blank (Spacing) (4)	
22	Continuation Record No. (1)	5.16
23 thru 27	VOR Frequency) (5)	5.34
28 thru 32	NAVAID Class (5)	5.35
33 thru 51	Blank (Spacing) (19)	
52 thru 55	TACAN Ident (4)	5.38
56 thru 64	TACAN Latitude (9)	5.36
65 thru 74	TACAN Longitude (10)	5.37
75 thru79	Station Declination (5)	5.66
80 thru 84	TACAN Elevation (5)	5.40
85	Figure of Merit (1)	5.149
86 thru 87	Blank (Spacing) (2)	
88 thru 90	Frequency Protection (3)	5.150
91 thru 93	Datum Code (3)	5.197
94 thru 123	TACAN Name (30)	5.71
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.33 Special Activity Area (ES)

The Special Activity Area (SAA) file contains details relating to operation that could be hazardous to aeronautical navigation around a specified location.

4.1.33.1 Special Activity Area Primary Record

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer /Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7	SAA Type (1)	5.278
8 thru 13	SAA Identifier (6)	5.279
14 thru 15	ICAO Code (2)	5.14
16 thru19	Airport Identifier (4)	5.6
20 thru 21	ICAO Code (2)	5.14
22	Continuation Record No. (1)	5.16
23	Blank (Spacing) (1)	
24 thru 32	Latitude (9)	5.36
33 thru 42	Longitude (10)	5.37
43 thru 45	SAA Size (3)	5.280
46 thru 51	Upper Limit (5)	5.121
52	Unit Indicator (1)	5.133
53	SAA Volume (1)	5.281
54 thru 56	Operating Times (3)	5.282
57	Public or Military (1)	5.177
58	Blank (Spacing) (1)	
59 thru 83	Controlling Agency (25)	5.140
84 thru 86	Communicating Type (3)	5.101
87 thru 93	Communication Frequency (7)	5.103
94 thru 123	Special Activity Area Name (30)	5.126
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.1.34 Communication Type Translation (TV)

Column	Field Name (Length)	Reference
1	Record Type (1)	5.3
2 thru 4	Reserved (3)	
5	Section (1)	5.4
6	Subsection (1)	5.5
7 thru 9	Communication Type (3)	5.101
10	Type Recognized By (1)	5.287
11 thru 90	Translation (80)	5.288
91	Used On (1)	5.289
92 thru 95	Communication Class (4)	5.283
96 thru 123	Blank (Spacing) (28)	
124 thru 128	File Record Number (1)	5.31
129 thru 132	Cycle Date (1)	5.32

4.2 Master Helicopter User File (HA)

This section contains record information unique to helicopter operations. In addition to the records identified in this Section, records identified in Section 4.1 Master Airline User File are used in the Master Helicopter User File.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.2.1 Heliport Records (HA)

This file will contain heliport information.

4.2.1.1 Heliport Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Heliport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 16	ATA/IATA Designator (3)	5.107
17 thru 21	PAD Identifier (5)	5.180
22	Continuation Record No. (1)	5.16
23 thru 27	Speed Limit Altitude (5)	5.73
28 thru 30	Datum Code (3)	5.197
31	IFR Indicator (1)	5.108
32	Blank (Spacing) (1)	
33 thru 41	Latitude (9)	5.36
42 thru 51	Longitude (10)	5.37
52 thru 56	Magnetic Variation (5)	5.39
57 thru 61	Heliport Elevation (5)	5.55
62 thru 64	Speed Limit (3)	5.72
65 thru 68	Recommended VHF Navaid (4)	5.23
69 thru 70	ICAO Code (2)	5.14
71 thru 75	Transition Altitude (5)	5.53
76 thru 80	Transition Level (5)	5.53
81	Public Military Indicator (1)	5.177
82 thru 84	Time Zone (3)	5.178
85	Daylight Indicator (1)	5.179
86 thru 91	Pad Dimension	5.176
92	Magnetic/True Indicator (1)	5.165
93	Reserved (Expansion) (1)	
94 thru 123	Heliport Name (30)	5.71
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.1.2 Heliport Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 92	Notes (69)	5.61
93 thru 123	Reserved (Expansion) (31)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.1.3 Heliport Flight Planning Continuation Records

This Continuation Record is used to indicate the FIR and UIR within which the Airport defined in the Primary Record resides, to provide the Start/End validity dates/times of the Primary Record, and to provide an indication if the Heliport defined in the Primary Record is associated with Controlled Airspace.

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record Number (1)	5.16
23	Application Type (1)	5.91
24 thru 27	FIR Identifier (4)	5.116
28 thru 31	UIR Identifier (4)	5.116
32	Start/End Indicator (1)	5.152
33 thru **66**	**Blank (Spacing) (34)**	
67	Controlled A/S Indicator (1)	5.217
68 thru 71	Controlled A/S Airport Identifier (4)	5.6
72 thru 73	Controlled A/S Airport ICAO (2)	5.14
74 thru 123	Blank (Spacing) (50)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.1.4 Heliport Flight Planning Continuation Records

Deleted by Supplement 19.

4.2.2 Heliport Terminal Waypoint Records (HC)

The Heliport Terminal Waypoint file contains all terminal waypoints and VFR waypoint within the geographical area of each heliport. Heliport Terminal Waypoints utilized by two or more heliports will be stored in the Enroute Waypoint file to eliminate duplication. Terminal Waypoints used jointly by an airport and a heliport are also stored in the Enroute waypoint file.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.2.2.1 Heliport Terminal Waypoint Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Heliport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 18	Waypoint Identifier (5)	5.13
19	Blank (Spacing) (1)	
20 thru 21	ICAO Code (2)	5.14
22	Continuation Record Number (1)	5.16
23 thru 26	Blank (Spacing) (4)	
27 thru 29	Waypoint Type (3)	5.42
30 thru 31	Waypoint Usage (2)	5.82
32	Blank (Spacing) (1)	
33 thru 41	Waypoint Latitude (9)	5.36
42 thru 51	Waypoint Longitude (10)	5.37
52 thru 74	Blank (Spacing) (23)	
75 thru 79	Dynamic Magnetic Variation (5)	5.39
80 thru 84	Reserved (Expansion) (5)	
85 thru 87	Datum Code (3)	5.197
88 thru 95	Reserved (Expansion) (8)	
96 thru 98	Name Format Indicator (3)	5.196
99 thru 123	Waypoint Name/Description (25)	5.43
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.2.2 Heliport Terminal Waypoint Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record Number (1)	5.16
23	Application Type) (1)	5.91
24 thru 92	Notes (69)	5.61
93 thru 123	Reserved (Expansion) (31)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.2.3 Heliport Terminal Waypoint Flight Planning Continuation Records

This Continuation Record is used to indicate the FIR and UIR within which the Waypoint defined in the Primary Records is located and to provide the Start/End validity dates/times of the Primary Record where applicable.

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record Number (1)	5.16
23	Application Type (1)	5.91
24 thru 27	FIR Identifier (4)	5.116
28 thru 31	UIR Identifier (4)	5.116
32	Start/End Indicator (1)	5.152
33 thru 43	**Blank (Spacing) (11)**	
44 thru 123	Reserved (Expansion) (80)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.2.4 Heliport Terminal Waypoint Flight Planning Continuation Records

Deleted by Supplement 19.

4.2.3 Heliport SID/STAR/Approach (HD/HE/HF)

Heliport SIDs, STARs and Approach Procedures are contained in three separate section/subsection groupings, using this single record format. Section/Subsection HD contains a sequential listing of those published Heliport Standard Instrument Departures that can be encoded according to this specification. Section/Subsection HE contains a sequential list of those published Heliport Standard Terminal Arrival Routes that can be encoded according to this specification. Section/Subsection HF contains a sequential listing of those published Heliport Standard Instrument Approach Procedures that can be encoded according to this specification.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.2.3.1 Heliport SID/STAR/Approach Primary Records

Column	Field Name (Length)		Reference
1	Record Type (1)		5.2
2 thru 4	Customer/Area Code (3)		5.3
5	Section Code (1)		5.4
6	Blank (Spacing) (1)		
7 thru 10	Heliport Identifier (4)		5.6
11 thru 12	ICAO Code (2)		5.14
13	Subsection Code (1)		5.5
14 thru 19	SID/STAR/APP Identifier (6)	Note 1	5.9, 5.10
20	Route Type (1)		5.7
21 thru 25	Transition Identifier (5)		5.11
26	Blank (Spacing) (1)		
27 thru 29	Sequence Number (3)		5.12
30 thru 34	Fix Identifier (5)		5.13
35 thru 36	ICAO Code (2)		5.14
37	Section Code (1)		5.4
38	Subsection Code (1)		5.5
39	Continuation Record Number (1)		5.16
40 thru 43	Waypoint Description Code (4)		5.17
44	Turn Direction (1)		5.20
45 thru 47	RNP (3)	Note 4	5.211
48 thru 49	Path and Termination (2)		5.21
50	Turn Direction Valid (1)		5.22
51 thru 54	Recommended Navaid (4)		5.23
55 thru 56	ICAO Code (2)		5.14
57 thru 62	ARC Radius (6)		5.204
63 thru 66	Theta (4)		5.24
67 thru 70	Rho (4)		5.25
71 thru 74	Magnetic Course (4)		5.26
75 thru 78	Route Distance/Holding Distance or Time (4)		5.27
79	Recommended Navaid Section (1)		5.4
80	Recommended Navaid Subsection (1)		5.5
81 thru 82	Reserved (Spacing) (2)		
83	Altitude Description (1)		5.29
84	ATC Indicator (1)		5.81
85 thru 89	Altitude (5)		5.30
90 thru 94	Altitude (5)		5.30
95 thru 99	Transition Altitude (5)		5.53
100 thru 102	Speed Limit (3)		5.72
103 thru 106	Vertical Angle (4)		5.70
107 thru 111	Center Fix or TAA Procedure Turn Indicator (5)		5.144 or 5.271
112	Multiple Code or TAA, Sector Identifier (1)		5.130 or 5.272
113 thru 114	ICAO Code (2)	Note 3	5.14
115	Section Code (1)	Note 3	5.4
116	Subsection Code (1)	Note 3	5.5
117	GNSS/FMS Indicator (1)		5.222
118	Speed Limit Description (1)		5.261
119	Apch Route Qualifier 1 (1)	Note 2	5.7
120	Apch Route Qualifier 2 (1)	Note 2	5.7
121 thru 123	**VSF (3)**		**5.293**
124 thru 128	File Record Number (5)		5.31
129 thru 132	Cycle Date (4)		5.32

Note 1: For approach route idents including "Multiple Indicators" see Section 5.10.

Note 2: Columns 119 and 120 (Approach Route Qualifier 1 and 1) are required to match the Primary Record to the Continuation Record(s). This non-standard column sorting sequence was selected to preserve the Primary Record Layout for SID/STAR/Approach Records as much as possible as these new fields were introduced in Supplement 14.

Note 3: When columns 107 thru 116 are providing a reference to a MSA or the center fix for an RF leg, all of the columns are used. When they are providing a reference to a TAA, only columns 107 thru 112 are used and 113 thru 116 are blank.

Note 4: If there is only one set of RNP criteria for the RNAV procedure, that criteria is provided in the RNP value field for Primary Record. Otherwise, the Primary Record contains one consistent set of RNP values for the least restrictive RNAV operating criteria and not a mix of RNP values for different RNP operating criteria.

4.2.3.2 Heliport SID/STAR/Approach Primary Extension Continuation Records

Column	Field Name (Length)		Reference
1 thru 38	Field as on Primary Records		
39	Continuation Record Number (1)		5.16
40	Application Type (1)		5.91
41 thru 44	CAT A Decision Height (4)		5.170
45 thru 48	CAT B Decision Height (4)		5.170
49 thru 52	CAT C Decision Height (4)		5.170
53 thru 56	CAT D Decision Height (4)		5.170
57 thru 60	CAT A Minimum Descent Height (4)		5.171
61 thru 64	CAT B Minimum Descent Height (4)		5.171
65 thru 68	CAT C Minimum Descent Height (4)		5.171
69 thru 72	CAT D Minimum Descent Height (4)		5.171
73 thru 75	Procedure TCH (3)		5.67
76	Localizer Altitude Description (1)		5.29
77 thru 81	Localizer Altitude (5)		5.30
82 thru 85	Localizer VNAV (4)		5.70
86 thru 89	**Procedure Category (4)**		**5.242**
90 thru 92	RNP (3)	Note 2	5.211
93 thru 97	**Procedure Design Mag Var (5)**	**Note 3**	**5.290**
98	**Procedure Design Mag Var Ind (1)**	**Note 3**	**5.291**
99 thru 106	Reserved (Expansion) **(8)**		
107 thru 110	**Reserved**		
111 thru 112	**ICAO Code (2)**		**5.14**
113 thru116	**Reserved**		
117 thru 118	**ICAO Code (2)**		**5.14**
119	Apch Route Qualifier 1 (1)	Note 1	5.7
120	Apch Route Qualifier 2 (1)	Note 1	5.7
121 thru 123	Blank (Spacing) (3)		
124 thru 128	File Record Number (5)		5.31
129 thru 132	Cycle Date (4)		5.32

Note 1: Columns 119 and 120 (Approach Route Qualifier 1 and 2) are required to match the Primary Record to the Continuation Record(s). This non-standard column sorting sequence was selected to preserve the Primary Record for SID/STAR/Approach as much as possible at the time the record was introduced in Supplement 14.

Note 2: For records used to define RNAV Procedures with multiple RNP values and for RNAV Procedures published with multiple RNP operating criteria, the least restrictive values

are carried in columns 45 thru 47 of the Primary record for each leg. Columns 90 thru 92 of this Primary Extension Continuation Record will carry the additional government source RNP values. There may be more than one Primary Extension Record provided when the procedure has more than two sets of distinct RNP values.

Note 3: When government source provides Procedure Design Mag Var at the procedure level, a single Primary Extension Continuation Record will be provided, associated to the first sequence in each transition and the Procedure Design Mag Var Indicator will be set to "P". This is consistent with the intent of this continuation record. When government source provides Procedure Design Mag Var the leg level, a Primary Extension Continuation Record will be provided associates with each sequence of each transitions and the Procedure Design Mag Var Indicator will be set to "L".

4.2.3.3 Heliport SID/STAR/Approach Flight Planning Continuation Records

This Continuation Record is used to indicate the Leg Distance for each segment of the Route and to provided the Start/End validity dates/times of the Primary Record where applicable.

Column	Field Name (Length)	Reference
1 thru 38	Fields as on Primary Records	
39	Continuation Record Number (1)	5.16
40	Application Type (1)	5.91
41	Start/End Indicator (1)	5.152
42 thru **74**	**Blank (Spacing) (33)**	
75 thru 78	Route Distance (4)	5.27
79 thru 123	Reserved (Expansion) (45)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.3.4 Heliport SID/STAR/Approach Flight Planning Continuation Records

Deleted by Supplement 19.

4.2.3.5 Heliport Procedure Data Continuation Record

The Heliport Procedure Data Continuation Record is used to provide information on SBAS authorization levels. This Continuation Record is provided once per procedure as a Continuation to Primary Approach Procedure Record that contains the encoding for Final Approach Fix (FAF) of the procedure.

Column	Field Name (Length)	Reference
1 thru 38	Fields as on Primary Record	
39	Continuation Record Number (1)	5.16
40	Application Type (1)	5.91
41	FAS Block Provided (1)	5.276
42 thru 51	FAS Block Provided Lev of Serv Name (10)	5.275
52	LNAV/VNAV Authorized for SBAS (1)	5.276
53 thru 62	LNAV/VNAV Level of Service Name (10)	5.275
63	LNAV Authorized for SBAS (1)	5.276
64 thru 73	LNAV Level of Service Name (10)	5.275
74 thru 118	Blank (Spacing) (45)	
119	Approach Route Type Qualifier 1 (1) Note 1	5.7
120	Approach Route Type Qualifier 2 (1) Note 1	5.7
121 thru 123	Blank (Spacing) (3)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: Columns 119 and 120 (Approach Route Type Qualifiers 1 and 2) are required to match the Primary Record to the Continuation Record(s). This non-standard column sorting sequence was selected to preserve the Primary Record for SID/STAR/Approach Records as much as possible as these new fields were introduced in Supplement 14.

4.2.4 Heliport MSA (HS)

The Heliport Minimum Sector Altitude (MSA) file contains details relating to available Sector Altitudes.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.2.4.1 Heliport MSA Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Heliport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 18	MSA Center (5)	5.144
19 thru 20	ICAO Code (2)	5.14
21	Section Code (1)	5.4
22	Subsection Code (1)	5.5
23	Multiple Code (1)	5.130
24 thru 38	Reserved (Expansion) (15)	
39	Continuation Record Number (1)	5.16
40 thru 42	Reserved (Spacing) (3)	
43 thru 48	Sector Bearing (6)	5.146
49 thru 51	Sector Altitude (3)	5.147
52 thru 53	Sector Radius (2)	5.145
54 thru 59	Sector Bearing (6)	5.146
60 thru 62	Sector Altitude (3)	5.147
63 thru 64	Sector Radius (2)	5.145
65 thru 70	Sector Bearing (6)	5.146
71 thru 73	Sector Altitude (3)	5.147
74 thru 75	Sector Radius (2)	5.145
76 thru 81	Sector Bearing (6)	5.146
82 thru 84	Sector Altitude (3)	5.147
85 thru 86	Sector Radius (2)	5.145
87 thru 92	Sector Bearing (6)	5.146
93 thru 95	Sector Altitude (3)	5.147
96 thru 97	Sector Radius (2)	5.145
98 thru103	Sector Bearing (6)	5.146
104 thru 106	Sector Altitude (3)	5.147
107 thru 108	Sector Radius (2)	5.145
109 thru 114	Sector Bearing (6)	5.146
115 thru 117	Sector Altitude (3)	5.147
118 thru119	Sector Radius (2)	5.145
120	Magnetic/True Indicator (1)	5.165
121 thru 123	Reserved (Expansion) (3)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.4.2 Heliport MSA Primary Record Extension

Column	Field Name (Length)	Reference
1 thru 38	Field as on Primary Records	5.2
39	Continuation Record No. (1)	5.16
40	Application Type (1)	5.91
41 thru 42	Reserved (Spacing) (3)	
43 thru 48	Sector Bearing (6)	5.146
49 thru 51	Sector Altitude (3)	5.147
52 thru 53	Sector Radius (2)	5.145
54 thru 59	Sector Bearing (6)	5.146
60 thru 62	Sector Altitude (3)	5.147
63 thru 64	Sector Radius (2)	5.145
65 thru 70	Sector Bearing (6)	5.146
71 thru 73	Sector Altitude (3)	5.147
74 thru 75	Sector Radius (2)	5.145
76 thru 81	Sector Bearing (6)	5.146
82 thru 84	Sector Altitude (3)	5.147
85 thru 86	Sector Radius (2)	5.145
87 thru 92	Sector Bearing (6)	5.146
93 thru 95	Sector Altitude (3)	5.147
96 thru 97	Sector Radius (2)	5.145
98 thru 103	Sector Bearing (6)	5.146
104 thru 106	Sector Altitude (3)	5.147
107 thru 108	Sector Radius (2)	5.145
109 thru 114	Sector Bearing (6)	5.146
115 thru 117	Sector Altitude (3)	5.147
118 thru 119	Sector Radius (2)	5.145
120 thru 123	Field as on Primary Records	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.4.3 Heliport MSA Continuation Records

Column	Field Name (Length)	Reference
1 thru 38	Fields as on Primary Records	
39	Continuation Record Number (1)	5.16
40	Application Type (1)	5.91
41 thru 109	Notes (69)	5.61
110 thru 123	Reserved (Expansion) (14)	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.5 Heliport Communications Records (HV)

These files will contain Heliport Communications Facilities.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.2.5.1 Heliport Communications Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Heliport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 15	**Blank (Spacing) (2)**	
16 thru 19	**Communication Class (4)**	**5.283**
20 thru 21	**Sequence Number (2)**	**5.12**
22	**Continuation Number (1)**	**5.16**
23 thru 25	**Communication Types (3)**	**5.101**
26 thru 32	**Transmit Frequency (7)**	**5.103**
33 thru 39	**Receive Frequency (7)**	**5.103**
40	**Frequency Units (1)**	**5.104**
41	**Radar Units (1)**	**5.102**
42	**H24 Indicator (1)**	**5.181**
43 thru 67	**Call Signs (25)**	**5.105**
68	**Muilt-Sector Indicator (1)**	**5.286**
69 thru 74	**Sectorization (6)**	**5.183**
75 thru 78	**Sector Facility (4)**	**5.185**
79 thru 80	**ICAO (2)**	**5.14**
81	**Section Code (1)**	**5.4**
82	**Subsection Code (1)**	**5.5**
83	**Altitude Description Code (1)**	**5.29**
84 thru 86	**Communication Altitude 1 (3)**	**5.184**
87 thru 89	**Communication Altitude 2 (3)**	**5.184**
90	**Distance Description Code (1)**	**5.187**
91 thru 92	**Communication Distance (2)**	**5.188**
93 thru 101	**Transmitter Latitude (9)**	**5.36**
102 thru 111	**Transmitter Longitude (10)**	**5.37**
112 thru 114	**Service Indicator (3)**	**5.106**
115	**Modulation (1)**	**5.198**
116	**Signal Emission (2)**	**5.199**
117 thru 123	**Blank (Spacing) (7)**	
124 thru 128	File Record Number (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.5.2 Heliport Communications Continuation Records

Column	Field Name (Length)	Reference
1 thru *21*	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	Application Type (1)	5.91
24 thru 27	**Remote Facility**	**5.200**
28 thru 29	**ICAO (2)**	**5.14**
30	**Section (1)**	**5.4**
31	**Subsection (1)**	**5.5**
32 thru 36	**Transmitter Site Mag Var (5)**	**5.39**
37 thru 41	**Transmitter Site Elevation (5)**	**5.92**
42 thru 47	**Additional Sectorization 1 (6)**	**5.183**
48	**Additional Sectorization 1 Altitude Description (1)**	**5.29**
49 thru 51	**Additional Sectorization 1 Altitude 1 (3)**	**5.184**
52 thru 53	**Additional Sectorization 1 Altitude 2 (3)**	**5.184**
54 thru 59	**Additional Sectorization 2 (6)**	**5.183**
60	**Additional Sectorization 2 Altitude Description (1)**	**5.29**
61 thru 63	**Additional Sectorization 2 Altitude 1 (3)**	**5.184**
64 thru 66	**Additional Sectorization 2 Altitude 2 (3)**	**5.184**
67	**Time Code (1)**	**5.131**
68	**NOTAM (1)**	**5.132**
69	**Time Indicator (1)**	**5.138**
70 thru 79	**Time of Operation (10)**	**5.195**
80 thru 89	**Time of Operation (10)**	**5.195**
90 thru 99	**Time of Operation (10)**	**5.195**
100 thru 109	**Time of Operation (10)**	**5.195**
110 thru 119	**Time of Operation (10)**	**5.195**
120 thru 123	**Blank (Spacing) (4)**	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.5.3 Heliport Communications Continuation Records

Column	Field Name (Length)	Reference
1 thru **21**	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23	**Application Type (1)**	**5.91**
24 thru 83	**Sectorization Narrative (60)**	**5.186**
84 thru 123	Reserved **(Spacing) (40)**	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.0 NAVIGATION DATA - RECORD LAYOUT

4.2.5.4 Heliport Communications Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No.(1)	5.16
23	Application Type (1)	5.91
24 thru 69	Blank (Spacing)(46)	
70 thru 79	Time of Operation (10)	5.195
80 thru 89	Time of Operation (10)	5.195
90 thru 99	Time of Operation (10)	5.195
100 thru 109	Time of Operation (10)	5.195
110 thru 119	Time of Operation (10)	5.195
120 thru 123	Reserved (Expansion) (4)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.5.5 Heliport Communications Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Field as on Primary Record	
22	Continuation Record Number (1)	5.16
23	Application Type (1)	5.91
24 thru 123	Time Narrative (100)	5.285
124 thru 128	File Record numbers (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.6 Heliport (TAA) (HK)

The Heliport Terminal Arrival Altitude (TAA) file contains details relating to TAA sectorization and sector altitudes.

4.2.6.1 Heliport TAA Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Heliport Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.14
13	Subsection Code (1)	5.5
14 thru 19	Approach Identifier (6)	5.10
20 thru 24	TAA IAF Waypoint (5)	5.273
25 thru 26	ICAO Code (2)	5.14
27	Section Code (1)	5.4
28	Subsection Code (1)	5.5
29	TAA Fix Position Indicator (1)	5.272
30	Continuation Record No. (1)	5.16
31 thru 32	Blank (2)	
33 thru 38	Sector Bearing (6)	5.146
39 thru 41	Sector Minimum Altitude (3)	5.147
42 thru 45	Sector Radius 1 (4)	5.274
46	Procedure Turn Indicator (1)	5.271
47 thru 52	Sector Bearing (6)	5.146
53 thru 55	Sector Minimum Altitude (3)	5.147
56 thru 59	Sector Radius 1 (4)	5.274
60	Procedure Turn Indicator (1)	5.271
61 thru 66	Sector Bearing (6)	5.146
67 thru 69	Sector Minimum Altitude (3)	5.147
70 thru 73	Sector Radius 1 (4)	5.274
74	Procedure Turn Indicator (1)	5.271
75 thru 80	Sector Bearing (6)	5.146
81 thru 83	Sector Minimum Altitude (3)	5.147
84 thru 87	Sector Radius 1 (4)	5.274
88	Procedure Turn Indicator (1)	5.271
89 thru 94	Sector Bearing (6)	5.146
95 thru 101	Sector Radius 1 (4)	5.274
102	Procedure Turn Indicator (1)	5.271
103 thru 108	Sector Bearing (6)	5.146
109 thru 111	Sector Minimum Altitude (3)	5.147
112 thru 115	Sector Radius 1 (4)	5.274
116	Procedure Turn Indicator (1)	5.271
117 thru 119	Blank (Spacing) (3)	
120	Mag/True Indicator (1)	5.165
121 thru 123	Blank (3)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.0 NAVIGATION DATA - RECORD LAYOUT

4.2.6.2 Heliport Terminal Arrival Altitude Continuation Records

Column	Field Name (Length)	Reference
1 thru 30	Fields as on Primary Records	
31	Users (1)	5.91
32	Blank (Spacing) (1)	
33 thru 38	Sector Bearing (6)	5.146
39 thru 41	Sector Minimum Altitude (3)	5.147
42 thru 45	Sector Radius 1 (4)	5.274
46	Procedure Turn Indicator (1)	5.271
47 thru 52	Sector Bearing (6)	5.146
53 thru 55	Sector Minimum Altitude (3)	5.147
56 thru 59	Sector Radius 1 (4)	5.274
60	Procedure Turn Indicator (1)	5.271
61 thru 66	Sector Bearing (6)	5.146
67 thru 69	Sector Minimum Altitude (3)	5.147
70 thru 73	Sector Radius 1 (4)	5.274
74	Procedure Turn Indicator (1)	5.271
75 thru 80	Sector Bearing (6)	5.146
81 thru 83	Sector Minimum Altitude (3)	5.147
84 thru 87	Sector Radius 1 (4)	5.274
88	Procedure Turn Indicator (1)	5.271
89 thru 109	Notes (21)	5.61
110 thru 123	Reserved (Expansion) (14)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.2.7 Helicopter Operations Company Route Records (RH)

This file contains company tailored route information for helicopter operations.

4.2.7.1 Helicopter Operations Company Route Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 11	From Airport Heliport/Fix (5)	5.75
12 thru 16	Helipad Ident (5)	5.180
17	Blank (Spacing) (1)	
18 thru 19	ICAO Code (2)	5.14
20	Section Code (1)	5.4
21	Subsection Code (1)	5.5
22 thru 26	To Airport/Heliport/Fix (5)	5.75
27 thru 31	Helipad Ident (5)	5.180
32	Blank (Spacing) (1)	
33 thru 34	ICAO Code (2)	5.14
35	Section Code (1)	5.4
36	Subsection Code (1)	5.5
37 thru 46	Company Route ID (10)	5.76
47 thru 49	Sequence No. (3)	5.12
50 thru 52	VIA Code (3)	5.77
53 thru 58	SID/STAR/App/Awy (6)	5.78
59	Section Code (1)	5.4
60	Subsection Code (1)	5.5
61	SID/STAR/App/Awy Route Type (1)	5.7
62	S/S/A Route Type Qualifier 1 (1)	5.7
63	S/S/A Route Type Qualifier 2 (1)	5.7
64 thru 66	Area Code (3)	5.3
67 thru 72	To Fix (6)	5.83
73 thru 74	ICAO Code (2)	5.14
75	Section Code (1)	5.4
76	Subsection Code (1)	5.5
77 thru 81	Runway/Helipad Transition ID (5)	5.84
82 thru 86	Enroute Transition ID(5)	5.85
87	Reserved (1)	
88 thru 92	Cruise Altitude (5)	5.86
93 thru 96	Terminal/Alternate Heliport (4)	5.87
97 thru 98	ICAO Code (2)	5.14
99	Section Code (1)	5.4
100	Subsection Code (1)	5.5
101 thru 104	Alternate Distance (4)	5.88
105 thru 107	Cost Index (3)	5.89
108 thru 111	Enroute Alternate Heliport (4)	5.148
112 thru 123	Reserved (Expansion) (12)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: This Helicopter Operations Company Route Record is defined for use with rotor wing aircraft operating at airports using helicopter operations procedures to/from runways, at heliports and from/to helipads at airports. Heliports referenced will be in Section/Subsection HA. Terminal Procedures referenced will be in Section/Subsection PD/PE/PF for helicopter operations from/to runways at airports (Section/Subsection PG) or in Section/Subsection HD/HE/HF for helicopter operations from/to heliports or helipads at airports (Section/Subsection HA). For fixed wing Aircraft Company routes, see Section 4.1.12.

4.0 NAVIGATION DATA - RECORD LAYOUT

4.0 NAVIGATION DATA - RECORD LAYOUT

ARINC 424 - 19 RECORD FORMAT

VHF NAVAID (D) 4.1.2.1 PRIMARY

5.2	5.3	5.4	5.5	5.6	5.14		5.33		5.14	5.16	5.34	5.35	5.36	5.37	5.38	5.36	5.37	5.66	5.40	5.149	5.90	5.150	5.197	5.71		5.297	5.277	5.31	5.32
5	10			15				20			25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120		125	130
S/T	CUST/ AREA	SEC CODE	SUB CODE	ARPT IDENT	ICAO CODE		VOR IDENT		ICAO CODE	CONT NR	VOR FREQ	VHF CLASS	VOR LATITUDE	VOR LONGITUDE	DME IDENT	DME LATITUDE	DME LONGITUDE	STA DECL	DME ELEV	MERIT	DME BIAS	FREQ PRO	DATUM CODE	VOR NAME (25)		INAPP DME DME OP SERV VOL		FILE RECORD NUMBER	CYCLE

NOTES:

VHF NAVAID (D) 4.1.2.2 CONTINUATION

SAME PARAGRAPH AS ABOVE																													
5	10			15				20			25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120		125	130
CONTINUATION RECORD SAME AS ABOVE										5.16 CONT NR	5.91 APPL TYPE	5.61 NOTES ON CONTINUATION RECORD (69)													RESERVED (31)					FILE RECORD NUMBER	CYCLE

NOTES:

VHF NAVAID (D) 4.1.2.3 SIMULATION CONTINUATION

SAME PARAGRAPH AS ABOVE																													
5	10			15				20			25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120		125	130
CONTINUATION RECORD SAME AS ABOVE										5.16 CONT NR	5.91 APPL TYPE		5.93 FAC CHAR								5.39 D MAG VAR	5.92 FAC ELEV	RESERVED (39)								FILE RECORD NUMBER	CYCLE

NOTES:

VHF NAVAID (D) 4.1.2.4 FLIGHT PLANNING CONTINUATION

SAME PARAGRAPH AS ABOVE																													
5	10			15				20			25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120		125	130
CONTINUATION RECORD SAME AS ABOVE										5.16 CONT NR	5.91 APPL TYPE	5.116 FIR IDENT	5.116 UIR IDENT	5.152 S/E IND			RESERVED (80)														FILE RECORD NUMBER	CYCLE

NOTES:

VHF NAVAID (D) 4.1.2.6 LIMITATION CONTINUATION

| SAME PARAGRAPH AS ABOVE | | | | | | | | | | | | | | | | | | |

4.0 NAVIGATION DATA - RECORD LAYOUT

ARINC 424 - 19 RECORD FORMAT

WAYPOINT (EA)(PC) 4.1.4.1 PRIMARY

5.2	5.3	5.4	5.5	5.41	5.14	5.5	5.13		5.14	5.16		5.42	5.82		5.36	5.37		5.39	5.XX X	5.197		5.196	5.43		5.31	5.32
5	10			15			20	25			30			35		40	45	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120	125	130
S/T	CUST/ AREA	SEC CODE	SUB CODE	REGN /ARPT CODE	ICAO CODE	SUB CODE	WAYPOINT IDENT		ICAO CODE	CONT NR		TYPE	USE/AGE		LATITUDE	LONGITUDE		D MAG VAR	WP ELEV (REST FOR SIM)	DATUM CODE		NAME IND	NAME/DESC (25)		FILE RECORD NUMBER	CYCLE

NOTES:

WAYPOINT (EA)(PC) 4.1.4.2 CONTINUATION

| SAME PARAGRAPH AS ABOVE | | | | | | | | | | | | | | | | | | | | | | | | | 5.16 | 5.91 | 5.61 | | | | | | | | | | | | | | | | | | |

4.0 NAVIGATION DATA - RECORD LAYOUT

ARINC 424-19 RECORD FORMAT

ENROUTE AIRWAYS (ER) 4.1.6.1 — PRIMARY

5.2	5.3	5.4	5.5		5.8		5.12	5.13	5.14	5.4	5.5	5.16	5.17	5.18	5.7	5.19	5.115	5.134	5.164	5.23	5.14	5.211		5.24	5.25	5.26	5.27	5.28		5.30	5.30	5.127	5.254	5.293	5.294	5.295		5.14			5.31	5.32
S/T	CUST/ AREA	SEC CODE	SUB CODE		ROUTE IDENT	SIX CHAR	SEQ NR	FIX IDENT	ICAO CODE	SEC CODE	SUB CODE	CONT NR	DESC CODE	BDY CODE	RT TYPE	LEVEL	DIRECT	TC IND	EU IND	RECD VHF	ICAO CODE	RNP		THETA	RHO	OB MAG CRS	ROUTE FROM DIST	IB MAG CRS		MIN ALTITUDE	MIN ALTITUDE	MAX ALTITUDE	FIX RADIUS	VERT SCALE FACTOR	RVSM MIN	VSF RVSM MAX	RESERVED (4)	ICAO CODE	RESERVED (4)		FILE RECORD NUMBER	CYCLE

NOTES:

ENROUTE AIRWAYS (ER) 4.1.6.2 — CONTINUATION

SAME PARAGRAPH AS ABOVE												5.16	5.91	5.61																											5.31	5.32
CONTINUATION RECORD SAME AS ABOVE												CONT NR	APPL TYPE	NOTES ON CONTINUATION RECORD (69)																											RESERVED (14)				FILE RECORD NUMBER	CYCLE

NOTES:

ENROUTE AIRWAYS (ER) 4.1.6.3 — FLIGHT PLANNING CONTINUATION

SAME PARAGRAPH AS ABOVE												5.16	5.91	5.152																													5.31	5.32
CONTINUATION RECORD SAME AS ABOVE												CONT NR	APPL TYPE	SIE IND																													FILE RECORD NUMBER	CYCLE

ICAO CODE	REST TYPE		RESTRICTIVE AIRSPACE DESIGNATION			MULTI CD	ICAO CODE	REST TYPE		RESTRICTIVE AIRSPACE DESIGNATION			MULTI CD	ICAO CODE	REST TYPE		RESTRICTIVE AIRSPACE DESIGNATION			MULTI CD	ICAO CODE	REST TYPE		RESTRICTIVE AIRSPACE DESIGNATION			MULTI CD	LK CONT												

NOTES:

AIRPORT (PA) 4.1.7.1 — PRIMARY

5.2	5.3	5.4		5.6	5.14	5.5	5.107			5.16	5.73	5.54	5.108	5.249	5.36	5.37	5.39	5.55	5.72	5.23	5.14	5.53	5.53	5.177	5.178	5.179	5.165	5.197		5.71										5.31	5.32
S/T	CUST/ AREA	SEC CODE		ARPT IDENT	ICAO CODE	SUB CODE	ATA/ IATA	RESERVED		CONT NR	SPEED LIMIT ALTITUDE	LONGEST RWY	IFR	LONG RWY	LATITUDE	LONGITUDE	MAG VAR	ELEV	SPEED LIMIT	RECD VHF	ICAO CODE	TRANS ALTITUDE	TRANS LEVEL	PUB/MIL	TIME ZONE	DAY TIME	M/T IND	DATUM CODE	RESVD (4)	AIRPORT NAME (30)										FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT (PA) 4.1.7.2 — CONTINUATION

SAME PARAGRAPH AS ABOVE																																								5.31	5.32
CONTINUATION RECORD SAME AS ABOVE																																									FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT (PA) 4.1.7.3 — FLIGHT PLANNING CONTINUATION

SAME PARAGRAPH AS ABOVE																																										5.31	5.32
CONTINUATION RECORD SAME AS ABOVE																																										FILE RECORD NUMBER	CYCLE

CONT NR	APPL TYPE	FIR IDENT		UIR IDENT		SIE IND															AS IND	AS ARPT IDENT	AS ICAO CODE	RESERVED (50)																		

NOTES:

4.0 NAVIGATION DATA - RECORD LAYOUT

ARINC 424 - 19 RECORD FORMAT

AIRPORT GATE (PB) 4.1.8.1 PRIMARY

5.2	5.3	5.4	5.6	5.14	5.5	5.56		5.16		5.36	5.37	5.XXX		5.60		5.31	5.32
S/T	CUST/ AREA	SEC CODE	ARPT IDENT	ICAO CODE	SUB CODE	GATE IDENT		CONT NR		LATITUDE	LONGITUDE	MAG HDG RESV FOR SIM	RESERVED (43)		NAME (25)	FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT GATE (PB) 4.1.8.2 CONTINUATION

SAME PARAGRAPH AS ABOVE				5.16	5.91	5.61				5.31	5.32
CONTINUATION RECORD SAME AS ABOVE				CONT NR	APPL TYPE	NOTES ON CONTINUATION RECORD (69)				FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT SIDS/STARS/APPROACH (PD/PE/PF) 4.1.9.1 PRIMARY

5.2	5.3	5.4	5.6	5.14	5.5	5.9 & 5.10	5.7	5.11		5.12	5.13	5.14	5.5	5.5	5.16	5.17	5.20	5.211	5.21	5.22	5.23	5.14	5.204	5.24	5.25	5.26	5.27	5.4	5.5	5.29	5.81	5.30	5.30	5.53	5.72	5.70	5.144 or 5.271	5.130 or 5.272	5.14	5.4	5.5	5.222	5.261	5.7	5.7	5.293	5.31	5.32
S/T	CUST/ AREA	SEC CODE	ARPT IDENT	ICAO CODE	SUB CODE	SID/STAR/APP IDENT	RT TYPE	TRANS IDENT		SEQ NR	FIX IDENT	ICAO CODE	SEC CODE	SUB CODE	CONT NR	DESC CODE	TURN DR	RNP	PATH TERM	TDV	RECD NAVAID	ICAO CODE	ARC RADIUS	THETA	RHO	MAG CRS	RTE DIST HOLD DIST/TIME	SEC CODE	SUB CODE	ALT DESC	ATC	ALTITUDE	ALTITUDE	TRANS ALTITUDE	SPEED LIMIT	VERT ANGLE	CENTER FIX OR TAA PT	MULTI CD	ICAO CODE	SEC CODE	SUB CODE	GNSS/FMS IND	SPD LMT	RTE QUAL1	RTE QUAL2	VERT SCALE FACTOR	FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT SIDS/STARS/APPROACH (PD/PE/PF) 4.1.9.2 CONTINUATION

SAME PARAGRAPH AS ABOVE																																																		
CONTINUATION RECORD SAME AS ABOVE																																																		

5.16	5.91	5.170	5.170	5.170	5.170	5.171	5.171	5.171	5.171	5.67	5.29	5.30	5.70	5.242	5.211	5.290	5.29	5.292	5.292	5.292	5.292		5.14		5.14	5.7	5.7		5.31	5.32
CONT NR	APPL TYPE	CAT A DH	CAT B DH	CAT C DH	CAT D DH	CAT A MDH	CAT B MDH	CAT C MDH	CAT D MDH	TCH	ALT DESC	LOC ALTITUDE	VERT ANGLE	PROC CAT	RNP	PROC DESIGN MAG VAR	PROCEDURES NAV/ NAV REQ	CAT A RADII	CAT B RADII	CAT C RADII	CAT D RADII	RESERVED	ICAO CODE	RESERVED	ICAO CODE	RTE QUAL1	RTE QUAL2		FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT SIDS/STARS/APPROACH (PD/PE/PF) 4.1.9.3 FLIGHT PLANNING CONTINUATION

SAME PARAGRAPH AS ABOVE																																																		
CONTINUATION RECORD SAME AS ABOVE																																																		

5.16	5.91	5.152																																																		
CONT NR	APPL TYPE	SIE IND																																																	

| 5.260 | | | | | | | | | | | | | | | | | | |

RUNWAY (PG) 4.1.10.1 — PRIMARY

5.2	5.3	5.4	5.6	5.14	5.5	5.46		5.16	5.57	5.58		5.36	5.37	5.212		5.225	5.68	5.69	5.67	5.109	5.270	5.44	5.80	5.79	5.44	5.80		5.59	5.31	5.32
S/T	CUST/ AREA	SEC CODE	ARPT IDENT	ICAO CODE	SUB CODE	RUNWAY IDENT		CONT NR	RUNWAY LENGTH	RUNWAY BEARING		LATITUDE	LONGITUDE	RWY GRAD		(LTP) ELLIPSOID HEIGHT	LNDG THRES ELEV	DSPLCD THR	TCH	WIDTH	TCH	LOC/MLS/ GLS IDENT	CAT/CLASS	STOPWAY	SEC LOC/ MLS/GLS IDENT	CAT/CLASS	RESERVED	RUNWAY DESCRIPTION (22)	FILE RECORD NUMBER	CYCLE

NOTES:

RUNWAY (PG) 4.1.10.2 — CONTINUATION

SAME PARAGRAPH AS ABOVE								5.16	5.91	5.61																			5.31	5.32
CONTINUATION RECORD SAME AS ABOVE								CONT NR	APPL TYPE	NOTES ON CONTINUATION RECORD (69)																	RESERVED (31)		FILE RECORD NUMBER	CYCLE

NOTES:

RUNWAY (PG) 4.1.10.3 — SIMULATION CONTINUATION

SAME PARAGRAPH AS ABOVE								5.16	5.91					5.94	5.95			5.98	5.97										5.31	5.32
CONTINUATION RECORD SAME AS ABOVE								CONT NR	APPL TYPE					TRUE BEARING	SOURCE			LOCATION	TDZ ELEV	RESERVED (52)									FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT & HELIPORT LOCALIZER & G.S. (PI) 4.1.11.1 — PRIMARY

5.2	5.3	5.4	5.6	5.14	5.5	5.44	5.80		5.16	5.45	5.46	5.36	5.37	5.47	5.36	5.37	5.48	5.49	5.50	5.51	5.52	5.66	5.67	5.74	5.33	5.14	5.4	5.5	5.59	5.31	5.32
S/T	CUST/ AREA	SEC CODE	ARPT IDENT	ICAO CODE	SUB CODE	LOC IDENT	CAT		CONT NR	FREQ	RUNWAY IDENT	LOC LATITUDE	LOC LONGITUDE	LOC BRG	G.S. LATITUDE	G.S. LONGITUDE	LOC FR RW END	+/-@	G.S. FR RW THRES	LOC WIDTH	G.S. ANGLE	STA DECL	TCH	G.S ELEV	SUPPORT FACILITY	ICAO CODE	SEC CODE	SUB CODE	RESERVED (13)	FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT & HELIPORT LOCALIZER & G.S. (PI) 4.1.11.2 — CONTINUATION

SAME PARAGRAPH AS ABOVE								5.16	5.91	5.61																			5.31	5.32
CONTINUATION RECORD SAME AS ABOVE								CONT NR	APPL TYPE	NOTES ON CONTINUATION RECORD (69)																	RESERVED (31)		FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT & HELIPORT LOCALIZER & G.S. (PI) 4.1.11.3 — SIMULATION CONTINUATION

SAME PARAGRAPH AS ABOVE								5.16	5.91		5.93			5.94	5.95				5.96	5.10	5.10	5.10	5.10	5.10			5.31	5.32
CONTINUATION RECORD SAME AS ABOVE								CONT NR	APPL TYPE		FAC CHAR			TRUE BEARING	SOURCE				G.S. BEAM WIDTH	APP IDENT 1	APP IDENT 2	APP IDENT 3	APP IDENT 4	APP IDENT 5			FILE RECORD NUMBER	CYCLE

NOTES:

COMPANY ROUTE (R) 4.1.12.1 — PRIMARY

5.2	5.3	5.4	5.5	5.75		5.14	5.4	5.5	5.75		5.14	5.4	5.5	5.76	5.12	5.77	5.78	5.3	5.83	5.14	5.4	5.5	5.84	5.85		5.86	5.87	5.14	5.88	5.89	5.148	5.7	5.7	5.7		5.31	5.32
S/T	CUST/ AREA	SEC CODE	SUB CODE	FROM ARPT/FIX		ICAO CODE	SEC CODE	SUB CODE	FROM ARPT/FIX		ICAO CODE	SEC CODE	SUB CODE	COMPANY ROUTE ID	SEQ NO.	VIA	S/S/A AWY	AREA	TO FIX	ICAO CODE	SEC CODE	SUB CODE	RUNWAY TRANS	ENRT TRANS		CRUISE ALTITUDE	TERM/ ALT ARPT	ICAO CODE	ALT DIST	COST INDEX	ENRT ALT ARPT	SID/STAR/APP	SID/STAR/APP	SID/STAR/APP		FILE RECORD NUMBER	CYCLE

NOTES:

4.0 NAVIGATION DATA - RECORD LAYOUT

ARINC 424 - 19 RECORD FORMAT

AIRPORT & HELIPORT LOCALIZER MARKERS/ LOCATORS (PM) 4.1.13.1 — PRIMARY

5.2	5.3	5.4	5.6	5.14	5.5	5.44	5.99	5.16	5.34	5.46	5.36	5.37	5.100	5.36	5.37	5.35	5.93	5.33		5.39		5.92		5.31	5.32
S/T	CUST/ AREA	SEC CODE	ARPT IDENT	ICAO CODE	SUB CODE	LOC IDENT	MKR TYPE	CONT NR	LCTR FREQ	RUNWAY IDENT	MKR LATITUDE	MKR LONGITUDE	MINOR AXIS TRUE BRG	LOCATOR LATITUDE	LOCATOR LONGITUDE	LCTR CLASS	LCTR FAC CHAR	LCTR IDENT		MAG VAR		FAC ELEV	RESERVED (21)	FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT & HELIPORT LOCALIZER (PM) 4.1.13.2 — CONTINUATION

SAME PARAGRAPH AS ABOVE								5.16	5.91															5.31	5.32
CONTINUATION RECORD SAME AS ABOVE								CONT NR	APPL TYPE	RESERVED (100)														FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT COMM (PV) 4.1.14.1 — PRIMARY

5.2	5.3	5.4	5.6	5.14	5.5		5.283	5.12	5.16	5.101	5.103	5.103	5.104	5.102	5.181	5.105	5.266	5.183	5.185	5.14	5.4	5.5	5.29	5.184	5.184	5.187	5.188	5.36	5.37	5.106	5.198	5.199		5.31	5.32
S/T	CUST/ AREA	SEC CODE	ARPT IDENT	ICAO CODE	SUB CODE		COMM CLASS	SEQ NR	CONT NR	COMM TYPES	TRANSMIT FREQ	RECEIVE FREQ	FREQ UNITS	RAD UNITS	H24 IND	CALL SIGNS	MIL SEC IND	SECTORIZATION	SEC FAC	ICAO	SEC CODE	SUB CODE	ALT DESC CODE	COMM ALT 1	COMM ALT 2	DIST DESC CODE	COMM DIST	TRANSMITTER LAT	TRANSMITTER LONG	SERV IND	MOD	SIGNAL EM		FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT COMM (PV) 4.1.14.2 — CONTINUATION

SAME PARAGRAPH AS ABOVE																																		5.31	5.32
FIELD AS ON PRIMARY RECORD									5.16	5.91	5.200	5.14	5.4	5.5	5.39	5.92	5.183	5.29	5.184	5.184	5.183	5.29	5.184	5.184	5.131	5.132	5.138	5.195	5.195	5.195	5.195	5.195		FILE RECORD NUMBER	CYCLE
									CONT REC	APPL TYPE	REMOTE FAC	ICAO	SEC	SUB	TRANS SITE MAG VAR	TRANS SITE ELEV	ADD SECTOR 1	ADD/SEC 2 ALT DESC	ADD SEC 1 ALT 1	ADD SEC 1 ALT 2	ADD SEC 2	ADD/SEC 2 ALT DESC	ADD SEC 2 ALT 1	ADD SEC 2 ALT 2	TIME CODE	NOTAM	TIME IND	TIME OF OP	TIME OF OP	TIME OF OP	TIME OF OP	TIME OF OP			

NOTES:

AIRPORT COMM (PV) 4.1.14.3 — CONTINUATION

SAME PARAGRAPH AS ABOVE																																		5.31	5.32
FIELDS AS ON PRIMARY RECORD																																		FILE RECORD NUMBER	CYCLE
									5.16	5.91	5.186																									
									CONT REC	APPL TYPE	SECTORIZATION NARRATIVE																									

CRUISING TABLES (TC) 4.1.16.1

PRIMARY

5.2	5.4	5.5	5.134	5.12		5.135	5.135	5.165	5.136	5.137	5.136	5.136	5.137	5.136	5.136	5.136	5.137	5.136		5.31	5.32
5				10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120	125	130
S/T	SEC CODE	SUB CODE	CRSE TBL IDENT	SEQ number		COURSE FROM	COURSE TO	M/T IND	CRUISE LEVEL FROM	VERTICAL SEPARATION	CRUISE LEVEL TO	CRUISE LEVEL FROM	VERTICAL SEPARATION	CRUISE LEVEL TO	CRUISE LEVEL FROM	VERTICAL SEPARATION	CRUISE LEVEL TO	CRUISE LEVEL FROM	VERTICAL SEPARATION	CRUISE LEVEL TO	RESERVED (24)	FILE RECORD NUMBER	CYCLE

NOTES:

FIR/UIR (UF) 4.1.17.1

PRIMARY

5.2	5.3	5.4	5.5	5.116	5.151	5.117	5.12	5.16	5.116	5.116	5.122	5.123	5.124	5.118	5.36	5.37	5.36	5.37	5.119	5.120	5.121	5.121	5.121	5.134	5.125	5.31	5.32
5				10		15		20	25		30			35	40	45	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120	125	130
S/T	CUST/ AREA	SEC CODE	SUB CODE	FIR/UIR IDENT	FIR/UIR ADDRESS	IND	SEQ NR	CONT NR	ADJ FIR IDENT	ADJ UIR IDENT	RUS	RUA	ENTRY	BDRY VIA	FIR/UIR LATITUDE	FIR/UIR LONGITUDE	ARC ORIGIN LATITUDE	ARC ORIGIN LONGITUDE	ARC DIST	ARC BNG	FIR UPPER LIMIT	UIR LOWER LIMIT	UIR UPPER LIMIT	TC IND	FIR/UIR NAME (25)	FILE RECORD NUMBER	CYCLE

NOTES:

FIR/UIR (UF) 4.1.17.2

CONTINUATION

SAME PARAGRAPH AS ABOVE			5.16	5.91			5.31	5.32
5	10	15	20	25	30	35	125	130
CONTINUATION RECORD SAME AS ABOVE			CONT NR	APPL TYPE	RESERVED (102)		FILE RECORD NUMBER	CYCLE

NOTES:

RESTRICTIVE AIRSPACE (UR) 4.1.18.1

PRIMARY

5.2	5.3	5.4	5.5	5.14	5.128	5.129	5.130	5.12	5.16	5.19	5.131	5.132		5.118	5.36	5.37	5.36	5.37	5.119	5.120		5.121	5.133	5.121	5.133	5.126	5.31	5.32
5				10		15	20		25				30	35	40	45	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120	125	130
S/T	CUST/ AREA	SEC CODE	SUB CODE	ICAO CODE	REST TYPE	RESTRICTIVE AIRSPACE DESIGNATION	MULTI CD	SEQ NR	CONT NR	LEVEL	TIME CD	NOTAM		BDRY VIA	LATITUDE	LONGITUDE	ARC ORIGIN LATITUDE	ARC ORIGIN LONGITUDE	ARC DIST	ARC BNG		LOWER LIMIT	UNIT IND	UPPER LIMIT	UNIT IND	RESTRICTIVE AIRSPACE NAME (30)	FILE RECORD NUMBER	CYCLE

NOTES:

RESTRICTIVE AIRSPACE (UR) 4.1.18.2

CONTINUATION

SAME PARAGRAPH AS ABOVE				5.16	5.91	5.131	5.132	5.138	5.195	5.195	5.195	5.195	5.195	5.195	5.195	5.195	5.140	5.31	5.32
5	10	15	20	25			30		35	40	45	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120	125	130
CONTINUATION RECORD SAME AS ABOVE				CONT NR	APPL TYPE	TIME CD	NOTAM	TIME IND	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	CONTROLLING AGENCY (24)	FILE RECORD NUMBER	CYCLE

NOTES:

RESTRICTIVE AIRSPACE (UR) 4.1.18.3

CONTINUATION

SAME PARAGRAPH AS ABOVE				5.16	5.91		5.152	5.153		5.31	5.32
5	10	15	20	25			30	35	40	125	130
CONTINUATION RECORD SAME AS ABOVE				CONT NR	APPL TYPE		SIE IND	START/END DATE	RESERVED (82)	FILE RECORD NUMBER	CYCLE

NOTES:

GRID MINIMUM OFF ROUTE ALTITUDE (MORA) RECORDS (AS) 4.1.19.1

PRIMARY

5.2	5.4	5.5		5.141	5.142		5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143	5.143		5.31	5.32
5			10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120											125	130
S/T	SEC CODE	SUB CODE		START LAT	START LONG		MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA	RESVD	FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT MINIMUM SECTOR ALTITUDE (MSA) (PS) 4.1.20.1

PRIMARY

5.2	5.3	5.4		5.6	5.14	5.5	5.144	5.14	5.4	5.5	5.130		5.16		5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.165		5.31	5.32
S/T	CUST/ AREA	SEC CODE		ARPT IDENT	ICAO CODE	SUB CODE	MSA CENTER	ICAO CODE	SEC CODE	SUB CODE	MULTI COD	RESERVED (15)	CONT NR		SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	MAG IND		FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT MINIMUM SECTOR ALTITUDE (MSA) (PS) 4.1.20.2

PRIMARY EXTENSION

SAME PARAGRAPH AS ABOVE													5.16	5.91	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145		5.31	5.32
CONTINUATION RECORD SAME AS ABOVE													CONT NR	APPL TYPE	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	FIELD AS ON PRIMARY RECORD		FILE RECORD NUMBER	CYCLE

NOTES:

AIRPORT MINIMUM SECTOR ALTITUDE (MSA) 4.1.20.3

CONTINUATION

SAME PARAGRAPH AS ABOVE															5.16	5.91	5.61			5.31	5.32
CONTINUATION RECORD SAME AS ABOVE															CONT NR	APPL TYPE	NOTES (69)		RESERVED	FILE RECORD NUMBER	CYCLE

NOTES:

ENROUTE AIRWAYS RESTRICTION RECORDS (EU) 4.1.21.1

ALTITUDE EXCLUSION PRIMARY

5.2	5.3	5.4	5.5	5.8	5.154	5.201	5.16	5.13	5.14	5.4	5.5	5.13	5.14	5.4	5.5	5.157	5.157	5.131	5.138	5.195	5.195	5.195	5.195	5.202	5.160	5.161	5.203	5.161	5.203	5.161	5.203	5.161	5.203	5.161	5.203	5.161	5.203	5.161	5.203	5.31	5.32
S/T	CUST/ AREA	SEC CODE	SUB CODE	ROUTE IDENT	SIX CHAR	REST IDENT	REST TYPE	CONT NR	START FIX IDENT	ICAO CODE	SEC CODE	SUB CODE	END FIX IDENT	ICAO CODE	SEC CODE	SUB CODE	START DATE	END DATE	TIME CD	TIME IND	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	EXC IND	UNITS ALT	REST ALT	BLCK IND	REST ALT	BLCK IND	REST ALT	BLCK IND	REST ALT	BLCK IND	REST ALT	BLCK IND	REST ALT	BLCK IND	REST ALT	BLCK IND	FILE RECORD NUMBER	CYCLE

NOTES:

ENROUTE AIRWAYS RESTRICTION RECORDS (EU) 4.1.21A.1

NOTE RESTRICTION PRIMARY

5.2	5.3	5.4	5.5	5.8	5.154	5.201	5.16	5.13	5.14	5.4	5.5	5.13	5.14	5.4	5.5	5.157	5.157	5.163			5.31	5.32
S/T	CUST/ AREA	SEC CODE	SUB CODE	ROUTE IDENT	SIX CHAR	REST IDENT	REST TYPE	CONT NR	START FIX IDENT	ICAO CODE	SEC CODE	SUB CODE	END FIX IDENT	ICAO CODE	SEC CODE	SUB CODE	START DATE	END DATE	RESTRICTION NOTES (69)			FILE RECORD NUMBER	CYCLE

NOTES:

ENROUTE AIRWAYS RESTRICTION RECORDS (EU) 4.1.21A.2

NOTE RESTRICTION CONTINUATION

SAME PARAGRAPH AS ABOVE							5.16	5.91											5.163					5.31	5.32
CONTINUATION RECORD SAME AS ABOVE							CONT NR	APPL TYPE	RESERVED (32)										RESTRICTION NOTES (69)					FILE RECORD NUMBER	CYCLE

NOTES:

ENROUTE AIRWAYS RESTRICTION RECORDS (EU) 4.1.21B.1

SEASONAL CLOSURE PRIMARY

5.2	5.3	5.4	5.5	5.8	5.154	5.201	5.16	5.13	5.14	5.4	5.5	5.13	5.14	5.4	5.5	5.157	5.157	5.131	5.138	5.195	5.195	5.195	5.195	5.134			5.31	5.32
S/T	CUST/ AREA	SEC CODE	SUB CODE	ROUTE IDENT	SIX CHAR	REST IDENT	REST TYPE	CONT NR	START FIX IDENT	ICAO CODE	SEC CODE	SUB CODE	END FIX IDENT	ICAO CODE	SEC CODE	SUB CODE	START DATE	END DATE	TIME CD	TIME IND	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	CRSE TBL IDENT			FILE RECORD NUMBER	CYCLE

NOTES:

ARINC 424 - 19 RECORD FORMAT

4.0 NAVIGATION DATA - RECORD LAYOUT

ENROUTE AIRWAYS RESTRICTION RECORDS (EU) 4.1.21C.1
CRUISE TABLE REPLACEMENT PRIMARY

5.2	5.3	5.4	5.5	5.8	5.154	5.201	5.16	5.13	5.14	5.4	5.5	5.13	5.14	5.4	5.5		5.157	5.157	5.131	5.138	5.195	5.195	5.195	5.195	5.134		5.31	5.32
S/T	CUST/ AREA	SEC CODE	SUB CODE	ROUTE IDENT	SIX CHAR	REST IDENT	REST TYPE	CONT NR	START FIX IDENT	ICAO CODE	SEC CODE	SUB CODE	END FIX IDENT	ICAO CODE	SEC CODE	SUB CODE		START DATE	END DATE	TIME CD	TIME IND	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	CRSE TBL IDENT		FILE RECORD NUMBER	CYCLE

NOTES:

ENROUTE AIRWAYS RESTRICTION RECORDS (EU) 4.1.21C.2
CRUISE TABLE REPLACEMENT CONTINUATION

SAME PARAGRAPH AS ABOVE																												
							5.16	5.91											5.131	5.138					5.134		5.31	5.32
CONTINUATION RECORD SAME AS ABOVE							CONT NR	APPL TYPE	RESERVED (32)										TIME CD	TIME IND	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	CRSE TBL IDENT		FILE RECORD NUMBER	CYCLE

NOTES:

ENROUTE AIRWAYS RESTRICTION RECORDS (EU) 4.1.21.2
ALTITUDE EXCLUSION CONTINUATION

| SAME PARAGRAPH AS ABOVE | | | | | | | | | | | | | | | | | | |

PREFERRED ROUTES (ET) 4.1.24.1

PRIMARY

5.2	5.3	5.4	5.5		5.8	5.220	5.12		5.16	5.83	5.14	5.4	5.5	5.77	5.78	5.3	5.19	5.7	5.194	5.14	5.4	5.5	5.194	5.14	5.4	5.5	5.30	5.127	5.131	5.221	5.115	5.29	5.30	5.30	5.7	5.7	5.7		5.31	5.32
S/T	CUST/ AREA	SEC CODE	SUB CODE		ROUTE IDENT	USE IND	SEQ NR		CONT NR	TO FIX IDENT	ICAO CODE	SEC CODE	SUB CODE	VIA	SID/STAR/AWY IDENT	CUST/ AREA	LEVEL	RTE TYPE	INT APT/ FIX	ICAO CODE	SEC CODE	SUB CODE	TERM APT/ FIX	ICAO CODE	SEC CODE	SUB CODE	MIN ALTITUDE	MAX ALTITUDE	TIME CD	ACFT USE GROUP	DIRECT	ALT DESC	ALTITUDE ONE	ALTITUDE TWO	2ND ALT RSTRCT TYPE	2ND ROUTE TYPE QUAL	2ND ROUTE TYPE QUAL	RESERVED (15)	FILE RECORD NUMBER	CYCLE

NOTES:

PREFERRED ROUTES (ET) 4.1.24.2

CONTINUATION

SAME PARAGRAPH AS ABOVE																													5.16	5.91	5.131	5.138		5.195	5.195	5.195	5.195	5.195	5.195		5.31	5.32
CONTINUATION RECORD SAME AS ABOVE																													CONT NR	APPL TYPE	TIME CD	TIME IND	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	TIME OF OPERATION	RESERVED (11)	FILE RECORD NUMBER	CYCLE

NOTES:

PREFERRED ROUTES (ET) 4.1.24.3

CONTINUATION

SAME PARAGRAPH AS ABOVE																																										5.16	5.91	5.61			5.31	5.32
CONTINUATION RECORD SAME AS ABOVE																																											CONT NR	APPL TYPE	NOTES ON CONTINUATION RECORD (69)			FILE RECORD NUMBER	CYCLE

NOTES:

CONTROLLED AIRSPACE (UC) 4.1.25.1

PRIMARY

5.2	5.3	5.4	5.5	5.14	5.213	5.214	5.4	5.5	5.215		5.130	5.12		5.16	5.19	5.131	5.132		5.118	5.36	5.37	5.36	5.37	5.119	5.120	5.211	5.121	5.133	5.121	5.133	5.216			5.31	5.32
S/T	CUST/ AREA	SEC CODE	SUB CODE	ICAO CODE	ARSP TYPE	AIRSPACE CENTER	SEC CODE	SUB CODE	AS CLASS		MULTI CD	SEQ NR		CONT NR	LEVEL	TIME CD	NOTAM		BDRY VIA	LATITUDE	LONGITUDE	ARC ORIGIN LATITUDE	ARC ORIGIN LONGITUDE	ARC DIST	ARC BRG	RNP	LOWER LIMIT	UNIT IND	UPPER LIMIT	UNIT IND	CONTROLLED AIRSPACE NAME (30)			FILE RECORD NUMBER	CYCLE

NOTES:

CONTROLLED AIRSPACE (UC) 4.1.25.2

CONTINUATION

| SAME PARAGRAPH AS ABOVE | | | | | | | | | | | | | | | | | | |

4.0 NAVIGATION DATA - RECORD LAYOUT

ARINC 424 - 19 RECORD FORMAT

FLIGHT PLANNING ARRIVAL/DEPARTURE (PR) 4.1.27.1 — PRIMARY

5.2	5.3	5.4	5.6	5.14	5.5	5.9, 5.10	5.230	5.11	5.13	5.14	5.4	5.5	5.231	5.13	5.14	5.4	5.5	5.231	5.11	5.13	5.14	5.4	5.5	5.231	5.12	5.16	5.232	5.233	5.234	5.235	5.236	5.131	5.237	5.238	5.239	5.26	5.29	5.240	5.240	5.72	5.134	5.261		5.31	5.32
S/T	CUST/ AREA	SEC CODE	ARPT IDENT	ICAO CODE	SUB CODE	SID/STAR/APRCH IDENT	PROC TYPE	RUNWAY TRANS IDENT	RUNWAY TRANS FIX	ICAO CODE	SEC CODE	SUB CODE	RUNWAY TRANS ATD	COMMON SEGMENT FIX	ICAO CODE	SEC CODE	SUB CODE	COMN SEG ATD	ENROUTE TRANS IDENT	ENROUTE TRANS FIX	ICAO CODE	SEC CODE	SUB CODE	ENRTE TRANS ATD	SEQ NR	CONT NR	NOE	TURBO FAN	RNAV	ATC WC	ATC IDENT	TIME CD	PROCEDURE DESCRIPTION	LTC	RPT	OB MAG CRS	ALT DESC	ALT	ALT	SPEED LIMIT	CRS TBL IDENT	SPD LMT		FILE RECORD NUMBER	CYCLE

NOTES:

FLIGHT PLANNING ARRIVAL/DEPARTURE (PR) 4.1.27.2 — CONTINUATION

SAME PARAGRAPH AS ABOVE																										5.16	5.91	5.13	5.14	5.4	5.5	5.231	5.241	5.13	5.14	5.4	5.5	5.231	5.241	5.13	5.14	5.4	5.5	5.231	5.241	5.13	5.14	5.4	5.5	5.231	5.241	5.31	5.32
CONTINUATION RECORD SAME AS ABOVE																										CONT NR	APPL	INTER FIX IDENT	ICAO CODE	SEC CODE	SUB CODE	ATD	FRT CODE	INTER FIX IDENT	ICAO CODE	SEC CODE	SUB CODE	ATD	FRT CODE	INTER FIX IDENT	ICAO CODE	SEC CODE	SUB CODE	ATD	FRT CODE	INTER FIX IDENT	ICAO CODE	SEC CODE	SUB CODE	ATD	FRT CODE	FILE RECORD NUMBER	CYCLE

NOTES:

FLIGHT PLANNING ARRIVAL/DEPARTURE (PR) 4.1.27.3 — CONTINUATION

| SAME PARAGRAPH AS ABOVE | | | | | | | | | | | | | | | | | | |

AIRPORT TAA (PK) 4.1.31.2 CONTINUATION

SAME PARAGRAPH AS ABOVE	5.91	5.146	5.147	5.274	5.271	5.146	5.147	5.274	5.271	5.146	5.147	5.274	5.271	5.146	5.147	5.274	5.271	5.61		5.31	5.32
CONTINUATION RECORD SAME AS ABOVE	APP TYPE	SECTOR BRG	SECTOR MIN ALT	SECTOR RAD 1	PROC TURN IND	SECTOR BEARING	SECTOR MIN ALT	SECTOR RAD 1	PROC TURN IND	SECTOR BEARING	SECTOR MIN ALT	SECTOR RAD 1	PROC TURN IND	SECTOR BEARING	SECTOR MIN ALT	SECTOR RAD 1	PROC TURN IND	NOTES (21)	RESERVED (EXPANSION) (14)	FILE RECORD NUMBER	CYCLE

NOTES:

TACAN-ONLY NAVAID (DT) 4.1.32.1 PRIMARY

5.2	5.3	5.4	5.5	5.6	5.14	5.33		5.16	5.34	5.35		5.38	5.36	5.37	5.66	5.40	5.149		5.150	5.197	5.71	5.31	5.32
S/T	CUST AREA CODE	SEC CODE	SUB CODE	AIRPORT ID	ICAO CODE	VOR ID		CONT REC NO	VOR FREQ	NAVAID CLASS		TACAN ID	TACAN LATITUDE	TACAN LONGITUDE	STATION DECL	TACAN ELEVATION	FIG OF MERIT		FREQ PROTECT	DATUM CODE	TACAN NAME	FILE RECORD NUMBER	CYCLE

NOTES:

SPECIAL ACTIVITY AREA (ES) 4.1.33.1 PRIMARY

5.2	5.3	5.4	5.5	5.278	5.279	5.14	5.6	5.14	5.16	5.36	5.37	5.280	5.121	5.133	5.281	5.282	5.177	5.140	5.101	5.103	5.126	5.31	5.32
S/T	CUST AREA CODE	SEC CODE	SUB CODE	SAA TYPE	SAA IDENT	ICAO CODE	AIRPORT ID	ICAO CODE	CONT REC NO	LATITUDE	LONGITUDE	SAA SIZE	UPPER LIMIT	UNIT IND	SAA VOL	OP TIMES	PUB OR MIL	CONTROLLING AGENCY	COMM TYPE	COMM FREQ	SPECIAL ACTIVITY AREA NAME	FILE RECORD NUMBER	CYCLE

NOTES:

COMMUNICATION TYPE TRANSLATION (TV) 4.1.34 PRIMARY

5.3		5.4	5.5	5.101	5.287	5.288		5.289	5.283		5.31	5.32
S/T	RES	SEC CODE	SUB CODE	COMM TYPE	TYPE REC BY	TRANSLATION		USED ON	COMM CLASS		FILE RECORD NUMBER	CYCLE

NOTES:

HELIPORT (HA) 4.2.1.1 PRIMARY

5.2	5.3	5.4	5.6	5.14	5.5	5.107	5.180	5.16	5.73	5.197	5.108	5.36	5.37	5.39	5.55	5.72	5.23	5.14	5.53	5.53	5.177	5.178	5.179		5.165	5.71	5.31	5.32
S/T	CUST/ AREA	SEC CODE	HEPT IDENT	ICAO CODE	SUB CODE	ATA/ IATA	PAD IDENT	CONT NR	SPEED LIMIT ALTITUDE	DATUM CODE	IFR	LATITUDE	LONGITUDE	MAG VAR	ELEV	SPEED LIMIT	RECD VHF	ICAO CODE	TRANS ALTITUDE	TRANS LEVEL	PLB/MIL	TIME ZONE	DAY TIME		M/T IND	RESERVED	HELIPORT NAME (30)	FILE RECORD NUMBER	CYCLE

NOTES:

HELIPORT (HA) 4.2.1.2 CONTINUATION

SAME PARAGRAPH AS ABOVE	5.16	5.91	5.61		5.31	5.32
CONTINUATION RECORD SAME AS ABOVE	CONT NR	APPL TYPE	NOTES ON CONTINUATION RECORD (69)	RESERVED (31)	FILE RECORD NUMBER	CYCLE

NOTES:

HELIPORT (HA) 4.2.1.3 FLIGHT PLANNING CONTINUATION

SAME PARAGRAPH AS ABOVE	5.16	5.91	5.116	5.116	5.152		5.217	5.6	5.14		5.31	5.32
CONTINUATION RECORD SAME AS ABOVE	CONT NR	APPL TYPE	FIR IDENT	UIR IDENT	S/E IND		AS IND	AS ARPT IND	AS ICAO CODE	RESERVED (50)	FILE RECORD NUMBER	CYCLE

NOTES:

HELICOPTER TERMINAL WAYPOINT (HC) 4.2.2.1 PRIMARY

5.2	5.3	5.4		5.41	5.14	5.5	5.13		5.14	5.16		5.42	5.82		5.36	5.37		5.39	5.XX X	5.197		5.196	5.43		5.31	5.32
S/T	CUST/ AREA	SEC CODE		HELIPORT IDENT	ICAO CODE	SUB CODE	WAYPOINT IDENT		ICAO CODE	CONT NR		TYPE	USEAGE		LATITUDE	LONGITUDE		D MAG VAR	WP ELEV (REST FOR SIM)	DATUM CODE		NAME IND	NAME/DESC (25)		FILE RECORD NUMBER	CYCLE
5	10		15		20			25		30		35		40	45	50	55	60	65	70	75	80	85	90	95	100	105	110	115	120	125	130

NOTES:

HELICOPTER TERMINAL WAYPOINT (HC) 4.2.2.2 CONTINUATION

| SAME PARAGRAPH AS ABOVE | | | | | | | | | | | | | | | | | | | | | | | | 5.16 | 5.91 | 5.61 | | | | | | | | | | | | | | | | | | |

ARINC 424 - 19 RECORD FORMAT

HELIPORT MINIMUM SECTOR ALTITUDE (MSA) (HS) 4.2.4.1 PRIMARY

5.2	5.3	5.4	5.6	5.14	5.5	5.144	5.14	5.4	5.5	5.130		5.16	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.165		5.31	5.32
S/T	CUST/ AREA	SEC CODE	ARPT IDENT	ICAO CODE	SUB CODE	MSA CENTER	ICAO CODE	SEC CODE	SUB CODE	MULTI CD	RESERVED (15)	CONT NR	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	MAG IND	RESRVD EXP (3)	FILE RECORD NUMBER	CYCLE

NOTES:

HELIPORT MINIMUM SECTOR ALTITUDE (MSA) (HS) 4.2.4.2 PRIMARY EXTENSION

SAME PARAGRAPH AS ABOVE												5.16	5.91	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145	5.146	5.147	5.145		5.31	5.32
CONTINUATION RECORD SAME AS ABOVE												CONT NR	APPL TYPE	RES	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	SEC BRG	SEC ALT	SEC RAD	FIELD AS ON PRIMARY	FILE RECORD NUMBER	CYCLE

NOTES:

HELIPORT MINIMUM SECTOR ALTITUDE (MSA) (HS) 4.2.4.3 CONTINUATION

| SAME PARAGRAPH AS ABOVE | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | 5.16 | 5.91 | 5.61 | | | | | | | | | | | | | | | | | | |

4.0 NAVIGATION DATA - RECORD LAYOUT

ARINC 424 - 19 RECORD FORMAT

HELIPORT TAA (HK) 4.2.6.1 PRIMARY

5.2	5.3	5.4		5.6	5.14	5.5	5.10	5.273	5.14	5.4	5.5	5.272	5.16		5.146	5.147	5.274	5.271	5.146	5.147	5.274	5.271	5.146	5.147	5.274	5.271	5.146	5.147	5.274	5.271	5.146	5.274	5.271	5.146	5.147	5.274	5.271	5.165		5.31	5.32
5				10			15	20				25	30		35		40	45		50		55	60		65		70	75		80		85		90	95	100	105		110	115	120		125	130
ST	CUST/ AREA	SEC CODE		HELI IDENT	ICAO CODE	SUB CODE	APPROACH IDENT	TAA IAF WAYPOINT	ICAO CODE	SEC CODE	SUB CODE	TAA FIX POS IND	CONT REC NO.		SECTOR BEARING	SECTOR MIN ALT	SECTOR RAD 1	PROC TURN IND	SECTOR BEARING	SECTOR MIN ALT	SECTOR RAD 1	PROC TURN IND	SECTOR BEARING	SECTOR MIN ALT	SECTOR RAD 1	PROC TURN IND	SECTOR BEARING	SECTOR MIN ALT	SECTOR RAD 1	PROC TURN IND	SECTOR BEARING	SECTOR RAD 1	PROC TURN IND	SECTOR BEARING	SECTOR MIN ALT	SECTOR RAD 1	PROC TURN IND	MAG/TRUE IND		FILE RECORD NUMBER	CYCLE

NOTES:

HELIPORT TAA (HK) 4.2.6.2 CONTINUATION

SAME PARAGRAPH AS ABOVE														5.91	5.146	5.147	5.274	5.271	5.146	5.147	5.274	5.271	5.146	5.147	5.274	5.271	5.146	5.147	5.274	5.271	5.61			5.31	5.32
5				10			15		20			25	30		35		40	45		50		55	60		65		70	75		80		85		90	95	100	105	110	115	120	125	130
CONTINUATION RECORD SAME AS ABOVE														USERS	SECTOR BEARING	SECTOR MIN ALT	SECTOR RAD 1	TAA FIX POS IND	SECTOR BEARING	SECTOR MIN ALT	SECTOR RAD 1	TAA FIX POS IND	SECTOR BEARING	SECTOR MIN ALT	SECTOR RAD 1	TAA FIX POS IND	SECTOR BEARING	SECTOR MIN ALT	SECTOR RAD 1	TAA FIX POS IND	NOTES (21)			RESERVED (14)		FILE RECORD NUMBER	CYCLE

NOTES:

HELICOPTER OPERATIONS COMPANY ROUTE 4.2.7.1 PRIMARY

5.2	5.3	5.4	5.5	5.75	5.180		5.14	5.4	5.5	5.75	5.180		5.14	5.4	5.5	5.76	5.12	5.77	5.78	5.4	5.5	5.7	5.7	5.7	5.3	5.83	5.14	5.4	5.5	5.84	5.85		5.86	5.87	5.14	5.4	5.5	5.88	5.89	5.148		5.31	5.32
5				10	15			20			25	30		35		40	45		50	55		60				65	70	75			80	85		90	95	100			105	110	115	120	125	130
ST	CUST/ AREA	SEC CODE	SUB CODE	FROM AIRPORT HELIPORT/FIX	HELIPAD ID		ICAO CODE	SEC CODE	SUB CODE	TO AIRPORT/ HELIPORT/ FIX	HELIPAD ID		ICAO CODE	SEC CODE	SUB CODE	COMPANY ROUTE ID	SEQ NO.	VIA CODE	SID/STAR/APP/ AWY	SEC CODE	SUB CODE	SS/AIA RT TYPE	SS/AIA RT Q 1	SS/AIA RT Q 2	AREA CODE	TO FIX	ICAO CODE	SEC CODE	SUB CODE	RWY/ HELIPAD TRANS ID	ENROUTE TRANS ID	RESERVED	CRUISE ALT	TERM/ALT HELIPORT	ICAO CODE	SEC CODE	SUB CODE	ALT DISTANCE	COST INDEX	ENROUTE ALT HELIPORT	RESERVED (EXP)	FILE RECORD NUMBER	CYCLE

NOTES:

ARINC 424 - 19 RECORD FORMAT

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.1 General

Chapter 5 sets forth definitions/descriptions and content for each type of field employed in the records discussed in Chapter 4. The following information is presented for each field:

(i) Field Name (paragraph heading)
(ii) Abbreviation used in proportional record layouts (Chapter 4) when different than Field Name (follows paragraph heading)
(iii) Field Definition/Description
(iv) Source/Content of each field
(v) Length of field, expressed in number of characters
(vi) Type of character allowed in each field, alpha or numeric or alpha/numeric
(vii) Examples of field content when appropriate and/or necessary

The following general rules apply to the format of all the fields:

(i) All numeric fields, and the numeric parts of latitude, longitude, magnetic variation, negative elevation and station declination fields will be right justified and filled with leading zeros.
(ii) All alpha and alpha/numeric fields will be left justified.
(iii) Allowable field content of blank is defined as alpha/numeric content.

5.2 Record Type (S/T)

Definition/Description: The "Record Type" field content indicates whether the record data are "standard," i.e., suitable for universal application, or "tailored," i.e. included on the master file for a single user's specific purpose (Section 1.2 of this Specification refers).

Source/Content: The field contains the letter "S" when the field data are "standard" and the letter "T" when they are "tailored."

Used On:	All records
Length:	1 character
Character Type:	Alpha

5.3 Customer/Area Code (CUST/AREA)

Definition/Description: The "Customer Area Code" field permits the categorization of standard records by geographical area and of tailored records by the airlines for whom they are provided in the master file. Several record types do not adhere to the established geographical boundaries. There is no "AREA" in such records.

Source/Content: "AREA" Codes should be derived from Figure 5-2. Airline codes should be derived from the standard list of abbreviated identifiers maintained and published in the IATA Airline Coding Directory. On Company Route and Preferred Route Records, an additional AREA field is used as a pointer to the AREA in which the Route Segment is located. For records, which do not follow geographical boundaries, the field is blank. For Preferred Routes, the field content is "PDR."

Used On:	All records with content as defined above.
Length:	3 characters max
Character Type:	Alpha
Examples:	Areas - USA, CAN, EUR Customer - UAL, DAL, TWA Preferred Routes - PDR

5.4 Section Code (SEC CODE)

Definition/Description: The "Section Code" field defines the major section of the navigation system data base in which the record resides.

Source/Content: Figure 5-1 shows the data base section encoding scheme.

Used On:	All records
Length:	1 character
Character Type:	Alpha

5.5 Subsection Code (SUB CODE)

Definition/Description: The "Subsection Code" field defines the specific part of the database major section in which the record resides.

Additionally, records that reference other records within the database use Section/Subsection Codes to make the reference, together with the record identifier. This is true for "fix" information in Holdings, Enroute Airways, Airport and Heliport SID/STAR/APPROACH, all kinds of Communications, Airport and Heliport MSA, Airport and Heliport TAA, Company Routes, Enroute Airway Restrictions, Preferred Routes and Alternate Records. The Section Code will define the major database section, the Subsection Code permits the exact section (file) to be identified and the "fix" (record) can then be located within this file.

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Source/Content: Figure 5-1 shows the database Subsection Encoding Scheme.

Used On: All records
Length: 1 character
Character Type: Alpha

Section Code	Section Name	Subsection Code	Subsection Name
A	MORA	S	Grid MORA
D	Navaid	Blank	VHF Navaid
		B	NDB Navaid
		T	**TACAN Duplicates**
E	Enroute	A	Waypoints
		M	Airway Markers
		P	Holding Patterns
		R	Airways and Routes
		T	Preferred Routes
		U	Airway Restrictions
		V	Communications
H	Heliport	A	Pads
		C	Terminal Waypoints
		D	SIDs
		E	STARs
		F	Approach Procedures
		K	TAA
		S	MSA
		V	Communications
P	Airport	A	Reference Points
		B	Gates
		C	Terminal Waypoints
		D	SIDs
		E	STARs
		F	Approach Procedures
		G	Runways
		I	Localizer/Glide Slope
		K	TAA
		L	MLS
		M	Localizer Marker
		N	Terminal NDB
		P	Path Point
		R	Flt Planning ARR/DEP
		S	MSA
		T	GLS Station
		V	Communications
R	Company Routes	Blank	Company Routes **(Master Airline File)**
		A	Alternate Records
		H	**Helicopter operation Routes (Master Helicopter File)**
T	Tables	C	Cruising Tables
		G	Geographical Reference
		N	RNAV Name Table
		V	**Communication Type**
U	Airspace	C	Controlled Airspace
		F	FIR/UIR
		R	Restrictive Airspace

Figure 5-1
Section and Subsection Encoding Scheme

5.6 Airport/Heliport Identifier (ARPT/HELI IDENT)

Definition/Description: The "Airport Identifier" and the "Heliport Identifier" fields contain the identification of the airport or heliport to which the data contained in the record relates.

Source/Content: The content of this field is derived from official government sources. It will be the four character ICAO Location Identifier of the airport or heliport when such is published. It will be the three or four character Domestic Identifier when published and no ICAO Location Identifier is available for the airport or heliport. When used on Airport or Heliport Flight Planning Continuation Records, it will be the Airport or Heliport Identifier owning the terminal controlled airspace referenced in that record.

Note: Within the continental United States and within Canada, in addition to using the published four character ICAO Location Identifiers, data suppliers append the character "K" for the USA or "C" for Canada to certain Domestic Identifiers to present an ICAO look-alike four character identifier.

COMMENTARY

Where no officially published identifier is available, a data supplier may create a unique, temporary and unofficial identifier. Airports or Heliports within such identifiers may supply Tailored Data only and with full knowledge and concurrence of the data user. Whenever possible, such temporary identifiers should be coordinated among the various data suppliers prior to its release.

The content of this Airport/Heliport Identifier should not be confused with the perhaps more familiar ATA/IATA two or three character identifiers often user by airlines for other than navigation purposes. These ATA/IATA identifiers are included in the ARINC 424 database in accordance with Section 5.107 of this specification.

Used On: Airport Identifier - VHF Navaid, NDB Navaid, Airport Terminal Waypoint, Airport, Airport Gate, Airport SID/STAR/Approach, Runway, Airport and Heliport Localizer, Airport and Heliport Localizer Marker, Holding Pattern, Airport Communications, Airport and Heliport MLS, GLS Airport MSA, Airport TAA**,** Path Point Flight Planning Arrival Departure Data, GLS Record, and Enroute Airway Restriction and Company Route to the Airport Identifier.

Heliport Identifier - VHF Navaid, NDB Navaid, Heliport Terminal Waypoint, Heliport, Heliport SID/STAR/Approach, Airport and Heliport Localizer, Airport and Heliport Localizer Marker, Holding Pattern, Heliport Communications, Airport and MLS, GLS Heliport MSA, Heliport TAA**,** Path Point Flight Planning Arrival/Departure Data, GLS Records, and Enroute Airway Restriction and Company Route to the Airport Identifier.

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Length:	4 characters maximum
Character Type:	Alpha/numeric
Examples:	KJFK, DMIA, 9Y9, CYUL, EDDF, 53Y, CA14

5.0 NAVIGATION DATA - FIELD DEFINITIONS

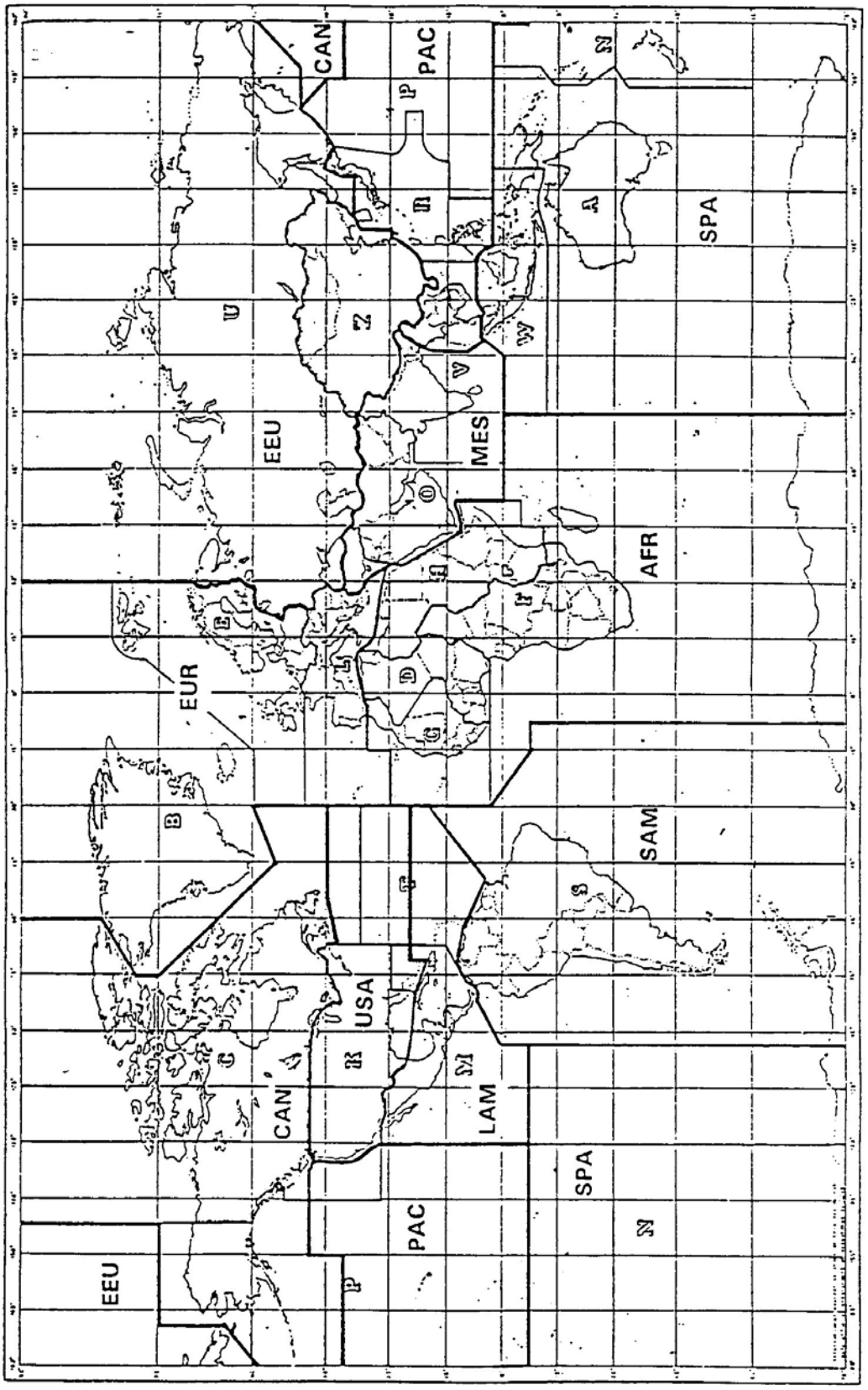


Figure 5-2
Geographical Area Codes

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.7 Route Type (RT TYPE)

Definition/Description: The "Route Type" field defines the type of Enroute Airway, Preferred Route, Airport and Heliport SID/STAR/Approach Routes of which the record is an element. For Airport and Heliport **SID/STAR/**Approach Routes, "Route Type" includes a "primary route type," and up to two "route type qualifiers."

Source/Content: The content of this field (for approach procedures) will be as indicated in the following tables:

Enroute Airway Records (ER)

Airway Type	Field Content
Airline Airway (Tailored Data)	A
Control	C
Direct Route	D
Helicopter Airways	H
Officially Designated Airways, except RNAV, Helicopter Airways	O
RNAV Airways	R
Undesignated ATS Route	S
TACAN Airway	**T**

Preferred Route Records (ET)

Route Type Description	Field Content
North American Routes for North Atlantic Traffic	C
Common Portion	
Preferential Routes	D
Pacific Oceanic Transition Routes (PACOTS**)**	J
TACAN Routes – Australia	M
North American Routes for North Atlantic Traffic – Non-common Portion	N
Preferred/Preferential Overflight Routes	O
Preferred Routes	P
Traffic Orientation System Routes (TOS)	S
Tower Enroute Control Routes (TEC)	T

Airport SID (PD) and Heliport SID (HD) Records

SID Route Type Description	Field Content
Engine Out SID	0
SID Runway Transition	1
SID or SID Common Route	2
SID Enroute Transition	3
RNP SID Run Way Transition	**R**
RNP SID or SID Common Route	**N**
RNP SID Enroute Transition	**P**
Vector SID Runway Transition	T
Vector SID Enroute Transition	V

Qualifier Description	**Qualifier 1 Field Content**	**Qualifier 2 Field Content**
DME Required	**D**	
GNSS Required	**G**	
Radar Required	**R**	
Helicopter SID from Runway	**H**	
VOR/DME RNAV		**C Note 1**
Data Base Supported RNAV		**D Note 2**
FMS Required		**F Note 3**
Conventional Departures		**G**

Note 1: Used for Conventional Area Navigation Departures, using RHO-RHO or RHO-THETA equipment. Coding Rules for such procedures are provided in Attachment 5.

Note 2: RNAV Departures which do not include references to ground based navaid associated information such as VOR/TACAN radials, NDB bearings, DME fixes/arcs, ILS courses, and MLS azimuths. Coding Rules for such procedures are provided in Attachment 5.

Note 3: Used when the government authority has designated a Departure as "FMS".

Airport STAR (PE) and Heliport STAR (HE) Records

STAR Route Type Description	Field Content
STAR Enroute Transition	1
STAR or STAR Common Route	2
STAR Runway Transition	3
RNP STAR Enroute Transition	**R**
RNP STAR or STAR Common Route	**N**
RNP STAR Runway Transitions	**P**

Qualifier Description	**Qualifier 1 Field Content**	**Qualifier 2 Field Content**
DME Required	**D**	
Radar Required	**R**	
RF Leg Capability Required	**F**	
GNSS Required	**G**	
Helicopter STAR to Runway	**H**	
Profile Descent STAR	**P**	
VOR/DME RNAV		**C**
Database Supported RNAV		**D**
FMS Required		**F Note 2**
Conventional Arrivals		**G**

Note 1: The RF Leg Capability Required Qualifier 1 of "F" is used only in conjunction with the non-RNP Route Types.

Note 2: Used when the government authority has designated an Arrival as "FMS."

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Airport Approach (PF) and Heliport Approach (HF) Records

Approach Route Type Description	Route Type Field Content
Approach Transition	A
Localizer/Backcourse Approach	B
VORDME Approach	D
Flight Management System (FMS) Approach	F
Instrument Guidance System (IGS) Approach	G
Area Navigation (RNAV) Approach with Required Navigation Performance (RNP) Approach	**H**
Instrument Landing System (ILS) Approach	I
GNSS Landing System (GLS)Approach	J
Localizer Only (LOC) Approach	L
Microwave Landing System (MLS) Approach	M
Non-Directional Beacon (NDB) Approach	N
Global Positioning System (GPS) Approach	P
Non-Directional Beacon + DME (NDB+DME) Approach	Q
Area Navigation (RNAV) Approach (Note 1)	R
VOR Approach using VORDME/VORTAC	S
TACAN Approach	T
Simplified Directional Facility (SDF) Approach	U
VOR Approach	V
Microwave Landing System (MLS), Type A Approach	W
Localizer Directional Aid (LDA) Approach	X
Microwave Landing System (MLS), Type B and C Approach	Y
Missed Approach	Z

The listing above for Approach Route Type is alphabetical and does not represent any kind of priority.

Airport Approach (PF) and Heliport (HF) Records – Route Type

Qualifier Description	Qualifier 1 Field Content (Note 1)	Qualifier 2 Field Content Note 2
DME Required for Procedure	D (Note 5)	
GPS (GNSS) required, DME/DME to RNP xx.x not authorized	J (Note 2)	
SAAAR/AR Procedure (RNAV RNP)	**F (Note 8)**	
Advanced (RNAV RNP, SAAAR/AR not required)	**A (Note 8)**	
GBAS Procedure	L (Note 2)	
DME Not Required for Procedure	N (Note 5)	
GNSS Required	P	
GPS (GNSS) or DME/DME to RNP xx.x required	R (Note 2)	
DME/DME Required for Procedure	T (Note 2)	
RNAV, Sensor Not Specified	U (Note 2)	
VOR/DME RNAV	V (Note 2)	
RNAV Procedure that Requires FAS Data Block	W (Note 4)	
Primary Missed Approach		A (Note 6)
Secondary Missed Approach		B (Note 6)
Engine Out Missed Approach		E (Note 6)
Procedure **with Circle–To-Land** Minimums		C (Note 3)
Helicopter with straight-in minimums		**H (Note 6)**
Helicopter with circle-to-land minimums		**I (Note 7)**
Helicopter with Helicopter Landing Minimums		**L (Note 6, 9)**
Procedure with Straight-in Minimums		S

Not all Qualifiers will apply to all Route Types, see notes below. Qualifier fields may be blank where their use is not required by source documentation.

Note 1: Qualifier 1 and 2 are carried on each sequence of every transition for Approach Procedure Coding (Approach Transition, Final Approach and Missed Approach) and will be identical for each sequence in a specific transition. Qualifier 2 will be different between Approach Transitions/Final approach coding where “S” or “C” will be used and Missed Approach where “A”, “B” or “E” will be used (See Note 6).

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Note 2: Route Type "R" is used for all types of RNAV procedure coding, excluding GLS **and RNP** Procedures that are coded as Route Type "J" **and "H"**. The type of RNAV Procedure is further defined through the content of Qualifier 1.

a. Conventional Area Navigation Approach Procedures using RHO-RHO or RHO-THETA equipment are coded as Route Type "R" and Qualifier 1 of "T" or "V".

b. GNSS based RNAV Approach Procedures are coded as Route Type "R" with Qualifier 1 set to "J", "L", "R" or "U" as required by source publications and mapped to this table.

Note 3: In Approach Transition and Final Approach Coding, Qualifier 2 is set **to indicate the type of minimums applicable to the coding as indicated in the table. A Qualifier 2 of "S" or "H" means the procedure has been coded as straight-in. There may also be circle-to-land minimums for the same procedure.** Qualifier 2 is required for all Route Types and is independent of the content of Qualifier 1.

Note 4: A Qualifier 1 value of "W" is used to indicate that the RNAV Procedure is authorized for SBAS only and requires the ARINC 424 Path Point with the Final approach Segment (FAS) Data Block. No other navigation senor is authorized for these procedures. Examples:

A RNAV (GPS) or RNAV (GNSS) Approach Procedure is authorized for SBAS navigation and requires the FAS Data Block. The Route Type would be coded as "R" and Qualifier 1 would be coded as "W". The associated GNSS/FMS Indicator (Section 5.222) would be set to indicate that SBAS is authorized. A Path Point Record carrying the FAS Data Block would be provided for the procedure. A Procedure Data Continuation Record would be provided and would be used to define the Levels of Service authorized and the official government source documentation Names for these Services.

A RNAV (GPS) or RNAV (GNSS) Approach Procedure is authorized for SBAS and also for single or multiple sensors other than SBAS. The Route Type would be coded as "R" and Qualifier 1 would be coded as "J" or "R", as appropriate. The setting of the GNSS/FMS Indicator would be appropriate to the level of authorization. A Path Point Record would or would not be provided, according to government source publications. A Procedure Data Continuation Record would be provided and would be used to define the Levels of Services authorized for SBAS and the official government source documentation Names for these Services.

A RNAV (GPS) or RNAV (GNSS) Approach Procedure is authorized for a single or multiple sensors other than SBAS; SBAS is not authorized. The Route Type would be coded as "R" and Qualifier 1 would be coded as "J", "R" "L", "U" or "P" as appropriate. The setting of the GNSS/FMS Indicator would be appropriate to the level of authorization. No Path Point Record would be provided. No Procedure Data Continuation Record would be provided.

Note 5: The Qualifier 1 codes of "D" and "N" are not used on RNAV Procedures (Route Type "R") of any kind. Additionally, these codes are not used in conjunction with Route Types that provide a "DME Option" of a procedure; specifically, they are not used in conjunction with the Route Types "D", "N", "Q", "S" or "V".

Note 6: The Qualifier 2 codes of "A", "B" and "E" can only be used in conjunction with a Route Type of "Z" = Missed Approach Coding. Qualifier 2 codes of "C", "S"**, "H", and "I"** can be used in conjunction with any Route Type except "Z".

Note 7: The Qualifier 2 code of "H" or "I" is only used with Approach (PF) Records.

Note 8: Used only for route type "H". The Qualifier 1 code of "F" indicates that the RNAV RNP procedure is a FAA Special Aircraft and Aircrew Authorization Required (SAAAR) or ICAO Authorization Required (AR) procedure. Qualifier 'A' indicates RNAV RNP without a SAAAR/AR requirement. Implied GNSS required.

Note 9: The Qualifier 2 code of "L" is used with Airport Procedure (PF) Records and Heliport Procedure (HF) Records and only for those government sources that provide Helicopter Minimums without specifying Straight-In or Circle-To-Land.

Used On:	Enroute Airways, Airport and Heliport SID/STAR/Approach, Preferred Route **and Company Route Records and Helicopter Operations Company Route Records**.
Length:	1 character for Enroute Airways, Airport and Heliport SID and STAR and Preferred Routes. 3 characters for Airport and Heliport Approach primary records.
Character Type:	Alpha/numeric
Approach Examples:	LDC = a Localizer-based procedure, for localizer only, no glide slope, with DME required, Circle-To-Land Minimums. LNC= A Localizer-based procedure, for localizer only, no glide slope, DME not required, Circle-To-Land Minimums S S = A VOR procedure, using VORDME or VORTAC Navaid, the DME is not required for the procedure, the minimums are straight-in.

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S C = A VOR procedure, using VORDME or VORTAC Navaid, the DME is not required for the procedure, the minimums are Circle-To-Land

D S = A VOR/DME procedure, using a VORDME or VORTAC Navaid, the DME is required for the procedure, the minimums are straight-in.

V S = A VOR procedure using VOR Navaid with only NAVAID, no DME installed, minimums are straight-in.

V C = A VOR procedure, using a VOR Navaid with no DME capability, the minimums are Circle-To-Land

N S = A NDB procedure, minimums are straight-in.

Q S = A NDB + DME procedure, the DME is required, the minimums are straight-in.

I H = ILS procedure, no DME requirements, procedure is designed for Helicopter operations to a runway at an airport, records are contents in Airport Approach (PF) file section.

I _ = ILS Procedure, no DME requirements, procedure is designed for Helicopter operations to a helipad at a heliport, records are contents in Heliport Approach (HF) file section.

5.8 Route Identifier (ROUTE IDENT)

Definition/Description: The "Route Identifier" field identifies a route of flight or traffic orientation, using the coding employed on aeronautical navigation charts and related publications.

Source/Content: For Enroute Airways, Route Identifier codes should be derived from official government publications. For Preferred Routes, Route Identifiers may or may not be provided in government publications. Where they are available, they will be used.

For North American Routes for North Atlantic Traffic, "Common Portion" and other similar route system, route identifier code shall be those published in government sources. For the European Traffic Orientation System or other similar route systems such as North American Routes for North Atlantic Traffic, "Non-common Portion," Preferred Routes and Preferential Routes published without official and/or flight plan identifiers, but published as between specific airports or other navigation fixes, route identifiers define the initial fix and the terminus fix idents according to the naming rules in Chapter 7. For routings which do not include a unique initial or terminus fix, rules on creating unique Route Identifiers are also contained in Chapter 7. Those rules have been developed with use of the Geographical Reference Tables (TG). Refer to Chapter 3, Section 3.2.7.2 and Chapter 4, Section 4.1.26 for more detail.

Used On:	Enroute Airway, Preferred Route Records and Geographical Reference Table
Length:	Enroute Airway - 5 character maximum Preferred Route - 10 character maximum
Character Type:	Alpha/numeric
Examples:	Enroute Airway - V216, C1150, J380, UA16, UB414 Preferred Routes - N111B, TOS13, TOS14WK, CYYLCYYC, ARTCOLAR, KZTLKSAV, SCNDICANRY

Refer to Section 7 for specific examples and their meaning.

5.9 SID/STAR Route Identifier (SID/STAR IDENT)

Definition/Description: "The SID/STAR Route Identifier" field contains the name of the SID or STAR, using the basic indicator, validity indicator and route indicator abbreviated to six characters with the naming rules in Chapter 7 of this document.

Source/Content: SID/STAR route identifier codes should be derived from official government publications describing the terminal procedures structure.

Used On:	Airport SID/STAR, Heliport SID/STAR and Flight Planning Arrival/Departure Data Records
Length:	6 characters max
Character Type:	Alpha/numeric
Examples:	DEPU2, SCK4, TRP7, 41M3, MONTH6

5.10 Approach Route Identifier (APPROACH IDENT)

Definition/Description: The "Approach Route Identifier" field contains the identifier of the approach route to be flown. To facilitate the provision of multiple approach procedures of the same type to a given runway, the field also is used to provide a "multiple indicator."

Source/Content:

Runway Dependent Procedure Ident

Column	Contents	
1	Type of Approach-Alpha Character, generally the same as the first column of field 5.7 Route Type **(Note 1)**	
2-3	Runway Identification- Numeric in tens of degrees, valid range 01-36	
4	Runway Designation	
	- (dash)	Place holder if other runway designation codes are not present and multiple indicators required.
	L	Left
	R	Right
	C	Center
	T	Runway oriented to True North
	Blank	Position 5 and 6 must also be Blank
5	Multiple Indicator Alphanumeric or Blank	
6	Blank	

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Circle-to-Land Procedures Identifier

Column	Contents	
1-3	Circling Procedure Ident (See below).	
4	- (dash)	Place holder if a multiple indicator is required
	A thru S, U thru Z	A government source provided procedure suffix that is not a multiple indicator
	T	Runway oriented to True North
	Blank	Position 5 and 6 must also be Blank
5	Multiple Indicator Alphanumeric or Blank	
6	Blank	

Circle-to-Land Route Type Identifier

Route Type Field Content (5.7)	1ST Three Characters of Circling Procedure Identifier
A	(Approach Transitions)
B	LBC
D	VDM
F	FMS
G	IGS
H	RNV
I	(No Circling ILS)
J	GLS
L	LOC
M	MLS
N	NDB
P	GPS
Q	NDM
R	RNV
S	VOR
T	TAC
U	SDF
V	VOR
W	MLS
X	LDA
Y	MLS
Z	(Missed Approach)

Helicopter Approach Procedures to Runways OR Final Approach Course Procedure Identifier

Column	Contents
1	Type of Approach-Alpha Character, generally the same as the first column of field 5.7 Route Type. (Note 1)
2-4	Three digit numeric character representing the runway designation or procedure final approach course, expressed in full degrees
5	Multiple Indicator Alphanumeric or "T" if the procedure approach course is oriented to true north or Blank
6	Blank

Helicopter Approach Procedures to Heliports and Coded to a Specific Pad Identifier

Column	Contents
1	Type of Approach-Alpha Character, generally the same as the first column of field 5.7 Route Type. (Note 1)
2-6	Pad Identification

Note 1: When the Route Type (Section 5.7) is equal to "H" and the column 1 of the Procedure Identifier is also equal to "H", the procedure is a RNAV RNP procedure. Such a procedure would normally have column 1 of the procedure identifier equal to "R." The character "H" is used to indicate that this RNAV procedure has been published with a "RNAV RNP" procedure title in the official government source documents. This allows the explicit identification of RNAV RNP titled procedures.

Used On: Airport and Heliport Approach Route Records, Flight Planning Arrival/Departure Data, Path Point and Airport, Heliport Localizer, Airport and Heliport TAA, and Simulation Continuation Records.

Length: 6 characters max.

Character Type: Alpha/numeric

Examples:

Runway Dependent: I26L, B08R, R29, V01L, N35 L16RA, L16RB, V08-A, V08-B I18L1, I18L2, N08T R35-Y, R35-Z

Circle-To-Land: **VOR, VDM, LOC**
VOR-A, VOR-B, NDB-1, NDB-2 (These are multiple indicators)
NDBB, VDMA, LOCD, NDBTA (There are source provided procedure suffixes

Helicopter to Runway: I13L, L040, V175, N175B

Helicopter to Helipad: IA127 = ILS Procedure to a pad designated A127
VBRAVO =VOR Procedure to a Pad designated BRAVO
N23 =NDB Procedure to a Pad designated 23
RWESTA RNAV Procedure to a Pad designated West Alpha

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.11 Transition Identifier (TRANS IDENT)

Definition/Description: The "Transition Identifier" field describes the type of transition to be made from the enroute environment into the terminal area and vice versa, and from the terminal area to the approach or from the runway or helipad to the terminal area.

Source Content: The content of the transition identifier field should be determined from the content of the Route Type field (See Section 5.7) in accordance with the rules set forth in Figure 5.3**.**

Record	Route Type	Field Content
Engine Out SID	0	Runway (RWY) or Pad Identifier
SID/RNAV SID	1	Runway (RWY) or Pad Identifier
	2	**Blank/RWY/PAD/ALL (Note 1 and 3)**
	3	**SID Enroute Transition Identifier (Note 5)**
Vector SID	T	Runway (RWY) or Pad Identifier
	V	Vector SID Enroute Transition Identifier
STAR/ RNAV STAR	1	STAR Enroute Transition Identifier **(Note 5)**
	2	**Blank RWY/PAD/ALL/ (Note 3)**
	3	**Runway (RWY) or Pad Identifier (Note 2)**
Approach Transitions	A	Approach Transition Identifier
Missed Approach	Z	Missed Approach Transition Identifier (Note 4)
Approach Procedure	All Other Codes Except A and Z (see Section 5.7)	Blank

Figure 5-3
Transition Identifier Field Content

Note 1: If there is no Route Type 1 for the SID, then the SID Records with the Route Type of 2 will have an entry in the Transition Identifier field. If there is a Route Type of 1 for the procedure, then the records with the Route Type of 2 will carry a "blank" transition identifier.

Note 2: If there is no Route Type 3 for the STAR, then the STAR record with the Route Type of 2 will have an entry in the Transition Identifier field. If there is a Route Type 3 for the procedure, then the Transition Identifier in the Route Type 2 will carry a "blank" transition identifier.

Note 3: The use of "ALL" in the Transition Identifier field indicates that the procedure is valid for two or more runways at an airport or all helipads at a heliport. The use of the character "B" along with the runway designation such as RW08B in the Transition Identifier field indicates that the transition is valid for two or more parallel runways, e.g. RW08L and RW08R.

Note 4: The Missed Approach Transition Identifier will only be used in missed approach procedure coding where the Qualifier Two field (5.7) has been set to "B". It will be the identifier of the Missed Approach Holding Fix or the last fix in the missed approach procedure coding if there is no missed approach holding fix. In cases where there are multiple instances of Qualifier Two set to "B" for a given approach procedure, the Missed Approach Transition Identifier will be modified according to the rules in Attachment 5, Section 7.5.

Note 5: Enroute Transition Identifiers are normally the identifier of the navaid or waypoint.

Transition Identifiers should be derived from official government sources, where provided.

Used On:	Airport and Heliport SID/STAR/ Approach, Flight Planning Arrival/Departure Data and Company Route Records
Length:	5 characters max.
Character Type:	Alpha/numeric
Examples:	9TU, ETX, KEENE, DEN, RW08R, Blank

5.12 Sequence Number (SEQ NR)

Definition/Description: For Route Type Records - A route of flight is defined by a series of records taken in order. The "Sequence Number" field defines the location of the record in the sequence defining the route of flight identified in the route identifier field. For Boundary Type Records - A boundary is defined by a series of records taken in order. The "Sequence Number" field defines the location of the record in the sequence defining a boundary. For Record Types requiring more than one primary record to define the complete content – In a series of records used to define a complete condition, the "Sequence Number" is used to define each primary record in the sequence. For Airport and Heliport TAA Records – Sequence Number 1 will always be assigned to the record based on the Center Fix upon which the Straight-In Area is predicated, Sequence Number 2 will always be assigned to the record based on the Center Fix upon which the Left Base Area is predicated, and Sequence Number 3 will always be assigned to the record based on the Center Fix upon which the Right Base Area is predicated. Therefore, if a TAA Record has a Straight-In Area and a Right Base Area, but no Left Base Area, only Sequence Numbers 1 and 3 will be used. If a TAA Record has a Straight-In Area and a Left Base Area but no Right Base Area, only Sequence Numbers 1 and 2 will be used.

Source/Content: Sequence numbers are assigned during the route, boundary or sequence definition phase of the data file assembly. Sequence numbers are assigned so as not be duplicated within the route, boundary or sequence assigned a unique identification/designation. For three or four digit Sequence Numbers, initially, an increment of ten should be maintained between the sequence numbers assigned to consecutive records. For one or two digit Sequence Numbers, the initial increment is one. In route or boundary records, should subsequent maintenance of the file necessitate the addition of a record or records, the new record(s) should be located in the correct position in the sequence and assigned a sequence number whose most significant characters are identical to those in the sequence

number of the preceding record in sequence. The unit character should be assigned a value midway between the units character values of the preceding and following record sequence numbers. For example, if it is desired to add one record to the sequence and the units characters of both the preceding and following records at the desired location are zeros (indicating no previous modification at this point), the units character or the inserted record's sequence number should be five (5). For records taken in sequence with one or two digit sequence numbers, additional data must be entered in the proper sequence and all subsequent records will be up numbered accordingly.

When an enroute airway crosses the boundary separating two geographical areas (Section 5.3), the airway fix lying on or closest to the boundary shall be coded twice, once for each geographical area, and should be assigned the same sequence number in each case. Record uniqueness in such cases is maintained through the "Boundary Code" (Section 5.18). Enroute airway record sequence numbers should be assigned in a manner which permits them to be arranged into continuous airway routes in flight sequence order when sorted according to the Route Identifier and Sequence Number only, without regard to their applicable Geographical Area Code.

When used on Enroute, Airport and Heliport Communications Primary and Continuation records, the Sequence Number is used as a "record counter" within a given Identifier and Communications Class for providing output file record uniqueness.

Used On:	Enroute Airways, Airport and Heliport MSA Records, Airport and Heliport TAA Records, Airport and Heliport SID/STAR/Approach, Company Route, Cruise Tables, FIR/UIR, Restrictive Airspace, Controlled Airspace, Preferred Routes, Flight Planning Arrival/Departure Data and VHF Navaid Limitation Continuation Records, **Helicopter Operations Company Routes**
Length:	4 characters - Enroute Airways, Preferred Routes, FIR/UIR and Restrictive Airspace 3 characters - SID/STAR/Approach and Company Routes 2 characters - VHF Navaid Limitation Continuation Records 1 character - MSA Table, TAA Table, Cruise Table
Character Type:	Numeric
Examples:	0010, 0135, 2076, 120, 030, 01, 84, 3

5.13 Fix Identifier (FIX IDENT)

Definition/Description: The "Fix Identifier" field contains the five-character-name-code, or other series of characters, with which the fix is identified. This includes Waypoint Identifiers, VHF NAVAID Identifiers, NDB NAVAID identifier, Airport Identifiers, and Runway Identifiers.

Source/Content: Officially published identifiers or identifiers derived in accordance with Chapter 7, Naming Conventions, of this document.

Used On:	Holding Patterns, Enroute Airways, Airport and Heliport SID/STAR/ Approach, Enroute Airway Restrictions, and Enroute Waypoints, Airport and Heliport Terminal Waypoints (Waypoint Ident) and Flight Planning Arrival/Departure Data Records.
Length:	5 characters max
Character Type:	Alpha/numeric (no embedded blanks)
Examples:	SHARP, DEN43, BHM, RW27L, KGRR

5.14 ICAO Code (ICAO CODE)

Definition/Description: The "ICAO Code" field permits records to be categorized geographically within the limits of the categorization performed by the "Area Code" field.

Source/Content: The code is to be employed in the ICAO code field may be found in ICAO Document No. 7910, "Location Indicators."

In order to permit sub-division of the United States into more easily manageable regions, the ICAO code for the USA (K) is followed by a numeric character obtained from Figure 5-5.

Used On:	All records except Cruising Tables and Grid MORA
Length:	2 characters max
Character Type:	Alpha/numeric
Examples:	K1,K7,PA,MM,EG,UT

5.15 Intentionally Left Blank

5.16 Continuation Record Number (CONT NR)

Definition/Description: When it is not possible to store all the information needed on a record within the 132 columns of the record itself, the so-called Primary Record; one or more continuation records may be used. The "Continuation Record Number" identifies the position of a continuation record in a sequence of such records.

Source/Content: Primary records contain the numeric "0" when no Continuation Records are included in the file for that Primary. The numeric "1" in this field of the Primary Record indicates that one or more Continuation Records follow the Primary Record. Continuation Records are numbered sequentially starting with the numeric "2" in the first continuation. If the information requirement goes beyond a Continuation Record with the numeric "9," the sequence is continued with alpha characters, starting with "A" and continuing through to "Z" as required.

Used On:	All records except Company Route, Airport Localizer Marker/Locator, Enroute Markers, Cruising Tables, FIR/UIR and Grid MORA
Length:	1 character
Character Type:	Alpha/numeric
Examples:	0, 1, 2 (through 9) A, B, C (through Z)

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.17 Waypoint Description Code (DESC CODE)

Definition/Description: **The "Waypoint Description" field facilitates the designation of the type, function, and attributes of a specific waypoint in Enroute Airway or Terminal Procedure segment coding.**

Source/Content: Valid contents for the "Waypoint Description Code" are contained in the **following** table**:**

Waypoint Description	**Enroute, STAR, APRCH for the line "Airport as Waypoint"**	**Column**	**Column**	**Column**	**Column**	**Remarks**
Type/ Function/ **Attribute**	Used On	**40**	**41**	**42**	**43**	
Airport as Waypoint	STAR, APCH	A				
Essential Waypoint	Enroute, SID, STAR, APCH	E				**Note 1**
Off Airway **Floating** Waypoint	Enroute	F				**Note 1**
Runway as Waypoint, Helipad as Waypoint	SID, STAR, APCH	G				
Heliport as Waypoint	STAR, APCH	H				
NDB Navaid as Waypoint	Enroute, SID, STAR, APCH	N				
Phantom Waypoint	SID, STAR, APCH	P				**Note 1**
Non-Essential Waypoint	Enroute	R				**Note 1**
Transition Essential Waypoint	Enroute	T				**Note 1**
VHF Navaid As Waypoint	Enroute, SID, STAR, APCH	V				
Flyover Waypoint, End**ing Leg**	SID, STAR, APCH		B			**Note 2**
End of **Continuous Segment**	Enroute, SID, STAR, APCH		E			**Note 2**
Uncharted Airway Intersection	Enroute		U			**Note 1**
Fly-Over Waypoint	APCH, SID, STAR,		Y			**Note 2**
Unnamed Stepdown Fix Final Approach **Segment**	APCH			A		
Unnamed Stepdown Fix **Intermediate Approach Segment**	APCH			B		
ATC Compulsory **Reporting Point**	SID, STAR, APCH Enroute			C		**Note 1**
Oceanic Gateway Waypoint	Enroute			G		**Note 1**
First Leg of Missed Approach Procedure	APCH			M		**Note 3**
Fix used for turning final approach (Course Change)	**APCH**			**R**		**Note 4**
Named Stepdown Fix	APCH			S		
Initial Approach Fix	APCH				A	**Note 1**
Intermediate Approach Fix	APCH				B	**Note 1**
Holding at Initial Approach Fix	APCH				C	
Initial Approach Fix **at FACF**	APCH				D	
Final End Point	APCH				E	**Note 1**
Final Approach Fix	APCH				F	**Note 1**
Source provided Enroute Waypoint without Holding	Enroute				G	
Source provided Enroute Waypoint with Holding	Enroute SID, STAR, APCH				H	
Final Approach Course Fix	APCH				I	**Note 1**
Missed Approach Point	**APCH**				M	**Note 1**
Engine Out SID Missed Approach Disarm Point	**SID (Engine Out), APCH**				N	**Note 5**

Figure 5-4 Waypoint Description

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Generic Note **There is a Waypoint Description field for each coding segment of an Enroute Airway or Terminal Procedure. For Enroute Airways, Column 40 will never be blank. For Terminal Procedures, Column 40 may be blank when the path terminator for that segment does not reference a fix. For details on path terminators and more information on sequence coding, refer to Attachments 5.**

Note 1: **For a definition of the waypoint type, function or attributes, refer to Section Two, Special Navigation Terms, of this Specification.**

Note 2: **The basic method of indicating that the government source has designated a specific fix as an Overfly Waypoint, meaning the fix must be overflown before commencing the maneuver defined in the subsequent leg, is to code a "Y" in Column 41. End of Continuous Segment indications are not source provided data, but rather an implementation of the translation of that source data based on this Specification. Column 41 End of Continuous indications of "E" will be provided in the last segment of an individual Terminal Procedure Transition and at the end of a specific airway. The indication is also provided in airway coding when the basic route designation continues beyond the fix, but there is a gap in the route definition. And the indication is provided in airway coding when there is a change in ARINC Area Code (Section 5.3) in the subsequent leg. In Terminal Procedure coding, when both conditions exist, i.e. the fix has been designated as an Overfly Waypoint and the End of Continuous Segment indication is required by the rules in this Specification, Column 41 is set to "B".**

Note 3: **The First Leg of Missed indication, "M" in column 42 is coded on the first leg of approach procedure coding that follows the designation of the Missed Approach Point (MAP) in Column 43.**

Note 4: **RF and only RF beginning and ending fixes also not designated as a procedure fix. The RF code will take precedence over a step-down fix code at the same fix.**

Notes 5: **An "N" in column 43 of an engine out SID or missed approach record designates the waypoint as the engine out SID (or missed approach) "disarm" point. For example if an engine failure is detected before this point, the engine out procedure is automatically loaded. If the engine failure is detected after this point, the engine out SID is not automatically loaded.**

Used On	Airport and Heliport SID/STAR/Approach, Enroute Airway Records
Length:	4 Characters
Character Type:	Alpha

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Figure 5-5
7 Subdivisions for United States

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5.18 Boundary Code (BDY CODE)

Definition/Description: Routes of flight frequently cross geographical boundaries. The "Boundary Code" field identifies the area into, or from which a continuous route passes when such a crossing occurs.

Source/Content: See Table 5-6.

Used On: Enroute Airways records
Length: 1 character
Character Type: Alpha/numeric

Area	Area Code*	Boundary Code
USA	USA	U
Canada and Alaska	CAN	C
Pacific	PAC	P
Latin America	LAM	L
South America	SAM	S
South Pacific	SPA	1
Europe	EUR	E
Eastern Europe	EEU	2
Middle East South Asia	MES	M
Africa	AFR	A

*From Figure 5-2

Table 5-6 - Boundary Codes

5.19 Level (LEVEL)

Definition/Description: The Level field defines the airway structure of which the record is an element.

Source/Content:

B	All Altitudes
H	High level Airways
L	Low Level Airways

Used On: Enroute Airway, Preferred Routes, Restrictive Airspace, and Controlled Airspace records
Length: 1 character
Character Type: Alpha

5.20 Turn Direction (TURN DIR)

Definition/Description: The "Turn Direction" field specifies the direction in which Terminal Procedure turns are to be made. It is also used to indication direction on course reversals, see Attachment 5 Path and Termination.

Source/Content: The field contains the alpha character "L" for Left turns, "R" for Right turns and "E" for turns in either direction.

Used On: Airport and Heliport SID/STAR/Approach records
Length: 1 character
Character Type: Alpha

5.21 Path and Termination (PATH TERM)

Definition/Description: The Path and Termination defines the path geometry for a single record of an ATC terminal procedure.

Source/Content: Attachment 5 to this document, "Path and Terminator," contains the various Path Term codes available for coding an ATC terminal procedure.

Used On: Airport and Heliport SID/STAR/Approach records
Length: 2 characters
Character Type: Alpha

5.22 Turn Direction Valid (TDV)

Definition/Description: This field is used in conjunction with Turn direction to indicate that a turn is required prior to capturing the path defined in a terminal procedure.

Source/Content: The field contains the alpha character "Y" when a turn is required prior to beginning the leg defined by the Path Term. The direction of the turn is specified in Section 5.20.

Used On: Airport and Heliport SID/STAR/Approach Records
Length: 1 character
Character Type: Alpha

5.23 Recommended NAVAID (RECD NAV)

Definition/Description: The "Recommended Navaid" field allows the reference facility for the waypoint in a given record "Fix Ident" field or for an Airport or Heliport to be specified. VHF, NDB (Enroute and Terminal), Localizer, TACAN, GLS and MLS Navaids may be referenced.

Source/Content: The 1, 2, 3 or 4 character identification of the Navaid appears in this field. Navaids recommended for waypoint reference in official government publications will be used when available. The following general rules on field content apply:

Procedures that use coding which require leg types referenced to specific navaids are covered by the procedure coding rules in Attachment 5 to this Specification.

a. A "VHF Navaid" may be any VOR, DME, VORDME, VORTAC, TACAN**,** Un-Biased ILSDME **or MLSDME** available in the database **following the specific rules in Table 5-7.**

b. A "NDB Navaid" may be any NDB or Locator **available in the Enroute Terminal** NDB **files in the database.**

c. Localizers, and MLS Azimuth are used as Recommended Navaids for procedures that reference those navaids, **including RNAV Transitions to these types of procedures.**

d. The Recommended Navaid in final approach coding will be the procedure reference facility**. As not all Final Approach Procedure reference a Navaid, i.e.**

RNAV and GPS, the Recommended Navaid is not provided in these types of procedure, see Attachment 5 for specific rules.

e. The Recommended Navaid in Airport and Heliport Records will be any VOR, VORDME, or VORTAC available in the database.

f. The Recommended Navaid in any Enroute Airway Record, **when provided,** will be any VORDME or VORTAC available in the database.

g. The Recommended Navaid in any Terminal Procedure Record other than the final approach coding will be the procedure reference facility of a type from the Definition/Description paragraph above and will be in accordance with the rules governing Recommended Navaids for Path Terminators and coding rule as defined in Attachment 5 of this Specification.

h. The rules for Recommended Navaids for Converging ILS Approach Procedures are the same as for ILS Approach Procedures.

i. The Recommended Navaid used in a GLS Approach Procedure will be the GLS Reference Path identifier appropriate to the runway and approach.

j. The use of non-collocated facilities of the types VORDME, VORTAC and Localizer/ILSDME or ILSTACAN as the recommended navaid in terminal procedure coding is limited to defined circumstances only. For a definition of "non-collocated", refer to Section 5.35 of this Specification. For the defined circumstances, refer to Figure 5.7 of this Specification.

Used On:	Enroute Airway Record, Airport and Heliport SID/STAR/Approach Records, Airport and Heliport Record
Length:	4 characters max.
Character Type:	Alpha/numeric
Examples:	P, PP, DEN, LAX, ILAX, MJFK

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Facility Type / Procedure User	SID/STAR	Approach Transitions	Path Terminator –CR, VR	Path Terminator –CD, VD	Path Terminator - FD	Missed Approach **Procedure**	Localizer Final Approach & Transitions of course or heading to intercept localizer	VORDME/VORTAC Final Approach	VOR Only Final Approach **Coding** & Single Record Transitions	NDB Only Final Approach **Coding**	NDB + DME Final Approach *Coding*	TACAN Final Approach **Coding**	Airports
Co-located VOR DME/VORTAC Theta-Rho	X X | X	X X | X	X X |	X |	X X | X	X X | X		X X | X	 |	 |	**[2]** | X	 |	X |
Non-Co-located VORDME/ VORTAC Theta/Rho	 |	X X | X	 |	 |	 |	X X | X	 |	X X | X	 |	 |	**[2]** | X	 |	X |
Localizer Theta/Rho	 |	X |	X |	 |	 |	X |	X X | X	 |	 |	 |	 |	 |	 |
VOR Theta/Rho	 |	X |	X X |	 |	 |	X |	 |	 |	X X |	 |	 |	 |	X |
DME Theta/Rho	 |	 |	 |	X |	X | X	X |	 |	 |	 |	 |	**[2]** | X	 |	 |
TACAN Theta/Rho	X X | X	X X | X	X |	X |	X | X	X X | X	 |	 |	 |	 |	**[2]** | X	X X | X	X |
NDB Theta/Rho	 |	X |	 |	 |	 |	X |	 |	 |	 |	X |	**[1]** |	 |	 |
ILSDME or ILSTACAN [3] **Theata/Rho**	 |	 |	 |	**X** |	**X** | **X**	**X** |	 |	 |	 |	 |	**[2]** | **X**	 |	 |

[1] On FACF and FAF Records

[2] On Runway/MAP Records Only

[3] ILSDMEs and ILSTACANs must be unbiased for use as a recommended navaid. They do not have to be collocated with the frequency paired localizer for use as a recommended navaid in the instances allowed.

Figure 5-7 Procedure Use

5.0 NAVIGATION DATA – FIELD DEFINITIONS

5.24 Theta (THETA)

Definition/Description: "Theta" is defined as the magnetic bearing to the waypoint identified in the record's "FIX Ident" field from the Navaid in the "Recommended Navaid" field.

Source/Content: Theta values are derived from official government sources when available. They are provided in degrees and tenths of a degree, with the decimal point suppressed. The content is controlled through requirements of the Path Terminator and coding rules contained in Attachment 5 of this specification.

Used On:	Airport and Heliport SID/STAR/Approach, Enroute Airway Records
Length:	4 characters
Character Type:	Alpha/numeric
Examples:	0000, 0756, 1217 1800

5.25 Rho (RHO)

Definition/Description: "RHO" is defined as the geodesic distance in nautical miles to the waypoint identified in the record's "Fix Ident" field from the NAVAID in the "Recommended NAVAID" field.

Source/Content: Rho values derived from official government sources will be used when available. They are entered into the field in nautical miles and tenths of a nautical mile, with the decimal point suppressed. The content is controlled through requirements of the Path Terminator and coding rules contained in Attachment 5 of this specification.

Used On:	Airport and Heliport SID/STAR/Approach, Enroute Airway Records
Length:	4 characters
Character Type:	Alpha/numeric
Examples:	0000, 0216, 0142, 1074

5.26 Outbound Magnetic Course (OB MAG CRS)

Definition/Description: "Outbound Magnetic Course" is the published outbound magnetic course from the waypoint identified in the record's "Fix Ident" field. In addition, this field is used for Course/Heading/Radials on SID/STAR Approach Records through requirements of the Path Terminator and coding rules contained in Attachment 5 of this specification.

Source/Content: Values from official government sources will be used when available. The field contains magnetic information expressed in degrees and tenths of a degree, with the decimal point suppressed. For route and procedure segments charted in "degrees true," the last character (tenths position) of the field will contain the character "T". See Section 5.165 of this document for more information on "degrees true" information.

Used On:	Airport and Heliport SID/STAR/Approach, Enroute Airway and Flight Planning Arrival/ Departure Data Records
Length:	4 characters
Character Type:	Alpha/numeric
Examples:	2760, 0231, 194T

5.27 Route Distance From, Holding Distance/Time (RTE DIST FROM, HOLD DIST/TIME)

Definition/Description: In Enroute Airways, "Route Distance From" is the distance in nautical miles from the waypoint identified in the records "Fix Ident" field to the next waypoint of the route. In SID, STAR and Approach Procedure records, the field may contain segment distances/along track distances/excursion distances/DME distances. The actual content is dependent on the Path and Termination. For more information on the content, refer to Table Three, Leg Data Fields, in Attachment 5 of this document.

Source/Content: The field contains distances, from official government source where available, expressed in nautical miles and tenths of nautical miles with the decimal point suppressed. For Holding Pattern Records and/or Path and Terminations defining holdings patterns, content may be "holding time" expressed in minutes and tenths of minutes preceded by the character "T" with the decimal point suppressed.

Used On:	Airport and Heliport SID/STAR/Approach, Enroute Airway Records
Length:	4 characters
Character Type:	Distance - Numeric; Time - Alpha/numeric
Examples:	1076, 2822, T010, 0208, 0016

5.28 Inbound Magnetic Course (IB MAG CRS)

Definition/Description: "Inbound Magnetic Course" is the published inbound magnetic course to the waypoint in the "Fix Ident" field of the records in which it is employed.

The "HX" group of Path Terminator codes is used to provide racetrack type course reversal flight paths. Government publications for these course reversal include an "inbound magnetic bearing." The SID/STAR/Approach Procedures records do not include a dedicated field for this inbound course. Instead, the information is included in the "Outbound Magnetic Course" field of such records.

Source/Content: Values from official government sources will be used when available. The field contains magnetic bearing in degrees and tenths of a degree, with the decimal point suppressed. For routes charted with true courses, the last character of this field will contain a "T" in place of tenths of a degree.

Used On:	Enroute Airways records
Length:	4 characters
Character Type:	Alpha/numeric
Examples:	2760, 0231, 194T

5.0 NAVIGATION DATA – FIELD DEFINITIONS

5.29 Altitude Description (ALT DESC)

Definition/Description: The Altitude Description field will designate whether a waypoint should be crossed "at," "at or above," "at or below" or "at or above to at or below" specified altitudes. The field is also used to designate recommended altitudes and cases where two distinct altitudes are provided at a single fix.

Source/Content: A code from the following table, selected based on official government source or coding rules in Attachment 5 to this document.

Field Content	Waypoint Crossing Description
+ (plus)	"At or above" altitude specified in first "Altitude" field. Also used with Localizer Only Altitude field.
- (minus)	"At or below" altitude specified in first "Altitude" field. Also used with Localizer Only Altitude field.
@ (blank)	"At" altitude specified in first "Altitude" field. Also used with Localizer Only Altitude field.
B	"At or above to at or below" altitudes specified in the first and second "Altitude" fields.
C	"At or above" altitude specified in second "Altitude" field. **Condition is which ever is earlier.**
D	**"At or above" altitude specified in second "Altitude" field. Condition is "which ever is later", which is operationally equivalent to the condition of "not before."**
G	Glide Slope altitude (MSL) specified in the second "Altitude" field and "at" altitude specified in the first "Altitude" field on the FAF Waypoint in Precision Approach Coding with electronic Glide Slope.
H	Glide Slope Altitude (MSL) specified in second "Altitude" field and "at or above" altitude specified in first "Altitude" field on the FAF Waypoint in Precision Approach Coding with electronic Glide Slope**.**
I	Glide Slope Intercept Altitude specified in second "Altitude" field and "at" altitude specified in first "Altitude" field on the FACF Waypoint in Precision Approach Coding with electronic Glide Slope**.**
J	Glide Slope Intercept Altitude specified in second "Altitude" field and "at or above" altitude specified in first "Altitude" field on the FACF Waypoint in Precision Approach Coding with electronic Glide Slope**.**
V	"At" altitude on the coded vertical angle in the second "Altitude" field and "at or above" altitude specified in first "Altitude" field on step-down fix waypoints.
X	"At" altitude on the coded vertical angle in the second "Altitude" field and "at" altitude specified in the first "Altitude" field on step-down fix waypoints.
Y	"At" altitude on the coded vertical angle in the second "Altitude" field and "at or below" altitude specified in the first "Altitude" field on step-down fix waypoints.

Note: The "B" entry may appear on any record type that includes altitude and altitude description data. The higher value will always appear first in the records with two altitudes or as the first three digits of the Altitude Limitation field. The "B" entry may be used on any sequence included in the Final Approach Coding as defined in this document except the missed approach point or final end point.

Note: The "C or D" entry is used to indicate that the leg has a conditional altitude termination, meaning the leg ends as specified in the Path and Terminator or at the altitude specified, under the condition indicated. These codes are limited to SID and Missed Approach Coding as the potential for an altitude termination exists and such a termination is only valid for ascending terminal procedure segments. See Attachment 5 of this Specification.

Note: In Final Approach Route Coding for Precision Approach Procedures with electronic Glide Slope the codes "@ (for blank)," "G," "H," "I," and "J," are applied as indicated in the table above. For more detail refer to procedure coding rules in Attachment 5 **(6.2.10.1, 6.4.2.1, 7.1.7, and 7.18)** of this Specification.

Note Codes "I" and "J" are only used when the first altitude for the FACF fix is not blank. If that altitude is blank, the Altitude Description is also blank, even when the second altitude is not blank. **Refer to Attachment 5 Rules 6.2.10.1, 6.4.2.1, 7.1.7, and 7.1.8 of this specification.**

Note: The codes "V," "X" and "Y" are used with all fixes defined in government source as step-down fixes and from the FACF inbound on final approach coding. There can be two altitudes provided on every fix in final approach coding. Altitude 1 is the altitude and constraint specified at the fix, the so-called "procedure altitude." Altitude 2 is the "at" altitude on the coded vertical path at the fix. See rules for fix altitude coding in Attachment 5 of this Specification.

Note: Localizer Only Altitude information is provided in the Approach Continuation Record for Precision Approach Procedures with electronic Glide Slope using the codes for "At," "At or Above" and "At or Below" appropriately based on government source publications. This Altitude is the non-precision altitude at the fix on which it is coded, provided on FAF and step-down fix waypoints**.**

Used On: Airport and Heliport SID/STAR/Approach, Primary and Continuation Records, Airport, Heliport and Enroute Communications, VHF NAVAID Limitation Continuation, Preferred Routes and Flight Planning Arrival/Departure Data Records.

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Length: 1 character
Character Type: Alpha

5.30 Altitude/Minimum Altitude

Definition/Description: The "Altitude/Minimum Altitude" field indicates the reference altitude associated with (1) Enroute Airways (MEA, MFA or other minimum altitudes as defined by source), (2) holding pattern path of Holding Pattern record, (3) altitudes at fixes in terminal procedures and terminal procedure path termination defined by the Path Terminator in the Airport or Heliport SID/STAR/Approach Record and (4) lowest altitude of the "blocked altitudes" for a Preferred Route.

Source/Content: Reference altitudes are determined during route definition. The values are derived from official government source when available. This specification includes specific rules for altitude provision and when those altitudes are not provided by source documents, they will be included by data suppliers according to those rules. The field may contain altitudes (all numeric) or flight level (alpha/numeric). The all-numeric fields will contain altitudes in feet with a resolution of one foot. The alpha/numeric fields will contain the alpha characters "FL" followed by the altitude expressed in hundreds of feet (three digits) or a code as indicated below.

On Airport and Heliport SID, STAR and Approach Route records, the first "Altitude" field will contain an altitude when "Altitude Description" field contains a plus (+), a minus (-), or one of the following characters: "B," "G," "H," or "V." The first "Altitude" field may contain an altitude when the "Altitude Description" field contains a "I" or "J" or when it is blank. The second "Altitude" field will contain an altitude when the "Altitude Description" field contains one of the following characters: "B," "C," **"D,"** "G," "H," "I," "J," or "V." In approach procedure coding, some fix "Altitudes" may be below sea level, **in the case of** altitudes at runway fixes when the runway threshold elevation is below sea level. In these cases, the "Altitude" will be expressed in feet with a minus (-) sign in the first character of the five-character field, see examples.

On Airport and Heliport SID/STAR/Approach Continuation Records, the "Localizer Only Altitude" field will contain an altitude when there is a non-precision altitude at the fix that is associated with using the ILS procedure as a Localizer Only (Glide Slope out) procedure. Such an altitude may be provided for the Final Approach Fix (FAF) or any step-down fix from the FACF inbound on the final approach course.

On Enroute Airway records, the first "Minimum Altitude" field will contain the MEA or MFA if the altitude is the same for both directions of flight and the second "minimum Altitude" will be blank. If the airway segment has directional MEAs/MFAs, the first "Minimum Altitude" field will contain the value for the direction of flight in which the airway is coded and the second "Minimum Altitude" field will contain the value for the opposite **route coding. When the official government authority provides different MEA/MFA values for a given airway segment that are based on the navigation sensor, e.g. Convention and RNAV, the MEA/MFA provided will be that value appropriate to the Route Type (Section 5.7) coded in that segment.** The first "Minimum Altitude" field may contain the alpha characters UNKNN when the MEA/MFA is unknown or the alpha characters NESTB when the MEA/MFA has not been established by the appropriate authority.

On Preferred Routes, the "Minimum Altitude" and the "Maximum Altitude" apply to the entire route and are a minimum and maximum block. Altitude 1 and Altitude 2 are fix related apply only to the fix in the sequence in which they occur and are defined by the Altitude Description field.

Used On: Airport and Heliport SID/STAR/Approach, Primary and Continuation Records**,** Holding Pattern, Enroute Airway, Preferred Routes.
Length: 5 characters
Character Type: Alpha/numeric
Examples: 05000, FL050, 18000, FL180 00600, -0012, 29000, FL290, UNKNN or NESTB (the last two on Enroute Airways only)

5.31 File Record Number (FRN)

Definition/Description: The "File Record Number" is a reference number assigned to the record for housekeeping purposes. Records are numbered consecutively, the first record on the file being assigned the number 00001, the second the number 00002, and so on through the final record on the file. File record numbers are subject to change at each file update.

Source/Content: File record numbers are assigned to records during the assembly of the data file. If the file reaches 99999, the next record number will start over with 00000.

Used On: All records
Length: 5 characters
Character Type: Numeric
Examples: 10640, 00420, 31462

5.32 Cycle Date (CYCLE)

Definition/Description: The "Cycle Date" field identifies the calendar period in which the record was added to the file or last revised. A change in any ARINC 424 field, except Dynamic Magnetic Variation, Frequency Protection, Continuation Record Number and File Record Number, requires a cycle date change. The cycle date will not change if there is no change in the data.

Source/Content: The first two digits of the field contain the last two digits of the year in which the addition or revision was made. The last two digits contain the numeric identity of the 28-day data update cycle during which the change occurred. Each calendar year contains 13 such cycles, however, on rare occasions 14 cycles will be encountered.

Used On: All records
Length: 4 characters
Character Type: Numeric

5.0 NAVIGATION DATA – FIELD DEFINITIONS

5.33 VOR/NDB Identifier (VOR IDENT/NDB IDENT)

Definition/Description: The "VOR/NDB Identifier" field identifies the VHF/MF/LF facility defined in the record.

Source/Content: The field contains the official government 1-, 2-, 3- and 4-character facility identification code.

Used On:	VHF NAVAIDs, NDB NAVAIDs, Airport Localizer Marker records, Airport and Heliport Localizer, and Airport and Heliport MLS records.
Length:	4 characters max
Character Type:	Alpha/numeric
Examples:	DEN, 6YA, PPI, TIKX

5.34 VOR/NDB Frequency (VOR/NDB FREQ)

Definition/Description: The "VOR/NDB Frequency" field specifies the frequency of the NAVAID identified in the "VOR/NDB Identifier" field of the record.

Source/Content: Frequencies are derived from official government sources. VHF NAVAID frequencies contain characters for hundreds, tens, units, tenths and hundredths of megahertz. NDB frequencies contain characters for thousands, hundreds, tens, units and tenths of kilohertz. The decimal point following the unit entry is suppressed in both cases.

Used On:	VHF NAVAID, NDB NAVAID, Airport Localizer Marker records
Length:	5 characters
Character Type:	Numeric
Examples:	VHF 11630, 11795 NDB 03620, 17040

5.35 NAVAID Class (CLASS)

Definition/Description: The "Navaid Class" field provides information in coded format on the type of navaid, the **coverage** of the navaid, information carried on the navaid signal and collocation of navaids in both an electronic and aeronautical sense. The field is made up of five columns of codes that define this information.

Source/Content: The information for the five columns is **transformed** from official government source. The mapping of the information codes to the output record columns for the various types of navaids is contained in the tables.

Used On:	Navaid Records (VHF, NDB and Airport/Heliport Localizer/Markers/Locators)
Length:	5 characters (including "blanks")
Character Type:	Alpha

VHF Navaid Record – Includes VOR, VORDME, VORTAC, TACAN, ILSDME and MLSDME type navaids, Output Record Section/Subsection D

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	Col 28	Col 29	Col 30	Col 31	Col 32	
Facility	Navaid Type 1	Navaid Type 2	Range/ Power	Add Info	Col-location	Explanation
VOR	V					
DME		D				
TACAN (channels 17-59 & 70-126)		T				
MIL TACAN (channels 1-16 & 60-69)		M				
ILS/DME		I				
ILS/TACAN		I				
MLS/DME/N		N				
MLS/DME/P		P				
Coverage						
Terminal			T			**Generally usable within 25NM of the facility and below 12000 feet**
Low Altitude			L			**Generally usable within 40NM of the facility and up to 18000 feet**
High Altitude			H			**Generally usable Within 130NM of the facility and up to 60000 feet**
Undefined			U			**Coverage not defined by government source**
ILS/TACAN			C			**Full TACAN facility frequency-paired and operating with the same identifier as an ILS Localizer. Coverage is Terminal**
Additional Information						
Biased ILSDME or ILSTACAN				D		**The zero range reading of the DME facility is not at the transmitting antenna site**
Automatic Transcribed Weather Broadcast				A		**The frequency of this Navaid is used for the continuous broadcast of some sort of automated weather system such as AWOS, ASOS, TWEB, AWIB, AWIS**
Scheduled Weather Broadcast				B		**The frequency of this Navaid is used for the scheduled, non-continuous broadcast of some sort of automated weather system such as VOLMET**
No Voice on Frequency				W		**The frequency of this Navaid is not used to support two-way communication between a ground station and aircraft**
Voice on Frequency				Blank		**The frequency of this Navaid is used to support two-way communication between a ground station and aircraft**
Collocation						
Collocated Navaids					Blank	**The latitude/longitude position of the VOR or Localizer portion and the DME or TACAN portion of a VORDME, VORTAC, ILSDME or ILSTACAN are identical. See also Note 1**
Non-Collocated Navaids					NNote 1	**The latitude/longitude position of the VOR or Localizer portion and the DME or TACAN portion of a VORDME, VORTAC, ILSDME or ILSTACAN are not identical. See also Note 1**

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NDB Navaid Record –NDBs and Terminal NDBs, Output Record Section/Subsection DB and PN

	Col 28	Col 29	Col 30	Col 31	Col 32	
Facility	Navaid Type 1	Navaid Type 2	Range/ Power	Add/ Info	Collocation	**Explanation**
NDB	H					
SABH	S					
Marine Beacon	M					
Inner Marker		I				**There is an Inner Marker beacon at this location**
Middle Marker		M				**There is a Middle Marker beacon at this location**
Outer Marker		O				**There is an Outer Marker beacon at this location**
Back Marker		C				**There is a Backcourse Marker at this location**
Coverage						
High-powered NDB			H			**Generally usable within 75NM of the facility at all altitudes**
NDB			Blank			**Generally usable within 50NM of the facility at all altitude**
Low-powered NDB			M			**Generally usable within 25NM of the facility at all altitude**
Locator			L			**Generally usable within 15NM of the facility at all altitudes**
Additional Information						
Automatic Transcribed Weather Broadcast				A		**The frequency of this Navaid is used for the continuous broadcast of some sort of automated weather system such as AWOS, ASOS, TWEB, AWIB, AWIS**
Scheduled Weather Broadcast				B		**The frequency of this Navaid is used for the scheduled, non-continuous broadcast of some sort of automated weather system such as VOLMET**
No Voice on Frequency				W		**The frequency of this Navaid is not used to support two-way communication between a ground station and aircraft**
Voice on Frequency				Blank		**The frequency of this Navaid is used to support two-way communication between a ground station and aircraft**
Collocation						
BFO Operation					B	**Use of Beat Frequency Oscillator type of equipment is required to received an aural identification signal**

Airport/Heliport Localizer Marker/Locator Record –NDB Locator and Marker Navaids, Output Record Section/Subsection PM

	Col. 75	Col. 76	Col. 77	Col. 78	Col. 79	
Facility	Navaid Type 1	Navaid Type 2	Range/ Power	Add Info	Collocation	
NDB	H					
SABH	S					
Marine Beacon	M					
Inner Marker		I				
Middle Marker		M				
Outer Marker		O				
Back Marker		C				
Coverage						
High-powered NDB			H			**Generally usable within 75NM of the facility at all altitudes**
NDB			Blank			**Generally usable within 50NM of the facility at all altitude**
Low-powered NDB			M			**Generally usable within 25NM of the facility at all altitude**
Locator			L			**Generally usable within 15NM of the facility at all altitudes**
Additional Information						
Automatic Transcribed Weather Broadcast				A		**The frequency of this Navaid is used for the continuous broadcast of some sort of automated weather system such as AWOS, ASOS, TWEB, AWIB, AWIS**
Scheduled Weather Broadcast				B		**The frequency of this Navaid is used for the scheduled, non-continuous broadcast of some sort of automated weather system such as VOLMET**
No Voice on Frequency				W		**The frequency of this Navaid is not used to support two-way communication between a ground station and aircraft**
Voice on Frequency				Blank		**The frequency of this Navaid is used to support two-way communication between a ground station and aircraft**
Collocation						
BFO Operation					B	**Use of Beat Frequency Oscillator type of equipment is required to receive an aural identification signal. See also Note 2**
Locator/Marker Collocated					A	**The latitude/longitude position of the Locator and Marker are identical. See also Note 1**
Locator/Middle Marker Not Collocated					N	**The latitude/longitude position of Locator and Marker are not identical. See also Note 1**

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Note 1: Collocations:

For VHF Navaid records, the character "N" in column 32 is entered if either the latitude and/or the longitude of the VOR and the Collocated DME or TACAN of a frequency paired VORDME or VORTAC differ by 1/10 arc minutes or more. Column 32 is "blank" on VHF Navaids where the difference in latitude or longitude is less than the 1/10-arc minutes. Column 32 of the VHF Navaid will also carry the "N" or "blank" meaning listed above for frequency paired ILSDMEs and ILSTACANs. Note that in this later case, the character is carried on the ILSDME or ILSTACAN record as the Localizer record is not part of the VHF Navaid Section.

For Airport/Heliport Localizer Marker/Locator records, the character "N" in column 79 is entered if either the latitude or longitude of a Marker and it aeronautically associated Locator differ by 1/10-arc minutes or more. The character "A" in column 79 is entered if the latitude or longitude of a Marker and its aeronautically associated Locator differ by less than 1/10-arc minutes. Column 79 is left "blank" when the latitude and longitude of the Marker and Locator are exactly the same.

Note 2: Airport/Heliport Localizer Marker/Locator Navaids, Operations/Collocation. Should both a collocation and a BFO operations requirement exist for one and the same Navaid Record, preference is given to the collocation characters.

5.36 Latitude (LATITUDE)

Definition/Description: The "Latitude" field contains the latitude of the navigational feature identified in the record.

Source/Content: Geographic positions whose latitudes must be included in the data base are defined during route design, many of them in official government publications. The field is constructed as follows. The first character position contains the alpha character "N" or "S" indicating whether the latitude is north or south of the equator. "N" is entered for latitudes falling on the equator. The following eight numeric characters define the latitude in degrees, minutes, seconds, tenths of seconds and hundredths of seconds. Degree, minute and second symbols and the decimal point are suppressed.

Note: Some RNAV system users may elect to round off latitude values to resolutions of less than one hundredth of a second prior to the entry of these data into the airborne computer.

The navigation reference points to be defined by latitude and longitude coordinates are listed in Figure 5-8.

Used On: NAVAID, Waypoint, **Airport Heliport,** Airport and Heliport ILS, Airport, Gate, Runway, Airport and Heliport Localizer Marker, Airport and Heliport MLS and GLS, Airport and Heliport MLS Continuation, Airway Marker, Airport and Heliport Communications, Enroute Communications, Heliport, Restrictive Airspace, FIR/UIR, Controlled Airspace, Path Point and GLS Records.
Length: 9 characters
Character Type: Alpha/numeric
Examples: N39513881

5.37 Longitude (LONGITUDE)

Definition/Description: The Longitude field contains the longitude of the geographic position of the navigational feature identified in the record.

Source/Content: Geographic positions whose longitudes must be included in the data base are defined during route design, many of them in official government publications. The field is constructed as follows: The first character position will contain the alpha character "E" or "W," indicating whether the longitude is east or west of the prime (zero degree) meridian. For longitudes falling on the 0 or 180 degree meridians, "E" is entered. The following nine numeric characters define the longitude in degrees, minutes, seconds, tenths of seconds and hundredths of seconds. Degree, minute and second symbols and the decimal point are suppressed.

Note: Some RNAV system users may elect to round off longitude values to resolutions of less than one hundredth of a second prior to the entry of these data into the airborne computer.

The navigation reference points to be defined by latitude and longitude coordinates are listed in Figure 5-8.

Used On: NAVAID, Waypoint, **Airport, Heliport** Airport and Heliport ILS, Airport, Gate, Runway, **Helipad**, Airport and Heliport Localizer Marker, Airport and Heliport MLS, GLS Airports and Heliport MLS Continuation, Airway Marker, Airport and Heliport Communications, Enroute Communications, Heliport, Restrictive Airspace, FIR/UIR, Controlled Airspace, Path Point and GLS Records.
Length: 10 characters
Character Type: Alpha/numeric
Examples: W104450794

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Record File	Lat/Long Field	Location Defined
Airport	Airport	Aerodrome Reference Point
Airport Comm	Comm (Note 7)	Antenna **Reference**
Enroute Comm	Comm (Note **8**)	Antenna **or Sector Reference**
Enroute Marker	Marker	Marker Antenna
FIR/UIR	FIR/UIR	Boundary Position
FIR/UIR	Arc Origin	Center of Arc
Gate	Gate	Gate
Heliport	Heliport	**Heliport Reference Point**
Heliport Comm	Comm **(Note 7)**	Antenna **Reference**
Localizer	Localizer	Localizer Antenna
Localizer	Glide Slope (Note 6)	Glide Slope Antenna
Marker/Locator	Marker Beacon	Marker Antenna
Marker/Locator	Locator	Locator Antenna
NDB Navaid	NDB	NDB Antenna
Restr. Airspace	Restr. Airspace	Boundary Position
Restr. Airspace	Arc Origin	Center of Arc
VHF Navaid	VOR (Note 1)	VOR Antenna
VHF Navaid	DME or TACAN (Note 2)	DME or TACAN Antenna
Runway	Runway (Note 5)	Runway Landing Threshold
Helipad	**Helipad (Note 9)**	**Helipad Reference Point**
Waypoint	Waypoint	Waypoint
MLS	Azimuth	Azimuth Antenna
MLS	Elevation	Elevation Antenna
MLS	Back Azimuth (Note 3)	Back Azimuth Antenna
MLS	Datum (Note 4)	MLS Reference Datum Point

Figure 5-8

Note 1: The VOR latitude and longitude fields are filled when the "NAVAID Class" field contains the letter "V" in column 28 of the record. If column 28 is blank, these fields are blank also.

Note 2: The DME or TACAN latitude and longitude fields are filled when the "NAVAID Class" field contains the letters "D," "I" "M," "N," "P" or "T" in column 29 of the record. If column 29 is blank, these fields are blank also.

Note 3: The MLS Back Azimuth latitude and longitude fields are to be left blank where no such facility exists.

Note 4: MLS Datum is the point on the runway center line closest to the phase center of the approach elevation antenna.

Note 5: The Runway latitude and longitude fields define the Runway Landing Threshold. This threshold can be the beginning of the landing runway pavement. It will be the displaced threshold (inward from the beginning of the landing runway pavement) when such is published by official government documentation.

Note 6: Localizer Glide Slope latitude and longitude may be blank then detail not available through source documentation.

Note 7: On Airport and Heliport Communications Records, the Latitude/Longitude defines the physical location of the transmitting antenna when this is provided in official government source. This may be a navaid or independent transmitter location. In cases where the physical location of the transmitting antenna is not provided in source, the Latitude/Longitude of the Aerodrome Reference Point will be provided. When the Latitude/Longitude provided are those of a navaid or the ARP, the content of the "Remote Facility" (5.200) will provide an indication of the reference made. In cases where the communications record is defining a digital service capability, the latitude/longitude will be blank.

Note 8: On Enroute Communications Records, the Latitude/Longitude defines the physical location of the transmitting antenna when this is provided in official government source. This may be a navaid or independent transmitter location. In cases where the physical location of the transmitting antenna is not provided in source but it is known to be at a specific airport, the Latitude/Longitude of the Aerodrome Reference Point will be provided. When the Latitude/Longitude provided are those of a navaid or the ARP, the content of the "Remote Facility" (5.200) will provide an indication of the reference made. In cases where the physical location of the transmitter is provided in source but the service/frequency contained in the record is assigned to a specific Enroute Communications Sector, the Latitude/Longitude defines the geographical center of that sector and not the physical transmitter location. In cases where none of the information defined above can be derived from official government sources, the Latitude/Longitude fields will be left blank to indicate the unknown position information. In these cases, the Position Narrative field will contain any such information available in the government sources. In cases where the communications record is defining a digital service capability, the latitude/longitude will be blank.

Note 9: The Helipad latitude and longitude fields are filled when the reference point or defining geographic coordinates of a particular helipad are provided by official government source. When no coordinates are provided, the field will be populated with the latitude and longitude of the heliport reference point.

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.38 DME Identifier (DME IDENT)

Definition/Description: The identification of a DME facility, a TACAN facility or the DME (or TACAN) component of a VORDME or VORTAC facility.

Source/Content: The "DME Identifier" field will contain the officially published 2-, 3-, or 4-character DME facility identifier. For VOR/DME and VORTAC facilities, if the identification codes of the VOR and DME components of the NAVAID defined in the record are the same, the field will be blank. If they are not the same, the VOR Identification will be as defined in Section 5.33 and the DME Identifier field will carry the identification of the DME component. The field is blank when the VHF Navaid facility in the reference record has no DME component. The field will always contain the DME Identifier for TACANs, DME Only NAVAIDS and Localizer or MLS DME facilities.

Used On: VHF NAVAID records
Length: 4 characters max
Character Type: Alpha/numeric
Examples: MCR, DEN, IDVR, DN, (Blank)

5.39 Magnetic Variation (MAG VAR, D MAG VAR)

Definition/Description: The "Magnetic Variation" field specifies the angular difference between True North and Magnetic North at the location defined in the record. "Dynamic Magnetic Variation" is a computer model derived value and takes location and date into consideration. For the "Station Declination" used in some record types, refer to Section 5.66.

Source/Content: Magnetic variations are obtained from official government data sources and other geographical magnetic variation source. A number of different terms are used in government documentation that have specific connotations for the information provided by that government. The most common is "Epoch Year Variation." In theory, this is a value determined by a government agency once every five years and published for general use. Along with Epoch Year Variation, some governments also publish an annual drift value. Data suppliers do not include annual drift derived figures in their databases but rather stay with the Epoch Year value. Another term encountered in source documentation is "Magnetic Variation of Record." This is generally an Epoch Year value. The difference here is that the government authority has established the value as valid for everything associated with a given location. For example, if a Magnetic Variation of Record is established for an airport location, everything referenced to that airport will use the same value. This is of interest as it means that Terminal Procedure design is also based on that value. Obvious differences can occur between a database supplied, semi-static value, and a value derived dynamically, either by the airborne systems or supplier ground systems. Dynamic Magnetic Variation, contained in the VHF Navaid Continuation Record and Enroute/Terminal Waypoint Primary Records, is a computed, earth model derived figure, and is updated dynamically on a schedule established by the data base supplier. Position one of the field contains an alpha character taken from the table below followed by the value of magnetic variation expressed in degrees and tenths of a degree, with the decimal point suppressed. When the first column is coded with the character "T," the value will be all zeros.

When used on Enroute, Airport, and Heliport Primary Extension Continuation records, the field contains the magnetic variation of the latitude/longitude position defined in the record. If that latitude/longitude represents the position of a navaid or airport (Figure 5.8 and Notes 7 and 8 of Section 5.37) the value provided will be identical to magnetic variation provided in the referenced record. If the latitude/longitude represents a stand-alone communications transmitter, the field will contain a government source provided value or the derived Dynamic Magnetic Variation when no source information is provided. If the latitude/longitude fields of the record are blank, the magnetic variation field will also be blank.

Field Content	Description
E	Magnetic variation is East of TRUE North
W	Magnetic variation is West of TRUE North
T	The element defined in the current record is oriented to TRUE North in an area in which the local variation is not zero.

Used On: Airport, NDB Navaid, Airport Localizer Marker, MLS, GLS, Airway Marker, Enroute/ Airport/ Heliport Communication, Heliport, Enroute Waypoint, Airport and Heliport Terminal Waypoint and GLS Primary Records and VHF Navaid Continuation Records.
Length: 5 characters
Character Type: Alpha/numeric
Examples: E0140, E0000, T0000

5.40 DME Elevation (DME ELEV)

Definition/Description: The "DME Elevation" field defines the elevation of the DME component of the NAVAID described in the record.

Source/Content: DME elevations specified in official government publications are entered into this field in feet with respect to MSL. When the elevation is below MSL, the first column of the field contains a minus (-) sign.

Used On: VHF NAVAID records
Length: 5 characters
Character Type: Alpha/numeric
Examples: 00530, -0140

5.41 Region Code (REGN CODE)

Definition/Description: The "Region Code" permits the categorization of waypoints and holding patterns as either enroute or terminal area waypoints. In the latter case the terminal area airport is identified in the field.

Source/Content: The field contains the alpha characters ENRT for enroute waypoints and airport identification code (Airport Ident) for terminal waypoints. In the holding pattern file, the content will match that of the holding fix, e.g. if the holding fix is an enroute waypoint or enroute Navaid, the content with be ENRT; if the holding fix is a terminal

5.0 NAVIGATION DATA - FIELD DEFINITIONS

waypoint or terminal NDB, the content will be the airport identification.

Used On: Waypoint and Holding Pattern records
Length: 4 characters
Character Type: Alpha/numeric
Examples: ENRT, KLAX, 9V9

5.42 Waypoint Type (TYPE)

Definition/Description: The "Waypoint Type" field identifies a number of data conditions.

1. **The first is whether or not the waypoint has been published in an official government source or created during database coding of routes or procedures.**
2. **The second is whether or not the waypoint is an intersection formed with reference to ground based navaids or is an RNAV Waypoint formed by the latitude and longitude.**
3. **The third is an indication of one or more functions assigned to that waypoint in terminal procedure coding.**
4. **The fourth is an indication of location of the waypoint with reference to airspace boundaries and/or grid lines.**
5. **The fifth is an indication of how ATC might be using the waypoint in operational clearances.**
6. **The sixth is an indication that the waypoint has been published for VFR use only.**
7. **Lastly, there is an indication of whether the waypoint is published for use in terminal procedure coding of a specific type, multiple types or not published at all.**

COMMENTARY

Users of this Specification should be aware that this section is intended for use in applications that do not use airway and terminal procedure records and that there is partial duplication of the information between this section and Section 5.17.

Source/Content: Valid contents for Waypoint type are contained in the table below. Unless specifically prohibited, all combinations of data from the three columns are valid.

ENROUTE AND TERMINAL WAYPOINTS				
Waypoint Type	Column 27	Column 28	Column 29	Use
ARC Center Fix	A	Note 3	Note 3	PC
Combined Named Intersection and RNAV Waypoint	C			EA, PC
Unnamed, Charted Intersection	I			EA, PC
Middle or Inner Marker as Waypoint	M			PC
NDB or Terminal NDB Navaid as Waypoint	N	Note 1	Note 2	EA, PC
Outer or Back Marker as Waypoint	O			PC
Named Intersection	R			EA, PC
Uncharted Airway Intersection	U			EA
VFR Waypoint	V	Note 4		EA, PC
RNAV Waypoint	W			EA, PC
Final Approach Fix		A		EA, PC
Initial Approach Fix and Final Approach Fix		B		EA, PC
Final Approach Course Fix		C		EA, PC
Intermediate Approach Fix		D		EA, PC
Off-Route Intersection		F		EA
Initial Approach Fix		I		EA, PC
Final Approach Course Fix and Initial Approach Fix		K		EA, PC
Final Approach Course Fix and Intermediate Approach Fix		L		EA, PC
Missed Approach Fix		M		EA, PC
Initial Approach Fix and Missed Approach Fix		N		EA, PC
Oceanic Gateway Fix		O		EA
Unnamed Stepdown Fix		P		PC
RF Leg Fix Not at Procedure Fix		R	Note 5	PC
Named Stepdown Fix		S		PC
FIR/UIR or Controlled Airspace Intersection		U		EA
Latitude/Longitude Fix, Half Degree of Latitude		V		EA
Latitude/Longitude Fix, Half Degree of Longitude		W		EA
Published for Use in SID			D	EA, PC
Published for Use in STAR			E	EA, PC
Published for Use in Approach Procedures			F	EA, PC
Published for Use in Multiple Terminal Procedure Types			Z	EA, PC
Source provided Enroute Waypoint			G	

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Used On: Enroute Waypoints, Airport and Heliport Terminal Waypoints.
Length: 3 characters
Character Type: Alpha

Note 1: Column 28 of the Enroute and Terminal Waypoint Types will always be blank when column 27 carries the "N" for NDB or Terminal NDB produced as Waypoints.

Note 2: Possible codes for column 29 are identical for both Enroute and Terminal Waypoints and are those carried in the third portion of the table. Column 29 will always be blank when column 27 carries the "N" for NDB or Terminal NDB produced as Waypoints.

Note 3: When column 27 equals "A" for ARC Center Fix Waypoint, columns 28 and 29 will always be blank.

Note 4: The code V in column 27 for VFR Waypoints is not used in conjunction with any codes from column 28 and 29.

Note 5 Column 28 content of R used only in conjunction with column 27 set to "C," R," or "W."

5.43 Waypoint Name/Description (NAME/DESC)

Definition/Description: The "Waypoint Name/Description" field sets the unabbreviated name of a named waypoint or a definition of an unnamed waypoint.

Source/Content: The name of a named waypoint is spelled out in full. Definitions for unnamed waypoints are described in Chapter 7 of this specification.

Used On: Enroute Waypoints, Airport and Heliport Terminal Waypoints.
Length: 25 characters max
Character Type: Alpha/numeric
Examples: FORT SMITH, LAX04026, LOS235/110, 6100N01234W (OCTA), OM RW26L ALTUR

5.44 Localizer/MLS/GLS Identifier (LOC, MLS, GLS IDENT)

Definition/Description: The "Localizer/MLS/GLS Identifier" field identifies the localizer, MLS facility or GLS Ref Path defined in the record. In the Runway Record, two "Landing Systems" may be defined.

Source/Content: The field contains the identification code of the Localizer or MLS facility or GLS Reference Path derived from official government sources. In the Runway Record, there are two fields labeled Localizer/MLS/GLS Reference Path identifier and second Localizer/MLS/ GLS Reference Path identifier to encode multiple Localizers, such as an ILS and a LDA associated with a single runway.

Used On: Runway, Localizer, Localizer Marker, MLS, MLS Continuation and GLS Record.
Length: 4 characters max
Character Type: Alpha/numeric
Examples: Localizer - IDEN, ISTX, IDU, PP MLS - MDEN, MSTX, MLAX GLS - LFBL, EGLC, KSAN

5.45 Localizer Frequency (FREQ)

Definition/Description: The "Localizer Frequency" field specifies the VHF frequency of the facility identified in the "Localizer Identifier" field.

Source/Content: The official government-source localizer frequency is entered into the field with a resolution of 50 kHz. The decimal point following the unit MHz entry is suppressed.

Used On: Airport and Heliport ILS Localizer records
Length: 5 characters
Character Type: Numeric
Examples: 11030, 11195

5.46 Runway Identifier (RUNWAY ID)

Definition/Description: The "Runway Identifier" field identifies the runways described in runway records and runways served by the ILS/MLS described in ILS/MLS records.

Source/Content: Runway identifiers are derived from official government sources and are shown in the following format:

The two letters "RW" are followed by two numerics, 01 thru 36, and may contain a fifth character designation of one of the following:

C = Center (Runway of three parallel runways)
L = Left (Runway of two or three parallel runways)
R = Right (Runway of two or three parallel runways)
T = (Runway and associated flight maneuvers referenced only in degrees true)

Any other designations (suffixes), such as North, South, East, West or STOL will not be included in the ARINC file.

Used On: Airport and Heliport ILS and MLS, GLS Runway, Airport and Heliport Localizer Marker, Path Point and GLS Records.
Length: 5 characters max
Character Type: Alpha/numeric
Examples: RW26L, RW08R, RW26C, RW05, RW17T

5.47 Localizer Bearing (LOC BRG)

Definition/Description: The "Localizer Bearing" field defines the magnetic bearing of the localizer course of the ILS facility/GLS approach described in the record.

Source/Content: Localizer courses, derived from official government sources, are entered into the field in degrees and tenths of a degree, with the decimal point suppressed. For localizer courses charted with true courses, the last

5.0 NAVIGATION DATA - FIELD DEFINITIONS

character of this field will contain a "T" in place of tenths of a degree.

Used On: ILS, GLS records
Length: 4 characters
Character Type: Alpha/numeric
Examples: 2570, 0147, 2910, 347T

5.48 Localizer Position (LOC FR RW END) Azimuth/Back Azimuth Position (AZ/BAZ FR RW END)

Definition/Description: The "Localizer/Azimuth Position" field defines the location of the facility antenna relative to one end of the runway.

Source/Content: The field contains the official government source distance, in feet, from the antenna to the runway end. The resolution is one foot.

Used On: ILS, MLS and MLS Continuation records
Length: 4 characters
Character Type: Numeric
Examples: 0950, 1000

5.49 Localizer/Azimuth Position Reference (@,+,-)

Definition/Description: The "Localizer/Azimuth Position Reference" field indicates whether the antenna is situated beyond the stop end of the runway, ahead of or beyond the approach end of the runway. The "Back Azimuth Position Reference" field indicates whether the antenna is situated ahead of the approach end of the runway, ahead of or beyond the stop end of the runway.

Source/Content: For Localizer and Azimuth positions the field is blank (@) when the antenna is situated beyond the stop end of the runway, it contains a plus (+) sign when the antenna is situated ahead of the approach end of the runway or a minus (-) sign when it is located off to one side of the runway. For Back Azimuth positions the field is blank (@) when the antenna is situated ahead of the approach end of the runway, it contains a plus (+) sign when the antenna is situated beyond the stop end of the runway or a minus (-) sign when it is located off to one side of the runway.

Used On: ILS, MLS and MLS Continuation records
Length: 1 character
Character Type: Alpha

5.50 Glide Slope Position (GS FR RW THRES) Elevation Position (EL FR RW THRES)

Definition/Description: The "Glide Slope/Elevation Position" field defines the location of the antenna with respect to the approach end of the runway.

Source/Content: The field contains four numeric characters indicating the distance in feet (to a resolution of one foot) from a line drawn at right angles to the runway at the antenna position to the threshold of the runway.

Used On: ILS and MLS records
Length: 4 characters max
Character Type: Numeric
Examples: 0980, 1417

5.51 Localizer Width (LOC WIDTH)

Definition/Description: The "Localizer Width" field specifies the localizer course width of the ILS facility defined in the record.

Source/Content: Localizer course widths from official government sources are entered into the field in degrees, tenths of a degree and hundredths of a degree with the decimal point suppressed.

Used On: ILS records
Length: 4 characters
Character Type: Numeric
Examples: 0500, 0400, 0350

5.52 Glide Slope Angle (GS ANGLE) Minimum Elevation Angle (MIN ELEV ANGLE)

Definition/Description: The "Glide Slope Angle" field defines the glide slope angle of an ILS facility/GLS approach. The "Minimum Elevation Angle" field defines the lowest elevation angle authorized for the MLS procedure.

Source/Content: Glide Slope and Elevation angles from official government sources are entered into the fields in degrees, tenths of a degree and hundredths of a degree with the decimal point suppressed.

Used On: ILS, GLS and MLS records
Length: 3 characters
Character Type: Numeric
Example: 275, 300

5.53 Transition Altitude/Level (TRANS ALTITUDE/LEVEL)

Definition/Description: The "Transition Altitude" field defines the altitude in the vicinity of an airport or heliport at or below which the vertical position of an aircraft is controlled by reference to altitudes (MSL). The "Transition Level" field defines the lowest flight level available for use above the transition altitude. Aircraft descending through the transition layer will use altimeters set to local station pressure, while departing aircraft climbing through the layer will be using standard altimeter setting (QNE) of 29.92 inches of mercury, 1013.2 millibars or 1013.2 hectopascals.

Source/Content: Transition Altitudes/Levels are derived from official government sources. For STAR and Approach records, the field defines the level, expressed in feet, at which the altimeter barometric setting is changed from standard to local values for the airport or heliport identified in the record. For SID records, the field will contain the Transition Altitude expressed in feet. The first leg of each Airport and Heliport SID/ STAR/Approach procedure shall contain the appropriate transition altitude with a resolution of one foot. If the transition altitude is unknown "by ATC," the field will be blank in procedure records. For Airport and Heliport records, the Transition Altitude and Transition Level will be entered into the appropriate fields, in feet with a resolution of one foot. If the Transition Altitude or Level is unknown, "by ATC" or has different values for varying procedures at the airport or heliport, the field will be blank.

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Used On: Airport and Heliport SID/STAR/Approach, Airport and Heliport Records
Length: 5 characters
Character Type: Numeric
Examples: 05000, 23000, 18000

5.54 Longest Runway (LONGEST RWY)

Definition/Description: The "Longest Runway" field permits airport to be classified on the basis of the longest operational hard-surface runway.

Source/Content: The longest runway will be derived from official government sources and entered in the field in hundreds of feet. This value will represent the longest hard-surfaced operational runway available without restriction at the airport. The value reflects overall pavement length declared suitable and available for the ground operations of aircraft. Where no hard-surfaced runway is available or those available do not meet criteria, the value will represent the longest operational runway at the airport.

Used On: Airport Records
Length: 3 characters
Character Type: Numeric
Examples: 040, 055, 098, 111

5.55 Airport/Heliport Elevation (ELEV)

Definition/Description: The elevation of the Airport/Heliport specified in the record is defined in the "Airport Elevation" and "Heliport Elevation" field.

Source/Content: Airport/Heliport elevations are to be derived from official government sources and entered into the field in feet to a resolution of one foot. For elevations above MSL, the field contains the numeric characters of the elevation only. For the below MSL elevations, the first character of the field is a minus (-) sign. Airport elevation is defined as the highest elevation of any landing surface on the airport.

Used On: Airport and Heliport records
Length: 5 characters
Character Type: Alpha/numeric
Examples: 02171, -0142, 05230

5.56 Gate Identifier (GATE IDENT)

Definition/Description: The airport gate defined in the record is identified in the "Gate Identifier" field.

Source/Content: Coded gate identity information is derived from official government sources and navigation system users.

Used On: Gate records
Length: 5 characters max
Character Type Alpha/numeric
Examples: C134B, 23, 30A, B12A

5.57 Runway Length (RUNWAY LENGTH)

Definition/Description: The "Runway Length" field defines the total length of the runway surface declared suitable and available for ground operations of aircraft for the runway identified in the records' Runway Identifier field.

Source/Content: Runway lengths are derived from official government sources and are entered in feet with a resolution of one foot. The value represents the overall length of the runway, with no regard for displaced thresholds. It does not include stopways, overruns or clearways. Available landing lengths and take-off runs are not necessarily identical to this runway length. Analysis of the content of Section 5.69, Displaced Threshold and 5.79, Stopway is required to determine these operational lengths. As the latitude/longitude information in the runway record reflects the Landing Threshold Point of the runway identified in the record, which may or may not be displaced, there is no direct correlation between the Runway Length and a value calculated based on these latitude/longitude values. For additional information on runway length analysis and operational length calculations, see Figure 5-9.

Used On: Runway Records
Length: 5 characters
Character Type: Numeric
Examples: 05000, 07000, 11480

5.58 Runway Magnetic Bearing (RWY BRG)

Definition/Description: The magnetic bearing of the runway identified in the "runway identifier" **or "pad ident"** field of the record is specified in the "Runway Magnetic Bearing" field.

Source/Content: Runway magnetic bearings derived from official government sources are entered into the field in degrees and tenths of a degree, with the decimal point suppressed. For runway bearings charted with true bearings, the last character of this field will contain a "T" in place of tenths of a degree. **When used on helipad records, it usually will contain the bearing of a former fixed-wing runway that has been converted to helicopter use only or when a specific bearing to approach a particular helipad has been provided by government source.**

Used On: Runway **and Helipad** Records
Length: 4 characters
Character Type: Alpha/numeric
Examples: 1800, 2302, 0605, 347T

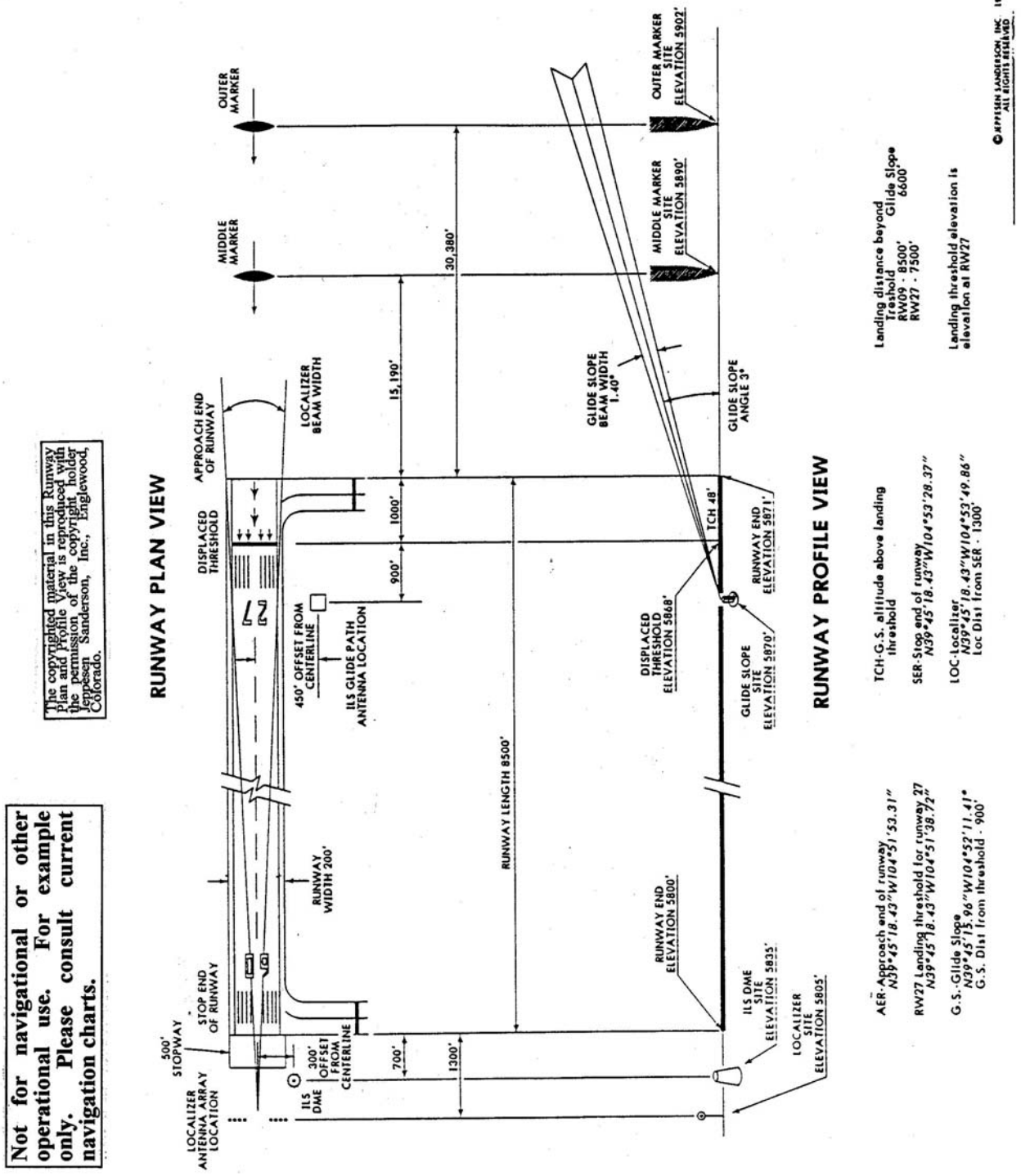
5.59 Runway Description (RUNWAY DESCRIPTION)

Definition/Description: If required, additional information concerning a runway can be included in a record in the "Runway Description" field.

Source/Content: Appropriate contents for the field will be determined when the record is assembled.

Used On: Runway records
Length: 22 characters max
Character Type: Alpha/numeric
Examples: GROOVED, SINGLE ENG. ONLY

5.0 NAVIGATION DATA - FIELD DEFINITIONS



The copyrighted material in this Runway Plan and Profile View is reproduced with the permission of the copyright holder Jeppesen Sanderson, Inc., Englewood, Colorado.

Not for navigational or other operational use. For example only. Please consult current navigation charts.

Figure 5-9

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.60 Name (NAME)

Definition/Description: The "Name" field defines the name commonly applied to the navigation entity defined in the record.

Source/Content: Appropriate contents for the field will be determined from official government or customer sources.

Used On: Gate and Holding Pattern records
Length: 25 characters max
Character Type: Alpha/numeric
Examples: HOLDING JIMEE MIAMI

5.61 Notes (Continuation Records) (NOTES)

Definition/Description: The "Notes" field (continuation record) is provided to accommodate any information that cannot be entered in the primary record.

Source/Content: Appropriate contents for the field will be determined at the time the primary record is assembled.

Used On: All except Company route records
Length: 70 characters max
Character Type: Alpha/numeric
Examples: EASTBOUND PREFERRED
090/0Z/230/0Z

5.62 Inbound Holding Course (IB HOLD CRS)

Definition/Description: The "Inbound Holding Course" field defines the inbound course to the holding waypoint.

Source/Content: Inbound holding courses derived from official government sources are entered into the field in degrees and tenths of a degree, with the decimal point suppressed. For holding courses charted with true bearings, the last character of this field contains a "T" in place of tenths of a degree.

Used On: Holding Pattern records
Length: 4 characters
Character Type: Alpha/numeric
Examples: 0456, 1800, 3034, 347T

5.63 Turn (TURN)

Definition/Description: The "Turn" field specifies the direction in which holding pattern turns are to be made.

Source/Content: The "Turn" field will always contain either L or R.

Used On: Holding Pattern records
Length: 1 character
Character Type: Alpha

5.64 Leg Length (LEG LENGTH)

Definition/Description: The "Leg Length" field specifies the distance between the point at which the aircraft rolls out on the inbound leg of the holding pattern and the fix at which the holding pattern is defined (Figure 5-10).

Source/Content: Leg length derived from official government sources is entered into the field in nautical miles and tenths of a nautical mile, with the decimal point suppressed.

Used On: Holding Pattern records
Length: 3 characters
Character Type: Numeric
Examples: 108, 055

5.65 Leg Time (LEG TIME)

Definition/Description: The "Leg Time" field specifies the length of the inbound leg of a holding pattern in units of time (Figure 5-10).

Source/Content: Leg time, derived from official government sources, is entered into this field in minutes and tenths of a minute, with the decimal point suppressed.

Used On: Holding Pattern records
Length: 2 characters
Character Type: Numeric
Examples: 10, 15, 20

5.66 Station Declination (STN DEC)

Definition/Description: For VHF NAVAIDS, the "Station Declination" field contains the angular difference between true north and the zero degree radial of the NAVAID at the time the NAVAID was last site checked. For ILS localizers, the field contains the angular difference between true north and magnetic north at the localizer antenna site at the time the magnetic bearing of the localizer course was established.

Source/Content: Station declinations are derived from official government sources. The field contains one of the alpha characters shown in the following table followed by the value of the declination in degrees and tenths of a degree, with the decimal point suppressed. When the first column of the Station Declination field is coded T or G, the remainder of the field should be coded all zeros.

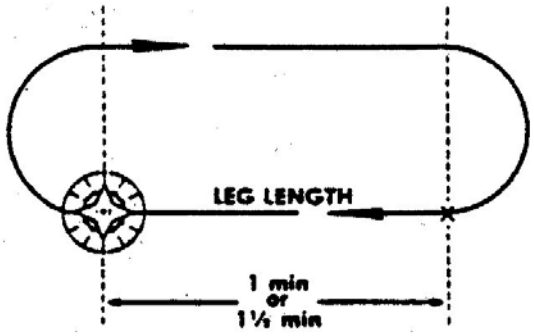
Column 1 Character	Declination Description
E	Declination is East of True North
W	Declination is West of True North
T	Station is oriented to True North in an area in which the local variation is not zero.
G	Station is oriented to Grid North

Used On: VHF NAVAID and ILS records
Length: 5 characters
Character Type: Alpha/numeric
Examples: E0072, E0000, T0000, G0000

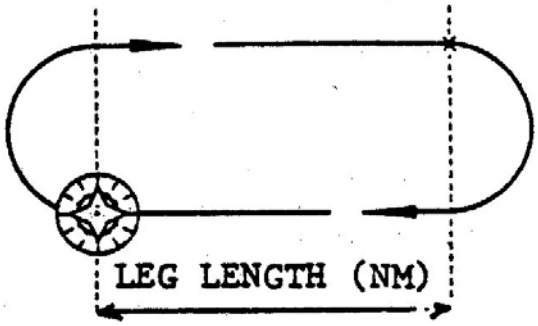
COMMENTARY

The appearance of the character "G" in column 1 of this field will alert users that although a NAVAID declination may not be zero, the fact that the grid reference is unknown prevents a value from being defined.

5.0 NAVIGATION DATA - FIELD DEFINITIONS



LEG LENGTH (TIME)



LEG LENGTH (DISTANCE)

Figure 5-10

HOLDING PATTERN LEG LENGTH

*Leg time diagram added by Supplement 1

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.67 Threshold Crossing Height (TCH)

Definition/Description: The "Threshold Crossing Height" specifies the height above the landing threshold on a normal glide path.

Source/Content: The Threshold Crossing Height will be derived from official government sources when available. As provided on Runway Records, the TCH value will be the Glide Slope Height at the landing threshold for runways with ILS or MLS approaches. If an ILS or MLS is not available and an RNAV approach is available it will be the published TCH for that procedure. When ILS/MLS or RNAV values are not available but a published VGSI with TCH is available, it will be used. If none of these values are available, it will be **40 or 50 feet based on the table below.** When used on Approach Continuation Records, the field will contain the published TCH for that procedure. When used on ILS or MLS Records, it will be the height of the glide slope at the landing threshold.

Content	Description
40 (feet)	**On Runway records for which all approach procedures are published for Category A and B aircraft only.**
	On Runway records with a length of less than 6000 feet and no published approach procedure.
50 (feet)	**On Runway records for which there is at least one approach procedure published for Category C or D aircraft.**
	On Runway records with a length of 6000 feet or greater and no published approach procedure.

COMMENTARY

Based on the information contained in the Source/Content paragraph, it should be noted that the single TCH value provided on the Runway Record may be different than the TCH value provided on the Approach Continuation Record for a procedure to that same runway. These differences may be significant. A comparison of procedure altitude data to threshold elevation and threshold crossing heights should only be made to the Approach Continuation Record. It should also be noted that a TCH associated with the VGSI for the same runway may be different than either value. Some government sources will provide information on these procedure to VGSI differences when they are three feet or greater

Used On:	Airport and Heliport ILS and MLS Runway**,** Airport, and Heliport Approach Continuation Records**.**
Length:	2 characters on Airport and Heliport ILS, MLS and Runway Records 3 characters on Airport and Heliport Approach Continuation Records
Character Type:	Numeric
Example	37, 50, 99 (ILS, MLS and Runway 044, 055, 102, (Approach)

5.68 Landing Threshold Elevation (LANDING THRES ELEV)

Definition/Description: The elevation of the landing threshold of the runway described in a runway record is defined in the "Landing Threshold Elevation" field.

Source/Content: Runway landing threshold elevations derived from official government sources are entered into this field in feet, to a resolution of 1 foot. For elevations above MSL, the field contains the numeric characters of the elevation only. For below MSL elevations, the first character of the field is a minus (-) sign.

Used On:	Runway records
Length:	5 characters
Character Type:	Alpha/numeric
Examples:	01250, -0150

5.69 Threshold Displacement Distance (DSPLCD THR)

Definition/Description: The distance from the extremity of a runway to a threshold not located at that extremity of that runway.

Source/Content: Threshold displacement distances derived from official government sources are entered into this field in feet.

Used On:	Runway records
Length:	4 characters
Character Type:	Numeric
Examples:	0485, 1260

5.70 Vertical Angle (VERT ANGLE)

Definition/Description: The "Vertical Angle" field defines the angular portion of vertical navigation path in STAR Route and Approach Procedure Route records. The Vertical Angle should cause the aircraft to fly at the last coded altitude and then descend on the VNAV path, projected back from the fix and altitude contained in the route sequence that contains the Vertical Angle.

Source/Content: Values from official government source documents will be used when available. In the coding of Precision Approach Procedures, the Vertical Angle is the angle assigned to the glide slope. In the coding of non-precision procedures, it will be the VNAV Path angular definition provided by the government source or a value computed based on the rules for such a computation documented in Attachment 5 of this Specification. Values greater than zero will be preceded by the minus sign (-) to indicate descending flight. When no government source value is available and none can be computed based on the rules in Attachment 5 of this Specification, the field is populated with all zeros, no minus sign. Vertical Angles are expressed in degrees, tenths and hundredths of degrees with the decimal point suppressed. The maximum value is 9.99 degrees.

Used On:	Airport and Heliport STAR **and** Approach **Route** Records
Length:	4 characters **(first character either (-) or blank)**
Character Type:	Alpha/numeric
Examples:	-300, -275, -542, **000**

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.71 Name Field

Definition/Description: This field will be used to further define the record by name.

Source/Content: Facility name will be derived from official government sources. A parenthetical name following the official name may be used to identify the location of the facility.

Used On:	Navaid, Airport, Heliport and Enroute Marker records
Length:	30 characters
Character Type:	Alpha

5.72 Speed Limit (SPEED LIMIT)

Definition/Description: The "Speed Limit" field defines a speed, expressed in Knots, Indicated (K.I.A.S.), for a fix in a terminal procedure or for an airport or heliport terminal environment.

Source/Content: The speed limit will be derived from official government source documentation and shown in Knots. When used on an Airport or Heliport Record, the field is an indication of the maximum allowed speed and applies to all flight segments departing or arriving that airport's or heliport's terminal area, at and below the specified Speed Limit Altitude (5.73). When used on Airport and Heliport SID/STAR/Approach Records, the field is an indication of a speed for a fix in the procedure description, used in conjunction with Speed Limit Description (5.261).

When applied on a leg coded with a Path and Termination (5.21) of HA, HF or HM in SID/STAR/Approach Coding, the speed limit is valid during execution of the leg.

On SID Procedure Records, the speed limit will apply to all legs up to and including the termination of the leg on which the speed is coded from the beginning of the procedure or a previous speed limit. If a different speed is coded on a subsequent leg, the limit will be applied for that leg and from that leg backwards to the previous terminator which contained a speed limit.

On STAR and Approach Procedure Records, the speed limit will be applied forward to the end of the arrival (e.g. throughout the procedure until the end of the Flight Plan) or until superceded by another speed limit.

The intent in both SIDs and STARs is to exclude speed changes inconsistent with the procedure

Used On:	Airport and Heliport SID/STAR/ Approach, Airport and Heliport, Flight Planning Arr/Dep Data Records
Length:	3 characters
Character Type:	Numeric
Examples:	250

5.73 Speed Limit Altitude

Definition/Description: "Speed Limit Altitude" is the altitude below which speed limits may be imposed.

Source/Content: The "Speed Limit Altitude" will be derived from official government sources in feet MSL or FL's.

Used On:	Airport and Heliport records
Length:	5 characters
Character Type:	Alpha/numeric
Examples:	10000, F125

5.74 Component Elevation (GS ELEV, EL ELEV, AZ ELEV, BAZ ELEV)

Definition/Description: The "Component Elevation" field defines the elevation of a given component in the Localizer, GLS and MLS records. The "Glide Slope Elevation (GS ELEV)" defines the elevation of the Glide Slope component in the Localizer Records. The "EL Elevation (EL ELEV)" defines the elevation of the Elevation component of the MLS Record, the "Azimuth Elevation (AZ ELEV)" defines the elevation of the Azimuth component of the MLS Record and the "Back Azimuth Elevation (BAZ ELEV)" defines the elevation of the Back Azimuth component of the MLS Record. The "GLS station elevation (GLS ELEV)" defines the elevation of the GLS ground station in the GLS record.

Source/Content: Elevations specified in official government publications are entered in this field with respect to MSL. When the elevation is below MSL, the first column of the field contains a minus (-) sign.

Used On:	Localizer, MLS and GLS Records and MLS Continuation Records
Length:	5 characters
Character Type:	Alpha/numeric
Examples:	00235, 01265, -0011

5.75 From/To – Airport/**Heliport**/Fix

Definition/Description: When used on Company Routes and Helicopter Operations Company Routes, the "From Airport/Heliport/Fix" is the fix from which the company route originates. The "To Airport/Heliport/Fix" is the fix at which the company route terminates. When used on Alternate Records, it is the Departure, Destination or Enroute Airport/Fix for which the alternate information is being provided.

Source/Content: The customer is responsible for defining points at which company routes originate and terminate and for defining which departure, destination or enroute points are to have alternate information. **On Company Routes and Helicopter Operations Company Routes, may reference airport, heliport, navaid or waypoint records. Which, will be further defined by ICAO, Section and Subsection data.**

Used On:	Company Route, **Helicopter Operations Company Route** and Alternate Records
Length:	5 characters max.
Character Type:	Alpha/numeric

5.76 Company Route Ident

Definition/Description: The "Company Route Ident" field identifies each unique route between origination and destination.

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Source/Content: This field is determined by the customer.

Used On: Company Route records **Helicopter Operations Company Routes**
Length: 10 characters
Character Type: Alpha/numeric

5.77 VIA Code

Definition/Description: The "VIA Code" field is used to define the type of route used in the SID/STAR/Approach/Airways field (Section 5.78) on Company Route records and defines the type of route used in the AWY Identifier on Preferred Route records. On the Preferred Route records, some codes define the use, or restriction to use, of a fix or routing.

Source/Content: The code to be entered must be selected from the table below:

Company Route Record (R)

VIA Field	Description
ALT	Alternate Airport
APP	Approach Route
APT	**Approach Transition**
AWY	Designated Airway
DIR	Direct to Fix
INT	Initial Fix
PRE	Preferred Route
SID	Standard Instrument Departure
SDE	Standard Instrument Departure - Enroute Transition
SDY	Standard Instrument Departure - Runway Transition
STR	Standard Terminal Arrival and Profile Descent
STE	Standard Terminal Arrival and Profile Descent - Enroute Transition
STY	Standard Terminal Arrival and Profile Descent - Runway Transition

Preferred Route Record (ET)

VIA Field	Description
AWY	Designated Airway
DIR	Direct to Fix
INT	Initial Fix
RVF	Route via Fix
RNF	Route via Fix not permitted
SID	Standard Instrument Departure
STR	Standard Terminal Arrival and Profile Descent

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Current	*Next Via*											
Via	***INT***	***DIR***	***SDY***	***SID***	***SDE***	***AWY***	***STE***	***STR***	***STY***	***APT***	***APP***	***ALT***
INT	*N*											
DIR	*N*											
SDY	*N*		*N*									
SID	*N*		*N*	*N*								
SDE	*N*		*N*	*N*	*N*							
AWY	*N*		*N*	*N*	*N*							
STE	*N*		*N*	*N*	*N*	*N*	*N*					
STR	*N*		*N*	*N*	*N*	*N*	*N*	*N*				
STY	*N*		*N*	*N*	*N*	*N*	*N*	*N*	*N*			
APT	*N*		*N*	*N*	*N*	*N*	*N*	*N*	*N*	*N*		
APP	*N*		*N*	*N*	*N*	*N*	*N*	*N*	*N*	*N*	*N*	
ALT	*N*	*N*	*N*	*N*	*N*	*N*	*N*	*N*	*N*	*N*	*N*	*N*

Note 1: "N" means sequence not allowed, "blank" means sequence is allowed.

Note 2: The "To Fix" must match the beginning fix of the following Via.

Used On: Company Route and Preferred Route records **and Helicopter Operations Company Routes**
Length: 3 characters
Character Type: Alpha/numeric

Note: Figure 5-11 illustrates how various fields are to be completed in the Company Route Record based on the various “VIA Codes” defined in this paragraph.

5.0 NAVIGATION DATA - FIELD DEFINITIONS

VIA	S/S/A AIRWAY	AREA	TO FIX	RWY TRANS	ENRT TRANS	CRUISE ALT	TERM/ ALT ARPT	ALT DIST
ALT	Blank	Area	Blank	Blank	Blank	ALT or Blank	Arpt or Heliport Ident	Dist in NM
APP	Apch Ident	Area	Optional	Blank	Tml Rte Ident or Blank	Blank	Arpt or Heliport Ident if TO FIXIdent is Terminal	Blank
APT	**Apch Ident**	**Area**	**Fix Ident**	**Blank**	**Trans Ident**	**Blank**	**Airport or Heliport ident if TO FIX Ident is Terminal**	**Blank**
AWY	Awy Ident	Area	Fix Ident	Blank	Blank	ALT or Blank	Blank	Blank
DIR, INT	Blank	Area	Fix Ident	Blank	Blank	ALT or Blank	Airport or Heliport Ident if TO FIX Ident is Terminal	Blank
SID	SID Ident	Area	Fix Ident	Rwy Ident/ All or Blank	Trans Ident or Blank	ALT or Blank	Airport or Heliport if TO FIX Ident is Terminal	Blank
SDE	SID Ident	Area	Fix Ident	Blank	Trans Ident	ALT or Blank	Airport or Heliport Ident if TO FIX Ident is Terminal	Blank
SDY	SID Ident	Area	Fix Ident	Rwy Ident	Blank	ALT or Blank	Airport or Heliport Ident if TO FIX Ident is Terminal	Blank
STR	STAR Ident	Area	Fix Ident of Blank	Rwy Ident/All	Trans Ident or Blank	ALT or Blank	Airport or Heliport Ident if TO FIX Ident is Terminal	Blank
STE	STAR Ident	Area	Fix Ident	Blank	Trans Ident	ALT or Blank	Airport or Heliport Ident if TO FIX Ident is Terminal	Blank
STY	STAR Ident	Area	Fix Ident	Rwy Ident	Blank	ALT or Blank	Airport or Heliport Ident if TO FIX Ident is Terminal	Blank
PRE	Pref Rte Ident	Area	Fix Ident	Blank	Blank	ALT or Blank	Airport or Heliport Ident if TO FIX Ident is Terminal	Blank

Figure 5-11
Company Route Record (R)
Field Content

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.78 SID/STAR/App/AWY (S/S/A/AWY)
SID/STAR/Awy (S/S/AWY)

Definition/Description: This field is used to **provide the identifier of the particular enroute airway or terminal** route to be flown as referenced by the "VIA" field (Section 5.77). **The identifier is further defined by the content of columns 95/96/97 of the Company Route or 106/107/108 of the Preferred Route which contain the Route Type and Route Type Qualifier data of the specific route or procedure.**

Source/Content: For Company Route records this field can contain the SID/STAR, Approach, Enroute Airway, or Preferred Route Identifier (Sections 5.8, 5.9, and 5.10). For Preferred Route records this field can contain the SID/STAR or Enroute Airway Route Identifier (Section 5.8). This field will be blank for certain records depending on the "VIA" field content (Section 5.77).

Used On: Company Route and Preferred Route Records**, and Helicopter Operations Company Routes**
Length: 6 characters
Character Type: Alpha/numeric
Examples:

VIA **Code**	S/S/A/AWY **Content**
SID	CUIT8
STR	LOCKE9
APP	I19L, R35-Z
AWY	J501

5.79 Stopway

Definition/Description: "Stopway" means the length of an area beyond the take-off runway, no less wide than the runway and centered upon the extended centerline of the runway, and designated for use in decelerating the airplane during an aborted takeoff.

Source/Content: The Stopway will be derived from official government sources and shown in feet (See Figure 5-3).

Used On: Runway records
Length: 4 characters
Character Type: Numeric
Examples: 0900, 1000

5.80 ILS/MLS/GLS Category (CAT)

Definition/Description: The Localizer/MLS/GLS Performance Categories have established operating minimums and are listed as Category I, II, and III. The level of Performance Category does not imply that permission exists to use the facility for landing guidance to that level and does not limit minimal using designated classification. This field is also used to define the classification, non-ILS/MLS/GLS, and localizer installation such as IGS, LDA, or SDF. As used in the runway record, there are two fields, one labeled Localizer/MLS/GLS Category/Classification and the other labeled Second Localizer/MLS/GLS Category/Classification.

Source/Content: The Localizer/MLS/GLS Category/Classification will be derived from official government sources and will be indicated by a value from the table below.

Definition	Category/ Classification
ILS Localizer Only, No Glideslope	0
ILS Localizer/MLS/GLS Category I	1
ILS Localizer/MLS/GLS Category II	2
ILS Localizer/MLS/GLS Category III	3
IGS Facility	I
LDA Facility with Glideslope	L
LDA Facility, no Glideslope	A
SDF Facility with Glideslope	S
SDF Facility, no Glideslope	F

Used On: Runway, Localizer, MLS and MLS Continuation Records, GLS Record.
Length: 1 character
Character Type: Alpha/numeric

5.81 ATC Indicator (ATC)

Definition/Description: The "ATC Indicator" field will be used to indicate that the altitudes shown in the altitude fields can be modified by ATC or the altitude will be assigned by ATC.

Source/Content: This field will contain the alpha character "A" when the official government source states that the altitude can be modified or assigned by ATC. This field will contain the alpha character "S" when the official government source states that the altitude will be assigned by ATC or if no altitude is supplied.

Used On: Airport and Heliport SID/STAR/ Approach Records
Length: 1 character
Character Type: Alpha

5.82 Waypoint Usage

Definition/Description: The waypoint usage field is employed to indicate the structure in which the waypoint is utilized.

Source/Content:

Usage	Record Column Content	
	30	31
HI and LO Altitude		B
HI Altitude		H
LO Altitude		L
Terminal Use Only		Blank
RNAV	R	

Used On: Enroute (EA) waypoint records
Length: 2 characters
Character Type: Alpha

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.83 To FIX

Definition/Description: The Company Route, **Helicopter Operation Company Route**, and Preferred Route "To Fix" field is used to terminate the route referenced in the SID/STAR/APCH/AWY field (Section 5.78), or terminate a "Direct" segment or start an "Initial" segment when no SID/STAR/APCH/AWY is referenced.

Source/Content: For Company Route records the field will contain Enroute Waypoint, Airport Terminal Waypoint, VHF NAVAID, NDB NAVAID, Terminal NDB NAVAID, Airport or Runway Identifier. For Helicopter Operations Company Route records the field will contain Enroute Waypoint, Helicopter Terminal Waypoint, VHF NAVAID, NDB NAVAID, Terminal NDB NAVAID, Airport, Heliport, Runway Identifier or Helipad Identifier. The customer will define where a particular route segment is to terminate. Terminal Fixes, Runway Identifiers or Helipad Identifiers must be for the From Airport/Heliport or To Airport/Heliport which must be consistent with the VIA Code. For Preferred Route records, the field will contain Enroute Waypoint, Terminal Waypoint, VHF NAVAID, NDB NAVAID or Terminal NDB NAVID, Airport Identifier.

Used On:	Company Route**, Helicopter Operations Company Route,** and Preferred Route Records
Length:	Company Route**/Helicopter Operations Company Route** - 6 characters max. Preferred Route - 5 characters max.
Character Type:	Alpha/numeric
Examples:	SHARP, BHM, DEN43, KDEN, RW35R

5.84 RUNWAY TRANS

Definition/Description: This field is used to identify the desired runway transition of the applicable SID or STAR. **Together with the Section/Subsection identified for the SID/STAR/App/AWY field, i**t is used to link directly to the SID/STAR procedure records depending on the Company Route**/Helicopter Operations Company Route** record "VIA" field (Section 5.77) and whether or not the SID/STAR has explicit runway transitions.

Source/Content:

VIA field contains "SID" or "STR":

If the applicable SID/STAR has explicit runway transitions as indicated by the Procedure Route Type, then this field uniquely identifies the desired runway transition. If the applicable SID/STAR has explicit runway transitions as indicated by the Procedure Route Type but no runway transition is desired in the Company Route, the field is blank. If the applicable SID/STAR does not have explicit runway transitions as indicated by the Procedure Route Type, this field is always non-blank and exactly matches the "TRANS IDENT" field of the SID/STAR procedure records. This is the case when a SID starts with Route Type 2 or a STAR ends with Route Type 2.

VIA field contains "SDY" or "STY":

In this situation the field contents are defined exactly as stated above (VIA field = "SID" or "STR") except that the field is always non-blank. This field is blank for all other contents of the VIA field.

Used On:	Company Route**, Helicopter Operations Company Route** Records
Length:	5 characters
Character Type:	Alpha/numeric
Examples:	RW08L, ALL, Blank **PADA1, NWPAD**

5.85 ENRT TRANS

Definition/Description: **Together with the Section/Subsection identified for the SID/STAR/App/AWY field, t**his field is used to identify the desired enroute transition of the applicable SID or STAR. It can also be used to identify the desired approach transition of an approach.

Source/Content:

VIA field contains "SID" or STR":

This field uniquely identifies the desired SID/STAR enroute transition. If no enroute transition is desired, the field is blank.

VIA field contains "SDE" or "STE":

In this situation the field contents are defined exactly as stated above (VIA field - "SID" or "STR") except that the field is always non-blank.

VIA field contain "APP":

This field uniquely identifies the desired approach transition. If no approach transition is desired, the field is blank.

The field is blank for all other contents of the VIA field.

Used On:	Company Route**, Helicopter Operations Company Route** Records
Length:	5 characters
Character Type:	Alpha/numeric
Examples:	ETS, KEENE, DEN

5.86 Cruise Altitude

Definition/Description: This field will be used to establish an Enroute Cruise Altitude. It will be entered on Company Route records as specified by the customer.

Source/Content: The customer will supply the Cruise Attitude in feet or flight level.

Used On:	Company Route**, Helicopter Operations Company Route** Records
Length:	5 characters
Character Type:	Alpha/numeric
Examples:	10000, 15000, FL090, FL240

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.87 TERMINAL/ALTERNATE Airport (TERM/ALT ARPT)

Definition/Description: This field has two uses depending on the "VIA" field and File Code for "To Fix." For "VIA" field content of "ALT" this field will contain the Alternate Airport Ident **or Heliport Ident** for this Company Route. If the file code for "To Fix" contains "P," this field will contain the Airport Ident for REGN CODE (Section 5.41) of Terminal Waypoints (PC records) and Runway (PG records). **If the file code for "To Fix" contains "H", this field will contain the Heliport Ident for REGN CODE (Section 5.41) of Helicopter Terminal Waypoints (HC records).**

Source/Content: See Section 5.6, Airport**/Heliport** Identifier.

Used On: **Company Route, Helicopter Operations Company** Route Records
Length: 4 characters
Character Type: Alpha/numeric
Examples: KDEN, EDDF

5.88 Alternate Distance (ALT DIST)

Definition/Description: **This field is used to supply the distance in nautical miles from the "To Airport/Heliport/Fix" to the Alternate Airport/Heliport**.

Source/Content: Values for this field will be supplied by the customer and must be equal to or greater than the great circle distance from the destination airport/fix to the alternate airport**/heliport**.

Used On: Company Route, Helicopter Operations Company Route Records
Length: 4 characters
Character Type: Numeric
Examples: 052, 0011, 0123

5.89 Cost Index

Definition/Description: The Cost Index field is used to define the relative value of fuel-related costs and time-related costs for a particular route.

Source/Content: Source will be by customer airline.

Used On: Company Route**, Helicopter Operations Company Route** Records
Length: 3 characters
Character Type: Numeric
Examples: 001, 011, 999

5.90 ILS/DME Bias

Definition/Description: This field is used to specify the DME offset.

Source/Content: The field contains a 2-digit bias term in nautical miles and tenths of a nautical mile with the decimal point suppressed. Field is blank for unbiased DME's.

Used On: VHF NAVAID Records containing ILS/DME or MLS/DME Facilities
Length: 2 characters
Character Type: Numeric
Examples: 13, 91

5.91 Continuation Record Application Type (APPL)

Definition/Description: This field indicates specific application of this continuation record.

Source/Content: The field will contain one of the following type codes:

Field Content	Description
A	A standard ARINC Continuation containing Notes or other formatted data **not covered by a define Continuation**
B	**Combined Controlling Agency/Call Sign and formatted Time of Operation**
C	Call Sign/Controlling Agency Continuation
E	Primary Record Extension
L	VHF Navaid Limitation Continuation
N	A Sector Narrative Continuation
T	A Time of Operations Continuation, "formatted time data"
U	A Time of Operations Continuation "**Narrative** time data"
V	A Time of Operations Continuation, **Start/End Date**
P	A Flight Planning Application Continuation
Q	A Flight Planning Application Primary Data Continuation
S	Simulation Application Continuation
W	An Airport or Heliport Procedure Data Continuation with SBAS use authorization information

Used On: Continuation Records
Length: 1 character
Character Type: Alpha

5.92 Facility Elevation (FAC ELEV)

Definition/Description: **The "Facility Elevation" field provides the elevation of navaids and communications transmitters.**

Source/Content: **Facility Elevation data is derived from official government source. It is provided in feet with a resolution of one foot. It is referenced to MSL. When the elevation is below MSL, the first character of the field will be a minus sign (-) indicating below sea level.**

Used On: ILS Marker, Airway Marker **Primary Records Enroute, Airport, and Heliport Primary Extension Continuation Records,**VHF Navaids and NDB Navaids **Simulation** Continuation Records
Length: 5 characters
Character Type: Alpha/numeric
Examples: 00530, -0014

5.93 Facility Characteristics (FAC CHAR)

Definition/Description: The "Facility Characteristics" field identifies the characteristics of the NAVAID facility.

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Source/Content:

Facility	28	29	30	31	32
VHF NAVAID, ILS & MLS					
Synchronous Asynchronous Unknown	S A U				
VHF NAVAID, NDB NAVAID and Locator					
Voice Ident No Voice Ident Undefined		Y N U			
NDB NAVAID					
Type of emission 400H 1020H Repetition Rate			Note 1	 4 1	 Note 2
ILS DME Location					
Collocated with Localizer Note 3 Collocated with Glide Slope Not cllocated with Localizer or Glide Slope					L G Blank
ILS Back Course					
Usable Unusable Restricted Undefined				Y N R U	
MLS, DME or DME/P Location					
Collocated with Azimuth Collocated with Elevation Not Collocated with Azimuth or Elevation					A E N
MLS Approach Azimuth Scan Rate				Note 4	

Used On: ILS Marker Primary records, VHF Navaid, NDB
Navaid and ILS/MLS continuation records
Length: 5 characters
Character Type: Alpha/numeric

Note 1: 0=A0, 1=A1, 2=A2

Note 2: Enter number of occurrences per minute if known. Leave blank if not known.

Note 3: Colocated means that the latitudes and longitudes of the two facilities differ by no more than 1 arc second.

Note 4: Where a high-rate approach azimuth guidance is available, enter "H," otherwise leave blank.

COMMENTARY

The NDB emission designators set forth in Note 1 above are being replaced with the new designators shown in the equivalency table below as the result of action taken at the 1979 ITU World Administrative Radio Conference.

Present Designator	New Designator	Description
A0	NON	Unmodulated Carrier
A1	A1A	Carrier keyed, bandwidth less than 0.1 kHz
A1	A1B	Carrier keyed, bandwidth greater than 0.1 kHz
A2	A2A	Tone keyed modulation

5.94 True Bearing (TRUE BRG)

Definition/Description: The "Magnetic Bearing" for ILS localizer, MLS Azimuth, MLS Back Azimuth and Runway records is given in the primary record. This field allows the true bearing to be entered independently of the magnetic variation.

Source/Content: True Bearings are entered into the field in degrees, tenths of a degree and hundredths of a degree, with the decimal point suppressed. See Section 5.95 for source description.

Used On: ILS Continuation, MLS Continuation and Runway Continuation records
Length: 5 characters
Character Type: Numeric
Examples: 19000, 23021, 06050

5.95 Government Source (SOURCE)

Definition/Description: The content of the source field indicates whether the "True Bearing" is derived from official government sources or from other sources.

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Source/Content: The field contains "Y" when the "True Bearing" is derived from official government sources and "N" when it is derived from other sources.

Used On: ILS, MLS, MLS continuation and runway continuation records
Length: 1 character
Character Type: Alpha

5.96 Glide Slope Beam Width (GS BEAM WIDTH)

Definition/Description: The "Glide Slope Beam Width" field specifies the glide path beam width of the Glide Slope defined in the record.

Source/Content: Glide Slope beam widths from official government sources are entered into this field in degrees, tenths of a degree and hundredths of a degree with the decimal point suppressed.

Used On: ILS continuation records
Length: 3 characters
Character Type: Numeric
Examples: 140, 180, 200

5.97 Touchdown Zone Elevation (TDZE)

Definition/Description: The "Touchdown Zone Elevation" is the highest elevation in the first 3,000 feet of the landing surface beginning at the threshold.

Source/Content: Touchdown zone elevations from official government sources will be used when available. If official source is not available, the runway threshold elevation will be entered. If the runway threshold elevation is not available, the Airport reference point elevation will be entered. (See TDZE Location, Section 5.98) The elevation will be entered in feet, to a resolution of 1 foot, with respect to MSL. For below MSL elevations, the first character of the field is a minus (-) sign.

Used On: Runway continuation records
Length: 5 characters
Character Type: Alpha/numeric
Examples: 02171, 05230, -0142

5.98 TDZE Location (LOCATION)

Definition/Description: The content of the "TDZE Location" field indicates whether the TDZ elevation was obtained from official government sources or from other sources.

Source/Content: The field will contain a "T" for official source or a "L" if the landing threshold elevation is used, or an "A" if the airport elevation is used.

Used On: Runway continuation records
Length: 1 character
Character Type: Alpha

5.99 Marker Type (MKR TYPE)

Definition/Description: The "Marker Type" field defines the type of marker.

Source/Content: The field contains the following information.

Type of Facility	Record Column Content		
	18	19	20
Inner Marker		I	M
Middle Marker		M	M
Outer Marker		O	M
Back Marker		B	M
Locator at Marker	L		

Used On: Airport Localizer Marker records
Length: 3 characters
Character Type: Alpha

5.100 Minor Axis Bearing (MINOR AXIS TRUE BRG)

Definition/Description: The "Minor Axis Bearing" field indicates the true bearing of the minor axis of marker beacons.

Source/Content: This field will contain the true bearing in degrees and tenths of a degree, with the decimal point suppressed.

Used On: Airport Localizer Marker records
Length: 4 characters
Character Type: Numeric
Examples: 0900, 2715

5.101 Communications Type (COMM TYPE)

Definition/Description: **The "Communications Type" is a three-character code indicating the type of communications service available on the frequency contained in the record. Decoding is available in the Communications Type Translation Table.**

Source/Content: **The field will be derived from official source or created by the data supplier. An indication of the origination of the code is contained in the Communications Type Translation Table.**

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Field Content	Description	Airport Heliport Comm Only	Enroute Comm Only	Both Comm Type
ACC	Area Control Center			X
ACP	Airlift Command Post	X		
AIR	Air to Air	X		
APP	Approach Control	X		
ARR	Arrival Control	X		
ASO	Automatic Surface Observing System (ASOS)	X		
ATI	Automatic Terminal Info Service (ATIS)	X		
AWI	Airport Weather Information Broadcast (AWIB)	X		
AWO	Automatic Weather Observing Service (AWOS)			X
AWS	Aerodrome Weather Information Services (AWIS)	X		
CBA	**Class B Airspace**	**X**		
CCA	**Class C Airspace**	**X**		
CLD	Clearance Delivery	X		
CPT	Clearance, Pre-Taxi	X		
CTA	Control Area (Terminal)	X		
CTF	**Common Traffic Advisory Frequencies (Note 2)**	**X**		
CTL	Control			X
DEP	Departure Control	X		
DIR	Director (Approach Control Radar)	X		
EFS	Enroute Flight Advisory Service (EFAS)		X	
EMR	Emergency			X
FSS	Flight Service Station			X
GCO	Ground Comm Outlet	X		
GND	Ground Control	X		
GTE	Gate Control	X		
HEL	Helicopter Frequency	X		
INF	Information			X
MBZ	**Mandatory Broadcast Zone (Note 2)**	**X**		
MIL	Military Frequency			X
MUL	Multicom			X
OPS	Operations	X		
PAL	Pilot Activated Lighting (Note 1)	X		
RDO	Radio			X
RDR	Radar			X
RFS	Remote Flight Service Station (RFSS)			X
RMP	Ramp/Taxi Control	X		
RSA	Airport Radar Service Area (ARSA)	X		
TCA	Terminal Control Area (TCA)	X		
TMA	Terminal Control Area (TMA)	X		
TML	Terminal	X		
TRS	Terminal Radar Service	X		
	Area (TRSA)			
TWE	Transcriber Weather Broadcast (TWEB)		X	
TWR	Tower, Air Traffic Control	X		
UAC	Upper Area Control		X	
UNI	Unicom	X		
VOL	Volmet		X	

Note 1: The Comm Type PAL is used only when the frequency(s) published are used exclusively for the activation of airport lighting. If the pilot activation of airport lighting is accomplished on a frequency that is also used for voice communications, the Pilot Controlled Lighting parameter of the Service Indicator is used.

Note 2: The Comm Types CTF and MBZ are used in Australia, New Zealand and East Timor only.

Used On: Enroute, Airport and Heliport Communications
Length: 3 characters
Character Type: Alpha

5.102 Radar (RADAR)

Definition/Description: **The "Radar" field indicates whether or not the communications unit identified in the record has access to and uses information derived from primary or secondary radars while performing the communications service indicated by the Communications Type. It is not an indication of an operational radar frequency.**

Source/Content: **The availability of radar information to a communications service provider is derived from official government sources. The field will be set to the character "R" if primary or secondary radar information is available to the service, the character "N" if the source documentation specifically states that the service does not have access to primary or secondary radar information or the character "U" if the source documentation does not provide details on radar information access for the service.**

Used On: Enroute, Airport and Heliport Communications records
Length: 1 character
Character Type: Alpha

5.103 Communications Frequency (COMM FREQ)

Definition/Description: **The "Communications Frequency" field specifies either transmit or receive frequency of the communications service, dependent on in which column the frequency is located. Each communications record will contain both transmit and receive frequencies unless the service is published as a Transmit Only or Receive Only service. The content will be identical if the service transmits and receives on the same frequency. The fields will be left blank when the service provided is a digital service.**

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Source/Content: Content is derived from official government sources. The following details apply:

HF frequencies are provided as five significant digits in kilohertz for 10 thousands, thousands, hundreds, tens and units. The remaining two positions of the seven-character field is zero filled.

Example: The HF frequency of 17955 kHz would be expressed as 1795500. The HF frequency of 8965 kHz would be expressed as 0896500.

VHF frequencies with 100, 50 or 25 kilohertz spacing are provided as three significant digits and three decimals in megahertz for hundreds, tens, units, tenths, hundredths and thousandths. The remainder of the seven-character field is zero filled.

Example: The VHF frequency of 118.50 MHz would be expressed as 0118500. The VHF frequency of 131.275 MHz would be expressed as 0131275.

UHF frequencies are provided as three significant digits and two decimals in megahertz for hundreds, tens, units, tenths and hundredths. The remainder of the seven-character field is zero filled.

Example: The UHF frequency of 267 MHz would be expressed as 0026700. The UHF frequency of 287.5 MHz would be expressed as 0028750.

VHF frequencies with 8.33 kHz spacing are provided as four significant digits and three decimals for the assigned channel number. The actual frequency (which would be three significant digits and four decimal places) is not provided.

Example: The VHF frequency of 132.0583 MHz will be provided as the channel number 132.060, expressed in seven digits as 0132060.

The decimal point is always suppressed. As all of these numeric expressions look alike, the “Frequency Units” field (Section 5.104) is provided to assist in actual frequency determination.

Used On: Enroute, Airport and Heliport Communications Records
Length: 7 characters
Character Type: Numeric

5.104 Frequency Units (FREQ UNIT)

Definition/Description: The “Frequency Units” field will designate the frequency spectrum area for the frequency in the “Communications Frequency” (Section 5.103) field as indicated in the table or will designate the content of the “Communications Frequency” field as a channel. **For VHF based units, the field will also designate the established frequency spacing required of the frequency for unambiguous use.**

Source/Content: This field contains the follow information.

Field Content	Description
L	**Low Frequency (Note 1)**
M	**Medium Frequency (Note 1)**
H	High Frequency (3000 kHz to 30,000 kHz)
K	**Very High Frequency 100 kHz spacing**
F	**Very High Frequency 50 kHz spacing**
T	**Very High Frequency 25 kHz spacing**
V	Very High Frequency (30,000 kHz to 200 MHz) **Non-standard spacing**
U	Ultra High Frequency (200 MHz to 3000 MHz)
C	**Very High Frequency** Communication Channel for 8.33kHz spacing
D	**Digital Service (Note 2)**

Note 1: The Codes L and M will only be used when the transmitting frequency is that of a LF/MF Navaid (NDB). If a receiving frequency is provided in the same communications record, it will be a VHF frequency.

Note 2: The Code D for Digital Service will be provided when the communications record contains a data link type service. In these cases, transmit and receive frequency columns will be blank.

Used On: Enroute, Airport and Heliport Communications records
Length: 1 character
Character Type: Alpha

5.105 Call Sign (CALL SIGN)

Definition/Description: **The “Call Sign” field contains the name of a communications service provider that is to be used when contacting that service/used by the service to identify itself when contacting aircraft on the frequencies contained in the record. The field is also used to provide the broadcast identification name of automated services.**

Source/Content: Call Signs and broadcast service identification names are derived from official government sources. The type of service may be omitted from the Call Sign field when it is the same as the service identified in the Communications Type (5.101).

Used On: Airport, Enroute **and Heliport** Communications Records
Length: 25 characters
Character Type: Alpha/numeric

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Examples: COMM TYPE CALL NAME
APP LION ("APPROACH" is omitted)
TWR LION ("TOWER" is omitted)
DEP LONDON APPROACH
ACC DENVER CENTER

5.106 Service Indicator (SERV IND)

Definition/Description: The "Service Indicator" field is used to further define the use of the frequency for the specified Communication Type (5.101).

Source/Content: The field may contain the following information:

AIRPORT **HELIPORT** COMMUNICATIONS RECORDS

Description	Column Content		
	27	28	29
Airport Advisory Service (AAS)	A		
Community Aerodrome Radio Station (CARS)	C		
Departure Service (Other than Departure Control Unit)	D		
Flight Information Service (FIS)	F		
Initial Contact (IC)	I		
Arrival Service (Other than Arrival Control Unit)	L		
Aerodrome Flight Information Service (AFIS)	S		
Terminal Area Control (Other than dedicated Terminal Control Unit)	T		
Aerodrome Traffic Frequency (ATF)		A	
Common Traffic Advisory Frequency (CTAF)		C	
Mandatory Frequency (MF)		M	
Secondary Frequency		S	
VHF Direction Finding Service (VDF)			D
Language other than English			L
Military Use Frequency			M
Pilot Controlled Light (PCL)			P

ENROUTE COMMUNICATIONS RECORDS

Description	Column Content		
	57	58	59
Aeronautical Enroute Information Service (AEIS)	A		
Flight Information Service (FIS)	F		
Air/Ground		A	
Discrete Frequency		D	
Mandatory Frequency		M	
Secondary Frequency		S	
VHF Direction Finding Service (VDF)			D
Language other than English			L
Military Use Frequency			M

Used On: Enroute, Airport and Helicopter Communications records
Length: 3 character
Character Type: Alpha

5.107 ATA/IATA Designator (ATA/IATA)

Definition/Description: The "ATA/IATA" field contains the Airport/Heliport ATA/IATA designator code to which the data contained in the record relates.

Source/Content: The content of this field should be derived from IATA Reservations Manual Part II, IATA Resolution 763/Location Identifiers.

Used On: Airport and Heliport records
Length: 3 characters
Character Type: Alpha
Examples: DEN, LHR, JFK

5.108 IFR Capability (IFR)

Definition/Description: The "IFR Capability" field indicates if the Airport/Heliport has any published Instrument Approach Procedures.

Source/Content: The field contains "Y" if there is an Official Government Instrument Approach Procedure published, otherwise the field will contain "N". (Note: The presence of "Y" in this field does not necessarily imply that the published instrument approach is coded in the data base.)

Used On: Airport and Heliport records
Length: 1 character
Character Type: Alpha

5.109 Runway Width (WIDTH)

Definition/Description: The width of the runway identified in the "Runway Identifier" field is specified in the "Runway Width" field.

Source/Content: Runway widths derived from Official Government Sources are entered into the field in feet, with a resolution of one foot. For runways of variable width, the minimum width encountered over the runway length will be entered.

Used On: Runway records
Length: 3 characters
Character Type: Numeric
Examples: 150, 300, 075

5.110 Marker Ident (MARKER IDENT)

Definition/Description: The "Marker Ident" field contains a unique computer ident assigned to each enroute marker.

Source/Content: A unique identifier will be created for each enroute marker since such idents are not designated by official sources. Marker idents will be established using the 2-character ICAO code followed by two numeric digits assigned to keep markers unique within a given ICAO region.

Used On: Enroute marker records
Length: 4 characters
Character Type: Alpha/numeric
Examples: EG01, EG02, K101, K102

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5.111 Marker Code (MARKER CODE)

Definition/Description: The "Marker Code" field contains the coded ident that provides an aural and visual indication of station passage in the cockpit. The code shall be keyed so as to transmit dots or dashes, or both, in an appropriate sequence on a radio frequency of 75 MHz. The frequency of the modulating tone is 3000 Hz.

Source/Content: The field contains the morse code ident (dots and dashes) derived from official government sources.

Used On: Enroute marker records
Length: 4 characters
Character Type: Alpha
Examples: -.-., ..., - - - -

5.112 Marker Shape (SHAPE)

Definition/Description: The "Shape" field defines the radiation pattern of an airways marker as being either "bone" or "elliptical."

Source/Content: The field contains the shape of the marker derived from official government sources when available. The character "B" will designate the "bone" shape and the character "E" will designate the elliptical shape. "E" will be entered when the source does not supply shape information.

Used On: Enroute airways marker records
Length: 1 character
Character Type: Alpha

5.113 High/Low (HIGH/LOW)

Definition/Description: The "High/Low" field indicates the power of the enroute marker.

Source/Content: The field contains the power derived from official government sources. The character "L" indicates low power for use at low altitudes. The character "H" indicates high power for general use.

Used On: Enroute marker records
Length: 1 character
Character Type: Alpha

5.114 Duplicate Indicator (DUP IND)

Definition/Description: The "Duplicate Identifier" field is used to further define holding patterns when official government source has designated more than one Holding Pattern on a Navaid or Waypoint.

Source/Content: Holding Patterns are derived from official government sources documents. That documentation will normally specify the airspace structure in which the holding is to be used. That documentation may also designate more than one Holding Pattern for a single Navaid or Waypoint. This field will contain details on airspace structure and multiple designations. More than one holding is designated on a single "fix" when one or more of the following elements are different for holdings within the same airspace structure.Inbound Holding Course, Turn Direction, Altitude, Leg Length or Leg Time, and Holding Speed.

If only one Holding Pattern is designated for a "fix" and the airspace structure in which that holding is to be used is not defined, the field will contain "00." If only one Holding Pattern is designated for a "fix" and the airspace structure in which that holding is to be used is defined or if the same holding is designated for more than one airspace structure, the first position of the "Duplicate Indicator" will contain a digit of 1 through 6 and the second position will contain a zero. If more than one holding is designated for a single "fix" in one type of airspace structure, the first position will contain a digit of 1 through 6 and the second position will contain a digit of **0** through 9, depending on the number of holdings on that "fix" within that airspace structure.

If multiple holdings are designated in official source documents for a single "fix" and the airspace structure is not defined for all holdings, those with "undefined airspace structure" will carry the digit 7 in position one and a digit of 0 through 9 in position two.

MULTIPLE HOLDING PATTERNS

Holding Pattern	Duplicate Indicator	
	Position One Airspace	Position Two Multiple
Undefined (None Defined)	0	See Note 1
High Altitude	1	See Note 1
Low Altitude	2	See Note 1
SID	3	See Note 1
STAR	4	See Note 1
Approach	5	See Note 1
Missed Approach	6	See Note 1
Undefined (with other defined)	7	See Note 1
All Altitude	8	See Note 1

Note 1: If there is only one holding pattern on a given fix within an airspace structure, position 2 will contain a "0." For additional (multiple) holdings on that same fix within the same airspace structure, position 2 will be incremented by 1, e.g. "0" for the first "1" for the second, etc.

Used On: Holding Pattern Records
Length: 2 characters
Character Type: Numeric
Examples: 00, 10, 61, 32

5.115 Directional Restriction

Definition/Description: The "Direction Restriction" field, when used on Enroute Airway records, will indicate the direction an Enroute Airway is to be flown. The "Direction Restriction" field, when used on Preferred Route records, will indicate whether the routing is available only in the direction of "from initial fix to terminus fix" or in both directions.

Source/Content: Direction Restrictions should be derived from official government sources. They will be coded and supplied as follows:

Enroute Airway Records

5.0 NAVIGATION DATA - FIELD DEFINITIONS

F = One way in direction route is coded (Forward).
B = One way in opposite direction route is coded (backward).
Blank = No restrictions on direction.

Preferred Route Records

F = Uni-directional Preferred Route, usable only from Initial Fix to Terminus Fix.
B = Bi-directional Preferred Route, usable from Initial Fix to Terminus Fix or from Terminus Fix to Initial Fix.

Used On: Enroute Airway and Preferred Route Records
Length: 1 character
Character Type: Alpha

5.116 FIR/UIR Identifier (FIR/UIR IDENT)

Definition/Description: The "FIR/UIR Identifier" field identifies the Flight Information Region and Upper Information Region of airspace with defined dimensions within which Flight Information Service and Alerting Service are provided. The Identifier is for the controlling Area Control Center or Flight Information Center.

Source/Content: FIR/UIR Identifiers will be derived from official government sources. This field contains the four character identifier assigned to the airspace. For those areas charted as "NO FIR," the identifier field will contain "XX plus a two digit numeric."

When used on Flight Planning Continuation records, the entry will be related to the altitude structure. For records that are classed or designated as high altitude, the FIR field will be blank. For areas assigned a FIR identifier only that is valid for both the low altitude and the high altitude structure, the UIR field will be blank. For detail records classed or designated as low altitude and high altitude, both the FIR and the UIR identifier will be entered.

Used On: FIR/UIR, VHF NAVAID, NDB NAVAID, Enroute, Terminal Waypoint, Airport Flight Planning Continuation and Heliport records
Length: 4 characters
Character Type: Alpha
Examples: DAAG, SGAS, XX02

5.117 FIR/UIR Indicator (IND)

Definition/Description: **When used on Enroute Communications Records, the content definition above for the FIR/UIR Record is to be applied when ever the FIR/RDO (5.190) field of the Enroute Communications Record contains an Information Region Identifier. In all other cases, the Indicator field of the Enroute Communications Record will be blank.**

Source/Content:

Type	Field Content
FIR	F
UIR	U
Combined FIR/UIR	B

Used On: FIR/UIR and Enroute Communications records
Length: 1 character
Character Type: Alpha

5.118 Boundary Via (BDRY VIA)

Definition/Description: The "Boundary VIA" defines the path of the boundary from the position identified in the record to the next defined position.

Source/Content: The path of the boundary will be determined from official government sources or the rule listed below and the "Boundary VIA" will be selected from the table below.

Field	Content	Description
Position 1	Position 2	
C		Circle
G		Great Circle
H		Rhumb Line
L		Counter Clockwise ARC
R		Clockwise ARC
	E	End of description, return to origin point

Application Rules:

1. Special Use Airspace designated as following rivers, country, state or other political boundaries will be averaged in coding by using a series of straight lines so that no path will be greater than two miles from the actual boundary. The Boundary VIA will be "G".

2. If there is a named waypoint on an airway which crossed an irregular FIR/UIR boundary, the waypoint coordinates will be used to define a point in the path defining that FIR/UIR boundary. The Boundary VIA will appropriate to the path definition.

3. Paths that follow lines of latitude will be coded with a Boundary Via of "H". Paths that follow lines of longitude may be coded with a Boundary Via of "G" or "H". Consistent use of one or the other with a single airspace is desired.
4. Other than for lines of latitude and longitude, the Boundary VIA of "H" shall only be used when specifically stated in the official government source. If not stated as "Rhumb Line" or not along latitude/longitude, all straight lines will be coded as "G".

Note: Refer to Figure 5-12 for sample coding of Boundary VIA Codes.

Used On: Controlled Airspace, FIR/UIR, and Restrictive Airspace records
Length: 2 characters
Character Type: Alpha

5.119 Arc Distance (ARC DIST)

Definition/Description: The "Arc Distance" field is used to define the distance in nautical miles from the "Arc

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Origin" position to the arc defining the lateral boundary of a FIR/UIR or Restrictive Airspace.

Source/Content: ARC distances should be derived from official government sources when available, in nautical miles and tenths of nautical mile, with the decimal point suppressed. The field will be entered only when "Boundary Via" is "A," "C," "L," or "R".

Used On: FIR/UIR, Restrictive Airspace, and Controlled Airspace records
Length: 4 characters
Character Type: Numeric
Examples: 0080, 0150, 1000

5.120 Arc Bearing (ARC BRG)

Definition/Description: The "Arc Bearing" field contains the true bearing from the "Arc Origin" position to the beginning of the arc.

Source/Content: Arc bearings should be derived from official government sources when available. The field contains true bearing in degrees and tenths of degree, with the decimal point suppressed. The field will only be entered when Boundary Via is "A," "C," "L" or "R".

Used On: FIR/UIR, Restrictive Airspace, and Controlled Airspace records
Length: 4 characters
Character Type: Numeric
Examples: 0900, 1800, 3450

5.121 Lower/Upper Limit

Definition/Description: Special Use Airspace is described by both lateral and vertical boundaries. The "Lower/Upper Limit" fields contain the lower and upper limits of the FIR/UIR or Restrictive Airspace being described.

Source/Content: Limits for the special use airspace should be derived from official government sources. The field may contain altitude (all numerics), flight levels (alpha/numerics) or an all alpha entry (see examples). The flight level entry will contain the alpha characters "FL" followed by the altitude in hundreds of feet. These fields will be entered on the first record only of each FIR/UIR or Restrictive Airspace being described.

Used On: FIR/UIR, Restrictive Airspace, and Controlled Airspace records
Length: 5 characters
Character Type: Alpha/numeric

Examples:
- All numeric: 05000, 25000
- Alpha/numeric: FL245, FL450
- All alpha: NOTSP (for Not Specified)
  UNLTD (for unlimited)
  GND (for Ground)
  MSL (for Mean Sea Level)
  NOTAM (for Restrictive Airspace only)

5.122 FIR/UIR ATC Reporting Units Speed (RUS)

Definition/Description: The "FIR/UIR ATC Reporting Units Speed" is used to indicate the units of measurement concerning True Air Speed used in the specific FIR/UIR to fulfill the requirements of ICAO flight plan.
Source/Content: FIR/UIR Reporting Units should be derived from official government publications. The field will be entered on the first record only for each FIR/UIR identifier.

Reporting Units	Field Entry
Not specified	0
TAS in Knots	1
TAS in Mach	2
TAS in Kilometers/hr	3

Used On: FIR/UIR records
Length: 1 character
Character Type: Numeric

5.123 FIR/UIR ATC Reporting Units Altitude (RUA)

Definition/Description: The "FIR/UIR ATC Reporting Units Altitude" field is used to indicate the units of measurement concerning the altitude used in the specific FIR/UIR to fulfill the requirements of ICAO flight plan.

Source/Content: FIR/UIR Reporting Units should be derived from official government publications. The field will be entered on the first record only for each FIR/UIR identifier.

Reporting Units	Field Entry
Not specified	0
ALT in Flight Level	1
ALT in Meters	2
ALT in Feet	3

Used On: FIR/UIR records
Length: 1 character
Character Type: Numeric

5.124 FIR/UIR Entry Report (ENTRY)

Definition/Description: The "FIR/UIR Entry Report" field is used to indicate whether an entry report on ICAO flight plan is required for that specific FIR/UIR.

Source/Content: FIR/UIR Entry Report should be derived from official government publications. "Y" in this field indicates Entry Report is required, "N" in this field indicates no Entry Report is required. The field will be entered on the first record only for each FIR/UIR identifier.

Used On: FIR/UIR records
Length: 1 character
Character Type: Alpha

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.125 FIR/UIR Name

Definition/Description: The "FIR/UIR Name" field contains the official name of the controlling agency of the FIR/UIR of which this record is an element.

Source/Content: The FIR/UIR name will be derived from official publications. The areas without a specific "FIR/UIR" designation will be labeled "NO FIR".

Used On:	FIR/UIR records
Length:	25 characters
Character Type:	Alpha/numeric
Examples:	ACCRA, FIR, ASUNCION FIR/UIR, NO FIR

5.126 Restrictive Airspace Name

Definition/Description: The "Restrictive Airspace Name" field will contain the name of the restrictive airspace when assigned.

Source/Content: Names will be derived from official government sources. The name, if assigned, will be entered in the first record only. If source does not assign a name, this field may be blank.

Used On:	Restrictive Airspace records
Length:	30 characters
Character Type:	Alpha/numeric
Examples:	RANDOLPH ONE MOA, SAMBURU GAME RESERVE

5.127 Maximum Altitude (MAX ALT)

Definition/Description: The "Maximum Altitude" field is used to indicate the maximum altitude allowed.

Source/Content: Maximum altitudes should be derived from official government publications describing the upper limit of the airway in feet or flight level.

Used On:	Enroute Airway, Holding Pattern and Preferred Route records
Length:	5 characters
Character Type:	Alpha/numeric
Examples:	
All numeric:	17999, 08000
Alpha/numeric:	FL100, FL450
All alpha:	UNLTD (for unlimited)

5.0 NAVIGATION DATA - FIELD DEFINITIONS

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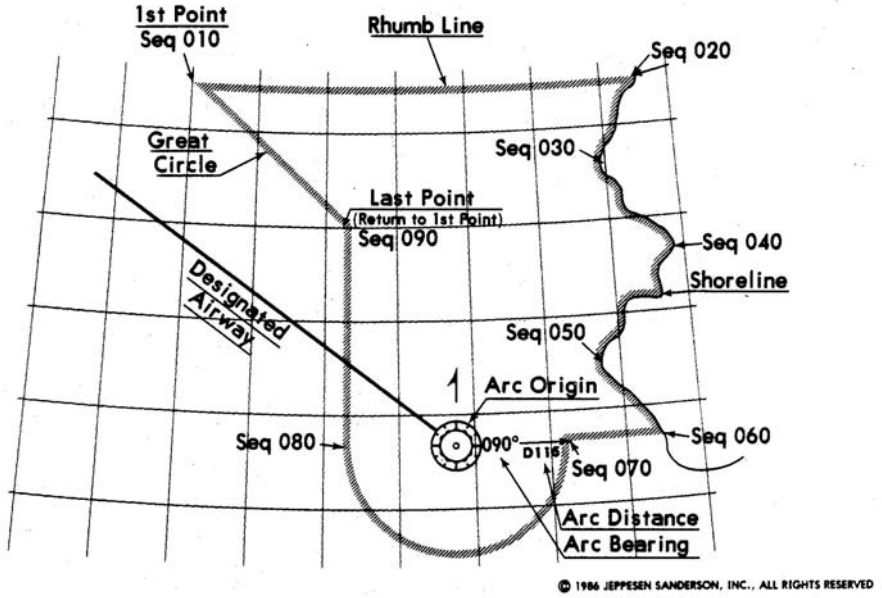
Not for navigational or other operational use. For example only. Please consult current navigation charts.

FIR/UIR AND RESTRICTIVE AIRSPACE



Seq No.	Boundary Via	Latitude	Longitude	Arc Origin Latitude	Arc Origin Longitude	Arc Dist	Arc Brg
010	H	N45-00-00	W060-00-00				
020	G	N45-00-00	W047-00-00				
030	G	N43-12-45	W048-05-00				
040	G	N41-18-24	W046-16-12				
050	G	N38-58-54	W048-30-36				
060	H	N37-20-15	W047-00-00				
070	R	N37-20-15	W049-31-00	N37-20-18	W052-30-30	115	090
080	H	N37-20-15	W055-30-00				
090	G	N42-00-00	W055-30-00				

Figure 5-12
Controlled and Restrictive Airspace and FIR/UIR Boundaries

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.128 Restrictive Airspace Type (REST TYPE)

Definition/Description: The "Restrictive Airspace Type" field is used to indicate the type of Airspace in which the flight of aircraft is prohibited or restricted. The restriction may be continuous or specified for certain times.

Source/Content: The "Restrictive Airspace Type" should be derived from official government publications.

Type	Field
Alert	A
Caution	C
Danger	D
Military Operations Area	M
National Security Area	**N**
Prohibited	P
Restricted	R
Training	T
Warning	W
Unspecified or Unknown	U

Used On: Restrictive Airspace and Enroute Airway Flight Planning Continuation records
Length: 1 character
Character Type: Alpha

5.129 Restrictive Airspace Designation

Definition/Description: The "Restrictive Airspace Designation" field contains the number or name that uniquely identifies the restrictive airspace.

Source/Content: The identifiers will be derived from official government sources. The field will contain a numeric number, or when designation is by name this field will contain the name up to 10 characters. When name is longer than 10 characters, the 10th position will contain an asterisk indicating the name field should be used for the full designator.

Used On: Restrictive Airspace and Enroute Airway Flight Planning Continuation records
Length: 10 characters
Character Type: Alpha/numeric

Field Content			
Charted Designator	ICAO	Type	Rest. Desig.
RJ(R)-116	RJ	R	116
R-2524	K2	R	2524
Crystal MOA	K4	M	Crystal
Randolph MOA One B	K4	M	Randolph*

5.130 Multiple Code (MULTI CD)

Definition/Description: The "Multiple Code" field will be used to indicate Restrictive Airspace Areas or MSA Centers having the same designator but subdivided or differently divided by lateral and/or vertical detail.

Source/Content: This field will be used when official government publications for Restrictive Airspace divides an area with the same designator into different areas of Activation, altitude or other defining characteristics. For MSA Centers, this provides different sectorization and altitudes for the MSA published with the same center. The field will contain an alpha character uniquely identifying each area or MSA. The first record affected could contain the character "A" and multiple primary records could contain the character "B," "C," "D," etc., as required.

Used On: Controlled Airspace, Restrictive Airspace, Airport and Heliport MSA Center, Airport and Heliport SID/STAR/Approach, and Enroute Airway Flight Planning Continuation Records.
Length: 1 character
Character Type: Alpha

5.131 Time Code (TIME CD)

Definition/Description: When used on the Primary or Primary Extension Continuation Record of the possible record types, with the exception of the Airway Restriction Records, the "Time Code" field is used to indicate that the data contained in the record is either available continuously or not continuously. **When "Time Code" is used in a Time of Operations Continuation Record, other that Airway Restriction Records, the field is used to indicate how to interpret Time of Operations Continuation Records. On Airway Restriction Primary and Continuation Records, the Time Code indicated either a continuous or non-continuous operation, the details of which are contained in the same record.**

Source/Content: Active times are derived from official government source. The field will contain an alpha character for which an associated description has been defined as indicated in the tables below.

Used On: **Primary Records: Restrictive Airspace, Preferred Route, Controlled Airspace – Used Primary Record Table portion.**

Primary Extension Continuation Records: Airport, Heliport and Enroute Communications Records – Use Primary Record Table portion.

Time of Operations Continuation Records: Restrictive Airspace, Preferred Route, Controlled Airspace, Airport, Heliport, Enroute Communications Records – Use Continuation Records Table portion.

	PRIMARY RECORDS
Field Content	Description
C	Active Continuously, including holidays
H	Active Continuously, excluding holidays
N	Active Non-Continuously, Refer to Continuation Record
P	**Active times announced by NOTAM**
U	**Active times are not specified in source documentation**

5.0 NAVIGATION DATA - FIELD DEFINITIONS

	CONTINUATION RECORDS
H	Active times are provided in Time of Operation format and exclude holidays
N	Activation Times are too complex for Time of Operation format and are provided in Note Form
T	Active times are provided in Time of Operation format and include holidays

Used On: Enroute Airway Restriction Primary and Continuation Records

	PRIMARY AND CONTINUATION RECORDS
Field Content	Description
C	Active Continuously, including holidays
H	Active Continuously, excluding holidays
S	Active times are provided in Time of Operation format and exclude holidays
T	Active times are provided in Time of Operation format and include holidays

Length: 1 character
Character Type: Alpha

5.132 NOTAM

Definition/Description: Restrictive Airspace areas may not have established active times and are activated by NOTAM or may be active by NOTAM in addition to established times.

Source/Content: Active times by NOTAM will be derived from official government source. When used on primary records, the area is active only by NOTAM and there will be no continuation record. When used on continuation records, the area is active by NOTAM in addition to the established times. The field will contain the alpha character "N" to indicate either condition, otherwise the field will be blank.

Used On: Controlled Airspace, Restrictive Airspace and Restrictive Airspace Continuation records
Length: 1 character
Character Type: Alpha

5.133 Unit Indicator (UNIT IND)

Definition/Description: Restrictive Airspace lower and upper limits are specified as "above mean sea level" (MSL) or "above ground level" (AGL). This field permits the unit of measurement to be indicated.

Source/Content: The units of lower and upper limits are derived from official government source. The alpha character "M" will indicate MSL and the alpha character "A" will indicate AGL.

Used On: Controlled Airspace, Restrictive Airspace records
Length: 1 character
Character Type: Alpha

5.134 Cruise Table Identifier (CRSE TBL IDENT)

Definition/Description: A standard cruising level table is established by ICAO and is to be observed except when, on the basis of regional air navigation agreements, a modified table of cruising levels is prescribed for use. This field permits the enroute airway record to identify the Cruise Table record that is to be used for cruise levels.

Source/Content: Cruise Levels will be derived from official government sources. For the standard ICAO cruise table this field will contain the alpha characters "AA". For those countries not using the standard ICAO table and having a modified table this field will contain the alpha characters "BB," "CC," etc. If a country uses the standard ICAO table or a Modified table but indicates that an airway or portion of an airway is to be flown opposite of the cruise table, the field will contain alpha/numeric characters that identify the table to be used.

Used On: Enroute Airway, FIR/UIR, Cruise Table and Flight Planning Arrival/ Departure Data Records
Length: 2 characters
Character Type: Alpha/numeric
Example:

Field Content	Description
AA	ICAO standard cruise table
AO	Exception to ICAO cruise table
BB - ZZ	Modified cruise table
BO - ZO	Exception to modified cruise table

5.135 Course FROM/TO

Definition/Description: The "Course From" field is used to indicate the lowest course for which a block of cruising levels are prescribed. The "Course To" field is used to indicate the highest course for which a block of cruising levels is prescribed.

Source/Content: The Courses will be derived from official government sources in degrees and tenths of degree with the decimal point suppressed. The Magnetic/True indicator field will be used to indicate True (T) or Magnetic (M) courses.

Used On: Cruising Table records
Length: 4 characters
Character Type: Numeric
Examples: 0000, 1790, 3590

5.136 Cruise Level From/To

Definition/Description: The "Cruise Level From" field is used to indicate the lowest cruising level prescribed for use within the Course From/To fields. The "Cruise Level To" field is used to indicate the highest cruising level prescribed for use within the Course From/To fields.

Source/Content: Cruise Levels will be derived from official government sources. When the level is entered in feet the field will be all numeric. When the level is entered in meters, the first column will contain the alpha character "M" followed by all numeric. If the "Level To" is unlimited, the field will contain the alpha characters "UNLTD."

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Used On: Cruising Table records
Length: 5 characters
Character Type: Alpha/numeric
Examples: 0200, M0600, M1585

5.137 Vertical Separation

Definition/Description: The "Vertical Separation" field is used to indicate the minimum separation prescribed to be maintained between the cruising levels.

Source/Content: Vertical Separation Values will be derived from official government sources and entered in feet or tens of meters with "M" in the first column.

Used On: Cruising Table records
Length: 5 characters
Character Type: Alpha/numeric
Examples: 01000, 02000, M0030, M0060

5.138 Time Indicator (TIME IND)

Definition/Description: The "Time Indicator" field is used to indicate whether the times shown in the "Time of Operations" field(s) are Local Time, Daylight Savings Time or Universal Coordinated Time.

Source/Content: Time contained in the affected record(s) is derived from official government sources. The "Time Indicator" will qualify those source derived times as indicated in the following table:

Field Content	Description
T	Times codes are Local Time
S	Times codes are to be adjusted for Daylight Savings Time
Blank	Times shown are Universal Coordinated Time (UTC)

Used On: Controlled Airspace, Restrictive Airspace Continuation, Referred Route Continuation, Enroute Airway Restriction, Airport and Heliport Communication Continuation and Enroute Communications Continuation Records
Length: 1 character
Character Type: Alpha

5.139 Intentionally Left Blank

5.140 Controlling Agency

Definition/Description: Some "Restrictive Airspace" areas are designated joint use and IFR operations in the area may be authorized by the controlling agency when it is not being utilized by the using agency.

Source/Content: The name of the Controlling Agency should be derived from official government sources and will be shown on the first record only. If no Controlling Agency is specified the field may be blank.

Used On: Controlled Airspace, Restrictive Airspace Continuation record
Length: 25 characters
Character Type: Alpha/numeric
Examples: LAX, ARTCC, Lumpur ACC, Butterworth APP

5.141 Starting Latitude

Definition/Description: The Grid MORA Table will contain records describing the MORA for each Latitude and Longitude block. Each record will contain thirty blocks and the "Starting Latitude" field defines the lower left corner for the first block of each record.

Source/Content: The "Starting Latitude" will be determined when the record is assembled.

Used On: Grid Mora record
Length: 3 characters
Character Type: Alpha/numeric
Examples: N00, N42, S20, S90

5.142 Starting Longitude

Definition/Description: The Grid MORA table will contain records describing the MORA for each Latitude and Longitude block. Each record will contain thirty blocks and the "Starting Longitude" field defines the lower left corner for the first block of each record.

Source/Content: The "Starting Longitude" will be determined when the record is assembled.

Used On: Grid Mora records
Length: 4 characters
Character Type: Alpha/numeric
Examples: E000, W150, E090, W180

5.143 Grid MORA

Definition/Description: Grid MORA Minimum Off-route Altitude (MORA) provides terrain and obstruction clearance within the section outlined by latitude and longitude blocks provided in the Starting Latitude and Starting Longitude fields.

Source/Content: Grid MORA values clear all terrain and obstructions by 1000 feet in areas where the highest elevations are 5000 feet MSL or lower. MORA values clear all terrain by 2000 feet in areas where the highest elevations are 5001 feet MSL or higher. The field will contain values expressed in hundreds of feet, for example, the value of 6000 feet is expressed as 060 and the value of 7100 feet is expressed as 071. For geographical sections that are not surveyed, the field will contain the alpha characters UNK for Unknown.

COMMENTARY

MORA values are generally not provided in government source and are calculated by the data supplier using the formula indicated in the Source/Content paragraph. There are, however, some governments that do provide off route altitude data and a data supplier may elect to use the government source values in their data services.

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Used On:	Grid MORA Records
Length:	3 characters
Character Type:	Alpha/numeric
Examples:	010, 071, 100, 123, UNK

5.144 Center Fix (CENTER FIX)

Definition/Description: When used on Airport and Heliport MSA Records, the "Center Fix" field represents the MSA Center; that point (Navaid or Waypoint) on which the MSA is predicated. When used on Terminal Procedure Records, it can be used in three ways:

1. **When the terminal procedure has an MSA defined, the field will contain the identifier of the fix on which the MSA is predicated. This will serve as a pointer to the specific MSA (PS) Record. This will be populated on the Final Approach Fix (FAF) Record only.**

2. **When the terminal procedure has a TAA defined, the field will contain the identifier of the fix on which the TAA Sector is predicated. This will serve as a pointer to the specific TAA (PK) Record. This will be populated on the first record for each approach transition.**

3. **When used in a terminal procedure record defined by an "RF" Path Terminator, the field will contain the fix that defines the center of the constant rate turn. If the RF Leg terminates at the FAF, where the MSA Center Fix information would normally be found, the RF Center Fix takes priority and the MSA Center Fix pointer will be moved to the FACF record.**

Source/Content: When used as "MSA Center," the field will contain the identification of the navigation facility, Enroute Waypoint, Terminal Waypoint, Runway or Airport Reference Point upon which the MSA coverage radius is predicated. Such content will be derived from official government sources. When used as a "TAA IAF Waypoint", the field contains the official identifier of the waypoint for which the TAA Sector is defined. They will be derived from official government sources. When used as "Radius Center," the field will contain the identification of the Terminal Waypoint used to define the center point of the RF turn.

Used On:	**Airport and Heliport MSA Records, Airport and Heliport TAA Records, Airport and Heliport Approach Procedure Records**
Length:	5 character max
Character Type:	Alpha/numeric

SEC CODE	SUB CODE	START LAT	START LONG	MORA	MORA	MORA	MORA	MORA	MORA	MORA	MORA
A	S	N00	E000	010	010	010	010	010	010	090	191
A	S	N01	E000	010	010	010	010	010	010	010	082
A	S	N02	E000	010	010	010	010	010	010	010	073
A	S	N03	E000	010	010	010	010	010	010	010	073
A	S	N04	E000	010	010	010	010	010	015	UNK	049
A	S	N05	E000	026	014	010	010	014	020	UNK	042
A	S	N06	E000	049	024	020	019	026	029	029	042
A	S	N07	E000	UNK	040	033	031	038	043	035	040
A	S	N08	E000	041	037	033	035	035	034	035	UNK
A	S	N09	E000	029	045	030	035	027	032	033	UNK
A	S	N10	E000	030	034	029	028	028	032	043	UNK
A	S	N11	E000	030	034	031	032	025	041	046	UNK
A	S	N12	E000	026	029	029	022	024	028	043	UNK
A	S	N13	E000	026	030	030	030	026	026	030	UNK
A	S	N14	E000	031	031	024	030	023	040	034	UNK
				E000	E001	E002	E003	E004	E005	E006	E029

Figure 5-13 GRID MORA Sample

The table shows a sample of the Grid Mora Table as it would appear in the file. The table starts at N00/E000 and ends at N14/E029, and is blocked at intervals of sixty minutes. The values shown in the Start Lat and Start Long fields are the lower left corner of a one degree Lat/Long box. The values shown at the bottom of the table are for illustration purpose only and show the Longitude of the lower corner for the MORA values in the table. The values from longitude E007 thru E028 have been omitted from this illustration.

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.145 Radius Limit

Definition/Description: The altitude shown in the "Sector Altitude" field provides a 1000 foot obstacle clearance with a specified radius from the navigational facility/fix. The "Radius Limit," field allows the radius to be specified.

Source/Content: Radius limits will be derived from official government sources. Values will be shown in whole nautical miles.

Used On:	Airport and Heliport MSA Records
Length:	2 characters
Character Type:	Numeric
Examples:	25, 30

5.146 Sector Bearing (SEC BRG)

Definition/Description: **When used on MSA Records, the "Sector Bearing" contains beginning and ending bearing values, referenced to the MSA Center, for each sector of the MSA. When used on TAA records, the "Sector Bearing" contains the beginning and ending bearings that define a TAA Area and are referenced to the TAA Initial Approach Fix (IAF) Waypoint.**

Source/Content: **Sector Bearing information will be derived from official government source. Each "Sector Bearing" field is made up of the start of sector bearing and the end of sector bearing. The values are provided in whole degrees. The first three digits define the start of the sector, the last three digits the end of the sector. For MSA, the values are sector dividing values and the end value of one sector is used as the start value of the next sector. For TAA, the values are inclusive. When multiple Sector Bearings are included in the same MSA or TAA record, they are provide starting with the lowest numbered values and in clockwise order. For MSA that include multiple radii and sector altitudes for the same sector, the Sector Bearings are repeated with the additional radius and altitude data before defining the next sector. For a MSA that is a provided in official government source as an un sectorized circle, both the start and the end sector bearing values are set to 180. Sector Bearing values may be magnetic or true bearings. The Mag/True Indicator in the MSA or TAA will provide this information. See Figure 5-14.**

Used On:	Airport and Heliport MSA and TAA Primary Records
Length:	6 characters
Character Type:	Numeric
Examples:	060140, a Sector that starts at 060 degrees and continues clockwise to end at 140 degrees. 140060, a Sector that starts at 140 degrees and continues clockwise to end at 060 degrees **180180, a sector that starts at 180 degrees and continues clockwise to end at 180, a full circle MSA, no sectorization**

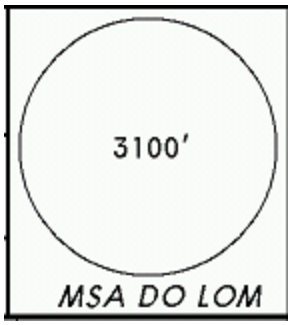
5.147 Sector Altitude (SEC ALT)

Definition/Description: When used on MSA records, the "Sector Altitude" provides a 1000 foot obstacle clearance within the specified sector. When used on TAA records, the "Sector Minimum Altitude" is the minimum altitude for that sector, providing obstacle clearance compatible with the instrument procedures with which the TAA is associated, generally 1000 feet or more as necessary in mountainous areas. Flight crews are expected to fly direct to the initial approach fix in the record at the appropriate sector altitude unless otherwise instructed by ATC.

Source/Content: Sector Altitude values are derived from official government source and are provided in hundreds of feet. **When the official government source does not provide a "Sector Altitude" for one or more sectors of a MSA, the value is provided as 999. See Figure 5-14.**

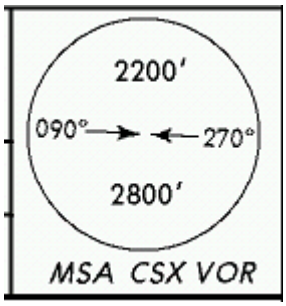
Used On:	Airport and Heliport MSA Records and TAA Primary Records
Length:	3 characters
Character Type:	Numeric
Examples:	010 = 1000ft, 025 = 2500, 100= 10,000 **999 = no sector altitude**

Un-Sectorized MSA:



Output Data: 18018003125, where 180180 represents the Sector Bearings, 031 the Sector Altitude and 25 the Sector Radius

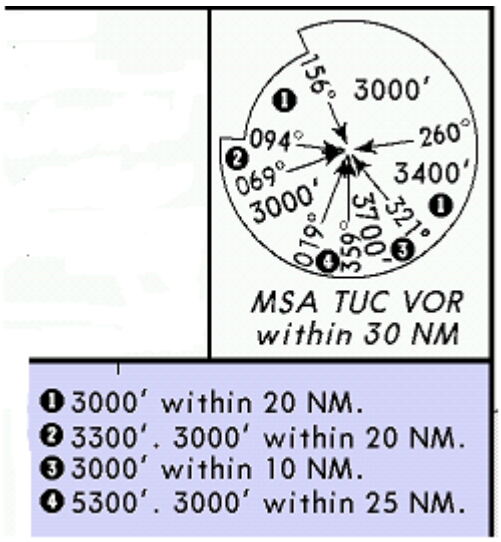
Sectorized MSA, Single Radius:



Output Data: 0902700222527009002825

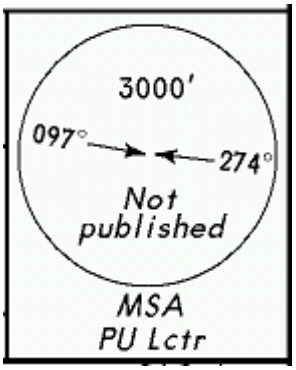
5.0 NAVIGATION DATA - FIELD DEFINITIONS

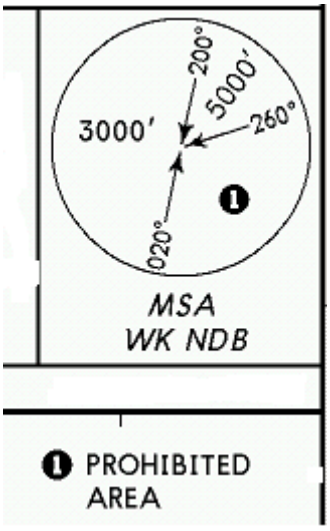
Sectorized MSA, Multiple Radius:



Output Data:
019069030300690940302006909403330094156030201562
600303026032103020260321034303213590301032135903
730359019030253590190533O

No Data Sectors:





Output Data:
097274030252740979992S
020200030252002600502526002099925

Figure 5-14 – MSA Data Examples

Note 1: The MSA example requires a total of 11 data sets of Bearing/Altitude/Radius to provide all of the information. The current MSA Primary Record only allows for 7 sets. The additional data sets would be provided in a continuation record that is formatted exactly the same as the Primary.

5.148 Enroute Alternate Airport/**Heliport** (EAA)

Definition/Description: The "Enroute Alternate Airport/Heliport" field identifies the most suitable emergency airport or heliport along a Company Route or Helicopter Operations Company Route.

Source/Content: This field is determined by the user airline and will contain the Airport or Heliport Ident.

Used On:	Company Route**, Helicopter Operations Company Route** records
Length:	4 characters
Character Type:	Alpha/numeric
Examples:	KDEN, EGKK, EDFF

5.149 Figure of Merit (MERIT)

Definition/Description: The "Figure of Merit" field is used to specify VHF Navaid facility usable ranges beyond that specified in the Class field. It is also used to specify when a VHF Navaid contained in the database is not available for operational use, i.e., is out of service and is used to flag a VHF Navaid that is not included in a civilian international NOTAM system.

Source/Content: Actual Field Entry Values are not contained in official government source but rather are derived values based on usage, class, availability etc. These may be further adjusted by input from actual users. The content will be as defined in the table below.

Field Content	Description
0	Terminal Use (generally within 25NM)
1	Low Altitude Use (generally within 40NM)
2	High Altitude Use (generally within 130NM)
3	Extended High Altitude Use (generally beyond 130NM)
7	Navaid not included in a civil international NOTAM system
9	Navaid Out of Service

Used On:	VHF Navaid Records
Length:	1 character
Character Type:	Numeric

5.150 Frequency Protection Distance (FREQ PRD)

Definition/Description: The "Frequency Protection Distance" field provides an indication of the distance to the next nearest NAVAID on the same frequency.

Source/Content: The distance to the next NAVAID will be computer generated values. Values will be entered on NAVAID with DME or TACAN equipped facilities only and will indicate the distance, in nautical miles, to the next nearest DME or TACAN equipped facility. Maximum relevant value will be 600 nautical miles.

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Used On : VHF Navaid records
Length: 3 characters
Character Type: Alpha/numeric
Examples: 030, 150, 600

5.151 FIR/UIR Address (ADDRESS)

Definition/Description: The "FIR/UIR Address" field contains the four character communications address of the FIR/UIR to supplement the FIR/UIR Ident.

Source/Content: When addressing ATS messages to the ATS Center in charge of a FIR or UIR, a three-letter designator followed by a filler of "X" or by a letter representing a department or division within the organization addressed should be used. The three-letter designators are to be those defined in ICAO Document 8585, Designators for Aircraft Operating Agencies, Aeronautical Authorities 2 and Services. ICAO Document 7910, Location Indicators, Address of Centers in charge of FIR/UIR, states that when addressing ATS messages to the ATS Center in change of a FIR or a UIR, one of the following designators should be added to the location indicator to complete the addressee indicator:

If the message is related to an IFR Flight—ZQZX
If the message is related to a VFR Flight—ZFZX

To satisfy this requirement, unless otherwise stipulated by the user, the following address codes will be used:

ZOZX if related to an Oceanic FIR/UIR.
ZRZX if related to all other FIR/UIRs.

When used on Enroute Communications Records, the content definition above for the FIR/UIR Record is to be applied when ever the FIR/RDO (5.190) field of the Enroute Communications Record contains an Information Region Identifier. In all other cases, the Address field of the Enroute Communications Record will be blank.

Used On: FIR/UIR and Enroute Communications records
Length: 4 characters
Character Type: Alpha
Examples: ZOZX, ZRZX

5.152 Start/End Indicator (S/E IND)

Definition/Description: The "Start/End Indicator" field is used to indicate if the "Start/End Date" field is the effective start date, effective end date or the effective date for a change of the Primary record.

Source/Content:

Field Content	Description
C	Change Date
E	End Date
S	Start Date

Note: If the Start/End Indicator field contains the alpha character "C" then the following continuation record is paired with it and the record will contain only the new values in the changed fields. All other fields will be blank.

Used On: Restrictive Airspace, VHF NAVAID, NDB NAVAID, Enroute, Terminal Waypoint, Airport Flight Planning Continuation and Heliport records
Length: 1 character
Character Type: Alpha

5.153 Start/End Date

Definition/Description: **The "Start/End Date" field provides the Start and End dates when the restrictions are in effect.**

Source/Content: **Start/End Dates will be derived from official government sources indicating the date/time in GMT of the effective times of the restriction. Default of the Start date is the current GMT date and default of the End date is forever.**

Used On: Restrictive Airspace Records
Length: 11 characters
Character Type: Alpha/numeric
Examples: 12JAN840000, 28SEP841200

5.154 Restriction Identifier (REST IDENT)

Definition/Description: The "Restriction Identifier" is used to assign a unique identifier to a restriction record and to multiple restrictions records for a particular route or route segment.

Source/Content: Restriction Identifiers are assigned during the data file assembly. Initially the identifier will be assigned in sequence with the first restriction assigned the numeric value "001," the second "002," the third "003," etc. If a restriction record is removed, only that record is deleted and there will be no effect on the other identifiers for that airway; i.e., if record "002" is deleted, records "001" and "003" will retain their identifiers. If a new restriction is added, within a few cycles of the deletion of "002," it will use the next higher number even if there are gaps in the sequence of identifiers.

Used On: Airway Restriction and Airway Restriction Continuation records
Length: 3 characters
Character Type: Numeric
Examples: 001, 002, 003

5.155 Intentionally Left Blank

5.156 Intentionally Left Blank

5.157 Airway Restriction Start/End Date (START/END DATE)

Definition/Description: The "Airway Restriction Start Date" field is used to indicate the earliest GMT date at which the restriction takes effect. The "Airway Restriction End Date" is used to indicate the latest GMT date at which the restriction is still in effect. This date information may be supplemented by "Time of Operation" information contained in an Airway Restriction Record, Type "AE" or "TC". When no "AE" or "TC" record exists for the

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Restriction Identifier, the Start time is 0000 GMT and the end time is 2359 GMT of the dates indicated.

Source/Content: When entered, start dates and end dates will be in the format DDMMMYY. If the YY portion is equal to blanks, the restriction is valid every year. When the start date is equal to blanks, the restriction is valid with immediate affect. When the end date is equal to blanks, the restriction is valid until further notice.

Used On: Enroute Airway Restriction records
Length: 7 characters
Character Type: Alpha/numeric
Examples: 15JAN92, 15 JAN (blank)

5.158 Intentionally Left Blank

5.159 Intentionally Left Blank

5.160 Units of Altitude (UNIT IND)

Definition/Description: The "Units of Altitude" field is used to indicate the units of measurement for the values in the "Restriction Altitude" fields.

Source/Content: The actual values are derived from official government sources and expressed as one of the following codes.

Field Content	Description
F	Restriction Altitudes are expressed in hundreds of feet
K	Restriction Altitudes are expressed in metric Flight Levels
L	Restriction Altitudes are expressed in feet Flight Levels
M	Restriction Altitudes are expressed in tens of meters

Used On: Airway Restriction records and Airway Restriction Continuation Records
Length: 1 character
Character Type: Alpha

5.161 Restriction Altitude (RSTR ALT)

Definition/Description: The "Restriction Altitude" fields are used to specify the altitude profile for a specific restriction.

Source/Content: Altitudes will be derived from official government sources and entered in hundreds of feet, tens of meters, standard or metric Flight Levels. The units used are determined through the "Units of Altitude" field. Altitudes are expressed in ascending order. All altitude fields after a "blank" altitude will also be blank.

Used On: Airway Restriction, Airway Restriction Continuation records
Length: 3 characters
Character Type: Numeric
Examples: 310 (standard FL310 or metric FL3199m or 31000 feet or 3100 meters)
090 (standard FL90 or metric FL900m or 9000 feet or 900 meters)

5.162 Step Climb Indicator (STEP)

Definition/Description: The "Step Climb Indicator" field is used to indicate if step climb up or down is permitted.
Source/Content:

Field Content	Description
B	Step climb up or down is permitted
D	Only step climb down is permitted
N	No step climb is permitted
U	Only step climb up is permitted

Used On: Airway Restriction and Airway Restriction Continuation records
Length: 1 character
Character Type: Alpha

5.163 Restriction Notes

Definition/Description: The "Restriction Notes" field may contain any restriction not otherwise covered by the altitude or time restriction.

Source/Content: Restriction notes will be derived from official government sources.

Used On: Airway Restriction continuation records
Length: 104 characters
Character Type: Alpha/numeric
Examples: AVAILABLE FOR WESTBOUND DEPARTURES FROM GATWICK. EASTBOUND AND OVER-FLIGHTS BY ATC ONLY. REROUTING MUST BE EXPECTED MON-FRI 1800-2400 DUE TO MILITARY TRAFFIC.

5.164 EU Indicator (EU IND)

Definition/Description: The "EU Indicator" field is used to identify those Enroute Airway records that have an Airway Restriction record without identifying the restriction.

Source/Content: The field will contain the alpha character "Y" when a restriction for the segment is contained in the restriction file or a blank when no restriction record exists.

Used On: Enroute Airways records
Length: 1 character
Character Type: Alpha

5.165 Magnetic/True Indicator (M/T IND)

Definition/Description: The "Magnetic/True Indicator" field is used to indicate if the "Course From" and "Course To" fields of the Cruise Table record and the "Sector Bearing" fields of the MSA and TAA record are in magnetic or true degrees. It is also used in the Airport Record to indicate that all detail and procedures for that airport are included in the data base with a reference to either Magnetic North or True North. The field is blank in Airport Record when the data base contains a mix of magnetic and true information for the airport.

Source/Content: Cruise Table Courses, MSA, and TAA Sector Bearings will be derived from official government

source. The field will contain the alpha character "M" if the Course From/To or Sector Bearings are magnetic. It will contain the alpha character "T" if the courses/bearings are true. In Airport Records, the field will contain the alpha character "M" if all detail and procedure for the airport are reported in magnetic bearing, the alpha character "T" if all detail and procedure for the airport are reported in true bearing. The field will be "blank" if details and procedures are provided in both magnetic and true for the airport. In such cases the individual detail or procedure records will contain the "magnetic" or "true" information.

Used On: Airport, Heliport, Cruise Table and Airport and Heliport MSA Records, and Airport and Heliport TAA Record
Length: 1 character
Character Type: Alpha

5.166 Channel

Definition/Description: The "Channel" field specifies the channel of the Azimuth, Elevation and Data transmissions for the MLS identified in the "MLS Identifier" field of the record.

Source/Content: Channels are derived from official government sources and range from 500 to 699.

Used On: MLS records
Length: 3 characters
Character Type: Numeric

5.167 MLS Azimuth Bearing (MLS AZ BRG)
MLS Back Azimuth Bearing (MLS BAZ BRG)

Definition/Description: The "MLS Azimuth Bearing" and the "MLS Back Azimuth Bearing" fields define the inbound magnetic final approach **course assigned to the center of the Azimuth or Back Azimuth Coverage (see Section 5.172).**

Source/Content: **The fields are populated with the inbound magnetic course information derived from official government source documents, generally the inbound course for a given approach procedure to a given runway considered the primary use of the MLS facility. The values are provided in degrees and tenths of degrees with the decimal point suppressed. Should the source value be available only in degrees true, the last character of the field will contain a "T" in place of the tenths of a degree value.**

Used On: MLS and MLS Continuation records
Length: 4 characters
Character Type: Numeric
Examples: 0550, 0155, 015T

5.168 Azimuth Proportional Angle Right/Left (AZ PRO RIGHT/LEFT)

Back Azimuth Proportional Angle Right/Left (BAZ PRO RIGHT/LEFT)

Definition/Description: The **MLS Azimuth and Back** Azimuth Proportional Angle fields define the limits of proportional guidance of the azimuth transmitter signal on the right and left side of the **MLS Azimuth bearing (Section 5.167).** The BAZ is identical to the AZ and also provides guidance for Missed Approach **Procedures** and departures. See figure under Section 5.172.

Source/Content: Azimuth Proportional angles will be derived from official government publications and entered in whole degrees.

Used On: MLS and MLS Continuation records
Length: 3 characters
Character Type: Numeric
Examples: 040, 025, 015

5.169 Elevation Angle Span (EL ANGLE SPAN)

Definition/Description: The "Elevation Angle Span" field defines the scan of the elevation transmitter signal between the lower and upper limits.

Source/Content: Elevation angle span limits will be derived from official government publications and entered in degrees and tenths of degrees with the decimal point suppressed.

Used On: MLS records
Length: 3 characters
Character Type: Numeric
Examples: 300, 150

5.170 Decision Height (DH)

Definition/Description: The "Decision Height" fields are used to specify a specific height in the precision approach at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.

Source/Content: Decision Height information is obtained from official government publications and will be the ILS CAT I barometric related value. The field will contain a numeric value expressed in feet above touchdown, with a resolution of one foot. This value is also referred to as "Height Above Touchdown" (HAT), a figure which, when added to the touchdown zone, runway end or threshold elevation, will give the Decision Altitude in feet above mean sea level. These fields will be shown on the first record of the final approach only and may be blank if Minimum Descent Altitude fields are entered.

Used On: Airport and Heliport Approach Continuation Records.
Length: 4 characters
Character Type: Alpha/numeric
Examples: 0200

5.171 Minimum Descent Height (MDH)

Definition/Description: The "Minimum Descent Height" fields specify the lowest height, expressed in feet, to which descent is authorized on final approach or during circle-to-land maneuvering in execution of a standard instrument approach procedure where no electric glide slope is provided.

Source/Content: Minimum Descent Height information is obtained from official government publications. The field will actually contain a figure expressed in feet above the

airport elevation, with a resolution of one foot. This value is also referred to as "Height Above Airport" (HAA), a figure which, when added to the airport elevation, will give the Minimum Descent Altitude in feet above mean sea level. These fields will be shown on the first record of the final approach only and may be blank if Decision Height fields are entered.

Used On: Airport and Heliport Approach Continuation records
Length: 4 characters
Character Type: Alpha/numeric
Examples: 0314, 0514

5.172 Azimuth Coverage Sector Right/Left (AZ COV RIGHT/LEFT)

Back Azimuth Coverage Sector Right/Left (BAZ COV RIGHT/LEFT)

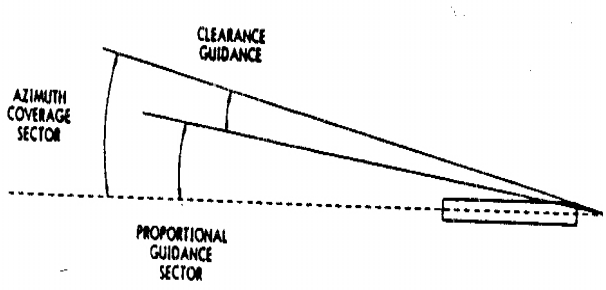
Definition/Description: The "Azimuth Coverage Sector" fields define the limit of the azimuth transmitter signal on the right and left side of the **MLS Bearing (Section 5.167).** The Back Azimuth Coverage Sector is identical to the Azimuth Coverage Sector and also provides guidance for Missed Approach **Procedures** and departures.

Source/Content: Azimuth Coverage Sectors will be derived from official government publications and entered in whole degrees.

Used On: MLS and MLS Continuation records
Length: 3 characters
Character Type: Numeric
Examples: 040, 062, 110

COMMENTARY

The Azimuth Coverage Sector includes the Proportional Guidance Sector and the Clearance Guidance Sector as illustrated in below.



Definition/Description: The "Pad Dimensions" field defines the landing surface dimensions of the helicopter landing pad. The pad may be described as a **runway, a** rectangle or a circle.

Source/Content: Pad dimensions will be derived from official government sources and entered into the field in feet with a resolution of one foot. **When the pad is defined as a runway, the first three digits define the length in hundreds of feet and the last three digits define the width, e.g. "125200" defines a runway 12,500 feet long by 200 feet wide. When defined as a rectangle, the first three digits define one side of the landing pad and the last three digits the other side of the pad, e.g "060120" defines a pad 60 feet by 120 feet. When the pad is circular, the first three digits define the diameter of the pad and the last three digits will be zeros, e.g. "080000" defines a pad that is 80 feet in diameter.**

Used On: Helipad record
Length: 6 characters
Character Type: Numeric
Examples: 125200, 060120, 080000

5.173 Nominal Elevation Angle (NOM ELEV ANGLE)

Definition/Description: The "Nominal Elevation Angle" field defines the normal glide path angle for the MLS installation.

Source/Content: Glide Path angles from official government sources are entered into the field in tens of degrees, tenths of a degree, and hundredths of a degree with the decimal point suppressed.

Used On: MLS records
Length: 4 characters
Character Type: Numeric
Examples: 1000, 0275

5.174 Restrictive Airspace Link Continuation (LC)

Definition/Description: The "Restrictive Airspace Link Continuation" field is used to indicate cases where it is not possible to store all Enroute Airway to Restrictive Airspace Links in the Flight Planning Continuation Record defined in 4.6.3 (more than four area links required).

Source/Content: When an additional Continuation Record (as defined in Section 4.1.6.4) is required to provide further Enroute Airway to Restrictive Airspace Links, this field will contain the alpha character "Y" to indicate that status.

Used On: Enroute Airway Flight Planning Continuation records
Length: 1 character
Character Type: Alpha

5.175 Holding Speed (HOLD SPEED)

Definition/Description: The "Holding Speed" will be the maximum speed in a holding pattern.

Source/Content: The speed limit will be derived from official government sources. If the value is different from the limit given with ICAO rules, it will be shown in knots, else the field will be blank.

Used On: Holding Pattern record
Length: 3 characters
Character Type: Numeric
Examples: 250, 015

5.176 Pad Dimensions

Definition/Description: The "Pad Dimensions" field defines the landing surface dimensions of the helicopter landing pad. The pad may be described as a rectangle or as a circle.

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Source/Content: Pad dimensions will be derived from official government sources and entered into the field in feet with a resolution of one foot. When the pad is rectangular, the first three digits define one side of the landing pad and the last three digits the other side of the pad, e.g. "060120" indicates the pad is 60 feet by 120 feet. When the pad is circular, the first three digits define the diameter of the pad and the last three digits will be zeros, e.g., "080000" indicates a pad that is 80 feet in diameter.

Used On: Heliport record
Length: 6 characters
Character Type: Numeric
Examples: 060060, 100050, 040040

5.177 Public/Military Indicator (PUB/MIL)

Definition/Description: Airports can be classified into **four** categories, airports open to the general public, military airports, **joint use civil and military,** and airports closed to the public. This field permits these airports to be categorized by their use.

Source/Content: Airport data is obtained from official government sources and their use is defined in these civil and or military publications. Airports that are considered joint use, both civil and military, will be shown as a civil airport.

Field Content	Description
C	Airport/Heliport is open to the public (civil)
M	Airport/Heliport is military airport
P	Airport/Heliport is not open to the public (private)
J	**Airport is joint Civil and Military**

Used On: Airport and Heliport records
Length: 1 character
Character Type: Alpha

5.178 Time Zone

Definition/Description: The standard time zone system is based on the division of world into 24 zones, each of 15 degrees longitude. The "zero" time zone is entered at Greenwich meridian with longitudes 7 degrees, 30 minutes West and 7 degrees, 30 minutes east, and there is no difference in the standard time of this time zone and Greenwich Mean Time. Time zones are designated by letters of the alphabet **and numbers** by which the standard time of each zone differs from that at Greenwich.

Source/Content: Time zones will be derived from official Time Zone Charts of the World**, or individual time zones can be published based on country**. The first character of the field indicates the time zone observed by the airport. Time zones are indicated by a letter of the alphabet **and numbers** according to the following table:

Field Cont	Diff to Zulu time	Lat/long Boundaries	Field Cont	Diff to Zulu time	Lat/Long Boundaries
Z	0	W007 30/E007 30	N	+1	W007 30/W022 30
A	-1	E007 30/E022 30	O	+2	W022 30/W037 30
B	-2	E022 30/E037 30	P	+3	W037 30/W052 30
C	-3	E037 30/E052 30	Q	+4	W052 30/W067 30
D	-4	E052 30/E067 30	R	+5	W067 30/W082 30
E	-5	E067 30/E082 30	S	+6	W082 30/W097 30
F	-6	E082 30/E097 30	T	+7	W097 30/W112 30
G	-7	E097 30/E112 30	U	+8	W112 30/W127 30
H	-8	E112 30/E127 30	V	+9	W127 30/W142 30
I	-9	E127 30/E142 30	W	+10	W142 30/W157 30
K	-10	E142 30/E157 30	X	+11	W157 30/W172 30
L	-11	E157 30/E172 30	Y	+12	W172 30/180 00
M	-12	E172 30/180 00	**1**	**+13**	**Phoenix Islands, Tonga**
			2	**+14**	**Kirbati Line Island**

The second and third characters indicate, in minutes, that the time observed by the airport/heliport must be adjusted from the hour by the number of minutes indicated.

When the 1st character is a 1 or 2, then the 2nd and 3rd characters will always be blank.

Used On: Airport and Heliport records
Length: 3 characters
Character Type: Alpha/numeric
Examples: India falls in the "E" (-5) and "F" (-6) time zones, however, the time zone observed in all of India is "E30" (-5 hours and 30 minutes). For any country falling into the "M" or "Y" time zone and observing a time equal to the next greater time zone, the adjustment of 1 hour will be indicated by "60" in the second and third positions.

5.179 Daylight Time Indicator (DAY TIME)

Definition/Description: The "Daylight" Time Indicator" field is used to indicate if the airport observes Daylight or Summer time when such time changes are in effect for the country or state the airport resides in.

Source/Content: Countries and states that observe Daylight time will be obtained from official publications and the field will contain the alpha character "Y" if airport observes Daylight or Summer time. The field will contain the alpha character "N" if the airport does not observe Daylight time or if it is unknown.

Used On: Airport and Heliport records
Length: 1 character
Character Type: Alpha

5.180 Pad Identifier (PAD IDENT)

Definition/Description: The "PAD Identifier" field identifies the helipad described in the heliport records, helipad field, or that pad served by ILS/MLS described in the Airport and Heliport ILS/MLS records. There is a unique Heliport Record for each Helipad at a given location.

Source/Content: PAD Identifiers will be derived from official government publications when available. If not

available from source, unique identifiers will be assigned by the data supplier.

Used On:	Heliport, Airport and Heliport ILS, **Helicopter Operations Company Routes** and MLS Records
Length:	5 characters max
Character Type:	Alpha/numeric
Examples:	Source Supplied - PADA1, NWPAD, ALPHA, A1 Data Supplier - HELO1, HELO2, HELO3

5.181 H24 Indicator (H24)

Definition/Description: The "24H Indicator" field is used to indicate **whether a communications service frequency is available for use on a continual, i.e. 24 hours a day, seven days a week, basis or not.**

Source/Content: **Hours of operation for a communications service frequency are derived from official government publications. The field will contain the character "Y" if the frequency is continually available, the character "N" if the frequency is not continually available and other Times of Operation are provided or the character "U" are unknown.**

If the field is set to "Y," the Time Code (5.131) in the Primary Extension Continuation Record for the frequency will be set to "C" or "H." If the field is set to "N," the Time Code in the Primary Extension Continuation Record for the frequency will be set to "N" or "P". If the field is set to "U," the Time Code will also be set to "U."

Used On:	Enroute/Airport **and Heliport** Communications records
Length:	1 character
Character Type:	Alpha

5.182 Guard/Transmit (G/T)

This section is withdrawn. The status of transmits only, receives only or both for a given frequency is provided by transmit and receive frequency columns of the communications records.

5.183 Sectorization (SECTOR)

Definition/Description: The "Sectorization" field is used to define the airspace sector a communication frequency is applicable for when an airport defines sectors by bearing from the same point.

Source/Content: **Source/Content: Sectors are derived from official government publication. Each Sectorization will contain two bearings, expressed in whole degrees, of the sector being defined. The first three numeric characters define the beginning bearing "from" the station and the last three characters define the ending bearing "from" the station, moving in a clockwise direction from start to end. If the sector is a complete circle, this field will be set to 180180. The radius of the circle will be provided as the Communications Distance, Section 5.188.**

Sectors which are defined by cardinal directions may be translated to bearings using the table below.

If the sectors are not defined by bearings, then the sectorization will be shown in narrative form in an Airport Communications Continuation Record.

Sector bearing data relates to the lat/long location of the Sector Facility (5.185). If no Section Facility is provided in the communications record, the sector bearing data relates to the lat/long included in the same communications record.

Source Cardinal Direction	**ARINC Sectorization**
Source Used Four Compass Points	
North	**3164045**
East	**046135**
South	**136225**
West	**226315**

Source Used Eight Compass Points	
North	**341025**
North East	**026070**
East	**071115**
South East	**116160**
South	**161205**
South West	**206250**
West	**251295**
North West	**296340**

If the sectors are not defined by bearings or cardinal directions, the sectorization will be shown in narrative form in a Continuation Record.

Used On:	Airport Communication records
Length:	6 characters
Character Type:	Alpha/numeric
Examples:	010189, 190009

5.184 Communication Altitude (COMM ALTITUDE)

Definition/Description: **The "Communications Altitude 1 and Altitude 2" fields are used to provide information on use of communications services frequencies with reference to specific altitudes. If the communications record in which Communications Altitude data is provided includes Sectorization data (5.183), the altitude data is valid only for the specific Sector.**

Source/Content: **Communications Altitude information will be derived from official government source documentation. The fields are to be processed in conjunction with the Communications Altitude Description field. The field will contain altitude expressed in hundreds of feet. The Communications Altitude 1 field will contain a value when the Communications Altitude Description contains the character + (plus) or - (minus). The Communications Altitude 1 field may contain a value when the Communications Altitude Description is blank, indicating that communications service/frequency is used at a specific altitude only. The Communications Altitude 1 and Altitude 2 fields will contain values when the Communications Altitude Description contains the character "B."**

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Used On: Enroute, Airport, **and Heliport Primary** Communications Records
Length: **3** characters
Character Type: Alpha/numeric
Examples: **050 (5000 feet), 245 (24500 feet)**

5.185 Sector Facility (SEC FAC)

Definition/Description: The "Sector Facility" field is used to define the Navaid or Airport upon which the information in the "Sector" (5.183) field is based.

Source/Content: Sector related facility information will be derived from official government sources. The field will contain the official Navaid or Airport identifier.

Used On: Airport and Heliport Communications Records
Length: 4 characters
Character Type: Alpha/numeric
Examples: IOC, COS, DEN, KJFK

5.186 **Sectorization** Narrative

Definition/Description: **The "Sectorization Narrative" field is used to define sectors of operations for communications services on specific frequencies in a narrative format when that data cannot be formatted in the Sectorization (5.183) field. The field may also be used to qualify the Sectorization information. This is usually the "and" situation, meaning the communications service/frequency is to be used in the defined sector and in some other defined situation that cannot be formatted such as Sectorization An example is "309127" in the Sectorization field and "When Departing Runway 31L/R" in the Narrative field.**

Source/Content: **Sector Narrative data will be derived from official government source.**

Used On: **Airport and Heliport Sector Narrative Continuation Records**
Length: 60 characters
Character Type: Alpha/numeric
Examples: **North Complex, Departures to North, When Rwy 09/27 is Active**

5.187 Distance Description (DIST DESC)

Definition/Description: The "Distance Description" field will designate whether a Communications frequency is to be used from the facility out to a specified distance or from a specified distance and beyond in the Airport Communications Record. In the VHF Navaid Limitation Continuation Record, the field is used to define whether the limitation applies from the navaid out to a specified distance or from a specified distance and beyond.

Source/Content: The field will contain the character "-" when the communications frequency or navaid limitation is "out to a specified distance." When the field content is "+", then the communications frequency is used or the navaid limitation applies "beyond" a specified distance. When the field is blank, no restrictions/limitations apply.

Used On: Airport Communications Records, VHF Navaid Limitation Continuation Records
Length: 1 character
Character Type: Alpha

5.188 Communications Distance (COMM DIST)

Definition/Description: The "Communications Distance" field is used to define the distance restriction a communication frequency is to be used within or beyond when such restrictions apply. This field is used in conjunction with the Distance Description field.

Source/Content: Distances restrictions are derived from official government publications and will contain a value in nautical miles from the communications facility. If the Distance Description field contains the character "-", then the frequency is to be used from the facility to the distance specified. If the Distance Description field contains the character "+" then the frequency is to be used from the distance specified and beyond. The field will be blank if no restrictions apply.

Used On: Airport Communications records
Length: 2 characters
Character Type: Numeric
Examples: 05, 10, 15

5.189 **Position Narrative**

Definition/Description: **The "Position Narrative" field is a textual description of the location of a communications transmitter. This may be the name of a Remote Communications Outlet, a Remote Communications Air/Ground Station or the place name of the geographical location of the transmitter site.**

Source/Content: **Position Narrative information will be derived from official government source. The field may be blank in cases where the source information is not available.**

Used On: Enroute Communications records
Length: 25 characters
Character Type: Alpha/numeric
Examples: CHEYENNE, ABBEVILLE

5.190 FIR/RDO Identifier (FIR/RDO)

Definition/Description: **The "FIR/RDO Identifier" field used on Enroute Communications records is the source provided identifier for a communication service as used in message addressing. For Information Regions (FIR/UIR) it is the four-character identifier assigned to the Information Region as published in ICAO Document 7910, Location Indicators. For Flight Service Stations, it is the three or four character identifier assigned to the station by the relevant authority. For other communications services established for enroute use and not addressable under the Information Region, Flight Service Station concept, it is the identifier assigned by the relevant authority to that station for the purpose of addressing message traffic.**

Source/Content: **The field content will be derived from official government source documentation as indicated**

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above. Only three or four character identifiers are to be used.

Used: Enroute Communications records
Length: 4 characters
Character Type: Alpha/numeric
Examples: KZDN, DEN

5.191 Triad Stations (TRIAD STA)

Deleted by Supplement 14.

5.192 Group Repetition Interval (GRI)

Deleted by Supplement 14.

5.193 Additional Secondary Phase Factor (ASF)

Deleted by Supplement 14.

5.194 Initial/Terminus Airport/Fix

Definition/Description: The "Initial Fix" and the "Terminus Fix" fields are used to define the departure airport or initial fix and the destination airport or terminus fix of a preferred route.

Source/Content: For preferred and preferential routes these fields will normally contain an airport identifier. For North America Routes for North Atlantic Traffic - Common portion routes, these fields may contain NAVAID or waypoint identifiers. For North America routes for North Atlantic Traffic - Non-common portion routes, these fields may contain airport, NAVAID or waypoint identifiers. These fields will be entered on the first sequence of a route only, except when the route serves more than one airport, in which case the additional airports are shown on succeeding sequences.

Used On: Preferred Route record
Length: 5 characters
Character Type: Alpha/numeric

Examples: KDEN, CYUL, DEN, YUL, COLOR

Entries for Metro Area "New York to Atlanta"
Seq 010 KJFK K6 KATL K7
Seq 020 KLGA K6
Seq 030KEWR K6
Entries for Atlanta to Metro Area "New York"
Seq 010KATL K7 KJFK K6
Seq 020LGA K6
Seq 030KEWR K6

5.195 Time of Operation

Definition/Description: The "Time of Operation" field is used to indicate the times of operation of a Facility or Restriction.

Source/Content: The times of operation are derived from official government source. Each "Time of Operation" group contains the definition of a daily period of operations within a calendar week.

The first two positions identify days of the week, with Monday equal to 1 and Sunday equal to 7. A single day, for example, Monday, is depicted as "01". A consecutive series of days, for example Monday through Friday, is depicted as "15". Non-consecutive days require multiple Time of Operation entries. The remaining 8 characters define a starting time of four characters and an ending time of four characters. These times are in the format HHMM (H= hours, M= minutes) using a 24 hour time system. For example, 00012350 starts at one minute after midnight and ends at 10 minutes before midnight. 07152000 starts at 07:15 hours and ends at 20:00 hours.

Times of Operation can also be expressed in terms of Sunrise (SR) and Sunset (SS). When a "Time of Operation" is defined as starting at or ending at Sunrise, that time is specified as "000R". When a "Time of Operation" is defined as starting at or ending at Sunset, that time is specified as "000S". When a "Time of Operation" is defined as starting at or ending at a certain number of hours/minutes before or after Sunrise or Sunset, those times are specified as in the following examples:

030R for 30 minutes before Sunrise or R030 for 30 minutes after Sunrise.
100R for 1 hour before Sunrise or R100 for 1 hour after Sunrise.
030S for 30 minutes before Sunset or S030 for 30 minutes after Sunset.
100S for 1 hour before Sunset or S100 for 1 hour after Sunset

Of the three digits associated with "R" or "S," the first is an expression of hours, the second and third an expression of minutes. 1 hour, 30 minutes would be 130, 2 hours, 15 minutes would be 215, etc.
When multiple definitions are required to fully define the "Time of Operation" for a given calendar week, these are coded as second and subsequent "Time of Operation" fields.

Examples:

A restriction valid on Mondays, Wednesdays and Fridays only, 0700 to 1700, would require three "Time of Operation" entries, one for 01 (Monday), one for 03 (Wednesday), one for 05 (Friday) and would be expressed as 0107001700 0307001700 0507001700.

A continuous restriction, starting on Monday at 0700 and ending on Friday at 1700 would require three "Time of Operation" entries, one for Monday of 0107002359, one for Tuesday through Thursday of 2400002359 and one for Friday of 0500001700.

When the times to be defined go over midnight, the second four characters of time information are valid on the actual ending day. For example, a "Time of Operation" of Monday through Friday, 1700 to 0300 actually ends on Saturday and would be shown as 1617000300, not 1517000300.

Used On: Enroute Airway Restriction Primary and the following Continuation Records - Airport/Heliport/Enroute Communications, Restrictive Airspace, Preferred Route, Enroute Airway Restrictions and Controlled Airspace

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Length: 10 characters
Character Type: Alpha/numeric

5.196 Name Format Indicator (NAME IND)

Definition/Description: The "Name Format Indicator" field is used to describe the format of the "Waypoint Name/Description" field (5.43). This field will be formatted according to the rules described in Chapter 7 of this Specification, Waypoint Naming Conventions.

Source/Content: Values for this field have no official government source and are adjusted by input from the following table. Code may not be used in combination between columns.

Record	Column	Content	Description
96	**97**	**98**	
A			Abeam Fix
B			Bearing and Distance Fix
D			Airport Name as Fix
F			FIR Fix
H		Note 1	Phonetic Letter Name Fix
I			Airport Ident as Fix
L			Latitude/Longitude Fix
M			Multiple Word Name Fix
N			Navaid Ident as Fix
P			Published Five - Letter - Name - Fix
Q			Published Name Fix, less than five letters
R			Published Name Fix, more than five letters
T			Airport/Rwy Related Fix (Note 2)
U			UIR Fix
	O		Localizer Marker with officially published five - letter identifier
	M		Localizer Marker without officially published five - letter identifier

Note 1: Column 98 is reserved for future expansion of the Name-Format-Indicator concept.

Note 2: The "T" indicator will be used with all fixes established in accordance with Chapter 7, Section 7.2.6, Terminal Waypoints, in this document.

Used On: Enroute Waypoints, Airport and Heliport Terminal Waypoints
Length: 3 characters
Character Type: Alpha

5.197 Datum Code (DATUM)

Definition/Description: The "Datum Code" field defines the Local Horizontal Reference Datum to which a geographical position, expressed in latitude and longitude, is associated.

Source/Content: Local Horizontal Reference Datums will be derived from official government documentation. The "Datum Code" field will contain a three letter code corresponding to that government publication. A listing of valid three letter codes is contained in Attachment 2 to this Specification.

Used On: VHF Navaid, NDB Navaid, Terminal NDB, Enroute Waypoint, Airport, Fan Marker, Heliport and GLS Transmitter Records
Length: 3 characters
Character Type: Alpha
Examples: AGD, NAS, WGA

5.198 Modulation (MODULN)

Definition/Description: The "Modulation" field will design the type of modulation for the frequency in the "Communication Frequency" (5.103) field.

Source/Content: The field contains the following information:

The field will be set to "A" unless the source documentation specifies otherwise.

Field Content	Description
A	Amplitude Modulated Frequency
F	Frequency Modulated Frequency

Used On: Enroute, Airport and Heliport Communication Records
Length: 1 character
Character Type: Alpha

5.199 Signal Emission (SIG EM)

Definition/Description: High Frequency (HF) signals used in aeronautical communications can be the complete signal or a portion of the signal, called a sideband. The "Signal Emission" field will designate for each HF Frequency what emission is used.

Source/Content: **The field will be set to "3" unless the source documentation specifies otherwise. The** field content contains the following information:

Note: The field is blank on records with frequencies that are not HF, see Section 5.104.

Field Content	Description
3	Double Sideband (A3)
A	Single sideband, reduced carrier (A3A)
B	Two Independent sidebands (A3B)
H	Single sideband, full carrier (A3H)
J	Single sideband, suppressed carrier (A3J)
L	Lower (single) sideband, carrier unknown
U	Upper (single) sideband, carrier unknown

Used On: Enroute, Airport and Heliport Communications Records
Length: 1 character
Character Type: Alpha/numeric

5.200 Remote Facility (REM FAC)

Definition/Description: **The "Remote Facility" field is used to identify a Navaid or Airport that has been used to provide the latitude/longitude of a communications transmitter, Figure 5-8 and Notes 7 and 8 in Section 5.37 of this Specification.**

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Source/Content: The field will contain the official identifier of the navaid or airport used, as derived from official government sources.

Used On: Enroute, Airport and Heliport Communications Records.
Length: 4 characters
Character Type: Alpha/numeric

5.201 Restriction Record Type (REST TYPE)

Definition/Description: The "Restriction Record Type" field is used to define what type of a restriction is contained in the Enroute Airway Restriction Record in question.

Source/Content: The content of this field should be selected from the following listing of possible codes:

AE = Altitude Exclusion. The record contains altitudes, normally available, that are excluded from use for the Enroute Airway Segment. May be further restricted by "Time of Operation" information.

TC = Cruising Table Replacement. The record contains only a reference to a Cruising Table Identifier. That Cruise Table will be in force, replacing the Cruise Table Identifier in the Enroute Airway segment records defined in the "Start Fix/End Fix" fields.

SC = Seasonal Restriction. Record is used to close an Airway or portion of an Airway on a seasonal basis.

NR = Note Restrictions. The record contains restrictions that do not fit the pattern of "formatted" information allowed by other "Restriction Record Types."

Used On: Enroute Airway Restriction Records
Length: 2 characters
Character Type: Alpha

5.202 Exclusion Indicator (EXC IND)

Definition/Description: The "Exclusion Indicator" field is an indication of how the altitudes contained in the Cruising Table record referenced by the Airway segment(s) are restricted. This is an "all altitude" restriction, further defined by direction of flight. These codes will not be used when certain altitudes remain available in a direction of flight.

Source/Content: The content of the field will be one of the codes from the following listing:

"A" = All altitudes in both directions of flight are restricted. This effectively closes the airway in both direction of flight.

"B" = All altitudes in the opposite direction in which the Enroute Airway is coded are restricted. This effectively closes the airway in one direction of flight i.e., the opposite direction from that in which the airway is coded.

"F" = All altitudes in the direction in which the Enroute Airway is coded are restricted. This effectively closes the airway in one direction of flight i.e., the direction in which the airway is coded.

(blank) = The restriction is not an "all altitude" restriction.

Used On: Enroute Airway Restriction Records
Length: 1 character
Character Type: Alpha

5.203 Block Indicator (BLOCK IND)

Definition/Description: The "Block Indicator" field is used to specify that the altitudes that follow in the restriction record are either "block" of altitudes that are restricted (not available for flight) or are individual altitudes that are restricted.

Source/Content: The field will either be set to "B" indicating an altitude block or "I" indicating individual altitudes. One or the other or both codes will appear in restriction records that are not "Exclusive" restrictions (see Section 5.201).

Used On: Enroute Airway Restriction, Enroute Airway Restriction Continuation Records
Length: 1 character
Character Type: Alpha
Examples: (using multiple columns of the record)

030B090 = all altitudes from 3000 feet to 9000 feet (inclusive) are not available
030I090 = the individual altitudes of 3000 feet and 9000 feet are not available
030I070B130 = the individual altitude of 3000 feet and all altitudes from 7000 feet to 13000 feet (inclusive) are not available

5.204 ARC Radius (ARC RAD)

Definition/Description: The "ARC Radius" field is used to define the radius of a precision turn. In Terminal Procedures, this is the "Constant Radius To A Fix" Path and Termination, for "RF" Leg. In Holding Patterns, this is the turning radius, inbound to outbound leg, for RNP Holding. The ARC Radius field is also used to specify the turn radius of RNP holding patterns included in SID, STAR, and Approach Records as HA, HF, and HM legs.

Source/Content: The content of the field will be derived from official source publications. It will be expressed in nautical miles, tenths, hundredths and thousandths of a nautical mile, with the decimal point suppressed. A conversion to feet of the resolution in nautical miles is equal to an accuracy of 6 feet.

Used On: SID, STAR and Approach Records, Holding Pattern Records
Length: 6 characters
Character Type: Numeric
Examples: 246868, 460820, 691231

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5.205 Navaid Limitation Code (NLC)

Definition/Description: The "Navaid Limitation Codes" field is used to define the type of limitation to be expected with a VHF Navaid.

Source/Content: The type of limitation will be derived from official government publications and entered using one of the codes defined in the table.

Content	Limitation Description
C	Coverage, the limitations are expressed as maximum reception reliability
F	Fluctuations, radial(s) are affected by course fluctuations.
G	Roughness, signal roughness experienced in the sector(s) defined.
N	Unreliable in the sector(s), at the altitude(s), at the distance(s) defined.
R	Restricted in the sector(s), at the altitude(s), at the distance(s) defined.
T	Unusable in the sector(s), at the altitude(s), at the distance(s) defined.
U	Out of Tolerance in the sector(s), at the altitude(s), at the distance(s) defined.

Used On: VHF Navaid Limitation Continuation Records
Length: 1 character
Character Type: Alpha

5.206 Component Affected Indicator (COMP AFFTD IND)

Definition/Description: The VHF Navaid File contains navaids that have one or two components - azimuth and/or distance. Published limitations may apply to one or both of the components. The "Component Affected Indicator" defines which component(s) are affected by the limitation.

Source/Content: The field content will be entered as indicated in the table based on official government publications. When different limitations apply to different components or components pairs, this will result in multiple Component Affected Indicators for a single navaid to cover the complete limitation. In these cases, the Sequence Number (Section 5.12) will start again with one (01) with each new Component Affected Indicator.

Content	Component Description
A	TACAN or VORTAC, TACAN azimuth component only affected.
B	VORDME, or VORTAC, both azimuth and distance component affected.
D	VORDME or DME, distance component only affected.
M	VORTAC or TACAN, TACAN azimuth and distance component affected.
T	TACAN or VORTAC, distance component affected.
V	VOR, VORDME or VORDME, VOR azimuth component affected.
Z	VORDME, VORTAC or TACAN, VOR and TACAN azimuth and distance component affected.

Used On: VHF Navaid Limitation Continuation Records
Length: 1 character
Character Type: Alpha

5.207 Sector From/Sector To (SECTR)

Definition/Description: The "Sector From/Sector To" field defines sectorization applicable to the range limited sectors of VOR/DME, VORTAC or TACAN facilities, using the sector letters from the table. Each sector is described by two characters and is to be interpreted as "from" the first character, clockwise "to" the second character.

Source/Content: Field content is derived through interpretation of official government publication information which may be in a variety of formats.

Sector Character	From (degrees true)	To (degrees true)
A	000	015
B	015	030
C	030	045
D	045	060
E	060	075
F	075	090
G	090	105
H	105	120
I	120	135
J	135	150
K	150	165
L	165	180
M	180	195
N	195	210
O	210	225
P	225	240
Q	240	255
R	255	270
S	270	285
T	285	300
U	300	315
V	315	330
W	330	345
X	345	000

Used On: VHF Navaid Limitation Continuation Records
Length: 2 characters
Character Type: Alpha
Examples: AB, TA, LW

5.208 Distance Limitation (DIST LIMIT)

Definition/Description: The "Distance Limitation" field is used to define the distance(s) from the navaid at which the limitation applies.

Source/Content: "Distance Limitations" are derived from official government publications. The field will contain one or two distances expressed in nautical miles from the facility. Used together with the "Distance Description" field, the distances can be provided as indicated in the table of examples. The field will be blank if there are no distances associated with the limitation.

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Used On: VHF Navaid Limitation Continuation Records
Length: 6 characters
Character Type: Alpha/numeric
Examples:

Distance Description	Distance Limit – First Three Digits	Ditance Limit – Second Three Digits	Description of Content
–	040	000	Limitation valid out to 40NM from the facility.
+	040	000	Limitation valid beyond 40NM from the facility.
B	100	040	Limitation valid between 40NM and 100NM.
Blank	040	000	Limitation valid at 40NM from the facility.

5.209 Altitude Limitation (ALT LIMIT)

Definition/Description: The "Altitude Limitation" field is used to define the altitude(s) at which the limitation applies.

Source/Content: "Altitude Limitations" are derived from official government publications. The field will contain one to two altitudes, expressed in hundreds of feet MSL. Used together with the "Altitude Description" field, the altitudes can be provided as indicated in the table of examples. The field will be blank if there are no altitudes associated with the limitation.

Used On: VHF Navaid Limitation Continuation Records
Length: 6 characters
Character Type: Alpha/numeric
Examples:

Altitude Description	Altitude Limit - First Three Digits	Altitude Limit - Second Three Digits	Description of Content
-	040	000	Limitation valid at or below 4000'/FL040.
+	040	000	Limitation valid at or above 4000'/Fl040.
B	100	040	Limitation valid from 4000'/FL040 to 10000'/FL100.
blank	040	000	Limitation valid at 4000'/FL040.

5.210 Sequence End Indicator (SEQ END)

Definition/Description: The "Sequence End Indicator" field is used to define the end of a set of sequences defining a given limitation to a given VHF Navaid Component or Component pair.

Source/Content: "Limitations" are derived from official government publications. The field will contain the character "E" in that sequence which is the end of a given limitation.

Used On: VHF Navaid Limitation Continuation Records
Length: 1 character
Character Type: Alpha

5.211 Required Navigation Performance (RNP)

Definition/Description: Required Navigation Performance (RNP) is a statement of the Navigation Performance necessary for operation within a defined airspace in accordance with ICAO Annex 15 and/or State published rules.

Source/Content: RNP values derived from official government source will be used when available. They are entered into the field in nautical miles (two digits) with a zero or negative exponent (one digit). The contents can be:

When used on Enroute Airway segments, RNP shall apply inbound to the fix when viewed in increasing sequence number order. The RNP applies only to the airway leg on which it is specified. If no RNP values is coded on a segment, there is not a database specified RNP for that segment.

When used on a SID, STAR and Approach Procedure records, the RNP shall apply to the segment on which it is coded. RNP will be coded on every segment where it is specified by source. Lack of a RNP value on a segment indicates no source supplied RNP value was available for that segment.

When used on Holding Patterns, the RNP applies to the holding pattern as defined in the record.

Note 1: The RNP concept will also be applied to defined airspaces, in addition to the specific flight paths as defined above. ARINC 424-13 addresses an "airspace record" that includes a reservation for RNP until actual content can be defined.

Note 2: There are no provisions for "Vertical RNP" in ARINC 424 at this time.

Used On: Airport and Heliport SID/STAR/Approach, **Enroute Airways, Airport and Heliport SID/STAR/Approach Continuation,** Controlled Airspace and Holding Pattern Records
Length: 3 character (see content paragraph)
Character Type: Numeric
Examples: 990 (equal to 99.0NM), 120 (equal to 12.0NM), 013

5.212 Runway Gradient (RWY GRAD)

Definition/Description: The Runway Gradient field indicates an overall gradient in percent, measured from the start of take-off roll end of the runway designated in the record. The gradient is expressed as a positive or negative

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gradient; positive being an upward and negative being a downward gradient.

Source/Content: The values will be derived from official government source. The first position will be either a "+" or a "-" sign indicating upward or downward gradient. Positions 2 through 5 indicate the gradient with the decimal point suppressed. The Maximum Gradient that can be expressed in this field is (+9.000 or -9.000).

Used On: Runway Records
Length: 5 characters
Character Type: Alpha/numeric
Examples: +0450, -0300

5.213 Controlled Airspace Type (ARSP TYPE)

Definition/Description: The "Controlled Airspace Type" field is used to indicate the type of controlled airspace, using codes from the table below.

Source/Content: The airspace type should be derived from official government publications. The table below shows the indicators used for the various types. For the USA, the previous applied designations such as TCA are supplied for ease of reference, they are longer officially published.

Field Content	Description
A	Class C Airspace (Was ARSA within the USA).
C	Control Area, ICAO Designation (CTA).
M	Terminal Control Area, ICAO Designation (TMA or TCA).
R	Radar Zone or Radar Area (Was TRSA within the USA).
T	Class B Airspace (Was TCA with the USA).
Z	Class D Airspace within the USA, Control Zone, ICAO Designation (CTR).

Used On: Controlled Airspace Records
Length: 1 character
Character Type: Alpha

5.214 Controlled Airspace Center (ARSP CNTR)

Definition/Description: The "Controlled Airspace Center" field is used to define the navigation element upon which the controlled airspace being defined is predicated, but not necessarily centered. Where the Airspace is not defined then the "Region Identifier" should be used. In this case, the Controlled Airspace Center will contain the ICAO Identification code for the Controlled Airspace to which the data contained in the record relates.

Source/Content: The Controlled Airspace Center will be determined during the construction of the records. As an example the New York Class B Airspace (formerly TCA) is centered on the JFK VOR, the LGA VOR and the Newark airport. The Controlled Airspace Center field could contain the Kennedy Airport identifier "KJFK" as the key for all records describing the New York Class B Airspace. The field may contain a Navaid, Enroute Waypoint, **Heliport** or Airport Identifier. A "Region Identifier" content should be derived from official government source where the controlling authority is published or from ICAO Document 7910, Location Indicators. In cases where no official identifier is published that can be used as the "Airspace Center" where the controlled airspace is used for more than one airport, the Region Identifier can be used.

COMMENTARY

It should be noted that during construction of a Controlled Airspace Center, no published Navaid, Enroute Waypoint, Airport Identifier or Region Identifier may be found to be suitable. Data suppliers may create a "center waypoint" for use in the "Airspace Center" field in such cases.

Used On: Controlled Airspace records
Length: 5 characters
Character Type: Alpha/numeric
Examples: OTR, FISHS, KJFK, EGTT

5.215 Controlled Airspace Classification (ARSP CLASS)

Definition/Description: The "Controlled Airspace Classification" field will contain an alpha character indicating the published classification of the controlled airspace, when assigned.

Source/Content: Classification codes will be derived from official government sources. If source does not provide a classification, the field will be blank.

Used On: Controlled Airspace records
Length: 1 character
Character Type: Alpha
Examples: A through G

5.216 Controlled Airspace Name (ARSP NAME)

Definition/Description: The "Controlled Airspace Name" field will contain the name of the controlled airspace when assigned.

Source/Content: Names will be derived from official government sources. The name, if assigned, will be entered in the first record only. If source does not assign a name, the field may be blank.

Used On: Controlled Airspace records
Length: 30 characters
Character Type: Alpha/numeric
Examples: DENVER CLASS B, OAKLAND OCTA

5.217 Controlled Airspace Indicator (CTLD ARSP IND)

Definition/Description: The "Controlled Airspace Indicator" field is used to indicate if an airport is associated with controlled airspace of a terminal type such as a Terminal Control Area (TMA or TCA) Radar Area or Class B or C Airspace within the USA.

Source/Content: Airports lying within or below terminal controlled airspace will be determined through the use of official government publications describing the lateral limits of such airspace. The Controlled Airspace Airport/ICAO fields identify the airport for which terminal controlled airspace has been included in the Controlled Airspace Section of the file. The Controlled Airspace

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Indicator field will contain one of the codes from the table below. If an airport is not associated with any terminal controlled airspace of the types in this table, the Controlled Airspace Indicator field will be blank. The Controlled Airspace Airport/ICAO may be identical to or different than the record airport. Although Control Zones (CTR) are provided as Controlled Airspace, no reference to them is made in this manner in the Airport Flight Planning Continuation Record.

Field Content	Description
A	The Airport is within or below the lateral limits of Class C Airspace.
C	The Airport is within or below the lateral limits of a CTA.
M	The Airport is within or below the lateral limits of a TMA or TCA.
R	The Airport is within or below the lateral limits Radar Zone.
T	The Airport is within or below the lateral limits of Class B Airspace.

Used On: Airport Flight Planning Continuation Records
Length: 1 character
Character Type: Alpha

5.218 Geographical Reference Table Identifier (GEO REF TBL ID)

Definition/Description: The "Geographical Reference Table Identifier" will be used to provide a unique identification for each Geographical Entity. As the "Geographical Entity" field is a large field with no established content, this two character code will act as a pseudo key for the record.

Source/Content: The content of this field will be determined by the data supplier using the rules below.

Position One - The first letter or other significant letter of the Geographical Entity.
Position Two - A numeric of 0 thru 9 for each multiple of the character in position one.

Used On: Geographical Reference Table records
Length: 2 characters
Character Type: Alpha/numeric
Examples: Scandinavia S1
Southern United Kingdom S2
Baleric Islands B0

5.219 Geographical Entity (GEO ENT)

Definition/Description: The Geographical Reference Table will be used to identify "Geographical Entities" not definable by other established encoding systems. For established systems refer to Section 7 of this document.

Source/Content: The content of the field will be derived from official government source documentation for preferred route systems of any kind.

Used On: Geographical Reference Table Records
Length: 29 characters
Character Type: Alpha/numeric

5.220 Preferred Route Use Indicator (ET IND)

Definition/Description: The "Preferred Route Use Indicator" provides information on whether the route in question is point-to-point and therefore usable for navigation, or area-to-area and usable only as advisory information which requires further processing. The field will also provide information on whether or not RNAV equipment is required to use the route.

Source/Content: The content of this field will be determined by the data supplier at the time the route is established. The two character field will be used to denote both the definition of the route initial/terminus nature and the RNAV equipment requirement. In position one, the field will contain the alpha character "P" if the route is point-to-point or "A" if the route is area-to-area. In position two, the field will contain the alpha character "R" if RNAV equipment is required and the alpha character "N" if RNAV equipment is not required.

Used On: Preferred Route and Geographical Reference Table Records
Length: 2 characters
Character Type: Alpha

5.221 Aircraft Use Group (ACFT USE GP)

Definition/Description: The "Aircraft Use Group" field provides information on what aircraft or groups of aircraft are permitted to use a certain route.

Source/Content: The raw information for this field will be derived from government sources and encoded according to the table below. The first column will contain the code valid for the routing. See Note One for the second column content.

Aircraft or Aircraft Group	Field Content	Field Content
All Aircraft	A	
All Aircraft, Cruise speed 250 kts or less	C	
Non-Jet and Turbo Prop	D	
Multi-Engine Props Only	E	
Jets and Turbo Props/Special, Cruise Speed 190 kts or greater	F	
Helicopter Only	H	
Jet Power	J	See
Turbo-Prop/Special, Cruise Speed 190 kts or greater	M	Note 1
Non-Jet, Non-Turbo Prop	N	
Non-Jet, Cruise speed 190 kts or greater	P	
Non-Jet, Cruise speed 189 kts or less	Q	
Aircraft as defined in a Notes Continuation Record	R	
Single Engine	S	
Twin Engine	T	

Note 1: When two routings have been defined between end fixes/areas for the sole purpose of separating aircraft groups of use, the first column will contain the code for the group that may use the routing and the second column will contain the code for the group that must use the alternative

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routing. If there is no alternative routing for aircraft group separation, the second column will be blank.

Used On: Preferred Route Records
Length: 2 characters
Character Type: Alpha
Examples: For a pair of routings established for aircraft group separation between Single Engine and Twin Engine, the Single Engine would carry the code of ST and the Twin Engine Route would be TS.

5.222 GNSS/FMS Indicator (GNSS/FMS IND)

Definition/Description: The "GNSS/FMS Indicator" field provides an indication of whether or not the responsible government agency has authorized the overlay of a conventional, ground based approach procedure with the use of a sensor capable of processing GNSS data or if the procedure may be flown with FMS as the primary navigation equipment. The field is also used to indicate when and RNAV procedure has been authorized for GNSS-based vertical navigation.

Source/Content: The Indicator will be selected from the table below.

Indicator Definition	Field Content
Procedure Not Authorized for GNSS or FMS Overlay **– Authorization not published.**	0
Procedure Authorized for GNSS Overlay, primary Navaids operating and monitored.**– Authorization not published.**	1
Procedure Authorized for GNSS Overlay, primary Navaids installed, not monitored **Authorization is published. Example: Procedure Title includes "(GPS)" or "(GNSS)"**	2
Procedure Authorized for GNSS Overlay, Procedure Title includes "GPS" or "GNSS"	3
Procedure Authorized for FMS Overlay	4
RNAV (GPS) or RNAV (GNSS) Procedure SBAS use authorized (Note **2**)	A **(Note 1)**
RNAV (GPS) or RNAV (GNSS) Procedure SBAS use not authorized *(Note **2**)*	B **(Note 1)**
RNAV (GPS) or RNAV (GNSS) Procedure, SBAS use not specified	C **(Note 3)**
Stand Alone GPS (GNSS) Procedure	P

Note 1: **WAAS and EGNOS are the only SBAS Systems approved at this time.**

Note 2: **The GNSS/FMS IND of "A" indicates that the RNAV (GPS) or RNAV (GNSS) or RNAV RNP procedure is authorized for GNSS-based vertical navigation. (LPV and LNAV/VNAV approaches). The GNSS/FMS IND of "B" indicates that the RNAV (GPS) or RNAV (GNSS) or RNAV RNP procedure is not authorized for GNSS-based vertical navigation. (LNAV only approaches).**

Note 3: **Procedure use of SBAS not published**.

Used On: Airport and Heliport Approach Procedure Records
Length: 1 character
Character Type: Alpha/numeric

5.223 Operation Type (OPS TYPE)

Definition/Description: The "Operation Type" field **indicates whether the operation is an approach procedure, an advanced operation or other operational to be defined later.**

COMMENTARY

Advanced operation can be straight-in approaches followed by a missed approach, a precision curved approach or departure procedures and roll-out and taxiing procedures .

Source/Content: This field should contain a value in the range of 0 to 15, defined as follows:

0 = straight-in approach procedure
1-15 = reserved for future definition

Used On: Path Point Record
Length: 2 characters
Character Type: Numeric

5.224 Route Indicator (RTE IND)

Definition/Description: The "Route Indicator" field is a single alpha character used to differentiate between multiple final approach segments to the same runway or helipad **contained in the Final Approach Coding.**

Source/Content: A code of "A" through "Z," (omitting "I" and "O").

Note: This single character is consistent with the "Multiple Approach Indicator" included as the fifth character of an Approach Procedure Identifier **as defined in Section 5.10 of this Specification.**

Used On: Path Point Record
Length: 1 character
Character Type: Alpha

5.225 Ellipsoidal Height

Definition/Description: The "Ellipsoidal Height" field is the height of a surveyed point in reference to the WGS-84 ellipsoid.

Source/Content: The Ellipsoidal Height will be an official publication value. It will be provided in meters with a resolution of one tenth. The decimal point is suppressed. When the published Height is below the Ellipsoid, the first position will carry a minus (-) sign to indicate this condition. Otherwise, this first position will be a plus (+) sign. When used on Path Point Records, the Ellipsoidal Height will be for the LTP Position defined in the Path Point Record. When used on Runway Records, the Ellipsoidal Height will be for the Landing Threshold defined in the Runway Record.

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Used On: Path Point Record, Runway Records
Length: 6 characters
Character Type: Alpha/numeric
Examples: +00356, +00051, +0**00**15 -00022, -01566

5.226 Glide Path Angle (GPA)

Definition/Description: The "Glide Path Angle" field is an angle, expressed in degrees, tenths and hundredths of degrees, measured at the Flight Path Control Point (FPCP) of those approach procedures that require the coding of a Path Point record. It establishes the intended descent gradient for the final approach flight path. For an illustration of the GPA and related points, see Figure 5-13.

Source/Content: The values will be derived from official government source.

Used On: Path Point Record
Length: 4 characters
Character Type: Numeric
Examples: 0275 (is equal to 2.75°), 1015 (is equal to 10.15°), 0300 (is equal to 3.00°)

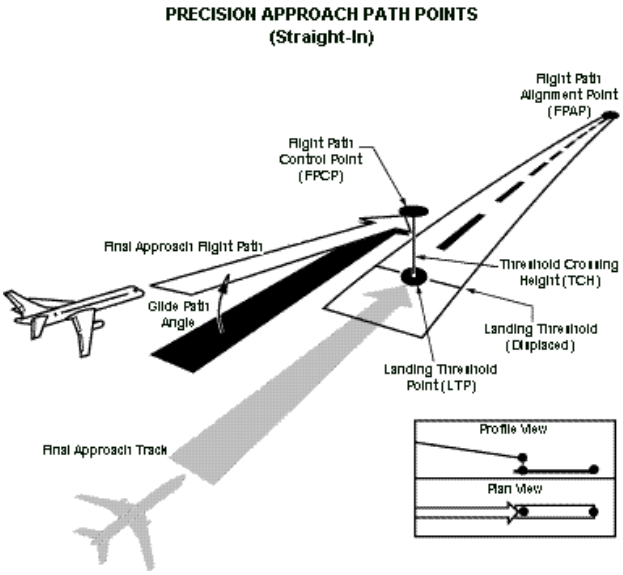

Figure 5-15 Precision Approach Path Points

5.227 Orthometric Height (ORTH HGT)

Definition/Description: The "Orthometric Height" field is the height of a surveyed point in reference to Mean Sea Level (MSL).

Source/Content: The Orthometric Height will be derived from official government source and entered with a resolution of a tenth a meter, with the decimal point suppressed. When the height is below MSL, the first position will carry a minus (-) sign, otherwise this position will be a plus (+) sign.

Used On: Path Point Record
Length: 6 characters
Character Type: Alpha/numeric
Examples: +00356, +00051, +01566, -00022, -01566

5.228 Course Width At Threshold (CRS WDTH)

Definition/Description: The "Course Width At Threshold" field defines the width of the lateral course at the Landing Threshold Point (LTP). This width, in conjunction with the location of the Flight Path Alignment Point (FPAP) defines the sensitivity of the lateral deviations throughout the approach.

Source/Content: The width will be derived from official government sources and entered in meters in the hundreds, tens, units, tenths and hundredths format with the decimal point suppressed. The value requires a data resolution of 0.25 meters and acceptable values will end in 00, 25, 50, and 75. When the procedure is to a helicopter alighting point, the value is 38 meters.

Used On: Path Point Records
Length: 5 characters
Character Type: Numeric
Examples: 08025, 14375, 03800

5.229 Final Approach Segment Data CRC Remainder (FAS CRC)

Definition/Description: The "Final Approach Segment Data CRC Remainder" field is an eight (8) character hexadecimal representation of the 32-bit CRC value provided by the source for the information contained in the aeronautical data fields being monitored for integrity. The value is calculated by a specific mathematical algorithm, which is both machine and man processible.

Source/Content: **For CRC calculation information refer to RTCA/DO-229 "Minimum Operational Performance Standards for Global Positioning System/Wide Area Augmentation System Airborne Equipment for Final Approach Segment (FAS) Data Block CRC standards.**

Used On: Path Point Record
Length: 8 characters
Character Type: Alpha/numeric
Examples: 243BC649, A6934B72

5.230 Procedure Type (PROC TYPE)

Definition/Description: The "Procedure Type" field used on Flight Planning Arrival/Departure Data Record is a single character code indication the type of procedure in the record, such as Arrival, Standard Instrument Arrival Route, Approach.

Source/Content: The Procedure Type code must be one of the following codes:

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Procedure Type Description	Procedure Type Code
Arrival Procedure, Available in Database	A
Arrival Procedure, Not Available in Database	B
Departure Procedure, Available in Database	C
Departure Procedure, Not Available in Database	D
Standard Terminal Arrival Route (STAR), Available in Database	E
Standard Terminal Arrival Route (STAR), Not Available in Database	F
Standard Instrument Departure (SID), Available in Database	G
Standard Instrument Departure (SID), Not Available in Database	H
Vector SID, Available in Database	I
Vector SID, Not Available in Database	J
Approach Procedure, Available in Database	K
Approach Procedure, Not Available in Database	L

Used On: Flight Planning Arrival/Departure Data Records
Length: 1 character
Character Type: Alpha

5.231 Along Track Distance (ATD)

Definition/Description: The "Along Track Distance" field used on Flight Planning Arrival/Departure Data Records is the total distance for a given transition, from the initial fix to the ending fix in the transition. A single occurrence of a Flight Planning Arrival/Departure Data record can contain up to three Along Track Distance fields, one for each of the transition types that can make up a single terminal route in the Primary Record and up to four possible intermediate fix points in each Continuation Record. Collectively, the values equal the along track distance from the first fix in the first transition to the last fix in the last transition.

Source/Content: The along track distances will be calculated by data suppliers using coded terminal procedures or uncoded terminal procedures derived from official government source and expressed in nautical miles with a 1NM resolution.

Used On: Flight Planning Arrival /Departure Data Records
Length: 3 characters
Character Type: Numeric

5.232 Number of Engines Restriction (NOE)

Definition/Description: The "Number of Engines Restriction" field used on Flight Planning Arrival/Departure Data Records is derived from government source and is included whenever a given procedure, normally departure, is restricted to, or designed for, aircraft with a specific number of engines.

Source/Content: The number of engines will be taken from official government source. The field will contain the character Y for each engine configuration position, 1, 2, 3 and 4, for which the procedure is authorized. Non-authorized configuration positions will contain the character N.

Used On: Flight Planning Arrival /Departure Data Records
Length: 4 characters
Character Type: Alpha
Examples: YYYY (1, 2 3 or 4 Engine aircraft may use procedure)
NNYY (3 and 4 Engine aircraft may use procedure)

5.233 Turboprop/Jet Indicator (TURBO)

Definition/Description: The "Turboprop/Jet Indicator" field used on Flight Planning Arrival/Departure Data Records is derived from government source and is included whenever a given procedure, normally departure, is restricted to, or designed for, aircraft with a specific kind of engines.

Source/Content: The indication of Turboprop, Jet or Both on the use restriction of given procedure will be taken from official government source. The field will indicate the use restriction with a character from the table below.

Aircraft or Aircraft Group	Field Content
All Aircraft	A
Jets and Turbo Props	B
All Aircraft, Cruise speed 250 kts or less	C
Non-jet and Turbo Prop	D
Multi-Engine Props Only	E
Jets	J
Non-Jet, Non-Turbo Prop	N
Turbo Props	P

Used On: Flight Planning Arrival /Departure Data Records
Length: 1 character
Character Type: Alpha

5.234 RNAV Flag (RNAV)

Definition/Description: The "RNAV Flag" field used on Flight Planning Arrival/Departure Data Records is derived from government source and is included whenever a given procedure included in the record is restricted to, or designed for, aircraft capable of flying RNAV Procedures.

Source/Content: The indication of RNAV, Yes or No, on a given procedure will be taken from official government source. The field will indicate Y for "Yes," the procedure is an RNAV procedure or N for "No," the procedure is not RNAV.

Used On: Flight Planning Arrival /Departure Data Records
Length: 1 character
Character Type: Alpha

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.235 ATC Weight Category (ATC WC)

Definition/Description: The "ATC Weight Category" field used on Flight Planning Arrival/Departure Data Records is derived from government source and is included whenever a given procedure included in the record is restricted to, or designed for, a specific aircraft weight grouping.

Source/Content: The indication of Heavy, Medium or Light aircraft on a given procedure will be taken from official government source. The field will be derived from that source to indicate:

H for Heavy, all aircraft types of 136,000kg (300000LB) or more.

M for Medium, aircraft types less than 136,000kg (300,000LB) and more than 7,000kg (155,000LB).

L for Light, aircraft types of 7,000kg (155,000LB) or less.

Used On: Flight Planning Arrival /Departure Data Records
Length: 1 character
Character Type: Alpha

5.236 ATC Identifier (ATC ID)

Definition/Description: The "ATC Identifier" field used on Flight Planning Arrival/Departure Data Records is the indication of the officially published procedure designation which is required for Flight Planning.

Source/Content: The ATC Identifier will be derived from official government source. This seven character field is required in addition to the six character identifier, the former is used in Flight Planning, the latter in accessing the database.

Used On: Flight Planning Arrival /Departure Data Records
Length: 7 characters
Character Type: Alpha/numeric

5.237 Procedure Description (PROC DESC)

Definition/Description: The "Procedure Description" field used on Flight Planning Arrival/Departure Data Records is the textual representation of the procedure name.

Source/Content: The Procedure Description will be derived from official government source. It will assist in matching flight plan content to charted procedures.

Used On: Flight Planning Arrival /Departure Data Records
Length: 15 characters
Character Type: Alpha/numeric

5.238 Leg Type Code (LTC)

Definition/Description: The "Leg Type Code" field used on Flight Planning Arrival/Departure Data Records is a simplification of the Path Terminator concept. It will provide the information on the path between intermediate waypoints as straight or curved and provide an indication of the change in direction of flight, expressed as left or right, at an intermediate waypoint.

Source/Content: The Leg Type Code will be derived from official government source. In this two character field, the first position will indicate with the character S, straight line point to point and with the character C, curved line flight track. The second position will be used as a turn indication, L for Left and R for Right when there is a turn requirement at an intermediate waypoint.

Used On: Flight Planning Arrival /Departure Data Records
Length: 2 characters
Character Type: Alpha

5.239 Reporting Code (RPT)

Definition/Description: The "Reporting Code" field used on Flight Planning Arrival/Departure Data Records is a simplification of the Waypoint Description concept. It will provide the information on intermediate waypoints as either Position Report Required (Compulsory Report) or Position Report Not Required (On-Request Report).

Source/Content: The Reporting Code will be derived from official government source. In this single character field, the code "C" will indicate Position Report Required and the code "X" Position Report Not Required.

Used On: Flight Planning Arrival /Departure Data Records
Length: 1 character
Character Type: Alpha

5.240 Altitude (ALT)

Definition/Description: The "Altitude" field used on Flight Planning Arrival/Departure Data Records is a simplification of the altitude concept used in the full procedure records. It will provide an altitude indication in hundreds of feet, no AGL, MSL, FL etc indication provided.

Source/Content: The Altitude will be derived from official government source and reduced to this flight planning resolution requirement.

Used On: Flight Planning Arrival /Departure Data Records
Length: 3 characters
Character Type: Numeric
Examples: FL100 = 100
10000 feet = 100
03500 feet = 035

5.241 Fix Related Transition Code (FRT Code)

Definition/Description: The "Fix Related Transition Code" is used on Flight Planning Arrival/Departure Data Continuation Records containing Intermediate Fix information and provides an indication, through use of the standard coding practices of separating the procedure into transitions, as to where in the procedure the intermediate fix is located.

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Source/Content: The field will contain a code an meaning as indicated in the table below.

Intermediate Fix is Located in Transition Type	Field Content
Fix Located in SID Runway Transition	1
Fix Located in SID Common Portion	2
Fix Located in SID Enroute Transition	3
Fix Located in STAR Enroute Transition	4
Fix Located in STAR Common Portion	5
Fix Located in STAR Runway Transition	6

Used On: Flight Planning Arrival/Departure Data Records
Length: 1 character
Character Type: Numeric

5.242 Procedure Category (PROC CAT)

Definition/Description: The Airport and Heliport SID/STAR/Approach Procedure Route Type supports the "All Sensor RNAV" Approach procedure. This kind of approach will have multiple sets of weather minimums (DH and NDA) associated with it. This field identifies the Procedure Categories for which these minimums apply.
Source Content: The field will contain a coded category from the following table:

Content	Procedure Category
LAAS	Local Area Differential Augmentation System
WAAS	Wide Area Differential Augmentation System
FMS	Flight Management System
GPS	Global Positioning System, no Augmentation
VDME	VORDME, VORTAC
CIRC	Circle-To-Land

Used On: Airport and Heliport SID/STAR/Approach Procedure Continuation Records
Length: 4 characters
Character Type: Alpha

5.243 GLS Station Identifier

Definition/Description: The "GLS Identifier" field defines the identification code for retrieval of such a transmitter from a database. This is not a transmitted identifier.

Source/Content: The content of this field will be the Airport or Heliport ICAO Location Identifier Code at which the transmitter is installed.

Used On: GLS Records
Length: 4 characters max
Character Type: Alpha/numeric

5.244 GLS Channel

Definition/Description: The "GNSS Channel Number" field identifies the channel that will be decoded to identify the augmentation system used.

Source/Content: The Channel Number is derived from official government sources. It consists of 5 numeric characters in the range 20001 to 99999 (numbers less then 20000 are reserved for ILS and MLS). Channel Numbers between 20001 and 39999 are reserved for GBAS (and SBAS if applicable) and 40000 to 99999 are reserved for SBAS.

Used On: GLS and Path Point Continuation Records
Length: 5 character
Character Type: Numeric
Examples: 20010, 56234

5.245 Service Volume Radius

Definition/Description: The service volume radius identifies the radius of the service volume around the transmitter in Nautical miles.

Source/Content: The value for this field will be derived from official government sources. If no source is provided, the default value will be blanked.

Used On: GLS Record
Length: 2 characters
Character Type Numeric
Examples: 05, 19

5.246 TDMA Slots

Definition/Description: The TDMA identifies the time slot(s) in which the ground station transmits the related approach. The high precision time source available through GPS permits utilization of Time division multiplexing or TDMA (Time Division Multiple Access), allowing multiple ground stations to share a common frequency by dividing it into eight time slots. An individual station may broadcast in one or more of eight slots.

Source/Content: The value for this field will be derived from official government sources. The range is 01 to FF. If no source is provided, the default value will be blank.

Used On: GLS Record
Length: 2 characters
Character Type: Alpha/numeric
Examples A2, 01, FF

5.247 Station Type

Definition/Description: The station type identifies the type of the differential ground station. The first character will be L for LAAS/GLS ground station, C for SCAT-1 station. The second and third character will be blank for the moment. They will indicate the interoperability standard to which the station conforms.

Source/Content: The value for this field will be derived from official government sources. If LAAS/GLS or SCAT-1 is not specified in source, the default value will be blank.

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Used on: GLS Record
Length: 3 characters
Character Type: Alpha/numeric
Examples: L, C

5.248 Station Elevation WGS84

Description/Definition: This field identifies the WGS84 elevation of the GLS ground station described in the record.

Source/Content: The value for this field will be derived from official government sources or entered into this field in feet with respect to the WGS84 ellipsoid. When elevation is below the WGS 84 ellipsoid, the first column of the field contains a minus (-) sign.

Used On: GLS Record
Length: 5 characters
Character Type: Alpha/numeric
Examples: 00530, -0140

5.249 Longest Runway Surface Code (LRSC)

Definition/Description: The "Longest Runway Surface Code" field is used to define whether or not there is a hard surface runway at the airport, the length of which is indicated in the Longest Runway field.

Source/Content: The content will be selected from the table below.

Field Content	Description
H	Hard Surface, for example, asphalt or concrete
S	Soft Surface, for example, gravel, grass or soil
W	Water Runway
U	Undefined, surface material not provided in source

Used On: Airport Records
Length: 1 character
Character Type: Alpha

5.250 Alternate Record Type (ART)

Definition/Description: The "Alternate Record Type" field identifiers the record as being applicable to the departure airport (take-off alternate), destination airport (arrival alternate) or a fix along the route (enroute alternate).

Source/Content: The "Alternate Record Type" will be selected from the following table:

Content	Description
AA	The Airport identifier in Columns 7 through 11 of the Primary Record are the identifier of the Arrival Airport.
DA	The Airport identifier in Columns 7 through 11 of the Primary Record are the identifier of the Departure Airport.
EA	The end fix of a Company Route is identified in Columns 7 through 15 of the Primary Record.

Used On: Alternate Records
Length: 2 characters
Character Type: Alpha

5.251 Distance To Alternate (DTA)

Definition/Description: The "Distance To Alternate" field defines either the direct (geodesic) distance from the Destination Airport or Fix to the Alternate Airport or the along track distance of an alternate Company Route.

Source/Content: When the Alternate Type field carries the character "A," the Distance to Alternate field carries the straight line (geodesic) distance in nautical miles between the Destination Airport or Fix and the Alternate Airport as listed in Alternate Identifier fields. When the Alternate Type field carries the character "C," the Distance to Alternate field carries the cumulative along track distance for the Alternate Company Route as listed in the Alternate Identifier fields.

Used On: Alternate Records
Length: 3 characters max.
Character Type: Numeric

5.252 Alternate Type (ALT TYPE)

Definition/Description: The "Alternate Type" field is an information processing indicator. The Alternate Destination can be defined as an airport or an airport and route to an airport. This field defines that an alternate airport or, an company route is defined in the Alternate Identifier fields.

Source/Content: The field will contain either the character "A" when an Airport is provided or the character "C" when a Company Route is provided.

Used On: Alternate Records
Length: 1 character
Character Type: Alpha

5.253 Primary and Additional Alternate Identifier (ALT IDENT)

Definition/Description: The Primary Alternate Identifier and the Additional Alternate Identifiers (two through five) uniquely identify either an Alternate Airport or an Alternate Company Route. The determination of whether the content is an Airport Identifier or a Company Route Identifier is accomplished through the Alternate Type field.

Source/Content: The content of this field is determined by the customer.

Used On: Alternate Records
Length: 10 characters max
Character Type: Alpha/numeric

5.254 Fixed Radius Transition Indicator (FIXED RAD IND)

Definition/Description: Indicates that a specific turn radius from the inbound course to the outbound course is required by the airspace controlling agency.

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Source/Content: When a fix radius turn is required a 3 digit numeric value will be entered in this field representing the radius of the turn to 1 decimal place (tenths, decimal point suppressed) in nautical miles. A blank entry in this field indicates that no fixed radius transition is required.

Used On: Enroute Airway Records
Length: 3 characters
Character Type: Numeric
Examples: 225=22.5nm, 150=15.0 nm

5.255 SBAS Service Provider Identifier (SBAS ID)

Definition/Description: The "SBAS Service Provider Identifier" field is used to associate the approach procedure to a particular satellite based approach system service provider.

Source/Content: A number from 00 to 15, the exact definition of which is currently under development by ICAO SBAS SARPS working groups.

Used On: Path Point Records
Length: 2 characters
Character Type: Numeric

5.256 Reference Path Data Selector (REF PDS)

Definition/Description: The "Reference Path Data Selector" field enables the automatic tuning of a procedure by Ground Based Augmentation Systems (GBAS) avionics.

Source/Content: A number from 00 to 48, the exact definition of which is currently under development by ICAO GBAS SARPS working groups.

Used On: Path Point Records
Length: 2 characters
Character Type: Numeric

5.257 Reference Path Identifier (REF ID)

Definition/Description: The "Reference Path Identifier" field is used to provide an identifier that can be used to confirm selection of the correct approach procedure.

Source/Content: The content will be derived from official government sources and analogous to the Morse code identifier on existing ILS approach Procedures.

See Section 5.243, GLS Station Identifier.

Used On: Path Point Records
Length: 4 characters
Character Type: Alpha/numeric
Examples: GDCA, SJK2

5.258 Approach Performance Designator (APD)

Definition/Description: The "Approach Performance Designator" field is used to indicate the type or category of approach.

Source/Content: A number between 0 and 7. The content will be derived from official government source documents. The ICAO GBAS SARPS working groups will provide the complete assignment.

Used On: Path Point Records
Length: 1 character
Character Type: **Numeric**
Example: 1 (Category I Approach)

5.259 Length Offset (OFFSET)

Definition/Description: The "Length Offset" field is the distance from the stop end of the runway (SER) to the FPAP. This distance defines the location where lateral sensitivity changes to the missed approach sensitivity. If the FPAP is located at the designated center of the opposite runway end, the distance is zero.

Source/Content: A value, expressed in meters, derived from official government sources (Explanation and details will appear in appropriate FAA and ICAO documents). The actual resolution is 8 meters.

Used On: Path Point Records
Length: 4 characters
Character Type: Numeric
Examples: 0000, 0432

5.260 Terminal Procedure Flight Planning Leg Distance (LEG DIST)

Definition/Description: The Terminal Procedure Flight Planning Leg distance is the along track distance required to complete any given leg. It is used to determine a cumulative track distance for a given terminal procedure for flight planning purposes, from the beginning of the take-off or arrival point to the termination point of the procedure.

Source/Content: The values will be determined during route definition of the procedure records. The content is controlled through requirements of the Path and Termination and coding rules in force with the data supplier. The values are expressed in nautical miles and tenths of nautical miles, with the decimal point suppressed.

Used On: Airport and Heliport SID, STAR and Approach Procedure Flight Planning Continuation Records
Length: 4 characters
Character Type: Numeric
Examples: 0176, 0822, 0208 0016, 0100

5.261 Speed Limit Description (SLD)

Definition/Description: The "Speed Limit Description" field will designate whether the speed limit coded at a fix in a terminal procedure description is a mandatory, minimum or maximum speed.

Source/Content: The content will be as defined in the table below.

5.0 NAVIGATION DATA - FIELD DEFINITIONS

Field Content Value	Description
@ (blank)	Mandatory Speed, Cross Fix AT speed specified in Speed Limit
+ (plus)	Minimum Speed, Cross Fix AT or ABOVE speed specified in Speed Limit
- (minus)	Maximum Speed, Cross Fix AT or BELOW speed specified in Speed Limit

Used On: Airport/Heliport SID/STAR/Approach Records
Length: 1 character
Character Type: Alpha

5.262 Approach Type Identifier (ATI)

Definition/Description: Identifies the approach types published on a given approach procedure which require path points.

Source/Content: Up to 10 characters representing the literal name of an approach with vertical guidance requiring path points, Horizontal Alert Limit (HAL) and Vertical Alert Limit (VAL). The name is derived from government source material.

Used On: Path Point Continuation Records
Length: 10 Characters
Character Type: Alpha/numeric
Examples: GLS, LPV, APV-II

5.263 HAL

Definition/Description: The Horizontal Alert Limit (HAL) is the radius of a circle in the horizontal plane (the local plane tangent to the WGS-84 ellipsoid), with its center being at the true position, which describes the region which is required to contain the indicated horizontal position with the required probability for a particular navigation mode assuming the probability of a GPS satellite integrity failure being included in the position solution is less than or equal to 10^{-4} per hour.

Source/Content: A value, expressed in meters to a resolution of tenths of meters with the decimal point suppressed, derived from official government sources.

Used On: Path Point Records
Length: 3 Characters
Character Type: Numeric
Examples: 400, 200

5.264 VAL

Definition/Description: The Vertical Alert Limit (VAL) is half the length of a segment on the vertical axis (perpendicular to the horizontal plane of WGS-84 ellipsoid), with its center being at the true position, which describes the region which is required to contain the indicated vertical position with a probability of $1\text{-}10^{-7}$ per approach, assuming the probability of a GPS satellite integrity failure being included in the position solution is less than or equal to 10^{-4} per hour.

Source/Content: A value, expressed in meters to a resolution of tenths of meters with the decimal point suppressed, derived from official government sources.

Used On: Path Point Records
Length: 3 Characters
Character Type: Numeric
Examples: 120, 500

5.265 Path Point TCH

Definition/Description: The Path Point TCH is the height above the runway threshold (LTP) or the helicopter alighting point. It is the same as the TCH defined in Section 5.67, but has greater resolution due to the required precision.

Source/Content: The value is derived from official government sources. The value may be expressed in feet to a resolution of tenths of feet, decimal point suppressed or expressed in meters to a resolution of hundredths of meters, decimal point suppressed. Whether or the value is in feet or meters can be determined from the TCH Units Indicator.

Used On: Path Point Records
Length: 6 characters
Character Type: Numeric
Examples: 566777, 356799

5.266 TCH Units Indicator

Definition/Description: The "TCH Units Indicator" field is used in Path Point Records to define the units, Feet or Meters for the Path Point TCH. This element is included in the "CRC Wrap" a possible metric source value cannot be converted to feet.

Source/Content: The field will contain the character "F" if the Path Point TCH is provided in source documentation in feet or the character "M" if that value is provided in meters.

Used On: Path Point Records
Length: 1 character
Character Type: Alpha

5.267 High Precision Latitude (HPLAT)

Definition/Description: The "High Precision Latitude" field contains the latitude of the navigation feature identified in the record.

Source/Content: The content of field is an expansion of the latitude defined in Section 5.36 to include degrees, minutes, tenths, hundredths, thousandths and tenths of thousandths of seconds to accommodate the high precision resolution of 0.0005 arc seconds.

Used On: Path Point Records
Length: 11 characters
Character Type: Alpha/numeric
Example: N3028422400

5.0 NAVIGATION DATA - FIELD DEFINITIONS

5.268 High Precision Longitude (HPLONG)

Definition/Description: The "High Precision Longitude" field contains the latitude of the navigation feature identified in the record.

Source/Content: The content of field is an expansion of the latitude defined in Section 5.36 to include degrees, minutes, tenths, hundredths, thousandths and tenths of thousandths of seconds to accommodate the high precision resolution of 0.0005 arc seconds.

Used On: Path Point Records
Length: 12 characters
Character Type: Alpha/numeric
Example: W08142030100

5.269 Helicopter Procedure Course (HPC)

Definition/Description: The "Helicopter Procedure Course" field is used on Path Point Continuation Records to define the final approach course of procedures designed to helipads and points in space.

Source/Content: The field will contain the full degree final approach course of a procedure designed to a helipad or point in space and be derived from official government source. It will be used in conjunction with the Approach Procedure Identifier and Runway/Helipad Identifier data in the Path Point Primary record to uniquely identify an approach procedure.

Used On: Path Point Continuation Records
Length: 3 Characters
Character Type: Numeric
Examples: 003, 013, 103, 310, 333

5.270 TCH Value Indicator (TCHVI)

Definition/Description: The "TCH Value Indicator" field will define which TCH value is provided in the runway record.

Source/Content: The field will contain a value from the following table:

Field Content	Description
I	TCH provided in Runway Record is that of the ILS or MLS Glide Slope
R	TCH provided in Runway Record is that of an RNAV procedure to the runway
V	TCH provided in the Runway Record is that of the VGSI for the runway
D	TCH provided in the Runway Record is the default value of **40 or** 50 feet **(See Section 5.67).**

Used On: Runway Records
Length: 1 character
Character Type: Alpha

5.271 Procedure Turn (PROC TURN)

Definition/Description: The "TAA Procedure Turn" field is used to indicate whether or a course reversal is necessary when flying within a particular TAA Area.

Source/Content: Official government source will carry an indication when the course reversal is not necessary. Generally, that indication is "NOPT". Otherwise, the execution of a course reversal is expected. When the course reversal is not necessary, this field will carry **an "N"**. When the course reversal is necessary, the field will carry **a "Y".** The indication is provided **for each sector on a particular TAA Initial Approach Fix.**

Used on: Airport or Heliport TAA Primary Record
Length: 1 character
Character Type: Alpha

5.272 TAA Sector Identifier

Definition/Description: **The Fix Position Indicator field contains an indication as to which TAA Initial Approach Fix (IAF) the data in the record applies.**

Source/Content: **Airport and Heliport Terminal Area Altitude (TAA) are published for each Initial Approach Fix (IAF) for some RNAV and GPS Approach Procedures. The field identifies the fix to which the data contained in the record applies. The content is derived from official government source and entered as indicated in the table below. The terminology "left", "right" and "center" refers to the position of the TAA fix to the final approach course. "Center" indicates on the final approach course. "Left" indicates left of the final approach course. "Right" indicates right of the final approach course. It could also be viewed as the direction of turn onto final approach an aircraft would make from the base leg defined by the fix. When used on Airport and Heliport Approach Procedure Records, it serves as a pointer to the specific Airport or Heliport TAA Record (PK) where the data pertaining to the fix resides.**

Field Content	**TAA Fix Position Indicator**
C	Straight-In or Center *Fix*
L	Left Base Area
T	Right Base Area

Used On: Airport and Heliport TAA Primary Records
Length: 1 character
Character Type: Alpha

5.273 TAA IAF Waypoint

Definition/Description: The "TAA IAF Waypoint" field contains the identifier of the Initial Approach Fix (IAF) associated with a given Terminal Area Altitude sector. There may be one, two or three such IAF waypoints defined for a single approach procedure. The TAA IAF Waypoint identified in the TAA Sector with the identifier of "C" (Section 5.272) is the fix to which all sector bearing information for the procedure TAA is defined. The TAA IAF Waypoint in the individual TAA Sector records is the fix from which radius distances are defined.

Source/Content: The field contains the official identifier of the waypoint for which the TAA Sector is defined. They will be derived from official government sources.

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Used On: Airport and Heliport TAA Records
Length: 5 character max
Character Type: Alpha/numeric

5.274 TAA Sector Radius

Definition/Description: The "Sector Radius" field in TAA records defines the start and end distances that define a TAA area. They are referenced to the TAA IAF Waypoint defined in that record. As TAA information is used towards that waypoint, the radius information is provided towards that waypoint. They enclose the sector defined in the record. The values are inclusive.

Source/Content: The Sector Radius information will be derived from official government source. Each TAA sector is made up of the start of sector radius and the end of sector radius. The values are provided in nautical miles. The first three digits define the radius for start of the sector, the second three digits the end of the sector, when flying towards the IAF Waypoint.

Used On: Airport and Heliport TAA Primary Records
Length: 4 characters
Character Type: Numeric
Examples: 3011, a Sector that starts at 30 nautical miles to the IAF Waypoint and ends at 11 nautical miles to the IAF Waypoint.

0500, a Sector that starts at 5 nautical miles to the IAF Waypoints and ends at that IAF Waypoint

5.275 Level of Service Name (LSN)

Definition/Description: The "Level of Service Name" field identifies the official procedure level of service based on published procedure operating minimums information for Approach Procedures authorized for SBAS **and/or RNP**.

Source/Content: The field will be derived from official government. The table below shows examples of Level of Service Names for SBAS in use at the time Supplement 18 was adopted.

Level of Service Name (Note 2)
LPV (Note 1)
LNAV
LNAV/VNAV

Used On: Procedure Data Continuation Records
Length: 10 characters (Note 3)
Character Type: Alpha

Note 1: At the time Supplement 18 was adopted, the only SBAS Level of Service published for which there is a "FAS Block provided" was LPV. If governments subsequently develop equivalent or different SBAS criteria for which there is a "FAS Block Provided", the published Level of Service Name would be provided for its associated "FAS Block Required" in the Airport or Heliport Procedure Data Continuation Record (Section 4.1.9.2 or 4.2.3.2).

Note 2: The Level of Service Names of LPV, LNAV/VNAV and LNAV are derived from FAA documentation in use force at the time Supplement 18 was adopted. Other government authorities may use other terminology to describe these procedures. It is possible for LNAV/VNAV and/or LNAV to be authorized for SBAS either with or without a "FAS Block provided."

Note 3: The field is "left justified" with the first character in the first (lowest number) column and any remaining columns filled with blanks. In the case that the field is not applicable because the associated Level of Service Authorized (Sections 4.1.9.5 or 4.2.3.5, and 5.276) is "N" for Not Authorized, the entire10-character Level of Service Name field should be blank.

Note 4: The RNP lever of service name fields are formatted per Section 5.211. In the case that the field is not applicable because the associated Level of Service Authorized (Sections 4.1.9.5 or 4.2.3.5, and 5.276) is "N" for Not Authorized, the RNP Level of Service will be populated regardless of the Level of Service Authorized (5.276) designation.

5.276 Level of Service Authorized

Definition/Description: The "Level of Service Authorized" field defines whether the Level of Service designated in an associated field (Section 5.275) is authorized or not authorized for a procedure.

Source/Content: The Level of Service Authorized can be derived from official government sources. It is a code selected from the table below.

Description	Field Content
Designated Level of Service is authorized for the procedure	A
Designated Level of Service is not authorized for the procedure	N

Used On: Procedure Data Continuation Records
Length: 1 characters
Character Type: Alpha

5.277 DME Operational Service Volume (D-OSV)

Definition/Description: The "DME Operational Service Volume" field is used to specify the service volume information of DME Navaids to support using DME-DME and DME-DME-IRU FMS capabilities in RNAV procedures and routes.

Source/Content: The information will be derived from official government source documentation and encoded based on the table below:

Field Content	D-OSV Description
A	40NM or less
B	70NM or less
C	130NM or less
D	Greater than 130NM
U	Unspecified

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Used On: VHF Navaid Primary Records
Length: 1 character
Character Type: Alpha

5.278 Activity Type

Definition/Description: The "Activity Type" is used to define the type of Special Activity that is occurring.

Source/Content: The "Activity Type" should be derived from official government publications.

Type	Field Content
Parachute Jumping Area	P
Glider Operations	G
Hang Glider Activities	H
Ultralight Activities	U

Used On: Special Activity Area records
Length: 1 character
Character Type: Alpha

5.279 Activity Identifier

Definition/Description: The "Activity Identifier" field contains the number or name that uniquely identifies the Special Activity Area.

Source/Content: The "Activity Identifier" is to be derived from official government publications. The field will contain an alphanumeric designation up to 6 characters.

Field Content			
Activity	Type	State/Nation	Activity Designator
Parachute Jumping Area	PP	TX	117
Glider Operations	G	VA	5
Hang Glider Activities	H	CA	45
Ultralight Activities	U	OR	99

Used On: Special Activity Area records
Length: 6 character
Character Type: Alpha/numeric
Examples: PTX117, GVA5, UOR99

5.280 Special Activity Area Size

Definition/Description: The "Special Activity Area Size" field contains the radius around the center point where the Special Activity is expected to occur.

Source/Content: The "Special Activity Area Size" is to be defined from official government publications when available. The radius is entered in nautical miles to a tenth of a nautical mile with the decimal point suppressed.

Used On: Special Activity Area records.
Length: 3 character
Character Type: Numeric
Examples: 020, 105, 050

5.281 Special Activity Area Volume

Definition/Description: The "Special Activity Area Volume" field contains the expected annual level of intensity of the Special Activity.

Source/Content: The "Special Activity Area Volume" is to be derived from official government publications when available.

Used On: Special Activity Area records
Length: 1 character
Character Type: Alpha/numeric

5.282 Special Activity Area Operating Times

Definition/Description: The "Special Activity Area Operating Times" field contains the annual expected operation schedule of the Special Activity.

Source/Content: The "Special Activity Area Operating Times" is to be derived from official government publications when available.

Field Content			
Description	Days	Holiday Qualifier	Time of Use
Continuous	D		
Weekdays	D		
Weekends	E		
Other	O		
Unknown	U		
Including Holidays		H	
Excluding Holidays		X	
Unknown		U	
SR-SS			D
Night Use			N
Continuous			C
Active by NOTAM			A

Used On: Special Activity Area records
Length: 3 character
Character Type: Alpha
Examples: DXD (Weekdays, Excluding Holidays from Sunrise to Sunset)

5.283 Communications Class (Comm Class)

Definition/Description: The "Communications Class" field will designate the major grouping of the Communications Types contained in the record.

Source/Content: The value will be selected from the options in the table below:

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Field Content	Description
LIRC	The Communications Type is that of one linked to an Information Region (FIR/UIR) for the purposes of providing control services to aircraft.
LIRI	The Communications Type is that of one linked to an Information Region (FIR/UIR) for the purposes of providing information services to aircraft.
USVC	The Communications Type is that of one used within an Information Region (FIR/UIR) for purposes other than control or information services and is not linked to that Region
ASVC	The Communications Type is that of one providing automated or broadcast services within an Information Region (FIR/UIR).
ATCF	The Communications Type is that of one providing ATC services to aircraft within the terminal area of an airport.
GNDF	The Communications Type is that of one providing ATC services to aircraft on the ground at an airport.
AOTF	The Communications Type is that of one providing services other than ATC functions on the ground or within the terminal area of an airport.
AFAC	The Communications Type is that of one provided automated or broadcast services to aircraft on the ground or with the terminal area of an airport.

Used On: Enroute, Airport and Heliport Primary and Continuation Communications Records and the Communications Type Translation Table Record
Length: 4 characters
Character Type: Alpha

5.284 Assigned Sector Name (ASN)

Definition/Description: The "Associated Sector Name" field is used to indicate the published name of an Enroute Communications Sector.

Source/Content: The content of the field will be derived from official government source.

Used On: Enroute Communications Records
Length: 25 characters max
Character Type: Alpha/numeric
Examples: West Sector, Mediterranean Sector, UR Sector, SE High Sector

5.285 Time Narrative

Definition/Description: The "Time Narrative" field is used to provide Time of Operations and/or Conditions of Operations in a narrative form when source information cannot be formatted in accordance with Section 5.195 of this Specification.

Source/Content: The field content will be derived from official government sources.

Used On: Enroute Airway Restriction, Enroute/Airport/Heliport Communications, Restrictive Airspace, Controlled Airspace and Preferred Route Continuation Records
Length: 100 characters max (per record)
Character Type: Alpha/numeric

5.286 Multi-Sector Indicator (MSEC IND)

Definition/Description: The "Multi-Sector Indicator" field is used to indicate that the communications service and frequency are used in more than one defined sector. The actual sector data will be contained in the primary and continuation records of the affected airport or heliport communications record set.

Source/Content: The field will be set to Y, indicating multi-sector data is published in official government source for the service and frequency or N, indicating that the official government source has provided only a single defined sector for the service and frequency. The field will be left blank if there is no defined sector data published for the service and frequency.

Used On: Airport and Heliport Communications Primary Records
Length: 1 character
Character Type: Alpha/numeric

5.287 Type Recognized By (TRB)

Definition/Description: The "Type Recognized By" field is used to provide an indication of the provider of a given Communications Type (5.101).

Source/Content: The field content will be derived from the official government source used to establish the Communications Type and will be selected from the table below:

Field Content	Description
I	The Communications Type is found in government source provided in accordance with ICAO standards.
F	The Communications Type is found in government source provided in accordance with US FAA standards.
B	The Communications Type is found in government source provided in accordance with both ICAO and US FAA standards.
C	The Communications type is found in government source provided by the country in which the communications is used.
O	The Communications type is found in government source provided by the country in which the communications is used.
S	The Communications Type has been established by the data supplier.

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Used On: Communications Type Translation Table Records
Length: 1 character
Character Type: Alpha

5.288 Translation

Definition/Description: The "Translation" field is used to provide a decoding of a three character Communications Type (5.101).

Source/Content: The content of the field will be derived from official government source documentation. There will be a listing for every Communications Type contained in the output file.

Note 1: The RNP level of service name fields are formatted per chapter 5 para 5.211. In the case that the field is not applicable because the associated Level of Service Authorized (Section 4.1.9.5 or 4.2.3.5, and 5.276) is "N" for Not Authorized, the RNP Level of Service will be populated regardless of the Level of Service Authorized (5.276) designation.

Used On: Communications Type Translation Table Records
Length: 80 characters max
Character Type: Alpha/numeric
Examples: TWR: ATC Control Tower
GCO: Ground Communication Outlet
ATI: Automated Terminal Services

5.289 Used On

Definition/Description: The "Used On" field provides an indication of what kind of communications records a particular Communications Type is used on.

Source/Content: The content will be derived from official government source and will be selected from the table below.

Field Content	Description
A	The Communications Type is used on Airport Communications Records only
E	The Communications Type is used on Enroute Communications Records only
H	The Communications Type is used on Heliport Communications Records only
B	The Communications Type is used on Airport, Heliport and Enroute Communications Records
C	The Communications Type is used on Airport and Heliport Communications Records

Used On: Communications Type Translation Table Records
Length: 1 character
Character Type: Alpha

5.290 Procedure Design Mag Var (PDMV)

Definition/Description: The "Procedure Design Mag Var" field specifies the angular difference between True North and Magnetic North at the location defined in the record. That "location" may be the airport for which the procedure was designed, the so-called Airport Magnetic Variation of Record, or may be the procedure leg defined in the record. Which "location" is intended can be determined from the content of the data coded to Section 5.291 (Procedure Design Mag Var Indicator).

Source/Content: Procedure Design Mag Var is obtained from official government procedure data sources and is understood to be the Epoch Year value used when the procedure last revised. This value may differ from magnetic variation data in the primary record of the airport for which the procedure was designed and from data for individual navaids or waypoints used in the procedure. Updating of this value is based only on procedure source data change. Position one of the field contains an alpha character taken from the table below. Positions 2 thru 5 carry the angular difference value expressed in degrees and tenths of a degree with the decimal point suppressed. When Position one is set to "T", Positions 2 thru 5 will be all zeros.

Field Content	Description
E	Procedure Designed based on Magnetic Variation (angular difference) that is East of True North
W	Procedure Designed based on Magnetic Variation (angular difference) that is West of True North
T	Procedure Designed based on True North

Length: 5 characters
Character Type: Alpha/numeric
Examples: E0140, E0007, T0000

5.291 Procedure Design Mag Var Indicator (PDMVI)

Definition/Description: The "Procedure Design Mag Var Indicator" field is an indication of how procedure design magnetic variation was provided in official source data for the procedure defined in the record/record set.

Source/Content: Procedure Design Mag Var (5.290) will be obtained from official government source procedure data. That data can be a value valid for the entire procedure or a series of values valid for individual legs of the procedure. The field will contain the alpha character "P" when the value applies to the entire procedure or the alpha character "L" with the value applies to the leg with which it is associated. With the exception of VOR radials and tracks in VORDME RNAV Approach procedures, Approach Procedures are designed using the airport magnetic variation of record and a single value will apply for the complete procedure. VOR radials use the establish station declination of the VOR. Tracks in VORDME

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RNAV procedures use the station declination of the procedure reference navaid.

Used On:	Airport and Heliport SID/STAR/Approach Primary Extension Continuation Records
Length:	One character
Character Type:	Alpha

5.292 Category Distance

Definition/Description: The "Category Radii" fields, expressed in tenths of nautical miles, specifies the obstacle clearance area for aircraft maneuvering to land on a runway which is not aligned with the FAC of the approach procedure. The limits of the circling area are defined to be an arc from the center of the end of each usable runway. The extremities of the adjacent arcs are joined by lines drawn tangent to the arcs. The area thus enclosed is the circling approach area.

Source/Content: Category radii are obtained from official government publications. The field will contain a figure expressed in nautical miles, with a resolution of 1/10. If the radii is not known or defined, the field is filled with 00.

Used On:	Airport and Heliport Approach Continuation Records
Length:	2 characters
Character Type:	Alpha/numeric
Examples;	00, 13, 15, 17, 23

5.293 Vertical Scale Factor (VSF)

Definition/Description: Vertical Scale Factor (VSF) is used to set the vertical deviation scale.

Source/Content: VSF values derived from official source will be used when available. They are entered into the field in feet (three digits). The content can be:

When used on Enroute Airway segments, VSF shall apply inbound to the fix when viewed in decreasing sequence number order. The VSF applies only to the airway leg on which it is specified. If no VSF value is coded on a segment, there is not a data base specified VSF for that segment.

When used on a SID, STAR, Approach Transition or Missed Approach record, the VSF shall apply to the balance of the procedure route unless superseded by another value of VSF on a subsequent record. Procedure route must be determined by the Route Type field (see Section 5.7).

When used on final approach records, VSF must apply to the waypoint referenced by the final approach record.

Used On:	Enroute Airways, SID, STAR and Approach Route and Controlled Airspace Records
Length:	3 character
Character Type:	Numeric
Examples:	250, 100, 050

5.294 RVSM Minimum Level

Definition/Description: RVSM Minimum Level is the lowest defined cruising level for an airway or holding pattern.

Source/Content: RVSM Minimum Levels are derived from official source when available. They are entered into the field as a three-digit numeric flight level.

Used On:	Enroute Airway Records, Holding Pattern Records
Length:	3 character
Character Type:	Numeric
Examples:	080,180,270

5.295 RVSM Maximum Level

Definition/Description: RVSM Maximum Level is the highest defined cruising level for an airway or holding pattern.

Source/Content: RVSM Maximum Levels are derived from official source when available. They are entered into the field as a three-digit numeric flight level.

Used On:	Enroute Airway Records, Holding Pattern Records
Length:	3 character
Character Type:	Numeric
Examples:	270,250,510

5.296 Pad Type

Definition/Description: The "Pad Type" field defines the type of landing surface used for helicopter operations at a specific heliport.

Source/Content: Pad Types will be derived from official government sources and expressed as one of the following codes:

Field Conent	Description
R	Runway (Note 1)
S	Square or Rectangular Helipad
C	Circular Helipad

Note 1: Many airports formerly used for fixed-wing traffic are being converted for use by helicopter traffic only. This code will be used when the runway has been kept intact and has not been divided into specific landing pads.

Used On:	Helipad record
Length:	1 character
Character Type:	Alpha

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5.297 RNP Level of Service (LSN)

Definition/Description: The Level of Service field identifies the official procedure level of service based on published procedure operating minimums information for Approach Procedures authorized for RNP.

Source/Content: The field will be derived from official government source and provided beginning with the least restrictive value. The table below shows examples of Level of Service for RNP.

Level of Service Name – RNP (Note 1)
031
152
112

Used On: Procedure Data Continuation Records
Length: 3 characters
Character Type: Numeric

Note 1: The RNP level of service name fields are formatted per Chapter 5, Section 5.211. In the case that the field is not applicable because the associated Level of Service Authorized (Section 4.1.9.5 or 4.2.3.5, and 5.276) is "N" for Not Authorized, the RNP Level of Service will be populated regardless of the Level of Service Authorized (5.276) designation.

5.298 Route Inappropriate Navaid Indicator

Definition/Description: A "Route Inappropriate Navaid Indicator" is used when a DME navaid has source provided information identifying the navaid as inappropriate for use in navigation solutions for RNAV 1 and RNAV 2 routes.

Source/Content: The content of the field is derived from official government sources. The field will be set to the character N when the DME navaid has not been published as being inappropriate for navigation solutions for RNAV 1 or RNAV 2 routes or the character Y when the DME navaid has been published as being inappropriate for navigation solutions for one or more RNAV 1 or RNAV 2 routes.

Used On: VHF Navaid Primary Records
Length: 1 character
Character Type: Alpha

6.0 ENCODING STANDARDS

6.1 General

This chapter sets forth the encoding standards to be employed for ARINC 424 and other master user data files. The data will be encoded with ASCII characters.

6.2 Header Records

There will be at least one 132-character header record for each data file. The header records contain information to uniquely identify each data file. Header record fields are considered required unless specified otherwise. Header records are identified by HDR in columns 1 through 3 of the record.

6.2.1 Header Record 1

Column	Field Name	Field Length	Reference
1 thru 3	Header Ident	3	Contains "HDR"
4 thru 5	Header Number	2	Contains decimal 01 to indicate this is the first Header Record.
6 thru 20	File Name	15	Contains the file name.
21 thru 23	Version Number	3	Contains 3 decimal numbers to uniquely identify revision of this file. Initially set to 001, but will be incremented if the file is created more than once in the same cycle.
24	Production/Test Flag	1	Contains "P" if this is a production data file. Contains "T" if this is a file created for test purposes.
25 thru 28	Record Length	4	Contains the decimal number 0132, i.e., the number of characters in each data record.
29 thru 35	Record Count	7	Contains the decimal count of the number of data records in the file.
36 thru 39	Cycle Date	4	Contains Cycle Date (5.32)
40 thru 41	Blank (spacing)	2	Contains blanks.
42 thru 52	Creation Date	11	Contains the date when the file was created. Format is DD-MMM-YYYY. Where DD is the two-digit decimal day of month, MMM is a three-character month abbreviation, and YYYY is the four-digit decimal year. (Example: 12-APR-2002)
53 thru 60	Creation Time	8	Contains the UTC time when the file was created. The format is two decimal digits each for hours, minutes, and seconds, separated by colons. (Example: 13:12:02 = 1:12:02 p.m.)
61	Blank (spacing)	1	Contains blank.
62 thru 77	Data Supplier Ident	16	Contains information to identify the data supplier. Content defined by the data supplier.
78 thru 93	Target Customer Ident	16	Contains information to identify the data user/customer (for example, the customer name(s), file codes). Content defined by the data supplier and/or customer. (Optional)
94 thru 113	Database Part Number	20	Unique part number for database. (Optional - Content TBD)
114 thru 124	Reserved	11	Contains blanks.
125 thru 132	File CRC	8	**This field contains the 32 bit CRC value for the ARINC data file (including data and header records).** **ARINC Report 665, Loadable Software Standards, Section 4.0 defines the use of CRC codes.** **The CRC Polynomial used to calculate the CRC of the ARINC 424 data file shall be the 32-bit CRC (0x04C11DB7), calculated as described in ARINC Report 665** For purposes of calculating a CRC value, Header record 1, Columns 125 through 132, shall be considered to contain zeros.

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6.2.2 Header Record 2

Column	Field Name	Field Length	Reference
1 thru 3	Header Ident	3	Contains "HDR"
4 thru 5	Header Number	2	Contains decimal 02 to indicate this is the second Header Record.
6 thru 16	Effective Date	11	If the Effective Date is the date associated with the Cycle Date, this field will contain blanks. If the file is created outside of standard cycles, this field will provide the effective date. The format is the same as the Creation Date. (Optional)
17 thru 27	Expiration Date	11	If the Expiration Date is the date associated with the Cycle Date, this field will contain blanks. If the file is created outside of standard cycles, this field will provide the expiration date. The format is the same as the Creation Date. (Optional)
28	Blank (spacing)	1	Contains blank.
29 thru 58	Supplier Text Field	30	Contains information specific to data supplier, contents to be defined by data supplier (For example, extract program version) (Optional)
59 thru 88	Descriptive Text	30	Contains description of file contents, or other information agreed upon by data supplier and customer. (For example, description of file content, test file notes, etc). (Optional)
89 thru 132	Reserved	43	Contains blanks.

6.2.3 Additional Header Records

Additional Header Records may be added.

6.3 Bit Density

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.4 Coding

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.5 Parity Convention

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.6 Reel-File Relationship

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.7 Labels

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.7.1 Volume Header Label (VOL)

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.7.2 Header 1 Label (HDR 1)

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.7.3 Header 2 Label (HDR 2)

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.7.4 End-of-File Trailer Label (EOF)

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.7.5 End-of-Volume Trailer Label (EOV)

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.8 Tape Marks

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.9 Summary of Tape Data Layout

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.9.1 One File, One Reel

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.9.2 One File, Multiple Reels

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.9.3 Multiple Files, One Reel

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.10 CRC Calculations

Section 6.10 and subsections were deleted by Supplement 19.

6.0 ENCODING STANDARDS

6.11 Application of CRC for Integrity Protection of Straight & Advanced Landing Approach Operations

Section 6.11 and subsections were deleted by Supplement 19.

7.0 NAMING CONVENTIONS

7.1 General

This chapter establishes the coding rules for Identifiers and Name fields when government source does not provide these Identifiers or Names within the rules established by ICAO Annex 11.

ICAO Annex 11 defines the international standards for coded designators of NAVAIDS, Waypoints, Airways, Standard Instrument Arrivals Routes, and Standard Instrument Departures.

7.2 Fix Identifiers

Fix identifiers will be assigned to all waypoints with the ground rules set forth in this chapter. Section 5.13 establishes the use and limits the field to five characters maximum.

7.2.1 VOR, VORDME, VORTAC, TACAN, and Non-Directional Beacons (NDB)

Waypoints located at any of the above types of facilities will take on the official 1-, 2-, 3-, or 4-character identifier of the facility in question.

Examples:

Facility	Fix Field Entry
Los Angeles VORTAC becomes	AX
Tyndall TACAN becomes	PAM
Ft. Nelson NDB becomes	YE

7.2.2 Non-Directional Beacons (NDB)

For systems employing the "NDB as Waypoint" concept, waypoints located at NDB's should be identified by the use of the station identifier followed by the alpha characters "NB."

Examples:

Facility	Fix Field Entry
Fort Nelson, Can becomes	YENB
Newark, NJ becomes	EWRNB

7.2.2.1 Navaid Waypoint

When the position of a navaid is used to create a waypoint such as during navaid outage or lack of complete navaid information, the waypoint identifier will be created using the navaid name, following the conventions of Section 7.2.3 for One Word Names and Multiple Word Names. For example, a waypoint established at the position of a navaid with the name "Uzgorod" would have an identifier of UZGOD. A waypoint established at the position of a navaid with the name of "Camp Henry" would have an identifier of CHENY.

7.2.2.2 Airport Waypoint

When the position of an airport is used to create a waypoint, the waypoint identifier will be created either from the airport identifier, if known, or from the airport name, following the conventions of Section 7.2.3 for One Word Names and Multiple Word Names. For example, a waypoint established at the position of an airport with the identifier of JHKD and a name of Juhankerd Airfield would used the airport identifier JHKD as the waypoint identifier. A waypoint established at the position of an airport without an identifier but with a name of Rondaville Airport would have an identifier of RONDE.

7.2.3 Named RNAV Waypoints, Intersections, and Reporting Points

In many countries these waypoints are assigned unique five-character names, and the identifier is the same as the name. For waypoints not so named, identifiers are developed using the following rules sequentially until five, or fewer, character groups remain.

ONE-WORD NAMES

A. If five or less characters are involved, use the full name.

Examples:

Facility	Fix Field Entry
DOT becomes	DOT
ACRA becomes	ACRA
LOGAN becomes	LOGAN

B. If the name is more than five characters, reduce to five characters with one or more of the following methods.

1. Eliminate double letters

Examples:

Waypoint Name	Fix Field Entry
KIMMEL becomes	KIMEL
COTTON becomes	COTON
RABBITT becomes	RABIT

2. Keep the first letter, first vowel, and last letter. Drop other vowels starting from right to left.

Examples:

Waypoint Name	Fix Field Entry
ADOLPH becomes	ADLPH
BAILEY becomes	BAILY
BURWELL becomes	BURWL

3. Drop consonants, starting from right to left

Examples:

Waypoint Name	Fix Field Entry
ANDREWS becomes	ANDRS
BRIDGEPORT becomes	BRIDT

MULTIPLE WORD NAMES

Use the first letter of the first word and shorten the last word using the rules for One-Word names to reduce it four characters, for a total of five characters.

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Examples:

Waypoint Name	Fix Field Entry
CLEAR LAKE becomes	CLAKE
ROUGH AND READY becomes	RREDY

PHONETIC LETTER NAMES

When an ICAO phonetic alpha character is used as a waypoint name (Alpha, Bravo, Charlie, etc.), use the rules established in One-Word Names. When more than one waypoint in a country has the same phonetic name, obtain uniqueness by applying Duplicate Identifier rules below.

Examples:

Waypoint Name	Fix Field Entry
ALPHA becomes	ALPHA
NOVEMBER becomes	NOVMR
CHARLIE becomes	CHARE

Two waypoints having the same Waypoint Identifiers within the same country two-letter ICAO, for example, CHARLIE, would become CHAR1 and CHAR2.

When a double phonetic, such as TANGO INDIA, is used as the waypoint name, use the rules established under Multiple Word Names. For example, TANGO INDIA becomes TINDA.

When a phonetic alpha character followed by a numeric and/or other alpha character (A1, A1N, B2, etc.), is used as the waypoint name, it will be coded in the data base the same as shown on aeronautical charts.

7.2.4 Unnamed Waypoints

Waypoints not assigned unique five-character names, but where a defined fix is required for charting and is to be included in navigation data bases, will have identifiers developed using the following guidelines:

A. Unnamed turn points, intersections, and bearing/distance waypoints

If the unnamed turn point, intersection, or bearing/distance is collocated with a named waypoint or NAVAID station on a different route structure (e.g., low level or approach), the name or identifier of the collocated waypoint should be used.

Example: Unnamed turn point on J2 between Lake Charles (LCH) and New Orleans (MSY) VORTACs is coincidental with the Lafayette (LFT) low level VORTAC. LFT should be used as the identifier code for the turn point.

Identifier codes for unnamed turn points, intersection, or bearing/distance waypoints that are not coincidental with named waypoints should be constructed by taking the identifier code of the reference NAVAID for the turn point/intersection/(bearing/ distance waypoint) (expected to be the nearest NAVAID serving the airway structure in which it is located) and the distance from the NAVAID to the turn point/intersection/(bearing/distance waypoint). If the distance is 99 nautical miles or less, the NAVAID identifier should be placed first, followed by the distance. If the distance is 100 nautical miles or more, the last two digits only should be used and placed ahead of the NAVAID identifier.

NAVAID	DISTANCE	CODE
INW	18	INW18
CSN	106	06CSN
TCS	89	TCS89

B. FIR, UIR, and Controlled Airspace Reporting Positions

In cases where the government authority does not provide unique, five-letter or less waypoint names and in cases where the government supplied name cannot be converted to a unique five-letter identifier using previous rules, the following rules should be applied in developing an identifier for such waypoints.

1. FIR - use the three characters “FIR” plus a numeric from 02 to 99. An identifier so developed is to be unique within the geographical area code.

2. UIR - use the three characters “UIR” plus a numeric from 02 to 99. An identifier so developed is to be unique within the geographical area code.

3. FIR/UIR - use “FIR” and a numeric as indicated above.

4. Controlled - use the three- letter characters for the Airspace type of controlled airspace plus a numeric from 02 to 99. If these are Terminal Waypoints, they are to be unique within the Terminal Area. If these are Enroute Waypoints, they are to be unique within the geographic area code. Examples of controlled airspace types are:

TMA Terminal Area
CTR Controlled Zone
ATZ Aerodrome Traffic Zone
CTA Controlled Area
TIZ Traffic Information Zone

7.2.5 Reporting Positions Defined by Coordinates

Entry, Exit and intermediate points within Oceanic Control Areas are often defined by waypoints which are “undesignated,” meaning there is no published five-letter-name-code. These points are quite often made available in source documentation as geographical coordinates, expressed in full degrees or half degrees of Latitude and full degrees of Longitude. When such waypoints are to be entered into a database, the following rules are to be applied:

A. Full Degree of Latitude

1. Positions in the northern hemisphere use the letters “N” and “E,” the southern hemisphere uses the letters “S” and “W.”

2. Latitude will always precede Longitude.

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3. Both will use numerics for latitude and longitude as follows:

   a. Latitude - use values provided by source.

   b. Longitude - use only the last two digits of the three digit longitude. Placement of the longitude value in reference to the identifier character (of "N," "S," "W" or "E," see below) will provide the information as to whether the longitude digit dropped was "0" or "1." That character will follow the longitude digits if the longitude is less than 100 degrees and precede the longitude digits if the longitude is equal to or greater than 100.

   c. Use of a single character to provide both latitude and longitude information:

      "N" = North Latitude and West Longitude
      "E" = North Latitude and East Longitude
      "S" = South Latitude and East Longitude
      "W" = South Latitude and West Longitude

   d. Examples:

      North Latitude/West Longitude, longitude less than 100 degrees

      N5200/W07500 - 5275N
      N5000/W04000 - 5040N
      N0700/W00800 - 0708N

      North Latitude/West Longitude, longitude equal to or greater than 100 degrees

      N7500/W17000 - 75N70
      N0700/W12000 - 07N20

      North Latitude/ East Longitude, longitude less than 100 degrees

      N5000/E02000 - 5020E
      N7500/E05000 - 7550E
      N0600/E00800 - 0608E

      North Latitude/East Longitude, longitude equal to or greater than 100 degrees

      N7500/E15000 - 75E50
      N0600/E11000 - 06E10

      South Latitude/West Longitude, longitude less than 100 degrees

      S5200/W07500 - 5275W
      S5000/W04000 - 5040W
      S0700/W00800 - 0708W

      South Latitude/West Longitude, longitude equal to or greater than 100 degrees

      S7500/W17000 - 75W70
      S0700/W12000 - 07W20

      South Latitude/East Longitude, longitude less than 100 degrees

      S5000/E02000 - 5020S
      S7500/E05000 - 7550S
      S0600/E00800 - 0608S

      South Latitude/East Longitude, longitude equal to or greater than 100 degrees

      S7500/E15000 - 75S50
      S0600/E11000 - 06S10

B. Half Degree of Latitude

1. Positions in the northern hemisphere use the letters "N" and "E," the southern hemisphere uses the letters "S" and "W."

2. Latitude will always precede Longitude.

3. Both will use numerics for latitude and longitude as follows:

   a. Latitude - use the full degree values provided by source. Placement of the latitude value in reference to the identifier character (of "N," "S," "W" or "E," see below) will provide the information as to whether the latitude is full degree or half degree.

   b. Longitude - use only the last two digits of the three digit longitude. Placement of the longitude value in reference to the identifier character (of "N," "S," "W" or "E," see below) will provide the information as to whether the longitude digit dropped was "0" or "1." That character will follow the longitude digits if the longitude is less than 100 degrees and precede the longitude digits if the longitude is equal to or greater than 100.

   c. Use of a single character to provide both latitude and longitude information:

      "N" = North Latitude and West Longitude
      "E" = North Latitude and East Longitude
      "S" = South Latitude and East Longitude
      "W" = South Latitude and West Longitude

   d. Examples:

      North Latitude/West Longitude, longitude less than 100 degrees

      N5630/W02000 - N5620
      N5030/W04000 - N5040
      N0730/W00800 - N0708

      North Latitude/West Longitude, longitude equal to or greater than 100 degrees

      N7530/W17000 - 7N570
      N0730/W12000 - 0N720
      North Latitude/ East Longitude, longitude less than 100 degrees

N5030/E02000 - E5020
N7530/E05000 - E7550
N0630/E00800 - E0608

North Latitude/East Longitude, longitude equal to or greater than 100 degrees

N7530/E15000 - 7E550
N0630/E11000 - 0E610

South Latitude/West Longitude, longitude less than 100 degrees

S5230/W07500 - W5275
S5030/W04000 - W5040
S0730/W00800 - W0708

South Latitude/West Longitude, longitude equal to or greater than 100 degrees

S7530/W17000 - 7W570
S0730/W12000 - 0W720

South Latitude/ East Longitude, longitude less than 100 degrees

S5030/E02000 - S5020
S7530/E05000 - S7550
S0630/E00800 - S0608

South Latitude/East Longitude, longitude equal to or greater than 100 degrees

S7530/E15000 - 7S550
S0630/E11000 - 0S610

7.2.6 Terminal Waypoints

The following rules should be applied in developing identifiers for waypoints used solely in terminal area procedures. Such waypoint identifiers will be unique only for the airport specified; a waypoint identifier used in a terminal area cannot be repeated in that terminal area but can be used in an enroute area encompassed by the same geographical area code. Terminal waypoint identifiers can be repeated in areas covered by different geographical codes. These identifier developing rules should only be applied when the waypoints in question have not been assigned official names/identifier by the government authority.

A. Airport/Heliport or Runway/Helipad related Terminal Waypoints

Single Approach Procedure for a given runway or helipad coded and Waypoints common to more than one approach.

The following two-character codes are to be added to the runway identifier or helicopter approach alignment bearing to create an airport related waypoint identifier when no named waypoint has been established by the government source for the fix type:

FF = Final Approach Fix
AF = Initial Approach Fix
IF = Intermediate Approach Fix
CF = Final Approach Course Fix
MA = Missed Approach Point Fix
SD = Stepdown Fix (when not using convention in paragraph “E”)
RW = Runway Fix
OM = Outer Marker Fix
MM= Middle Marker Fix
IM = Inner Marker Fix
BM = Backcourse Marker Fix
TD = Touchdown Fix inboard of runway threshold
HC = Helipad Fix
EP = Final End Point

Multiple Approach Procedures for a given runway or helipad coded for which common waypoints cannot be established:

The following two-character codes are to be added to the runway identifier to create an airport-related waypoint identifier when no named waypoint has been established by the government source for the fix type:

Fx = Final Approach Fix, where “x” equals the “Route Type” (Section 5.7) for the procedure in question
Ax = Initial Approach Fix, where “x” equals the “Route Type” (Section 5.7) for the procedure in question
Ix = Intermediate Approach Fix, where “x” equals the “Route Type” (Section 5.7) for the procedure in question
Cx = Final Approach Course Fix, where “x” equals the “Route Type” (Section 5.7) for the procedure in question
Mx = Missed Approach Point Fix, where “x” equals the “Route Type” (Section 5.7) for the procedure in question
Sx = Step-Down Fix Note: if multiple step-down fix waypoints need to be created, replace “D” with another character, retain the “S.”
Rx = Runway Centerline Fix, where “x” equals the “Route Type” (Section 5.7) for the procedure in question
Tx = Touchdown Fix inboard of runway threshold, where “x” equals the “Route Type” (Section 5.7) for the procedure in question

The convention for Multiple Approaches/Multiple Waypoints is contained in Table 7-1.

Note: “C-T-L” is “Circle-To-Land” Approach

The prefixes indicated in the table above assume that a unique geographical position (Latitude/Longitude) is required for each Waypoint and the “common waypoint” idea cannot be used. Should a single waypoints' geographical position be such that it will serve as the same waypoint type for more than one coded approach procedure, a “common waypoint”; the Single Approach/Common Waypoint convention should be used.

Note on prefixes for FMS(F) Approach Waypoints:

As the majority of the prefixes generated using the standard convention and the Route Type “F” produced duplicates or two character codes that would be easily

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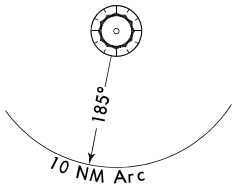
confused with other coded, the numeric/alpha/runway identifier concept is used.

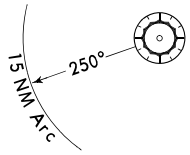
B. Bearing and Distance Waypoints

Identifiers should be developed by the application of the following rules:

1. The first character of the fix identifier should be "D."

2. Character two through four should simplify the Navaid course on which the waypoint lies.
3. The last character should be the distance of the radius defining the position of the waypoint. This radius should be expressed as the equivalent letter of the alphabet, i.e., A = 1nm, G = 7nm, O = 15nm, etc.





4. If the arc radius is greater than 26 NM, then use the convention for unnamed Turn Points, Intersections, and Bearing/Distance Waypoints.

5. If the arc radius is provided in official government source as nautical miles and tenths of nautical miles, the letter of the alphabet will reflect values rounded to full nautical miles, i.e., 10.5nm = 11nm or "K," 10.4nm = 10nm or "J." All values between 0.1 and 1.4 will be character "A."

C. Along Track Distance Waypoints

Along Track Distance Waypoints are expressed in government source documentation as being "x" number of nautical miles from a named waypoint/fix. On aeronautical charts, they are normally identified as "xx.x NM from Named Waypoint."

When not provided by the source document, identifiers for such waypoints should be developed from the along track distance portion of the source information, in two parts:

Part One - the distance in nautical miles and tenths of nautical miles when the tenths is greater than zero, with the decimal point suppressed. Tenths values equal to zero are dropped.

Part Two - the suffix "NM" if the value is equal to or less than 9.9 or a prefix of "NM" if the value is greater than 9.9.

Examples:

3.0 NM from DOOTY should be expressed as 3NM.
2.8 NM from CHASS should be expressed as 28NM.
11.0 NM from BACUP should be expressed as NM11.
13.8 NM from KITTY should be expressed as NM138.

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Waypoint Type	Waypoint codes based on the procedure route type.					
	ILS (I)	ILS (L)	ILS (B)	VOR (V)	NDB (N)	MLS (M)
IAF	AI	AL	AB	AV	AN	AM
IF	II	IL	IB	IV	IN	IM
FACF	CI	CL	CB	CV	CN	CM
FAF	FI	FF	FB	FV	FN	FM
MAP	MI	ML	MB	MV	MN	MM
TDP	TI	TL	TB	TV	TN	TM
Step-down	SI	SL	SB	SV	SN	SM
FEP	EI	EL	EB	EV	EN	EM
	RNAV (R)	TACAN (T)	IGS (G)	LDA (X)	SDF (U)	GPS (P)
IAF	AR	AT	AG	AX	AU	AP
IF	IR	IT	IG	IX	IU	IP
FACF	CR	CT	CG	CX	CU	CP
FAF	FR	FT	FG	FX	FU	FP
MAP	MR	MT	MG	MX	MU	MP
TDP	TR	TT	TG	TX	TU	TP
Step-down	SR	ST	SG	SX	SU	SP
FEP	ER	ET	EG	EX	EU	EP
	MLS (W)	MLS (Y)	**NDB+DME(Q)**	FMS (F)	**GLS (J)**	**VORDME (D)**
IAF	AW	AY	**AQ**	1F	**AJ**	**AD**
IF	IW	IY	**IQ**	2F	**IJ**	**ID**
FACF	CW	CY	**CQ**	3F	**CJ**	**CD**
FAF	FW	FY	**FQ**	4F	**FJ**	**FD**
MAP	MW	MY	**MQ**	5F	**MJ**	**MD**
TDP	TW	TY	**TQ**	6F	**TJ**	**TD**
Step-down	SW	SY	**SQ**	7F	**SJ**	**SD**
FEP	EW	EY	**EQ**	8F	**EJ**	**ED**
	VOR (S)					
IAF	**AS**					
IF	**IS**					
FACF	**CS**					
FAF	**FS**					
MAP	**MS**					
TDP	**TS**					
Step-down	**SS**					
FEP	**ES**					

Table 7-1 Multiple Approaches/Multiple Waypoints

D. Constant Radius to a Fix Waypoint

The “Constant Radius to a Fix” Path Terminator (RF LEG) has available a constellation of three fixes to assist in defining the arc. These are the ARC Center Fix, the Initial Fix, and the Ending Fix. As the waypoints in question will be related to a specific terminal procedure or set of procedures for an airport, these waypoints are defined as Terminal Waypoints.

When not provided by the source document, identifiers for such waypoints should be developed from their use in the arc definition, in two parts:

Part One - a three character alpha code indicating position within the constellation:

ARC = ARC Center Waypoint
AIF = ARC Initial Waypoint
AEF = ARC Ending Waypoint

Part Two - a two character numeric that ensures a unique waypoint within the set of terminal waypoints for a given airport.

Examples: ARC01, AIF01, AEF01

E. Unnamed Step-down Fix Waypoints

The majority of published, unnamed step-down fix waypoints are defined by DME distances from a DME associated with procedure reference facility. The naming convention for these points makes use of that general standard. The convention will still be used for unnamed step-down fix waypoints even if they are not DME defined.

1. Two digits to identify the distance.
2. Three characters to identify the procedure type.
3. Position digits to identify decimal or full nautical miles.

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Examples. An unnamed step-down fix at 0.5 DME from an ILS DME = 05ILS

An unnamed step-down fix at 1.7 DME from a LOC DME = 17LOC

An unnamed step-down fix at 3.5 GPS = 35GPS

An unnamed step-down fix at 12 DME from a VORDME = DME12

7.3 Waypoint Name/Description

The waypoint Name field is assigned to all waypoints in accordance with the ground rules set forth in this section. ICAO Document 4444 defines an international standard for the name of both officially assigned and non-assigned designators at significant points along a route of flight. These rules are in accordance with that standard.

7.3.1 Named Waypoints

UNIQUE FIVE-LETTER

The name field will contain the same five-letter name as the Waypoint Identifier field.

Examples:

Fix Ident	Fix Name
LOGAN	LOGAN

NAME WITH MORE THAN FIVE LETTERS

The name field will contain the full name of the fix.

Examples:

Fix Ident	Fix Name
RABIT	RABBITT
RREDY	ROUGH AND READY

NAMES WITH MORE THAN FIVE LETTERS AND AN ASSIGNED FIX IDENTIFIER

The name field will contain the full name of the fix with the assigned identifier in parenthesis.

Examples:

Fix Ident	Fix Name
SPH	SEA PERCH (SPH)
CRP	CARP (CRP)

7.3.2 Unnamed Waypoints

UNNAMED TURN POINTS, INTERSECTIONS, AND BEARING/DISTANCE WAYPOINTS

The name field for unnamed waypoints whose identifiers are established under rule 7.2.4 will have a description of the waypoint to assist in finding the location on aeronautical charts. This description will use the forming NAVAID identifier and bearing/distance information.

Examples:

Fix Ident	Fix Name	
ABC12	ABC090012	ABC 090 degrees, 12 nm
81ABC	ABC090181	ABC 090 degrees, 181 nm
AB13	AB180013	AB 180.3 degrees 12.8

Decimal values, round up for .5 or greater and round down for .4 or less

D185J	ABC185010	ABC 185 degrees, 10 nm

If duplications result from this convention, the duplicate and subsequent waypoint idents will remove the 3rd character of the navaid ident (if necessary) and add a duplicate alpha indicator suffix after the distance information beginning with "A". When the distance information is at the beginning of the waypoint ident, the duplicate indicator will prefix the distance information.

Examples:

Fix Ident	Fix Name	
SD40A	**SDG313040**	**SDG 313 degrees, 40 nm**
SD40B	**SDG318040**	**SDG 318 degrees, 40 nm**
SD40C	**SDG327040**	**SDG 327 degrees, 40 nm**
A81SD	**SDG090181**	**SDG 090 degrees, 181 nm**
B81SD	**SDG100181**	**SDG 100 degrees, 181 nm**
C81SD	**SDG110181**	**SDG 110 degrees, 181 nm**

Example from ST PIERRE, REUNION (FMEP):

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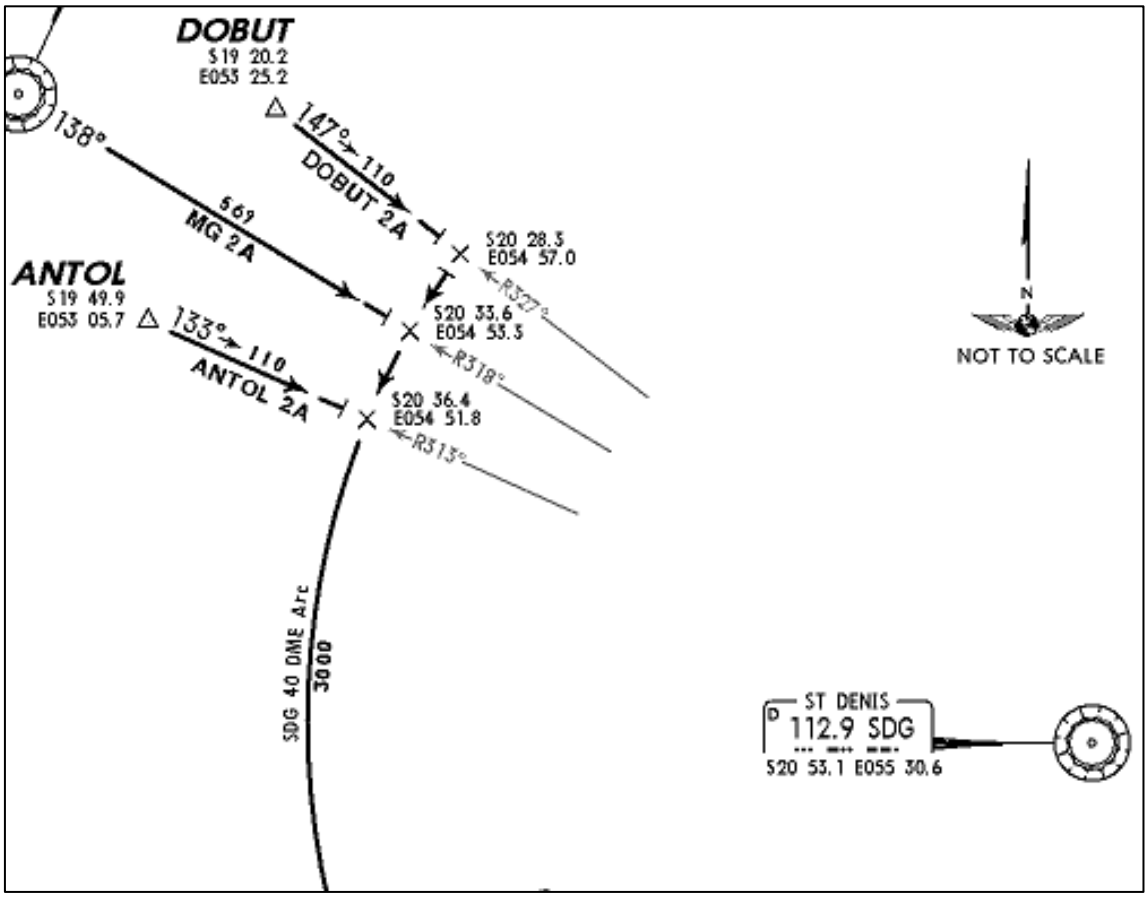

7.3.3 Airport-Related Waypoints

MARKERS AS TERMINAL WAYPOINTS

For Markers that are shown as Terminal Waypoints, the runway with which the marker is associated will be included in the name field.

Pos 1 and 2: OM, IM, MM or BM
Pos 3: blank
Pos 4 thru 8: runway identifier
Pos 9: blank
Pos 10 thru 25: additional name when required

Examples:

Fix Ident	Fix Name
OM18	OM RW18
ALTUR	OM RW26L ALTUR

7.3.4 Navaid Waypoint

The Name/Description field of a waypoint established at the position of a navaid will contain the full navaid name, and navaid type when known.

Examples:	Fix Ident	Fix Name
UZGOD		Uzgorod NDB
CHENY	Camp Henry	VORTAC

7.3.5 Airport Waypoint

The Name/ Description field of a waypoint established at the position of an airport will contain the full airport name and the published "airport term."

Examples:	Fix Ident	Fix Name
	JHKD	Juhankerd Airfield
	RONDE	Rondaville Airport

7.3.6 VFR Waypoints

If a VFR Waypoint is to be included in the Enroute (EA), Airport Terminal (PC) or Heliport Terminal (HC) Waypoint files for which the government authority has not provided a five-letter-name-code, one is to be created using the following conventions:

Positions 1 and 2 = use the characters VP or VC, alternatively VF or VS

Positions 3 thru 5 = numeric

The identifiers so created are to be unique within a given ARINC Area Code.

Examples: VP001
VC101

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"VP" or "VC" are the preferred position 1 and 2 codes. VF and VS are to be used after all numeric, 001 to 999, for given ARINC Area Code have been used.

7.4 SID/STAR Procedure Identifiers

Naming Rules

When source documents for procedure identifiers published by the controlling agency include computer abbreviations, they will be used in the appropriate fields. When the source provide codes designations are not compatible with the requirements of an aeronautical database, modifications are required. In such cases, SID/STAR identifiers are assigned to all procedures in accordance with the ground rules set forth in this Chapter. The SID/STAR identifier must be limited to a maximum of 6 characters in length. Current international standards for assigning coded designators permit up to 7 characters (ICAO Annex 11, Appendix 3). These seven characters normally consist of a basic indicator, validity indicator, and, when required, a route indicator. The basic indicator names the significant point where the departure terminates or the arrival begins. The ICAO validity indicator publicizes the current edition of the arrival or departure. This is a numeric character from one to nine. The ICAO route indicator is an alpha character which is added, as necessary, to distinguish between more than one departure terminating at the same significant point or arrival beginning at the same point.

A. For a published SID or STAR identifier not derived from the ending/beginning NAVAID or waypoint follow these rules:

If the identifier is:

1. Alphanumeric, then shorten the published name down to six characters by simply dropping characters from the name from right to left. If such a SID or STAR uses numeric or alpha detail, always retain that suffix (validity/route indicator) detail, dropping an additional number of characters from the name as required.

   Examples:

   POGO Departure, no waypoint named POGO, would be POGO

   North Departure (or Departure to North), would be NORTH

   Military One Arrival would be MILIT1

   Noise Abatement Six Departure would be NOISE6
   Arrival Seven would be ARRIV7

2. Either all numeric or a runway identifier, then add the characters DEP for Departure or ARR for Arrival to the identifier, dropping letters (on DEP or ARR) from right to left where required.

   Examples:

   One Departure would be DEP1
   31 Arrival would be ARR31
   131 Departure would be DEP131
   311 Arrival would be ARR311
   1001 Departure would be DE1001
   2000 Arrival would be AR2000

   Runway 07 Departure would be DEP07
   Runway 25 Arrival would be ARR25
   Runway 01L Departure would be DEP01L

B. For a published SID or STAR identifier derived from a NAVAID or Waypoint follows these rules:

If the SID (departure) terminates or STAR (arrival) begins with a

1. NAVAID, then use the ident of the NAVAID in all cases, even when the NAVAID "name" is five characters or less.

   Examples:

   Bucks Seven Arrival from Bucks VOR "BKS" would be BKS7

   Kellogg Five Alpha Departure to Kellogg VOR "WDK" would be WDK5A

   Fink Two Delta Arrival from Fink VOR "FNK" would be FNK2D

2. Published Waypoint with 5-character name, then if:

   a. No validity indicator or route indicator has been published, then retain the basic name as published.

   b. Only a validity indicator has been published, then retain the basic name and the validity indicator as published.

   c. Only a route indicator has been published, then retain the basic name and the route indicator as published.

      Examples:

      ALLAN Departure to ALLAN waypoint would be ALLAN

      CAROL One Departure to CAROL waypoint would be CAROL1

      STEVE Alpha Arrival from STEVE waypoint would be STEVEA

   d. Both a validity indicator and a route indicator have been published, then drop the last (5th) character of the basic name and retain the validity indicator and route indicator.

      Examples:

      DAVIS Five Bravo Departure to DAVIS waypoint would be DAVI5B

      ANITA Six Delta Arrival from ANITA waypoint would be ANIT6D

e. The Waypoint name contains double letters and both validity indicator and route indicator are published, then drop the first double letter (instead of the 5th character of the waypoint name) and retain the validity indicator and route indicator.

Examples:

WITTY One Alpha Departure to WITTY waypoint would be WITY1A

MASSA Two Charlie Arrival from MASSA waypoint would be MASA2C

3. Published Waypoint with more than 5 characters, then reduce the name to 5 characters using the established waypoint rules from this chapter and then apply rule B.2 above.

Examples:

COTTON One Departure to COTTON waypoint would be COTON1

BURWELL Bravo Arrival from BURWELL waypoint would be BURWLB

CLEAR LAKE Three Golf Departure to CLEAR LAKE waypoint would be CLAK3G

4. Published Duplicate Waypoint (as identified by WAYPOINT IDENTIFIERS, Section 7.2.3) then, drop the digit added to provide uniqueness (unless the digit is necessary because of the procedures existing at the same airport).

Examples:

CHARLIE Departure to a waypoint in the data base as CHAR1 or CHAR2 would be CHAR (retain only these four characters)

CHARLIE One Departure to a waypoint in the data base as CHAR1 or CHAR2 would be CHAR1

CHARLIE One Alpha Departure to a waypoint in the data base as CHAR1 or CHAR2 would be CHAR1A

SHAWNEE Departure to a waypoint in the data base as SHA1E as there are more than nine points named SHAWNEE within the ICAO would be SHAE (as SHAE1 through SHAE9 came before SHA1E)

SHAWNEE One BRAVO Departure to a waypoint in the data base as SHA1E as there are more than nine points named SHAWNEE within the ICAO would be SHAE1B.

5. Unpublished Waypoint name, then use the rule for the points as described in WAYPOINT IDENTIFIERS, Section 7.2.4 (unnamed waypoints), and apply the rule in B.2 above.

6. Unpublished SIDs or STARs without any name or identifier are currently not included in the aeronautical navigation data base and, hence, are not currently provided for in these naming rules for SID/STAR identifiers.

For "Engine Out SIDs," use an identifier provided by source documentation when such is available. If an Engine Out SID is to be coded that does not have a source provided identifier, an identifier will be created by adding the prefix "EO" to the Runway Designator. For example, an Engine Out SID for Runway 07L would be designated "EO07L." Note that with this convention, only one Engine Out SID per runway can be included in a master airline user file.

7.5 Preferred Route Identifiers

7.5.1 North American Routes

For North American Routes for North Atlantic Traffic, "Non-Common" portion and other Preferred or Preferential Routes without published identifiers but with unique initial and terminus fix points, the route identifier will be developed using the initial and terminus fix identifiers, as indicated in the table below.

Fix Type	Create Identifier Using
Airport	Three or Four Character Airport Identifier
Navaid	Navaid Identifier
Waypoint	Waypoint Identifier (five character max)

Examples: From Airport to Airport - CYYLCYYC
From Airport to Navaid - CYYLART
From Navaid to Waypoint - ARTCOLAR

7.5.2 Multiple Routes - Same Fix

If there is more than one routing without a published identifier between the same to fixes and the rules in Section 7.5.1 is being used to create the route identifier, then add numerics to indicate the multiple routings.

Examples: For two routes between the airports CYYL and CYYC, CYYLCYYC1 and CYYLCYYC2

7.5.3 Preferred or Preferential Routes

For Preferred or Preferential Routes without a published identifier and not between unique initial and terminus fix points but rather from areas or regions such as Terminal Control Areas, FIRs or Geographical Entities, the route identifier will be derived from commonly understood elements such as communications center identifiers, country or region abbreviations and the like. Note that if one end of the routing is a unique fix, the rules in Section 7.5.1 apply for that fix.

Area or Region	Create Identifier Using
FIR, ARTCC	Four Character ICAO Identifier of FIR or Center
Terminal Area	Three or Four Character Identifier of owning airport
Geographical Entity	Commonly used abbreviations
ICAO Region	Two character ICAO Region Code

7.0 NAMING CONVENTIONS

Examples: From Terminal Area to Airport - CYULCYYC
From FIR to FIR - ENBOGCCC
From Center to Airport - KZTLKRDM
From Geographical Entity to Terminal Area SCANDIGCCC

7.5.4 Multiple Routes - Same Points/Areas/Regions

If there is more than one routing without a published identifier between the same two points/areas/regions and the rules in Section 7.5.3 is being used to create the route identifier, add numerics to indicate the multiples.

Examples: For two routes between KZTL and KRDM
KZTLKRDM1 and KZTLKRDM2

7.5.5 Preferred or Preferential Overfly Routings

For Preferred or Preferential Overfly Routings, routings that are not designed to serve a initial departing airport/terminal area or terminus arriving airport/terminal area, the route identifier will be derived from the fix, area or region to be overflown and a direction of overflight prefix or a direction of origin in reference to the direction of overflight suffix, according to the table below. If the route is an overflight route and no directional restrictions apply, the character "O" for "overfly" is used instead of the directional indication.

Fix/Area/ Region	Direction	Create Identifier Using
Airport		Three or 4-character Airport Identifier
Navaid		Navaid Identifier
Waypoint		Waypoint Identifier
FIR, ARTCC		Four Character FIR/Center ICAO Identifier
Terminal Area		Identifier of owning Airport
Geographical Entity		Commonly used abbreviations
ICAO Region	North	Two character ICAO Region Code
	South	The character "N"
	East	The character "S"
	West	The character "E"
	Overfly	The character "W"
		The character "O"

Note: The direction codes shown in the table are provided for guidance only. Any published direction may be indicated by the use of a 1, 2 or three character prefix/suffix.

Examples:

Overflying Terminal Area Eastbound - ECYUL
Overflying FIR Southbound - SENBO
Overflying from West of a Center - KZTLW
Overflying Center Southwestbound - SWKZDV
Overflying Terminal Area (no direction specified) OEGLL

7.5.6 Multiple Routes - Overfly

If there is more than one routing without a published identifier between the fix/area/region and the rules in Section 7.5.5 are being used to create the identifier, add a numeric to indicate the multiples.

Examples: SENBO1 and SENBO2
OEGLL1,OEGLL2,0EGLL3

7.5.7 Preferred Weekday/Weekend

For Preferred or Preferential Routings that are published with a "weekday" and a "weekend" version, the rules for "multiples" are replaced with a two character suffix (replacing the numerics). "WK" is used for "weekday" and "WE" is used for "weekend." This rule applies to routes both with and without published identifiers.

Examples: Published Identifiers -
TOS1WK and TOS1WE

Unpublished Identifiers -
SENBOWK and SENBOWE

7.5.8 Weekday/Weekend

If there is more than one routing published as "weekday" or "weekend" and the rules in Section 7.5.7 are being used to create the identifier, add a numeric to indicate the multiples.

Examples: TOS1WK1 and TOS1WK2

7.5.9 Geographical Routings

For Preferred or Preferential Routings that are published as being between large areas not definable with aeronautical terms, a convention of Geographical Entity abbreviations is used to create Route Idents. As these Route Idents will have reduced the entity name down considerably, the Geographical Entity Reference Table is used to provide a link between the Route Ident and the full entity name. While the Route Ident is 10 characters long and normal split five and five between the initial and terminus points of the route, that split does not have to be applied when creating Route Idents based on Geographical Entities.

Examples: Routing between UK North and Greece West - UKNOGRECW or NOUKWGRECE

7.5.10 Multiple Routes - Geographical

If there is more than one routing without a published identifier between geographical entities and the rules in Sections 7.5.8 are being used to create the identifier, add a numeric to indicate the multiples.

Examples: UKNOGRECW1 and UKNOGRECW2

7.5.11 Off Load Route

For Preferred or Preferential Routings that are published with an "off load route," the rules for "multiples" are replaced with a three character suffix (replacing the numerics). "OLR" is used for the "off load route." The standard route would not use a suffix. This rule applies to routes both with and without published identifiers.

7.0 NAMING CONVENTIONS

Examples: Published Identifiers -
TOS1 and TOS1OLR

Unpublished Identifiers -
SENBO and SENBOOLR

7.5.12 Multiple Routes - Off Load

If there is more than one routing published as "off load route" and the rules in Section 7.5.11 are being used to create the identifier, add a numeric to indicate the multiples.

Examples: SENBOOLR1 and SENBOOLR2

7.6 Transition Identifiers

SID/STAR Runway Transition Identifiers will be established using the characters RW (for runway) followed by the runway number. If a transition applies identically to parallel runways, the transition need only be coded once for the parallel runways. In those cases, it should be identified with a suffix of "B" after the runway number.

Example: Transitions for Runway 08L and 08R are identical. They can be coded once as RW08B rather than twice as RW08L and RW08R.

Approach Transition Identifiers are normally the identifier of the NAVAID or Waypoint at which the transition starts or as published in official government source. These identifiers may be modified to ensure a unique identifier is available for each transition.

SID/STAR Runway Transition Identifiers for procedures from or to runways will be established using the characters RW (for runway) followed by the runway designation. If a transition applies identically to parallel runways, the transition need only be coded once for the parallel runways. In those cases, it will be identified with the suffix "B" after the runway number, replacing the Left or Right indications in the runway designation.

Example: Transitions for Runway 08L and 08R are identical for a given SID or STAR. They can be coded once as RW08B.

SID/STAR Runway Transition Identifiers for procedures from or to helipads will be established using the helipad identifier, followed by an additional character when there is more than one Runway Transition from the same pad.

Examples: Transitions for a Helipad with the identifier of MAIN in multiple directions would have the Transition Identifiers of MAIN1, MAIN2 or MAINN, MAINS.

When Approach Transition Identifiers are not published in source documentation, they are created using the identifier of the Navaid or Waypoint at which the transition starts. If multiple Approach Transitions start at the same Navaid or Waypoint, the Navaid or Waypoint is supplemented with an additional character to ensure a unique identifier for each transition.

Example: When two transitions start at a Navaid ABC, one would be given the identifier of ABC1 and the other ABC2, or ABCA and ABCB. When two transitions start at a Waypoint ABCDE, drop the last character and assign a number such as ABCD1 and ABCD2.

When multiple Missed Approach Procedures are required, a Missed Approach Transition Identifier is also required. The identifiers will be developed based on the identifier of the missed approach holding fix, Navaid or Waypoint, or the last fix coded in the Missed Approach Procedure if no missed approach holding fix is provided. If multiple Missed Approach Procedures end at the same missed approach holding or ending fix, the Navaid or Waypoint identifier is supplemented with an additional character to ensure a unique identifier for each transition.

Example: When two transitions end at a Navaid ABC, one would be given the identifier of ABC1 and the other ABC2, or ABCA and ABCB. When two transitions end at a Waypoint ABCDE, drop the last character and assign a number such as ABCD1 and ABCD2.

Unless a transition identifier, compatible with the maximum five-character length for such identifiers, are provided in official government source data, transition identifiers are created by the data supplier based on the rules in this section.

Transition Identifiers are required for Approach Transitions, SID and STAR Enroute Transitions, SID and STAR Runway Transitions and certain types of Missed Approach Transitions. See Section 5.11 of this document.

When official government source provides a Transition Name, that Name will be used as the Transition Identifier, as long as it is compatible with the maximum five-character length for this data field.

When official government source provides a Transition Name that exceeds the maximum character length of five characters, that Name will be shortened to five characters for the Transition Identifier, using the same conventions that are applied to Waypoint Names and Identifiers as defined in Section 7.2.3.

Data suppliers will create Transition Identifiers in those cases where a source provided Identifier or Name is not available. These identifiers will be based on the identifier assigned to a fix as follows:

For Approach Transitions and STAR Enroute Transitions, the identifier of the fix at which the transition begins.

For SID Enroute and some Missed Approach Transitions, the identifier of the fix at which the transition ends.

7.0 NAMING CONVENTIONS

For SID and STAR Runway Transitions, the identifier of the runway from which (SID) or to which (STAR) the transition is associated using the convention detailed in Section 7.2.6.A of this Specification, i.e. RW plus the runway designation.

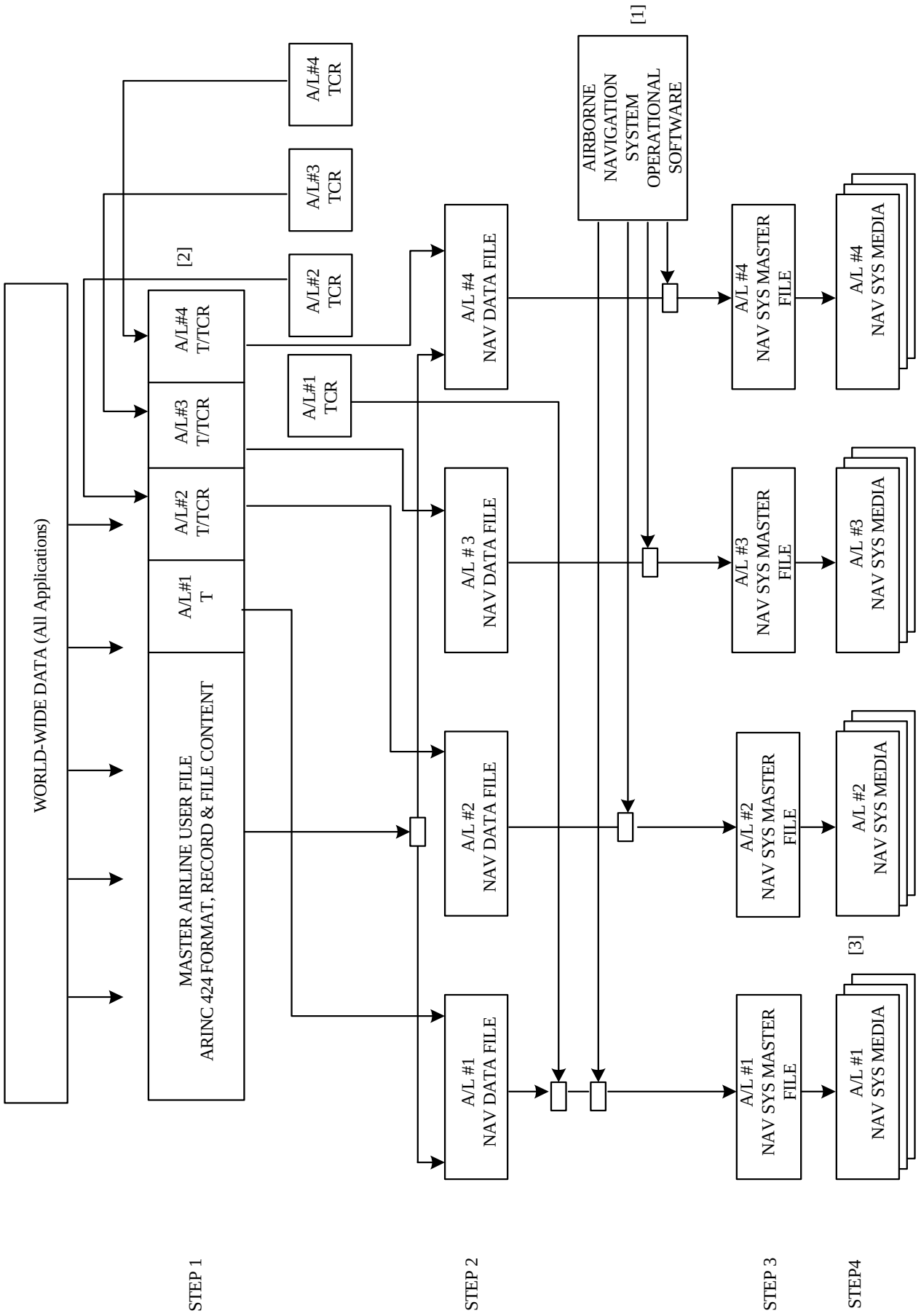
If a SID or STAR Runway Transition applies identically to parallel runways, the transition is coded once and the Transition Identifier is created by replacing the parallel indications of L (Left) and R (Right) with the alpha character B (Both). For example, transitions for Runways (RW) 08L and 08R are identical. The transition can be coded once with an identifier of RW08B.

If the fix involved in Rules 7.6.5.1 and 7.6.5.2 is a waypoint and the identifier of that waypoint has been created or modified by the data supplier using the conventions contained in Sections 7.2.2.1, 7.2.2.2, 7.2.3, 7.2.4, 7.2.5, and 7.2.6 of this section, the exact same convention is applied to the creation or modification of the Transition Identifier.

If there are multiple transitions starting at the same fix (Approach and STAR) or ending at the same fix (SID, some Missed Approach), one Transition Identifier established using the rules above. A second or subsequent is modified by adding a numeric character as a suffix when the identifier is four characters or less or replacing the last character with a numeric suffix when the identifier is five characters in length.

In cases where the multiple transitions starting or ending at the same fix are based on aircraft category such and A&B, C&D, the transition valid for C&D aircraft will carry the standard identifier and the A&B will carry the modified identifier.

ATTACHMENT 1
FLOW DIAGRAM



See next page for Notes 1 and 2.

ATTACHMENT 1
FLOW DIAGRAM

Notes Concerning the Flow Diagram

1. The flow diagram shows alternate paths to individual airline master files for tailored company route (TCR) data in the individual airline boxes of step 1 level.

2. Airline tailored records (denoted by the letter T in the individual airline boxes of the Step 1 level) will be formatted according to the standards set forth in this document.

ATTACHMENT 2
LOCAL HORIZONTAL REFERENCE DATUM NAME, DATUM CODE, ELLIPSOID LIST, AND OGP REFERENCE

Attachment 2 represents the ARINC 424 standard for the horizontal datum values found in Section 5.197 Datum Code, in Chapter 5.

Although Attachment 2 is a representation of the ARINC 424 standard, the tables within the attachment were created using the following documentation:

NIMA TR8350.2 3rd edition, January 3, 2000. Supplementing the NIMA document were updated versions of datum documentation from the International Association of Oil and Gas Producers (OGP), ESRI datum code list, University of Colorado Geography Dept, ISO 1911 Geographic Information – Spatial referencing by coordinates, and web searches for specific state provided information.

Attachment 2 will not be maintained based on revision activity of these docs.

Of the various columns in Attachment 2, only the ARINC Datum Code column data is used to populate the 5.197 Datum Code field.

Attachment 2A also includes Datum Names that do not have an associated Datum Code that can be used for reference purposes.

ATTACHMENT 2
LOCAL HORIZONTAL REFERENCE DATUM NAME, DATUM CODE, ELLIPSOID LIST, AND OGP REFERENCE

ARINC Datum Name	ARINC Datum Code	Associated Ellipsoid Name	Associated Ellipsoid Code	Typical Location Use	OGP DATUM CODE	OGP Ellipsoid Code	OGP Description/Notes
Adindan	ADI	Clarke 1880	CD	Burkina Faso, Cameroon, Ethiopia, Mali, Senegal, Sudan	6201	7012	
Afgooye	AFG	Krassovsky 1940	KA	Somalia	6205	7024	
Ain El Abd 1970	AIN	International 1924	IN	Bahrain Island, Saudi Arabia	6204	7022	
American Samoa 1962	AMA	Clarke 1866	CC	American Samoa Islands	6199	7008	
Anna 1 Astro 1965	ANO	Australian National	AN	Cocos Islands	6708	7003	
Antigua Island Astro 1943	AIA	Clarke 1880 (RGS)	CD	Antigua, Leeward Islands	6601	7012	
Arc 1950	ARF	Clarke 1880 (Arc)	CD	Botswana, Burundi, Lesotho, Malawi, Swaziland, Zaire, Zambia, Zimbabwe	6209	7013	
Arc 1960	ARS	Clarke 1880 (RGS)	CD	Kenya, Tanzania	6210	7012	
Ascension Island 1958	ASC	International 1924	IN	Ascension Island	6712	7022	
Astro Beacon “E” 1945	ATF	International 1924	IN	Iwo Jima	6709	7022	Iwo Jima 1945
Astro DOS 71/4	SHB	International 1924	IN	St. Helena Island	6710	7002	St. Helena 1971
Astro Tern Island (Frig) 1961	TRN	International 1924	IN	Tern Island	6707	7002	Tern Island 1961
Astronomical Station 1952	ASQ	International 1924	IN	Marcus Island	6711	7002	Marcus Island 1952
Australian Geodetic 1966	AUA	Australian National	AN	Australia and Tasmania	6202	7003	Australian Geodetic Datum 1966 - Alias AGD66
Australian Geodetic 1984	AUG	Australian National	AN	Australia and Tasmania	6203	7003	Australian Geodetic Datum 1984 - Alias AGD84
Austria NS	ANS	International 1924	IN				
Ayabelle Lighthouse	PHA	Clarke 1880 (RGS)	CD	Djibouti	6713	7012	Alias - Base SW, Graciosa, Azores Central 1948
Bellevue (IGN)	IBE	International 1924	IN	Efate and Erromango Islands	6714	7022	Bellevue - Alias Bellevue (IGN)
Bermuda 1957	BER	Clarke 1866	CC	Bermuda Islands	6216	7008	
Bissau	BID	International 1924	IN	Guinea-Bissau	6165	7022	Bissau
Bogota Observatory	BOO	International 1924	IN	Colombia	6218	7022	Alias is 'Bogota'
Bukit Rimpah	BUR	Bessel 1841	BR	Bangka and Belitung Islands (Indonesia)	6219	7004	
Camp Area Astro	CAZ	International 1924	IN	Camp McMurdo Area, Antarctica	6715	7022	

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ARINC Datum Name	ARINC Datum Code	Associated Ellipsoid Name	Associated Ellipsoid Code	Typical Location Use	OGP DATUM CODE	OGP Ellipsoid Code	OGP Description/Notes
Campo Inchauspe 1969	CAI	International 1924	IN	Argentina	6221	7022	Campo Inchauspe
Canton Astro 1966	CAO	International 1924	IN	Phoenix Islands	6716	7022	Phoenix Islands 1966
Cape	CAP	Clarke 1880 (Arc)	CD	South Africa	6222	7013	Alias - South Africa
Cape Canaveral	CAC	Clarke 1866	CC	Florida (East) and Bahamas	6717	7008	
Carthage	CGE	Clarke 1880 (IGN)	CD	Tunisia	6223	7011	
Chatham Island Astro 1971	CHI	International 1924	IN	Chatham Island (New Zealand)	6672	7022	Superceded by Chatham Islands Datum 1979
Chua Astro	CHU	International 1924	IN	Paraguay	6224	7022	Chua
Co-Ordinate System 1937 of Estonia	EST	Bessel 1841	BR	Estonia			
Corrego Alegre	COA	International 1924	IN	Brazil	6225	7022	
Dabola	DAL	Clarke 1880 (IGN)	CD	Guinea	6155	7011	Dabola 1981
Danish Geodetic Institute 1934 System	DAN	Danish 1876	DA	Denmark		7051	
Deception Island	DID	Clarke 1880 (RGS)	CD	Deception Island, Antarctica	6736	7012	
Djakarta (Batavia)	BAT	Bessel 1841	BR	Sumatra (Indonesia)	6211	7022	Alias - GENUK
DOS 1968	GIZ	International 1924	IN	Gizo Island (New Georgia Islands)	6722	7022	South Georgia 1968
Easter Island 1967	EAS	International 1924	IN	Easter Island	6719	7022	
European 1950	EUR	International 1924	IN	Austria, Belgium, Channel Islands, Cyprus, Denmark, Egypt, England, Finland, France, Federal Republic of Germany (West Germany), Gibraltar, Greece, Iran, Iraq, Ireland, Israel, Italy, Jordan, Kuwait, Lebanon, Luxembourg, Malta, Netherlands, Norway, Portugal, Sardinia, Saudi Arabia, Scotland, Shetland Islands, Sicily, Spain, Sweden, Switzerland, Syria, Tunisia	6230	7022	European Datum 1950 - Alias - ED50

ATTACHMENT 2
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ARINC Datum Name	ARINC Datum Code	Associated Ellipsoid Name	Associated Ellipsoid Code	Typical Location Use	OGP DATUM CODE	OGP Ellipsoid Code	OGP Description/Notes
European 1979	EUS	International 1924	IN	Austria, Finland, Netherlands, Norway, Spain, Sweden, Switzerland	6668	7022	European Datum 1979 - Alias ED79
Fort Thomas 1955	FOT	Clarke 1880	CD	Nevis, St. Kitts, Leeward Islands	6605	7012	St. Kitts 1955
Gan 1970	GAA	International 1924	IN	Republic of Maldives	6684	7022	
Gandajika Base	GAN	International 1924	IN	Zaire	6259	7022	Malongo 1987 - Alias - Mhast
Geodetic Datum 1949	GEO	International 1924	IN	New Zealand	6272	7022	New Zealand Geodetic Datum 1949 - Alias - GD49
Graciosa Base SW 1948	GRA	International 1924	IN	Faial, Graciosa, Pico, Sao Jorge and Terceira Islands (Azores)	6183	7022	Azores Central Islands 1948 - Alias - Base SW or Graciosa
Greek Geodetic Reference System 1987	GRX	GRS 80	RF	Greece	6121	7019	Alias - GGRS87
Guam 1963	GUA	Clarke 1866	CC	Guam	6675	7008	
Gunuung Segara	GSE	Bessel 1841	BR	Kalimantan (Indonesia)	6613	7004	Alias - P2 Exc, P2 Exc-T9, Samboja
GUX 1 Astro	DOB	International 1924	IN	Guadalcanal Island	6718	7022	Solomon 1968
Herat North	HEN	International 1924	IN	Afghanistan	6255	7022	
Hermannskogel	HER	Bessel 1841	BR	Yugoslavia (Prior to 1990), Slovenia, Croatia, Bosnia and Herzegovina, and Serbia	6312	7004	Militar-Geograp hische Institut - Alias - MGI, HR1901, D48
Hjorsey 1955	HJO	International 1924	IN	Iceland	6658	7022	
Hong Kong 1963	HKD	International 1924	IN	Hong Kong	6738	7022	Alias - HK63
Hu-Tzu-Shan	HTN	International 1924	IN	Taiwan	6236	7022	
Indian	IND	Everest	EA	Bangladesh			
			EC	India, Nepal			
			EF	Pakistan			
Indian 1954	INF	Everest 1830 (1937 Adjustment)	EA	Thailand	6239	7015	
Indian 1960	ING	Everest 1830 (1937 Adjustment)	EA	Vietnam	6131	7015	
Indian 1975	INH	Everest 1830 (1937 Adjustment)	EA	Thailand	6240	7015	
Indonesian 1974	IDN	Indonesian 1974	ID	Indonesia	6238	7021	Indonesian Datum 1974 - Alias - ID74
Ireland 1965	IRL	Modified Airy	AM	Ireland	6300	7002	Geodetic Datum of 1965 - Alias - TM65
ISTS 061 Astro 1968	ISG	International 1924	IN	South Georgia Island	6722	7022	South Georgia 1968

ATTACHMENT 2
LOCAL HORIZONTAL REFERENCE DATUM NAME, DATUM CODE, ELLIPSOID LIST, AND OGP REFERENCE

ARINC Datum Name	ARINC Datum Code	Associated Ellipsoid Name	Associated Ellipsoid Code	Typical Location Use	OGP DATUM CODE	OGP Ellipsoid Code	OGP Description/Notes
ISTS 073 Astro 1969	IST	International 1924	IN	Diego Garcia	6724	7022	Diego Garcia 1969
Johnston Island 1961	JOH	International 1924	IN	Johnston Island	6725	7022	
Kandawala	KAN	Everest 1830 (1937 Adjustment)	EA	Sri Lanka	6244	7015	
Kerguelen Island 1949	KEG	International 1924	IN	Kerguelen Island	6698	7022	IGN 1962 Kerguelen - From Kerguelen Island 1949 Datum
Kertau 1948	KEA	Everest	EE	West Malaysia, Singapore	6245	7018	Kertau 1968 - Alias - Malaysia Revised Triangulation 1968
Kusaie Astro 1951	KUS	International 1924	IN	Caroline Islands, Fed. States of Micronesia	6735	7022	Kusaie 1951
L.C. 5 Astro 1961	LCF	Clarke 1866	CC	Cayman Brac Island	6726	7008	Little Cayman 1961
Leigon	LEH	Clarke 1880	CD	Ghana	6250	7012	
Liberia 1964	LIB	Clarke 1880	CD	Liberia	6251	7012	
Luzon	LUZ	Clarke 1866	CC	Philippines	6253	7008	Luzon 1911 Superceded by Philippine Reference System 1992 - Alias - PRS92
M'Poraloko	MPO	Clarke 1880	CD	Gabon	6266	7011	Datum - Clarke 1880 (IGN)
Mahe 1971	MIK	Clarke 1880	CD	Mahe Island	6256	7012	
Manchurian Principal System	MCN	Bessel 1841	BR	Manchuria			
Massawa	MAS	Bessel 1841	BR	Eritrea (Ethiopia)	6262	7004	
Merchich	MER	Clarke 1880 (IGN)	CD	Morocco	6261	7011	
Midway Astro 1961	MID	International 1924	IN	Midway Islands	6727	7022	Midway 1961
Minna	MIN	Clarke 1880 (RGS)	CD	Cameroon, Nigeria	6263	7012	
Montjong Lowe	MOL	Bessel 1841	BR	Sulawesi (Indonesia)			
Montserrat Island Astro 1958	ASM	Clarke 1880 (RGS)	CD	Montserrat, Leeward Islands	6604	7012	Montserrat 1958
Nahrwan	NAH	Clarke 1880 (RGS)	CD	Masirah Island (Omna), United Arab Emirates, Saudi Arabia	6270	7012	Nahrwan 1967 - Datum Clarke 1880 (RGS)
Nanking 1960	NAN	International 1924	IN	China			
Naparima, BWI	NAP	International 1924	IN	Trinidad and Tobago	6271	7022	Naparima 1972

ATTACHMENT 2
LOCAL HORIZONTAL REFERENCE DATUM NAME, DATUM CODE, ELLIPSOID LIST, AND OGP REFERENCE

ARINC Datum Name	ARINC Datum Code	Associated Ellipsoid Name	Associated Ellipsoid Code	Typical Location Use	OGP DATUM CODE	OGP Ellipsoid Code	OGP Description/Notes
North American 1927	NAS	Clarke 1866	CC	Bahamas, Canada, Canal Zone, Caribbean, Central America, Greenland, Mexico, United States	6267	7008	Several versions of NAD27 exist. Care must be taken when converting NAD27 to include the country and in some cases a region within the country for coordiante transformation.
North American 1983	NAR	GRS 1980	RF	Canada, Central America, Mexico, United States	6269	7019	
North Sahara 1959	NSD	Clarke 1880	CD	Algeria	6307	7012	Nord Sahara 1959 - Datum Clarke 1880 (RGS)
Nouvelle Triangulation de France (FRANCE)	IGF	Clarke 1880	CD				
Observatorio Meteorologico 1939	FLO	International 1924	IN	Corvo and Flores Islands (Azores)	6182	7022	Azores Occidental Islands 1939 - Alias - Observatario Flores, Observatorio 1966, Azores Occidental 1939
Old Egyptian 1907	OEG	Helmert 1906	HE	Egypt	6229	7020	Egypt 1907 - Alias Old Egypt superceded by Egypt 1930 - Alias - New Egypt
Old Hawaiian	OHA	Clarke 1866	CC	Hawaiian Islands	6135	7008	
Oman	FAH	Clarke 1880	CD	Oman	6134	7012	PDO Survey Datum 1993 - Datum Clarke 1880 (RGS)
Ordnance Survey of Great Britain 1936	OGB	Airy 1830	AA	England, Isle of Man, Scotland, Shetland Islands, Wales	6277	7001	OSGB 1936 - Updated with OS (SN) 1980 (6279)
Palmer Astro	PAM	International 1924	IN	Antarctica			
Parametrop Zemp 1990 (English translation) The Parameters of the Earth 1990	RPE	PZ-90	PE	Russian Federation Alias – PE90 and SGS90	6740	7054	Used for GLONASS
Pico de las Nieves	PLN	International 1924	IN	Canary Islands	6728	7022	
Pitcairn Astro 1967	PIT	International 1924	IN	Pitcairn Island	6729	7022	Pitcairn 1967
Point 58	PTB	Clarke 1880	CD	Burkina Faso, Niger	6620	7012	Datum - Clarke 1880 (RGS)
Point Noire 1948	PTN	Clarke 1880	CD	Congo	6282	7011	Congo 1960 Pointe Noire - Datum Clarke 1880 (IGN)
Porto Santo 1936	POS	International 1924	IN	Porto Santo and Madeira Islands	6663	7022	Porto Santo 1936 (6615) updated to Porto Santo 1995 (6663)
Portuguese Datum 1973	PRD	International 1924	IN				
Potsdam	PDM	Bessel 1841	IN	Germany	6746	7004	Muliple versions of the Potsdam datum exist, i.e. Postdam Datum/83
Provisional South American 1956	PRP	International 1924	IN	Bolivia, Chile, Colombia, Ecuador, Guyana, Peru, Venezuela	6248	7022	Alias - PSAD56
Provisional South Chilean 1963 (also known as Hito XVIII 1963)	HIT	International 1924	IN	Southern Chile (near 53°S)	6254	7022	Also called Hito XVIII 1963
Puerto Rico	PUR	Clarke 1866	CC	Puerto Rico and Virgin Islands	6139	7008	

ATTACHMENT 2
LOCAL HORIZONTAL REFERENCE DATUM NAME, DATUM CODE, ELLIPSOID LIST, AND OGP REFERENCE

ARINC Datum Name	ARINC Datum Code	Associated Ellipsoid Name	Associated Ellipsoid Code	Typical Location Use	OGP DATUM CODE	OGP Ellipsoid Code	OGP Description/Notes
Pulkovo 1942	PUK	Krassovsky 1940	KA	Russia	6200	7024	Pulkovo 1942 (6284) updated to Pulkovo 1995 (6200)
Qatar National	QAT	International 1924	IN	Qatar	6614	7022	Qatar 1974 (6285) updated to Qatar National Datum 1995 (6614)
Qornoq	QUO	International 1924	IN	South Greenland	6194	7022	Qornoq 1927
Reunion	REU	International 1924	IN	Mascarene Islands	5156	7022	Reunion 1947 (6626) - Alias Piton des Neiges - updated to Reseau Geodesique de la Reunion 1992 (6627)
Rome 1940	MOD	International 1924	IN	Sardinia	6806	7022	Monte Mario (6265) - Alias Rome 1940 updated to Monte Marion (Rome) (6806)
RT90	RTS	Bessel 1841	BR	Sweden - Onshore & Offshore	6124	7004	Rikets Koordinatsystem 1990 (RT90) - Central Meridan = Stockholm
S-42 (Pulkovo 1942)	SPK	Krassovsky 1940	KA	Albania, Czechoslovakia (prior to 1 January 1993), Hungary, Kazakhstan, Latvia, Poland, Romania	6178	7024	Pulkovo 1942/58 (6179) updated to Pulkovo 1995 (6178) - Alias - S42
Santo (DOS) 1965	SAE	International 1924	IN	Espirito Santo Island	6730	7022	Santo 1965 - Alias - Santo (DOS)
Sao Braz	SAO	International 1924	IN	Sao Miguel, Santa Maria Islands (Azores)	6664	7022	Datum Azores Oriental Islands 1940 (6184) updated to Azores Oriental Islands 1995 (6664)
Sapper Hill 1943	SAP	International 1924	IN	East Falkland Island	6292	7022	
Schwarzeck	SCK	Bessel Namibia (GLM)	BN	Namibia	6293	7046	Ellip - Bessel Namibia (GLM)
Selvagem Grande 1938	SGM	International 1924	IN	Salvage Islands	6616	7022	Datum - Alias - Marco Astro, Selvagem Grande
Sierra Leone 1960	SRL	Clarke 1880	CD	Sierra Leone	6175	7012	Ellip - Clarke 1889 (RGS)
S-JTSK	CCD	Bessel 1841	BR	Czechoslovakia (prior to 1 January 1993)	6818	7004	S-JTSK (Ferro)
South American 1969	SAN	GRS 1967 (SAD69)	SA	Argentina, Bolivia, Brazil, Chile, Colombia, Ecuador, Baltra and Galapagos Islands, Guyana, Paraguay, Peru, Trinidad and Tobago, Venezuela	6618	7050	
South Asia	SOA	Modified Fischer 1960	FA	Singapore			
Stockholm 1938 (RT38)	STO	Bessel 1841	BR	Sweden - Onshore & Offshore	6308	7004	Stockholm 1938 (6308) Prime Meridian = Greenwich, superseded by Rikets koordinatsystem 1990 (RT90) (6124)
Sydney Observatory	SYO	Clarke 1858	CA	New South Wales, Australia			
Tananarive Observatory 1925	TAN	International 1924	IN	Madagascar	6297	7022	Tananarive 1925 (6297)

ATTACHMENT 2
LOCAL HORIZONTAL REFERENCE DATUM NAME, DATUM CODE, ELLIPSOID LIST, AND OGP REFERENCE

ARINC Datum Name	ARINC Datum Code	Associated Ellipsoid Name	Associated Ellipsoid Code	Typical Location Use	OGP DATUM CODE	OGP Ellipsoid Code	OGP Description/Notes
Timbalai 1948	TIL	Everest	EB	Brunei and East Malaysia (Sarawak and Sabah)	6298	7016	Ellip - Everest 1830 (1967 Definition)
Tokyo	TOY	Bessel 1841	BR	Japan, Okinawa, South Korea	6301	7004	
Trinidad Trigonometrical Survey	TRI	Clarke 1858	CA	Trinidad and Tobago	6302	7007	Trinidad 1903
Tristan Astro 1968	TDC	International 1924	IN	Tristan da Cunha Island Group	6734	7022	Tristan 1968
Unknown	U						
Viti Levu 1916	MVS	Clarke 1880 (International Foot)	CD	Viti Levu Island (Fiji Islands)	6752	7055	Viti Levu 1916 also know as Viti Levu 1912
Voirol 1874	VOI	Clarke 1880	CD	Tunisia, Algeria	6304	7011	
Voirol 1960	VOR	Clarke 1880	CD	Algeria			
Wake Island Astro 1952	WAK	International 1924	IN	Wake Atoll	6733	7022	Wake Island 1952
Wake-Eniwetok, 1960	ENW	Hough 1960	HO	Marshall Islands	6732	7053	Marshall Islands 1960
World Geodetic System 1960	WGA	WGS-60	WS	Global			
World Geodetic System 1966	WGB	WGS-66	WC	Global			
World Geodetic System 1972	WGC	WGS-72	WD	Global	6322	7043	Alias WGS72
World Geodetic System 1984	WGE	WGS-84	WE	Global	6326	7030	WGS84
Yacare	YAC	Intl 1924	IN	Uruguay	6309	7022	
Zanderij	ZAN	Intl 1924	IN	Suriname	6311	7022	

ATTACHMENT 2A
LOCAL HORIZONTAL REFERENCE DATUM NAME, DATUM CODE, ELLIPSOID LIST, AND OGP REFERENCE

The following is a list of datum names that do not have an assigned datum code.

ARINC Datum Name	ARINC Datum Code	Associated Ellipsoid Name	Associated Ellipsoid Code	Typical Location Use	OGP DATUM CODE	OGP Ellipsoid Code	OGP Description/Notes
Australian Antarctic Datum 1998	Not Assigned	GRS 1980	RF	Antarctica - Australian Sector	6176	7019	Alias - AAD98
Azores Central Islands 1995	Not Assigned	International 1924	IN	Central Azores - Gracisoa, Terceira, Sao Jorge, Pico, Faial	6665	7002	Alias - Base SW, Graciosa, Azores Central 1995
Azores Occidental Islands 1939	Not Assigned	International 1924	IN	Western Azores - Flores, Corvo	6182	7002	Alias - Observatario Flores, Azores Occidental 1939, Observatorio 1966
Azores Oriental Islands 1940	Not Assigned	International 1924	IN	Eastern Azores - Sao Miguel, Santa Maria	6184	7002	Alias - Sao Bras, Azores Oriental 1940
Azores Oriental Islands 1995	Not Assigned	International 1924	IN	Eastern Azores - Sao Miguel, Santa Maria	6664	7002	Alias - Sao Bras, Azores Oriental 1995
Chatham Island Datum 1979	Not Assigned	Internatioanl 1924	IN	Chatham Island (New Zealand)	6673	7022	
Estonia 1992	Not Assigned	GRS 1980	RF	Estonia	6133	7019	Alias -EST92
Estonia 1997	Not Assigned	GRS 1980	RF	Estonia	6180	7019	Alias - EST97
Fiji 1956	Not Assigned	International 1924	IN	Fiji - Viti Levu, Vanua Levu, Taveuni	6721	7022	Supercedes Vitu Levu 1912 and Vanua Levu 1915.
Fiji Geodectic Datum 1986	Not Assigned	WGS 1972	WD	Fiji - Viti Levu, Vanua Levu, Taveuni, Yasawa Group, Kadavu Group, Lau Islands, Rotuma Islands	6720	7043	Supercedes Vitu Levu 1912 and Vanua Levu 1915 and Fiji 1956 - Alias - Fiji 1986, FGD 1986
Geocentric Datum of Australia 1994	Not Assigned	GRS 1980	RF	Mainland Australia+F120	6283	7019	Geocentric Datum of Australia 1994 - Alias - GDA94
Geodetic Datum of Malaysia 2000	Not Assigned	GRS 1980	RF		6751	7019	Supercedes all Malayasian Datums
Hong Kong 1980	Not Assigned	International 1924	IN	Hong Kong	6611	7022	Alias - HK80
International Terrestrial Reference Frame 1988	Not Assigned	GRS 1980	RF		6647	7019	ITRF88
International Terrestrial Reference Frame 1989	Not Assigned	GRS 1980	RF		6648	7019	ITRF89
International Terrestrial Reference Frame 1990	Not Assigned	GRS 1980	RF		6649	7019	ITRF90
International Terrestrial Reference Frame 1991	Not Assigned	GRS 1980	RF		6650	7019	ITRF91
International Terrestrial Reference Frame 1992	Not Assigned	GRS 1980	RF		6651	7019	ITRF92

ATTACHMENT 2A
LOCAL HORIZONTAL REFERENCE DATUM NAME, DATUM CODE, ELLIPSOID LIST, AND OGP REFERENCE

The following is a list of datum names that do not have an assigned datum code.

ARINC Datum Name	ARINC Datum Code	Associated Ellipsoid Name	Associated Ellipsoid Code	Typical Location Use	OGP DATUM CODE	OGP Ellipsoid Code	OGP Description/Notes
International Terrestrial Reference Frame 1993	Not Assigned	GRS 1980	RF		6652	7019	ITRF93
International Terrestrial Reference Frame 1994	Not Assigned	GRS 1980	RF		6653	7019	ITRF94
International Terrestrial Reference Frame 1995	Not Assigned	GRS 1980	RF		6654	7019	ITRF95
International Terrestrial Reference Frame 1996	Not Assigned	GRS 1980	RF		6655	7019	ITRF96
International Terrestrial Reference Frame 1997	Not Assigned	GRS 1980	RF		6656	7019	ITRF97
International Terrestrial Reference Frame 2000	Not Assigned	GRS 1980	RF		6657	7019	ITRF2000
International Terrestrial Reference Frame 2005	Not Assigned	GRS 1980	RF		6658	7019	ITRF2005
Libyan Geodetic Datum 2006	Not Assigned	International 1924	IN		6754		Alias - LGD2006 - 5 stations tied to ITRF2000
Monte Mario (Rome)	Not Assigned	International 1924	IN	Italy	6806	7022	Alias - Rome 1940 (Rome)
New Zealand Geodetic Datum 1949	Not Assigned	International 1924	IN	NZGD49	6272	7022	Alias - GD49, NZGD49
New Zealand Geodetic Datum 2000	Not Assigned	GRS 1980	RF	NZGD2000	6167	7019	Alias - NZGD2000
North American 1983 (High Accuracy Regional Network)	Not Assigned	GRS 1980	RF	USA, South Pacific Islands, Caribbean Islands			Alias - NAD83(HARN), NAD83(HPGN)
Egypt 1930	Not Assigned	Internatioanl 1924	IN	Egypt	6199	7022	Egypt 1930 - Alias - New Egypt - supercedes Old Egypt 1907
Ordnance Survey of Great Britain 1970 (SN)	Not Assigned	Airy 1830	AA	UK	6278	7001	Alias - OSGB70
Philippine Reference System 1992	Not Assigned	Clarke 1866	CC	Philippines - Onshore and Offshore.	6683	7008	Replaced Luzon 1911
Potsdam Datum/83	Not Assigned Not Assigned	Bessel 1841	BR	Germany	6746	704	Alias - PD/83
Pulkovo 1942/58	Not Assigned	Krassovsky 1940	KA	Poland	6179	7024	

ATTACHMENT 2A
LOCAL HORIZONTAL REFERENCE DATUM NAME, DATUM CODE, ELLIPSOID LIST, AND OGP REFERENCE

The following is a list of datum names that do not have an assigned datum code.

ARINC Datum Name	ARINC Datum Code	Associated Ellipsoid Name	Associated Ellipsoid Code	Typical Location Use	OGP DATUM CODE	OGP Ellipsoid Code	OGP Description/Notes
Pulkovo 1942/83	Not Assigned	Krassovsky 1940	KA	Former East Germany	6178	7024	
Pulkovo 1995	Not Assigned	Krassovsky 1940	KA	Russian Federation	6200	7022	
Qatar National Datum 1995	Not Assigned	International 1924	IN	Qatar	6614	7022	Alias - QND95
Reseau Geodesique de la Reunion 1992	Not Assigned	GRS 1980	RF	Reunion Islands	6627	7019	Alias - RGR92 - Supercedes Reunion 1947 -
Sierra Leone 1968	Not Assigned	Clarke 1880 (RGS)	CD	Sierra Leone	6175	7012	
SIRGAS 1995	Not Assigned	GRS 1980	RT	Central and South America	6170	7019	Tied to ITRF94 - Superceded by SIRGAS 2000
SIRGAS 2000	Not Assigned	GRS 1980	RT	Central and South America	6674	7019	Also called Sistema de Referencia Geocentrico para America del Sur 2000, tied to ITRF2000
Stockholm 1938 (Stockholm) (RT38 - Stockholm)	Not Assigned	Bessel 1841	BR	Sweden - Onshore	6814	7004	Rikets Koordinatsystem 1938 Stockholm - Prime Meridan = Stockholm
SWEREF99	Not Assigned	GRS 1980	RT	Sweden - Onshore & Offshore	6619	7019	Densification of ETRS89 - Equilv to WGS84
Porto Santo 1995	Not Assigned	International 1924	IN	Porto Santo and Madeira Islands	6663	7022	Update of Porto Santo 1936 - Alias - Base SE, Madeira SE Base 1995

ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

The following pages show a sample computer printout of an ARINC 424 data file. Each record in an ARINC 424 file is included. Relationships between various record types within the data file has been maintained, e.g., the Navaids and Waypoints used on the Enroute Airways are available in the appropriate Sections/Subsections.

The following ARINC 424 data file is for example only.

ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

MINIMUM OFF ROUTE ALTITUDE (MORA) (AS)

S	AS	N36W120	10516813509514310410411611210512211710713915511611107305204604603403003303703503803002702300018708
S	AS	N36W150	01001001001001001001001001001001001001001001001001001001001001001001001001001001001001001000028508
S	AS	N37W120	15516612911811811712713713113713716516616416716009507505304904103703703703703203104708390200038902
S	AS	N37W150	01001001001001001001001001001001001001001001001001001001001001001001001001001001001001001000046057082
S	AS	N38W120	14813614114313915412114514013915112916716816514708407205304504004104203202803002803102602800058708
S	AS	N38W150	01001001001001001001001001001001001001001001001001001001001001001001001001001001001001001000046071036127
S	AS	N39W120	13111113812913114414412413125119132165168166117086074057049042036031037037032025029030023000788708
S	AS	N39W150	01001001001001001001001001001001001001001001001001001001001001001001001001001001001001001000098095116
S	AS	N40W120	11312212112013712810913414115914612014714616608907805905004604604404203102802802702320230240009987708
S	AS	N40W150	01001001001001001001001001001001001001001001001001001001001001001001001001001001001001001000054113109128103
S	AS	N41W120	10911712112713213112311712312111911913414412009107906005604904504104903904003104102603403200118708
S	AS	N41W150	01001001001001001001001001001001001001001001001001001001001001001001001001001001001001001000041113165104122

CRUISING TABLE (TC)

S	TCA01	36001790M	020000200028000280000300031000310000400OUNLTD	000358706
S	TCA02	18003590M	010000200029000290000400OUNLTD	000368706
S	TCA11	36003590M	29000020004300O	000378810
S	TCA21	36003590M	28000020004200O	000388810
S	TCA31	36001790M	30000020004000O	000398810
S	TCA32	18003590M	29000020004100O	000408810
S	TCAA1	36001790M	010000200029000290000400OUNLTD	000418706
S	TCAA2	18003590M	020000200028000280000300031000310000400OUNLTD	000428706
S	TCBB1	03002090M	010000200029000290000400OUNLTD	000438810
S	TCBB2	21000290M	020000200028000280000300031000310000400OUNLTD	000448903
S	TCC11	36000890M	010000200029000330000400OUNLTD	000458810
S	TCC12	09001790M	015000200027500300000400OUNLTD	000468810
S	TCC13	18002690M	020000200028000310000400OUNLTD	000478810
S	TCC14	27003590M	025000200028500320000400OUNLTD	000488810
S	TCCC1	36001790M	010000200029000290000400OUNLTD	000498810
S	TCCC2	18003590M	020000200028000280000300031000310000400OUNLTD	000508810

The following ARINC 424 data file is for example only.

ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

VHF NAVAID (D)
(With Simulation and Flight Planning Continuations)

SUSAD	ACV	K211020VDTA	N40585370W124062570	E0170001910	256NASARCATA		015638502
SUSAD	ACV	K22 TB030100+040000B100040					015648502
SUSAD	ACV	K23PKZSEKZSE					015658613
SUSAD	ACV	K24S UY		E017500191			015668502
SUSAD	AHC	K2110900VDTW	N40160466W120090328	E0170040080	206NASAMEDEE (HERLONG)		015678811
SUSAD	AHC	K22PKZOAKZOA					015698613
SUSAD	AHC	K23S UN		E016604008			015708201
SUSAD	ALW	K1111640VDLA	N46051360W118172930	E0200011501	NASWALLA WALLA		015718413
SUSAD	ALW	K12PKZSEKZSE					015738613
SUSAD	ALW	K13S UY		E018401150			015748413
SUSAD	AST	K1111400VDLA	N46094270W123524480	E0190000101	330NASASTORIA		015758809
SUSAD	AST	K12PKZSEKZSE					015778613
SUSAD	AST	K13S UY		E019700010			015788406
SUSAD	AVE	K2111710VTH	N35384925W119583948	E0160007102	567NASAVENAL		015798401
SUSAD	AVE	K22PKZLAKZLA					015818613
SUSAD	AVE	K23S UY		E015000710			015828401
SUSAD	BTG	K1111660VTHA	N45445270W122352520	E0210002502	451NASBATTLE GROUND		015998801
SUSAD	BTG	K12PKZSEKZSE					016018801
SUSAD	BTG	K13S UY		E019300250			016028801
SUSAD	ELN	K1111790VTHA	N47012830W120272620	E0210017702	451NASELLENSBURG		016398304
SUSAD	ELN	K12PKZSEKZSE					016418613
SUSAD	ELN	K13S UY		E019401770			016428304
SUSAD	ENI	K2111230VTHA	N39031200W123162310	E0160029802	411NASMENDOCINO		016438807
SUSAD	ENI	K22PKZOAKZOA					016458613
SUSAD	ENI	K23S UY		E016702980			016468409
SUSAD	EPH	K1111260VTHA	N47224100W119252230	E0210012502	394NASEPHRATA		016478411
SUSAD	EPH	K12PKZSEKZSE					016498613
SUSAD	EPH	K13S UY		E019301250			016508411
SUSAD	HOD	K2111460VDH	N37373880W120572460	E0170000902	571NASMODESTO		017398608
SUSAD	HOD	K22PKZOAKZOA					017418613
SUSAD	HOD	K23S UY		E015800090			017428608
SUSAD	MVA	K2111510VTHA	N38335535W118015484	E0170078602	476NASMINA		017438306
SUSAD	MVA	K22PKZOAKZOA					017458613
SUSAD	MVA	K23S UY		E015507860			017468306
SUSAD	NUQ	K2111760 TL	NUQ	N37255680W122032320E0170000041	314NASNAVY MOFFETT FIELD		017758806
SUSAD	NUQ	K22PKZOAKZOA					017778806
SUSAD	NUQ	K23S UY		E0159			017788806
SUSAD	NUW	K1111380 TH	NUW	N48211838W122393600E0200000502	325NASNAVY WHIDBEY ISLAND		017798806
SUSAD	NUW	K12PKZSEKZSE					017818806
SUSAD	NUW	K13S UY		E0205			017828806
SUSAD	OAK	K2111680VTH	N37433360W122132100	N37433360W122132100E0170000103	477NASOAKLAND		017838110
SUSAD	OAK	K22PKZOAKZOA					017858613
SUSAD	OAK	K23S UY		E016000010			017868110
SUSAD	OED	K1111360VTHA	N42284710W122544250	N42284710W122544250E0190020802	410NASMEDFORD		017918411
SUSAD	OED	K12PKZSEKZSE					017938613
SUSAD	OED	K13S UY		E017902080			017948411

The following ARINC 424 data file is for example only.

ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

VHF NAVAID (D)
(With Simulation and Flight Planning Continuations)
(Continued)

```
SUSAD     UBG  K111174OVTHA N45211220W122583700    N45211220W122583700E0210014402  493NASNEWBERG                019078110
SUSAD     UBG  K12PKZSEKZSE                                                                                    019098613
SUSAD     UBG  K13S    UY                                    E0191014440                                      019108110
SUSAD     YKM  K111160OVTHA N46341350W120263640    N46341350W120263640E0210009802  525NASYAKIMA                 019158502
SUSAD     YKM  K12PKZSEKZSE                                                                                    019178613
SUSAD     YKM  K13S    UY                                    E0192009800                                      019188502
SUSAD KSEAK1 ISZI K1111170 IT          ISZIN47260947W122183980      003660  588NASSEATTLE-TACOMA INTL            019588808
SUSAD KSEAK1 ISZI K12PKZSEKZSE                                                                                 019598713
SUSAD KSEAK1 ISZI K13S    UN  L                               E0200000366                                      019608901
```

NDB NAVAID (DB)
(With Simulation and Flight Planning Continuations)

```
SUSADB    ARU  K2102150H MW N41281600W120332500              E0180        NASALTURAS                          019768110
SUSADB    ARU  K22S    U21                                                                                    019788110
SUSADB    ARU  K23PKZSEKZSE                                                                                   019798110
SUSADB    CAN  K1102740H MW N47243880W122501510              E0200        NASCARNEY (BREMERTON)               019848308
SUSADB    CAN  K12S    U21                                                                                    019868308
SUSADB    CAN  K13PKZSEKZSE                                                                                   019878308
SUSADB    CC   K2103350HOMW N38024740W122015640              E0170        NASKANAN                            019888807
SUSADB    CC   K22S    U21                                                                                    019908807
SUSADB    CC   K23PKZOAKZOA                                                                                   019918807
SUSADB    F    K2103140H  W N37415400W123001200              E0170        NASFARALLON ISLAND                  020088110
SUSADB    F    K22S    U21                                                                                    020108110
SUSADB    F    K23PKZOAKZOA                                                                                   020118110
SUSADB    MOG  K2103820H  A N41433840W122285050              E0200        NASMONTAGUE                         020768110
SUSADB    MOG  K22S    U21                                      02620                                       020788110
SUSADB    MOG  K23PKZSEKZSE                                                                                   020798110
```

The following ARINC 424 data file is for example only.

ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

ENROUTE WAYPOINT (EA)
(With Flight Planning Continuations)

```
SUSAEAENRT  26FLW K21  I D   N36442340W121282270          E0156  NAS        B  FLW306/D126          021528110
SUSAEAENRT  26FLW K22PKZOAKZOA                                                                    021538701
SUSAEAENRT  ALFOR K11    R F L N44183310W123090510       E0187  NAS        P  ALFOR               021648207
SUSAEAENRT  ALFOR K12PKZSEKZSE                                                                    021658613
SUSAEAENRT  ALTAM K21    R Z L N37484410W121444580       E0160  NAS        P  ALTAM               021668613
SUSAEAENRT  ALTAM K22PKZOAKZOA                                                                    021678701
SUSAEAENRT  BRINY K21    RF  L N37181740W122393800       E0159  NAS        P  BRINY               022408303
SUSAEAENRT  BRINY K22PKZOAKZOA                                                                    022418701
SUSAEAENRT  BTG32 K11    I   L N45335140W121524960       E0190  NAS        B  BTG089/DLS234       022488802
SUSAEAENRT  BTG32 K12PKZSEKZSE                                                                    022498802
SUSAEAENRT  BTG51 K11    I   L N44522830W122361410       E0189  NAS        B  BTG160/EUG010        022508802
SUSAEAENRT  BTG51 K12PKZSEKZSE                                                                    022518802
SUSAEAENRT  LOFAL K11    R Z L N47503790W122401980       E0203  NAS        P  LOFAL               025738409
SUSAEAENRT  LOFAL K12PKZSEKZSE                                                                    025748613
SUSAEAENRT  MOGNB K21    N   B N41433840W122285050       E0175  NAS        Q  MONTAGUE            026278110
SUSAEAENRT  MOGNB K22PKZOAKZOA                                                                    023858613
SUSAEAENRT  ODESS K11    R Z H N47081310W117582330       E0188  NAS        P  ODESS               026708304
SUSAEAENRT  ODESS K12PKZSEKZSE                                                                    026718613
SUSAEAENRT  SHOEY K21    R Z L N36444462W122075863       E0157  NAS        P  SHOEY               028438904
SUSAEAENRT  SHOEY K22PKZOAKZOA                                                                    028448701
```

ENROUTE MARKER (EM)

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SUSAEM      K101 K10.-.  EL  N46123100W123575100129O    E0210  NAS        FORT STEVENS          029988706
SUSAEM      K104 K10.-.  EL  N43072800W123205410177O    E0200  NAS        WINSTON               029998613
```

HOLDING (EP)

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SUSAEPENRT          10AVE  K2D 01300R 1518000FL450                                  AVENAL          030038904
SUSAEPENRT          10ODESSK1EA02530R 10      17999                                 ODESS           030048904
SUSAEPENRT          20ALTANK2EA01770L 10050001799916O                               ALTAM           030058904
SUSAEPENRT          20BTG  K1D 01490R 10      17999                                 BATTLE GROUND   030068904
SUSAEPENRT          20DED  K1D 03370R 10      17999                                 MEDFORD         030078904
SUSAEPENRT          20UBG  K1D 00030L 10      17999                                 NEWBERG         030088904
SUSAEPENRT          21BRINYK2EA00570R 10      17999                                 BRINY           030098904
SUSAEPENRT          22BRINYK2EA02370R 10      17999                                 BRINY           030108904
SUSAEPENRT          30EPH  K1D 02460R 10      17999                                 EPHRATA         030118904
SUSAEPENRT          40EPH  K1D 02460R 10      17999                                 EPHRATA         030128904
SUSAEPENRT          40UBG  K1D 00030L 10      17999                                 NEWBERG         030148904
SUSAEPENRT          50MOD  K2D 02440L 10      17999                                 MODESTO         030158904
SUSAEPENRT          50ELN  K1D 02500R 10      17999                                 ELLENSBURG      030168904
SUSAEPKSEAK1        40FINNYK1PC00230R 05001500090001 50                             FINNY           030178904
SUSAEPKSEAK1        50FINNYK1PC00230R 05001500090001 50                             FINNY           030188904
```

The following ARINC 424 data file is for example only.

ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

ENROUTE AIRWAY (ER)

SUSAER	C1415	0010FOT K2D 0V CB AA		252012400000	UNKNN	FL450				030308803
SUSAER	C1415	0020REDOOK2EAOEEC CB		000000002510						030318702
SUSAER	C1416	0010FOT K2D 0V CB AA		305013300000	UNKNN	FL450				030328704
SUSAER	C1416	0020DAASHK2EAOEEC CB		000000003050						030338902
SUSAER	C1418	0010HQM K1D 0V CB AA		210013800000	UNKNN	FL450				030348704
SUSAER	C1418	0020SEDARK1EAOEEC CB		000000002100						030358110
SUSAER	C1419	00100NP K1D 0V CB AA		216013700000	UNKNN	FL450				030368704
SUSAER	C1419	0020HEMLOK1EAOEEC CB		000000002160						030378110
SUSAER	C1486	0010ENI K2D 0V C CB AA		285019600000	UNKNN	FL450				030388803
SUSAER	C1486	0020REDOOK2EAOEEC CB		000000002850						030398803
SUSAER	J1	05700AK K2D 0V C OH AA		343014203010	1800022000	FL450				030408704
SUSAER	J1	0580RBL K2D 0V C OH AA		330014603420	1800022000	FL450				030418704
SUSAER	J1	0590QED K1D 0V C OH AA		345019703290	1800022000	FL450				030428704
SUSAER	J1	0600BTG K1D 0V OH AA		345006503430	1800022000	FL450				030438801
SUSAER	J1	0610ALDERK1EAOT H OH AA		345003703450	1800022000	FL450				030448704
SUSAER	J1	0620SEA K1D 0VEC OH		000000003450						030458207
SUSAER	J110	00100AK K2D 0V C OH AA		138007000000	18000	FL450				030468704
SUSAER	J110	0030SNS K2D 0V OH AA		064008801380	18000	FL450				030478704
SUSAER	J110	0040CZQ K2D 0V C OH AA		086002600670	24000	FL450				030488704
SUSAER	J110	0050PINNIK2EAOE OH AA		086003400860	24000	FL450				030498704
SUSAER	J110	0055MITELK2EAOR OH AA		086011700860	24000	FL450				030508704
SUSAER	J20	0210PDT K1D 0V C OH AA		290008202750	18000	FL450				030828704
SUSAER	J20	0220YKM K1D 0V OH AAY		284005402880	18000	FL450				030838704
SUSAER	J20	0225RADDYK1EAOR H OH AAY		281003902840	18000	FL450				030848704
SUSAER	J20	0230SEA K1D 0VEC OH		000000002810						030858110
SUSAER	J3	00100AK K2D 0V C OH AA		343014200000	18000	FL450				030868704
SUSAER	J3	0020RBL K2D 0V C OH AAY		010016303420	18000	FL450				030878704
SUSAER	J3	0030LKV K1D 0V C OH AAY		356013400100	18000	FL450				030888704
SUSAER	J3	0040IMB K1D 0V C OH AA		006019503550	18000	FL450				030898704
SUSAER	V105	0220YERINK2EAOE OL AA		299003102990	10000	17999				031678704
SUSAER	V105	0230CHIMEK2EAOE OL AA		299001502990	10000	17999				031688704
SUSAER	V105	0240FMG K2D 0VEC OL		000000002990						031698603
SUSAER	V107	0120CITIEK2EAOR OL AA		313002003130	07000	17999				031708704
SUSAER	V107	0130PXN K2D 0V C OL AA		296003403130	07000	17999				031718704
SUSAER	V107	0140CATMEK2EAOE OL AA		296002602960	07000	17999				031728704
SUSAER	V107	0150VINCOK2EAOR OL AA		294000502940	06000	17999				031738704
SUSAER	V107	0160MABRYK2EAOR OL AA		294000602940	05500	17999				031748704
SUSAER	V107	0170WISOMK2EAOT OL AA		294000502940	04500	17999				031758704
SUSAER	V107	0180INPLYK2EAOT OL AA		294000502940	04500	17999				031768704
SUSAER	V107	0190DECOTK2EAOT OL AA		294001102940	04500	17999				031778704
SUSAER	V107	02000AK K2D 0V C OL AA		288001602940	05000	17999				031788704
SUSAER	V107	0210COWMOK2EAOE OL AA		288000802880	05000	17999				031798704
SUSAER	V107	0220WICRAK2EAOT OL AA		288001302880	05000	17999				031808704
SUSAER	V107	0230PYE K2D 0V C OL AA		289001502880	05000	17999				031818704
SUSAER	V107	0240BOARSK2EAOEE OL		000000002890						031828206

ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

The following ARINC 424 data file is for example only.

ENROUTE AIRWAY (ER)
(Continued)

SUSAER	V108	0050STS	K2D 0V	0L AA	118002900000 04500	17999			031838704
SUSAER	V108	0060SGD	K2D 0V	0L AA	114001101170 03000	17999			031848704
SUSAER	V108	0070CROITK2EA0E		0L AA	079000701140 03000	17999			031858704
SUSAER	V108	0080CCR	K2D 0V	0L AA	071000700790 03000	17999			031868704
SUSAER	V108	0090PITTSK2EA0E		0L AA	071001100710 03500	17999			031878902
SUSAER	V108	0100OAKEYK2EA0E		0L AA	071001600710 02000	17999			031888902
SUSAER	V108	0110LOODIK2EA0E		0L AA	071001500710 02000	17999			031898704
SUSAER	V108	0120LIN	K2D 0VEC	0L	000000000710				031908110

ENROUTE AIRWAY RESTRICTION RECORD (EU)

SUSAEUJ20	001TCOYKM	K1D SEA	K1D 31JAN9031MAR90CS1707001700			A0	031918913
SUSAEUJ3	001NR1RBL	K2D IMB	K1D 01JAN9031DEC90AVBL ONE WAY RBL TO IMB DURING LAKEVIEW AFB				031928913
SUSAEUJ3	001NR2		MORNING AND EVENING SCRAMBLE ACTIVITY				031938913

ENROUTE COMMUNICATIONS (EV)

SUSAEVALW	MOSES LAKE		FSS0012240 V1	R	A	N47123960W119185640E0193	Y			039188810
SUSAEVALW	MOSES LAKE		FSS0012240 V2					WALLA WALLA		039198810
SUSAEVALW	MOSES LAKE		TWE0011500 V1		A	N47123960W119185640E021001177Y			MWH K1D	039208810
SUSAEVALW	MOSES LAKE		TWE0011500 V2							039218810
SUSAEVKZSEZRFTHE	DALLES		ACC0011965 V1	DGRA	N45425000W121055900E0190	Y				041568810
SUSAEVKZSEZRFTHE	DALLES		ACC0011965 V2					SEATTLE		041578810
SUSAEVKZSEZRFYAKIMA			ACC0012030 V1	GRA	N46314000W120304600E0192	Y				041628810
SUSAEVKZSEZRFYAKIMA			ACC0012030 V2					SEATTLE		041638810
SUSAEVKZSEZRFYAKIMA			ACC0013260 V1	DGRA	N46314000W120304600E0192	Y				041648810
SUSAEVKZSEZRFYAKIMA			ACC0013260 V2					SEATTLE		041658810
SUSAEVKZSEZRUYAKIMA			ACC0012030 V1	GRA	N46314000W120304600E0192	Y				041948810
SUSAEVKZSEZRUYAKIMA			ACC0012030 V2					SEATTLE		041958810
SUSAEVKZSEZRUYAKIMA			ACC0013475 V1	GRA	N46314000W120304600E0192	Y				041968810
SUSAEVKZSEZRUYAKIMA			ACC0013475 V2					SEATTLE		041978810
SUSAEVSEA	THE DALLES		FSS0012200 V1E	R	A	N45425000W121055900E021003220N			DLS K1D	045168810
SUSAEVSEA	THE DALLES		FSS0012200 V2	H	1507002300			SEATTLE		045178810
SUSAEVSEA	THE DALLES		FSS0012265 V1	R	A	N45425000W121055900E021003220Y			DLS K1D	045188810
SUSAEVSEA	THE DALLES		FSS0012265 V2					SEATTLE		045198810
SUSAEVSEA	THE DALLES		TWE0011230 V1		A	N45425000W121055900E021003220N			DLS K1D	045208810
SUSAEVSEA	THE DALLES		TWE0011230 V2	N						045218810
USUAEVSEA	THE DALLAS		TWE0011230 V3		SPRING AND FALL PERIODS ONLY					045228810
SUSAEVSEA	YAKIMA		FSS0012250 V1	R	A	N46313300W120315200E0192	Y			045288810
SUSAEVSEA	YAKIMA		FSS0012250 V2					SEATTLE		045298810
SUSAEVSEA	YAKIMA		TWE0011600 V1		A	N46341350W120263640E021000980N			YKM K1D	045308810
SUSAEVSEA	YAKIMA		TWE0011600 V2							045318810

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ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

```
                                    HELIPORT (HA)
                            (With Flight Planning Continuations)
SUSAH KKENK1AKAHHEL01110000NASY N47440610W12215330OE01930099625OSEA K11800018000CUOOY050100  KENMORE AIR TERMINAL      045328807
SUSAH KKENK1AKAH     2PKZSEKZSE                                                                                          045338807

                                 HELIPORT COMMUNICATIONS (HV)
SUSAP KKENK1VTWR0011990 V0     A N47265700W122182910E019900429Y          00      KENMORE                                 045348810
SUSAP KKENK1VUNI0012295 V0     A N47265700W122182910E019900429Y          00      KENMORE TERMINAL                        045358811

                                      AIRPORT (PA)
                            (With Flight Planning Continuations)
SUSAP KSEAK1ASEA     110000119Y N47265700W122182910E019900429250SEA K11800018000CUOOY NAS    SEATTLE-TACOMA INTL      045698808
SUSAP KSEAK1ASEA     2PKZSEKZSE                                                                                          045718808

                                 TERMINAL WAYPOINT (PC)
                            (With Flight Planning Continuations)
SUSAP KSEAK1CANVIL K11    RCF   N47370820W122183010          E0201          P  ANVIL                                  045728804
SUSAP KSEAK1CANVIL K12PKZSEKZSE                                                                                       045738613
SUSAP KSEAK1CBEAVR K11    R D   N44371085W122183020          E0201          P  BEAVR                                  045748803
SUSAP KSEAK1CBEAVR K12PKZSEKZSE                                                                                       045758613
SUSAP KSEAK1CBISSL K11    R E   N47103130W121315750          E0197          P  BISSL                                  045768304
SUSAP KSEAK1CBISSL K12PKZSEKZSE                                                                                       045778613
SUSAP KSEAK1CDONDO K11    D F   N47215090W122182790          E0200          O  OM RW34R DONDO                         045788110
SUSAP KSEAK1CDONDO K12PKZSEKZSE                                                                                       045798613
SUSAP KSEAK1CFACTS K11    R Z   N47090850W122183020          E0199          P  FACTS                                  045808304
SUSAP KSEAK1CFACTS K12PKZSEKZSE                                                                                       045818613
SUSAP KSEAK1CFF16L K11    IAF   N47315580W122183020          E0201          T  SEA338/D5.8                            045828804
SUSAP KSEAK1CFF16L K12PKZSEKZSE                                                                                       045838613
SUSAP KSEAK1CFF34L K11    IAF   N47215010W122183020          E0200          T  SEA158/D4.3                            045848804
SUSAP KSEAK1CFF34L K12PKZSEKZSE                                                                                       045858613
SUSAP KSEAK1CFINNY K11    R E   N46374050W122183020          E0196          P  FINNY                                  045868304
SUSAP KSEAK1CFINNY K12PKZSEKZSE                                                                                       045878613
SUSAP KSEAK1CGRAME K11    R E   N46560520W122183020          E0198          P  GRAME                                  045888304
SUSAP KSEAK1CGRAME K12PKZSEKZSE                                                                                       045898613
SUSAP KSEAK1CMILLT K11    RCF   N47150990W122183020          E0199          P  MILLT                                  045908804
SUSAP KSEAK1CMILLT K12PKZSEKZSE                                                                                       045918613
SUSAP KSEAK1CPARKK K11    RAF   N47315720W122182060          E0201          P  PARKK                                  045928804
SUSAP KSEAK1CPARKK K12PKZSEKZSE                                                                                       045938613
SUSAP KSEAK1CTHUNN K11    R E   N47060860W122183020          E0198          P  THUNN                                  045948811
SUSAP KSEAK1CTHUNN K12PKZSEKZSE                                                                                       045958613
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The following ARINC 424 data file is for example only.

ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

STANDARD INSTRUMENT DEPARTURES (SIDs) (PD)
(With Flight Planning Continuations)

SUSAP	KSEAK1DMOUNT12ALL	010ELN	K1D	1VE	IF		18000	045968507
SUSAP	KSEAK1DMOUNT12ALL	010ELN	K1D	2P		0000		045978801
SUSAP	KSEAK1DMOUNT13GEG	010ELN	K1D	1V	IF		18000	045988507
SUSAP	KSEAK1DMOUNT13GEG	010ELN	K1D	2P		0000		045998801
SUSAP	KSEAK1DMOUNT13GEG	020HAMURK1EA1E			TF			046008413
SUSAP	KSEAK1DMOUNT13GEG	020HAMURK1EA2P				0630		046018613
SUSAP	KSEAK1DMOUNT13GEG	030GEG	K1D	1VE	TF			046028413
SUSAP	KSEAK1DMOUNT13GEG	030GEG	K1D	2P		0630		046038613
SUSAP	KSEAK1DMOUNT13MLP	010ELN	K1D	1V	IF		18000	046048507
SUSAP	KSEAK1DMOUNT13MLP	010ELN	K1D	2P		0000		046058801
SUSAP	KSEAK1DMOUNT13MLP	020HAMURK1EA1E			TF			046068413
SUSAP	KSEAK1DMOUNT13MLP	020HAMURK1EA2P				0630		046078613
SUSAP	KSEAK1DMOUNT13MLP	030DESSK1EA1E			TF			046088413
SUSAP	KSEAK1DMOUNT13MLP	030DESSK1EA2P				0400		046098613
SUSAP	KSEAK1DMOUNT13MLP	040MLP	K1D	1VE	TF			046108413
SUSAP	KSEAK1DMOUNT13MLP	040MLP	K1D	2P		0970		046118613
SUSAP	KSEAK1DMOUNT13ODESS	010ELN	K1D	1V	IF		18000	046128507
SUSAP	KSEAK1DMOUNT13ODESS	010ELN	K1D	2P		0000		046138801
SUSAP	KSEAK1DMOUNT13ODESS	020HAMURK1EA1E			TF			046148413
SUSAP	KSEAK1DMOUNT13ODESS	020HAMURK1EA2P				0630		046158613
SUSAP	KSEAK1DMOUNT13ODESS	030DESSK1EA1EE			TF			046168413
SUSAP	KSEAK1DMOUNT13ODESS	030DESSK1EA2P				0400		046178613

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ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

STANDARD INSTRUMENT ARRIVALS (STARs) (PE)
(With Flight Planning Continuations)

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SUSAP KSEAK1EELN2 1GEG  010GEG  K1D 1V      IF                    18000                                  046328508
SUSAP KSEAK1EELN2 1GEG  010GEG  K1D 2P                  0000                                           046338804
SUSAP KSEAK1EELN2 1GEG  020HAMURK1EA1E  TF                                                             046348508
SUSAP KSEAK1EELN2 1GEG  020HAMURK1EA2P                 0630                                           046358613
SUSAP KSEAK1EELN2 1GEG  030ELN  K1D 1VE H  TF                                                          046368508
SUSAP KSEAK1EELN2 1GEG  030ELN  K1D 2P                 0630                                           046378613
SUSAP KSEAK1EELN2 1HAMUR 010HAMURK1EA1E  IF                    18000                                  046388508
SUSAP KSEAK1EELN2 1HAMUR 010HAMURK1EA2P                0000                                           046398804
SUSAP KSEAK1EELN2 1HAMUR 020ELN  K1D 1VE H  TF                                                         046408508
SUSAP KSEAK1EELN2 1HAMUR 020ELN  K1D 2P                0630                                           046418613
SUSAP KSEAK1EELN2 1MLP  010MLP  K1D 1V      IF                 18000                                  046428508
SUSAP KSEAK1EELN2 1MLP  010MLP  K1D 2P                 0000                                           046438804
SUSAP KSEAK1EELN2 1MLP  020DESSK1EA1E  H  TF                                                           046448613
SUSAP KSEAK1EELN2 1MLP  020DESSK1EA2P                  0970                                           046458613
SUSAP KSEAK1EELN2 1MLP  030HAMURK1EA1E  TF                                                             046468508
SUSAP KSEAK1EELN2 1MLP  030HAMURK1EA2P                 0400                                           046478613
SUSAP KSEAK1EELN2 1MLP  040ELN  K1D 1VE H  TF                                                          046488508
SUSAP KSEAK1EELN2 1MLP  040ELN  K1D 2P                 0630                                           046498613
SUSAP KSEAK1EELN2 1DDESS 010DDESSK1EA1E  H  IF                 18000                                  046508613
SUSAP KSEAK1EELN2 1DDESS 010DDESSK1EA2P                0000                                           046518804
SUSAP KSEAK1EELN2 1DDESS 020HAMURK1EA1E  TF                                                            046528508
SUSAP KSEAK1EELN2 1DDESS 020HAMURK1EA2P                0400                                           046538613
SUSAP KSEAK1EELN2 1DDESS 030ELN  K1D 1VE H  TF                                                         046548508
SUSAP KSEAK1EELN2 1DDESS 030ELN  K1D 2P                0630                                           046558613
SUSAP KSEAK1EELN2 2RW34B 010ELN  K1D 1V  H  IF                 18000                                  046568508
SUSAP KSEAK1EELN2 2RW34B 010ELN  K1D 2P                0000                                           046578804
SUSAP KSEAK1EELN2 2RW34B 020BISSLK1PC1E  TF                                                            046588508
SUSAP KSEAK1EELN2 2RW34B 020BISSLK1PC2P                0450                                           046598613
SUSAP KSEAK1EELN2 2RW34B 030         1      VD SEA K1      26000300          250                      046608508
SUSAP KSEAK1EELN2 2RW34B 030         2P                    0060                                        046618613
SUSAP KSEAK1EELN2 2RW34B 040         1      VD SEA K1      26000250  10000                             046628508
SUSAP KSEAK1EELN2 2RW34B 040         2P                    0050                                        046638613
SUSAP KSEAK1EELN2 2RW34B 050KSEA K1PA1AE    VM             2600                                        046648508
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The following ARINC 424 data file is for example only.

ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

INSTRUMENT APPROACH PROCEDURES (PF)
(With Flight Planning Continuations)

```
SUSAP KSEAK1F116R APAE  010PAE  K1D 1V      FC PAE K1  00000000016100131 + 02000   18000                                    047128504
SUSAP KSEAK1F116R APAE  010PAE  K1D 2P                                 0131                                                047138613
SUSAP KSEAK1F116R APAE  020ANVILK1PC1EE     CF ISZIK1  33830110161000040 + 02000                                          047148401
SUSAP KSEAK1F116R APAE  020ANVILK1PC2P                                  0040                                                047158613
SUSAP KSEAK1F116R I     010ANVILK1PC1E  I   IF ISZIK1  33830110          0000 I 020000190018000                          047168504
SUSAP KSEAK1F116R I     010ANVILK1PC2P                                  0000                                            047178804
SUSAP KSEAK1F116R I     020PARKKK1PC1E  F   CF ISZIK1  33970058158000052 G 0190001803      000PARKK K1PC                   047188904
SUSAP KSEAK1F116R I     020PARKKK1PC2P                                  0052                                            047198613
SUSAP KSEAK1F116R I     030RW16RK1PG1G      CF ISZIK1  33830017158000041 01809                                           047208506
SUSAP KSEAK1F116R I     030RW16RK1PG2P                                  0041                                            047218613
SUSAP KSEAK1F116R I     040OONDOK1PC1E  M   CF SEA K1  15770043157200060 01800                                           047228506
SUSAP KSEAK1F116R I     040OONDOK1PC2P                                  0060                                            047238613
SUSAP KSEAK1F116R I     050OONDOK1PC1EE HR  HM         33807010                                                         047248506
SUSAP KSEAK1F116R I     050OONDOK1PC2P                                  0000                                            047258506
SUSAP KSEAK1FV16L APAE  010PAE  K1D 1V      FC PAE K1  00000000016200123 + 02100   18000                                    047448709
SUSAP KSEAK1FV16L APAE  010PAE  K1D 2P                                 0123                                                047458613
SUSAP KSEAK1FV16L APAE  020FF16LK1PC1EE     CF PAE K1  16030223158000100 + 02100                                        047468709
SUSAP KSEAK1FV16L APAE  020FF16LK1PC2P                                  0100                                            047478613
SUSAP KSEAK1FV16L ASEA  010SEA  K1D 1V      IF                                18000                                    047488602
SUSAP KSEAK1FV16L ASEA  010SEA  K1D 2P                                  0000                                            047498804
SUSAP KSEAK1FV16L ASEA  020FF16LK1PC1E      CF SEA K1  33800058339000058 + 02100                                        047508709
SUSAP KSEAK1FV16L ASEA  020FF16LK1PC2P                                  0058                                            047518613
SUSAP KSEAK1FV16L ASEA  030FF16LK1PC1E  R   PI SEA K1  33800058293000100 + 02100                                        047528709
SUSAP KSEAK1FV16L ASEA  030FF16LK1PC2P                                  0100                                            047538613
SUSAP KSEAK1FV16L ASEA  040FF16LK1PC1EE     CF SEA K1  33800058158001000 + 01800                                        047548304
SUSAP KSEAK1FV16L ASEA  040FF16LK1PC2P                                  0100                                            047558613
SUSAP KSEAK1FV16L V     020FF16LK1PC1E  F   IF SEA K1  33800058          0000  01800   18000      SEA  K1D                047568904
SUSAP KSEAK1FV16L V     020FF16LK1PC2P                                                                                   047578804
SUSAP KSEAK1FV16L V     030RW16LK1PG1G      CF SEA K1  34070016158000041 00478                    -310                    047588507
SUSAP KSEAK1FV16L V     030RW16LK1PG2P                                  0041                                            047598613
SUSAP KSEAK1FV16L V     040OONDOK1PC1E  M   CF SEA K1  15770043158500059 + 01800                                        047608711
SUSAP KSEAK1FV16L V     040OONDOK1PC2P                                  0059                                            047618613
SUSAP KSEAK1FV16L V     050OONDOK1PC1EE HR  HM         33807010                                                         047628507
SUSAP KSEAK1FV16L V     050OONDOK1PC2P                                  0000                                            047638804
```

RUNWAY (PG)
(With Simulation Continuations)

```
SUSAP KSEAK1GRW16L 1119001604 W4727454W122182351      00428049050150   0000                                            047888808
SUSAP KSEAK1GRW16L 2S                  18040N         L00428                                                           047908808
SUSAP KSEAK1GRW16R 1094251604 W4727503SW122183511     00426000055150 ISZI 0000                                        047918808
SUSAP KSEAK1GRW16R 2S                  18030N         L00426                                                           047938808
SUSAP KSEAK1GRW34L 1094253404 W4726173SW122183593     00359000050150   0000                                            047948808
SUSAP KSEAK1GRW34L 2S                  00030N         L00359                                                           047968808
SUSAP KSEAK1GRW34R 1119003404 W4725528GW122182451     00343000064150 ISEA 0000                                        047978808
SUSAP KSEAK1GRW34R 2S                  00040N         L00343                                                           047998808
```

ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

The following ARINC 424 data file is for example only.

```
                                  LOCALIZER/GLIDE SLOPE (PI)
                                (With Simulation Continuations)
SUSAP KSEAK1I1SEA1  111030RW34RN47275488W12218234233380N47260403W12218185904641134033127SE02206400352 048008808
SUSAP KSEAK1I1SEA1  2S    U Y                 36000N                                          140       048018505
SUSAP KSEAK1I1SZI2  111170RW16RN47260944W12218360015800N47273932W12218410307991119039530OE02205500421 048028313
SUSAP KSEAK1I1SZI2  2S    U N                 18000N                                          140       048038313

                                             MLS (PL)

SUSAP KSEAK1WWSZ11  1516  RW34RN47275430W12218233033800N47260400W12218190004601130040040040040120E02200550027533S 048048808
SUSAP KSEAK1WWSZ11  2     S  HEN47260940W12218360015800N47270510W12218384004600200200200201700Y3500Y56  048058808

                                        LOCALIZER MARKER (PM)

SUSAP KSEAK1WISEA MW 0     RW34RN47251830W12218248000040                                  E0220     048068808
SUSAP KSEAK1WISEALOM 002240RW34RN47215090W12218279000040472150909W12218279OHOMW  U21 SE E0220     048078802
SUSAP KSEAK1WISZI IM 0     RW16RN47275920W12218346018030                                  E0220     048088613
SUSAP KSEAK1WISZI MW 0     RW16RN47282060W12218351018030                                  E0220     048098613
SUSAP KSEAK1WISZILOM 002810RW16RN47315720W12218206018030472315720W12218206OHOMW  U21 S2 E0220     048108802

                                   MINIMUM SECTOR ALTITUDE (MSA) (PS)

SUSAP KSEAK1SDOWDOK1PC          0 25180062270071360034                                     048118612
SUSAP KSEAK1SPARKKK1PC          0 25180062360045                                           048128704
SUSAP KSEAK1SSEA  K1D           0 25180062360034                                           048138612

                                   AIRPORT COMMUNICATIONS (PV)

SUSAP KSEAK1VAPP0011920 V0   RA N47265700W12218291OE01990042 9Y070158B040001000OKSEAK1PA 00 SEATTLE              048148810
SUSAP KSEAK1VAPP0011950 V0L DRA N47265700W12218291OE01990042 9Y261306      KSEAK1PA 00 SEATTLE              048158811
SUSAP KSEAK1VAT10012800 V0     A N47265700W12218291OE01990042 9Y                       00                    048168810
SUSAP KSEAK1VCPT0012800 V0P    A N47265700W12218291OE01990042 9Y                       00 SEATTLE            048178810
SUSAP KSEAK1VDEP0011920 V0   RA N47265700W12218291OE01990042 9Y070158  KSEAK1PA 00 SEATTLE              048188811
SUSAP KSEAK1VDEP0011950 V0 S RA N47265700W12218291OE01990042 9Y261306  KSEAK1PA 00 SEATTLE              048198810
SUSAP KSEAK1VGND0012170 V0     A N47265700W12218291OE01990042 9Y                       00 SEATTLE            048228810
SUSAP KSEAK1VGTE0012625 V0     A N47265700W12218291OE01990042 9Y                       00 GATE CONTROL       048238810
SUSAP KSEAK1VTCA0011920 V1   RA N47265700W12218291OE01990042 9Y                       00 SEATTLE APPROACH   048248904
SUSAP KSEAK1VTCA0011920 V2 RW16 070-140, RW34 280-069                                      048258904
SUSAP KSEAK1VTCA0011950 V1   RA N47265700W12218291OE01990042 9Y141279 KSEAK1PA 00 SEATTLE APPROACH   048268811
SUSAP KSEAK1VTCA0011950 V2 RWY 34                                                          048278810
SUSAP KSEAK1VTWR0011990 V0     A N47265700W12218291OE01990042 9Y                       00 SEATTLE            048328810
SUSAP KSEAK1VUN10012295 V0     A N47265700W12218291OE01990042 9Y                       00 SEATTLE-TACOMA INTL 048338811
```

The following ARINC 424 data file is for example only.

ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

AIRPORT (PA)
(With Flight Planning Continuations)

```
SUSAP KYKMK1AYKM     110000076Y N46340610W12032330OE01930109525OYKM K11800018000MUOOY NAS    YAKIMA AIR TERMINAL 054488807
SUSAP KYKMK1AYKM     2PK2SEKZSE                                                                                  054508807
```

TERMINAL WAYPOINT (PC)
(With Flight Planning Continuations)

```
SUSAP KYKMK1CC127    K11    ICF   N46290320W120123080          E0191          T  IYKM089/015.5                 054518903
SUSAP KYKMK1CC127    K12PK2SEKZSE                                                                                054738903
```

TERMINAL NDB NAVAID (PN)
(With Simulation and Flight Planning Continuations)

```
SUSAPNKSEAK1 SE    K11033350HOMW N47215090W122182790          E0220          NASSEATTLE                     019888807
SUSAPNKSEAK1 SE    K12S     U21                                                                                019908807
SUSAPNKSEAK1 SE    K13PKZSEKZSE                                                                               019918807
SUSAPNKSEAK1 S2    K11031400HOMW N47315720W122182060          E0220          NASSEATTLE                     020088110
SUSAPNKSEAK1 S2    K12S     U21                                                                               020108110
SUSAPNKSEAK1 S2    K13PKZSEKZSE                                                                               020118110
```

FIR/UIR (UF)

```
                                                                  17999          AA SEATTLE
SUSAUFKZSEZQZXF00100C2VR  13N G N48200000W128000000                                                           059648809
SUSAUFKZSEZQZXF00200C2VR    H N48300000W125000000                                                             059658804
SUSAUFKZSEZQZXF00300C2VR    G N48300000W124450000                                                             059668809
SUSAUFKZSEZQZXF00400C2VR    G N48132015W123314105                                                             059678809
SUSAUFKZSEZQZXF00500C2VR    G N48163607W123150796                                                             059688809
SUSAUFKZSEZQZXF00600C2VR    G N48251928W123065473                                                             059698809
SUSAUFKZSEZQZXF00700C2VR    G N48412300W123151982                                                             059708809
SUSAUFKZSEZQZXF00800C2VR    G N48455181W122595674                                                             059718809
SUSAUFKZSEZQZXF00900C2VR    G N48491916W122595780                                                             059728809
SUSAUFKZSEZQZXF01000C2VR    H N49000000W123182562                                                             059738804
SUSAUFKZSEZQZXF01050C2VR    H N49000000W121300000                                                             059748804
SUSAUFKZSEZQZXF01100C2VR    H N49000000W121000000                                                             059758804
SUSAUFKZSEZQZXF01200C2VR    H N49000000W120300000                                                             059768804
SUSAUFKZSEZQZXF01300C2VR    H N49000000W120000000                                                             059778804
SUSAUFKZSEZQZXF01400C2VR    H N49000000W119300000                                                             059788804
SUSAUFKZSEZQZXF01500C2VR    H N49000000W119000000                                                             059798804
SUSAUFKZSEZQZXF01600C2VR    H N49000000W118300000                                                             059808804
SUSAUFKZSEZQZXF01700C2VR    H N49000000W118000000                                                             059818804
SUSAUFKZSEZQZXF01800C2VR    H N49000000W117300000                                                             059828804
```

The following ARINC 424 data file is for example only.

ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

FIR/UIR (UF)
(Continued)

```
SUSAUFKZSEZQZXF01900CZVR H N49000000W117000000                                059838804
SUSAUFKZSEZQZXF02000CZVR H N49000000W116330000                                059848804
SUSAUFKZSEZQZXF02100CZVR H N49000000W116300000                                059858804
SUSAUFKZSEZQZXF02200CZVR H N49000000W116000000                                059868804
SUSAUFKZSEZQZXF02300CZEG H N49000000W115300000                                059878804
SUSAUFKZSEZQZXF02400CZEG H N49000000W115000000                                059888804
SUSAUFKZSEZQZXF02500CZEG G N49000000W114400000                                059898809
SUSAUFKZSEZQZXF02600KZLC G N48250000W115000000                                059908804
SUSAUFKZSEZQZXF02700KZLC H N45200000W115000000                                059918804
SUSAUFKZSEZQZXF02800KZLC G N45200000W117450000                                059928809
SUSAUFKZSEZQZXF02900KZLC G N44510000W118270000                                059938809
SUSAUFKZSEZQZXF03000KZLC G N43380000W119170000                                059948809
SUSAUFKZSEZQZXF03100KZLC G N42400000W119000000                                059958809
SUSAUFKZSEZQZXF03200KZLC H N41000000W119300000                                059968804
SUSAUFKZSEZQZXF03300KZOA G N41000000W121150000                                059978809
SUSAUFKZSEZQZXF03400KZOA H N41200000W122250000                                059988804
SUSAUFKZSEZQZXF03500KZOA H N41200000W123000000                                059998804
SUSAUFKZSEZQZXF03600KZOA G N41200000W123320000                                060008804
SUSAUFKZSEZQZXF03700KZOA G N40231500W123320000                                060018809
SUSAUFKZSEZQZXF03800KZOA H N40130000W123500000                                060028804
SUSAUFKZSEZQZXF03900KZOA G N40130000W125200000                                060038809
SUSAUFKZSEZQZXF04000KZOA G N40590000W126540000                                060048809
SUSAUFKZSEZQZXF04050KZOA G N43180800W126404600                                060058812
SUSAUFKZSEZQZXF04100KZOA G N45000000W126300000                                060068809
SUSAUFKZSEZQZXF04200KZOA GEN45302800W126425900                                060078812
```

The following ARINC 424 data file is for example only.

ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

FIR/UIR (UF)
(Continued)

18000FL600 AA SEATTLE

```
SUSAUFKZSEZQZXU00100 CZVR11N G N48200000W128000000                    060088809
SUSAUFKZSEZQZXU00200 CZVR   H N48300000W125000000                    060098804
SUSAUFKZSEZQZXU00300 CZVR   G N48300000W124450000                    060108809
SUSAUFKZSEZQZXU00400 CZVR   G N48132015W123314105                    060118809
SUSAUFKZSEZQZXU00500 CZVR   G N48163607W123150796                    060128809
SUSAUFKZSEZQZXU00600 CZVR   G N48251928W123065473                    060138809
SUSAUFKZSEZQZXU00700 CZVR   G N48412300W123151982                    060148809
SUSAUFKZSEZQZXU00800 CZVR   G N48455181W122595674                    060158809
SUSAUFKZSEZQZXU00900 CZVR   G N48491916W122595780                    060168809
SUSAUFKZSEZQZXU01000 CZVR   H N49000000W123182562                    060178804
SUSAUFKZSEZQZXU01100 CZVR   H N49000000W121300000                    060188804
SUSAUFKZSEZQZXU01200 CZVR   H N49000000W121000000                    060198804
SUSAUFKZSEZQZXU01300 CZVR   H N49000000W120300000                    060208804
SUSAUFKZSEZQZXU01400 CZVR   H N49000000W120000000                    060218804
SUSAUFKZSEZQZXU01500 CZVR   H N49000000W119300000                    060228804
SUSAUFKZSEZQZXU01600 CZVR   H N49000000W119000000                    060238804
SUSAUFKZSEZQZXU01700 CZVR   H N49000000W118300000                    060248804
SUSAUFKZSEZQZXU01800 CZVR   H N49000000W118000000                    060258804
SUSAUFKZSEZQZXU01900 CZVR   H N49000000W117300000                    060268804
SUSAUFKZSEZQZXU02000 CZVR   H N49000000W117000000                    060278804
SUSAUFKZSEZQZXU02100 CZVR   H N49000000W116300000                    060288804
SUSAUFKZSEZQZXU02200 CZVR   H N49000000W116000000                    060298804
SUSAUFKZSEZQZXU02300 CZEG   H N49000000W115300000                    060308804
SUSAUFKZSEZQZXU02400 CZEG   H N49000000W115000000                    060318804
SUSAUFKZSEZQZXU02500 CZEG   G N49000000W114400000                    060328809
SUSAUFKZSEZQZXU02600 KZLC   G N48250000W115000000                    060338809
SUSAUFKZSEZQZXU02700 KZLC   H N45200000W115000000                    060348804
SUSAUFKZSEZQZXU02800 KZLC   G N45200000W117450000                    060358809
SUSAUFKZSEZQZXU02900 KZLC   G N44510000W118270000                    060368812
SUSAUFKZSEZQZXU03000 KZLC   G N43380000W119170000                    060378809
SUSAUFKZSEZQZXU03100 KZLC   G N42400000W119000000                    060388809
SUSAUFKZSEZQZXU03200 KZOA   H N41000000W119300000                    060398804
SUSAUFKZSEZQZXU03300 KZOA   H N41000000W121150000                    060408809
SUSAUFKZSEZQZXU03400 KZOA   H N41200000W122250000                    060418804
SUSAUFKZSEZQZXU03500 KZOA   H N41200000W123000000                    060428812
SUSAUFKZSEZQZXU03600 KZOA   G N41200000W123320000                    060438812
SUSAUFKZSEZQZXU03700 KZOA   G N40231500W123320000                    060448812
SUSAUFKZSEZQZXU03800 KZOA   H N40130000W123500000                    060458804
SUSAUFKZSEZQZXU03900 KZOA   G N40130000W125200000                    060468809
SUSAUFKZSEZQZXU04000 KZOA   G N40590000W126540000                    060478809
SUSAUFKZSEZQZXU04100 KZOA   G N43180800W126404600                    060488812
SUSAUFKZSEZQZXU04200 KZOA   G N45000000W126300000                    060498809
SUSAUFKZSEZQZXU04300 KZOA   GEN45302800W126425900                    060508812
```

The following ARINC 424 data file is for example only.

ATTACHMENT 3
NAVIGATION DATA FILE RELATIONSHIPS

```
                                RESTRICTIVE AIRSPACE (UR)

SUSAURK1A680     A00101LN   CE                N48110000W1223800000030     GND   03000MA-680                                  060518807
SUSAURK1A680     A00102T T1512002400150001020O                                                                               060528808
SUSAURK1A680     B00101LN   CE                N48110000W1223800000030     03000M10000MA-680                            060538807
SUSAURK1A680     B00102T T1512002400150001013O                                                                               060548808
SUSAURK1MCHINOOK A 00101LN   G N48060300W122371500                        00300M05000MCHINOOK A                        060558807
SUSAURK1MCHINOOK A 00102H T1707001700                                                          FAA SEATTLE ARTCC         060568811
SUSAURK1MCHINOOK A 00200     G N48055500W122341600                                                                      060578806
SUSAURK1MCHINOOK A 00300     G N47520700W122363100                                                                      060588806
SUSAURK1MCHINOOK A 00400     GEN47522100W122393000                                                                      060598806
SUSAURK1R5704      00101LN   G N45520000W119290000                        GND   03999MR-5704                           062498807
SUSAURK1R5704      00102T S1509001700                                                                                      062468807
SUSAURK1R5704      00200     H N45500000W119290000                                                                      062478807
SUSAURK1R5704      00300     G N45500000W119303000                                                                      062488807
SUSAURK1R5704      00400     HEN45520000W119303000                                                                      062498807
SUSAURK1W460A      00101B N  G N46430000W128490000                        GND   UNLTD W-460A                          064108807
SUSAURK1W460A      00102                                                                       FAA OAKLAND ARTCC        064118807
SUSAURK1W460A      00200     G N47013000W127230000                                                                      064128807
SUSAURK1W460A      00300     G N46080000W127000000                                                                      064138807
SUSAURK1W460A      00400     GEN45500000W128270000                                                                      064148807

                                                                 GATES (PB)

TXYZP KSEAK1BABCDE 0         N47263000W122180600                                CENTER CONCOURSE B737-300064158813
TXYZP KSEAK1BNORTH 0         N47274200W122180600                                NORTH APRON NOSE IN-STAND064168813
```

ATTACHMENT 4
AIRWAY MINIMUM ALTITUDES AND REQUIRED NAVIGATION PERFORMANCE (RNP)

Airway Minimum Altitude Coding

A. An ARINC 424 Data Base may contain three "levels" of Enroute Airways. These are "High," "Low" and "Both" Level routes. The following descriptions apply:

1. High Altitude Airways, Airway Level code of "H," shall contain:

   a. Routes that exist only in the upper airspace as officially designated by the appropriate authority.

   b. Routes that are officially designated as "Upper" or "High" even though the structure in which they exist has not been officially established as "Upper Airspace."

   c. Routes that, by virtue of the assigned MEA or MFA, must be charted as high level routes.

2. Both Altitude Airways. Airway Level code of "B," shall contain:

   a. Routes that are not specifically defined into either the upper or lower airspace in a structure that does recognize these airspace divisions, for example the "Control Routes" in the USA and CAN coverages.

   b. Routes that exist without a "level designator" that are in a structure that does recognize the division of Upper and Lower Airspace.

   c. Routes that exist in a structure that has Upper and Lower Airspace when such routes have a MEA or MFA assigned lower than the upper limit of Lower Airspace and a MAA above the upper limit of Lower Airspace.

3. Low Altitude Airways. Airway Level code of "L," shall contain:

   a. Routes that exist only the lower airspace as officially established by the appropriate authority.

   b. Routes that, by virtue of the published MAA, must be charted in lower airspace only.

4. Enroute Airway Sequencing.

   Airways changing from one level to another level will be sequenced in order as any airway in the same level. The Airway Level Code is not used to sort airways in an ARINC 424 data base.

   When an airway changes from Airway Level Code "B" to two separate airways that are coded as "L" and "H," the point of change will carry the "B" in the level field.

B. High Altitude Airways.

   The altitude information shown on "High Level" records will be established with the following criteria:

1. The altitude information included for High Altitude Airways will be derived from official government source. The values entered for "Minimum Altitude" will be published MEAs (Minimum Enroute Altitude) or MFAs (Minimum Flight Altitude). If neither of those two values are available through source documentation, the lower limit of the designated upper airspace will be entered.

2. There are two Minimum Altitude fields. The second of these is only used when an Enroute Airway has been published with "Directional MEAs" or "Directional MFAs." "Directional" information is considered to exist when the difference in altitude in opposing flight directions is higher than would be indicated by normal separation standards.

3. For Enroute Airways published with non-standard separation or blocked altitudes, the first Minimum Altitude field will contain the lowest altitude available. The non-standard separation and/or blocked altitude information will be available in the "Cruise Table" referenced in the Enroute Airway Record.

4. The Maximum Altitude field will contain the highest useable altitude for the Enroute Airway Segment. This will be equal to the Upper Limit of the Designated Upper Airspace unless a lower altitude, a "MAA" or Maximum Authorized Altitude, has been published in the official government source.

ATTACHMENT 4
AIRWAY MINIMUM ALTITUDES AND REQUIRED NAVIGATION PERFORMANCE (RNP)

C. Low Level and "Both" Level Airways.

The altitude information shown on "Both Level" and "Low Level" records will be established with the following criteria:

1. The altitude information included for "Both" Altitude and "Low" Altitude Airways will be derived from official government source. The values entered for "Minimum Altitude" will be published MEAs (Minimum Enroute Altitude) or MFAs (Minimum Flight Altitude) when such are available. If neither of those two values are available through source documentation, a code indicating one of the following two conditions will be used:

NESTB - MEA/MFA not established in source documentation. Used when the source does not establish minimum altitudes as a general rule. Also used when source documentation does provide minimum altitude information as a general rule and has explicitly not established a value for a specific route segment or segments.

UNKNN - MEA/MFA Minimum Altitude was unknown at the time the data base was produced but the source documentation does provide MEA or MFA as a general rule. The data base supplier expects that future source documentation will provide some minimum altitude information.

2. There are two Minimum Altitude fields. The second of these is only used when an Enroute Airway has been published with "Directional MEAs" or "Directional MFAs." "Directional" information is considered to exist when the difference in altitude in opposing flight directions is higher than would be indicated by normal separation standards. "Directional" altitudes will not be provided for "NESTB" and "UNKNN."

3. For Enroute Airways published with non-standard separation or blocked altitudes, the first Minimum Altitude field will contain the lowest altitude available. The non-standard separation and/or blocked altitude information will be available in the "Cruise Table" referenced in the Enroute Airway Record.

4. The Maximum Altitude field will contain the highest useable altitude for the Enroute Airway Segment. This will be equal to the highest available Flight Level in the Designated Airspace in which the route is available (Low Altitude Structure or "Both" Altitude Structure) unless a lower altitude, a "MAA" or Maximum Authorized Altitude, has been published in the official government source.

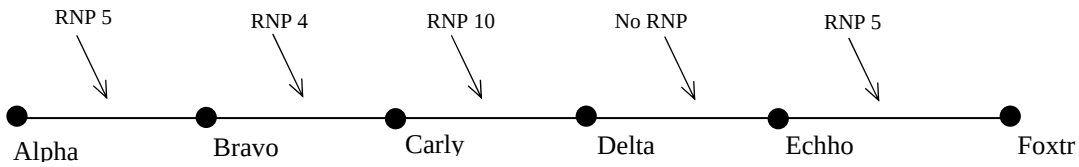
RNP Coding

RNP values are supplied inbound to the fix in the airway sequence record, when viewed in increasing sequence number order. The RNP applies only to the airway leg on which it is supplied. Like-values in subsequent sequences will be repeated in the airway record. If no RNP values is supplied on a segment, there is not a database specified RNP for that segment.

Examples of coding:

Airway Sequence Number	Airway Fix Ident	Airway Segment RNP
010	ALPHA	Blank
020	BRAVO	050
030	CARLY	040
040	DELTA	100
050	ECHHO	Blank
060	FOXTR	050

ATTACHMENT 4
AIRWAY MINIMUM ALTITUDES AND REQUIRED NAVIGATION PERFORMANCE (RNP)



Explanation:

Sequence 010 has a “blank” RNP field. In the example, sequence 010 is the first sequence of the airway and there is no “inbound to the fix” data

Sequence 020 has a coded RNP value of 050, meaning that a RNP of 5.0NM applies to the segment defined by the waypoints ALPHA and BRAVO, regardless of the direction flown.

Sequence 030 has a coded RNP value of 040, meaning that a RNP of 4.0NM applies to the segment defined by the waypoints BRAVO and CARLY, regardless of the direction flown

Sequence 040 has a coded RNP value of 100, meaning that a RNP of 10.0NM applies to the segment defined by the waypoints CARLY and DELTA, regardless of the direction flown

Sequence 050 has a “blank” RNP field, meaning there is no published RNP for the segment defined by the waypoints DELTA and ECHHO, regardless of the direction flown

Sequence 060 has a coded RNP value of 050, meaning that a RNP of 5.0NM applies to the segment defined by the waypoints ECHHO and FOXTR, regardless of the direction flown. In this example, sequence 060 is the last sequence of the airway.

ATTACHMENT 5
PATH AND TERMINATOR

Throughout this attachment many rules and standards for the preparation of coding for Terminal Procedures (SIDs/STARs/Approaches) from official government source documentation into the ARINC Navigation Database format have been documented. These rules and standards use the words “must” and “will” as defined below:

“MUST” = Obligation, no other choice.
“WILL” = Desired, decision by data authority implied.

The Path and Terminator concept is a means to permit coding of Terminal Area Procedures, SIDs, STARs and Approach Procedures, without proliferating the number of named waypoints required to support such procedures. Although it is the intent of this attachment to provide consistent rules, where a contradiction exists between a general rule and a specific rule, the specific rule must be used. The Path and Terminator concept includes a set of defined codes referred to as Path Terminators. Each Code defines a specific type of flight path and a specific type of termination of that flight path. Path Terminators are assigned to all SIDs, STARs and Approach Procedure segments in accordance with the rules set forth in this Attachment. This Attachment also includes rules regarding leg data fields associated with each Path Terminator.

It is desirable that all navigation systems be designed to accept all leg types defined in this Attachment. However, as this Attachment has been dynamic, with new leg types being added, it may be required or desirable not to implement all leg types in any given navigation airborne system. An example of this is the heading (VX) legs versus the course (CX) legs, which were added later. A given system may not have the CX legs (with the exception of the CF leg) implemented. Coding by database suppliers must be accomplished using Path Terminators most appropriately reflecting the official government source documentation.

COMMENTARY

> The use of XA legs is required to maintain consistency with published instrument procedure instructions. It is recognized that the length of XA legs (CA, FA, or VA) is highly dependent upon the aircraft performance and the altitude situation and therefore may cause unusual path generation cases.

Unless otherwise specifically stated, all of the rules, information and guidelines in this Attachment apply equally to fixed-wing and rotor-wing terminal procedures.

The RF Leg type, added with Supplement 11, was introduced with the guidelines listed below.

The RF Leg is to be used only in the following cases:

1. When coding procedure types which were designed with the RF Constant Radius Turn capability as a design criteria.

2. When coding procedure types which were not designed with the RF Leg capability as a criteria as long as both the original coding and the RF Leg specific coding are available and uniquely identified.

3. When coding procedure types which were not designed with the RF Leg capability as a criteria but which cannot be coded using other path terminators.

In order to achieve these coding rule goals, and to ultimately simplify the path terminator matrix currently required to define present-day terminal area procedures, it is in the interest of all user airlines to prevail upon their government agencies and ATC authorities to:

A. Permit FMS-equipped airplanes to fly tracks instead of procedural headings and,

B. Design Terminal area procedures to be compatible with the capabilities of the increasing number of FMS-Equipped airplanes entering service.

The Path and Terminator concept is to accommodate the performance capabilities of various fixed-wing aircraft types. Airmass Path and Terminator constraints are generally for fixed-wing aircraft only. In order to accomplish this, certain values are established for coding the Path and Termination for fixed-wing aircraft. These values have been established to allow data base suppliers to code turn and distance fields to a single set of rules. If an official government source specifies values other than these established values, source data will be used.

1. Distance to Calculation

   A speed of 210 knots, ground speed, will be used to compute distance based on time (3.5 NM per minute). On “Course Reversal” Path Terminators, if no time or distance is specified, a minimum distance of 4.3 NM will be used prior to turning inbound.

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PATH AND TERMINATOR

2. Bank Angle

   A maximum bank angle of 25 degrees will be used to compute turn radius. A full 180 degree turn would require a minimum of 4 nautical miles in diameter at 250 kt. ground speed.

3. Climb Rate

   A climb rate of 500 feet per nautical mile will be used for computations. For missed approach, the climb rate must begin at the Missed Approach Point. For departure procedures, the climb rate must begin at the take-off end of the runway unless otherwise specified by source.

4. Outbound Leg Length for Teardrop Procedures

   If no distance limit is given, or if a time is given, use the following table to determine the length of the outbound leg.

Angle of Divergence	Nautical Miles	Outbound Time
18	10.5	2:45
20	9.5	2:30
22	8.6	2:15
24	7.9	2:00
26	7.3	1:55
28	6.8	1:45
30	6.3	1:40
32	5.9	1:30
34	5.6	1:28
36	5.3	1:23
38	5.0	1:18
40	4.7	1:14
42	4.5	1:10
44	4.3	1:07

This table is based on a speed of 210 knots and a Density Altitude of 5000 Feet. Any procedure that does not fall within this table would not be coded.

5. Intercept Angles

   When the government source does not specify the intercept angle, the following angles must be used:

   A. Use the angle determined in accordance with Rule 6.3.6 of this attachment on approach transitions to intercept the localizer approach path.

   B. Use 30 to 45 degrees on all other procedures.

   C. For procedures other than approach transitions to intercept the localizer approach path, use a VI Path Terminator and 30 to 45 degrees intercept if there is a fix termination in the current leg followed by a 3NM or greater gap between start of turn and the track in the leg to be intercepted.

1.0 General Rules

1.1 NAVAID related Leg Types

Specific leg types require a reference Navaid. The details of which leg types and which Navaid types are to be used can be seen in Section 5.23 of the main document and Table 3 of this Attachment.

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PATH AND TERMINATOR

1.2 Beginning and Ending Leg Types

The Beginning and Ending Leg of a SID, STAR or Approach Route will be selected from the following table.

Note: In general, the same Beginning and Ending Leg types will be used for Helicopter SIDs, STARs and Approaches as listed for fixed-wing aircraft. No additional types are authorized and use of the authorized types should be consistent with helicopter flight capabilities.

Procedure	Beginning Leg	Ending Leg
SID Runway Transition Route Type 0, 1,and T	CA, CD, CF, CI, CR, DF, FA, FC, FD FM, IF, VA, VD, VI, VM, VR	AF, CF, DF, FM, HA, HM[2], RF, TF, VM
SID Route Type 2,	CA[3], CD[3], CF[3], CI[3], CR[3], DF[3], FA, FC, FD, FM, IF , VA[3], VD[3], VI[3], VM[8], VR[3]	AF, CF, DF, FM, HA, IF[4], TF, RF, VM
SID Enroute Transition Route Type 3, or V	FA, FC, FD, IF	AF, CF, DF, HA, RF, TF,
STAR Enroute Transition Route Type 1,	FC, FD, IF	AF, CF, DF, HM, RF, TF
STAR Route Type 2,	FC, FD, IF	AF, CF, DF, FM, IF[4], RF, TF, VM
STAR Runway Transition Route Type 3	FC, FD, HF, IF	AF, CF, FM, HF, HM, RF, TF, VM, **IF[9]**
Approach Transition	FC, FD, HF, IF, PI	AF, CF, CI[5], HF, HM, PI, RF, TF, VI[5]
Final Approach **Coding**	IF	CF, RF, TF[6],
Missed Approach	AF[7], CA, CD, CF, CI, CR[7] DF, FA, FC, FD, FM, HA, HM, RF[7], TF, VA, VD, VI, VM, VR	AF, CA, CF, DF, FM, HM, RF, TF, VA, VM,

Explanation of Notes in Table

1 When followed by a CF or DF leg or when Route Type is "T," Vector SID.

2 When Route Type is "0," Engine Out SID.

3 When SID Procedure has NO Runway Transitions

4 When "IF" leg is the one and only record in the SID/STAR route.

5 When Approach Transition is localizer based.

6 When Final Approach **coding** is GPS or some types of MLS or other specific cases where it has been determined that a "TF" is more satisfactory than a "CF."

7 When "AF" or "RF" are published to begin at the published Missed Approach Point.

8 When preceded by runway coded as an IF leg for SID runway transition.

9 For a STAR that has explicit runway transitions for only a subset of the runways to which it applies then: For other applicable runways, the data supplier may code single leg runway transitions consisting of an IF leg at the common point of the defined runway transitions

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1.3 Leg Sequencing

The following table defines the permitted leg sequences within the individual procedures. A shaded space indicates that the "current leg/next leg" sequence is not permitted within individual procedure routes.

CURRENT LEG \ NEXT LEG	AF	CA	CD[2]	CF[1]	CI	CR	DF	FA[2]	FC	FD	FM[3]	HA	HF	HM	IF	PI	RF	TF	VA[2]	VD	VI	VM[3]	VR
AF																							
CA																							
CD		@	@	@	@	@	@	@	@	@	@				@				@	@	@	@	@
CF		@	@	@	@	@	@&	@	@	@	@								@	@	@	@	@
CI				@				@	@	@	@				@								
CR		@	@	@	@	@	@	@	@	@	@				@				@	@	@	@	@
DF							&																
FA																							
FC		@	@	@	@	@	@&												@	@	@	@	@
FD		@	@	@	@	@	@												@	@	@	@	@
FM																							
HA																							
HF																							
HM																							
IF								*	*	*	*	*	*	*		*							
PI																							
RF																							
TF							&																
VA															@								
VD		@	@	@	@	@	@	@	@	@	@				@				@	@	@	@	@
VI				@				@	@	@	@												
VM																							
VR		@	@	@	@	@	@	@	@	@	@				@				@	@	@	@	@

* = The IF leg is coded only when the altitude constraints at each end of the "FX," "HX" or "PI" leg are different.

& = A CF/DF, DF/DF, TF/DF, or FC/DF sequence should only be used when the termination of the first leg must be overflown, otherwise alternative coding should be used. See Rule 3.1 in this attachment.

@ = Leg combinations for which the first leg is allowed to be coded with a conditional altitude termination, See Rule 1.3.1 in this attachment and Section 5.29.

[1] = TF is the preferred leg type in RNAV Terminal Procedures that are not using ground-based navaid references. The CF leg remains available for these procedures based on rules contained elsewhere in this Attachment.

[2] = The altitude termination leg types on RNAV Terminal Procedures that are not using ground-based navaid references are primarily used as the first leg ofa SID or SID Runway Transition and Missed Approach.

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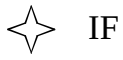
[3] = The manual termination leg types on RNAV Terminal Procedures that are not using ground based navaid references are primarily used as the ending legs of STARs and Missed Approach Procedures and in the definition of SID intial climb-out with ATC intervention.

1.3.1 Conditional Leg Sequencing

The Altitude Description Code (Section 5.29) defines a "conditional altitude termination" of a leg. In such cases, the Leg Type and Leg Sequencing used must respect both the coded leg and leg combination rules and the conditional altitude termination leg combination rules. The "conditional altitude termination" may only be used on Cx leg, Vx legs, and on distance terminated legs (FD and FC). The "conditional altitude termination" may not be used on Hx or xM legs. The result is that the "conditional termination may only be used when the coded leg type can be translated to an altitude termination leg (other than Hx and xM). This means that the "conditional altitude termination" may not be used with certain leg types. For example, there is no "arc to an altitude termination leg" in the valid legs table, Section 1.4, therefore the "conditional altitude termination" may not be used on AF legs, in any combination. This also means that the leg sequencing used must be valid for both the coded leg and the "conditional altitude termination" leg. For example, while CD/AF is a valid coded leg combination, it is not valid when the "conditional altitude termination" is applied on the CD leg as the resulting CA/AF combination is invalid. Given the purpose and intent of the "conditional altitude termination" coding, it is also invalid on any leg for which the primary termination is "altitude".

1.4 Leg Type Descriptions

The following illustrations on the various Leg Types are provided to assist coding and decoding of the Path and Terminator concept.

Leg Code	Example Path	Description
IF		Figure 1: Initial Fix or IF Leg. Defines a database fix as a point in space.
TF		Figure 2: Track to a Fix or TF Leg. Defines a great circle track over ground between two known databases fixes.
CF		Figure 3: Course to a Fix or CF Leg. Defines a specified course to a specific database fix.

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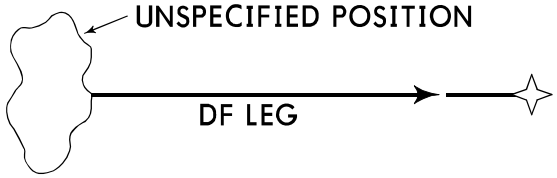
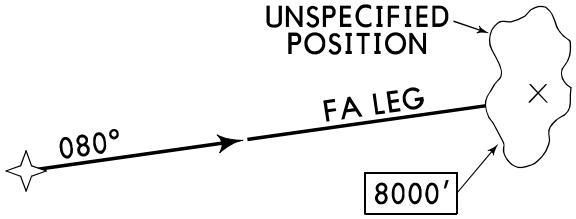
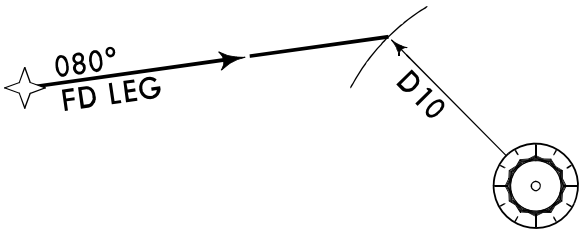
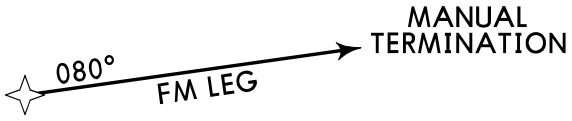
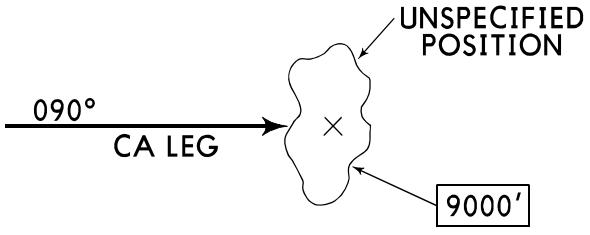
DF		Figure 4: Direct to a Fix or DF Leg. Defines an unspecified track starting from an undefined position to a specific database fix. Note: See also Table 1.3, Leg Sequencing, for other uses of the DF Leg.
FA		Figure 5: Fix to an Altitude or FA Leg. Defines a specified track over ground from a database fix to a specified altitude at an unspecified position.
FC		Figure 6: Track from a Fix from a Distance or FC Leg. Defines a specified track over ground from a database fix for a specific distance.
FD		Figure 7: Track from a Fix to a DME Distance or FD Leg. Defines a specified track over ground from a database fix to a specific DME Distance which is from a specific database DME Navaid.
FM		Figure 8: From a Fix to a Manual termination or FM Leg. Defines a specified track over ground from a database fix until Manual termination of the leg.
CA		Figure 9: Course to an Altitude or CA Leg. Defines a specified course to a specific altitude at an unspecified position.

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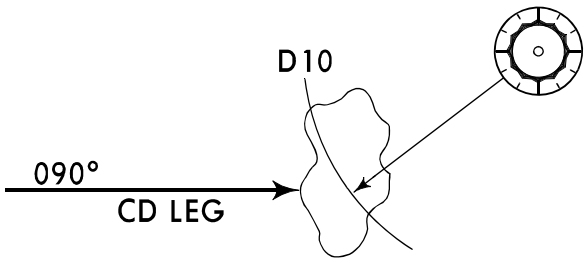
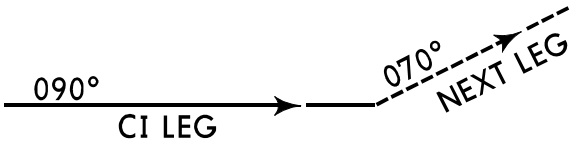
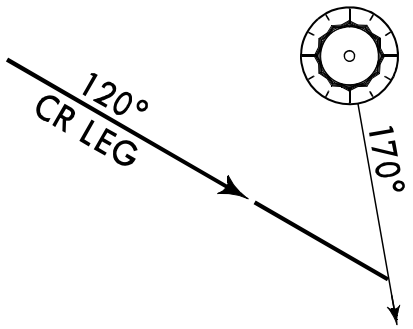
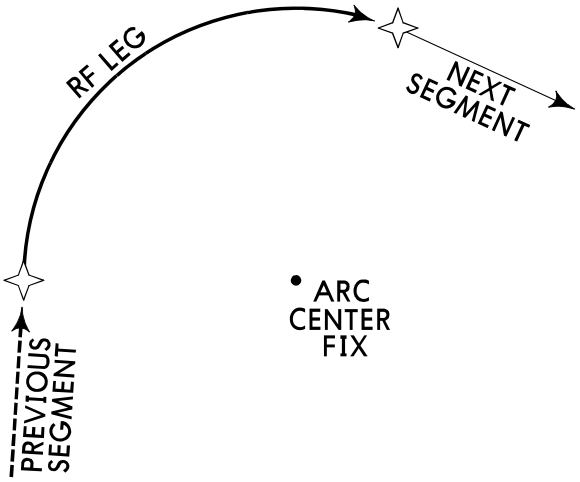
CD		Figure 10: Course to a DME Distance or CD Leg. Defines a specified course to a specific DME Distance which is from a specific database DME Navaid.
CI		Figure 11: Course to an Intercept or CI Leg. Defines a specified course to intercept a subsequent leg.
CR		Figure 12: Course to a Radial termination or CR Leg. Defines a course to a specified Radial from a specific database VOR Navaid.
RF		Figure 13: Constant Radius Arc or RF Leg. Defines a constant radius turn between two database fixes, lines tangent to the arc and a center fix. Note: While the arc initial point, arc ending point and arc centerpoint are all available as database fixes, implementation of this leg type may not require them to be available as fixes.

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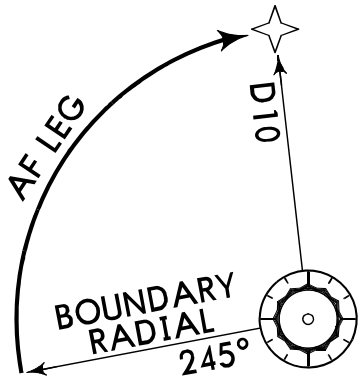
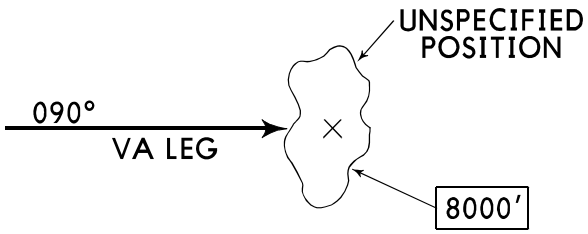
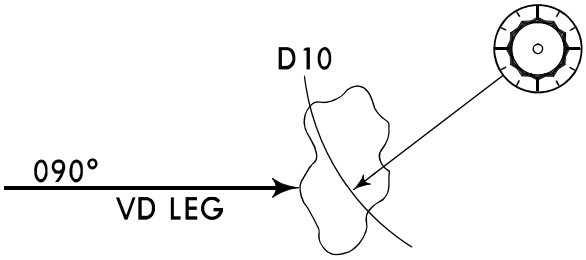
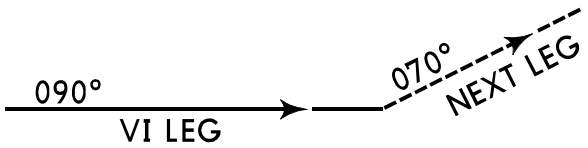
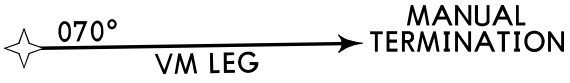
AF		Figure 14: Arc to a Fix or AF Leg. Defines a track over ground at specified constant distance from a database DME Navaid.
VA		Figure 15: Heading to an Altitude termination or VA Leg. Defines a specified heading to a specific Altitude termination at an unspecified position.
VD		Figure 16: Heading to a DME Distance termination or VD Leg. Defines a specified heading terminating at a specified DME Distance from a specific database DME Navaid.
VI		Figure 17: Heading to an Intercept or VI Leg. Defines a specified heading to intercept the subsequent leg at an unspecified position.
VM		Figure 18: Heading to a Manual termination or VM Leg. Defines a specified heading until a Manual termination.

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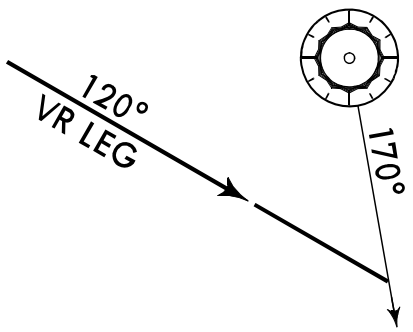
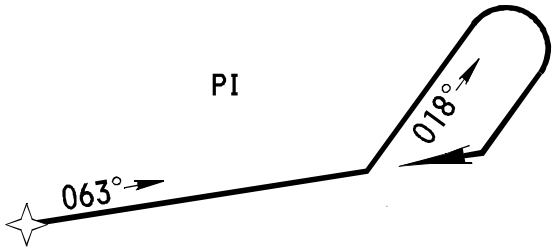
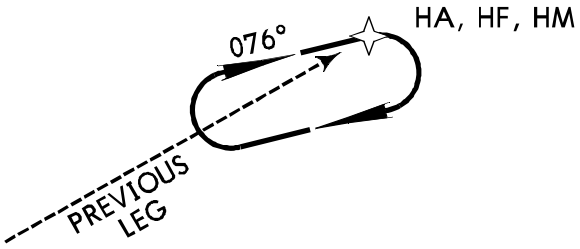
VR		Figure 19: Heading to a Radial termination or VR Leg. Defines a specified heading to a specified radial from a specific database VOR Navaid.
PI		Figure 20: 045/180 Procedure Turn or PI Leg. Defines a course reversal starting at a specific database fix, includes Outbound Leg followed by a left or right turn and 180 degree course reversal to intercept the next leg. A Maximum excursion Time or Distance is included as a data field.
HA, HF, HM		Figure 21: Holding in lieu of Procedure Turn (HF) for Approach Procedures and Mandatory Holds (HA, HM) in SID/STAR and Missed Approach coding. The HA, HF, and HM Leg Types define a holding pattern in lieu of procedure turn course reversal or a terminal procedure referenced mandatory holding pattern at a specified database fix. Leg time or distance is included as a data field**.** The three codes indicate different path termination types: HA = Altitude Termination HF = Single circuit terminating at the fix. HM = Manual Termination.

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1.5 Leg Data Fields

The following table provides detail on “Required” and “Optional” parameters used to define each leg type. An “O” in the table indicates that the parameter is considered optional and may be omitted as required in individual cases. All other entries indicate some type of a “required” situation for leg definition.

Leg Data Fields Table 3

PT	W/P ID	OVR FLY	HLD	TD	TDV	RMD NAV	THETA	RHO	OBD MAG CRS	TM/ DST	ALT ONE	ALT TWO	SPD LMT (1)	VRT ANG	ARC CTR	COMMENTS
AF	X	O	O	X		X	X	X	R		O	O	O			OB MAG CRS=BNDY RDL, THETA=FIX RDL
CA				O	O				C		+		O			ALT TERM WILL BE “AT OR ABOVE”
CD				O	O	X			C	D	O	O	O			
CF	X	B	O	O	O	X	X	X	C	F	O	O	O	O		OB MAG CRS IS CRS TO SPECIFIED FIX
CI		O		O	O	O			C		O	O	O			
CR		O		O	O	X	X		C		O	O	O			
DF	X	B	O	O		O	O	O			O	O	O			
FA	X			O	O	X	X	X	C		+		O			ALT TERM WILL BE “AT OR ABOVE”
FC	X	B		O	O	X	X	X	C	P	O	O	O			
FD	X	O		O	O	X	X	X	C	D	O	O	O			
FM	X			O	O	X	X	X	C		O		O			
HA	X	O		X		O	O	O	C	X	+		O			ALT TERM WILL BE “AT OR ABOVE”
HF	X	O		X		O	O	O	C	X	O		O			
HM	X	O		X		O	O	O	C	X	O		O			
IF	X		O			O	O	O			O	O	O			
PI	X			X		X	X	X	C	P	X		O			DIST IS EXCURSION DIST FROM FIX
RF	X		O	X		O	**O**		**O**	A	O	O	O	O	X	
TF	X	B	O	O	O	O	O	O	O	O	O	O	O	O		
VA				O	O				H		+		O			ALT TERM WILL BE “AT OR ABOVE”
VD				O	O	X			H	D	O	O	O			
VI		O		O	O	O			H		O	O	O			
VM	O			O	O				H		O		O			FOR W/P ID SEE STAR CODING RULES
VR		O		O	O	X	X		H		O	O	O			

LEGEND:

X = REQUIRED FIELD
R = BOUNDARY RADIAL
D = DME DISTANCE
A = ALONG TRACK DISTANCE
C = COURSE
+ = “AT OR ABOVE” ONLY
O = OPTIONAL FIELD
H = HEADING
SHADED = NOT APPLICABLE FIELD
P = PATH LENGTH
B = “REQUIRED” FOR CF/DF, DF/DF, TF/DF OR FC/DF COMBINATIONS, OTHERWISE “OPTIONAL”

(1) When the optional Speed Limit data is provided, it is valid for the Fix in all fix-related leg types and for the Leg in all non-fix-related leg types. The exception to this rule is the HA, HF and HM leg types. Although these leg types have a Fix, the Speed Limit applies to the Leg.

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2.0 Coding Rules Applicable To All Procedures

2.1 All Procedures must be coded to provide guidance specified by source documentation.

2.2 Vertical angles are referenced to the terminating fix.

2.3 Use of a "C" in the Altitude Description field (5.29) may only be used in SID records and there only with the following leg types:

CD, CF, CR, FC, FD, TF, VD, VR

The conditional termination altitude can be coded in columns 90 through 94 of the SID record. If a "+," "-" or "blank" is coded in the Altitude Description field, input of a second altitude must imply a condition altitude termination.

2.4 Altitude terminations must not be used in descent procedures.

2.5 Lost Communication Procedures may be coded in place of "Vector Legs" if the procedure defines a complete route of flight to the end of a SID or STAR.

2.6 The "Turn Direction" and "Turn Direction Valid" leg data fields are used in combination to force a particular turn direction whenever the track/heading change exceeds 135 degrees. If the turn direction is indicated with the "L" or "R," **and the turn exceeds 135 degrees,** then the turn direction valid character must be set to "Y." If the turn direction is indicated with "E," then the turn direction valid field must always be blank. **For legs that are "turning legs" such as the AF or RF legs, the "Turn Direction" is always required.**

The Turn Direction/Turn Direction Valid combination is used to indicate that turn in the specified direction must be executed prior to intercepting the path defined in the record. Turn Direction must always be indicated whenever the turn is 90 degrees or more.

2.7 The first leg of each procedure will contain the appropriate transition "altitude." If the transition altitude is 18,000 feet, it may be omitted. See Section 5.53 for specifics on the appropriate altitude for each type of Terminal Procedure.

2.8 Non essential and transition essential waypoint codes are not used in the waypoint description field on terminal procedures. All waypoints must be considered as essential in these procedures.

2.9 Required Navigation Performance (RNP) must be applied to segment on which the value is coded. RNP will be coded on every segment where it is specified by source. Lack of a RNP value on a segment indicated that no source RNP value is available for the segment.

Examples of RNP Coding:

Standard Instrument Departure (SID)

Route Type	Trans Ident	Sequence Number	Path Terminator	Fix Ident	Segment RNP
1	RW08	010	CA		Blank
1	RW08	020	DF	ALPHA	Blank
1	RW26	010	CA		Blank
1	RW26	020	DF	ALPHA	Blank
2	Blank	010	IF	ALPHA	Blank
2	Blank	020	TF	CARLY	010
3	BRAVO	010	IF	CARLY	Blank
3	BRAVO	020	TF	BRAVO	010
3	DELTA	010	IF	CARLY	Blank
3	DELTA	020	TF	INTER	010
3	DELTA	030	TF	DELTA	050
3	ECHHO	010	IF	CARLY	Blank
3	ECHHO	020	TF	ECHHO	040

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Explanation:

There is no defined RNP for the Runway Transitions. The common segment has a RNP of 010 or 1.0NM from ALPHA to CARLY. The Enroute Transition from CARLY to BRAVO is RNP 010 or 1.0NM. The Enroute Transition to DELTA is RNP 010 or 1.0NM from CARLY to INTER and 050 or 5.0NM from INTER to DELTA. The Enroute Transition from CARLY to ECHHO is RNP 040 or 4.0NM. The "initial fix" segments carry no RNP values.

Instrument Approach Procedure

Route Type	Trans Ident	Sequence Number	Path Terminator	Fix Ident	Segment RNP
A	EAST	010	IF	EAST	Blank
A	EAST	020	TF	ALPHA	Blank
A	WEST	010	IF	WEST	Blank
A	WEST	020	TF	ALPHA	010
R	Blank	010	IF	ALPHA	Blank
R	Blank	020	TF	BRAVO (FAF)	031
R	Blank	030	TF	RW18 (MAP)	031
Z	Blank	040	DF	CARLY	010
Z	Blank	050	TF	DELTA	010
Z	Blank	060	TF	ECHHO	010
Z	Blank	070	HF	ECHHO	010

Explanation:

There is no defined RNP for the EAST transition. The WEST transition has a RNP of 010 or 1.0NM from WEST to ALPHA. The Final Approach **Coding** from ALPHA to Runway 18 (RW18) has a RNP of 031 or 0.3 NM. The Missed Approach **Procedure** from the runway through the holding pattern at ECHHO has a RNP of 010 or 1.0 NM.

3.0 Path and Termination Related Rules Valid For All Procedure Types **Except RNAV Terminal Procedures That Do Not Reference Ground-based Navaids.**

3.1 DF legs must be used to start from an unknown position such as an altitude or from DME or Distance terminations. A DF leg may follow a CF leg only when the CF leg fix must be overflown, otherwise a TF leg should be used. A DF may also be followed by another DF leg. In these cases, the fix at the end of the first DF leg must be overflown. When DF legs follow DME or Distance Terminations, that termination must be overflown. For distance terminations, the overfly parameter must be set, otherwise the combination is not permitted.

3.2 The distance leg data field must be completed on all CF legs. When the CF is preceded by an intercept, the no wind intercept distance will be provided. If the CF leg is the first leg of a missed approach, the distance entered will be from the approach runway fix or missed approach fix, whichever applies.

3.3 When a leg terminating at a fix ("XF" leg) is followed by a PI leg, the PI fix must be the same fix as the terminating fix on the preceding leg.

3.4 Rules specific to "arc" legs, leg type AF:

3.4.1 When an AF-AF leg sequence is coded, both legs must use the same "Recommended VHF Navaid" facility and the DME distance must be the same for both legs.

3.4.2 When any holding leg ("HX") or fix termination ("XF") is followed by an AF leg, the preceding termination fix must lie on the arc defined in the AF leg.

3.4.3 When a FD leg is followed by an AF leg type, the fix in the FD leg must have the same "Recommend VHF Navaid" as that defining the AF leg.

3.4.4 When a CD or VD leg type is followed by an AF leg type, both legs must have the same Recommended VHF Navaid. The DME distance must be the same for both legs.

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3.4.5 When a CI leg type is followed by an AF leg type, the "course to" must be to the "Recommended VHF Navaid" which defines the AF arc.

3.5 When an AF, CF, DF, **IF**, RF, TF or "HX" leg is followed by any course-from leg type ("FX" legs), the "FX" leg must be from the same point as the preceding termination.

3.6 Leg types of CD, CR, FD, VD and VR "overfly" the terminator point. If "turn anticipation" is required to reflect the source, alternate leg types must be used.

3.7 When the leg data type "Recommended Navaid" is coded in a CI or VI leg, it must be the same as the Recommended Navaid in the leg to be intercepted.

3.8 The TF leg type will be coded in preference to the CF leg type in all cases where the resulting path will be the same, except when coding some types of final approach procedure legs, see Rules in Section 6.0, 7.0 and 8.0 of this Attachment.

3.9 The FC leg type must be used when the distance in the "Time/Distance" field is the path length and is measured from the fix entered in the Waypoint Identifier field.

The FD leg type must be used when the distance in the "Time/Distance" field is the "DME" distance from the Navaid entered in the Recommended Navaid field.

3.10 FC or FD legs will not be used if the distance is greater than 60NM and are followed by a CF leg.

3.11 A PI leg is used to make a 180 degree course reversal when a holding or a tear drop turn is not specified. The course must be coded as 45 degrees from the reciprocal of the inbound course, unless otherwise specified by government sources. The turn direction is the direction made during the 180 reversal within the PI leg. A one minute outbound leg is implied from the fix to the initial 45 degree turn.

3.12 The IF leg type will normally be used in an initial sequence of a procedure. The IF leg type, followed by a TF leg type will be used in other than the first sequence if such is required to correctly code the procedure as published by the source documentation when one or more of the following criteria are met -

- there is no VHF Navaid available for use as the Recommended Navaid that would permit coding with other leg types.
- the leg to be intercepted will have a distance of more than 60 NM between the point of intercept and the terminating fix.

This will allow a segment to be constructed, from one fix to the next fix, using an "intercept" where coding would otherwise not be possible. See the sample use of this rule below.

3.13 When coding "arc paths," when the source defined ARC Center is a VHF Navaid of the types VORDME or VORTAC and the path is defined as a "DME ARC," the AF leg must be used instead of the RF leg.

3.14 Deleted by Supplement 19.

3.15 Use of a single RF leg is limited to turns of equal to or greater than 2 degrees and equal to or less than 300 degrees.

3.16 Phantom Waypoints. These database waypoints are established during procedure coding, used to facilitate more accurate navigation under certain circumstances.

3.17 Path and Termination Related Rules Valid For All RNAV Terminal Procedures That Do Not Reference Ground-based Navaids

3.17.1 Waypoint flyby/flyover requirements.

3.17.1.a Setting Position Two of the Waypoint Description field (Chapter Five, Section 5.17) to "Y" or "B" indicates a Flyover Waypoint; the fix in the record is to be over flown before flying the next leg. Absence of the "Y" or "B" indicates a Flyby Waypoint; turn anticipation may be used to acquire the next leg.

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3.17.1.b **The coding requirement of the Flyby or Flyover condition is derived from official government source.**

3.17.1.c **The RF leg overflys the terminating fix in all cases.**

3.18 **The RF Leg is to be used only in the following cases:**

3.18.1 **When coding procedure types, which were, designed with the RF Constant Radius ARC capability as a design criteria.**

3.18.2. **When coding procedure types which were not designed with the RF Leg capability as a criterion but for which the RF leg coding is determined to correctly reflect source intent.**

3.18.3 **Use of a single RF leg is limited to turns of equal to or greater than 2 degrees and equal to or less than 358 degrees.**

3.18.4 **The RF leg, either within the transition being coded, or in conjunction with previous or subsequent transitions, may comply with the design criteria of the RF leg, which is tangential track inbound to start of arc and tangential track outbound from the end of the arc, but do not have to.**

3.19 **The CF leg is available as a leg type in RNAV Terminal Procedures only when specifically called out in the government source documentation. When this is the case, the leg data will include a reference from which the magnetic variation for use in flying the CF Leg can be determined. This reference will be provided in the Recommended Navaid field of the procedure record.**

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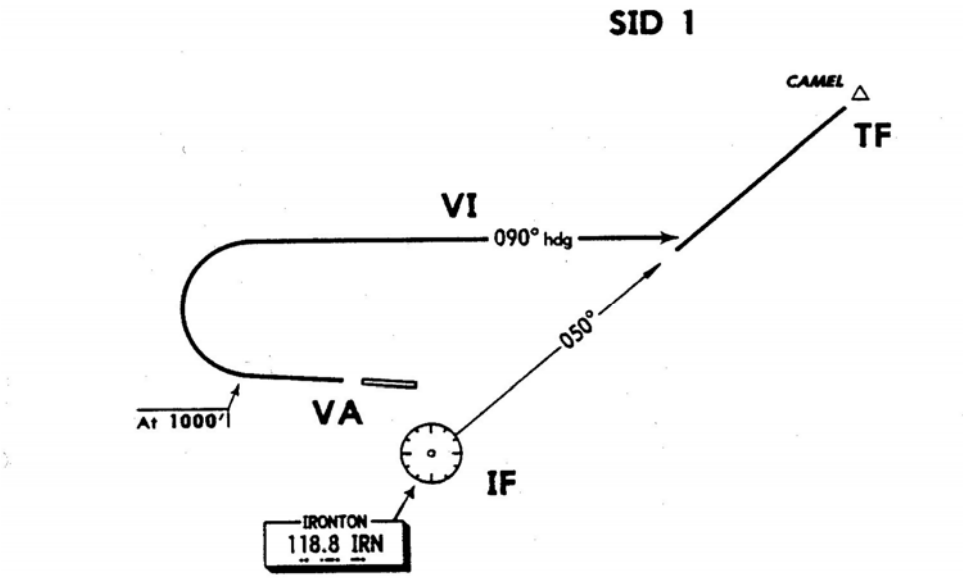
Illustration of Coding Rule 3.12



Normal SID Coding if IRN were a VORDME

SID IDENT	ROUTE TYPE	TRANS IDENT	WAYPOINT IDENT	DESC CODE	PATH TERM
SID 1	2	RW29			VA
SID 1	2	RW29			VI
SID 1	2	RW29	CAMEL	EE	CF

SID Coding required when IRN is VOR only

SID IDENT	ROUTE TYPE	TRANS IDENT	WAYPOINT IDENT	DESC CODE	PATH TERM
SID 1	2	RW29			VA
SID 1	2	RW29			VI
SID 1	2	RW29	IRN	V	IF
SID 1	2	RW29	CAMEL	EE	TF

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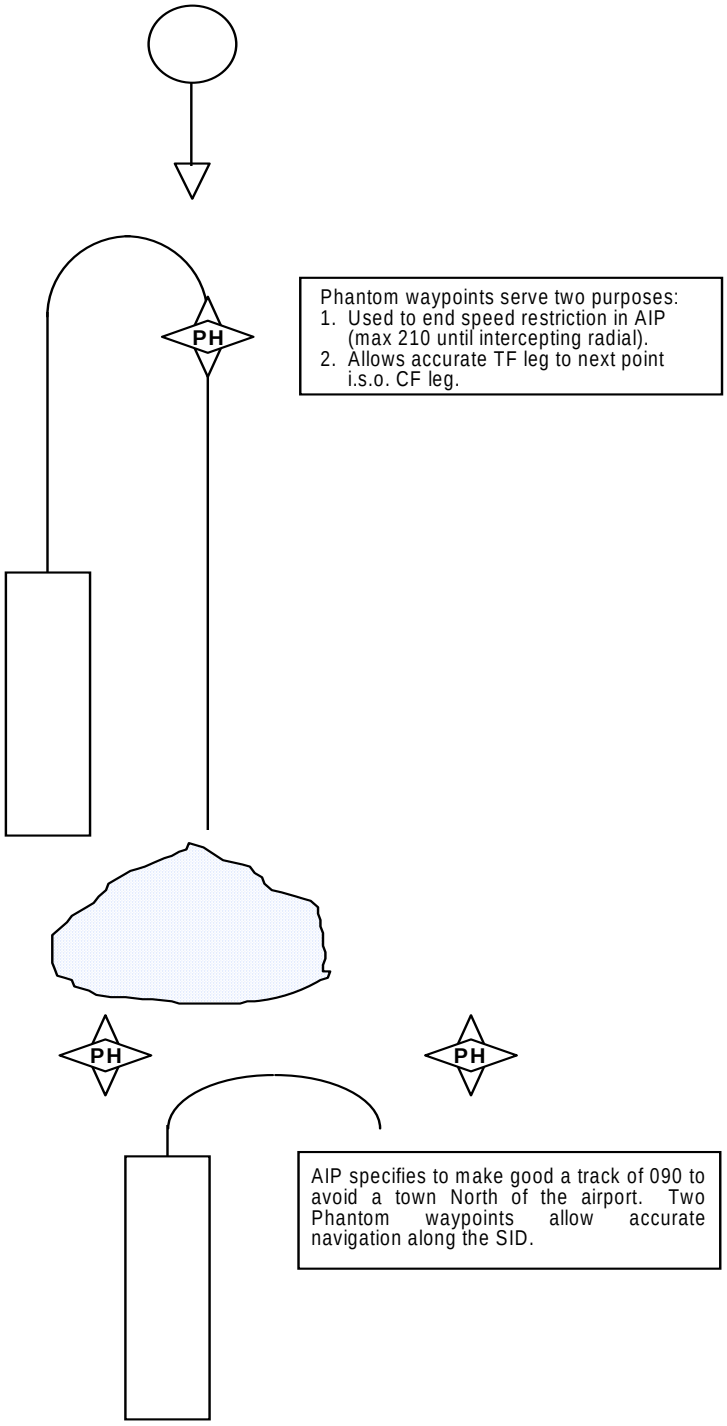

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PATH AND TERMINATOR

4.0 STANDARD INSTRUMENT DEPARTURE (SID) CODING RULES

4.1 The following rules cover the altitude coding requirements of the initial leg of a departure:

If a published take-off requires a turn of greater than 15 degrees from the runway bearing, code a CA, VA or **IF**/FA leg on the runway bearing/heading to an altitude of 400 feet above the airport elevation (AFE).

If a published take-off is straight ahead or requires a turn of less than 15 degrees and is to a fix such as a waypoint, Navaid, or DME distance termination, code an altitude on that fix when included in the government source, otherwise no altitude is coded.

If a published take-off is straight ahead or requires a turn of less than 15 degree and is to a specified altitude termination, code the appropriate leg type (VA, **IF/**FA, CA) to that altitude.

The altitude description on a coded altitude, either a termination or at a fix, may be "at," "at or above," "at or below" and "at or below to at or above."

4.2 For the first leg of a SID, course legs ("CX" or "FX") are preferred over heading legs unless the source requires that heading legs be coded.

4.2.1 For RNAV SIDs that do not reference ground-based navaids, the following first leg rules apply.

4.2.2 Unless the initial leg of a departure is to a Fix, IF/FA is the preferred tale-off leg combination, except as follow:

a. the government source prescribes the specific use of a CA or VA leg.
b. the CA leg is suitable as the first leg on take-off if there is a possibility of aircraft position or runway survey uncertainty.
c. the VA leg is suitable as first leg on takeoff when procedure design includes air mass track dispersion for noise or ATC procedures and therefore requires aircraft on heading legs.

4.3 The use of FM or VM leg types in the first leg of a SID Runway Transition for Route Types of 0 **or** 1is permitted when the initial heading is defined in the source**.**

4.4 If a SID ends in vectors, the heading for the FM or VM leg must be based on source documentation.

4.5 If the last fix of a SID sequence or SID Enroute Transition sequence is a fix on an Enroute Airway, the Waypoint Description Code in the Enroute Airway record for that fix must designate that fix as "Essential" or "Transition Essential."

4.6 SID Enroute Transition legs will be coded using TF legs where possible to simplify coding. This rule does not apply if the use of TF legs would require the creation of additional waypoints.

4.7 SID Enroute Transitions published in source documentation but wholly contained in other SID Enroute Transitions will not be coded separately.

4.8 All **enroute** waypoints used in SID common or Vector SIDs must be in the Enroute Waypoint File as an Off-Route Intersection unless the waypoint is also used for Enroute Airway Coding.

4.9 A SID which consists of a single path from an origination fix to a termination fix will be coded as a Route Type 2.

4.10 A SID which consists of Enroute Transitions only can be coded with a single IF leg as a Route Type 2, followed by the required Route Type 3 coding. The fix on which the IF leg is coded must be the first fix in all of the Enroute Transitions. The Transition Identifier must be coded in accordance with Chapter Five, Section 5.11. **In the cases where all the Enroute Transitions do not begin at the same fix, but where most begin at the same fix then a partial SID may be coded.**

4.11 For Vector SIDs which consist of Enroute Transitions only, the coding must be a Runway Transition Route Type "**T**," followed by the Enroute Transition(s), Route Type **"V."** The Enroute Transition(s) must be an IF/DF leg combination with the Airport as the fix in the IF leg and the first fix of the Enroute Transition as the fix in the DF leg. The DF leg must carry a distance value **in the continuation record,**equal to the total distance between the Airport and the fix along the approximated path.

4.12 When a SID Route or portion of a SID Route is repeated with different Runway Identifiers in the Transition Identifier, it must be coded as a Runway Transition Route Type of 1 or T (Vector SID). When a SID Route is

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repeated with different fix identifiers in the Transition Identifier, it must be coded as an Enroute Transition, Route Type of 3 or *V* (Vector SID).

4.13 "Engine Out" SIDs must be coded as Route Type "0" only. Route Type "0" is not used in combination with other SID Route Types. The Runway Transition Identifier must contain a specific Runway Identification or Helipad Identification. All other rules for Route Type 1 apply in the coding of Route Type "0."

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SID CODING EXAMPLE 1

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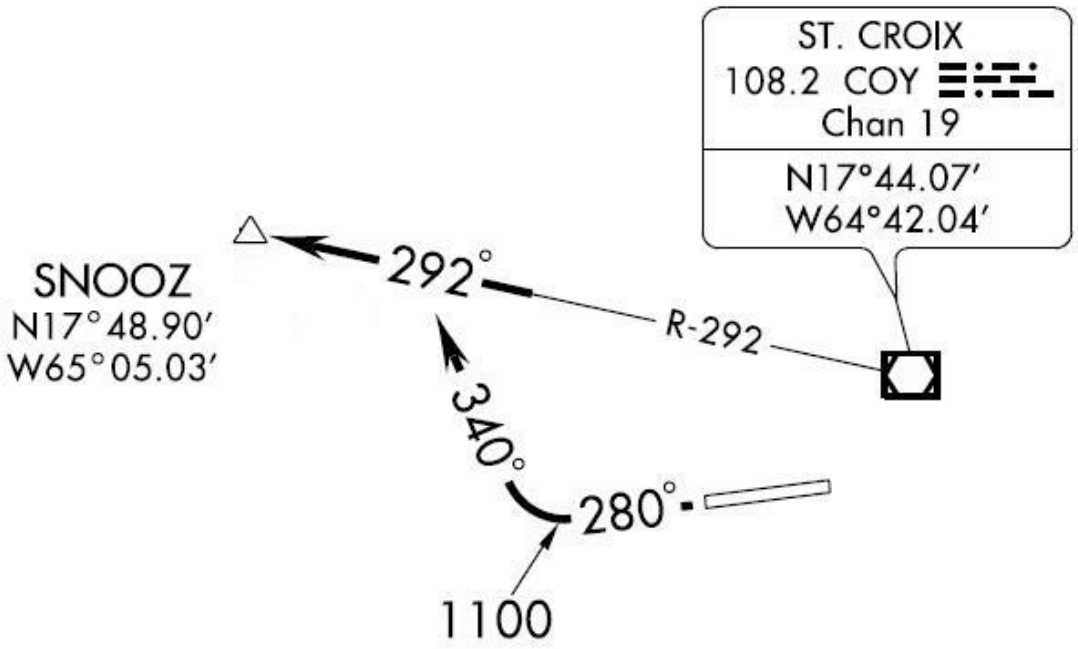
SNOOZ TWO DEPARTURE (SNOOZ2.SNOOZ)



Normal SID Coding when COY is a VORDME

SID IDENT	ROUTE TYPE	TRANS IDENT	SEQ	WAYPOINT	DESC CODE	PATH TERM
SNOOZ2	2	RW28	010		_ _ _ _	VA
SNOOZ2	2	RW28	020		_ _ _ _	VI
SNOOZ2	2	RW28	030	SNOOZ TI EA	E E _ _	CF

SID Coding required if COY were a VOR only

SID IDENT	ROUTE TYPE	TRANS IDENT	SEQ	WAYPOINT	DESC CODE	PATH TERM
SNOOZ2	2	RW28	010		_ _ _ _	VA
SNOOZ2	2	RW28	020		_ _ _ _	VI
SNOOZ2	2	RW28	030	COY TI D	V _ _ _	IF
SNOOZ2	2	RW28	040	SNOOZ TI EA	E E _ _	TF

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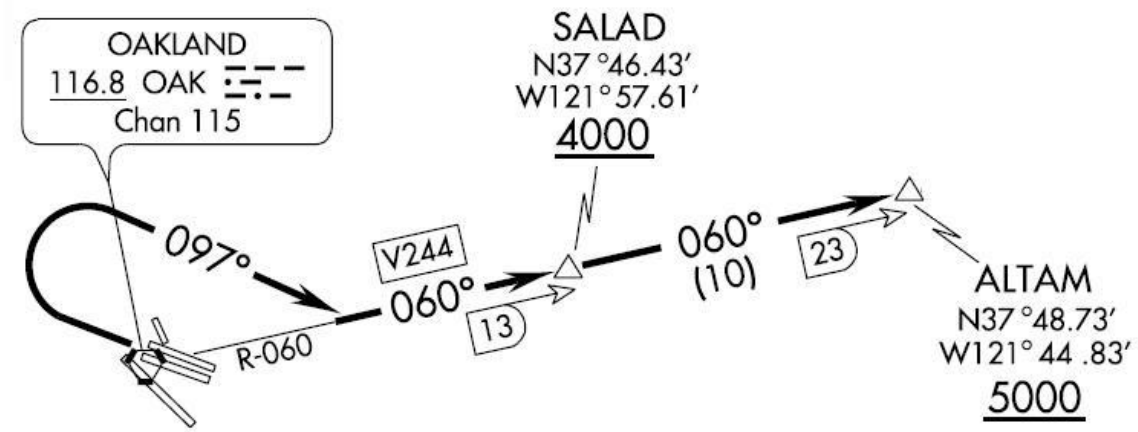
SID CODING EXAMPLE 2

Not for navigational or other operational use. For example only. Please consult current navigation charts.

SALAD ONE DEPARTURE (SALAD1.ALTAM)



SID IDENT	ROUTE TYPE	TRANS IDENT	SEQ	WAYPOINT	DESC CODE	PATH TERM
SALAD1	2	RW27B	010		_ _ _ _	VA
SALAD1	2	RW27B	020		_ _ _ _	VI
SALAD1	2	RW27B	030	SALAD K2 EA	E _ _ _	CF
SALAD1	2	RW27B	040	ALTAM K2 EA	E E _ _	TF

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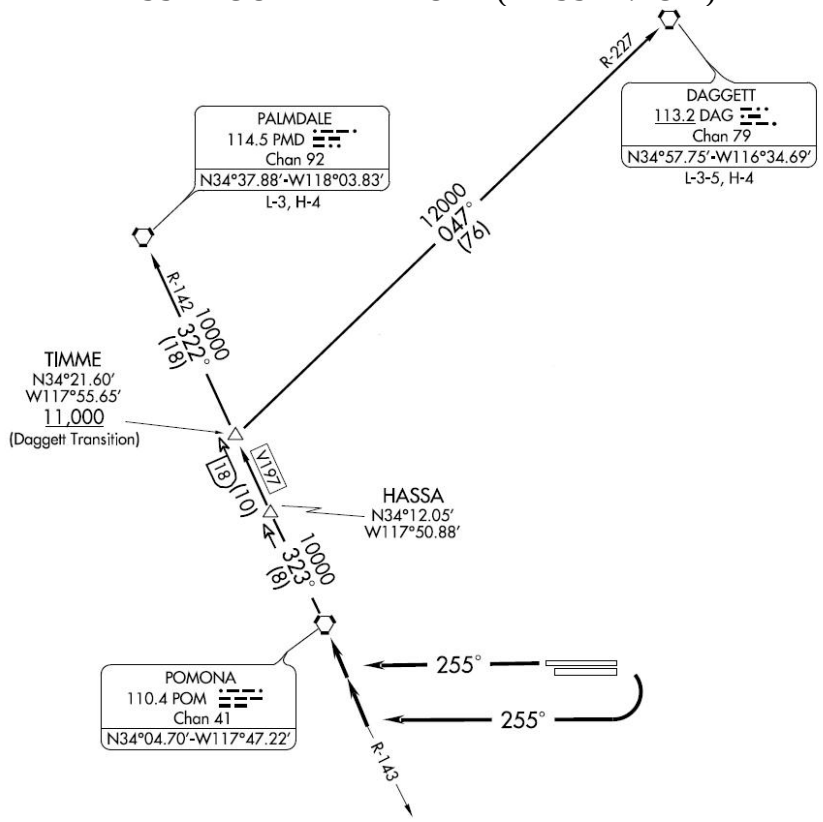
SID CODING EXAMPLE 3

Not for navigational or other operational use. For example only. Please consult current navigation charts.

HASSA FOUR DEPARTURE (HASSA4.POM)



SID IDENT	ROUTE TYPE	TRANS IDENT	SEQ	WAYPOINT		DESC CODE	PATH TERM
HASSA4	1	RW08B	010			_ _ _ _	VA
HASSA4	1	RW08B	020			_ _ _ _	VI
HASSA4	1	RW08B	030	POM	K2 D	V E _ _	CF
HASSA4	1	RW26B	010			_ _ _ _	VI
HASSA4	1	RW26B	020	POM	K2 D	V E _ _	CF
HASSA4	2		010	POM	K2 D	V _ _ _	IF
HASSA4	2		020	HASSA	K2 EA	E _ _ _	TF
HASSA4	2		030	TIMME	K2 EA	E E _ _	TF
HASSA4	3	DAG	010	TIMME	K2 EA	E _ _ _	IF
HASSA4	3	DAG	020	DAG	K2 D	V E _ _	TF
HASSA4	3	PMD	010	TIMME	K2 EA	E _ _ _	IF
HASSA4	3	PMD	020	PMD	K2 D	V E _ _	TF

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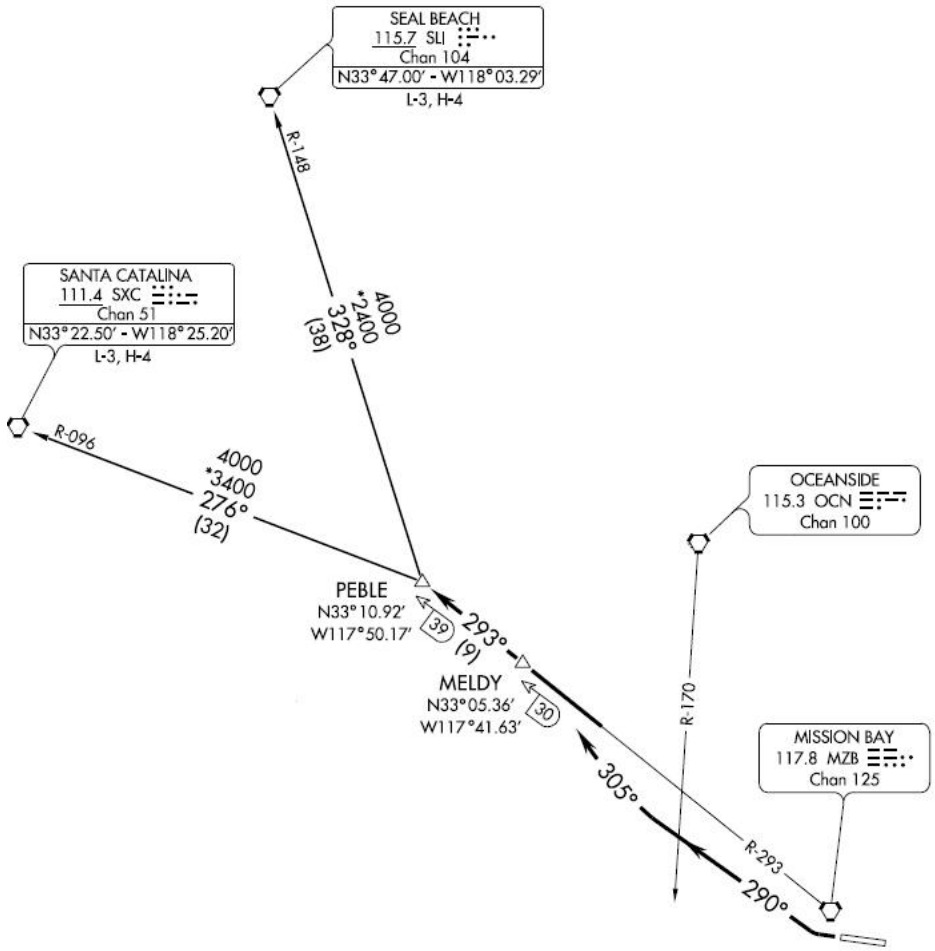
SID CODING EXAMPLE 4

Not for navigational or other operational use. For example only. Please consult current navigation charts.

PEBLE THREE DEPARTURE (PEBLE3.PEBLE)



SID IDENT	ROUTE TYPE	TRANS IDENT	SEQ	WAYPOINT		WAYPOINT	PATH TERM
PEBLE3	2	RW27	010			_ _ _ _	VA
PEBLE3	2	RW27	020			_ _ _ _	VR
PEBLE3	2	RW27	030			_ _ _ _	VI
PEBLE3	2	RW27	040	MELDY	K2 EA	E _ _ _	CF
PEBLE3	2	RW27	050	PEBLE	K2 EA	E E _ _	TF
PEBLE3	3	SLI	010	PEBLE	K2 EA	E _ _ _	IF
PEBLE3	3	SLI	020	SLI	K2 D	V E _ _	TF
PEBLE3	3	SXC	010	PEBLE	K2 EA	E _ _ _	IF
PEBLE3	3	SXC	020	SXC	K2 D	V E _ _	TF

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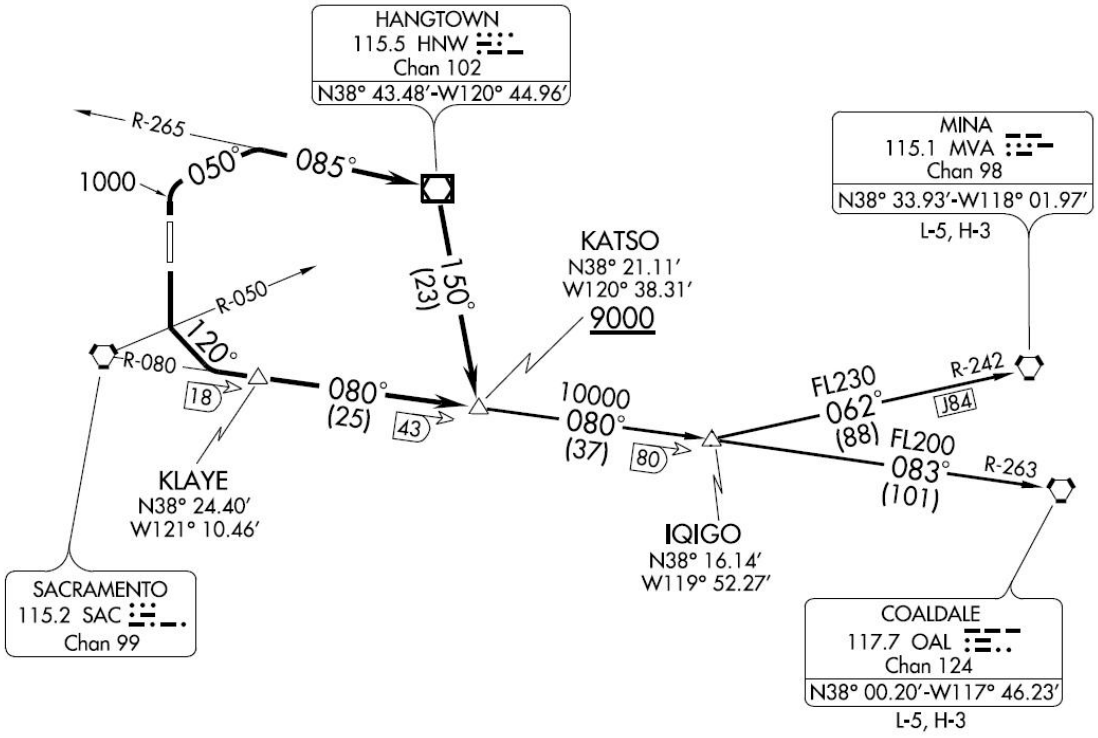
SID CODING EXAMPLE 5

Not for navigational or other operational use. For example only. Please consult current navigation charts.

KATSO TWO DEPARTURE (KATSO2.KATSO)



SID IDENT	ROUTE TYPE	TRANS IDENT	SEQ	WAYPOINT		DESC CODE	PATH TERM
KATSO2	1	RW16	010			_ _ _ _	VR
KATSO2	1	RW16	020			_ _ _ _	VI
KATSO2	1	RW16	030	KLAYE	K2 EA	E _ _ _	CF
KATSO2	1	RW16	040	KATSO	K2 EA	E E _ _	TF
KATSO2	1	RW34	010			_ _ _ _	VA
KATSO2	1	RW34	020			_ _ _ _	VI
KATSO2	1	RW34	030	HNW	K2 D	V _ _ _	CF
KATSO2	1	RW34	040	KATSO	K2 EA	E E _ _	TF
KATSO2	2		010	KATSO	K2 EA	E _ _ _	IF
KATSO2	2		020	IQIGO	K2 EA	E E _ _	TF
KATSO2	3	MVA	010	IQIGO	K2 EA	E _ _ _	IF
KATSO2	3	MVA	020	MVA	K2 D	V E _ _	TF
KATSO2	3	OAL	010	IQIGO	K2 EA	E _ _ _	IF
KATSO2	3	OAL	020	OAL	K2 D	V E _ _	TF

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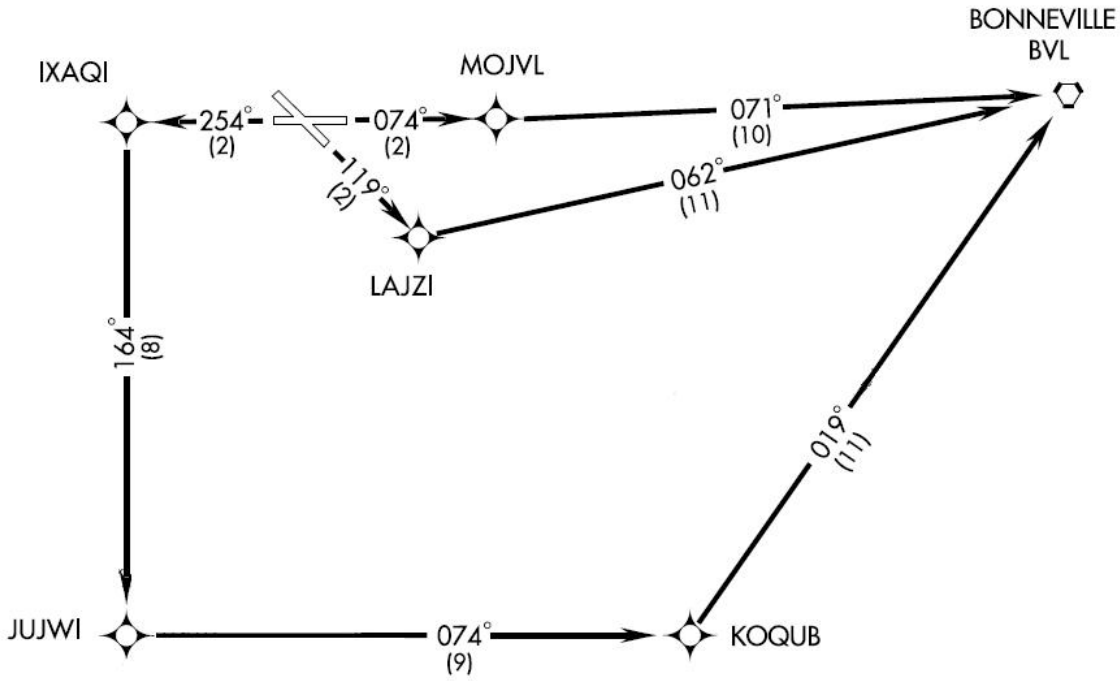
SID CODING EXAMPLE 6

Not for navigational or other operational use. For example only. Please consult current navigation charts.

BONNEVILLE ONE DEPARTURE (ENV1.BVL)



SID IDENT	ROUTE TYPE	TRANS IDENT	SEQ	WAYPOINT		DESC CODE	PATH TERM
ENV1	1	RW08	010	MOJVL	K2 EA	E _ _ _	CF
ENV1	1	RW08	020	BVL	K2 D	V E _ _	TF
ENV1	1	RW12	010	LAJZI	K2 EA	E _ _ _	CF
ENV1	1	RW12	020	BVL	K2 D	V E _ _	TF
ENV1	1	RW26	010	IXAQI	K2 EA	E _ _ _	CF
ENV1	1	RW26	020	JUJWI	K2 EA	E _ _ _	TF
ENV1	1	RW26	030	KOQUB	K2 EA	E _ _ _	TF
ENV1	1	RW26	040	BVL	K2 D	V E _ _	TF

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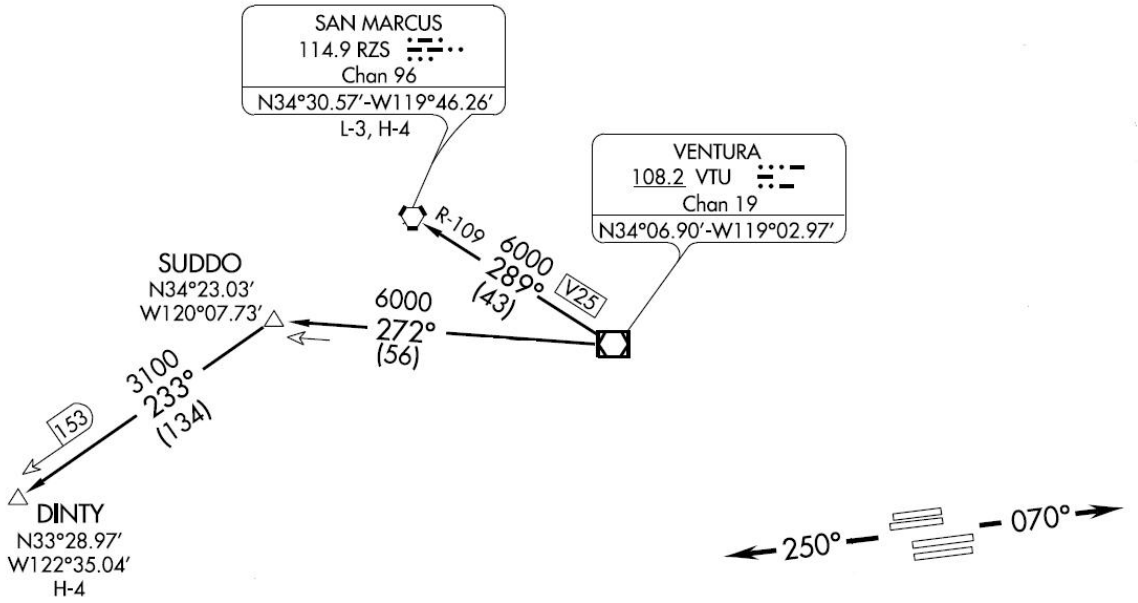
SID CODING EXAMPLE 7

Not for navigational or other operational use. For example only. Please consult current navigation charts.

VENTURA FIVE DEPARTURE (VTU5.VTU)



SID IDENT	ROUTE TYPE	TRANS IDENT	SEQ	WAYPOINT		DESC CODE	PATH TERM
VTU5	1	RW06B	010			_ _ _ _	VM
VTU5	1	RW06B	020	VTU	K2 D	V E _ _	DF
VTU5	1	RW24B	010			_ _ _ _	VM
VTU5	1	RW24B	020	VTU	K2 D	V E _ _	DF
VTU5	3	RZS	010	VTU	K2 D	V _ _ _	IF
VTU5	3	RSZ	020	RZS	K2 D	V E _ _	TF
VTU5	3	DINTY	010	VTU	K2 D	V _ _ _	IF
VTU5	3	DINTY	020	SUDDO	K2 EA	E _ _ _	TF
VTU5	3	DINTY	030	DINTY	K2 EA	E E _ _	TF

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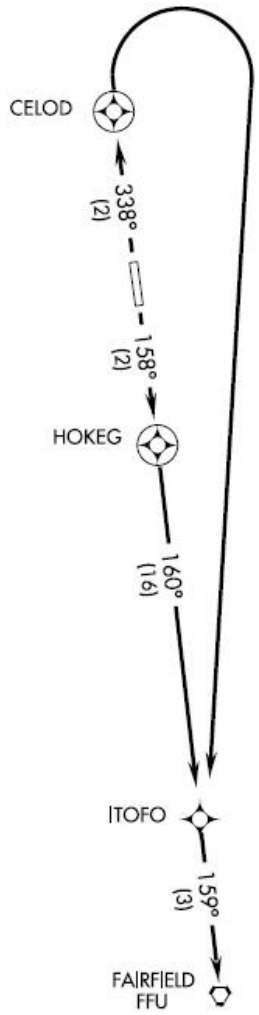
SID CODING EXAMPLE 8

Not for navigational or other operational use. For example only. Please consult current navigation charts.

ITOFO ONE DEPARTURE (ITOFO1.FFU)



SID IDENT	ROUTE TYPE	TRANS IDENT	SEQ	WAYPOINT		DESC CODE	PATH TERM
ITOFO1	1	RW16	010	HOKEG	K2 EA	E Y _ _	CF
ITOFO1	1	RW16	020	ITOFO	K2 EA	E E _ _	TF
ITOFO1	1	RW34	010	CELOD	K2 EA	E Y _ _	CF
ITOFO1	1	RW34	020	ITOFO	K2 EA	E E _ _	DF
ITOFO1	2		010	ITOFO	K2 EA	E _ _ _	IF
ITOFO1	2		020	FFU	K2 D	V E _ _	TF

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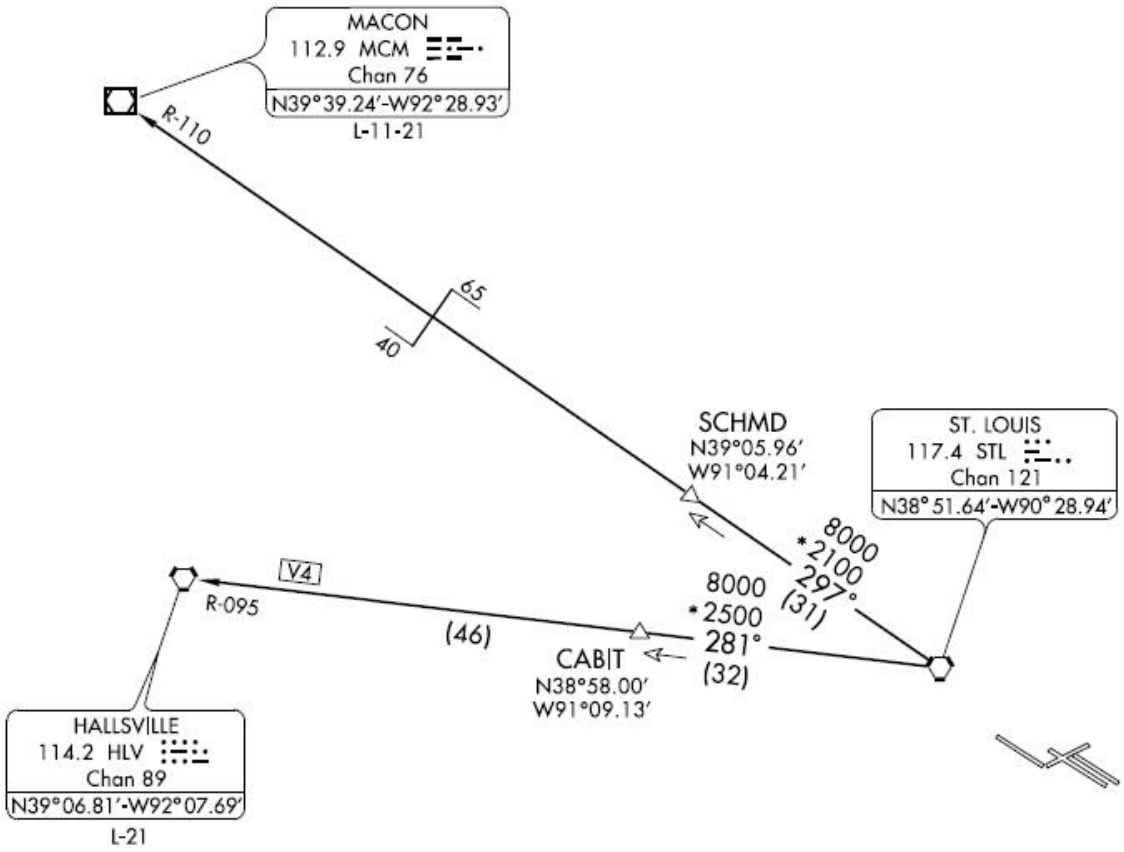
SID CODING EXAMPLE 9

Not for navigational or other operational use. For example only. Please consult current navigation charts.

SNOOZ TWO DEPARTURE (SNOOZ2.SNOOZ)



SID IDENT	ROUTE TYPE	TRANS IDENT	SEQ	WAYPOINT		DESC CODE	PATH TERM
SNOOZ2	2	ALL	010	STL	K2 D	V E _ _	IF
SNOOZ2	3	HLV	010	STL	K2 D	V _ _ _	IF
SNOOZ2	3	HLV	020	CABIT	K2 EA	E _ _ _	TF
SNOOZ2	3	HLV	020	HLV	K2 D	V E _ _	TF
SNOOZ2	3	MCM	010	STL	K2 D	V _ _ _	IF
SNOOZ2	3	MCM	020	SCHMD	K2 EA	E _ _ _	TF
SNOOZ2	3	MCM	030	MCM	K2 D	V E _ _	TF

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5.0 Standard Terminal Arrival Route (STAR) Coding Rules

5.1 If a STAR ends in vectors to a final approach (VM leg), the Airport Record **or Heliport Record for which the procedure is established** will be coded in the Waypoint Ident field of the STAR Record.

5.2 Deleted by Supplement 19.

5.3 If a STAR does not begin at a fix in the source documentation, the closest named fix along the STAR track must be assigned as the initial fix (IF leg) for the STAR.

5.4 If no crossing altitudes are specified on intermediate fixes of a STAR or Profile Descent, a "vertical angle" will be coded in the last leg of the procedure. This angle will be computed, based on the altitudes specified at the end fixes, to provide a constant descent path through all intermediate fixes. The angle provided will ensure compliance with minimum enroute altitudes for those segments without assigned altitudes.

5.5 A STAR which consists of a single path from an origination fix to a termination fix will be coded as a Route Type 2.

5.6 When a STAR Route or portion of a STAR Route is repeated with different Runway Identifiers or different Helipad Identifiers in the Transition Identifier it must be coded as a Runway Transition Route Type 3. When a STAR Route/Profile Descent Route or portion of a STAR Route is repeated with different Fix Identifiers in the Transition Identifier, it must be coded as an Enroute Transition Route Type 1.

5.7 When an Arrival Route serves the same runway or helipad as an Approach Route and the Arrival Route overlaps an Approach Transition, both the Arrival Route and the Approach Transition will be coded in their entirety in accordance with source documentation.

5.8 A STAR which consists of Enroute Transitions only can be coded with the required Route Type 3 coding, followed by a single IF leg as a Route Type 2. The fix on which the IF leg is coded must be the last fix in all of the Enroute Transitions. The Transition Identifier must be coded in accordance with Chapter Five, Section 5.11. In the cases where all the Enroute Transitions do not end at the same fix, but where most end at the same fix then a partial STAR may be coded.

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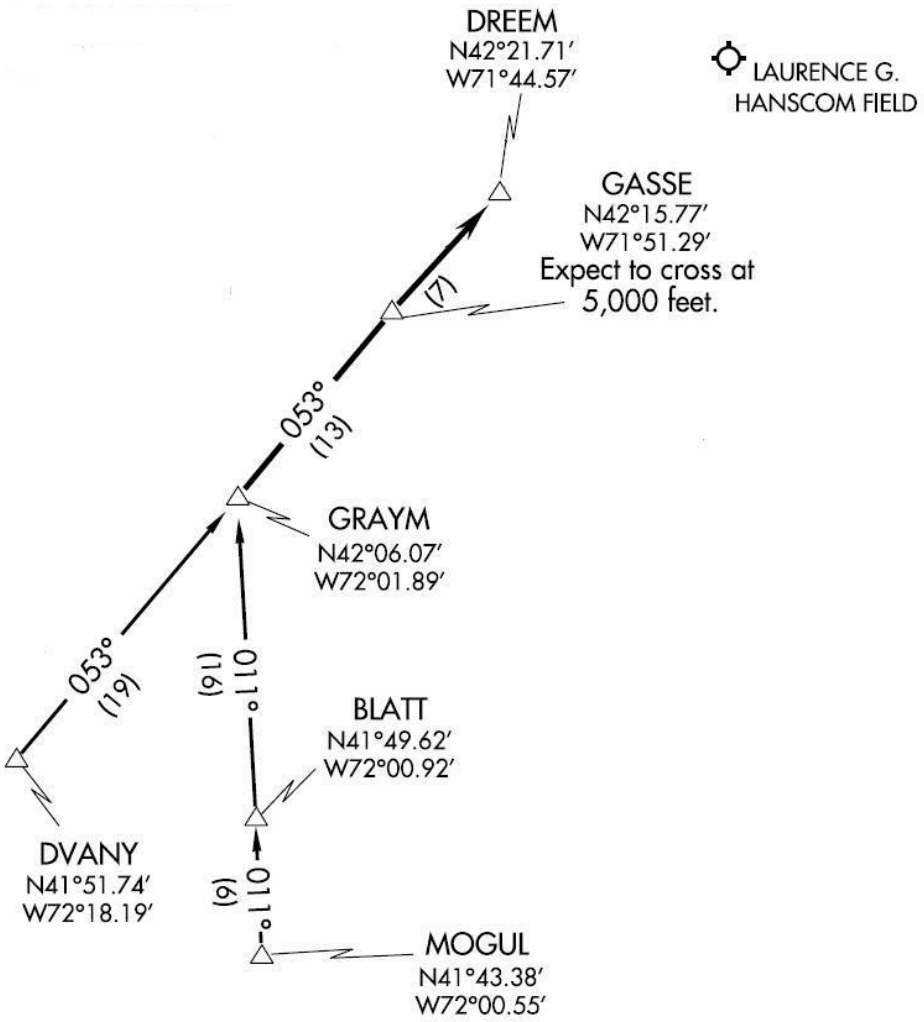
STAR CODING EXAMPLE 1

Not for navigational or other operational use. For example only. Please consult current navigation charts.

GRAYM TWO ARRIVAL (GRAYM.GRAYM2)



STAR IDENT	ROUTE TYPE	TRANS IDENT	SEQ	WAYPOINT	DESC CODE	PATH TERM
GRAYM2	1	DVANY	010	DVANY K6 EA	E _ _ _	IF
GRAYM2	1	DVANY	020	GRAYM K6 EA	E E _ _	TF
GRAYM2	1	MOGUL	010	MOGUL K6 EA	E _ _ _	IF
GRAYM2	1	MOGUL	020	BLATT K6 EA	E _ _ _	TF
GRAYM2	1	MOGUL	030	GRAYM K6 EA	E E _ _	TF
GRAYM2	2	ALL	010	GRAYM K6 EA	E _ _ _	IF
GRAYM2	2	ALL	020	GASSE K6 EA	E _ _ _	TF
GRAYM2	2	ALL	030	DREEM K6 EA	E E _ _	TF

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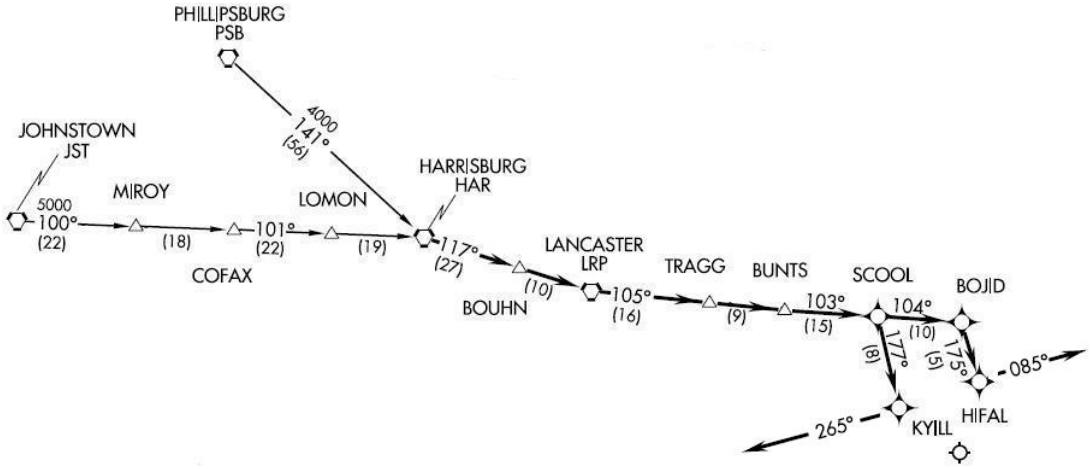
STAR CODING EXAMPLE 2

Not for navigational or other operational use. For example only. Please consult current navigation charts.

BOJID ONE ARRIVAL (BOJID.BOJID1)



STAR IDENT	ROUTE TYPE	TRANS IDENT	SEQ	WAYPOINT		DESC CODE	PATH TERM
BOJID1	1	JST	010	JST	K6 D	V _ _ _	IF
BOJID1	1	JST	020	MIROY	K6 EA	E _ _ _	TF
BOJID1	1	JST	030	COFAX	K6 EA	E _ _ _	TF
BOJID1	1	JST	040	LOMON	K6 EA	E _ _ _	TF
BOJID1	1	JST	050	HAR	K6 D	V E _ _	TF
BOJID1	1	PSB	010	PSB	K6 D	V _ _ _	IF
BOJID1	1	PSB	020	HAR	K6 D	V E _ _	TF
BOJID1	2		010	HAR	K6 D	V _ _ _	IF
BOJID1	2		020	BOUHN	K6 EA	E _ _ _	TF
BOJID1	2		030	LRP	K6 D	V _ _ _	TF
BOJID1	2		040	TRAGG	K6 EA	E _ _ _	TF
BOJID1	2		050	BUNTS	K6 EA	E _ _ _	TF
BOJID1	2		060	SCOOL	K6 EA	E E _ _	TF
BOJID1	3	RW09B	010	SCOOL	K6 EA	E _ _ _	IF
BOJID1	3	RW09B	020	KYILL	K6 EA	E _ _ _	TF
BOJID1	3	RW09B	030	KPHL	K6 PA	A E _ _	VM
BOJID1	3	RW27B	010	SCOOL	K6 EA	E _ _ _	IF
BOJID1	3	RW27B	020	BOJID	K6 EA	E _ _ _	TF
BOJID1	3	RW27B	030	HIFAL	K6 EA	E _ _ _	TF
BOJID1	3	RW27B	040	KPHL	K6 PA	A E _ _	VM

NOTE: If a STAR route ends with a Vector heading, the Airport ident is entered in the waypoint ident field.

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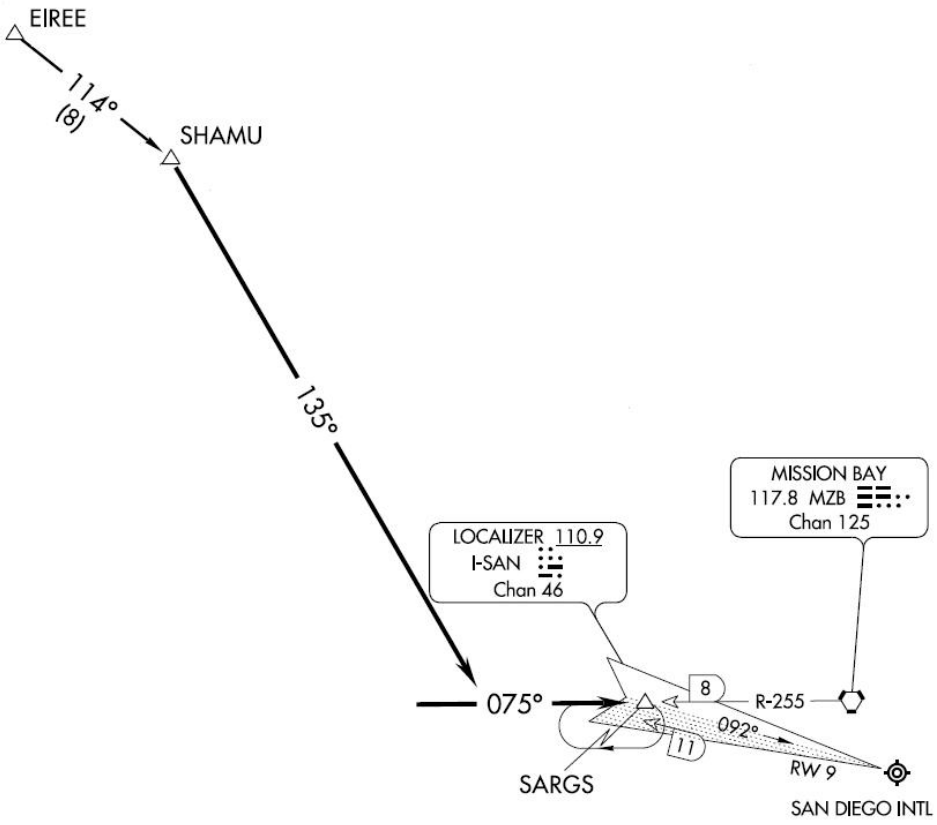
STAR CODING EXAMPLE 3

Not for navigational or other operational use. For example only. Please consult current navigation charts.

SHAMU ONE ARRIVAL (SHAMU.SHAMU1)



STAR IDENT	ROUTE TYPE	TRANS IDENT	SEQ	WAYPOINT	DESC CODE	PATH TERM
SHAMU1	2	RW09	010	EIREE K2 EA	E _ _ _	IF
SHAMU1	2	RW09	020	SHAMU K2 EA	E _ _ _	TF
SHAMU1	2	RW09	030		_ _ _ _	VI
SHAMU1	2	RW09	040	SARGS K2 EA	E E _ _	CF

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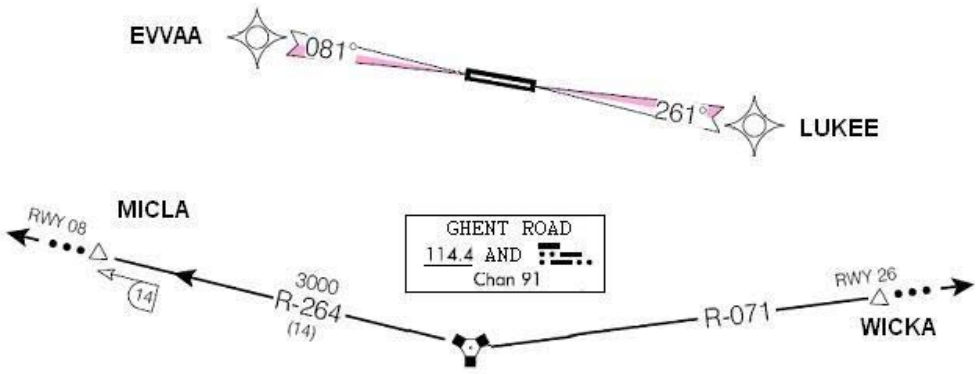
STAR CODING EXAMPLE 4

Not for navigational or other operational use. For example only. Please consult current navigation charts.

GHENT ONE ARRIVAL (AND.GHENT1)



STAR IDENT	ROUTE TYPE	TRANS IDENT	SEQ	WAYPOINT		DESC CODE	PATH TERM
GHENT1	3	RW08	010	AND	K3 D	V _ _ _	IF
GHENT1	3	RW08	020	MICLA	K3 EA	E _ _ _	TF
GHENT1	3	RW08	030	MICLA	K3 EA	E _ _ _	FM
GHENT1	3	RW08	040	EVVAA	K3 EA	E E _ _	CF
GHENT1	3	RW26	010	AND	K3 D	V _ _ _	IF
GHENT1	3	RW26	020	WICKA	K3 EA	E _ _ _	TF
GHENT1	3	RW26	030	WICKA	K3 EA	E _ _ _	FM
GHENT1	3	RW26	040	LUKEE	K3 EA	E E _ _	CF

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6.0 Approach Procedure Rules Valid For All Procedure Types

6.1 Multiple Approach Procedure Coding

6.1.1 Multiple Approach Procedure Definition

1. ILS Localizer
2. IGS Localizer
3. LDA Localizer
4. SDF Localizer
5. Localizer (Only)
6. Localizer Backcourse
7. MLS (all types)
8. RNAV **(Area Navigation)**
9. VORDME
10. VORTAC
11. VOR (no DME)
12. TACAN
13. NDB + DME
14. NDB
15 *FMS*
16 *GPS*
17 GLS

Notes: GPS, GLS, RNAV, are not facility types but rather an equipment classification. RNAV procedures use VORDME or VORTAC navaids along with the RNAV equipment. For the purpose of these rules, RNAV is to be considered a facility type. This will allow coding of a RNAV and VORDME or VORTAC procedure to the same runway or helipad. For GPS, GLS, a GPS or GLS sensor input to the equipment is required.

It is possible to have multiple RNAV-type approach procedures to the same runway due to official government source procedure designations, such as RNAV and GPS. These procedure designations are handled according to the rules for Route Type in Section 5.7. It is also possible to have multiple RNAV Terminal Procedures of the same type to the same runway, due to official government source procedure designations such as RNAV Y Rwy 27 and RNAV Z Rwy 27. These multiple procedures of the same type are handled according to the rules for Approach Procedure Idents in Section 5.10

Circle to Land minimum version of the various approach sensors are covered through the Approach Route Qualifier (see Section 5.7). Circle-To-Land is not a facility type but rather a weather minimum criteria. For the purpose of these rules, Circle-To-Land is to be considered an equal to the procedure reference facility.

There are three types of MLS Approach, each with a unique Route Type. Normally, there will only be one approach referencing MLS to any given runway or helipad.

GLS provides the necessary corrections to ensure precision navigation for alignment and descent during final approach. GLS-based approaches are consequently considered precision approach procedures. For GLS-based procedures, a GNSS and GLS sensor input is required.

6.1.2 Multiple Approach Procedure Identifiers

Multiple approach procedures are identified by unique procedure identifiers and unique route types (refer to Sections 5.7 and 5.10 of this specification).

6.1.3 Multiple Approach Procedure Waypoints

Multiple approach procedures to one and the same runway or helipad may require multiple Final Approach **Coding** waypoints of the same category such as FACF, FAF and MAP. Where such waypoints are not established with unique identifiers through source documentation, **they must be created following the rules for identifiers in** Section 7.2.6 of this specification.

6.1.4 Multiple Approach Procedure Detail

Specific details of approach procedures such as speed, altitudes and vertical angles are considered unique for the procedure and must be coded in those records where they apply, including duplication of such detail where appropriate.

6.1.5 Transitions in Multiple Approach Procedure Coding

6.1.5.1 Approach transitions are coded to be used together with specific approach procedures. As such, a transition route must be unique to a given approach, “multiple use” with more than one approach cannot be coded. Transition routes required for more than one approach must be coded multiple times.

6.1.5.2 If an approach transition route is be coded multiple times, it must be coded with an identifier that is unique to the approach procedure for which it is to be used.

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6.2 General Rules, applicable to all Approach Route coding.

6.2.1 "Altitudes" used in approach route coding between the final approach course fix (FACF) and the runway or helipad or missed approach point will be coded in combination with Altitude Description Codes as detailed in Section 5.29 of this specification and in accordance with government source documents. This coding rule is intended to match the altitude publishing methods in official government sources, which may specify altitudes as minimum, maximum, mandatory, recommended, or between altitudes, defined with a minimum and a maximum altitude.

These definitions include two kinds of altitudes. One is "procedural altitudes" information on mandatory or minimum altitudes at designated fixes along the final approach path. The other is the altitude related to the electronic Glide Slope or published vertical angle with recommended altitudes. The ordering of Procedural and Glide slope altitudes in the first and second altitude fields is accomplished in accordance with Section 5.29.

6.2.2 All fixes associated with the lateral and vertical path of approach procedures must be coded, including step-down fixes, both before and after the Final Approach Fix.

6.2.3 With the exception of the NDB + DME Approach or a Helicopter version of a NDB + DME Approach **Procedure**, the recommended Navaid must be the same facility for all **sequences in the** final approach **coding, excluding the** missed approach **procedure** sequences, **which** require a recommended facility. The recommended **facility** must be the procedure reference facility. For Approach Transitions, the recommended Navaid **must** be the procedure reference facility or a VORDME or VORTAC facility. When a VORDME or VORTAC is coded as the recommended Navaid in **Final** Approach Coding, the Navaid will be within 40 NM of the fix in which it is coded. GPS approach procedures do not include a recommended Navaid. GLS approach procedures **must** reference the GLS facility. For specific rules on recommended Navaid for NDB + DME Approach Procedures, see Rule 6.8.1.4.b of this Attachment.

6.2.4 If a PI leg is from the FAF waypoint and the distance between the FACF and the FAF is less than 6.0 NM, code a CF leg after the PI, with the FACF as the fix in the CF. The route distance leg data field on the CF leg will be the difference between the distance coded on the PI leg and the distance between the FACF and FAF waypoints.

6.2.5 Approach Procedure Fix Requirements:

- All **Final** Approach Coding requires a Final Approach Fix (FAF) and a Missed Approach Point Fix (MAP).
- The coding of Final Approach Course Fix (FACF) must be accomplished using the following rules:
  - A FACF Waypoint is required on all localizer based approaches, including, ILS, LOC, SDF, LDA, IGS and LOC Backcourse.
  - A FACF Waypoint is required **in the Final** Approach Procedure coding **of other approach procedure types when:**
    - At a published Intermediate Fix
    - **A FACF is** published **in** source **documentation**
    - A named fix **is the** common ending point for transitions
    - A transitions end in the intercept of a track rather than to a fix.
    - The course or track inbound to the FAF is different than the course or track from the FAF to the MAP. This requirement will be met by the addition of data supplier created FACF waypoint as indicated in Rule 6.2.5.2.

6.2.5.1 Transitions may end at a FAF fix regardless of whether a FACF is coded for the procedure or not.

6.2.5.2 If no waypoint is established by source documentation for the final approach course fix and one is required by the requirements in Rule 6.2.5, one must be computed by the data supplier. For Localizer based approach procedures, the computer FACF will be on the localizer course at a distance of 2 to 8NM from the Final Approach Fix (FAF). For all other approach types, the FACF will be computed on the published course to the FAF at a distance not less than 2NM to the FAF. Section 2, Subsection 2.3, Special Navigation Terms, Altitude coding for this fix is defined in Rule 6.2.10.

6.2.5.3 If no waypoint is established by source documentation for the Final Approach Fix (FAF), one must be computed on the final approach course, using the initial approach altitude and the vertical descent angle (source or computed). For non-precision approach procedures such as VOR or NDB, the minimum distance between the FAF and the runway threshold or helipad alighting point (or MAP) will be 4 nautical miles. For Localizer and GLS based approach procedures, establish the FAF, when none is provided by source, at the nominal outer marker position.

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6.2.5.4 **Except as indicated by specific rules within this Attachment, the source provide published Missed Approach Point (MAP) is always coded as part of the approach procedure. This MAP may be a runway threshold (Landing Threshold Point, LTP), a helipad alighting point or a dedicated missed approach point fix. The published missed approach point may be replaced by a database provider established fix when it can be determined that the published fix was intended to be the runway threshold or helipad alighting point. For more detail, see section 8.10 of this Attachment.**

6.2.6 Removed by Supplement 19.

6.2.7 Removed by Supplement 19.

6.2.8 Intentionally left blank

6.2.9 Lateral Coding Rules

All approach procedure coding must be to the published Missed Approach Point, as indicated below. Missed Approach Procedure coding must begin at that point. For missed approach procedure coding, refer to Section Nine of this Attachment. For the rules that follow, the term "runway threshold" is meant to refer either to the landing threshold point (LTP) of an actual runway or to a helipad alighting point (HAP), when the procedure is coded to a helipad.

6.2.9.1 If the published Missed Approach Point is a fix prior to the runway threshold, lateral coding is to that published Missed Approach Point.

6.2.9.2 If the published Missed Approach Point is the runway threshold, lateral coding is to the runway threshold as the published Missed Approach Point.

6.2.9.3 If the published missed approach point is beyond the runway threshold and the runway threshold will be coded as a fix in the lateral path that fix will be on the established path, with no course changes.

6.2.9.4 **If the published Missed Approach Point is beyond the runway threshold and a runway threshold fix cannot be inserted as defined in Rule 8.10, a Final End Point fix is to be inserted into the final approach coding sequence. For complete details, see Rule 8.10.**

6.2.9.5 If the published Missed Approach Point is abeam the runway threshold, lateral coding must be to the published Missed Approach Point.

Refer to the Examples 1 through 15 at the end of this section for a visual depiction of these rules.

6.2.10 Vertical Coding Rules, Procedure Fix Altitudes

Vertical Approach Procedure Coding is provided through two elements, Procedure Fix Altitudes and a Vertical Angle. This section covers the Fix Altitude. Sections 7 and 8 cover the Vertical Angle for Precision and Non-Precision Approach Procedures.

6.2.10.1 Procedure Fix Altitudes, Final Approach Course Fix and Final Approach Fix.

Procedure Fix Altitudes for the Final Approach Course Fix will be coded according to official government sources and will be left blank when no altitude data is provided by the source. When coded, these altitudes will be assigned altitude descriptions codes indicating the altitude as mandatory, minimum or recommended (see Section 5.29 of this Specification). When the coded Final Approach Fix is established by the government source procedure data, the altitudes for this fix will also be coded according to official government sources. These altitudes will be assigned altitude descriptions codes indicating the altitude as mandatory, minimum or recommended (see Section 5.29 of this Specification). For government supplied altitude data, both Altitude 1 and Altitude 2 may be provided. If the Final Approach Fix is as an established fix rather than a published fix, the altitude for this fix must be computed using the procedures detailed in Sections 7 and 8 of this Attachment. For calculated altitude data, only Altitude 1 will be provided.

6.2.10.2 Procedure Fix Altitudes for the published Missed Approach Point, a runway threshold fix prior to the published Missed Approach Point or a Final End Point prior to the Missed Approach Point must be as indicated below.

6.2.10.2.a For a published Missed Approach Point prior to the runway threshold, an "at" altitude equal to the computed altitude at the published Missed Approach Point must be coded in Altitude 1. (See example 7)

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6.2.10.2.b For a published Missed Approach Point at the runway threshold, an "at" altitude equal to the runway threshold elevation plus the published TCH must be coded in Altitude 1. If **no procedure** TCH is specified by source then use **40 or** 50 **feet (See Section 5.67of this standard).**

6.2.10.2.c For a published Missed Approach Point beyond the runway threshold and where a landing threshold point fix has been inserted into the final approach coding by the data supplier, based on the rules in Rule 8.10 of this Attachment, code an "at" altitude equal to the runway threshold elevation plus the published TCH (if no procedure TCH is specified by source, then use 40 or 50 feet) in Altitude 1 of the landing threshold point fix record. See Rule 8.10 of this attachment

6.2.10.2.d For a published Missed Approach Point beyond the runway threshold and where a final end point fix has been inserted into the final approach coding by the data supplier, based on the rules in Rule 8.10 of this Attachment, code an "at" altitude equal to the runway threshold elevation plus the published TCH (if no procedure TCH is specified by source, then use 40 or 50 feet) in Altitude 1 of the final end point fix record. See also Rule 8.10 of this attachment.

6.2.10.2.e For a published Missed Approach Point abeam the runway threshold, code the altitude equal to the runway threshold elevation plus the published TCH. If **no** TCH is specified by source then use **40 or** 50 feet **(See Section 5.67of this standard).**

6.2.10.3 Step down fixes will have altitude codes according to the government source documentation. These altitudes will be assigned altitude descriptions codes indicating the altitude as mandatory, minimum or recommended and altitudes on the vertical path (see Section 5.29 of this Specification). Both Altitude 1 and Altitude 2 will be used on step-down fixes.

6.2.10.4 For RF Leg Fixes not at Procedure Fix locations, Waypoint Description Code, 3rd position of "R", will have altitudes coded according to government source documentation. If no altitude information is provided for these fixes in source data, Altitude 1 and 2 will be left blank.

6.2.11 Vertical angle information is in Section 7 and 8 of Attachment 5 in this document.

6.2.12 Missed Approach Point

In general, the design of missed approach procedures require that the runway, helipad or missed approach point be overflown prior to commencing any turn. In these cases, to ensure procedure coding reflects design specific intentions, the Overfly Indication must be coded into the Waypoint Description field. However, certain types of approach procedures design do require a turn prior to the runway, helipad or missed approach point. In these cases, to ensure procedure coding reflects design specific intentions, the Overfly Indication will not be set in the Waypoint Description field of the appropriate record.

6.3 Approach Transition Route Coding Rules

6.3.1 Recommended coding on approach transitions that end in leg to fix ("XF") is that the fix in the ending leg must be either the Final Approach Course Fix or the Final Approach Fix. If this is not the case, for example HF leg type transitions on fixes off-set from the final approach path, a series of legs must be substituted representing the original flight path, but ending with a CF or TF[1] leg type to one of these two fixes. If neither of these two coding recommendations can be followed, such as in cases where the "XF" would terminate at the missed approach waypoint or a step-down fix not associated with the lateral guidance of the final approach, the transition must be omitted.

6.3.2 When a holding pattern used for course reversal or a procedure turn is part of an approach route, it will be included in an approach transition route.

6.3.3 If an approach transition for a specific runway or helipad is common to more than one approach, that transition must be coded for each approach, with a transition identifier that must correspond to the approach procedure identifier.

6.3.4 Rule Deleted by Supplement 16

6.3.5 Transitions of VOR based approach procedures, TACAN based approach procedures, and RNAV **(VOR DME)** approach procedures.

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[1] In general "CF" legs are used in final approach coding. "TF" legs are used in FMS and GPS Approach Procedures, some types of MLS Procedures and in other procedure types where the determination has been made that a TF will work better than a CF.

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6.3.5.1 Any recommended navaid used in coding must be a VOR, VORDME, VORTAC, TACAN, DME, NDB or Un-Biased ILS DME, see Section 5.23 of this document.

6.3.6 Transitions for Localizer Based Approach Procedures

6.3.6.1 The ending leg of all localizer-based transitions will either –

- end at the FACF (AF, CF, RF, TF, HF, HM)
- end in an intercept of the localizer inside the FACF (PI, CI or VI)
- end in a course reversal, normally at the FAF (HF, HM)

6.3.6.2 The ending leg of all localizer-based transitions will contain a recommended Navaid –

- if CF, RF, TF, CI or VI, the recommended Navaid will be the procedure reference localizer
- if AF, HF, HM or PI, the recommended Navaid will be a VORDME or VORTAC or TACAN
- the HF, HM and PI leg may use the procedure reference localizer when a VORDME, VORTAC or TACAN is not available.

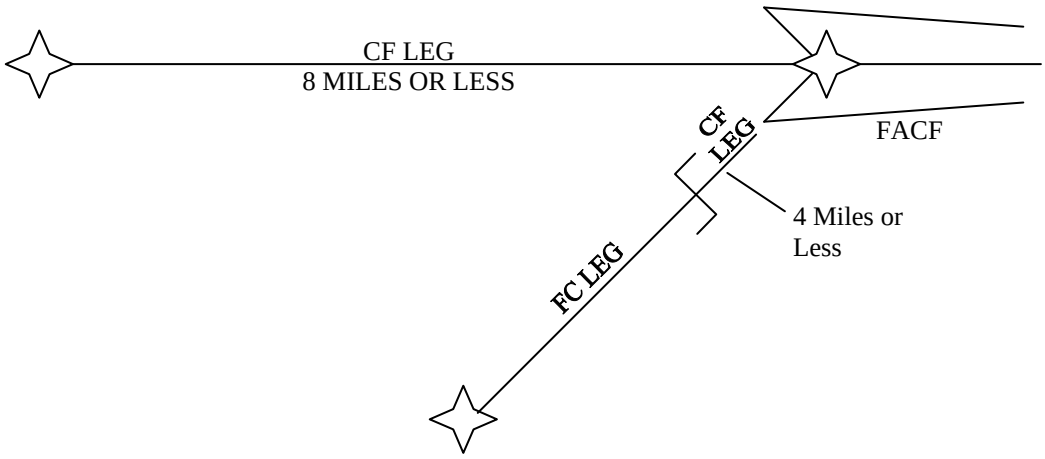
6.3.6.3 Deleted by Supplement 17.

6.3.6.4 Legs ending in an intercept will ideally be at angles of 30 degrees to the track intercepted. Angles between 10 and 90 degrees may be coded as required by source documentation, provided the resulting intercept is within the reception area of the localizer.

6.3.6.5 When a CF leg is used as the ending leg of a transition to a localizer-based procedure, the maximum leg distance will be within 8NM of the FACF or within the reception area of the localizer as depicted in Figure A5-6-1.

Note: **For precision approaches relying on an electronic glide slope, an** FC/CF is preferred over a TF as illustrated in Figure A5-6-1 for those legs ending at a fix.

Figure A5-6-1



6.3.6.6 When a CI or VI leg is used as the ending leg of a transition to a localizer-based procedure, the intercept will be between the FACF and the FAF, at no less than 2 NM to the FAF.

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6.3.7 Transitions for NDB Based Approach Procedures:

6.3.7.1 Transitions for NDB based approach procedures may use a NDB Navaid as the recommended Navaid, except for transitions that are DME Arcs.

6.3.8 Transitions for MLS/GLS Approach Procedures:

6.3.8.1 MLS/GLS approach procedure coding is such that the rules can be identical to those used for Localizer based procedure coding. **See Rule 6.3.6 above.**

6.3.9 Transitions for Circle-To-Land Approach Procedures

6.3.9.1 If the Circle-To-Land approach procedure is runway or helipad dependent, the rules on transition route coding are identical to those of the reference facility procedure type, e.g. for a VOR Circle-To-Land that is runway dependent, follow the VOR based approach procedure rules for approach transition route coding.

6.3.9.2 If the Circle-To-Land approach procedure is not runway or helipad dependent, being valid for more than one landing direction, the rules for coding approach transitions routes are as follows.

6.3.9.3 Recommended navaids used in coding all legs except ending legs must be a VOR, VORDME, VORTAC, **TACAN,** DME or Un-Biased ILSDME. For the ending leg sequences, the recommended navaid, where required, must be the procedure reference facility, see Section 5.23 of this document.

6.4 Localizer Based Approach Procedure Coding

6.4.1 **The following rules apply to the Final Approach Coding of Localizer-based approach procedures**. **These procedures may include Full ILS (Localizer and Glide Slope), Localizer Only, Instrument Guidance System (IGS), Localizer Directional Aid (LDA) and Simplified Directional (SDF) procedures.**

6.4.1.1 All Localizer based approach procedures must begin at the FACF. They must consist of a FACF, FAF and runway Fix (precision approach) or missed approach point fix (non-precision approach).

6.4.1.2 The FACF is defined as a fix located on the localizer beam center, 8NM or less from the FAF or within the reception range of the Localizer. This may be a source document provided fix or a fix created using these positioning rules.

6.4.1.3 The FACF is coded as an IF leg with an altitude assigned, based on the source document or equal to the altitude of a procedure turn or the altitude of the last transition leg.

6.4.1.4 The track from the FACF to the FAF is coded as a CF or a TF leg with altitude constraints as indicated for the specific procedure types below.

6.4.1.5 The recommended navaid will be the procedure reference localizer. Theta and Rho will be provided from the localizer for each sequence of the final approach, including the runway fix and/or missed approach point.

6.4.1.6 The “Outbound Magnetic Course” field in all sequences will be equal to the localizer magnetic bearing, rounded to the nearest whole degree, derived from official government source.

6.4.1.7 No Localizer based approach procedure may include a Final End Point Fix (FEP). For this reason, the Rules 6.2.9.3, 6.2.9.4, 6.2.10.2.c, 6.2.10.2.d do not apply.

6.4.2 Full ILS (Localizer and Glide Slope) Precision Approach Procedure

6.4.2.1 For full ILS procedures **and GLS procedures**, code the glide slope intercept altitude in the altitude 2 field of the FACF **and FAF** record**s**,

6.4.2.2 Rule removed in Supplement 19.

6.5 MLS Approach Procedure Coding

MLS Approach Procedures utilizing raw azimuth and elevation data **will be coded as** localizer equivalent. The rules **applied** for the Final Approach **Coding must** be identical with those stated in Section 6.4 **of this attachment.** The Route Type of such **procedures** will be coded as “M”. Approach procedures predicated on the use of MLS Area Navigation (MLS/RNAV) will be coded with a “W” or “Y” described below.

There are three types of MLS/RNAV approaches, listed in increasing level of complexity, computed lateral/raw

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vertical guidance, computer lateral and vertical guidance and curved path.

6.5.1 Approaches using computed lateral path and raw vertical path guidance, also referred to as Type "A," will be used primarily where the MLS azimuth transmitter cannot be located on the extended runway centerline, but the elevation transmitter is sited normally abeam the touchdown point. All legs will be straight and aligned with the inbound course. **There** will be codes with Route Type "W" in column 20 of the primary approach record. Path definition will be the equivalent of a full ILS approach (Rule 6.4.2) with the exception that the leg from the PFAF inbound will be a "TF" leg, terminating at the runway waypoint, with the published final approach source in the Outbound Magnetic Course field. The PFAF will be coded as the Final Approach Fix in the Waypoint Description field and the first fix prior to the PFAF will be coded as the Final Approach Course Fix.

6.5.2 Approach using computed lateral and vertical guidance by no curved legs, also referred to as Type "B," will be coded as Route Type "Y" in column 20 of the primary approach record. All legs will be straight and aligned with the inbound course. Path definition will be the equivalent of the full ILS approach (Rule 6.4.2) with the exception that the leg from the PFAF inbound will be a "TF" leg, with the published final approach course in the Outbound Magnetic Course field. The altitude of the PFAF and all waypoints inbound from it will be the glide path altitude at that point. The PFAF will be coded as the Final Approach Fix in Waypoint Description field and the first fix prior to the PFAF will be coded as the Final Approach Course Fix.

6.5.3 MLS/RNAV approaches using curved legs, also referred to as Type "C," will be used for a variety of reasons, including parallel sidestep approaches, separation of different categories of aircraft, noise abatement, etc. These will always be precision approaches. They will be coded a with a Route type of "Y" in column 20 of the primary approach record. The following rules apply:

6.5.3.1 The first leg of an MLS/RNAV approach with curved legs will be an "IF/TF" leg combination. All other straight legs will be coded as "TF" legs. All "TF" legs in an MLS/RNAV with curved legs procedure will have the published course included in the Outbound Magnetic Course field.

6.5.3.2 All curved legs will be coded as "RF" legs.

6.5.3.3 The initial portion of a MLS/RNAV approach with curved legs may be an "IF/RF" combination provided a straight leg approach transition is coded to the point in the "IF" and the rules in Section 6.5.2 are complied with.

6.5.3.4 The PFAF will be coded as the Final Approach Fix in the Waypoint Description field and the first fix prior to the PFAF will be coded as the Final Approach Course Fix. If there is not a fix at the glide path intercept, then the first fix after the intercept will be the PFAF. There must be one and only one PFAF for each MLS/RNAV approach with curved legs.

6.5.3.5 The last leg of an approach transition prior to an MLS/RNAV approach will be one of the following types CF, CI, HF, PI, RF or TF, except as indicated in Section 6.5.3.3. If the leg type is CF, CI, RF or TF, then the Recommended Navaid will contain the identifier of the MLS used for the approach. If the leg type is PI or HF, then the Recommended Navaid will contain the VHF Navaid the defines the PI or HF leg.

6.5.3.6 If the last leg prior to the approach is a "CI" leg, the intercept angle will be 300 or less, and the intercept point will be between the first and second terminator fixes in the approach, but no closer that 2NM to the second fix.

6.5.3.7 The PFAF and the FACF altitudes will be coded according to the rules outlined for Precision Approach Procedures in Rule 6.4.2.

6.5.4 The PFAF will be used in precision MLS/RNAV approaches. It is defined as that fix along the lateral path where the published barometric altitude intercepts the glide slope. Prior to the PFAF, the aircraft is expected to fly barometric altitude to intercept the glide path. All waypoints up to the PFAF should be coded using the published barometric crossing altitude. The PFAF and all waypoints after it should be coded using the true altitude of the glide path at those points.

6.6 VOR, VORDME, VORTAC and **Area Navigation** Approach Procedure Coding

The following rules apply to the **Final Approach** Coding of all VOR based approach procedures, regardless of the reference facility type, and to RNAV **(VOR DME) Approach Procedures.**

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6.6.1 Reference Facility Specific Rules

The following rules apply to the **Final Approach** Coding of specific reference facility VOR based approach procedures.

6.6.1.1 When the reference facility is VOR only or there is no DME collocated with VOR (see VOR coding examples 1, 3 and 8), the following applies:

6.6.1.1.a Final Approach **Coding** will be **accomplished** using IF and CF or TF legs only.

6.6.1.1.b Final Approach **Coding** must include either a FAF and a runway fix or FAF and missed approach point fix.

6.6.1.1.c The recommended navaid will be the procedure reference VOR. Theta values will be provided from that facility in all Final Approach **Coding** sequences.

6.6.1.2 When the reference facility is VORDME or VORTAC (see VOR coding examples 2 and 6), the following applies:

6.6.1.2.a Final Approach **Coding** will be **accomplished** using IF and CF or TF legs only.

6.6.1.2.b Final Approach **Coding** must include FAF and either a runway fix or missed approach point fix. The Final Approach **Coding** will include a FACF when one is required by Rule 6.2.5.

6.6.1.2.c The recommended navaid will be the procedure reference VORDME or VORTAC. Theta and Rho values will be provided from that facility in all Final Approach **Coding** sequences.

6.6.2 Examples of VOR Coding

6.6.2.1 Example of missed approach point before the runway threshold, refer to VOR coding examples 7 and 8.

6.6.2.2 Example of missed approach point at the runway threshold, refer to VOR coding example 1 and 2.

6.6.2.3 Example of missed approach point beyond the runway threshold and the final course passes over the runway threshold, refer to VOR coding examples 3 and 4.

6.6.2.4 Example of missed approach point beyond the runway threshold and the final approach course does not cross runway threshold, refer to VOR coding examples 5 and 6.

6.6.3 RNAV- **Area Navigation** Procedures

RNAV-Area Navigation procedures in the context of this attachment are the so called RNAV Procedures that require referencing ground –based navigational aids. These procedures are also referred to as VOR DME RNAV Procedures.

6.6.3.1 All RNAV- **Area Navigation** approach procedures will be coded to a runway threshold **as the last leg in the Final Approach Coding sequence. The runway threshold may be a source defined name waypoint.**

6.6.3.2 The recommended navaid for VORDME RNAV approaches will be the procedure VORDME or VORTAC. Theta and Rho values will be provided from that facility in all Final Approach **Coding** sequences.

6.6.3.3 Final Approach **Coding** will be **accomplished** using IF and CF or TF legs only. RF legs may be used in the final approach **coding**, **FACF to FAF and FAF to MAP sequence.**

6.6.3.4 Final Approach **Coding** must include either a FAF and a runway fix or FAF and missed approach point fix.

6.7 TACAN Approach Procedure Coding

The following rules apply to the Final Approach Coding of all TACAN based approach procedures, regardless of the reference facility type.

6.7.1 Final Approach **Coding** will be **accomplished** using IF and CF or TF legs only.

6.7.2 Final Approach **Coding** must include, FAF and either a Runway Fix or Missed Approach Point Fix. **The Final Approach Coding must also include a FACF when one is required by the rules in 6.2.5.**

6.7.2.1 Deleted by Supplement 19.

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6.7.2.2 Rule Deleted by Supplement 19.

6.7.3 The Recommended Navaid will be the procedure reference TACAN **or VORTAC.**Theta and Rho values will be provided from that facility in all Final Approach **Coding** sequences.

6.7.4 Vertical Angle Rules

Vertical angle will be coded per the Rules in Section 6.2.11.

6.8 NDB Approach Procedure Coding

The following rules apply to the **Final Approach** Coding all NDB based approach procedures. NDB based approach procedures include procedures using a NDB or Locator as the reference facility and procedures using a NDB or Locator and a DME (NDB + DME) as reference facilities. NDB approach procedures not requiring DME but using the DME for reduced minimums will be coded as NDB + DME procedures.

6.8.1 Specific Reference Facility Rules

The following rules apply to the **Final Approach** Coding of specific reference facility NDB based approach procedures.

6.8.1.1 NDB **Final** Approach **Coding** must include at least FAF and a runway fix or missed approach point fix.

6.8.1.1.a Rule deleted by Supplement 17.

6.8.1.2 NDB + DME **Final** Approach **Coding** must include an FAF and runway fix or missed approach point fix. The Final Approach Coding will include a FACF when one is required by Rule 6.2.5.

6.8.1.3 **Final Approach** Coding will **be accomplished** use IF and CF or TF legs through to the runway fix or missed approach point fix. The IF leg will be at the FAF or **at the** FACF when **one is** coded.

6.8.1.4 Recommended Navaid Requirements

6.8.1.4.a On NDB *P*rocedures **Final Approach Coding**, the recommended navaid information will be provided on the FACF (where coded), the FAF and the Missed Approach Fix record. It will be the procedure reference NDB or Locator. Theta and Rho information will not be provided on any sequence.

6.8.1.4.b On NDB + DME procedures **Final Approach Coding**, the recommended navaid information will be provided on all sequences. On the FACF (when coded) and FAF, this navaid will be the procedure reference NDB or Locator. On the runway fix or missed approach point fix, the recommended navaid will be the procedure reference DME. A procedure reference DME may be any navaid with DME, including **unbiased** ILSDMEs. The Theta and Rho information will not be provided in any sequence of the NDB + DME final approach except in the runway fix or missed approach point fix sequence. That sequence will include Rho information from the procedure reference DME Navaid.

6.8.2 Examples of NDB Coding

6.8.2.1 Example of missed approach point before the runway threshold, refer to NDB coding example 4.

6.8.2.2 Example of missed approach point at the runway threshold, refer to NDB coding example 1.

6.8.2.3 Example of missed approach point beyond the runway threshold and the final course passes over the runway threshold, refer to NDB coding example 2.

6.8.2.4 Example of missed approach point beyond the runway threshold and the final approach course does not cross runway threshold, refer to NDB coding example 3.

6.9 **RNAV Procedure Coding**

This section provides coding guidelines for RNAV Approach Procedures. The term RNAV Approach Procedure is defined as an approach procedure that was designed to provide lateral and vertical path guidance using criteria that do not rely on radial, bearing, arcs or glide paths from ground-based navaids. As such, a RNAV Approach Procedure will not include references to ground-based navaid associated information such as VOR radial, NDB bearings, DME distances, ILS course, MLS Azimuth and ILS or MLS glide slopes. Exceptions to this standard may be outlined in the rules that follow. Navaids may be referenced as waypoints in the coding of the RNAV Approach Procedures.

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6.9.1 Recommended Navaids

RNAV Approach Procedures do not require referencing a specific ground-based navaid. The exception to this rule is that in those cases where the official government source a specified a Path and Termination (leg type) that requires a Recommended Navaid (see Table 1.5, Leg Data Fields), the navaid will be included in the coding of the RNAV Approach Procedure as a Recommended Navaid. Including a Recommended Navaid does not imply that this navaid is required to be used in the navigation solution.

6.9.2 The Final Approach Coding must include a Final Approach Fix (FAF) and a Missed Approach Point fix (Runway Threshold or Missed Approach Point). The Final Approach Coding must include a Final Approach Course Fix (FACF when such a fix is required by the rules in Section 6.2.5.

6.9.3 The Final Approach Coding of RNAV Approach Procedures covered by this Section 6.9 do not require a Recommended Navaid except as governed by other rules in this attachment.

6.9.4 RNAV Approach Procedures coded as GLS Procedure according to Section 5.7 must reference the GLS facility as a Recommended Navaid.

6.9.5 The track from the FACF (when code) to the FAF and the FAF to MAP must be coded as TF or RF legs. The RF leg is not allowed as the first leg of the Final Approach Coding according to the Beginning/Ending Leg Table. The preferred coding when an approach starts with a precision arc is IF at the intermediate fix (FACF), followed by RF to the FAF.

6.10 Coding of Approach Procedures with Circle-To-Land Minimums

6.10.1 Any source provided approach procedure with circle-to-land minimums may be coded, regardless of sensor type through correct application of Route Type and Qualifier, see Section 5.7 of this specification.

6.10.2 Approach Procedures with circle-to-land minimums may include the runway designation in the procedure identifier when the straight-in alignment criteria are met or may be a unique identifier indicating non-straight-in alignment of the final approach course. See Section 5.10 of this specification.

6.10.2.1 Approach Procedures with circle-to-land minimums that meet straight-in alignment criteria and include the runway designation in the procedure identifier are coded following the same rules sets found elsewhere in this attachment for the appropriate sensor type.

6.10.2.2 Approach Procedures with circle-to-land minimums that do not meet straight-in alignment criteria and use the unique procedure identifiers without runway designation will be coded using the rule set **defined in Rule 6.10.3 of this attahcment.**

6.10.2.3 If a runway is in alignment, the first altitude field in the missed approach point fix record will be coded as the LTP elevation + published TCH (if no procedure TCH is specified by source use 40 or 50 feet. See Section 5.67 of this specification.

6.10.3 Coding Rules, Approach Procedures, Circle-To-Land Minimums, Non-straight-in Alignment

6.10.3.1 The final approach coding must include a FAF and a Missed Approach Point Fix.

6.10.3.2 If no runway is specified, the altitude 1 field in the missed approach point fix record will be coded with the airport elevation.

6.10.3.3 Altitude constraints and vertical angles will be coded as indication in Section 8 of this attachment, specifically Rules 8.6.4, 8.9.3, 8.9.4 and 8.9.8

6.10.3.4 Final approach segment coding will use IF and CF or TF legs only.

6.11 Lateral Guidance Rules

For FMS and GPS procedures, a missed approach point beyond the runway is not allowed, therefore Rules 6.2.9.3, 6.2.9.4, 6.2.9.5, 6.2.10.2.c, 6.2.10.2.d, and 6.2.10.2.e do not apply.

6.11.1 The track from the FACF to the FAF, where an FACF exists, is coded with TF or RF legs. The RF leg is not allowed as the first leg of the approach coding according to the Beginning/Ending Leg Table. The preferred coding when an approach starts with a precision arc is "IF" at the FACF, followed by "RF" to the FAF.

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6.12 Helicopter Procedure Coding

The following rules apply to the **Final Approach** Coding of of all Helicopter Approach Procedure. These rules cover Helicopter Approach Procedures which may be coded to Airports and Runways included in Sub-sections PA and PG only.

6.12.1 Helicopter Approach Procedures will be coding using the rules in Section 6 of this attachment, appropriate to the type of sensor required for the procedure, such as VORDME or ILS or RNAV- **Area Navigation**. This includes rules for Recommended Navaid, FACF requirements and leg types.

6.12.2 The Lateral Path Rules for the sensor related procedure coding reference a “runway fix” as a missed approach point or a missed approach point. Those same rules apply to helicopter procedures. For procedures designed with a dedicated helipad as the missed approach point, a Terminal Waypoint will be established and used as the missed approach point fix.

6.12.3 The Vertical Path Rules in Section 6 apply without exception, using the rules appropriate for the sensor type.

6.12.4 Missed Approach code will be accomplished according to the rule in Section 7 of this attachment, appropriate for the sensor type.

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VOR CODING EXAMPLE 1

Excerpted from Jeppesen Chart Adrar, Algeria, VOR Rwy 04 Approach Chart by permission of Jeppesen Sanderson, Inc.

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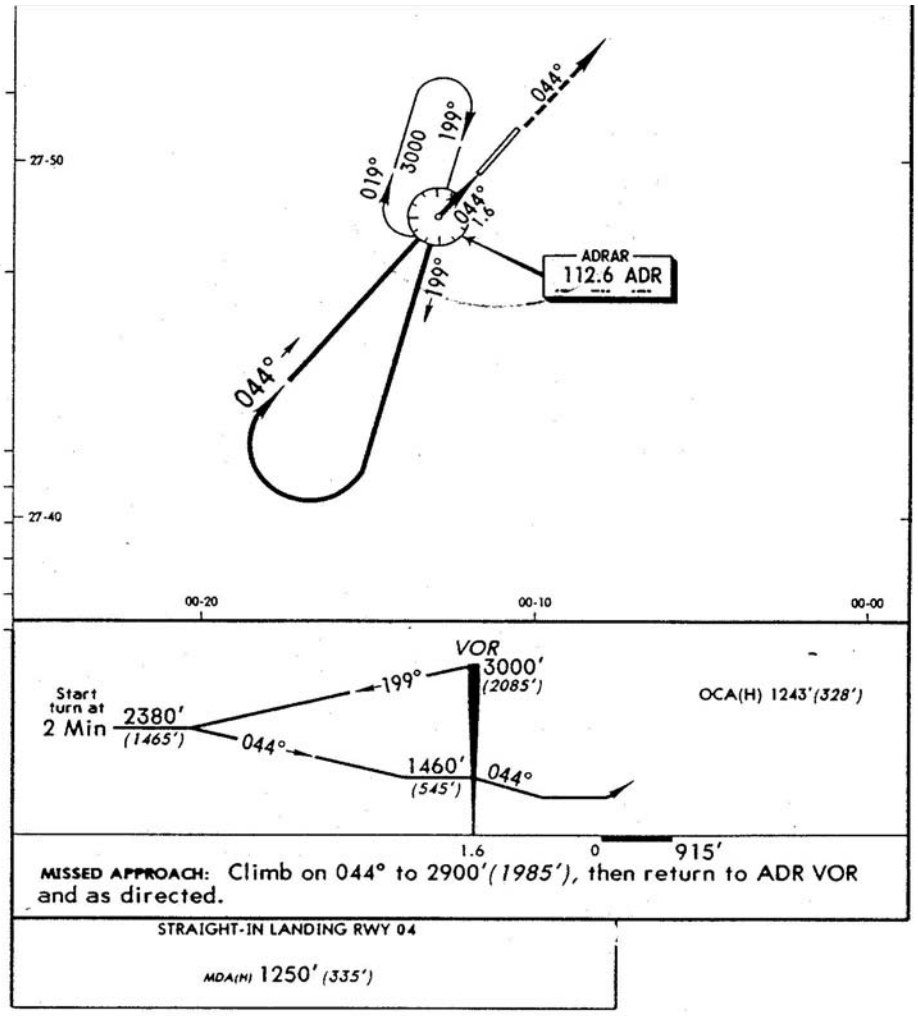

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V04	020	ADR	CF	ADR	V__F	0000	0440	0034	01460	
V04	030	RW04	CF	ADR	G___	0016	0440	0016	00965	-300
V04	040		VA		__M_		0440		02900	
V04	050	ADR	DF		VE__					

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VOR CODING EXAMPLE 2

Excerpted from Jeppesen Chart Agadir, Morocco, VOR Rwy 28 Approach Chart by permission of Jeppesen Sanderson, Inc.

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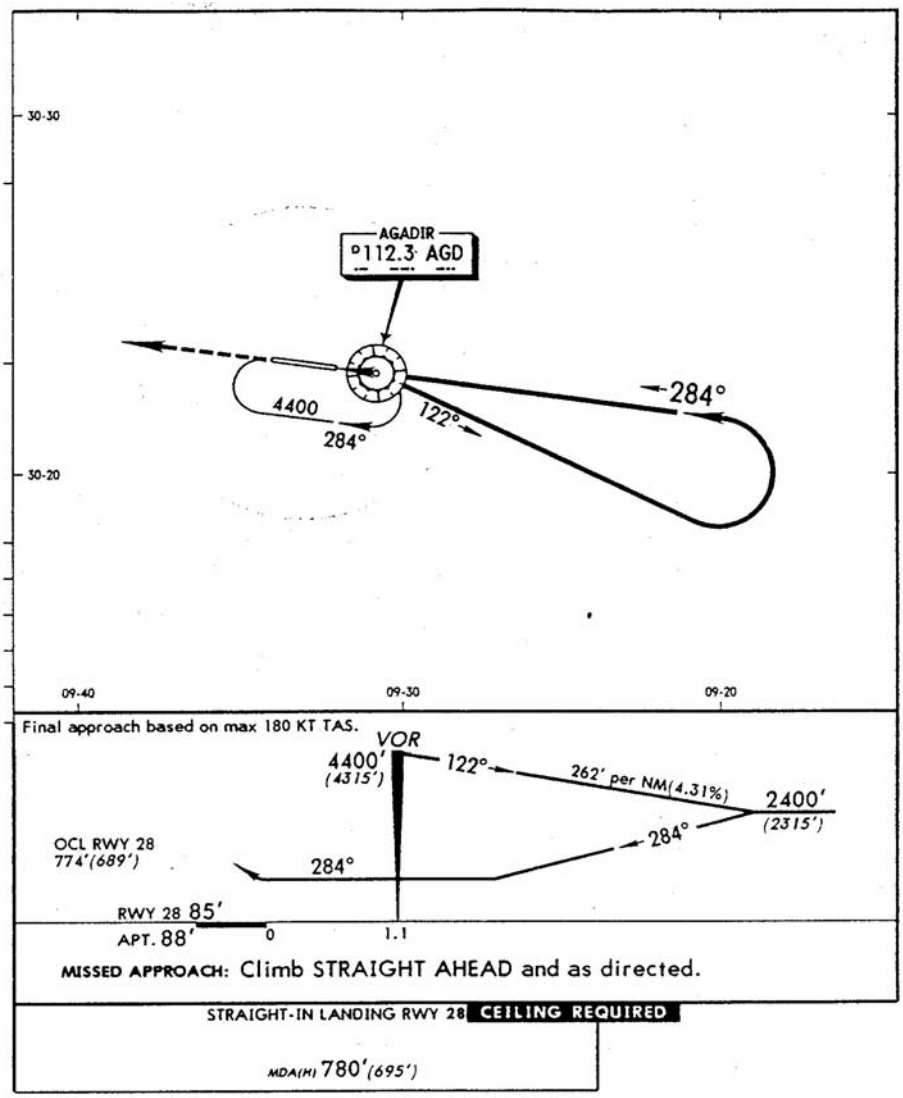

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V28	020	AGD	CF	AGD	V__F	0000	2840	0076	00486	
V28	030	RW28	CF	AGD	G___	0011	2840	0011	00135	-300
V28	040		VA		__M_		2840		00488	
V28	050		VM		EE__		2840			

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VOR CODING EXAMPLE 3

Excerpted from Jeppesen Chart Annaba, Algeria, VOR Rwy 28 Approach Chart by permission of Jeppesen Sanderson, Inc.

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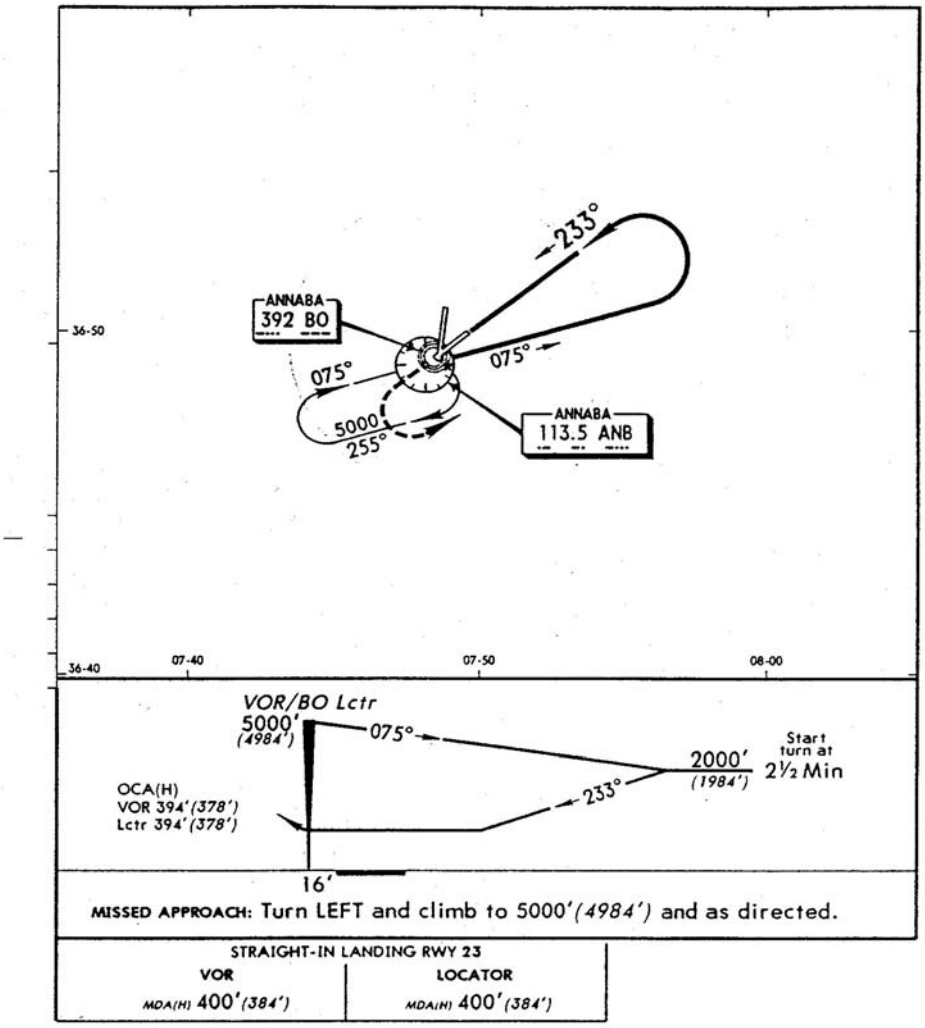

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V23	030	RW23	CF	ANB	G___	0017	2330	0063	00066	-301
V23	040	ANB	CF	ANB	V__M	0000	2330	0017	00400	
V23	050		VM		_E__		2330		05000	

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VOR CODING EXAMPLE 4

Excerpted from Jeppesen Chart Annaba, Algeria, VOR Rwy 19 Approach Chart by permission of Jeppesen Sanderson, Inc.

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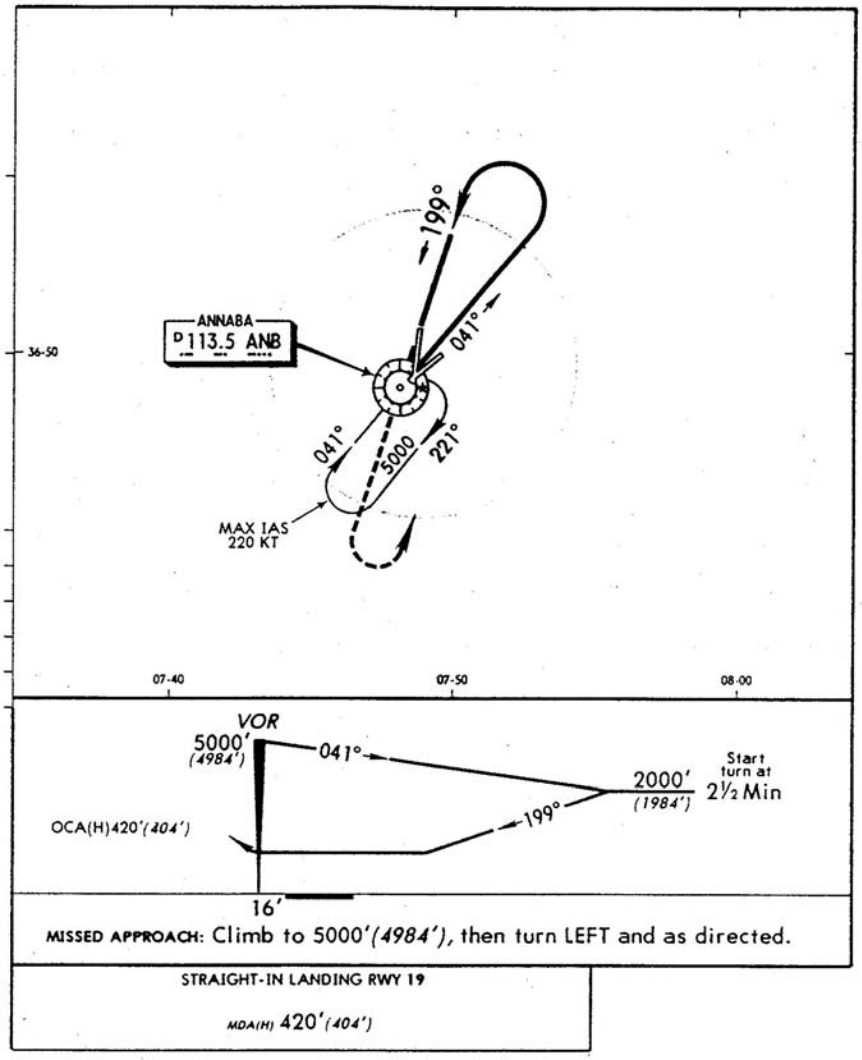

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V19	040	ANB	CF	ANB	V_M_	0000	1990	0020	00420	
V19	050		VM		_E__		1990		05000	

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VOR CODING EXAMPLE 5

Excerpted from Jeppesen Chart Alicante, Spain, VOR Rwy 29 Approach Chart by permission of Jeppesen Sanderson, Inc.

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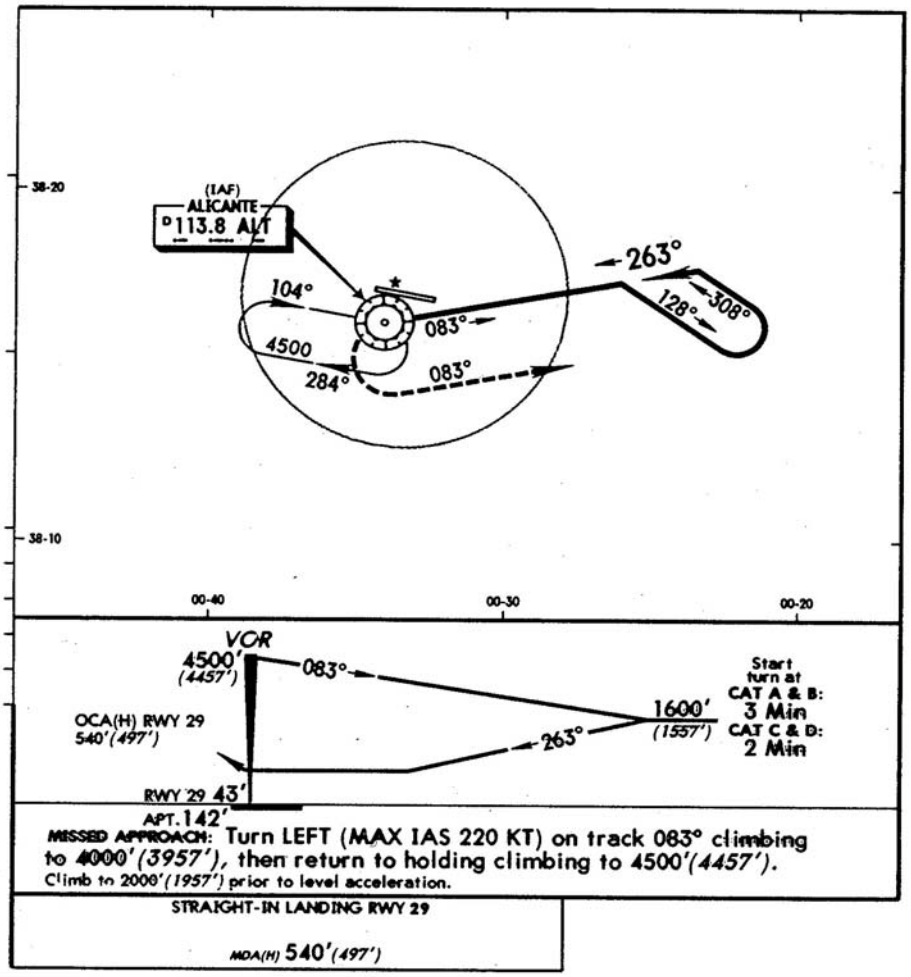

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V29	030	ALT	CF	ALT	V__M	0000	2630	0026	00540	000
V29	040		VA		__M_		2630		00542	
V29	050		VA				0830		04000	
V29	060	ALT	DF		V___				04500	
V29	070	ALT	HM		VE_H		1040	001T	04500	

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VOR CODING EXAMPLE 6

Excerpted from Jeppesen Chart Baldonnel, Ireland, VORDME Rwy 11 Approach Chart by permission of Jeppesen Sanderson, Inc.

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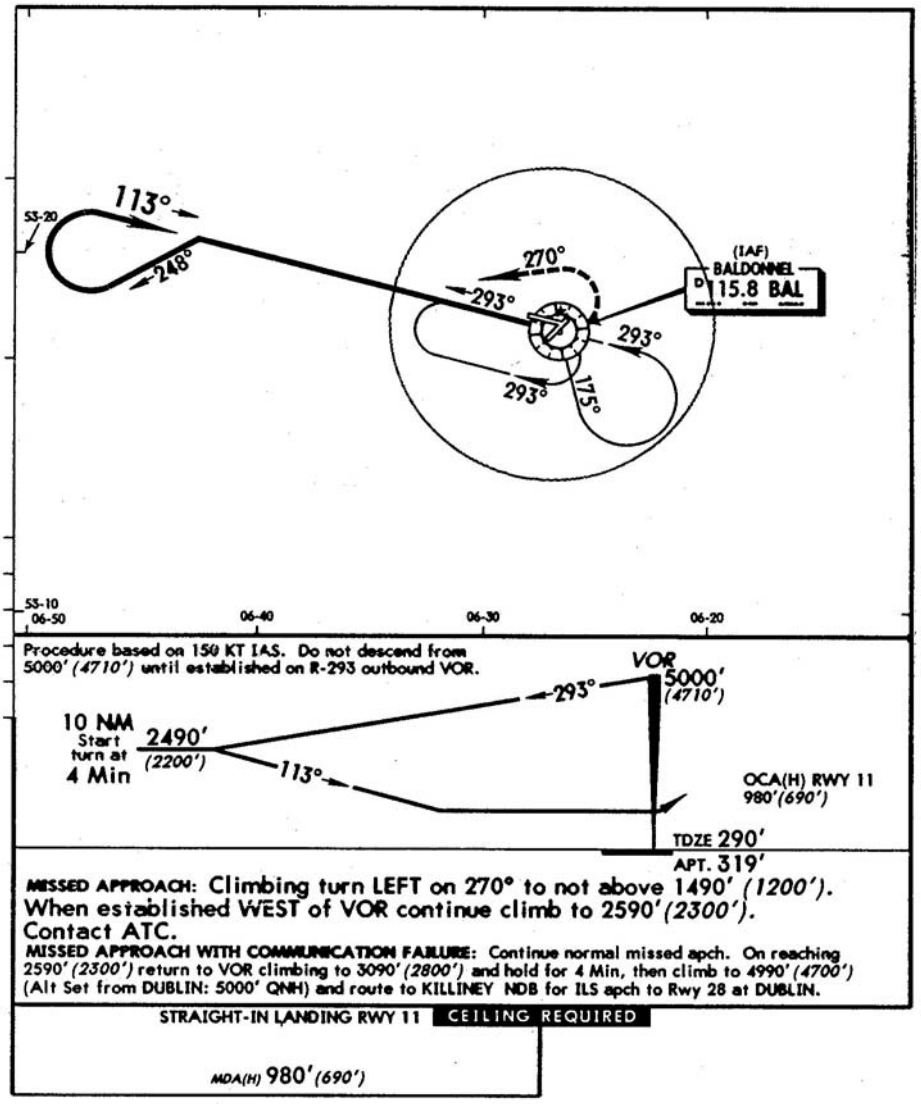

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V11	025	RC11	CF	BAL	R__L	0029	1130	0035	00974	
V11	030	BAL	CF	BAL	V__M	0000	1130	0015	00980	000
V11	040		VA		__M_		1130		00980	
V11	050		VM		_E__		2700		01490	

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VOR CODING EXAMPLE 7

Excerpted from Jeppesen Chart Aleppo, Syria, A. R. VORDME Rwy 27 Approach Chart by permission of Jeppesen Sanderson, Inc.

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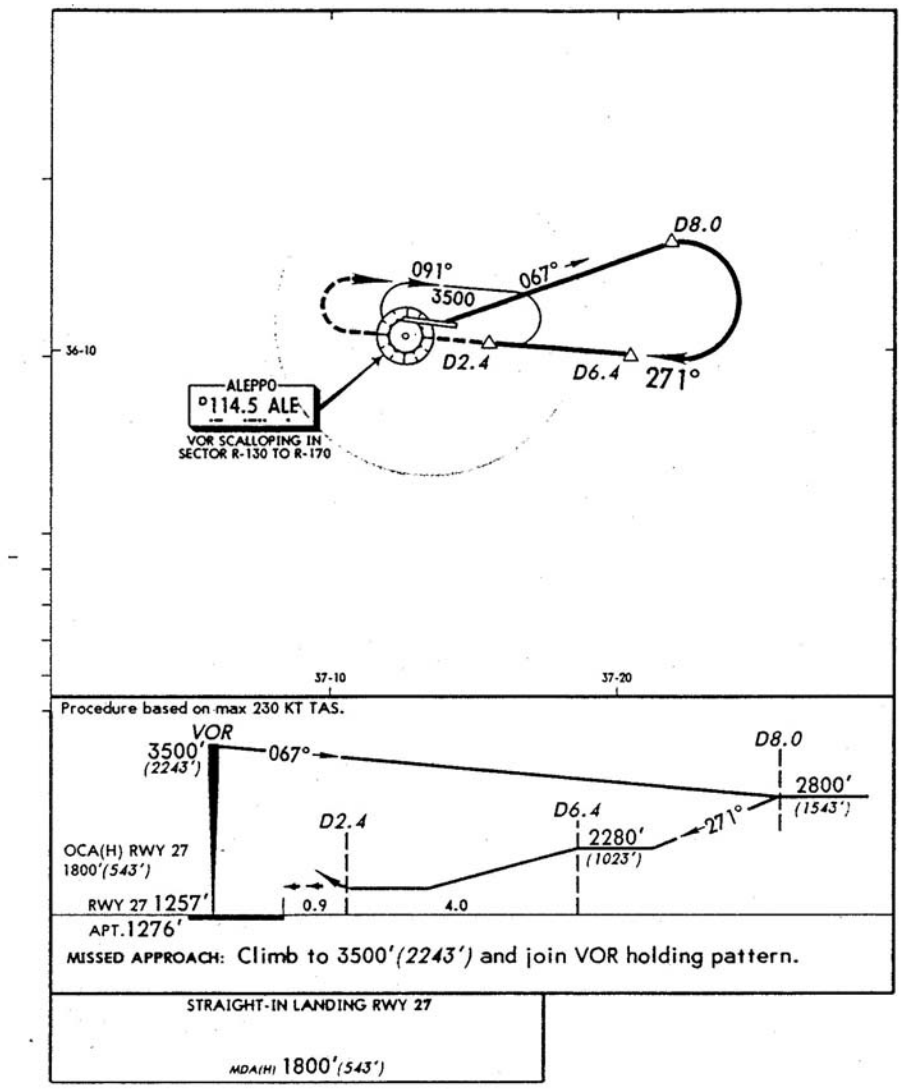

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V27	040		VA		__M_		2710		03500	
V27	050	ALE	DF	ALE	V___	0000				
V27	060	ALE	HM	ALE	VE_H	0000	2710	001T	03500	

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VOR CODING EXAMPLE 8

Excerpted from Jeppesen Chart Al Hoceima, Morocco, VOR Rwy 18 Approach Chart by permission of Jeppesen Sanderson, Inc.

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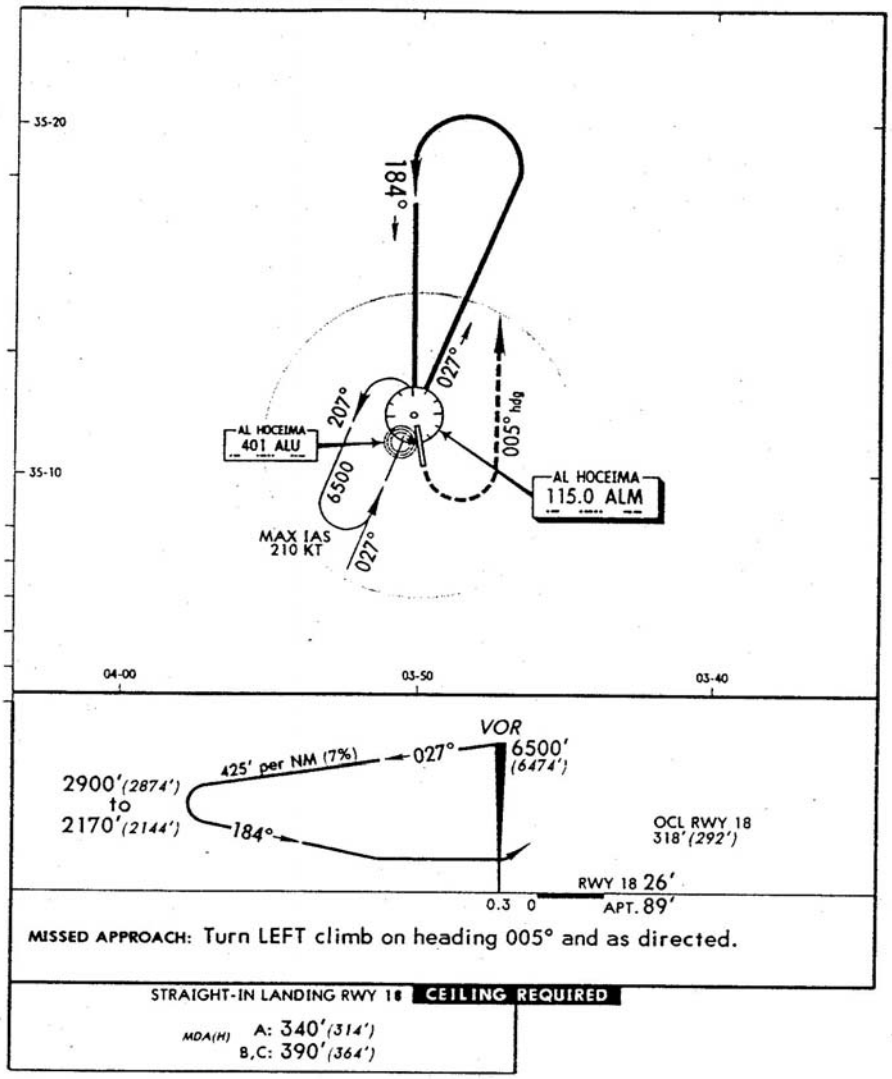

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V18	040		VA		__M_		1840		00489	
V18	050		VM		_E__		0050			

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VOR CODING EXAMPLE 9

Excerpted from Jeppesen Chart Elgin AFB, Illinois, USA, TACAN Rwy 01 Approach Chart by permission of Jeppesen Sanderson, Inc.

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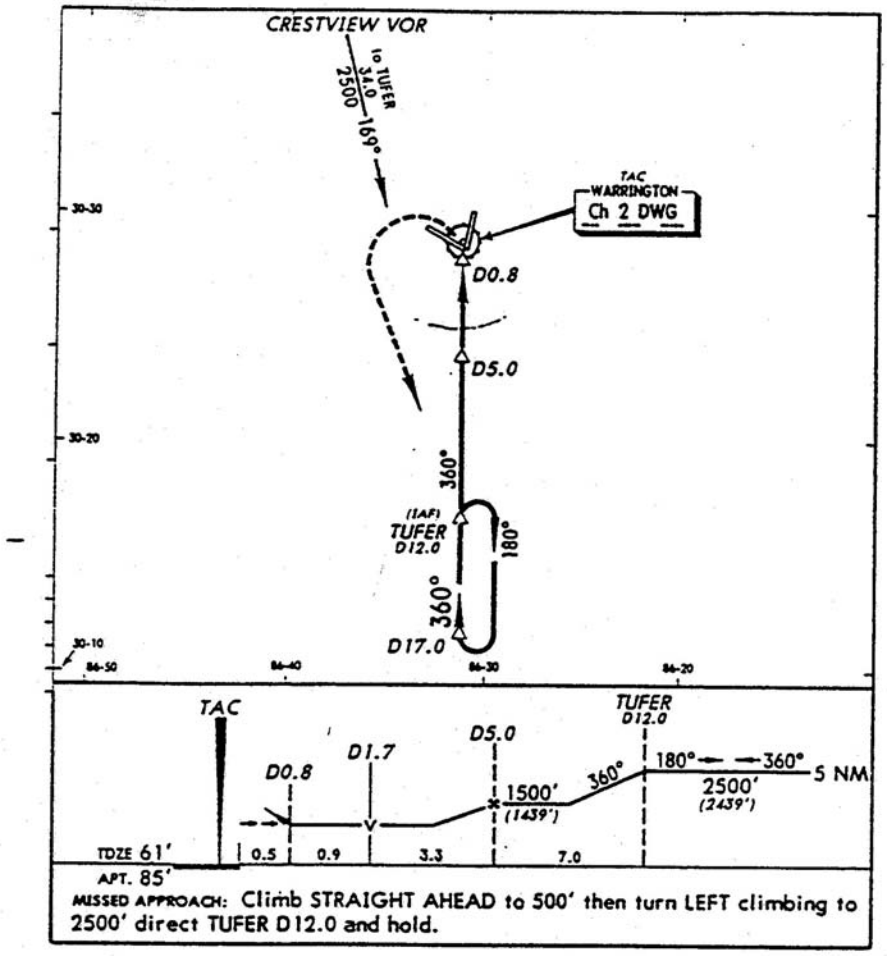

APP ID	SEQ NR	FIX ID	P/T	RECD NAV	W/P DESC	RHO	MAG CRS	DIST	ALT	VERT ANG
T01	010	TUFER	IF	DWG	E__I	0120			02500	
T01	020	FF01	CF	DWG	E__F	0050	3600	0070	01500	
T01	030	RW01	CF	DWG	G___	0008	3600	0042	00111	-300
T01	040		VA		__M_		3600		00500	
T01	050	TUFER	DF		E___				02500	
T01	060	TUFER	HM	DWG	EE_H	0120	3600	001T	02500	

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NDB CODING EXAMPLE 1

Excerpted from Jeppesen Chart Majuro Intl, Marshall Island, NDB Rwy 25 Approach Chart by permission of Jeppesen Sanderson, Inc.

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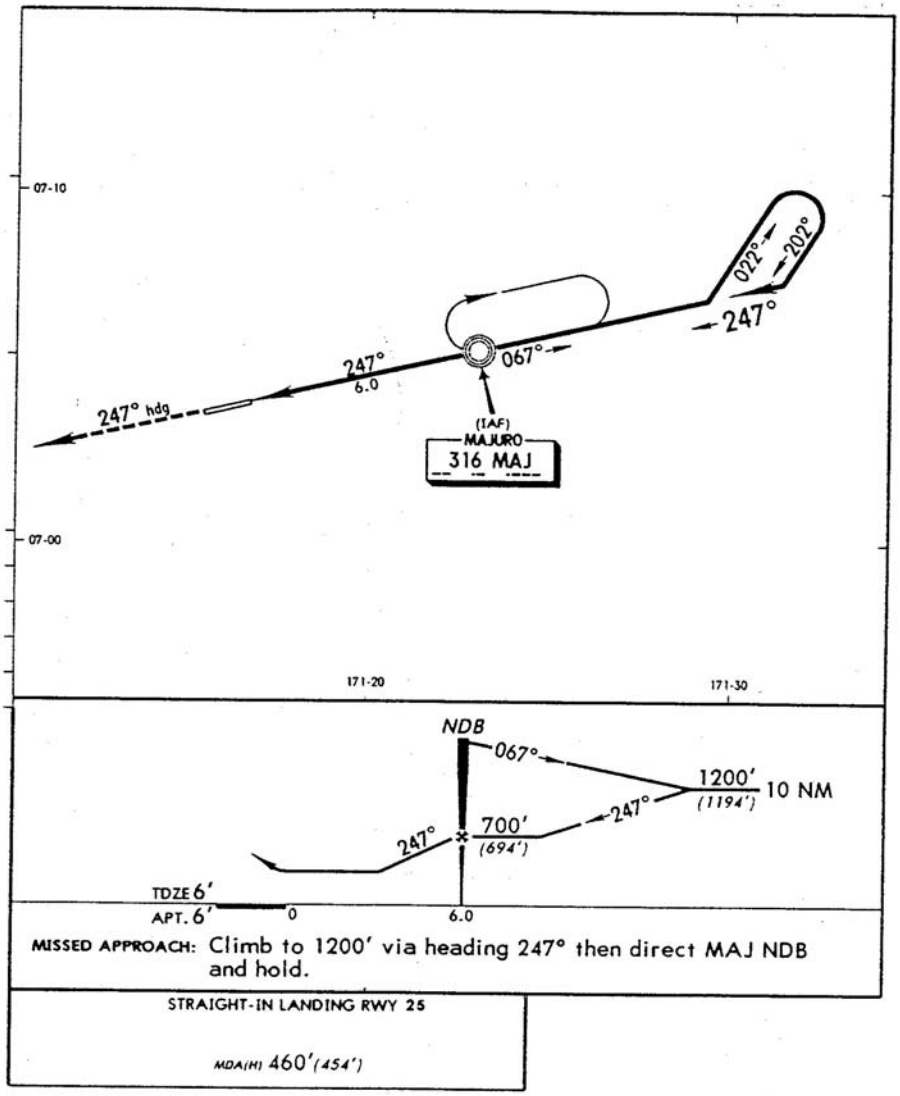

APP ID	SEQ NR	FIX ID	P/T	RECD NAV	W/P DESC	RHO	MAG CRS	DIST	ALT	VERT ANG
N25	010	CF25	IF		E__I				01200	
N25	020	MAJ	CF	MAJ	E__F		2470	0035	00700	
N25	030	RW25	CF		G___		2470	0060	00056	-300
N25	040		CA				2470		01200	
N25	050	MAJ	DF		E___					
N25	060	MAJ	HM		EE_H		2470	001T		

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NDB CODING EXAMPLE 2

Excerpted from Jeppesen Chart Edmonton, Alberta, Canada, NDB Rwy 11 Approach Chart by permission of Jeppesen Sanderson, Inc.

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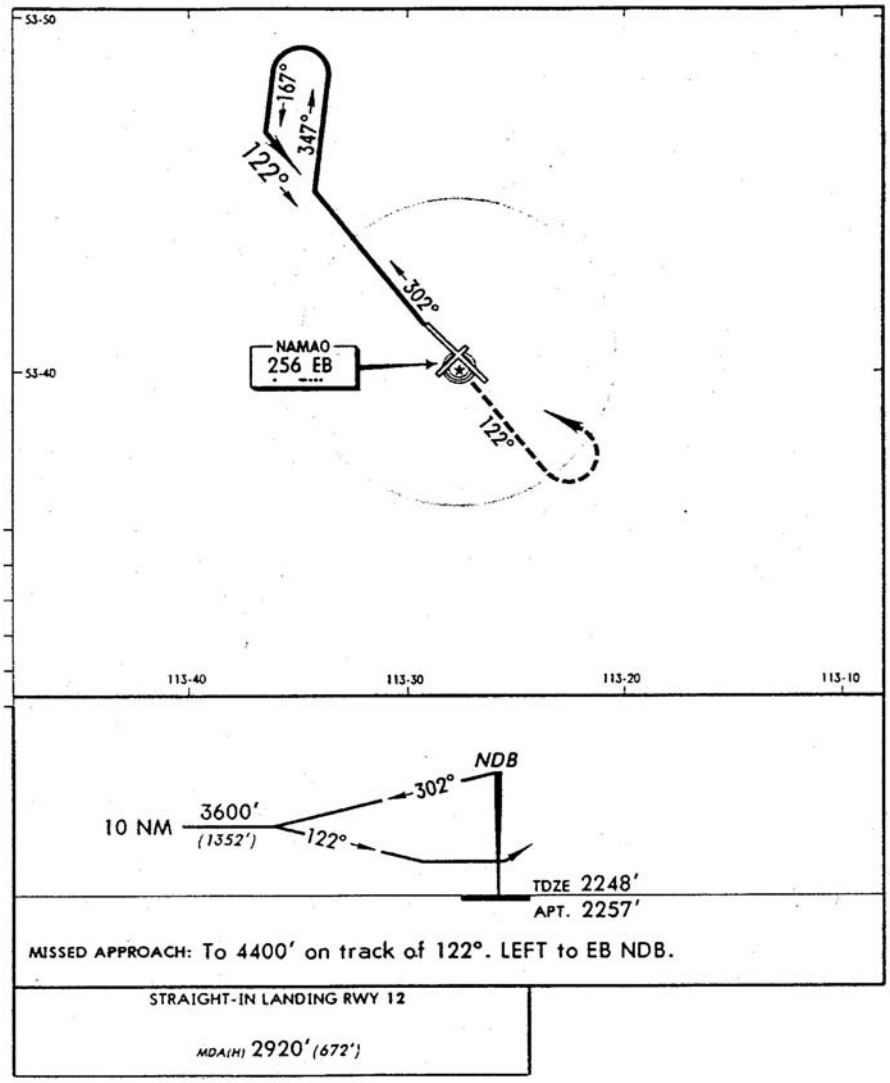

APP ID	SEQ NR	FIX ID	P/T	RECD NAV	W/P DESC	RHO	MAG CRS	DIST	ALT	VERT ANG
N11	020	FF11	IF	EB	E__F				03600	
N11	030	RW11	CF		G___		1220	0055	02298	-300
N11	040	EB	CF	YEG	E_M_	0325	1220	0015	02657	
N11	050		CA				1220		04400	
N11	060	EB	DF	EB	EE__					

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NDB CODING EXAMPLE 3

Excerpted from Jeppesen Chart Sioux Lookout, Ontario, Canada, NDB Rwy 34 Approach Chart by permission of Jeppesen Sanderson, Inc.

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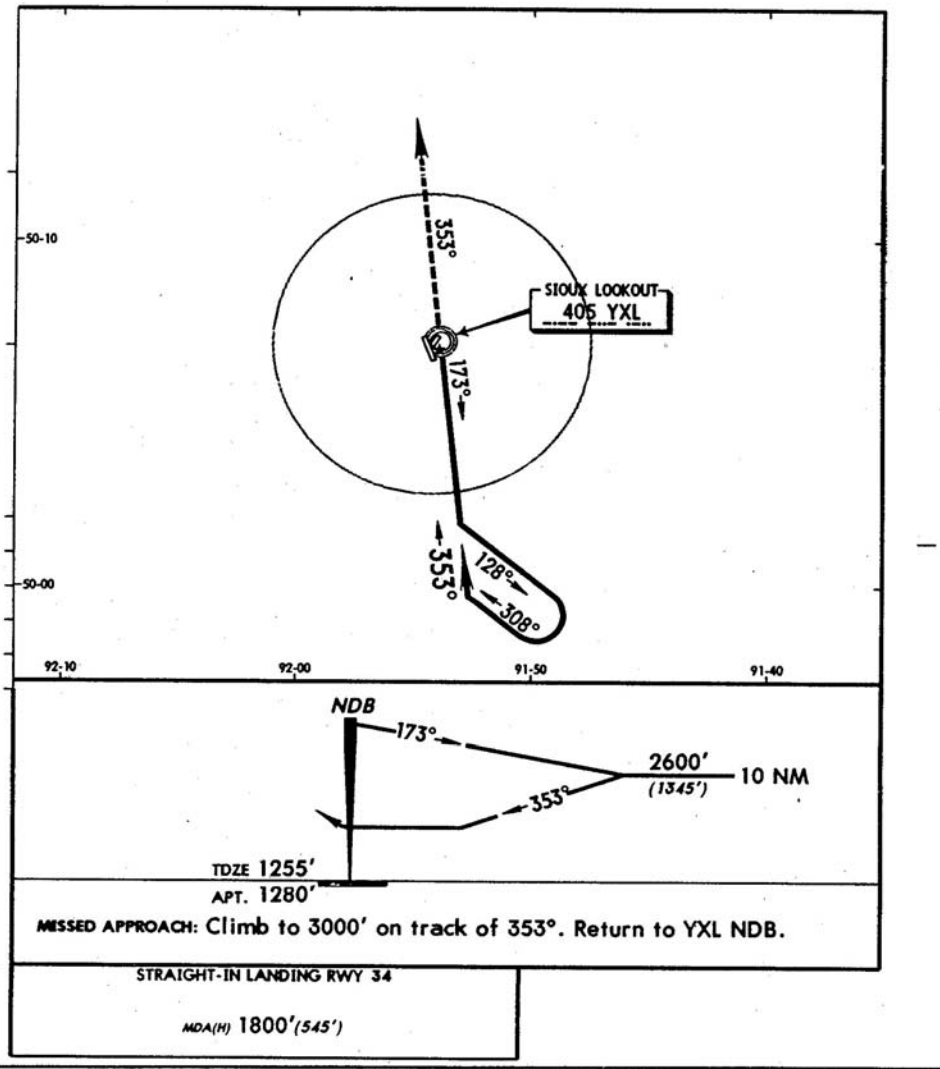

APP ID	SEQ NR	FIX ID	P/T	RECD NAV	W/P DESC	RHO	MAG CRS	DIST	ALT	VERT ANG
N34	020	FF34	IF	YXL	E__F				02600	
N34	025	RC34	CF	YXL	R L		353	0045	01621	
N34	030	YXL	CF		E__M		3530	0015	01800	000
N34	040		CA		__M_		3530		03000	
N34	050	YXL	DF	YXL	EE__					

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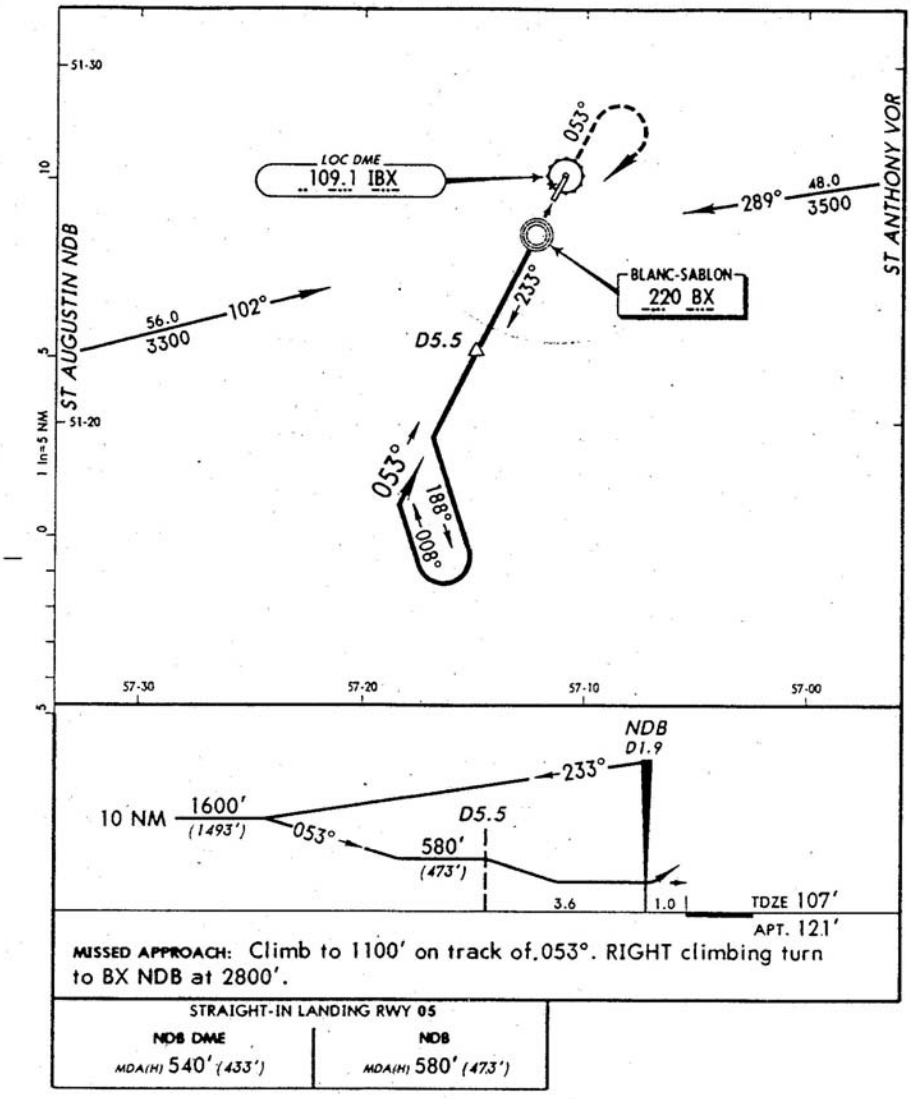
NDB CODING EXAMPLE 4

Excerpted from Jeppesen Chart Blano-Sablon, Quebec, Canada, NDB Rwy 05 Approach Chart by permission of Jeppesen Sanderson, Inc.

Not for navigational or other operational use. For example only. Please consult current navigation. charts.



APP ID	SEQ NR	FIX ID	P/T	RECD NAV	W/P DESC	RHO	MAG CRS	DIST	ALT	VERT ANG
N05	020	FF05	IF	BX	E__F				00580	
N05	030	BX	CF		E__M		0530	0036	00448	-300
N05	040		VA		__M_		0530		01100	
N05	050	BX	DF		EE__					

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PATH AND TERMINATOR

APPROACH AND APPROACH TRANSITION CODING RULES

7.0 Precision Approach Procedure Coding

7.1 Final Approach Segment

The following rules apply to the **Final Approach** Coding of full ILS Localizer based approach procedures. These procedures may include full ILS (localizer and GS), converging ILS, and those IGS (Instrument Guidance System) that are full ILS equivalent. **These rules will be applied to the final approach coding of LDA and SDF procedures when those procedures include reference to an electronic glide slope, to all types of MLS and GLS Approach Procedures.**

7.1.1 All such approach procedures must begin at the FACF. They must consist of a FACF, FAF and missed approach point fix and all step-down fixes published in the vertical path.

7.1.2 For localizer based procedures, the FACF is defined as a fix located on the localizer beam center, 8NM or less from the FAF or within the reception range of the Localizer. This may be a source document provided fix or a fix created using these positioning rules.

7.1.3 The FACF is coded as an IF leg. An altitude will not be assigned to the FACF unless specified in government source documents.

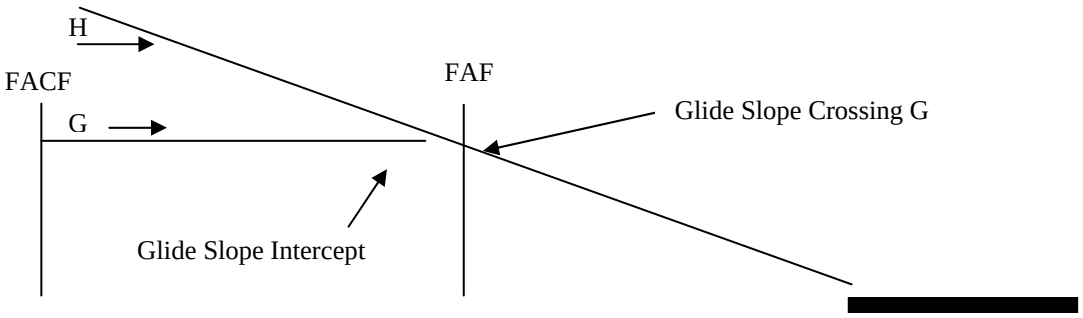
7.1.4 The track from the FACF to the FAF is coded as a CF or TF leg with altitude constraints as indicated for the specific procedure types below.

7.1.5 The recommended navaid must be the procedure reference localizer. Theta and Rho must be provided from that navaid for each sequence of the Final Approach **Coding**, including any step-down fixes, the runway or helipad fix and/or missed approach point.

7.1.6 The "Outbound Magnetic Course" field in all sequences must be equal to the localizer magnetic bearing or MLS course, derived from official government source.

7.1.7 For approach procedures with an electronic glide slope, the vertical angle must be coded in both the Final Approach Fix and the fix, which carries the missed approach point coding. The FAF record carries the Final Approach Fix waypoint description code of "F" in position four of that field. The missed approach point fix carries the waypoint description code of "M" in this position four. The vertical angle will be the published glide slope angle for the installation or procedure. If the altitude 1 and 2 constraints for the FAF are "at" (G) and equal to the FACF altitude whatever its altitude description of (I **or J**), the vertical angle is omitted from the FAF record.

7.1.8 The Coding of FAF and FACF Altitude description fields are as follows:



	Altitude 1	**Altitude 2**	**Alt Desc** [2]
FAF	G/S Intercept Altitude is "at" G/S Intercept Altitude is "at or above"	Glide Slope Crossing Altitude is "at" Glide Slope Crossing Altitude is "at"	G H
FACF	State Defined Altitude is "at" [3] State Defined Altitude is "at or above" [3]	Glide Slope Intercept [1] Glide Slope Intercept [1]	I J

Notes:
[1] Glide Slope intercept altitudes are not specified as either "at" or "at or above."
[2] Reference Section 5.29
[3] FACF Altitude 1 may be blank. When it is blank, the Altitude Description is also blank, even when Altitude 2 is not blank.

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APPROACH AND APPROACH TRANSITION CODING RULES

7.3 GLS Precision Approach Procedure Coding

7.3.1 The rules for coding GLS Approach Procedures are understood to be identical to those of Localizer coding as found in Section 7.1 of this attachment.

7.4 MLS Approach Procedure Coding

MLS Approach Procedure Code utilizing raw azimuth and elevation data is limited to those procedures that are designed as a localizer equivalent. If such a procedure is coded, the rules for the Final Approach **Coding** are to be identical with those stated in Rule 7.1 above. The Route Type of such approaches must be coded as "M" in column 20 of the primary approach record. Approach procedures predicated on the use of MLS Area Navigation (MLS/RNAV) must be coded with a "W" or "Y" in column 20 of the primary approach record. MLS/RNAV approaches are coded as described below.

There are three types of MLS/RNAV approach, listed in increasing levels of complexity, computed lateral/raw vertical guidance, computed lateral and vertical guidance and curved path.

7.4.1 Approaches using computed lateral path and raw vertical path guidance, also referred to as Type "A," will be used primarily where the MLS azimuth transmitter cannot be located on the extended runway centerline, but the elevation transmitter is sited normally abeam the touchdown point. All legs will be straight and aligned with the inbound course. They must be codes with Route Type "W" in column 20 of the primary approach record. Path definition will be the equivalent of a full ILS approach (Rule 6.4.2) with the exception that the leg from the PFAF inbound will be a "TF" leg, terminating at the runway or helipad waypoint, with the published final approach source in the Outbound Magnetic Course field. The PFAF will be coded as the Final Approach Fix in the Waypoint Description field and the first fix prior to the PFAF will be coded as the Final Approach Course Fix.

7.4.2 Approach using computed lateral and vertical guidance but no curved legs, also referred to as Type "B," must be coded as Route Type "Y" in column 20 of the primary approach record. All legs will be straight and aligned with the inbound course. Path definition will be the equivalent of the full ILS approach (Rule 6.4.2) with the exception that the legs from the PFAF inbound will be a "TF" leg, with the published final approach course in the Outbound Magnetic Course field. The altitude of the PFAF and all waypoints inbound from it must be the glide path altitude at that point. The PFAF will be coded as the Final Approach Fix in Waypoint Description field and the first fix prior to the PFAF will be coded as the Final Approach Course Fix.

7.4.3 MLS/RNAV approaches using curved legs, also referred to as Type "C," will be used for a variety of reasons, including parallel sidestep approaches, separation of different categories of aircraft, noise abatement, etc. These must always be precision approaches. They must be coded a with a Route type of "Y" in column 20 of the primary approach record. The following rules apply:

7.4.3.1 The first leg of an MLS/RNAV approach with curved legs must be an "IF/TF" leg combination. All other straight legs must be coded as "TF" legs. All "TF" legs in an MLS/RNAV with curved legs procedure must have the published course included in the Outbound Magnetic Course field.

7.4.3.2 All curved legs will be code as "RF" legs.

7.4.3.3 The initial portion of a MLS/RNAV approach with curved legs may be an "IF/RF" combination, provided a straight leg approach transition is coded to the point in the "IF" and Rule 7.4.2 are complied with.

7.4.3.4 The PFAF will be coded as the Final Approach Fix in the Waypoint Description field and the first fix prior to the PFAF will be coded as the Final Approach Course Fix. If there is not a fix at the glide path intercept, then the first fix after the intercept will be the PFAF. There must be one and only one PFAF for each MLS/RNAV approach with curved legs.

7.4.3.5 The last leg of an approach transition prior to an MLS/RNAV approach must be one of the following types CF, CI, HF, PI, RF or TF, except as indicated in Rule 6.5.3.3. If the leg type is CF, CI, RF or TF, then the Recommended Navaid must contain the identifier of the MLS used for the approach. If the leg type is PI or HF, then the Recommended Navaid must contain the VHF Navaid that defines the PI or HF leg.

7.4.3.6 If the last leg prior to the approach is a "CI" leg, the intercept angle will be 30° or less, and the intercept point must be between the first and second terminator fixes in the approach, but no closer that 2NM to the second fix.

7.4.3.7 The PFAF and the FACF altitudes must be coded according to Precision Approach Procedures Rule 6.4.2.

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7.4.4 The PFAF will be used in precision MLS/RNAV approaches. It is defined as that fix along the lateral path where the published barometric altitude intercepts the glide slope. Prior to the PFAF, the aircraft is expected to fly barometric altitude to intercept the glide path. All waypoints up to the PFAF should be coded using the published barometric crossing altitude. The PFAF and all waypoints after it should be coded using the true altitude of the glide path at those points.

8.0 Non-precision Approach Procedure Coding

8.1 General

8.1.1 For approach procedures without an electronic glide slope, the Final Approach Fix will be that designated by government source. If no FAF is established in the government source, one will be computed according to Rule 6.2.5.*3* of this attachment. The fix, whether published or established, must carry the Final Approach Fix Waypoint Description code of "F" in position four of that code field. Note that only one record in a coded approach procedure can carry the "F" in position four of the Waypoint Description. Altitudes for this fix are coded in accordance with Rule 6.2.10.1 of this attachment.

8.1.2 **A vertical Angle must be coded in the Missed Approach Point, Runway Threshold or Final End Point, which ever occurs first, for each approach procedure. A Vertical Angle may be coded in the Final Approach Fix Segment for each approach that includes a FACF.** Vertical Angles will be from official government source or computed. This Vertical Angle will only be repeated on all step-down fixes on the segment **FAF to MAP. The government source Vertical Angle will also be repeated on fixes associated with an RF Leg as the start or end of the arc, when these fixes are not at procedure fix locations and the RF Leg is in the FAF to MAP portion of the final approach coding.**

8.1.3 Missed Approach Point (MAP) Location. **The MAP location will be as published on the non-precision approach procedure by the appropriate government authority**. **See also Rule 8.10 of this attachment.**

Note: If the source document states that the MAP and the LTP are not at the same location even if the distance is 0.1 mile or less, the MAP will not be placed at the LTP.

8.2 Final Approach Path Coding - Localizer-based Procedures

The following rules apply to the **Final Approach** Coding of non-precision Localizer-based approach procedures. These procedures may include Localizer Only, IGS (Instrument Guidance System) LDA, Localizer Directional Aid) and SDF (Simplified Directional Aid) procedures.

8.2.1 All such approach procedures must begin at the FACF. They must consist of a FACF, FAF and missed approach point fix and all step-down fixes published in the vertical path.

8.2.2 The FACF is defined as a fix located on the localizer beam center, 8NM or less from the FAF or within the reception range of the Localizer. This may be a source document provided fix or a fix created using these positioning rules.

8.2.3 The FACF is coded as an IF leg with an altitude only when assigned by government source.

8.2.4 The track from the FACF to the FAF is coded as a CF or TF leg with altitude constraints as indicated for the specific procedure types below.

8.2.5 The recommended navaid must be the procedure reference localizer. Theta and Rho must be provided from the localizer for each sequence of the Final Approach **Coding**, including any step-down fixes, the runway or helipad fix and/or missed approach point.

8.2.6 The "Outbound Magnetic Course" field in all sequences must be equal to the localizer magnetic bearing, derived from official government source.

8.3 Final Approach Path Coding - VOR-based Procedures

The following rules apply to the **Final Approach** Coding of non-precision VOR-based approach procedures. These procedures may include VOR, VORDME, VORTAC, **TACAN, and** RNAV **Area Navigation** procedures.

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8.3.1 Final Approach **Coding** must be coded using IF and CF or TF[1] legs only.

8.3.2 Final Approach **Coding** must include both a FAF and missed approach fix. The missed approach fix may be a runway or helipad fix or a designated missed approach point. Coding of a FACF is defined in Rule 6.2.5.

8.3.3 The recommended navaid must be the procedure reference VOR or TACAN. Theta values must be provided from that facility in all Final Approach **Coding** sequences, including any step-down fixes that are included.

8.3.4 When the reference facility is VORDME or VORTAC or TACAN, the following applies:

8.3.4.1 Final Approach **Coding** must be **accomplished** using IF and CF or TF[1] legs only.

8.3.4.2 Final Approach **Coding** must include a FAF and a missed approach point **that** may be a missed approach point fix, a runway fix or a helipad fix, and all step-down fixes published in the vertical path. The Final Approach **Coding** will include an FACF when one is required by Rule 6.2.5.

8.3.4.3 The recommended navaid must be the procedure reference VORDME or VORTAC or TACAN. Theta and Rho values must be provided from that facility in all Final Approach **Coding** sequences, including any step-down fixes that are included.

8.3.5 When the procedure reference is RNAV (**VORDME**), the following applies:

8.3.5.1 Final Approach **Coding** must be **accomplished** using IF and CF or TF[1] legs only.

8.3.5.2 All RNAV- **Area Navigation** approach procedure missed approach points must be at or prior to a runway threshold or helipad alighting point. These points may be a source defined named waypoint.

8.3.5.3 The recommended navaid is not required unless specified **by source** or if required by the leg type. The recommended navaid must be the procedure reference VORDME or VORTAC. Theta and Rho values must be provided from that facility in all final approach sequences **where the recommended navaid is coded, including any step-down fixes. The Recommended Navaid, Theta and Rho are never provided on fixes associated with the start and end of a precision arc (RF) leg used in the final approach coding sequences.**

8.3.6 Rule deleted by Supplement 17.

8.3.6.1 Rule deleted by Supplement 17.

8.3.6.2 Rule deleted by Supplement 17.

8.3.6.3 Rule deleted by Supplement 17.

8.4 Final Approach Path Coding - NDB-Based Procedures

The following rules apply to the **Final Approach** Coding of all NDB based approach procedures. NDB based approach procedures include procedures using a NDB or Locator as the reference facility and procedures using a NDB or Locator and a DME (NDB + DME) as reference facilities. NDB approach procedures not requiring DME but using the DME for reduced minimums will be coded as NDB + DME procedures.

8.4.1 **NDB Final Approach Coding must include a FAF and a missed approach point that may be a missed approach point fix, a runway fix or a helipad fix, and all step-down fixes published in the vertical path. The Final Approach Coding must include a FACF when one is required by Rule 6.2.5.**

8.4.2 Rule deleted by Supplement 17.

8.4.3 NDB + DME **Final** Approach **Coding** must include a FAF and runway or helipad fix or missed approach point fix and all step-down fixes published in the vertical path. The Final Approach **Coding** will include a FACF when one is required by Rule 6.2.5.

---

[1] In general “CF” legs are used in final approach coding. “TF” legs are used in FMS and GPS Approach Procedures, some types of MLS Procedures and in other procedure types where the determination has been made that a TF will work better than a CF.

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APPROACH AND APPROACH TRANSITION CODING RULES

8.4.4 Coding must use IF and CF or TF[1] legs only through to the runway or helipad fix or missed approach point fix. The IF leg must be at the FAF (or at the optional FACF) for NDB procedures or at the FACF for NDB + DME procedures.

8.5 Intentionally Left Blank

8.6 Final Approach Path Coding - Circle-to-Land Procedures

Procedures that only have Circle-To-Land operating minimums may be included in the database. When they are included, they are identified with the Route Type (Section 5.7) appropriate to the reference facility and with the Route Type Qualifier Two set to "C." The following rules apply to such approach procedures:

8.6.1 The last segment in the Final Approach **Coding** must be the missed approach point fix.

8.6.2 For Circle-To-Land Procedures that are to a runway or helipad, **or published missed approach point**, all the rules listed above in Sections 6, 7 and 8 apply, as well as the rules for Missed Approach Procedure coding in Section 9.

8.6.3 For Circle-To-Land Procedures that are not to a runway or helipad, **or published missed approach point**, the missed Approach Point will be **established at** the center of the airport or heliport.

8.6.4 **A Vertical Angle must be provided on the missed approach point fix , or, when the procedure meets the criteria specified in Rule 6.10.2.1, on the Final End Point fix when such as fix is coded for the approach. Vertical Angle information must be in accordance with the rules in Section 6, 7 and 8 of this attachment for the type of reference facility on which the procedure is based.**

8.7 Final Approach Path Coding – RNAV Procedure

This section provides coding guidelines for RNAV Approach Procedures. The term RNAV Approach Procedure is defined as an approach procedure that was designed to provide lateral and vertical path guidance using criteria that do not rely on radial, bearing, arcs or glide paths from ground-based navaids. As such, a RNAV Approach Procedure will not include references to ground-based navaid associated information such as VOR radial, NDB bearings, DME distances, ILS course, MLS Azimuth and ILS or MLS glide slopes. Exceptions to this standard may be outlined in the rules that follow. Navaids may be referenced as waypoints in the coding of the RNAV Approach Procedures.

8.7.1 **The Final Approach Coding must include a Final Approach Fix (FAF) and a Missed Approach Point fix (Runway Threshold or Missed Approach Point). The Final Approach Coding must include a Final Approach Course Fix (FACF) when such a fix is required by Rule 6.2.5.**

8.7.2 The FAF (or optional FACF) is coded as an IF leg with an altitude only when assigned by government source.

8.7.3 The track from the FACF (when coded) to the FAF is coded with TF or RF legs. The RF leg is not allowed as the first leg of the approach coding according to the Beginning/Ending Leg Table. The preferred coding when an approach starts with a precision arc is the use of an "IF" leg at the FACF or FAF, followed by "RF" to the FAF or MAP. According to the rules on "RF" legs, this must require that a straight line, fix terminated, approach transition to the FACF or FAF has been included. The track in the transition must be tangent to the arc and the fix at the end of the transition must be overflown. The rule also does not exclude the use of an RF leg in between FAF and the final TF leg to the missed approach point. Such RF legs must be coded with the 4th character of the Waypoint Description field blank.

8.7.4 The Final Approach Coding of RNAV Approach Procedures covered by this Section 8.7 do not require a Recommended Navaid except as governed by other rules in this attachment.

8.7.5 RNAV Approach Procedures coded as GLS Procedure according to Section 5.7 must reference the GLS facility as a Recommended Navaid.

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[1] In general "CF" legs are used in final approach coding. "TF" legs are used in FMS and GPS Approach Procedures, some types of MLS Procedures and in other procedure types where the determination has been made that a TF will work better than a CF.

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APPROACH AND APPROACH TRANSITION CODING RULES

8.8 Final Approach Path Coding - Helicopter Approach Procedures

Helicopter Approach Procedures will be coding using the rules in Sections 6, 7 and 8 and 9 of this attachment, appropriate to the type of sensor required for the procedure, such as VORDME or ILS or RNAV or GNSS. This includes rules for Recommended Navaid, FACF requirements, beginning and ending leg types, etc.

The Lateral Path Rules for the sensor related procedure coding reference a "missed approach point fix," a "runway fix" or a "helipad fix" as the missed approach point. Those same rules apply to the coding of helicopter procedure.

8.9 Vertical Navigation Path (VNAV Path) or Descent Gradient Considerations

If the government source provides a vertical angle and/or TCH, it will be used. The only exception is when the source provides more than one angle for the Final Approach **Coding** segment **FAF to MAP.** If more than one angle is provided for **this** segment, the highest angle will be used.

The following guidelines have been developed for the coding of the vertical angles on the Final Approach **Coding** when the vertical angle is not provided by the government sources. **Rule 8.9.1 through 8.9.8 applies to the FAF to MAP segment only. Rule 8.9.9 applies to the FACF to FAF segment only.**

8.9.1 **The descent angle is to be calculated from Landing Threshold Point elevation plus the published procedure TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification]) to the altitude at the Final Approach Fix (FAF). The curvature of the earth should not be used in the calculations of the descent angle. Refer to example 1, 7, and 8. The descent angle must always be rounded up to the nearest one hundredth of a degree.**

Examples of Rounding:

3.111 = 3.12
3.346 = 3.35

8.9.1.1 **The published TCH for a procedure used in these calculations is the procedure TCH (if no procedure TCH is specified by source use 40 or 50 feet). TCH is further defined in Chapter Five, Sections 5.67 and 5.265 of this document.**

8.9.1.2 The lateral distance used in the descent angle calculation istobe the along track distance of the path.

8.9.2 If the calculated angle is less than three degrees, it will be raised to a minimum of three degrees.

8.9.3 If a step-down fix is included in the Final Approach **Coding** segment **to FAF and MAP** and it is determined that the calculated descent angle will be above the step-down fix altitude, that calculated descent angle will be used. If it is determined that the step-down fix altitude is above the descent path, the descent angle will be calculated from the LTP plus 50 feet to the altitude at the step-down fix. Refer to Examples 3 and 4.

8.9.4 The descent angle should not be less than the Visual Glide Slope Indicator (VGSI) associated with the landing runway. If the calculated descent angle is less than the VGSI, the descent angle should be raised to match the VGSI or to 3.77.

8.9.5 **If the final approach course does not cross over the runway threshold, a position abeam the landing threshold point on the final approach course will be calculated and coded as a Final End Point. The descent angle must be calculated using distances that start at the FEP fix and an altitude equal to the LTP elevation plus the published procedure TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification]) to the altitude at the Final Approach Fix (FAF). Refer to Examples 6, 9, and 10.**

8.9.6 **If the missed approach point is prior to the runway threshold, the descent angle will be computed from the LTP elevation plus the published procedure TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification]) to the altitude at the FAF. Altitude will be specified at the MAP fix and will be the altitude where the calculated descent angle passed through the MAP. Refer to Examples 7 and 8.**

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APPROACH AND APPROACH TRANSITION CODING RULES

8.9.7 When circling-to-land minimums are the only landing minimums and the runway is in alignment with the Final Approach **Coding** segment **FAF and MAP**, a descent angle will be provided. **The descent angle will be computed from the LTP elevation plus the published procedure TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification])to the altitude at the FAF.** Refer to Examples 11 and 12.

8.9.8 When circling-to-land minimums are the only landing minimums and the runway is not in alignment with the Final Approach **Coding** segment **FAF and MAP**, a descent angle will be provided. The descent angle will be **computed** from a point on the final approach **course** Abeam the LTP of the nearest landing runway to the altitude at the FAF. Refer to Example 13.

8.9.9 **If one or more step-down fixes are published in the official government source in the intermediate approach segment of the procedure, the fixes and the appropriate altitudes will be included as part of the Final Approach Coding. A vertical angle will be coded on the FAF waypoint that will ensure that any step-down fix altitude in the Final Approach Coding, FACF, generally the Intermediate Fix (IF) waypoint, to the FAF is cleared by that angle. Vertical angle information is not provided in initial approach segments, IAF to IF, which are generally coded as Approach Transitions and not included in the Final Approach Coding. However, any such fix will be included in the coding, along with the government source supplied altitudes**.

8.10 VNAV Coding of Non-Precision Approach Procedures With Missed Approach Points - Other Than Landing Threshold

Coding standards for Non-Precision Approach Procedures that have a published missed approach beyond the Landing Threshold Point have been developed. These standards are included in the three VNAV coding scenarios in the following paragraphs. In these scenarios, the term "final approach course crosses over the landing threshold" is used as a qualification for the three scenarios. This attachment does not define with any precision what is meant by this qualification. The intent of this wording is that the published final approach course will lead to the landing threshold without course changes or corrections.

8.10.1 **Missed Approach Point beyond the landing threshold and the published Final Approach Course crosses the landing threshold. The ARINC 424 rules for this case call for inserting the Landing Threshold Point as a fix in procedure coding.** See VNAV Coding Example A.

8.10.2 **Missed Approach Point beyond the landing threshold and the published Final Approach Course does not cross the landing threshold. The ARINC 424 rules for this case call for inserting a Final End Point as a fix in the procedure coding**. See Coding Example B.

8.10.3 **Missed Approach Point is a Navaid beyond the landing threshold. The published final approach course does not cross over the landing threshold but the navaid is located equal to or less than 0.10NM from that threshold. The ARINC 424 rules for this case call for coding the navaid as the missed approach point, no insertion of an additional fix. See Coding Example C.**

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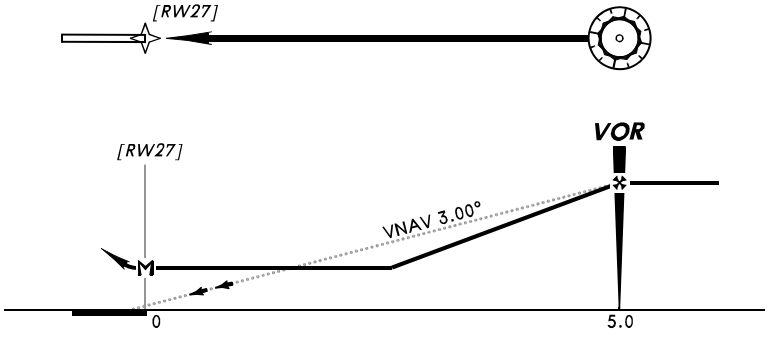
NON-PRECISION APPROACH CODING EXAMPLE 1

This example shows a Final Approach Fix (FAF) to Missed Approach Point (MAP) at Landing Threshold Point (LTP) final approach **course** with Straight-In landing alignment. The VNAV **Path** angle, when not provided in official government source, is calculated from LTP **elevation + TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification])** to the FAF altitude and coded in the MAP sequence.



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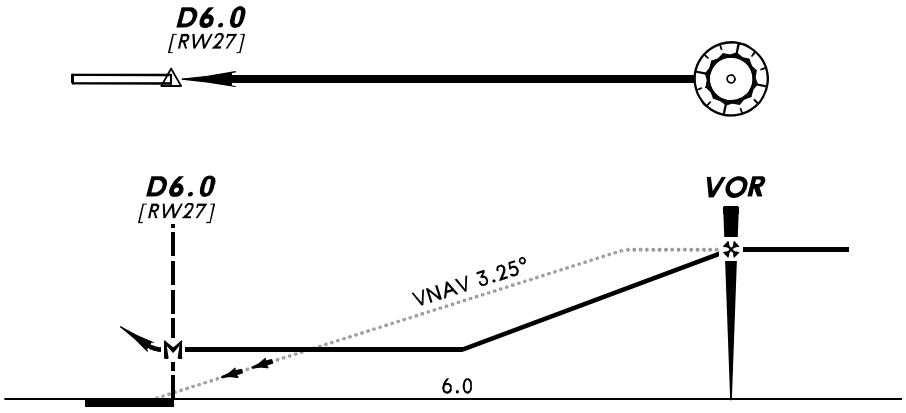
NON-PRECISION APPROACH CODING EXAMPLE 2

This example shows a Final Approach Fix (FAF) to Missed Approach Point (MAP) at Landing Threshold Point (LTP) final approach **course** with Straight-In landing alignment. The VNAV **Path** angle is calculated from LTP **+ TCH elevation (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification])** to the FAF altitude, and raised to an optimum angle of 3.00 degrees or to an angle matching the VASI/PAPI angle. This adjusted angle is coded in the MAP sequence



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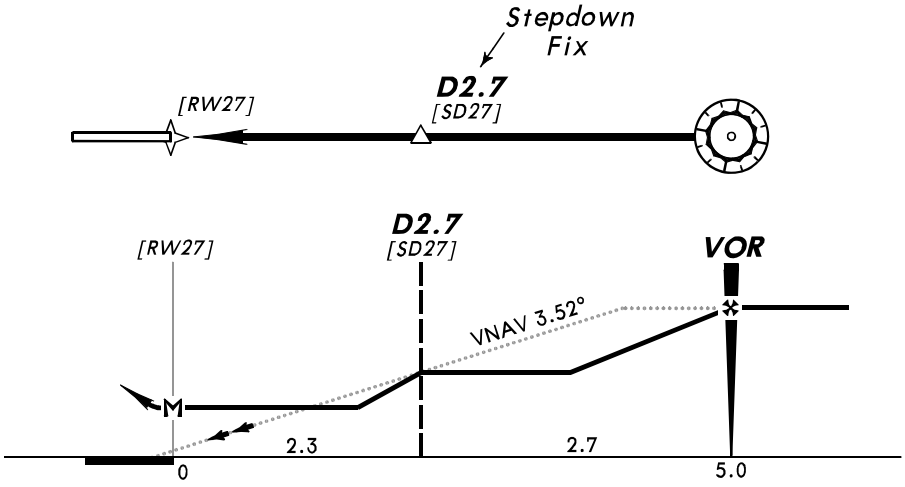
NON-PRECISION APPROACH CODING EXAMPLE 3

This example shows Final Approach Fix (FAF) to Missed Approach Point (MAP) at Landing Threshold Point (LTP) via Step-down Fix final approach **course** with Straight-In landing alignment. The VNAV **Path** angle is calculated from LTP **+ TCH elevation (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification])** to the FAF altitude and coded in the MAP sequence. If the Step-down penetrates VNAV path **calculated from the** LTP **elevation + TCH (or LTP elevation** + 50 feet **if no TCH is published)** to FAF altitude, as shown in this example, the VNAV angle is raised to clear step-down and this revised angle is used for entire Final Approach **Coding**.



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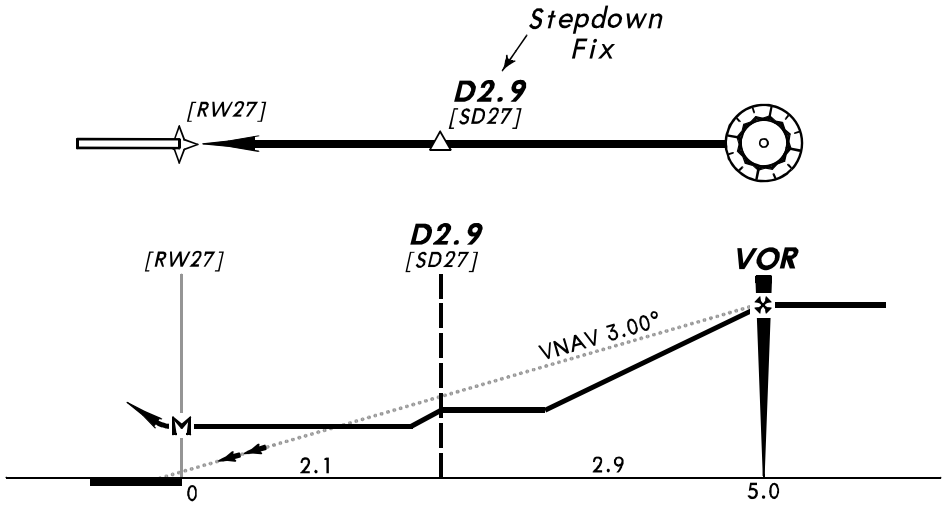
NON-PRECISION APPROACH CODING EXAMPLE 4

This example shows a Final Approach Fix (FAF) to Missed Approach Point (MAP) at Landing Threshold Point (LTP) via Step-down Fix final approach **course** with Straight-In landing alignment. The VNAV **Path** angle is calculated from LTP **+ TCH elevation (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification])** to the FAF altitude and coded in the MAP sequence. When the Step-down does not penetrate VNAV path of LTP + 50 feet to FAF altitude, there is no requirement to raise angle.



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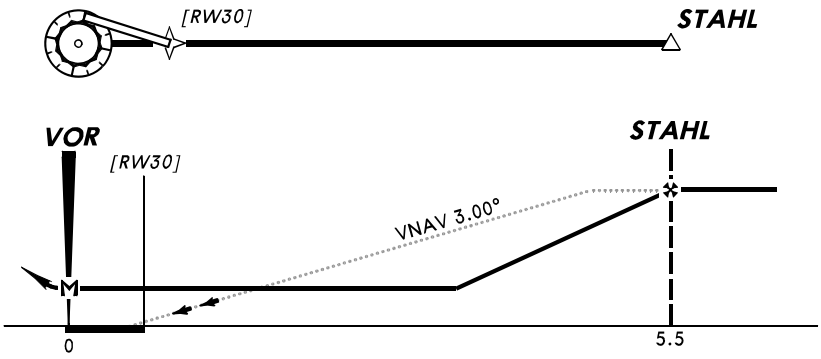
NON-PRECISION APPROACH CODING EXAMPLE 5

This example shows a Final Approach Fix (FAF) to Missed Approach Point (MAP) beyond Landing Threshold Point (LTP) final approach **course** with Straight-In landing alignment. The MAP position is the officially published government source position and is beyond the LTP. A "LTP Fix" waypoint (runway) is required in the correct coding of this example. The VNAV **Path** angle is calculated from LTP **+ TCH elevation (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification])** not the MAP, to the FAF altitude, and coded in the LTP sequence.



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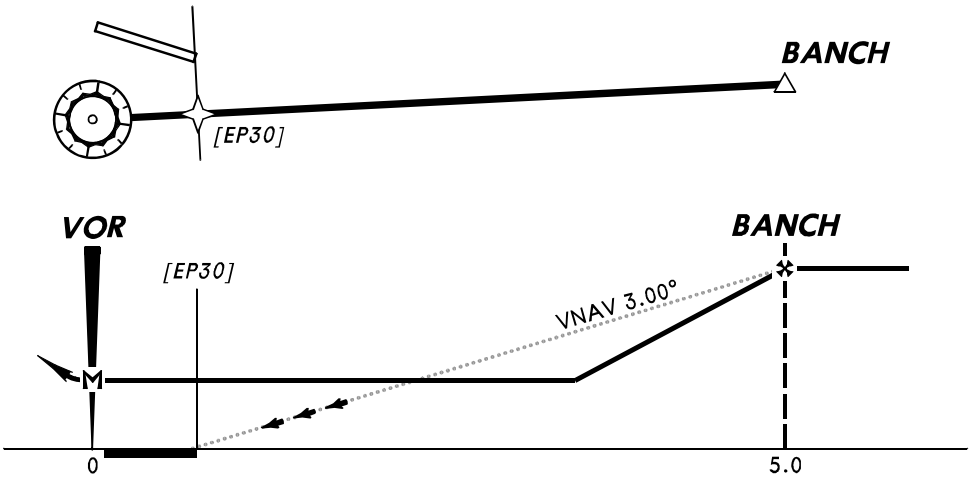
NON-PRECISION APPROACH CODING EXAMPLE 6

The example shows a Final Approach Fix (FAF) to Missed Approach Point (MAP) beyond Landing Threshold Point (LTP) final approach course. This final approach course does not cross the landing threshold. The MAP position is the officially published government source position. A Final End Point (FEP) waypoint is required. The procedure does meet straight-in alignment criteria. The VNAV Path angle is calculated from the LTP elevation + TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification]) to the FAF altitude, using the FEP position and along track distance in determining the distance used in the calculation. The VNAV Path angle is coded in the FEP sequence.



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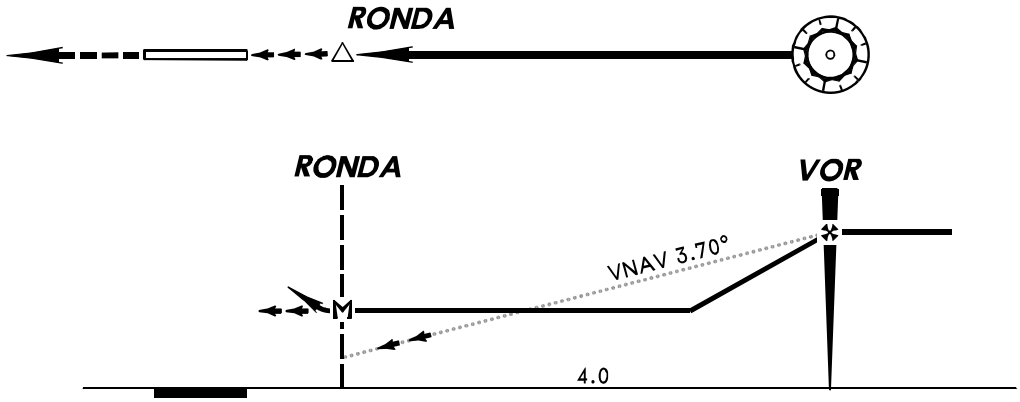
NON-PRECISION APPROACH CODING EXAMPLE 7

This example shows a Final Approach Fix (FAF) to Missed Approach Point (MAP) before Landing Threshold (LTP) final approach **course** with Straight-in landing alignment. The MAP position is the officially published government source position. The VNAV **Path** angle is calculated from the LTP **+ TCH elevation (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification])** to the FAF altitude, and coded in the MAP sequence. The Altitude in the MAP sequence is assigned based on computation, using the calculated angle. The LTP is not included in the coding as a waypoint. The Missed Approach Procedure is coded as straight ahead over runway.



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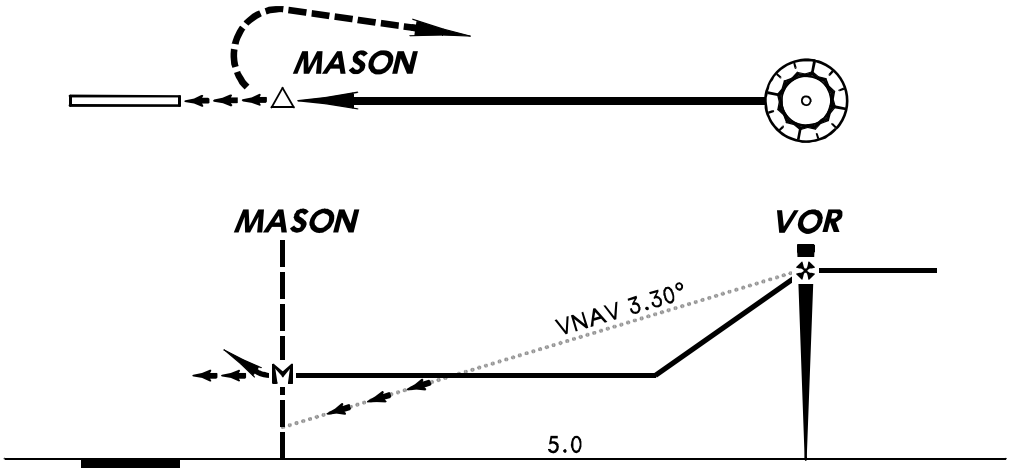
NON-PRECISION APPROACH CODING EXAMPLE 8

This example shows a Final Approach Fix (FAF) to Missed Approach Point (MAP) final approach **course** meeting Straight-in landing alignment criteria with the MAP before Landing Threshold Point (LTP). The MAP position is the officially published government source position. The VNAV **Path** angle is calculated from the LTP **+ TCH elevation (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification])** to the FAF altitude, and coded in the MAP sequence. The Altitude in the MAP is assigned based on computation, using the angle. The LTP is not included in the coding as a waypoint. The Missed Approach Procedure includes immediate turn at the MAP**.**



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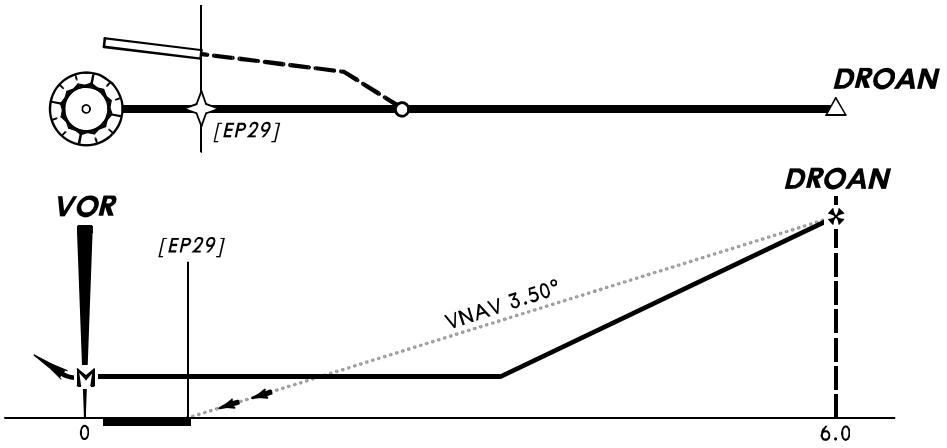
NON-PRECISION APPROACH CODING EXAMPLE 9

The example shows a Final Approach Fix (FAF) to Missed Approach Point (MAP) beyond Landing Threshold Point (LTP) final approach course. This final approach course does not cross the landing threshold. The MAP position is the officially published government source position. A Final End Point (FEP) waypoint is required. The procedure does meet straight-in alignment criteria. The VNAV Path angle is calculated from the LTP elevation + TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification]) to the FAF altitude, using the FEP position and along track distance in determining the distance used in the calculation. The VNAV Path angle is coded in the FEP sequence.



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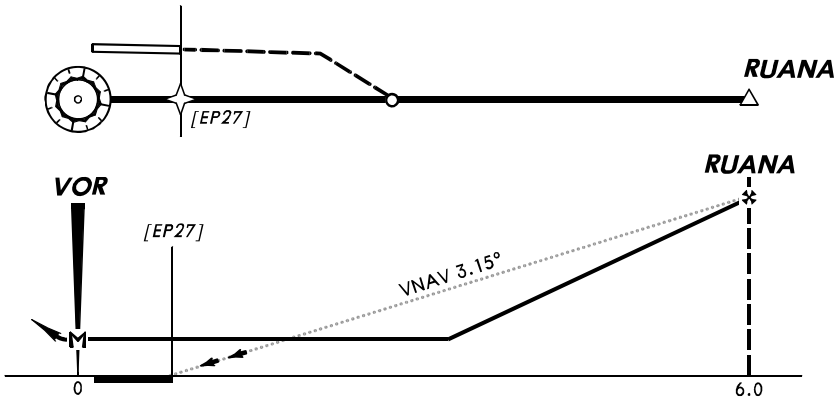
NON-PRECISION APPROACH CODING EXAMPLE 10

The example shows a Final Approach Fix (FAF) to Missed Approach Point (MAP) beyond Landing Threshold Point (LTP) final approach course. This final approach course is parallel/near parallel to the runway alignment. The MAP position is the officially published government source position. A Final End Point (FEP) waypoint is required. The VNAV Path angle is calculated from the LTP elevation + TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification]) to the FAF altitude, using the FEP position and along track distance in determining the distance used in the calculation. The VNAV Path angle is coded in the FEP sequence.



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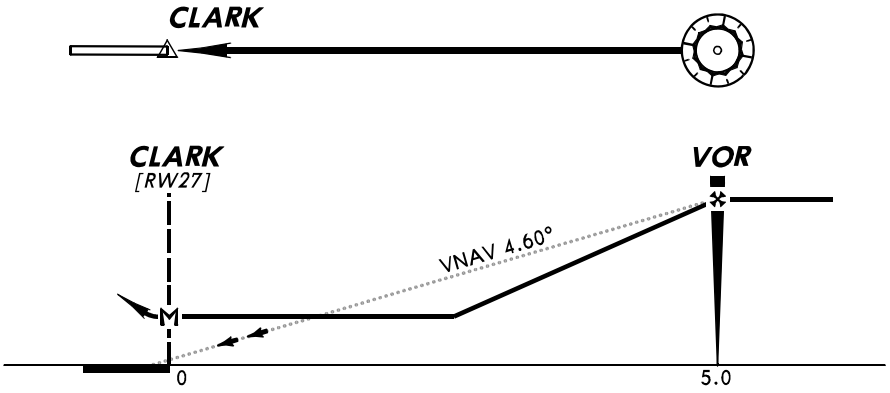
NON-PRECISION APPROACH CODING EXAMPLE 11

This example shows a Final Approach Fix (FAF) to Missed Approach Point (MAP) at Landing Threshold Point (LTP) final approach **course**. The procedure is published with Circle-To-Land weather minimums although the straight-in landing alignment criteria are met. The MAP position is the officially published government source position. The MAP is a published waypoint at the LTP. The VNAV **Path** angle is calculated from the LTP **+ TCH elevation (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification])** to the FAF altitude, and coded in the MAP sequence.



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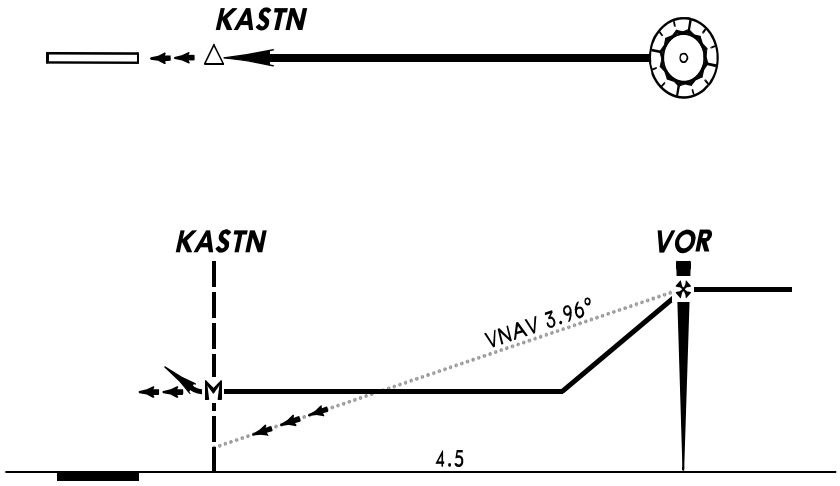
NON-PRECISION APPROACH CODING EXAMPLE 12

This example shows a Final Approach Fix (FAF) to Missed Approach Point (MAP) final approach **course** where the MAP is before the Landing Threshold Point (LTP).

The procedure is Circle-To-Land weather minimums although Straight-in landing alignment criteria are met. The MAP position is the officially published government source position. The LTP is not included in the coding. The VNAV **Path** angle is calculated from *a* LTP **+ TCH elevation (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification])** to the FAF altitude and coded in the MAP sequence.



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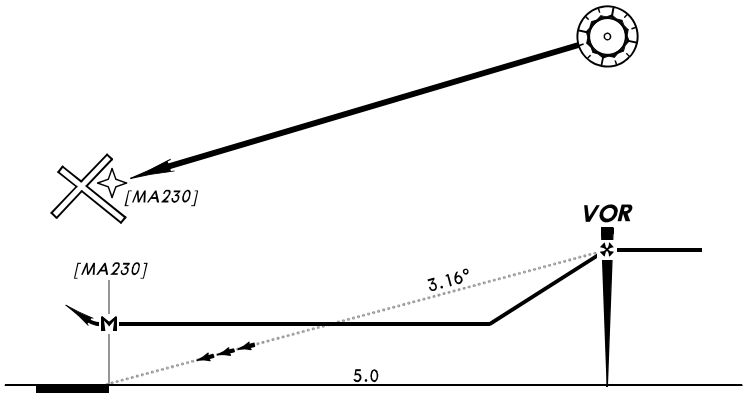
NON-PRECISION APPROACH CODING EXAMPLE 13

This example shows a Final Approach Fix (FAF) to Missed Approach Point (MAP) that is other than the Landing Threshold Point (LTP), and is not before, beyond or abeam the Landing Threshold Point. The procedure is published with Circle-To-Land weather minimums. Straight-in landing alignment criteria are not met.

The MAP position is the officially government source position or, if not published, established as a point abeam the nearest landing threshold **on the Final Approach Course.** The VNAV angle is calculated from the airport elevation to the FAF altitude and coded in the MAP sequence.



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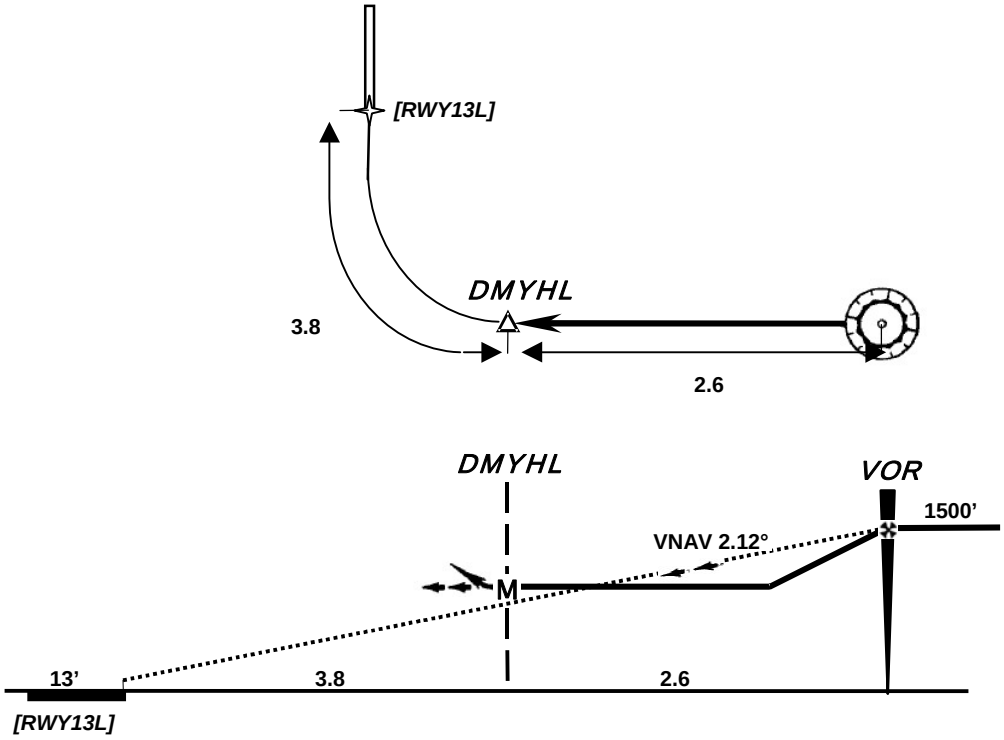
NON-PRECISION APPROACH CODING EXAMPLE 14

The example shows a Final Approach Fix (FAF) to Missed Approach Point (MAP) before the Landing Threshold Point (LTP) final approach course. This final approach course does not cross the landing threshold. The MAP position is the officially published government source position. The procedure does not meet straight-in alignment criteria. The VNAV Path angle is calculated from the LTP elevation + TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification]) to the FAF altitude, using the LTP position and along track distance from FAF in determining the distance used in the calculation. The altitude at the MAP is assigned based on computation, using the calculated VNAV angle. The VNAV Path angle is coded in the MAP sequence.



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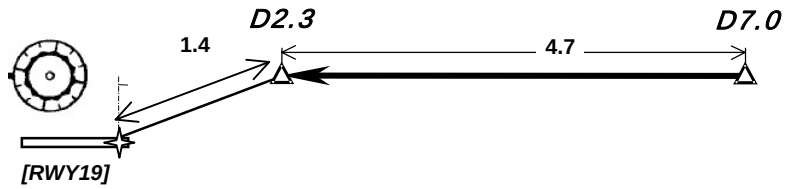
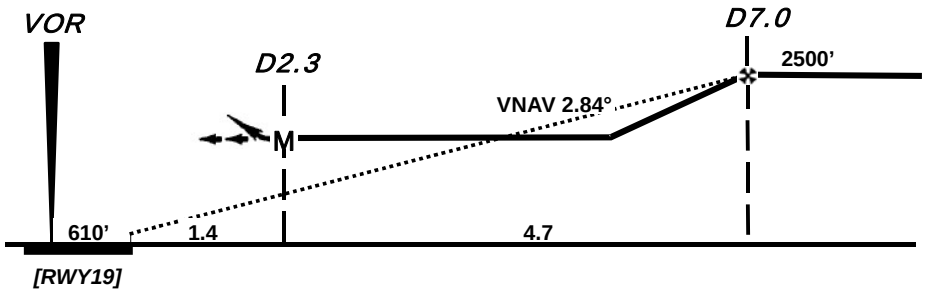
NON-PRECISION APPROACH CODING EXAMPLE 15

The example shows a Final Approach Fix (FAF) to Missed Approach Point (MAP) before the Landing Threshold Point (LTP) final approach course. This final approach course does not cross the landing threshold. The MAP position is the officially published government source position. The procedure does not meet straight-in alignment criteria. The VNAV Path angle is calculated from the LTP elevation + TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification]) to the FAF altitude, using the LTP position and along track distance from FAF in determining the distance used in the calculation. The altitude at the MAP is assigned based on computation, using the calculated VNAV angle. The VNAV Path angle is coded in the MAP sequence.



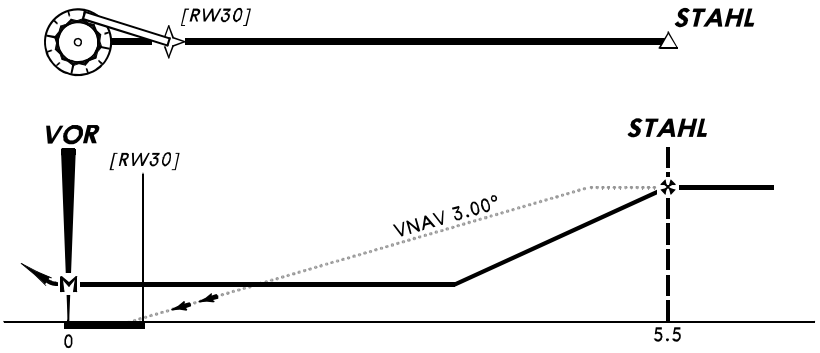


ATTACHMENT 5
PATH AND TERMINATOR

CODING EXAMPLE A
VNAV APPROACH CODING EXAMPLE

Inserted Runway Fix (Rule 8.10.1)

This example shows a procedure published as Final Approach Fix (FAF) to a Missed Approach Point (MAP) beyond the Landing Threshold Point (LTP). The Final Approach Course (FAC) crosses over the landing threshold. The landing alignment is straight-in. As the officially published MAP position is beyond the LTP, a runway fix waypoint [RW30] is inserted as an additional waypoint into the Final Approach Coding of this example. The VNAV Path angle is calculated from the LTP elevation + TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification])using the LTP and the FAF to determine the distance used in the calculation, to the FAF altitude and is coded in the Runway Fix sequence. The missed approach procedure is not included in the graphic. It is included in the coding sequence example. The inserted runway fix is coded as the MAP. The coded first leg of the missed approach path is a continuation of the FAC from the inserted Runway Fix to the officially published MAP. The first leg of the published missed approach path is a climb on the FAC to an altitude of airport elevation plus 400 feet, or as specified by source, followed by a direct to a fix at the FAF.



Inserted Runway Fix Coding:

APP ID	SEQ NR	FIX ID	P/T	WAYPOINT DESCRIPTION				MAG COURSE	DIST	ALT DESC/ALT	VERT ANGLE	ARINC REF.
D30	020	STAHL	IF	E			F	0.0	0.0	At or Above Procedure Altitude		Attachment 5, Rule 8.1.1
D30	030	RW30	CF	G	Y		M	Published FAC	4.7	**At LTP + published TCH (if no procedure TCH is specified by source use 40 or 50 feet)**	-3.00	Attachment 5, Rule 8.1.2 Rule 6.2.9.3 Rule 6.2.10.2.c
D30	040	VOR	CF	V		M		Published FAC	0.8			Attachment 5, Rule 6.2.10.2.c
D30	050		CA					Published FAC		At or Above Airport Plus 400'		Attachment 5, Rule 9.4.1.4
D30	050	STAHL	DF	E	E					At or Above Procedure Altitude		

Waypoint Description:
Column One - Fix Type: E = Waypoint, G = Runway, V = VHF Navaid
Column Two - E = End of Final Approach Coding, Y = Flyover waypoint
Column Three - M = First Leg of Missed Approach Procedure
Column Four - Fix Function in Coding: F = FAF, M = Missed Approach Fix

Note: FAF Altitude Description may be “at” when this is prescribed by source documentation. The altitude in sequence 050 may be a source provided value for the first leg of a missed approach or may be regionally adjusted to 500 feet above the airport. The At or Above Airport Plus 400' is the minimum requirement.

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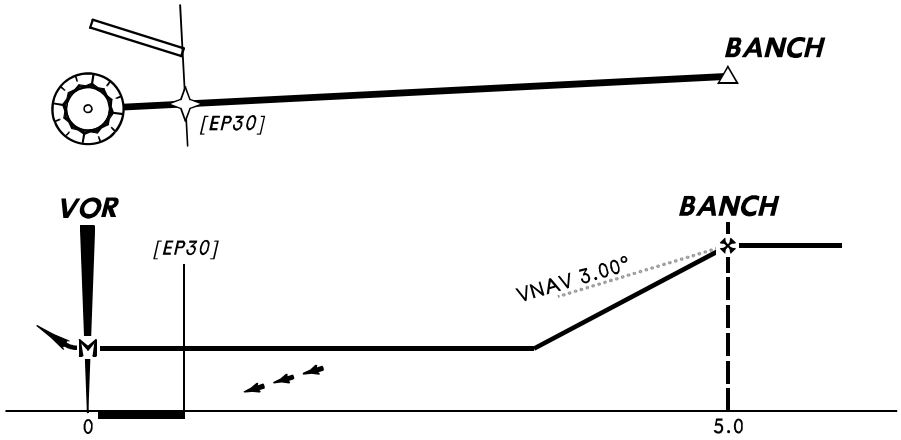
CODING EXAMPLE B
VNAV APPROACH CODING EXAMPLE

Inserted Final End Point Fix (Rule 8.10.2)

This example shows a procedure published as Final Approach Fix (FAF) to a Missed Approach Point (MAP) beyond the Landing Threshold Point (LTP). The Final Approach Course (FAC) does not cross over the landing threshold. The landing alignment is straight-in. As the MAP is located beyond the landing threshold, a Final End Point Fix (FEP) waypoint [EP30] is inserted as an additional waypoint in the Final Approach Coding of this example. The VNAV Path angle is calculated from the LTP elevation + TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this specification])using the FEP and the FAF to determine the distance used in the calculation, to the FAF altitude and is coded in the FEP Fix sequence. The missed approach path is not included in the graphic. It is included in the coding sequence example. The inserted FEP Fix is coded with its designated unique code in the last position of the Waypoint Description on the FEP sequence. The coded first leg of the missed approach path is a continuation of the FAC to the officially published MAP. The first leg of the published missed approach path is on the FAC to an altitude of airport elevation plus 400 feet, or as specified by source, followed by a direct to a fix at the FAF.



Inserted Final End Point Fix Coding:

APP ID	SEQ NR	FIX ID	P/T	WAYPOINT DESCRIPTION				MAG COURSE	DIST	ALT DESC/ALT	VERT ANGLE	ARINC REF.
D30	020	BANCH	IF	E			F	0.0	0.0	At or Above Procedure Altitude		Attachment 5 Rule 8.1.1
D30	030	EP30	CF	E	Y		E	Published FAC	4.1	At LTP + published TCH (if no procedure TCH is specified by source use 40 or 50 feet)	-3.00	Attachment 5 Rule 8.1.2 Rule 6.2.9.4 Rule 6.2.10.2.d
D30	040	VOR	CF	V			M	Published FAC	0.9	At or Above Airport Plus 400'		Attachment 5 Rule 6.2.10.2.d
D30	050		CA			M		Published FAC		At or Above Airport Plus 400'		Attachment 5 Rule 9.4.1.4
D30	060	BANCH	DF	E	E					At or Above Procedure Altitude		

Waypoint Description:
Column One - Fix Type: E = Waypoint, V = VHF Navaid
Column Two - E = End of Final Approach Coding, Y = Flyover Waypoint
Column Three - M = First Leg of Missed Approach Procedure
Column Four - Fix Function in Coding: F = FAF, E, M; E = Final End Point

Note: FAF Altitude Description may be “at” when this is prescribed by source documentation. The altitude in sequence 040 and 050 may be a source provided value for the first leg of a missed approach or may be regionally adjusted to 500 feet above the airport. The At or Above Airport Plus 400' is the minimum requirement.

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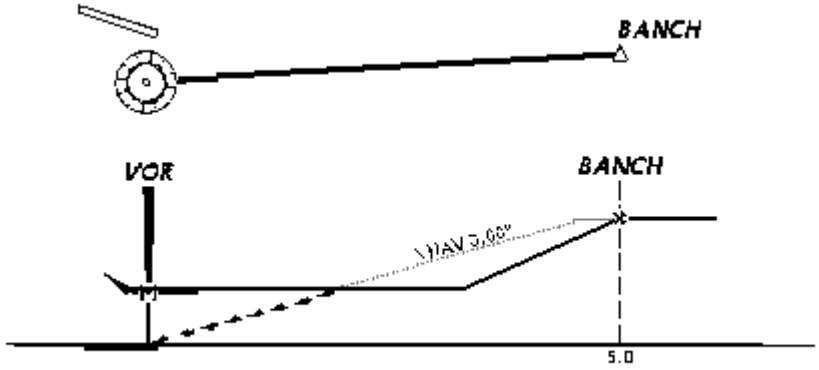
CODING EXAMPLE C
VNAV APPROACH CODING EXAMPLE

MAP is Navaid within 0.10NM of LTP (Rule 8.10.3)

This example shows a procedure published as Final Approach Fix (FAF) to Missed Approach Point (MAP) beyond the Landing Threshold Point (LTP). The Final Approach Course (FAC) does not cross over the LTP. The landing alignment is straight-in. As the MAP is a Navaid located equal to or less than 0.10NM from the LTP, coding is to the Navaid as the MAP and no waypoints are inserted into the Final Approach Coding. The VNAV Path angle is calculated from the MAP to the FAF altitude so as to ensure an LTP elevation + TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification])) over the threshold, references the MAP position for distance portion of the calculation and is coded in the MAP sequence. The missed approach path is not included in the graphic. It is included in the coding sequence example. Final Approach Coding is to the officially published MAP. The coded first leg of the missed approach path is a continuation of the FAC as a climb on that course to an altitude of airport elevation plus 400 feet, or a specified by source, followed by a direct to fix at the FAF.



No inserted fixes, MAP is Navaid, beyond threshold but within 0.14NM Coding:

APP ID	SEQ NR	FIX ID	P/T	WAYPOINT DESCRIPTION				MAG COURSE	DIST	ALT DESC/ALT	VERT ANGLE	ARINC REF.
S29	020	BANCH	IF	E			F	0.0	0.0	At or Above Procedure Altitude		Attachment 5, Rule 8.1.1
S29	030	VOR	CF	V	Y		M	Published FAC	5.0	See Note Below	-3.00	Attachment 5, Rule 8.1.2, 6.2.5.3
S29	040		CA			M		Published FAC		At or Above Airport Plus 400'		Attachment 5, Rule 9.4.1.4
S29	050	BANCH	DF	E	E					At or Above Procedure Altitude		

Waypoint Description:
Column One - Fix Type: E = Waypoint, V = VHF Navaid
Column Two - E = End of Final Approach Coding, Y = Flyover Waypoint
Column Three - M = First Leg of Missed Approach Procedure
Column Four - Fix Function in Coding: F = FAF, M = Missed Approach Fix

Altitude/Altitude Description of MAP Sequence: Will be an "At" altitude, calculated on the path that continues over the LTP at the published procedure TCH (or 50 feet if no TCH is published). The altitude for the MAP sequence will not be less that the LTP elevation. The FAF Altitude Description may be "at" when this is prescribed by source documentation. The altitude in sequence 040 (first leg of missed approach path) may be a source provided value or may be regionally adjusted to 500 feet above the airport elevation. The "At or Above Airport Elevation Plus 400 Feet" is the minimum requirement.

ATTACHMENT 5
PATH AND TERMINATOR

COMMENTARY

Final End Point (FEP)
Output File Delivery Options

The purpose of all of the rules on vertical navigation coding for non-precision approach procedures contained in the Attachment 5 to ARINC 424 is to ensure that standards are established that will allow for the provision of a VNAV Path Angle for every approach. These standards have been broken into four groups that are the result of the location of the missed approach point for the procedures as published in official government source documentation. These groups are defined in Rule 8.10 of this attachment and are illustrated as Coding Examples 1 through 15 following that paragraph. The primary group is "Missed Approach Point at Landing Threshold Point" and "Missed Approach Point Prior to Landing Threshold Point". The other three groups cover the situations when the published missed approach point is beyond the Landing Threshold Point. These three groups are further illustrated in Coding Examples "A", "B" and "C" associated with Rule 8.10.

One of these three groups is the situation where the published missed approach point is beyond the Landing Threshold Point and the published Final Approach Course does not pass through a defined area related to the Landing Threshold Point. This situation is commonly referred to as Inserted Final End Point Coding, Coding Example "B".

Coding Example "B" represents the best possible solution for providing the intent of the government source procedure and encoding the data necessary for execution of the procedure within the avionics. During the iterations required for the development of this encoding standard and its rules, it was identified that not all existing avionics would be able to process this new data in a manner compatible with their operational software. The required modifications to that operational software would more than likely result in a need to re-certify that software. As it is not within the scope of ARINC 424 to refined new avionics requirements or impose data configurations that would result in new avionics requirements; methods were reviewed that would ensure that the vertical navigation benefits of FEP Coding could be made available to the broadest possible user base.

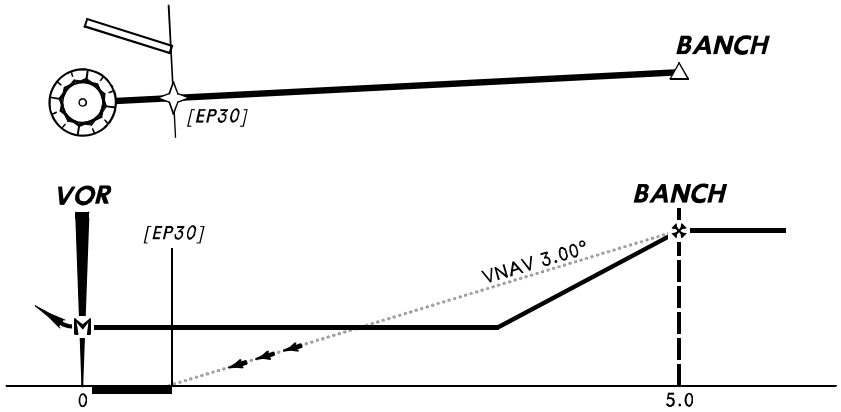
This review effort resulted in identifying that four data encoding versions for non-precision approach procedures that were designed such that they resulted in the FEP situation would make FEP data available to the largest number of users and negate the impacts of a single solution on operational software. One of these was selected as the ARINC 424 standard. The other options were to be used when specified between avionics vendor and data provider. To ensure that the process and result of any option specification would result in an identical implementation by all data providers to any given vendor, it was agreed that all four encoding versions would be incorporated into ARINC 424 as "Delivery Format Options One through Four". An illustration of the four FEP Format Delivery Options is provided on the pages that follow this Commentary.

ATTACHMENT 5
PATH AND TERMINATOR

FEP Delivery Format One – ARINC 424 Standard Coding
VNAV Approach Coding Example Inserted Final End Point Fix

This example shows a procedure published as Final Approach Fix (FAF) to a Missed Approach Point (MAP beyond the Landing Threshold Point (LTP). The Final Approach Course (FAC) does not cross over the landing threshold. The landing alignment is straight-in. As the MAP is located beyond the LTP and the FAC does not cross over the landing threshold, a Final End Point Fix (FEP) waypoint is inserted as an additional waypoint in the final approach coding of this example. The VNAV Path angle is calculated from the LTP elevation + TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification]) to the FAF altitude. It is included in the coding sequence example. The inserted "FEP Fix" is coded with its designated unique code in the last position of the Waypoint Description on the FEP sequence. The coded first leg of the missed approach path is a continuation of the FAC to the officially published MAP. There is an indication in the coding of the officially published missed approach point. The leg from the FEP Fix to the MAP is not coded as the first leg of the missed approach path, meaning no indication in position three of the Waypoint Description. The first leg of the missed approach path is the published first leg and is a climb on the FAC to an altitude of airport elevation plus 400 feet, or as specified by source, followed by a direct to a fix at the FAF.

Note – Add Y to column two on the FEP Fix sequence to indicate flyover required and update the Waypoint Description information to include the Y = Flyover Waypoint.



Inserted Final End Point Fix Coding:

APP ID	SEQ NR	FIX ID	P/T	WAYPOINT DESCRIPTION				MAG COURSE	DIST	ALT DESC/ALT	VERT ANGLE	ARINC REF.
D30	020	BANCH	IF	E			F	0.0	0.0	At or Above Procedure Altitude		Attachment 5 Rule 8.1.1
D30	030	EP30	CF	E			E	Published FAC	4.1	At LTP + published TCH (if no procedure TCH is specified by source use 40 or 50 feet)	-3.00	Attachment 5 Rule 8.1.2Rule 6.2.9.4Rule 6.2.10.2.d
D30	040	VOR	CF	V			M	Published FAC	0.9	At or Above Airport Plus 400'		Attachment 5, Rule 6.2.10.2.d
D30	050		CA			M		Published FAC		At or Above Airport Plus 400'		Attachment Five, Rule 9.4.1.4
D30	060	BANCH	DF	E	E					At or Above Procedure Altitude		

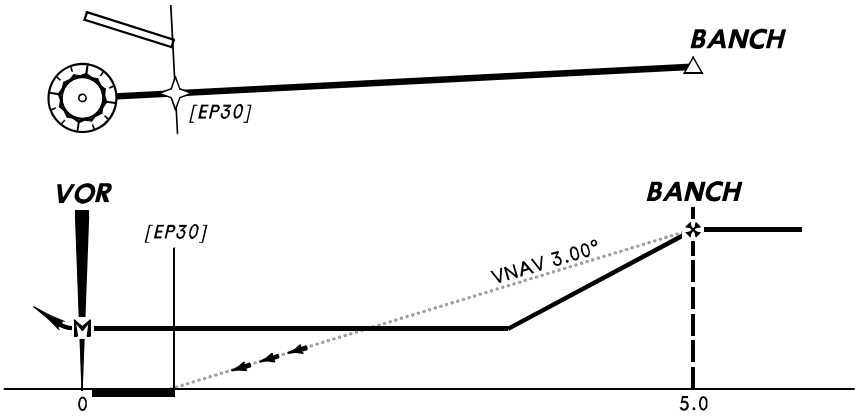
Waypoint Description:
Column One - Fix Type: E = Waypoint, V = VHF Navaid
Column Two - E = End of Final Approach Coding
Column Three - M = First Leg of Missed Approach Procedure
Column Four - Fix Function in Coding: F = FAF, E = Final End Point, M = Missed Approach Point
Note: FAF Altitude Description may be "at" when this is prescribed by source documentation. The altitude in sequence 040 and 050 may be a source provided value for the first leg of a missed approach or may be regionally adjusted to 500 feet above the airport. The At or Above Airport Plus 400' is the minimum requirement.

ATTACHMENT 5
PATH AND TERMINATOR

FEP Delivery Format Two - FEP Coded as MAP
VNAV Approach Coding Example Inserted Final End Point Fix

This example shows a procedure published as Final Approach Fix (FAF) to a Missed Approach Point (MAP) beyond the Landing Threshold Point (LTP). The Final Approach Course (FAC) does not cross over the landing threshold. The landing alignment is straight-in. As the MAP is located beyond the LTP and the FAC does not cross over the landing threshold, a Final End Point Fix (FEP) waypoint is inserted as an additional waypoint in the final approach coding of this example. The VNAV Path angle is calculated from the LTP elevation + TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification]) to the FAF altitude, references the FEP position and is coded in the FEP sequence. The missed approach path is not included in the graphic. It is included in the sequence example. The inserted "FEP Fix" is coded with the missed approach point code in the last position of the Waypoint Description. The path from the inserted FEP fix is coded as the first leg of the missed approach path and is a continuation of the FAC to the officially published missed approach point. There is no indication that the fix in sequence 030 is an FEP or that the fix in sequence 040 is the officially published MAP. Although no code indication is provided, the first leg of the published missed approach path is coded from the published missed approach point and is a climb on the FAC to an altitude of airport elevation plus 400 feet, or as specified by source, followed by a direct to a fix at the FAF.

Note – Add Y to column two on the MAP (FEP) Fix sequence to indicate flyover required and update the Waypoint Description information to include the Y = Flyover Waypoint.



Inserted Final End Point Fix code as Missed Approach Point Coding:

APP ID	SEQ NR	FIX ID	P/T	WAYPOINT DESCRIPTION				MAG COURSE	DIST	ALT DESC/ALT	VERT ANGLE	ARINC REF.
D30	020	BANCH	IF	E			F	0.0	0.0	At or Above Procedure Altitude		Attachment 5 Rule 8.1.1
D30	030	EP30	CF	E			M	Published FAC	4.1	At LTP + published TCH (if no procedure TCH is specified by source use 40 or 50 feet)	-3.00	Attachment 5 Rule 8.1.2 Rule 6.2.9.4 Rule 6.2.10.2.d
D30	040	VOR	CF	V		M		Published FAC	0.9	At or Above Airport Plus 400'		Attachment 5, Rule 6.2.10.2.d
D30	050		CA					Published FAC		At or Above Airport Plus 400'		Attachment Five, Rule 9.4.1.4
D30	060	BANCH	DF	E	E					At or Above Procedure Altitude		

Waypoint Description:
Column One - Fix Type: E = Waypoint, V = VHF Navaid
Column Two - E = End of Final Approach Coding
Column Three - M = First Leg of Missed Approach Procedure
Column Four - Fix Function in Coding: F = FAF, M = Missed Approach Point
Note: FAF Altitude Description may be "at" when this is prescribed by source documentation. The altitude in sequence 040 and 050 may be a source provided value for the first leg of a missed approach or may be regionally adjusted to 500 feet above the airport. The At or Above Airport Plus 400' is the minimum requirement.

ATTACHMENT 5
PATH AND TERMINATOR

FEP Delivery Format Three - FEP Coded as MAP
FEP Waypoint Indication Provided

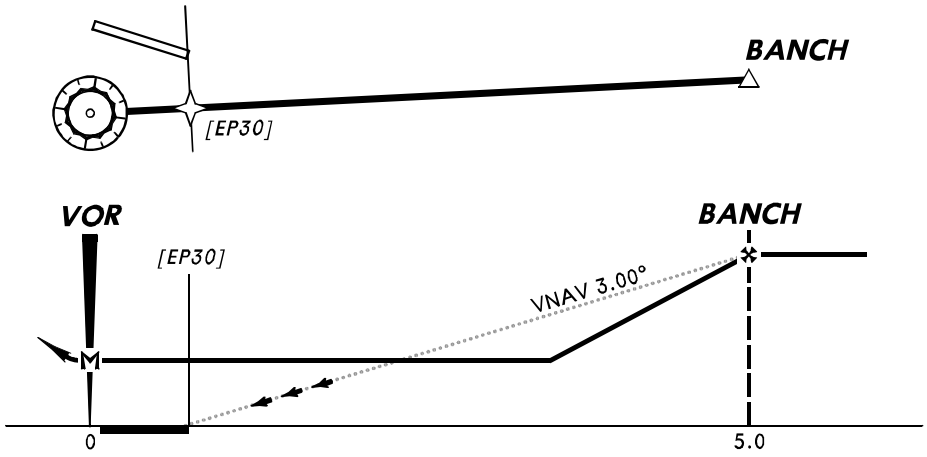
VNAV Approach Coding Example Inserted Final End Point Fix

This example shows a procedure published as Final Approach Fix (FAF) to a Missed Approach Point (MAP) beyond the Landing Threshold Point (LTP). The Final Approach Course (FAC) does not cross over the landing threshold. The landing alignment is straight-in. As the MAP is located beyond the LTP and the FAC does not cross over the landing threshold, a Final End Point Fix (FEP) waypoint is inserted as an additional waypoint in the final approach coding of this example. The VNAV Path angle is calculated from the LTP elevation + TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification])) to the FAF altitude, references the FEP position and is coded in the FEP sequence. The missed approach path is not included in the graphic. It is included in the sequence example. The inserted "FEP Fix" is coded with the missed approach point code in the last position of the Waypoint Description and a unique waypoint type code in the first position of the Waypoint Description, indicating that this "MAP" is at an "FEP" location. The path from the inserted FEP fix is coded as the first leg of the missed approach path and is a continuation of the FAC to the officially published missed approach point. There is an indication provided that the fix in sequence 030 is really an FEP but none that the fix in sequence 040 is the officially published MAP. Although no code indication is provided, the first leg of the published missed approach path is coded from the published missed approach point and is a climb on the FAC to an altitude of airport elevation plus 400 feet, or as specified by source, followed by a direct to a fix at the FAF.

Modify the column one of the Waypoint Description to read:

> **Column One – Fix Type: E = Waypoint, V = VHF Navaid, F = FEP Fix Waypoint.**
> **The code "F" for FEP Fix Waypoint is not included in paragraph 5.17 of ARINC 424 as a valid code. It is used only in this FEP Delivery Format option to indicate that the fix coded as the MAP is not the officially published MAP but rather a fix located at the FEP position.**

Note – Add Y to column two on the MAP (FEP) Fix sequence to indicate flyover required and update the Waypoint Description information to include the Y = Flyover Waypoint.



Inserted Final End Point Fix coded as Missed Approach Point Coding:

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APP ID	SEQ NR	FIX ID	P/T	WAYPOINT DESCRIPTION				MAG COURSE	DIST	ALT DESC/ALT	VERT ANGLE	ARINC REF.
D30	020	BANCH	IF	E			F	0.0	0.0	At or Above Procedure Altitude		Attachment 5 Rule 8.1.1
D30	030	EP30	CF	F			M	Published FAC	4.1	At LTP + published TCH (if no procedure TCH is specified by source use 40 or 50 feet)	-3.00	Attachment 5 Rule 8.1.2 Rule 6.2.9.4 Rule 6.2.10.2.d
D30	040	VOR	CF	V		M		Published FAC	0.9	At or Above Airport Plus 400’		Attachment 5, Rule 6.2.10.2.d
D30	050		CA					Published FAC		At or Above Airport Plus 400’		Attachment 5, Rule 9.4.1.4
D30	060	BANCH	DF	E	E					At or Above Procedure Altitude		

Waypoint Description:
Column One - Fix Type: E = Waypoint, V = VHF Navaid, F = FEP Waypoint
Column Two - E = End of Final Approach Coding
Column Three - M = First Leg of Missed Approach Procedure
Column Four - Fix Function in Coding: F = FAF, M = Missed Approach Point
Note: FAF Altitude Description may be “at” when this is prescribed by source documentation. The altitude in sequence 040 and 050 may be a source provided value for the first leg of a missed approach or may be regionally adjusted to 500 feet above the airport. The At or Above Airport Plus 400’ is the minimum requirement.

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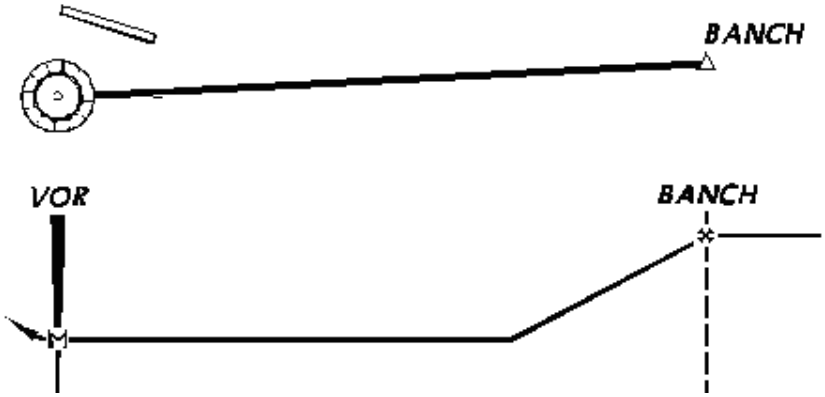
Delivery Format Four - No FEP Provided
No VNAV Provided

(The ARINC 424 Standard prior to FEP Coding Rules)
VNAV APPROACH CODING EXAMPLE

This example shows a procedure published as Final Approach Fix (FAF) to a Missed Approach Point (MAP) beyond the Landing Threshold Point (LTP). The Final Approach Course (FAC) does not cross over the landing threshold. The landing alignment is straight-in. The MAP is located beyond the LTP and the FAC does not cross over the landing threshold, no VNAV data is provided. The missed approach path is not included in the graphic. It is included in the coding sequence example. The officially published MAP is provided with the appropriate code in the last position of the Waypoint Description. The coded first leg of the missed approach path is the officially published missed approach path and is from the published missed approach point. It is a climb on the FAC to an altitude of airport elevation plus 400 feet, or a specified by source, followed by a direct to a fix at the FAF.



Inserted Final End Point Fix Coding:

APP ID	SEQ NR	FIX ID	P/T	WAYPOINT DESCRIPTION				MAG COURSE	DIST	ALT DESC/ALT	VERT ANGLE	ARINC REF.
D30	020	BANCH	IF	E			F	0.0	0.0	At or Above Procedure Altitude		Attachment 5 Rule 8.1.1
D30	030	VOR	CF	V			M	Published FAC	5.0	Note 1	0.00	
D30	040		CA			M		Published FAC		At or Above Airport Plus 400’		Attachment 5, Rule 9.4.1.4
D30	050	BANCH	DF	E	E					At or Above Procedure Altitude		

Waypoint Description:
Column One - Fix Type: E = Waypoint, V = VHF Navaid
Column Two - E = End of Final Approach Coding
Column Three - M = First Leg of Missed Approach Procedure
Column Four - Fix Function in Coding: F = FAF, M = Missed Approach Point
Note: FAF Altitude Description may be “at” when this is prescribed by source documentation. The altitude in sequence 040 may be a source provided value for the first leg of a missed approach or may be regionally adjusted to 500 feet above the airport. The At or Above Airport Plus 400’ is the minimum requirement.

Note One: For a published Missed Approach Point beyond the landing threshold and no Landing Threshold Fix or Final End Point Fix has been include in the lateral path, the altitude in the MAP sequence will be an “at” altitude equal to the lowest MDA published for the procedure. The Vertical Angle for this Delivery Option will be provided on the MAP sequence and will be 0.00.

Note – Add Y to column two on the MAP Fix sequence to indicate flyover required and update the Waypoint Description information to include the Y = Flyover Waypoint

ATTACHMENT 5
PATH AND TERMINATOR

APPROACH AND APPROACH TRANSITION CODING RULES

9.0 Missed Approach Procedure Rules Valid For All Procedure Types

Missed Approach Procedure coding must be accomplished as an integral part of the Approach Procedure Coding and will be provided for each approach procedure contained in the data base. Specific coding must be incorporated to facilitate identification of where the Missed Approach Coding starts within any given approach procedure.

The structure of the procedure records included in this Specification as defined in this attachment is such that multiple missed approach procedure paths may be coded for a single approach procedure. This will accommodate those procedures with alternative missed approach paths based on aircraft climb performance. Coding for multiple missed approach paths for a single approach procedure must commence at the same missed approach point. Identification of multiple missed approach procedures, when coded, will be accomplished through the coding of a specific Transition Identifier which closely aligns with published information.

9.1 Missed Approach Point

9.1.1 All Approach Procedure coding must have a segment that identifies the Missed Approach Point Fix. Such a fix must be the published Missed Approach Point, either a IFR Landing Threshold or a Helipad Alighting Point or a dedicated Missed Approach Point (MAP Fix).

9.1.2 Identification of the fix within a sequence of procedure records and the type of fix, must be accomplished through code in the "Waypoint Description" field (see Section 5.17).

9.1.2.1 When the Missed Approach Point Fix is a Runway or Helipad Fix, Waypoint Description Position One, must carry a character "G" and the "M" in Position Four.

9.1.2.2 When the Missed Approach Point Fix is MAP Fix, Waypoint Description Position One will carry a code equal to the type of fix such a Navaid or waypoint and must carry a character "M" in Position Four.

9.2 First Leg of Missed Approach Procedure

9.2.1 The first sequence of the Missed Approach Procedure must always be coded with the character "M" in Position Three of the "Waypoint Description" field.

9.2.2 Coding of the Missed Approach Procedure assumes that the procedure will be flown commencing at the Missed Approach Point Fix.

9.3 Vertical Path of Missed Approach Procedure

9.3.1 If the published Missed Approach Point Fix is a Runway fix or a Helipad Fix, then the following rules apply:

9.3.1.1 The Altitude 1 value in the Missed Approach Point sequence must be equal to Runway Threshold or Helipad Alighting Point Elevation plus the published TCH. **If no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification].**

9.3.1.2 The first leg of a Missed Approach Procedure will contain any government source provided altitude constraints. These altitudes may be provided as "At," "At or Above," "At or Below," "At or Below to At or Above," dependent on the government source documentation and coded in accordance with Section Five, Subsection 5.29 of this Specification. Except as indicated below, if the government source does not provide an altitude constraint for the first leg of the Missed Approach Procedure, none will be coded

9.3.1.3 **In precision approach procedures such as Full ILS or GLS, the Decision Height (DH) value at which the Missed Approach Path would normally be commenced is not included as part of the Final Approach Coding. Other portions of this attachment define what altitude data is included in the coding of the procedure for the missed approach point and the first leg of the missed approach path.**

9.3.1.4 **In non-precision approach procedure, the Minimum Descent Altitude (MDA) value at which the missed approach decision would normally be made is not included as part of the Final Approach Coding. . Other portions of this attachment define what altitude data is included in the coding of the procedure for the missed approach point and the first leg of the missed approach path.**

ATTACHMENT 5
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APPROACH AND APPROACH TRANSITION CODING RULES

9.3.1.5 When an immediate turn is specified in an ILS, MLS, or GLS missed approach, or if the source describes a turn greater than 15 degrees from the final approach **course**, without an altitude specified before the turn, as the first leg of a missed approach, a course from or heading to an altitude (CA, FA, VA) leg must be coded as the first leg of the missed approach **and must include a command to climb before the turning leg**, using the **final** approach course for the leg heading or course. The altitude will be coded as a minimum altitude, at least 400 feet above the airport elevation and the leg will terminate at that altitude.

9.4 Rule deleted by Supplement 17.

9.4.1 Rule deleted by Supplement 17.

9.4.1.1 Rule deleted by Supplement 17.

9.4.1.2 Rule deleted by Supplement 17.

9.4.1.3 Rule deleted by Supplement 17.

9.4.1.4 Rule deleted by Supplement 17.

9.4.1.5 Rule deleted by Supplement 17.

9.5 Other Missed Approach Procedure Considerations.

9.5.1 Opposite end runway or helipad fixes must not be used in the coding of missed approach procedures.

9.5.2 The procedure reference Navaid must be used as the Recommended Navaid when required, with the following exceptions:

9.5.2.1 The first leg of a Missed Approach Procedure may be a CD, FD or VD leg. For these legs a DME may be used as the Recommended Navaid, with the Theta field left blank and the DME distance entered in the Rho field.

9.5.2.2 Rule deleted by Supplement 17.

9.5.2.3 The first leg of the Missed Approach Procedure may be a CR or a VR leg. For these legs, a VOR facility (without or without DME) may be used as the Recommended Navaid. When the facility has no DME, the Rho field is left blank and the VOR radial is entered in the Theta field.

9.5.2.4 When the first leg of a Missed Approach Procedure is coded as a “HA or HM” leg, the coding will also include a government source provided altitude.

ATTACHMENT 5
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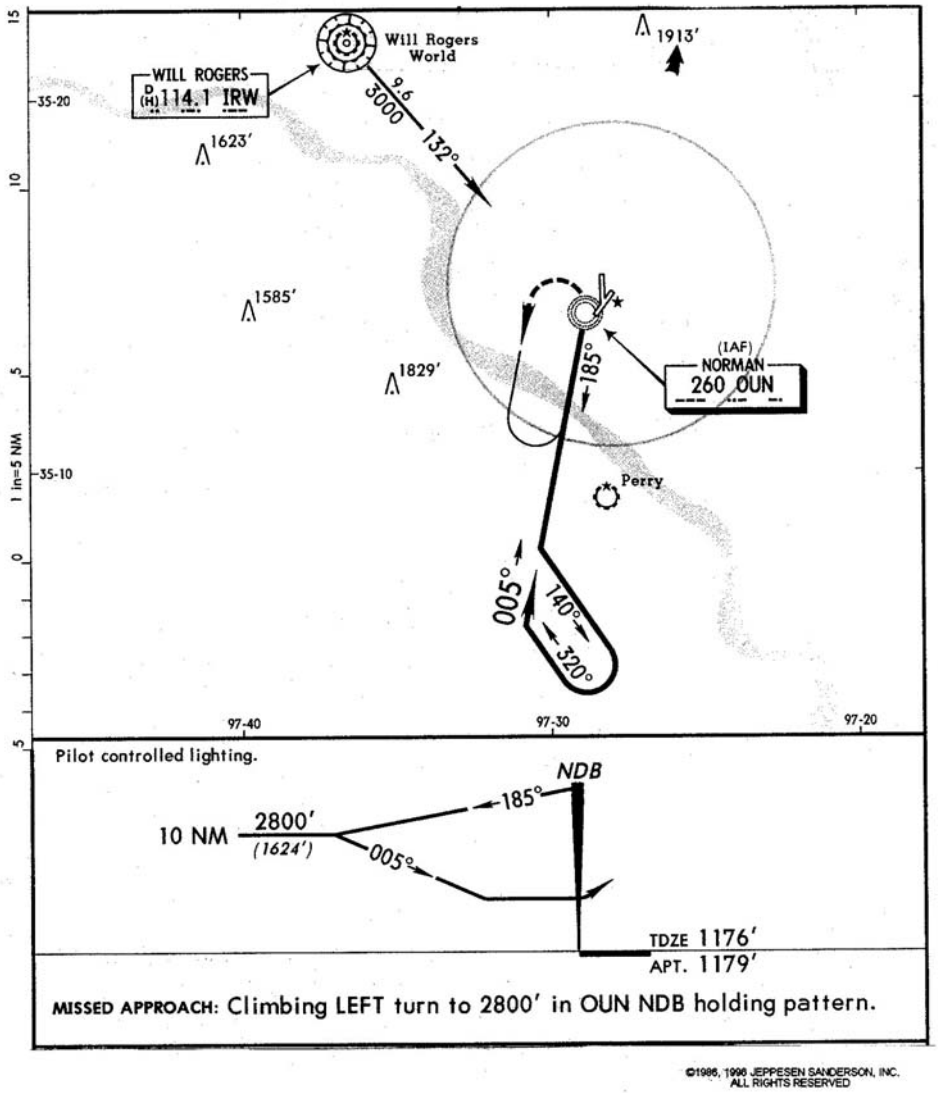
MISSED APPROACH CODING EXAMPLE 1

The copyrighted material in the following Missed Approach Coding Examples is reproduced with the permission of the copyright holder Jeppesen Sanderson, Inc, Englewood, Colorado.

Not for navigational or other operational use. For example only. Please consult current navigation charts.



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APP ID	SEQ NR	FIX ID	P/T	RECD NAV	W/P DESC	RHO	MAG CRS	DIST	ALT	VERT ANG
N03	020	FN03	IF	OUN	E__F				01680	
N03	030	OUN	CF		E__M		0050	0040	01680	000
N03	040	OUN	HM		EEMH		0050	001T	02800	

ATTACHMENT 5
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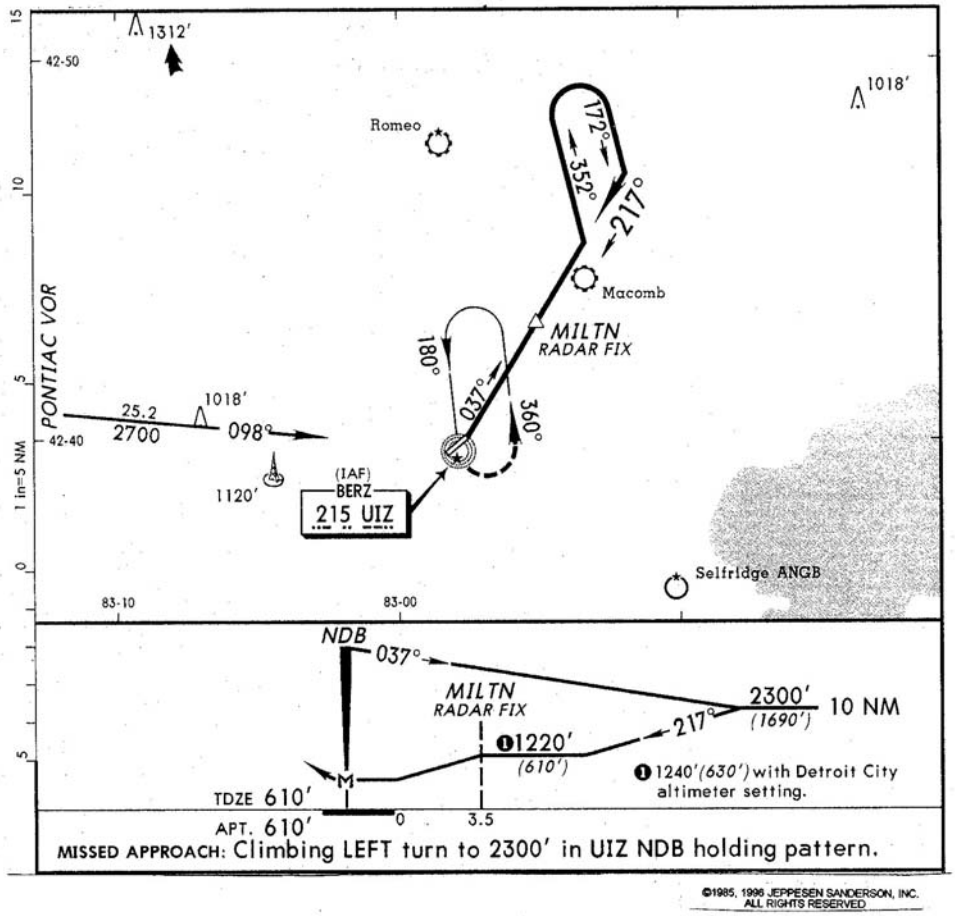
MISSED APPROACH CODING EXAMPLE 2

The copyrighted material in the following Missed Approach Coding Examples is reproduced with the permission of the copyright holder Jeppesen Sanderson, Inc, Englewood, Colorado.

Not for navigational or other operational use. For example only. Please consult current navigation charts.



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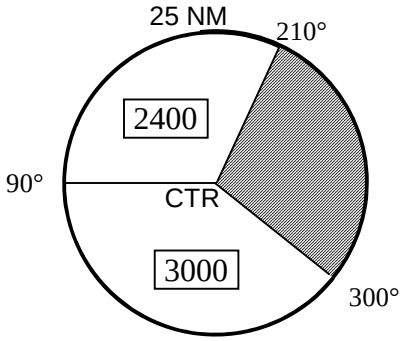
APP ID	SEQ NR	FIX ID	P/T	RECD NAV	W/P DESC	RHO	MAG CRS	DIST	ALT	VERT ANG
N22	020	MILTN	IF	UIZ	E__F				01220	
N22	030	UIZ	CF		E__M		2170	0040	01110	000
N22	040	UIZ	HM		EEMH		1800	001T	02300	

ATTACHMENT 5
PATH AND TERMINATOR

10.0 MSA Coding

The purpose of this attachment is to define the coding rules for MSA sectors, especially when the whole circle is not defined and when different radii are used.

10.1 MSA With one Radius Only



The MSA record must contain the minimum altitudes for all sectors. If a sector is forbidden or not drawn on the chart, its altitude must be coded '999' (99 900 ft), as the altitude field must be numeric. This value is consistent with forbidden areas as the aircraft cannot fly such an altitude.

The sectors must be coded in the clockwise order, starting with the sector having the smallest start bearing.

Coding Example:

Sector #	1	2	3	4	5	6	7
Start bearing	090	210	300				
End bearing	210	300	090				
Altitude	024	999	030				
Radius	25	25	25				

Corresponding PS record (col 43 to 75) : "090210024252103009992530009003025"

APPENDIX 1
CHRONOLOGY AND BIBLIOGRAPHY

1. Chronology

The AEEC Area Navigation Subcommittee established a working group at its September 1973 meeting in Los Angeles to look into the possibilities of standardizing RNAV system reference data format and encoding characteristics. The Subcommittee had observed that different RNAV system manufacturers were taking divergent paths with respect to reference data organization, and believed that, unless this trend was halted, the airline industry would be faced with the very high costs of supporting the production of data files in several different formats. Obvious economic benefits would result if this could be avoided.

The working group met for the first time in January 1974. It examined the approaches to reference data organization being taken by RNAV system manufacturers and, with the help of Jeppesen and Co., looked at the data whose characteristics needed definition. It established the philosophy concerning standardization stated in Section 1.2 of this Specification, and determined the path to follow to implement it. This was the vital first step in this activity, as it established where in the overall process of producing FDSU cassettes and ADEU cards the application of standardization would and would not yield cost benefits.

At its second and third meetings, held in March and May of 1974 respectively, the working group concentrated on defining characteristics for data elements. These included field content, record structure and file organization. Sufficient progress was made for a first draft of this Specification to be prepared for presentation to the RNAV Subcommittee for review.

The Subcommittee considered this first draft at its September 1974 meeting in Washington, D.C. It endorsed the principles established by the working group for the preparation of the draft. The Subcommittee concluded, however, that it should take on the rest of the specification-writing itself, building on the foundation represented by the draft. The working group was, therefore, disbanded. The Subcommittee's in-depth review of the draft produced numerous amendments and proposals for addition which, following the meeting, were incorporated into the document to produce the second draft.

The Subcommittee reviewed the second draft at its January 1975 meeting in Washington, D.C. Incorporation of the amendments developed at this meeting was felt by the Subcommittee to be all that was necessary to complete the Specification. The third draft, incorporating them, was approved by AEEC for publication at the General Session held in Washington, D.C. in the Spring of 1975.

2. Bibliography

The following is a list of AEEC letters associated with the preparation of ARINC Specification 424. A list of letters related to the RNAV Subcommittee's activities as a whole may be found in ARINC Report 299, "AEEC Letter Index."

AEEC LETTER NO.	DATE	SUBJECT
73-124/RNAV-87	Oct 8, 1973	Report of the Area Navigation Subcommittee Meeting held in Los Angeles, California, September 11th, 12th and 13th, 1973.
74-017/RNAV-89	Apr 2, 1974	Progress Report on the Activities of the RNAV Subcommittee's Data Format Standardization Working Group.
74-029/RNAV-90	Jun 17, 1974	Circulation of Draft No. 1 of Project Paper 424, "Area Navigation System Data Base Specification."
74-046/RNAV-94	Aug 7, 1974	Thomson CSF Comments on ARINC Characteristics 583 Synchro Excitation Provisions and on RNAV Data Format Standardization.
74-058/RNAV-95	Sep 26, 1974	Report of the Area Navigation Subcommittee Meeting Held September 17th, 18th and 19th, 1974 in Washington, D.C.
74-071/RNAV-97	Nov 14, 1974	Circulation of Draft No. 2 of Project Paper 424, "Area Navigation System Data Base Specification."
75-013/RNAV-99	Feb 27, 1975	Circulation of Draft No. 3 of Project Paper 424, "Area Navigation System Data Base Specification."
75-014/RNAV-100	Mar 7, 1975	Report of the Area Navigation Subcommittee Meeting held in Washington, D.C., January 21st, 22nd and 23rd, 1975.
75-018/RNAV-101	Mar 25, 1975	Additions to Project Paper 424 proposed by Jeppesen.

3. Meeting Attendees

The following people constituted the RNAV Subcommittee's Data Format Standardization Working Group and attended Subcommittee meetings held in September 1974 and January 1975.

APPENDIX 1
CHRONOLOGY AND BIBLIOGRAPHY

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APPENDIX 2
STRAIGHT-IN CRITERIA

Deleted by Supplement 19.

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SUPPLEMENT 1

TO

ARINC SPECIFICATION 424

(AREA) NAVIGATION SYSTEM DATA BASE

Published: September 8, 1980

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: June 17, 1980

A. PURPOSE OF THIS SUPPLEMENT

This Supplement overhauls and extends the coverage of Specification 424 to enable the data base defined therein to support the navigation functions of ARINC 702 and similar flight management computers.

B. ORGANIZATION OF THIS SUPPLEMENT

The first part of this document, printed on goldenrod paper, contains descriptions of the changes introduced into the Specification by this Supplement. The second part consists of replacement white pages for the Specification modified to reflect these changes. The modified and added material on each replacement page is identified with c-1 symbols in the margins.

Existing copies of Specification 424 may be updated by inserting the replacement white pages where necessary and destroying the pages they displace. The goldenrod pages should be inserted inside the rear cover of the Specification.

Copies of the Specification bearing the number 424-1 already contain this Supplement and thus do not require revisions by the reader.

C. CHANGES TO ARINC SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This Section presents a complete tabulation of the changes and additions to the Specification introduced by this Supplement. Each change or addition is entitled by the section number and title currently employed in the Specification, or by the section number and title that will be employed when the Supplement is eventually incorporated. In each case there is included a brief description of the addition or change and, for other than very minor revisions, any text originally contained in the Specification is reproduced for reference.

TITLE AND CHAPTER HEADINGS

Title of document changed from "Area Navigation System Data Base" to "Navigation System Data Base." "RNAV" changed to "Navigation" in chapter headings and in text.

1.2 Data Format Standardization Philosophy

Final two sentences of third paragraph of section amended to refer to new format standards for tailored company route data introduced into the document by this Supplement. Commentary added following first paragraph.

ORIGINAL TEXT FOLLOWS

(Final two sentences of third paragraph of 1.2)

A tailored record may contain data for which formatting rules may be found in this document, or data for which no such rules have been established. In the latter case, an arbitrary format will be used.

1.3 Organization of this Document

Second paragraph of section deleted to reflect deletion of ADEU card standards from the Specification by this Supplement.

ORIGINAL TEXT FOLLOWS

(Second paragraph of 1.3)

The document closes with a consideration of the standards applicable to the data cards employed with airborne Automatic Data Entry Units (ADEUs).

1.4 Reference Documents

ARINC Characteristic 702, "Flight Management Computer System" added to list of navigation systems with which the Specification 424 data base may be used.

2.1 Data Processing Terms

Definition for "card" deleted. Definition for "subsection" modified.

ORIGINAL TEXT FOLLOWS

Card — A data storage medium in the form of a thin flexible board on which information is encoded magnetically, by a pattern of punched holes, by a pattern of visible marks or by other means.

Subsection — A collection of records of functionally similar RNAV data items. The records for high level enroute airways form a subsection of the data base.

3.2.3 Navaid Section (D) NDB Subsection (DB)

New section introduced by this Supplement to introduce new master file subsection.

3.2.4 Enroute Section (E)

Section 3.2.3 of Specification 424 amended to include holding patterns and a new approach to airways classification. original Section 3.2.4 deleted.

ORIGINAL TEXT FOLLOWS

3.2.3 Enroute Section (E)

The Enroute section of the master airline user file should contain:

(i) all waypoints for EB, EH and EL airways,

(ii) all government-designated airways.

3.2.4 Holding Pattern Section (H)

The Holding Pattern section of the master airline user file should contain all holding patterns designated by governments for charting.

3.2.8 Airport Section (P) – Standard Instrument Departures (SIDs) Subsection (PD)

Second and third sentences of paragraph deleted.

ORIGINAL TEXT FOLLOWS

(Second and third sentences of 3.2.8)

Evaluation and conventional SIDs are not included in the master airline user file. They are, however, available in data banks in standard record format for those airlines that desire them.

3.2.9 Arrival Section (P) – Standard Terminal Arrival Routes (STARs) Subsection (PE)

Second and third sentences of paragraph deleted.

ORIGINAL TEXT FOLLOWS

Second and third sentences of 3.2.9)

Evaluation and conventional STARs are not included in the master airline user file. They are, however, available in data banks in standard record format for those airlines that desire them.

3.2.10 Airport Section (P) – Approach Routes Subsection (PF)

Section revised. Referenced figure (3-2) deleted.

ORIGINAL TEXT FOLLOWS

The Approach Route subsection of the master airline user file should contain all published RNAV approach routes to the airports referenced in Section 3.2.5 of this document. As illustrated in Figure 3-2, an approach route starts at the intermediate fix (IF) waypoint and ends at the missed approach holding waypoint. The published waypoints between these two waypoints are included, except for step-down fixes. Transitions from the enroute structure that proceed to the intermediate fix are included. Transitions from the enroute structure that proceed to the final approach fix (FAF) and require a course reversal are not included.

3.2.11 Airport Section (P) – Runways Subsection (PG)

Section revised to call for runway records to be included in the master airline user file.

ORIGINAL TEXT FOLLOWS

Runway records are not included in the master airline user file since they are not required for basic RNAV system operation. If needed for other operations, runway information is available in standard file and record format, and may be ordered by those persons needing it.

3.2.13 Company Route Section (R)

Section amended to delete statement that company route information is not included in the master airline user file.

ORIGINAL TEXT FOLLOWS

Company route information is available only as tailored records, and thus will not be included in the master airline user file. Formats for these tailored records are not specified in this document.

4.1 General

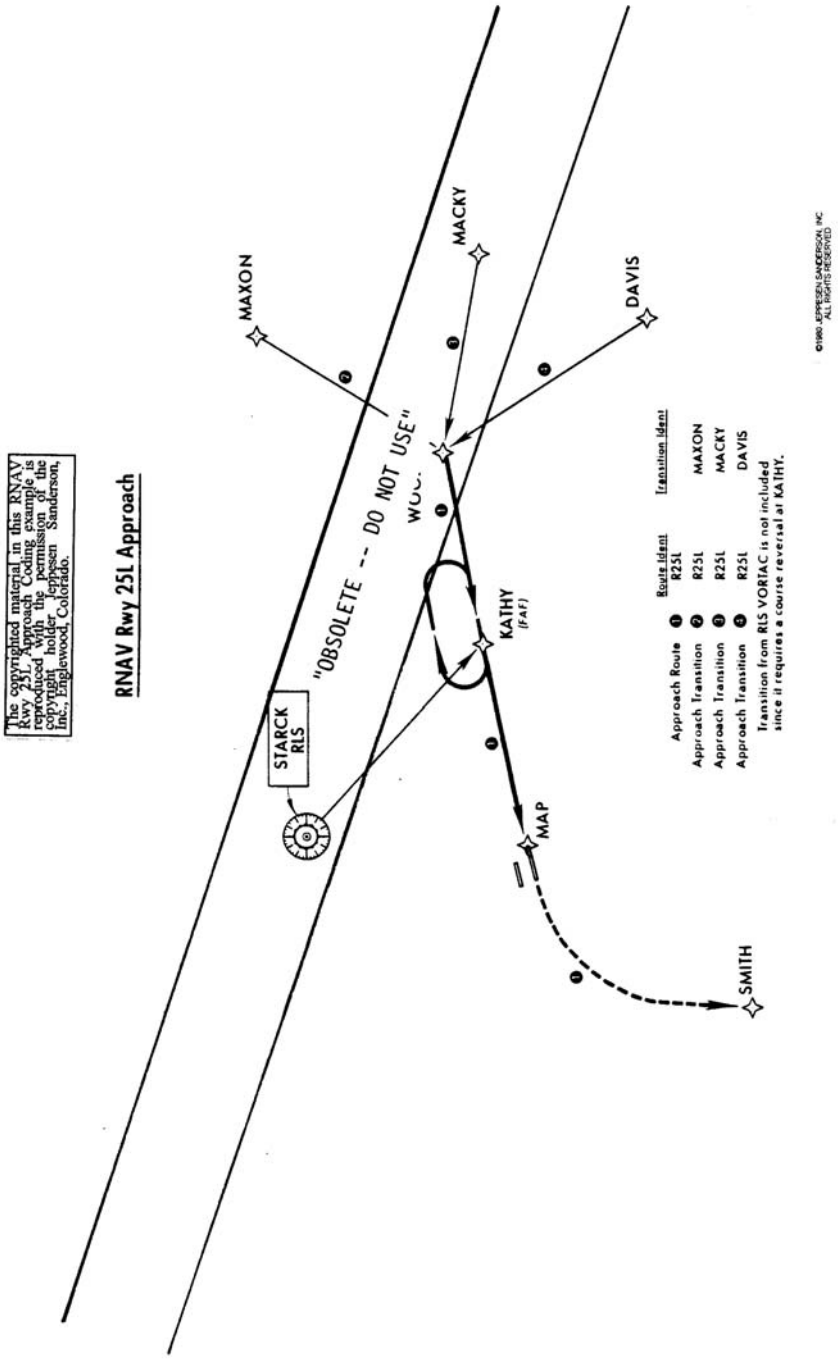
Section paragraph amended to indicate that the standard record length is increased from 108 to 132 columns.

ARINC STAFF NOTE: This record length increase affects every record layout description in Chapter 4. Also, changes in Chapter 5 introduced by this Supplement, yet to be described, further affect many of them. We are therefore presenting the whole of the original Chapter 4 as “ORIGINAL TEXT” for reference, marked “Obsolete – do not use.”

4.12 Company Route Records

New section added by this Supplement.

3.0 RNAV DATA – ORGANIZATION AND CONTENT (cont'd)



The copyrighted material in this RNAV Rwy 25L Approach Coding example is reproduced with the permission of the copyright holder Jeppesen Sanderson, Inc., Englewood, Colorado.

Not for navigational or other operational use. For example only. Please consult current navigation charts.

Figure 3-2

4.0 RNAV DATA – RECORD LAYOUT

4.1 General

Each record is made up of combinations of the fields described in Chapter 5 of this document. This chapter sets forth the standard layout of each type of record found in the data base. These layouts are also presented diagrammatically in Figure 4-1.

Each record contains 108 character positions or columns. Not all of these are used in every record. Some are left blank to permit like information to appear in the same columns of different records and others are reserved for the possible future expansion of the record's content. In the tables that follow, the former are identified by the term "Blank (Spacing)" under the "Field" heading. The latter are identified by the term "Reserved", followed by the function for which the reservation is made (where it can specifically be stated).

The tables show the record columns occupied by each field. For convenience, the number of characters in each field is shown in brackets following the field name. Also, the paragraph numbers in Chapter 5 of this document wherein individual fields are defined are referenced. Each table appears under a paragraph heading that is followed by the data base Section and Sub-section codes employed in the record described.

4.2 Enroute Airways Records (EB, EH or EL)

The Enroute Airways file will contain the sequential listing of Both Level (EB), High Level (EH) and Low Level (EL) airways by geographic areas. ("Both Level" airways are airways not designated as either high level or low level). This file will also contain enroute off-airway waypoints by geographic area and altitude, without sequence numbers or airway idents.

4.2.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Sub-Section Code (1)	5.5
7 thru 10	Blank (Spacing) (4)	
11	Route Type (1)	5.7
12 thru 16	Route Identifier (5)	5.8
17	Reserved (6th char. Rte. Ident) (1)	Note 1
18 thru 23	Blank (Spacing) (6)	
24 thru 27	Sequence Number (4)	5.12
28 thru 32	Waypoint Identifier (5)	5.13
33	Reserved (6th char. Wpt. Ident) (1)	Note 1
34 thru 35	ICAO Code (2)	5.14
36	File Code (1)	5.15
37	Continuation Record No. (1)	5.16
38 thru 41	Waypoint Description Code (4)	5.17

"OBSOLETE -- DO NOT USE"

4.2.1 Primary Records (cont'd)

Column	Field Name (Length)	Reference
42	Boundary Code (1)	5.18
43 thru 45	Forward Change-Over Point (To) (3)	5.19
46 thru 48	Backward Change-Over Point (From) (3)	5.20
49 thru 52	Recommended VHF NAVAID (4)	5.21
53 thru 56	First Reserved VHF NAVAID (4)	5.22
57 thru 60	Second Reserved VHF NAVAID (4)	5.23
61 thru 64	Theta (4)	5.24
65 thru 68	Rho (4)	5.25
69 thru 72	Outbound Magnetic Course (4)	5.26
73 thru 76	Route Distance From (4)	5.27
77 thru 80	Inbound Magnetic Course (4)	5.28
81 thru 99	Reserved (Expansion) (19)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

Note 1: The standard lengths for the Route Identifier and the Waypoint Identifier fields are each five characters. Some users envisage the need for six-character fields in certain instances. These reserved columns will permit this usage.

4.2.2 Continuation Records

Column	Field Name (Length)	Reference
1 thru 36	Fields as on Primary Records	
37	Continuation Record No. (1)	5.16
38 thru 80	Notes (43)	5.61
81 thru 99	Reserved (Expansion) (19)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

4.3 SID/STAR Records (PD or PE)

SID/STAR records comprise two files, one (PD) for SID's and the other (PE) for STAR's. The SID file contains the sequential listing of published Standard Instrument Departures. The STAR file contains the sequential listing of published Standard Terminal Arrival Routes.

4.3.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport ICAO Identifier (14)	5.6
11	Sub-Section Code (1)	5.5

4.0 RNAV DATA – RECORD LAYOUT (cont'd)

4.3.1 Primary Records (cont'd)

Column	Field Name (Length)	Reference
12 thru 16	SID/STAR Identifier (5)	5.9
17	Reserved (6th char. of SID/STAR Ident.) (1)	Note 1
18	Route Type (1)	5.7
19 thru 23	Transition Identifier (5)	5.11
24	Reserved (6th char. of Trans. Ident. (1)	Note 1
25 thru 27	Sequence Number (3)	5.12
28 thru 32	Waypoint Identifer (5)	5.13
33	Reserved (6th char of Wpt. Ident) (1)	Note 1
34 thru 35	ICAO Code (2)	5.14
36	File Code (1)	5.15
37	Continuation Record No. (1)	5.16
38 thru 41	Waypoint Description Code (4)	5.17
42	Blank (Spacing) (1)	
43 thru 45	Forward Change-Over Pt. (To) (3)	5.19
46 thru 47	Reserved (Path Termination) (2)	Note 2
48	Blank (Spacing) (1)	
49 thru 52	Recommended VHF NAVAID (4)	5.21
53 thru 56	First Reserved VHF NAV-AID (4)	5.22
57 thru 60	Second Reserved VHF NAV-AID (4)	5.23
61 thru 64	Theta (4)	5.24
65 thru 68	Rho (4)	5.25
69 thru 72	Outbound Magnetic Course (4)	5.26
73 thru 76	Route Distance From (4)	5.27
77 thru 80	Inbound Magnetic Course (4)	5.28
81	Altitude Descript (1)	5.29
82 thru 86	Altitude (5)	5.30
87 thru 91	Altitude (5)	5.30/Note 3
92 thru 99	Reserved (Expansion) (8)	
100 thru 104	File Record No. (5)	5.31
104 thru 108	Cycle Date (4)	5.32

Note 1: The standard lengths of the SID/STAR Identifier, the Transition Identifier and the Waypoint Identifier fields are five characters each. Some users envisage the need for six-character fields in certain cases. These reserved columns will permit this usage.

Note 2: This two-character field is reserved for the possible inclusion in the record of a Path Termination code. This code could be associated with the use of the airborne RNAV system to fly terminal ATC procedures not formulated in RNAV terms.

Note 3: The second altitude field (columns 87 thru 91) will be filled only when the Altitude Descript field contains the entry "B". Section 5.29 of this document refers.

"OBSOLETE -- DO NOT USE"

4.3.2 Continuation Records

Column	Field Name (Length)	Reference
1 thru 36	Fields as on Primary Records	
37	Continuation Record No. (1)	5.16
38 thru 80	Notes (43)	5.61
81 thru 99	Reserved (Expansion) (19)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

4.4 Approach Route Records (PF)

The Approach Routes file contains the sequential listing of Standard Instrument Approaches.

4.4.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport ICAO Identifier (4)	5.6
11	Sub-Section Code (1)	5.5
12 thru 16	Approach Identifier (5)	5.10
17	Reserved (6th char. of App. Ident) (1)	Note 1
18	Route Type (1)	5.7
19 thru 23	Transition Identifier (5)	5.11
24	Reserved (6th char. of Trans. Ident) (1)	Note 1
25 thru 27	Sequence No. (3)	5.12
28 thru 32	Waypoint Identifier (5)	5.13
33	Reserved (6th char. of Wpt. Ident) (1)	Note 1
34 thru 35	ICAO Code (2)	5.14
36	File Code (1)	5.15
37	Continuation Record No. (1)	5.16
38 thru 41	Waypoint Description Code (4)	5.17
42	Blank (Spacing) (1)	
43 thru 45	Forward Change-Over Pt. (To) (3)	5.19
46 thru 47	Reserved (Path Termination) (2)	Note 2
48	Blank (Spacing) (1)	
49 thru 52	Recommended VHF NAVAID (4)	5.21
53 thru 56	First Reserved VHF NAVAID (4)	5.22
57 thru 60	Second Reserved VHF NAVAID (4)	5.23
61 thru 64	Theta (4)	5.24
65 thru 68	Rho (4)	5.25
69 thru 72	Outbound Magnetic Course (4)	5.26
73 thru 76	Route Distance From (4)	5.27
77 thru 80	Inbound Magnetic Course (4)	5.28
81	Altitude Descript (1)	5.29
82 thru 86	Altitude (5)	5.30
87 thru 99	Reserved (Expansion) (13)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

4.0 RNAV DATA – RECORD LAYOUT (cont'd)

4.4.1 Primary Records (cont'd)

Note 1. The standard length of the Approach Identifier, the Transition Identifier and the Waypoint Identifier fields are five characters each. Some users envisage the need for six characters in certain cases. These reserved columns will permit this usage.

Note 2: This two-character field is reserved for the possible inclusion in the record of a Path Termination Code. This code could be associated with the use of the airborne RNAV system to fly terminal ATC procedures not formulated in RNAV terms.

4.4.2 Continuation Records

Column	Field Name (Length)	Reference
1 thru 36	Fields as on Primary Records	
37	Continuation Record No (1)	5.16
38 thru 80	Notes (43)	5.61
81 thru 99	Reserved (Expansion)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

4.5 VHF NAVAID Records (D)

The VHF NAVAID file contains details of all VOR, VOR/DME, VORTAC and TACAN stations in the geographic area of interest. For non-frequency paired VOR and TACAN stations having the same identifier the TACAN is stored.

4.5.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6 thru 11	Blank (Spacing) (6)	
12 thru 15	VOR Identifier (4)	5.33
16 thru 17	Blank (Spacing) (2)	
18 thru 19	ICAO Code (2)	5.14
20	Continuation Record No. (1)	5.16
21 thru 25	VOR Frequency (5)	5.34
26 thru 30	NAVAID Class (5)	5.35
31 thru 39	VOR Latitude (9)	5.36
40 thru 49	VOR Longitude (10)	5.37
50 thru 53	DME Ident (4)	5.38
54 thru 62	DME Latitude (9)	5.36
63 thru 72	DME Longitude (10)	5.37
73 thru 77	Station Declination (5)	5.66
78 thru 82	DME Elevation (5)	5.40
83	Reserved (NAVAID Priority No.) (1)	Note 1
84	Reserved (NAVAID Figure of Merit No.) (1)	Note 2
85 thru 99	Reserved (Expansion) (15)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

4.5.1 Primary Records (cont'd)

Note 1: "NAVAID Priority No.", although assigned by the FAA in the U.S. and used by them in facility flight test procedures, has no immediate applications in RNAV operations. The field is reserved, however, in case this situation changes.

Note 2: "NAVAID Figure of Merit No.", similarly, has no RNAV function. Should official approval be forthcoming, however, for the use of NAVAID's beyond ranges specified in the class field, this field could indicate the limitations of such use.

4.5.2 Continuation Records

Column	Field Name (Length)	Reference
1 thru 19	Fields as on Primary Records	
20	Continuation Record No. (1)	5.16
21 thru 40	Notes (20)	5.61
41 thru 99	Reserved (Expansion) (59)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

"OBSOLETE -- DO NOT USE"

Waypoint Records (EA or PC)

The Enroute Waypoint file contains all enroute on-airway and off-airway waypoints within a designated geographic area. The Terminal Waypoint file contains all terminal waypoints within the geographic area of each airport.

4.6.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Sub-Section Code (En-Route) (1)	5.5/Note 1
7 thru 10	Region Code (4)	5.41/ Note 2
11	Sub-Section Code (Terminal) (1)	5.5/Note 1
12 thru 16	Waypoint Identifier (5)	5.13
17	Reserved (6th char. Wpt. Ident) (1)	Note 3
18 thru 19	ICAO Code (2)	5.14
20	Continuation Record No. (1)	5.16
21 thru 24	Blank (Spacing) (4)	
25 thru 27	Waypoint Type (3)	5.42
28 thru 30	Blank (Spacing) (3)	
31 thru 39	Waypoint Latitude (9)	5.36
40 thru 49	Waypoint Longitude (10)	5.37
50 thru 74	Waypoint Name/Description (25)	5.43
75 thru 99	Reserved (Expansion) (25)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

4.0 RNAV DATA – RECORD LAYOUT (cont'd)

4.6.1 Primary Records (cont'd)

Note 1: In enroute waypoint records, the sub-section code occupies column 6 and column 11 is blank. In terminal waypoint records, the sub-section code occupies column 11 and column 6 is blank.

Note 2: In terminal waypoint records, the region code field contains the airport ICAO identification code.

Note 3: The standard length for the waypoint identifier field is 5 characters. Some users envisage the need for 6 characters in certain cases. This reserved column will permit this usage.

4.6.2 Continuation Records

Column	Field Name (Length)	Reference
1 thru 19	Fields as on Primary Records	
20	Continuation Record No. (1)	5.16
21 thru 40	Notes (20)	5.61
41 thru 99	Reserved (Expansion) (59)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

4.7 ILS (Localizer and Glide Slope) Records (PI)

This file will contain a sequential listing of all localizer/glide slope systems.

4.7.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport ICAO Identifier (4)	5.6
11	Sub-Section Code (1)	5.5
12 thru 15	Localizer Identifier (4)	5.44
16 thru 17	Blank (Spacing) (2)	
18 thru 19	ICAO Code (2)	5.14
20	Continuation Record No. (1)	5.16
21 thru 25	Localizer Frequency (5)	5.45
26 thru 30	Runway Identifier (5)	5.46
31 thru 39	Localizer Latitude (9)	5.36
40 thru 49	Localizer Longitude (10)	5.37
50 thru 53	Localizer Bearing (4)	5.47
54 thru 62	Glide Slope Latitude (9)	5.36
63 thru 72	Glide Slope Longitude (10)	5.37
73 thru 76	Localizer Position (4)	5.48
77	Localizer Position Reference (1)	5.49
78 thru 81	Glide Slope Position (4)	5.50
82 thru 85	Localizer Width (4)	5.51

"OBSOLETE -- DO NOT USE"

4.7.1 Primary Records (cont'd)

Column	Field Name (Length)	Reference
86 thru 88	Glide Slope Angle (3)	5.52
89 thru 93	Station Declination (5)	5.66
94 thru 95	Glide Slope Height at Landing Threshold (2)	5.67
96 thru 99	Reserved (Expansion) (4)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

4.7.2 Continuation Records

Column	Field Name (Length)	Reference
1 thru 19	Fields as on Primary Records	
20	Continuation Record No. (1)	5.16
21 thru 99	Notes (79)	5.61
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

Any future expansion of the file needed to accommodate landing guidance aids other than ILS localizers and glide slopes (e.g. marker beacons) will be handled by the continuation record capability specified above.

4.8 Airport Records (PA)

This file contains airport information.

4.8.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport ICAO Identifier (4)	5.6
11	Sub-Section Code (1)	5.5
12 thru 17	Blank (Spacing) (6)	
18 thru 19	ICAO Code (2)	5.14
20	Continuation Record No. (1)	5.16
21 thru 25	Transition Altitude (5)	5.53
26 thru 27	Airport Class (2)	5.54
28 thru 30	Blank (Spacing) (3)	
31 thru 39	Airport Reference Pt. Latitude (9)	5.36
40 thru 49	Airport Reference Pt. Longitude (10)	5.37
50 thru 74	Notes (25)	5.60
75 thru 79	Magnetic Variation (5)	5.39
80 thru 84	Airport Elevation (5)	5.55
85 thru 99	Reserved (Expansion) (15)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

4.0 RNAV DATA – RECORD LAYOUT (cont'd)

4.8.2 Continuation Records

Column	Field Name (Length)	Reference
1 thru 19	Fields as on Primary Records	
20	Continuation Record No. (1)	5.16
21 thru 40	Notes (20)	5.61
41 thru 99	Reserved (Expansion) (59)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

4.9 Gate Records (PB)

This file contains passenger gate information.

4.9.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport ICAO Identifier (4)	5.6
11	Sub-Section Code (1)	5.5
12 thru 16	Gate Identifier (5)	5.56
17	Blank (Spacing) (1)	
18 thru 19	ICAO Code (2)	5.14
20	Continuation Record No. (1)	5.16
21 thru 30	Blank (Spacing) (10)	
31 thru 39	Gate Latitude (9)	5.36
40 thru 49	Gate Longitude (10)	5.37
50 thru 74	Notes (25)	5.60
75 thru 99	Reserved (Expansion) (25)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

4.9.2 Continuation Records

Column	Field Name (Length)	Reference
1 thru 19	Fields as on Primary Records	
20	Continuation Record No. (1)	5.16
21 thru 40	Notes (20)	5.61
41 thru 99	Reserved (Expansion) (59)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

4.10 Runway Records (PG)

This file contains runway information.

4.10.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	

"OBSOLETE -- DO NOT USE"

4.10.1 Primary Records (cont'd)

Column	Field Name (Length)	Reference
7 thru 10	Airport ICAO Identifier (4)	5.6
11	Sub-Section Code (1)	5.5
12 thru 16	Runway Identifier (5)	5.46
17	Blank (Spacing) (1)	
18 thru 19	ICAO Code (2)	5.14
20	Continuation Record No. (1)	5.16
21 thru 25	Runway Length (5)	5.57
26 thru 29	Runway Magnetic Bearing (4)	5.58
30	Blank (Spacing) (1)	
31 thru 39	Runway Latitude (9)	5.36
40 thru 49	Runway Longitude (10)	5.37
50 thru 51	Reserved (ECS* use) (2)	
52 thru 57	Reserved (ECS* use) (6)	
58	Blank (Spacing) (1)	
59 thru 64	Reserved (ECS* use) (6)	
65 thru 86	Runway Description (22)	5.59
87 thru 91	Landing Threshold Elevation (5)	5.68
92 thru 99	Reserved (Expansion) (8)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

* ECS = Electronic Chart System

4.10.2 Continuation Records

Column	Field Name (Length)	Reference
1 thru 19	Fields as on Primary Records	
20	Continuation Record No. (1)	5.16
21 thru 40	Notes (20)	5.61
41 thru 99	Reserved (Expansion) (59)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

4.11 Holding Pattern Records (H)

The holding pattern file contains the holding patterns recommended by the officical government source for inclusion on aeronautical navigation charts.

4.11.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Region Code (4)	5.41
11 thru 27	Blank (Spacing) (17)	
28 thru 32	Waypoint Identifier (5)	5.13
33	Reserved (6th character of Wpt. Ident) (1)	Note 1
34 thru 35	ICAO Code (2)	5.14
36	File Code (1)	5.15

4.0 RNAV DATA – RECORD LAYOUT (cont'd)

4.11.1 Primary Records (cont'd)

Column	Field Name (Length)	Reference
37	Continuation Record No. (1)	5.16
38 thru 41	Inbound Holding Course (4)	5.62
42	Turn (1)	5.63
43 thru 45	Leg Length (3)	5.64
46 thru 47	Leg Time (2)	5.65
48 thru 52	Altitude (5)	5.30
53 thru 76	Notes (24)	5.60
77 thru 99	Reserved (Expansion) (23)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

Note 1: The standard length of the Waypoint Identifier field is five characters. Some users envisage the need for a six-character field in certain cases. This reserved column will permit this usage.

4.11.2 Continuation Records

Column	Field Name (Length)	Refere
1 thru 36	Fields as on Primary Records	
37	Continuation Record No. (1)	5
38 thru 57	Notes (20)	
58 thru 99	Reserved (Expansion) (42)	
100 thru 104	File Record No. (5)	5.31
105 thru 108	Cycle Date (4)	5.32

"OBSOLETE -- DO NOT USE"

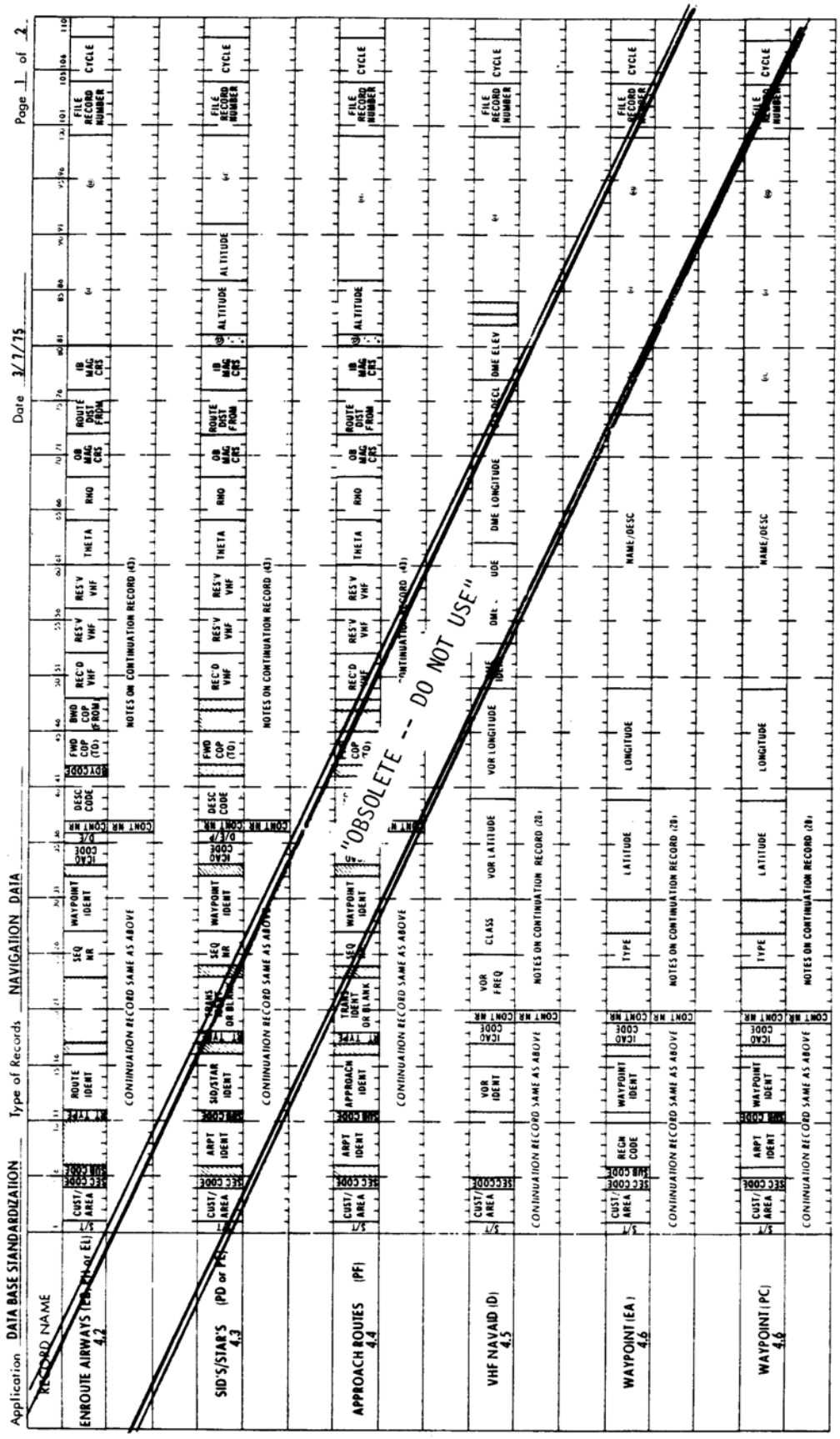

PROPORTIONAL RECORD LAYOUT FORM
Application DATA BASE STANDARDIZATION
Type of Records NAVIGATION DATA
Date 3/1/75
Page 1 of 2
RECORD NAME
ENROUTE AIRWAYS (EA or EU) 4.2
SID'S/STAR'S (PD or PE) 4.3
APPROACH ROUTES (PF) 4.4
VHF NAVAID (D) 4.5
WAYPOINT (EA) 4.6
WAYPOINT (PC) 4.6
"OBSOLETE -- DO NOT USE"

Figure 4-1

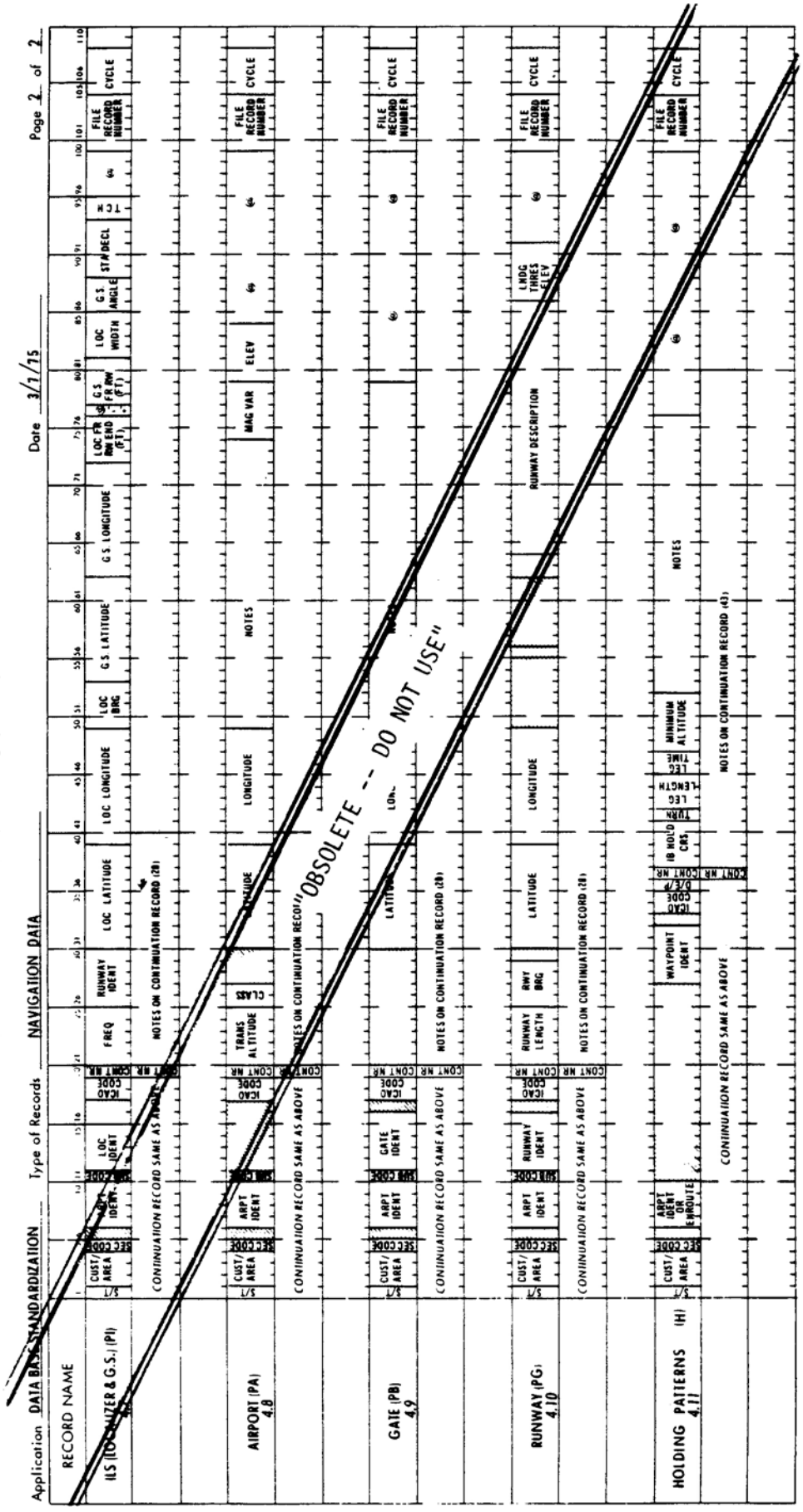


Figure 4-1 (cont'd)

AERONAUTICAL RADIO, INC.
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SUPPLEMENT 2

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: February 5, 1982

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: December 9, 1981

A. PURPOSE OF THIS SUPPLEMENT

This Supplement corrects typographical and editorial errors in Specification 424-1 and introduces a number of clarifying amendments.

B. ORGANIZATION OF THIS SUPPLEMENT

The first part of this document, printed on goldenrod paper, contains descriptions of the changes introduced into the Specification by this Supplement, and where appropriate, extracts from the original text for comparison purposes. The second part consists of replacement white pages for the Specification modified to reflect these changes. The modified and added material on each replacement page is identified with c-2 symbols in the margins.

Existing copies of Specification 424-1 may be updated by inserting the replacement white pages where necessary and destroying the pages they displace. The goldenrod pages should be inserted inside the rear cover of the Specification, following Supplement 1.

C. CHANGES TO ARINC SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This Section presents a complete tabulation of the changes and additions to the Specification introduced by this Supplement. Each change or addition is identified using the section number and title currently employed or the section number and title that will be employed when the Supplement is eventually incorporated. In each case there is included a brief description of the addition or change.

1.2 Data Format Standardization Philosophy

Commentary added.

2.2 Special Navigation Terms

The words "terminal structure" are substituted for the words "approach structure" in definition for Transition Essential Waypoints.

3.2.2 VHF NAVAID Section (D)

New sentence added to existing text.

3.2.5 Airport Section (P) – Airport Reference Point Subsection (PA)

Runway length and instrument approach restriction deleted.

ORIGINAL TEXT FOLLOWS

3.2.5 Airport Section (P) – Airport Reference Point Subsection (PA)

The Airport Reference Point Subsection of the master airline user file should contain reference points for all airports having at least one hard-surfaced runway of 4000 feet length or greater, for which an approved instrument approach procedure is published.

3.2.12 Airport Section (P) – ILS NAV DATA Subsection (PI)

Sentence added to existing text. Commentary added.

4.4.1 Primary Records

Columns 30 and 31 (previously blank for spacing reasons) assigned to "Waypoint Usage" coding described in Section 5.82. Figure 4-1 amended to reflect this change.

5.7 Route Type (RT TYPE)

Line added to Enroute Airways Record table defining field content "B" and "Both (High/Low)" airway type.

STAR Record table amended to add word "Transition" to "Profile Descent Enroute" entry.

5.10 Approach Route Identifier (APPROACH IDENT)

Field length reduced from 5 characters max. to 4 characters max. Example "B8R" changed to "B08R."

5.11 Transition Identifier (TRANS IDENT)

Table (ii) revised completely.

Example "TRAIL" changed to "RW08R."

ORIGINAL TEXT FOLLOWS

Record	"Route Type" Field Content	"Transition Identifier" Field Content
SID/STAR	1 or 4	SID runway transition identifier
	2 or 5	Blank/RWY/All/Other
	3 or 6	SID Enroute transition identifier
	1, 4, or 7	STAR enroute transition identifier
	2, 5, or 8	Blank/RWY/All/Other
	3, 6, or 9	STAR runway transition identifier
Approach Routes	A	Waypoint identifier of first waypoint of transition
	All other possible entries	Blank

Table (ii)
Transition Identifier Field Content

5.17 Waypoint Description Code (DESC CODE)

Table (iii) modified.

ORIGINAL TEXT FOLLOWS

Waypoint Description	Record Column Content			
	40	41	42	43
Airport W/P	A			
Essential W/P*	E			
Off-Airway W/P*	F			
Runway W/P	G			
Non-Essential W/P*	R			
Transition Essential W/P*	T			
VOR, VORDME, VORTAC	V			
End of Continuous Airway		E		
Overfly**		Y		
ATC Compulsory W/P*			C	
Gateway Fix			G	
First Leg of Missed Approach			M	
Holding Fix				H

* See Chapter 2 for definitions.

** Fix must be overflown before turning.

Note: Column 40 may be blank on some SID/STAR/APPROACH records.

Table (iii)
Waypoint Description Codes

5.22 Turn Direction Valid (TDV)

Definition/Description amended.

ORIGINAL TEXT FOLLOWS

Definition/Description: This field is used to indicate that a turn is required prior to executing the Path Term defined in a terminal procedure.

5.23 Recommended VHF NAVAID (REC'D VHF)

Note added after "Definition/Description" paragraph.

5.29 Altitude Descript (ALT DESC)

Field contents "G" and "I" and their associated waypoint crossing definitions added to Source/Content table. Second sentence added to first note.

5.32 Cycle Data (CYCLE)

Two sentences added to "Definition/Description" paragraph.

5.35 NAVAID Class (CLASS)

Sentence added to note under table (v).

5.54 Airport Class (CLASS)

Definition/Description paragraph and Source/Content table revised.

ORIGINAL TEXT FOLLOWS

5.54 Airport Class (CLASS)

Definition/Description: The "Airport Class" field permits airports to be classified on the basis of available runway length.

Source/Content: The first character of the field is selected from the following table. The second character position is blank.

Available Runway	Field Entry
6000 feet and over	A
5000 to 5999 feet	B
4000 to 4999 feet	C

Used On: Airport records
Length: 2 characters max.
Character Type: Alpha

5.64 Leg Length (LEG LENGTH)

Definition/Description revised to define leg length (distance) as the distance between the point at which the airport rolls out on the inbound leg and the fix at which the holding pattern is defined. Figure 5-4 amended to reflect this change.

ORIGINAL TEXT FOLLOWS

Definition/Description: The "Leg Length" field specifies the diagonal distance from the holding waypoint to the end of the outbound leg of a holding pattern.

5.66 Station Declination (STN DEC)

"G" entry added to Source/Content description.

5.70 Vertical Angle (VERT ANGLE)

Definition/Description and Source/Content paragraphs modified. Figure 5-5 revised to increase range of vertical angle and to correct formula for vertical angle computation.

ORIGINAL TEXT FOLLOWS

5.70 Vertical Angle (VERT ANGLE)

Definition/Description: "Vertical Angle" will be specified on the runway waypoint of approaches. The angle should cause the aircraft to fly level, then descend.

Source/Content: Values from official government sources will be used when available. The value may be computed from the FAF to a point 50 feet above the runway threshold. The range shall be 2.40 to 3.77 degrees (3.00 degrees nominal), with decimal point suppressed. (See Figure 5-5.)

Used On: Approach Routes records, STAR records, Runway records
Length: 4 characters
Character Type: Numeric
Examples: -250, -300, 360

5.70 Vertical Angle (VERT ANGLE) (cont'd)

Note: Vertical angle will be included in runway records only when runway has published straight-in landing minimums.

5.82 Waypoint Usage

New section added by this Supplement.

Attachment 2 – Waypoint Identifiers

Back course marker prefix in Section F1 changed from "SM" to "BM."

Attachment 3 – Navigation Chart/File Date Relationship

New version of computer print-out of the master user tape file data equivalent to the chart examples in this attachment substituted for that existing.

ARINC STAFF NOTE: Since this material is illustrative only, we are not reproducing the original version for reference in this Supplement.

Attachment 5 – Path and Terminator

New version of this attachment introduced by this Supplement. The pages of the original version of Attachment 5 follow, marked "Obsolete – Do Not Use."

ATTACHMENT 5 *

PATH AND TERMINATOR

Path Terminators are assigned to all SID/STAR/Approach records in accordance with the rules set forth in this Attachment.

GENERAL RULES

A. The following leg types are NAVAID oriented and the defining parameters will be found in the RECD VHF, RHO, THETA, OB MAG CRS, and WAYPOINT IDENT fields:

AF	CI	FC	FM	VD
CF	FA	FD	PI	VR

B. The first and last legs of a SID/STAR/Approach will be selected from the following leg types:

Procedure	Beginning Leg	Ending Leg
SID	CF, DF, FA, FC, [illegible]F VA, VD, VI, VR [illegible]r VM followed by C[illegible] [illegible]F	AF, CF, DF, HA, IF
STAR	AF, FC, FD	AF, CF, DF, FM, TF, VM
SID Route Transition	AF, FA, FD, IF, TF	AF, CF, DF, TF
STAR Route Transition	AF, FC, FD, IF	AF, CF, DF, HA, HM, TF
SID Runway Transition	AF, FA, FC, FD, IF	AF, CF, HF, HM, TF
STAR Runway Transition	AF, FA, FD, HF, IF, PI	AF, CF, HF, HM, PI, TF, VI (LOC only)
Approach	IF	CF
Missed Approach	CF, FA, FC, FD, FM, VA, VD , VI, VM, VR	AF, CF, DF, FM, HM, TF, VA, VM

NOTE: Approach transitions and Profile Descent runway transitions will be coded with the same set of beginning and ending legs as a STAR Runway Transition.

C. Table 1 defines permitted leg sequences. A shaded square indicates that the current leg/next leg sequence is not allowed.

D. Table 2 illustrates the leg types available for coding.

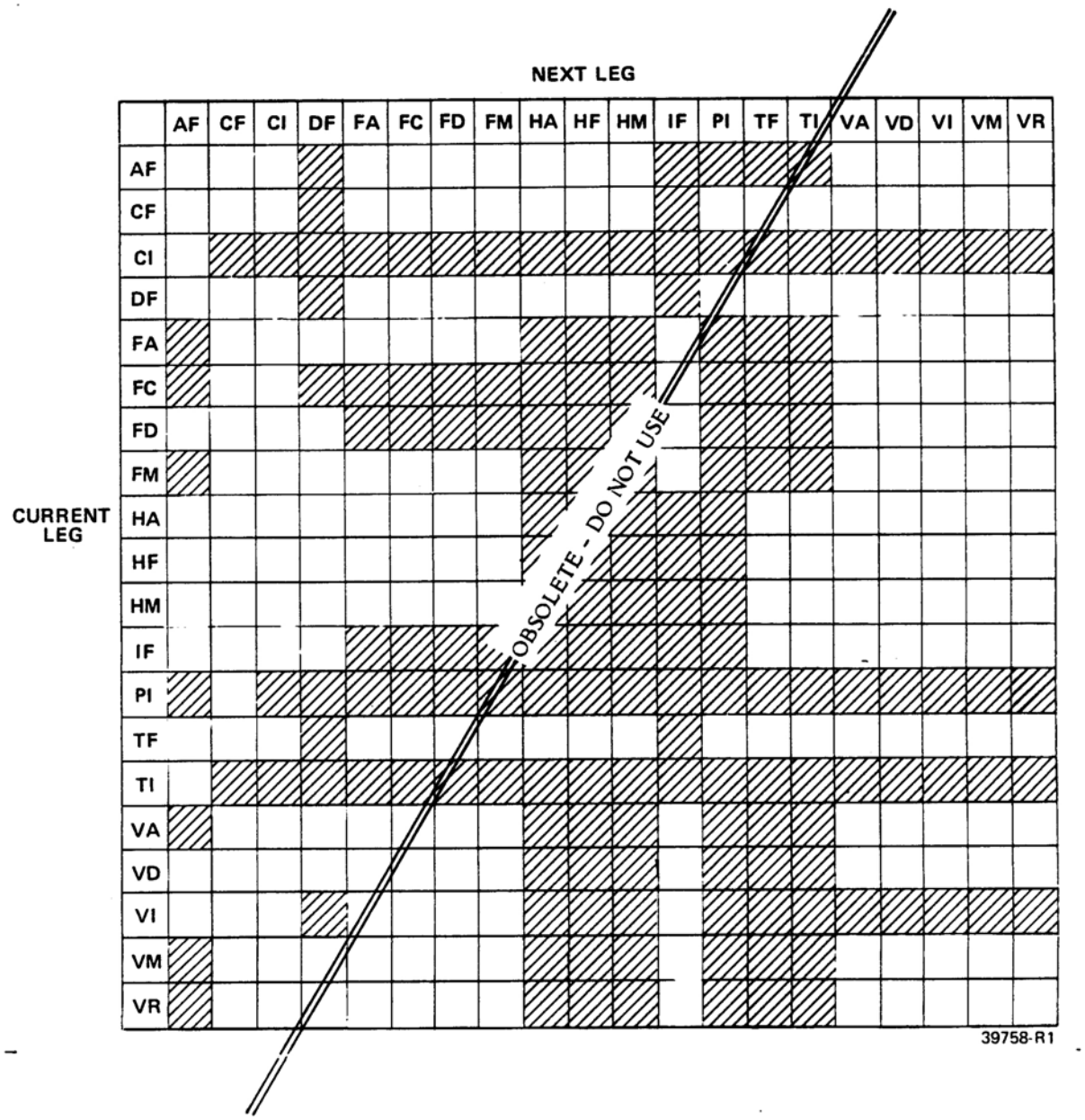
OBSOLETE - DO NOT USE

* THIS ATTACHMENT WAS INTRODUCED INTO SPECIFICATION 424 BY SUPPLEMENT NO. 1.

ATTACHMENT 5 (cont'd)

PATH AND TERMINATOR

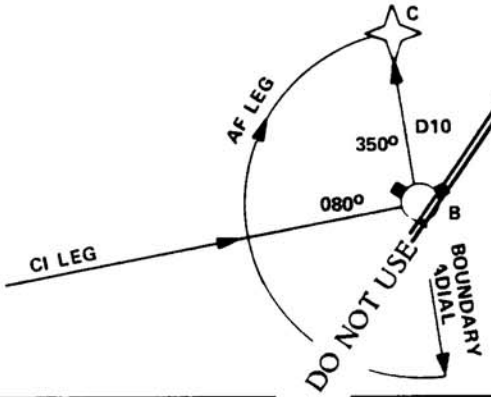
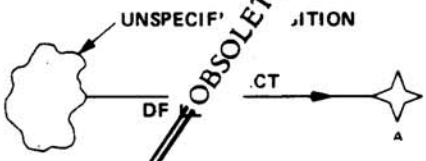
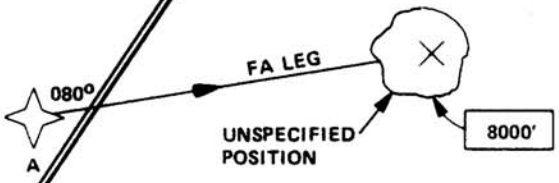
GENERAL RULES



Leg Sequence
Table 1

ATTACHMENT 5 (cont'd)

PATH AND TERMINATOR
GENERAL RULES

Leg(s)	Example	Description
CF		Course to a fix
CI-AF		Course to a fix terminating at the intercept (CI leg) of the next leg, followed by a DME arc to a fix (AF leg). The DME arc is defined by a boundary radial, a turn direction, and a fix on the arc. Waypoint is not required at intersection of CI and AF leg.
DF		Computed track direct to a fix
FA		Course from a fix to an altitude
FC		Course from a fix to a distance

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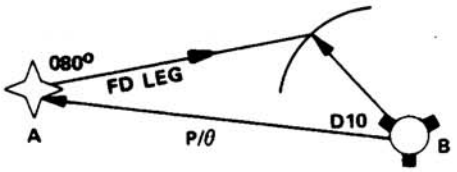
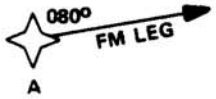
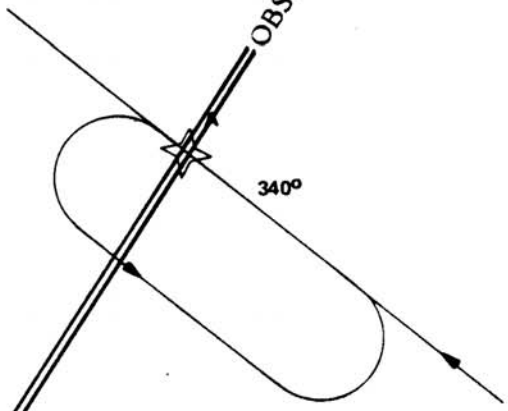
OBSOLETE - DO NOT USE

Leg Type Illustrations
Table 2

ATTACHMENT 5 (cont'd)

PATH AND TERMINATOR

GENERAL RULES

Leg(s)	Example	Description
FD		Course from a fix to a DME distance
FM		Course from a fix to a manual termination
HF HA HM		Holding pattern terminating: automatically at the fix after one full circuit (HF) automatically at a fix after reaching an altitude (HA) manually (HM)
IF		Initial fix

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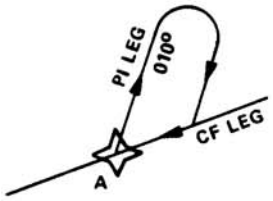
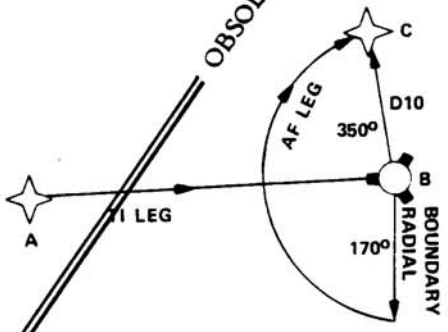
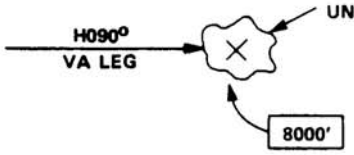
OBSOLETE - DO NOT USE

Leg Type Illustrations
Table 2 (cont)

ATTACHMENT 5 (cont'd)

PATH AND TERMINATOR
GENERAL RULES

Leg(s)	Example	Description
PI-CF		Procedure turn (PI) followed by a course to a fix (CF)
TF		Track between two fixes (great circle)
TI-AF		Track to a next leg (TI) followed by a constant DME arc to a fix (AF). Waypoint is not required at intersection of TI and AF leg.
VA		Heading to an altitude (position unspecified)

OBSOLETE - DO NOT USE

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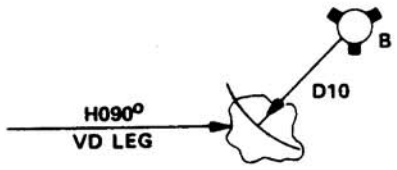
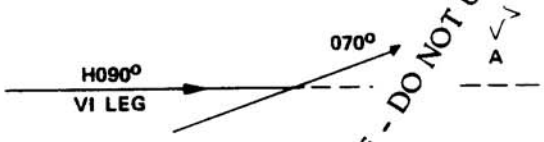
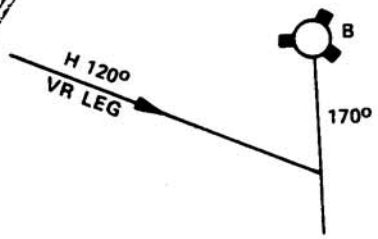
Leg Type Illustrations
Table 2 (cont)

ATTACHMENT 5 (cont'd)

PATH AND TERMINATOR
GENERAL RULES

Leg(s)	Example	Description
VD		Heading to a DME distance
VI-CF		Heading to a next leg (VI) followed by a course to a fix (CF) Intercept point undefined.
VM		Heading to a manual termination
VR		Heading to a radial termination Intercept point undefined.

OBSOLETE - DO NOT USE

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Leg Type Illustrations
Table 2 (cont)

ATTACHMENT 5 (cont'd)

E. Table 3 lists the required and optional parameters used to define each leg type. An "0" in Table 3 indicates that the parameter may be omitted or included as required for an individual case. All other entries in the table indicate that the field is required for leg definition.

F. All procedures will be coded to provide guidance specified by source documents.

G. DF legs will be used to start from unknown positions such as altitude or DME terminations.

H. Vertical angles are referenced to the next fix.

I. Use of a "C" in the Altitude Description field is restricted to SID records with the following path and terminator codes:

CF	FD	VR
FC	VD	TF

The conditional termination altitude will be coded in columns 90 thru 94 of the SID record. If a "+", "-", or blank is entered in the Altitude Description field, entry of a second altitude will imply a conditional altitude termination. Conditional altitude termination is not allowed when a "B" is entered in the Altitude Description field.

J. Altitude terminations will not be used in descending procedures.

K. Lost communication procedures will be coded in place of vector legs if the procedure defines a complete route to the end of a SID or STAR.

L. Localizer facilities will not be used to define SID or STAR waypoints.

M. RHO and THETA for terminating fix on AF legs will be provided.

N. The turn and turn direction valid fields will be used to force a particular turn direction whenever the track/heading change exceeds 135°.

OBSOLETE - DO NOT USE

SID CODING RULES

A. If on take off, there is a turn greater than 15° without an altitude specified before the turn, a course from or a heading to an altitude (FA or VA) leg will be coded before the turn using runway heading for the VA leg terminating at 400 ft above the airport elevation. This altitude may vary with local controlling agency requirements.

B. If a SID ends in vectors to a fix, a VM leg followed by a CF or DF leg to that fix will be used. The heading for the VM leg (if any) will be based on the source document.

ATTACHMENT 5 (cont'd)

PATH AND TERMINATOR GENERAL RULES

PT	W/P IDENT	OVER FLY	PASSED APPR	HOLDING	TURN DR	TURN DR VLD	RECD VHF	THETA	RHO	OB MAG CRS	DIST/TIME	ALT 1	ALT 2	SPEED LIMIT	VERT ANGLE	COMMENTS
AF	X	O			R		X	X	X	R		O	O	O		OB MAG CRS IS BOUNDARY RADIAL, THETA IS FIX RADIAL
CF	X	O	O		O	O	X	X	X	C	D	O	O	O	O	OB MAG CRS IS CRS TO SPECIFIED FIX
CI	X	O			O	O	X	X	X	C	D	O		O		OPTIONAL ALT WILL BE "AT" IF SUPPLIED
DF	X	O			O	O	O	O	O			O	O	O		
FA	X		O		O	O	X	X	X	C	D	+	O	O	O	ALT TERMINATION WILL BE "AT OR ABOVE"
FC	X	O	O		O	O	X	X	X	C	D	O	O	O		
FD	X	O	O		O	O	X	X	X	C	D	O	O	O		
FM	X		O		O	O	X	X	X	C		O		O		
HA	X			X	O		O	O	O	O	X	+		O		ALT TERMINATION WILL BE "AT OR ABOVE"
HF	X			X	O		O	O	O	C	X	O		O		
HM	X			X	O		O	O	O	O	X	O		O		
IF	X						O	O	O			O	O			
PI	X				X		X	X	X	C	D	X		O		DIST IS FROM SPECIFIED FIX TO OUTER BOUNDARY OF TURN
TF	X	O			O	O	O	O	O	C	D	O	O	O	O	
TI	X	O			O	O				O		O		O		OPTIONAL ALT WILL BE "AT" IF SUPPLIED
VA			O		O	O				H		+		O		ALT TERMINATION WILL BE "AT OR ABOVE"
VD			O		O	O	X			H	D	O	O	O		
VI			O		O	O				H		O	O	O		
VM			O		O	O				H		O		O		
VR			O		O	O	X	X		H		O	O	O		

OBSOLETE - DO NOT USE

LEGEND:
X = REQUIRED
O = OPTIONAL
R = BOUNDARY RADIAL
C = COURSE
H = HEADING
D = DISTANCE
+ = "AT OR ABOVE" ONLY
SHADED = NOT APPLICABLE

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Leg Data Fields
Table 3

ATTACHMENT 5 (cont'd)

STAR CODING RULES

A. If a STAR ends in vectors to final approach (VM leg), the airport reference point will be coded in the WAYPOINT IDENT field.

B. If a STAR or Profile Descent does not begin at a fix, the closest named fix along the STAR/Profile Descent track will be assigned as the initial fix (IF) for the procedure.

C. If no crossing altitudes are specified on intermediate fixes of a STAR/Profile Descent, a vertical angle will be assigned to the last fix. This angle will be computed based on the altitude specified at the end fixes to provide a constant descent path through all fixes. The angle provided will ensure compliance with minimum enroute altitudes for those segments without assigned altitudes.

APPROACHES AND APPROACH TRANSITIONS

A. Semi-precision approaches

1. Approaches with circle to land minimums only will not be coded.

2. When a holding pattern use course reversal or a procedure turn is part of an approach, it will be inc in a transition.

3. If a runway transition common to more than one approach, it will be coded each time with the IDE corresponding to the approach IDENT.

4. Approach step down fixes will not be coded if a single CF leg to the runway waypoint with a vertical angle will satisfy intermediate altitude requirements. Minimum descent angle is 3 degrees when not specified by source document.

5. The CF leg is the preferred leg type for approaches.

6. The recommended VHF Navaid will be the same for all approach legs (missed approach legs not included).

7. Runway waypoint altitude will be 50 ft above the threshold unless otherwise specified by the controlling agency.

8. Transitions which are wholly contained in another transition will not be coded separately.

B. Precision approaches

1. ILS procedures will consist of Final Approach Course Fix (FACF), FAF, runway fix, and missed approach. The FACF is a fix located on the localizer beam center 8 NM or less from the outer marker. The FACF is coded as an IF leg with an altitude assigned based on the source document or equal to the altitude of the procedure turn or the altitude of the last transition leg.

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ATTACHMENT 5 (cont'd)

APPROACHES AND APPROACH TRANSITIONS

2. Recommended VHF for ILS approach will be the appropriate localizer. RHO and THETA will be provided for each leg to the runway fix.

3. ILS approaches will use an initial fix (IF leg) at the FACF followed by CG legs to the runway fix.

4. The altitude for the last leg of an ILS approach will be the glideslope intercept altitude.

5. The recommended VHF for the last CF leg of an ILS transition will be the localizer.

C. Approach transitions for ILS approaches will be as follows:

1. Last leg in approach transition will

a. Course to FACF (CF leg).

b. Heading to intercept (VI

C. Procedure turn (PI le

2. If case 1.a. above (CF le en:

a. The last leg fix RP MD VHF will be localizer identifier.

b. The CF leg length will be 8 miles or less and within the reception range of the localizer. If the last transition fix is more than 8 miles from the FACF, a combination of an FC and CF leg will be used with the CF leg length less than or equal to 4 miles. (See Figure 1.)

c. The approach course intercept angle will be 30 degrees or less.

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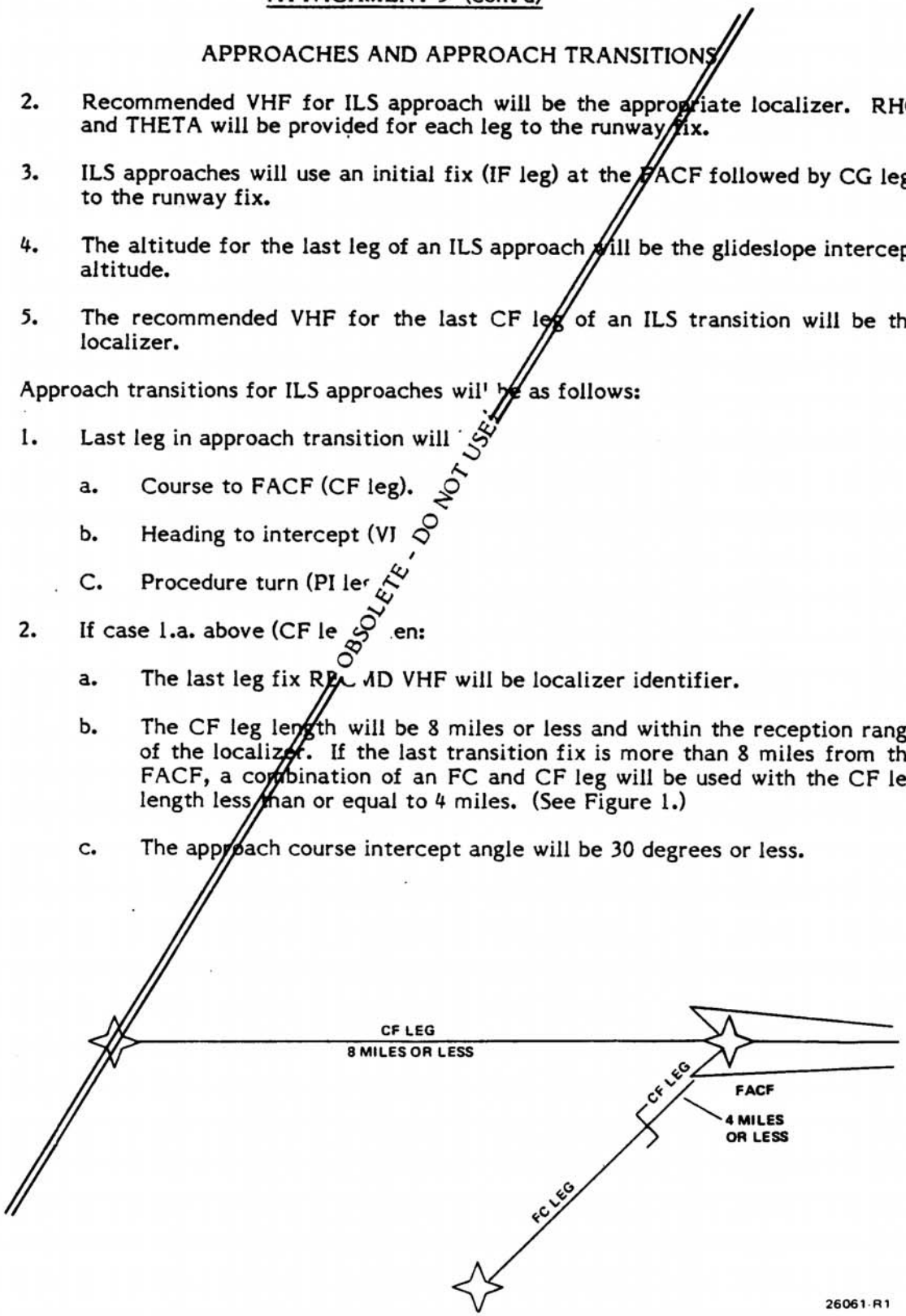

ILS Transitions
Figure 1

ATTACHMENT 5 (cont'd)

APPROACHES AND APPROACH TRANSITIONS

3. If case 1.b. above (VI leg), then:

   a. Heading will intercept ILS approach course between FACF and FAF.

   b. Intercept angle will be 30 degrees or less or per source document, and no less than 2 miles from FAF.

D. Missed Approaches

1. Missed approaches will be coded as part of each approach. The first missed approach record will contain an "M" in column 42 of the DESC CODE field.

2. The first leg of the missed approach will contain an altitude to command a climb. If a turn in excess of 15° from runway heading is required, then a VA or FA leg on the runway heading to at least above the airport elevation will be the first leg of the missed approach followed by the required turn.

3. Opposite end runway fixes will be used on missed approaches.

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AERONAUTICAL RADIO, INC.
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SUPPLEMENT 3

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: January 17, 1983

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: November 4, 1982

A. PURPOSE OF THIS SUPPLEMENT

This Supplement restructures the format of the company route record, adds guidance on coding VOR approaches in Attachment 5 and corrects previously undiscovered editorial and typographical errors. Additionally, it introduces a small number of clarifying amendments into the Specification.

B. ORGANIZATION OF THIS SUPPLEMENT

The first part of this document, printed on goldenrod paper, contains descriptions of the changes introduced into the Specification by this Supplement, and where appropriate, extracts from the original text for comparison purposes. The second part consists of replacement white pages for the Specification modified to reflect these changes. The modified and added material on each replacement page is identified with c-3 symbols in the margins.

Existing copies of Specification 424-2 may be updated by inserting the replacement white pages where necessary and destroying the pages they displace. The goldenrod pages should be inserted inside the rear cover of the Specification, following Supplement 2.

C. CHANGES TO ARINC SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This Section presents a complete tabulation of the changes and additions to the Specification introduced by this Supplement. Each change or addition is identified using the section number and title currently employed or the section number and title that will be employed when the Supplement is eventually incorporated. In each case there is included a brief description of the addition or change.

2.2 Special Navigation Terms

The term "Off-Airway Floating Waypoint" changed to "Off-Route Floating Waypoint" because such waypoints may exist in terminal areas as well as in the enroute environment.

4.2.1 Primary Records (VHF NAVAIDS)

New field "DME Bias (2)" added in columns 86 and 87. "Reserved (Expansion) (8)" field in columns 86 through 98 changed to "Reserved (Expansion) (6)" columns 88 through 93. Reference to Section 5.90 added for "DME Bias (2)" field.

4.6.1 Primary Records Enroute Airways (ER)

"Blank (Spacing) (4)" field in columns 22 through 25 changed to "Blank (Spacing) (6)" in columns 20 through 25. "Route Type (1)" moved from column 20 to column 45. "Level (B) (L) (H) (I)" moved from column 21 to column 46.

4.9.1 Primary Records

"Inbound Magnetic Course (4)" field in columns 79 through 82 changed to "Blank (Spacing) (4)." Reference to Note 2 for "Altitude field in columns 90 through 94 deleted. Note 2 deleted.

ORIGINAL TEXT FOLLOWS

Note 2: The second altitude field (columns 90 through 94) will be filled only when the Altitude Descript field contains the entry "B." Section 5.20 of this document refers.

4.10.1 Primary Records

"Vertical Angle (4)" field in columns 78 through 81 changed to "Blank (Spacing) (4)" and reference to Section 5.70 deleted.

4.12 Company Route Records (R)

This section revised completely.

ORIGINAL TEXT FOLLOWS

4.12 Company Route Records (R)

This file contains company tailored route information.

4.12.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Reserved (Spacing) (1)	
7 thru 11	From Airport/Fix (5)	5.75
12 thru 13	ICAO Code (2)	5.14
14	File Code (1)	5.15
15 thru 19	To Airport/Fix (5)	5.75
20 thru 21	ICAO Code (2)	5.14
22	File Code (1)	5.15
23 thru 27	Company Route Identifier (5)	5.76
28 thru 31	Reserved (4)	Note 1
32 thru 36	Via Identifier (5)	5.77
37 thru 39	Sequence No. (3)	5.12
40 thru 123	Company Route Description (84)	5.78
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

Note 1: Although the Company Route Identifier field is five characters in length, some users envisage the need for nine characters in certain cases. These reserved columns will permit this usage.

ARINC STAFF NOTE: The foregoing changes in Chapter 4 necessitated revisions in Figure 4-1 in Specification 424-2.

5.3 Customer/Area Code

New sentence added to Source/Content paragraph to describe use of Area Code on Company Route records.

5.11 Transition Identifier (TRANS IDENT)

Each occurrence of "type 1" in Note 1 replaced with "type 1 or 4" and each occurrence of "type 2" replaced with "type 2 or 5." Note 3 added.

5.17 Waypoint Description Code (DESC CODE)

"Uncharted Airway Intersection +," with corresponding "U" in column 41, and "Missed Approach Point ++,"

with corresponding "M" in column 43, added to table (iii). "+" and "++" notes added below table (iii).

5.24 Theta (THETA)

New final sentence added to "Definition/Description paragraph.

5.26 Outbound Magnetic Course (OB MAG CRS)

New final sentence added to "Definition/Description" paragraph.

5.27 Route Distance From, Holding Distance/Time

New material added to "Definition/Description" paragraph to describe use of "Route Distance From" in path terminator codes.

5.28 Inbound Magnetic Course (IB MAG CRS)

New final sentence added to Definition/Description" paragraph.

5.35 NAVAID Class (CLASS)

"Biased DME" added to VHF Navaid section of table (v) with corresponding "D" entry in column 31. "Longitude" changed to "longitudes" in second line of note.

5.42 Waypoint Type (TYPE)

"Unnamed Intersection" entry in table changed to "Unnamed, Charted Intersection," "Speed Limit Point" and entry "S" in column 27 deleted, "Off-Airway Intersection" entry changed to "Off-Route Intersection" and "Uncharted Airway Intersection," with "U" entry in column 27, added.

5.53 Transition Altitude (TRANS ALTITUDE)

"Definition/Description" paragraph revised completely. "18,000" deleted from examples.

ORIGINAL TEXT FOLLOWS

5.53 Transition Altitude (TRANS ALTITUDE)

Definition/Description: The "Transition Altitude" field defines the altitude at which the altimeter barometric setting is changed from 1012.25 mb (29.92 ins of mercury) to the local value for the airport identified in the record.

Source/Content: Transition altitudes are derived from official government sources.

Used On:	SID/STAR, Approach Route records
Length:	5 characters
Character Type:	Numeric
Examples:	18000, 05000, 23000

5.63 Turn (TURN)

"Source/Content" paragraph added.

5.66 Station Declination (STN DEC)

New final sentence added to "Source/Content" paragraph. "T000" and "G000" added to examples. Commentary added.

5.70 Vertical Angle (VERT ANGLE)

"Definition/Description" and "Source/Content" paragraphs revised completely. "Runway records" deleted from "Used on" list and "360" from examples. Note deleted.

ORIGINAL TEXT FOLLOWS

5.70 Vertical Angle (VERT ANGLE)

Definition/Description: "Vertical Angle" will be specified on the runway waypoint of approaches. The angle should cause the aircraft to fly level, then descend. A 0° vertical angle should cause the aircraft to descend and then fly level. On runway records, the vertical angle is used to define the glideslope angle for the ILS defined in the LOC IDENT field.

Source/Content: Values from official government sources will be used when available. On runway records thru range is –2.00° through –4.00° inclusive. On approach records the range is 0.00° and –2.00° through –4.00° except for the runway waypoint where the range is –2.00° through -4.00°. For semi-precision approaches the vertical angle may be computed from the FAF altitude constraint to 50 feet above the runway threshold using a minimum value of –3.00°.

Used On:	Approach Routes records, STAR records, Runway records
Length:	4 characters
Character Type:	Numeric
Examples:	-250, -300, 360

Note: Vertical angle will be included in runway records only when runway has published straight-in landing minimums.

ARINC STAFF NOTE: Vertical Angle limits also changed from "2.00° thru 4.00°" to "–3.00° to –3.77°" on Figure 5-5.

5.76 Company Route Ident

Length changed from 5 characters to 10 characters. Note explaining provisions for extension from 5 characters to 9 characters deleted.

5.77 Via Identifier (VIA IDENT)

Section revised completely.

ORIGINAL TEXT FOLLOWS

5.77 Via Identifier (VIA IDENT)

Definition/Description: The "Via Identifier" field, is used to further define the company route.

5.77 Via Identifier (VIA IDENT) (cont'd)

Source/Content: This field is determined by the customer.

Used On: Company Route records
Length: 5 characters
Character Type: Alpha/Numeric

5.78 SID/STAR/App/Awy

Section renamed and revised completely.

ORIGINAL TEXT FOLLOWS

5.78 Company Route Description

Definition/Description: The "Company Route Description" field is a flight plan type route description of airways and fixes from point of origin to destination. SIDs, STARs, and Runway information may also be included.

Source/Content: This field is determined by the customer, and each route may contain more than one record. (See Figure 5-6).

Used On: Company Route Records
Length: 84 characters
Character Type: Alpha/Numeric

5.83 To Fix

New section added by this Supplement.

5.84 Runway Transition

New section added by this Supplement.

5.85 Enroute Transition

New section added by this Supplement.

5.86 Cruise Altitude

New section added by this Supplement.

5.87 Terminal/Alternate Airport

New section added by this Supplement.

5.88 Alternate Distance

New section added by this Supplement.

5.89 Cost Index

New section added by this Supplement.

5.90 DME Bias

New section added by this Supplement.

Figure 5-6 - Company Route Record Example

Figure revised completely. See page 5 of this Supplement for the original (now obsolete) figure.

Attachment 2 - Waypoint Identifiers

E. Duplicate Identifiers

Words "or letter" added to end of first sentence. Example "SHAWNEE (FLA)" changed to "SHAWNEE (CAL)." Following examples added:

CPR 29 CPR 29A

CPR 29 CPR 29B

F. Terminal Waypoints

1. Airport-Related Waypoints

Prefix "CF Final Approach Course Fix" added to list.

3. Bearing/Distance Waypoints

Title changed (was "DME Arc Waypoints") and subsection "d" added.

Attachment 3 – Navigation Chart/File Data Relationship

Pages 50 through 59 re-ordered to present file data in record order.

Attachment 5 – Path and Terminator

General Rule B: "IF (route type 2 or 5 only)" added to "Ending Leg" list for SID/STAR Runway Transition procedure.

Note following VI Ending Leg descriptor for Approach Transition procedure expanded from "(LOC only)."

Table I: "+" added at CF row/DF column intersection on diagram. "+" note added below diagram and "+" note revised. (Original Words: Sequence permitted only if altitude of IF leg is different than leg termination altitude.)

General Rule G: Second and third sentences added.

General Rule I: Second paragraph added.

General Rule N: Revised completely. (Original Words: The turn and turn direction valid fields will be used to force a particular turn direction whenever the track/heading change exceeds 135°. These two fields are used together to indicate that a turn in the specified direction must be executed prior to intercepting the path defined in the record.

Table 3: Path Length designator "P" added to legend. "P" substituted for "D" in DIST/TIME column of table at CF, CI, FC and PI row intersections.

General Rule AA: New material added by this Supplement.

General Rule BB: New material added by this Supplement.

STAR Coding Rule E: Route types "2 or 5" added.

Approaches and Approach Transitions: This section extensively revised. See pages 5-7 of this Supplement for the original (now obsolete) material.

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ATTACHMENT 5 (cont)

PATH AND TERMINATOR

APPROACHES AND APPROACH TRANSITIONS

A. General

All approach altitudes between the FACF and the runway inclusive will be coded as "AT".

B. Semi-precision approaches

1. Approaches with circle-to-land minimums only will not be coded.
2. When a holding pattern used for course reversal or a procedure turn is part of an approach, it will be included in a transition.
3. If a runway transition is common to more than one approach, it will be coded each time with the IDENT corresponding to the approach IDENT.
4. Approach step-down fixes will not be coded if a single CF leg to the runway waypoint with a vertical angle will satisfy intermediate altitude requirements. Minimum descent angle is 3° when not specified by source document.
5. The CF leg is the preferred leg type for approaches.
6. The recommended VHF Navaid will be the same for all approach legs (missed approach legs not included).
7. Runway waypoint altitude will be 50 feet above the threshold unless otherwise specified by the controlling agency.
8. Transitions which are wholly contained in another transition will not be coded separately.

C. Precision approaches

1. ILS procedures will consist of Final Approach Course Fix (FACF), FAF, runway fix, and missed approach. The FACF is a fix located on the localizer beam center 8 NM or less from the outer marker or within reception range of the localizer. The FACF is coded as an IF leg with an altitude assigned, based on the source document or equal to the altitude of the procedure turn or the altitude of the last transition leg.
2. Recommended VHF for ILS approach will be the appropriate localizer. RHO and THETA will be provided for each leg to the runway fix.
3. ILS approaches will use an initial fix (IF leg) at the FACF, followed by CF legs to the runway fix.

													(SID – NO TRANS. STAR AND TRANS)		
T	P	R	KSFO	K2	P	KDFW	K4	P	01021	BCE	001	<LNDNS (RW28B), MELTS + K2/USA/... SONNY + K2. J92, MVA + K2..BCE + K2 @ FL 270. I58, SPS + K4> BOIDS 4-S	12345	7912
T	P	R	KSFO	K2	P	KDFW	K4	P	01021	BCE	002	PS (RW17B) "KIAH + K4" "	12348	7912
												(ROUTE FOR ALTERNATE AIRPORT ABOVE)		
T	P	R	KDFW	K4	P	KIAH	K4	P	02001	BILEE	001	/USA/.. DFW + K4. J87, BILEE + K4> CUGAR 2 – BILEE (ALL)"	12960	8003
												(SID AND TRANS. NO STAR)		
T	P	R	KSFO	K2	P	CYYC	CY	P	C34B	RBL	001	< SCD 3 – RBL (RW01B), RBL + K2 @ FL 350/USA/. J1, SEA + K1. J308, ONSET + K1/CAN/. HL505, YYC + CY"	12345	7912
												(NO SID (VECTOR) AND TRANS. NO STAR. ENDS AT NAVAID)		
T	P	R	KPHX	K2		MZB	K2	D	CS1	GBN	004	< MOBIE 3 – GBN (RW26B), GBN + K2/USA/. J2, MCB + K2@ FL 350"	14235	7912
												(GREAT CIRCLE ROUTE WITH SID AND STAR)		
T	P	R	KSFO	K2	P	KMIA	K7	P	GC1		001	< PORTE 2 – FAT (ALL), FAT + K2/USA/... 01 HEC + K2 @ FL 390 ... 05BEC + K2 ... 64PHX + K2 ... SJN35 + K2 ... LVS	15234	7912
T	P	R	KSFO	K2	P	KMIA	K7	P	GC1		002	64 + K2 .. TMA + K4 ... SP585 + K4 ... SLR78 + K4 ... LCH 52 + K4 ... LEV42 + K7 ... 30CEW + K7 .. SRQ + K7 LEILA1 –	15235	7912
T	P	R	KSFO	K2	P	KMIA	K7	P	GC1		003	SRQ"	15236	7912
												(RNAV WITH NO SID OR STAR. ENDS AT A WAYPOINT)		
T	P	R	KDEN	K2	P	BEBOP	K2	E	A1R	J802R	001	/USA/ ... BLAND + K2 . J802R, OAL + K2 @ FL 350. J964R, BEBOP + K2"	16235	8001

LEGEND			
.	Via Airway	/ /	Area Coverage
..	Via Direct VHF NAVAID	*	End of Route
...	Via Direct Waypoint	–	SID & STAR Separator from Transition
<	Via SID	+	Ident & ICAO Separator
>	Via STAR	@	At Altitude
( )	Runway	" "	Alternate Airport
,	To (Separator)		

OBSOLETE
DO NOT USE

Figure 5-6
COMPANY ROUTE RECORD (R) EXAMPLE

OBSOLETE
DO NOT USE

ATTACHMENT 5 (cont)

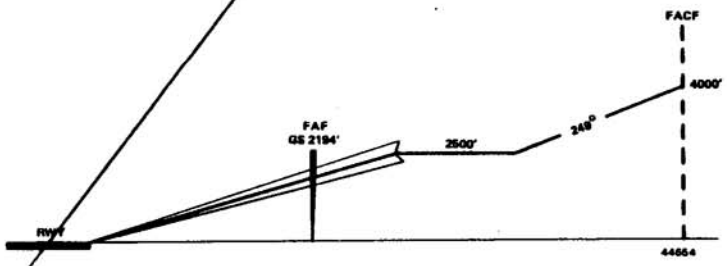
PATH AND TERMINATOR

APPROACHES AND APPROACH TRANSITIONS

C. 4. The altitude for the last leg of an ILS approach will be assigned according to the FAF rules in paragraph C.6.

5. The recommended VHF for the last CF or VI leg of an ILS transition will be the localizer.

6. ILS Glideslope captures are classified into three types: A, B, and C. In type A (Figure 1) descent is authorized immediately after passing the FACF to a level intercept segment which terminates at or near the FAF. To code this approach, the FACF crossing altitude will be coded as AT 4000'. The glideslope intercept altitude of 2500' will be coded in altitude 1 on the FAF record. The glideslope altitude at the FAF, 2194', will be entered in altitude 2 with the altitude description field containing a "G". The vertical angle of -0.00° will be coded on the FAF record to indicate that descent to the intercept altitude is authorized immediately after passing the FACF. The GS intercept altitude (2500') will be coded in ALT 2 of the FACF record with an ALT DESC of "I".



Type A Descent
Figure 1

OBSOLETE
DO NOT USE

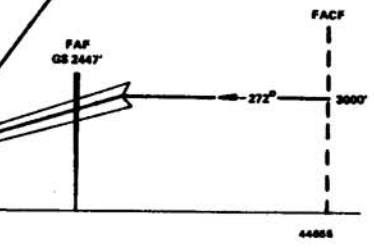
ATTACHMENT 5 (cont)

PATH AND TERMINATOR

APPROACHES AND APPROACH TRANSITIONS

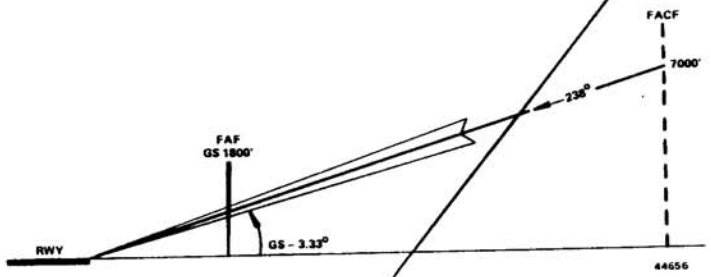
C. 6. (cont)

In Type B (Figure 2), descent below the FACF crossing altitude is not authorized until interception of the glideslope or its extension. In this case the FACF record will contain AT 3000' as the altitude constraint in ALT 1 and 3000' in ALT 2 with an "I" in the ALT DESC field. The FAF record will contain the glideslope altitude at the FAF in altitude 1 (AT 2447') and the glideslope angle (-3.22°) in the vertical angle field. This coding will cause the aircraft to fly level at 3000' until intercepting the vertical angle.



Type B Descent
Figure 2

The Type C descent is illustrated in Figure 3. In this case the FACF is the first point on a constant descent path. The approach is coded with the FACF crossing altitude of AT 7000' in ALT 1 and 7000' in ALT 2 with an "I" in the ALT DESC field. The FAF record will contain the GS altitude at the FAF (1800') and the GS angle of -3.33° will be coded in the vertical angle field as "D3.33". The "D" indicates that a constant descent path is to be flown between the FACF and the FAF.



Type C Descent
Figure 3

C. 7. The OB MAG CRS field will contain the localizer bearing rounded to the nearest whole degree to concur with the official source document course when the course is nominally the final approach course.

D. Approach transitions for ILS approaches will be as follows:

1. Last leg in approach transition will be:

   a. Course to FACF (CF leg).

   b. Heading to intercept (VI leg).

   c. Procedure turn (PI leg).

2. If case 1.a. above (CF leg), then:

   a. The last leg fix RECMD VHF will be localizer identifier.

   b. The CF leg length will be 8 miles or less or within the reception range of the localizer. If the last transition fix is more than 8 miles from the FACF, a combination of an FC and CF leg will be used with the CF leg length less than or equal to 4 miles. (See Figure 4.)

   c. The approach course intercept angle will be 30° or less.

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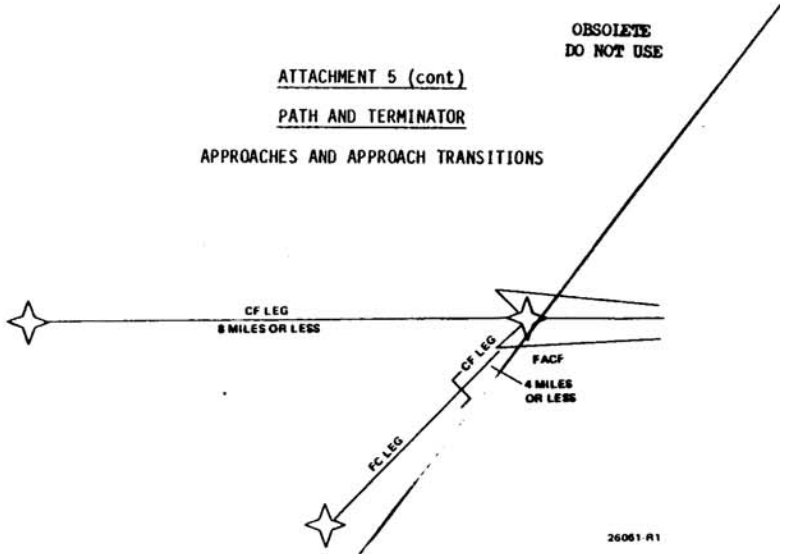
DO NOT USE

ATTACHMENT 5 (cont)

PATH AND TERMINATOR

APPROACHES AND APPROACH TRANSITIONS



ILS Transitions
Figure 4

D. 3. If case 1.b. above (VI leg), then:

   a. Heading will intercept ILS approach course between FACF and FAF.

   b. Intercept angle will be 30° or less or per source document, and no less than 2 miles from FAF.

   c. The VI leg will contain the localizer ident in the recommended VHF field.

E. Missed Approaches

1. Missed approaches will be coded as part of each approach. The first missed approach record will contain an "M" in column 42 of the DESC CODE field.

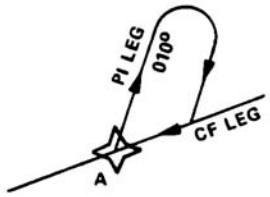
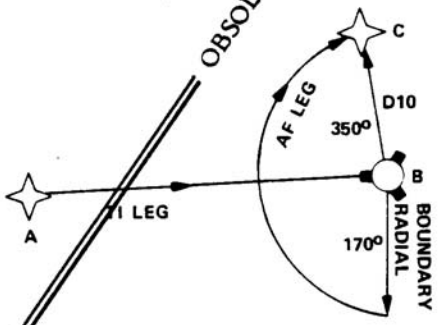
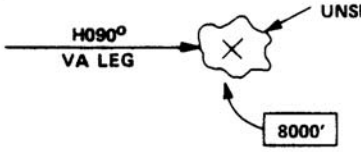
2. The first leg of the missed approach will contain an altitude to command a climb. If a turn in excess of 15° from runway heading is required, then a VA or FA leg on the runway heading to 400 feet above the airport elevation will be the first leg of the missed approach followed by the required turn.

3. Opposite end runway fixes will not be used on missed approaches.

ATTACHMENT 5 (cont'd)

PATH AND TERMINATOR
GENERAL RULES

OBSOLETE - DO NOT USE

Leg(s)	Example	Description
PI-CF		Procedure turn (PI) followed by a course to a fix (CF)
TF		Track between two fixes (great circle)
TI-AF		Track to a next leg (TI) followed by a constant DME arc to a fix (AF). Waypoint is not required at intersection of TI and AF leg.
VA		Heading to an altitude (position unspecified)

39624 (3) R1

Leg Type Illustrations
Table 2 (cont)

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SUPPLEMENT 4

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: November 10, 1983

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: October 12, 1983

A. PURPOSE OF THIS SUPPLEMENT

This Supplement extends the scope of Specification 424 to cover the navigation data base needs of airline flight simulators and introduces a number of corrections for typographical and other minor errors.

B. ORGANIZATION OF THIS SUPPLEMENT

The first part of this document, printed on goldenrod paper, contains descriptions of the changes introduced into the Specification by this Supplement, and where appropriate, extracts from the original text for comparison purposes. The second part consists of replacement white pages for the Specification modified to reflect these changes. The modified and added material on each replacement page is identified with c-4 symbols in the margins.

Existing copies of Specification 424-3 may be updated by inserting the replacement white pages where necessary and destroying the pages they displace. The goldenrod pages should be inserted inside the rear cover of the Specification, following Supplement 3.

C. CHANGES TO ARINC SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This Section presents a complete tabulation of the changes and additions to the Specification introduced by this Supplement. Each change or addition is identified using the section number and title currently employed or the section number and title that will be employed when the Supplement is eventually incorporated. In each case there is included a brief description of the addition or change.

1.1.1 Coverage of Flight Simulator Needs

New section added by this Supplement.

3.2.14 Airport Section (P) – ILS Markers Subsection (PM)

New section added by this Supplement.

3.2.15 Airport Section (P) – Airport Communication Subsection (PV)

New section added by this Supplement.

3.2.16 Enroute Section (E) – Airways Marker Subsection (EM)

New section added by this Supplement.

4.2.1 Primary Records

Columns 86 through 87 – field name changed from "DME Bias" to "ILS/DME Bias."

4.2.2 Continuation Records

Column 23 – reassigned from "Notes" to "Reserved (Spacing)." Subsequent "Notes" field reduced in length by one character. Note added below table describing reason of column 23 reservation.

4.2.3 Simulation Continuation Records

New section added by this Supplement.

4.3.1 Primary Records

Columns 75 through 79 – field name changed from "Station Declination" to "Magnetic Variation" and reference changed from 5.66 to 5.39.

4.3.2 Continuation Records

Identical change to that described above for Section 4.2.2.

4.3.3 Simulation Continuation Records

New section added by this Supplement.

4.5.2 Continuation Records

Column 40 – reassigned from "Notes" to "Reserved (Spacing)." Subsequent "Notes" field reduced in length by one character. Note added below table describing reason for Column 40 reservation.

4.6.2 Continuation Records

Identical change to that described above for Section 4.5.2.

4.7.1 Primary Records

Columns 14 through 16 – reassigned from "Blank (Spacing)" to "ATA/IATA Designator" and 5.107 reference added. Columns 17 through 18 – reassigned from "Blank (Spacing)" to "Reserved (Expansion)." Column 30 – reassigned from "Blank (Spacing)" to "IFR Capability" and 5.108 reference added.

4.7.2 Continuation Records

Identical change to that described above for Section 4.2.2.

4.8.2 Continuation Records

Identical change to that described above for Section 4.2.2.

4.9.2 Continuation Records

Identical change to that described above for Section 4.5.2.

4.10.1 Primary Records

Columns 52 through 59 – reassigned from "Reserved (Electronic Chart System Use)" to "Blank (Spacing)." Columns 61 through 66 – reassigned from "Reserved (Electronic Chart System Use)" to "blank (Spacing)." Columns 78 through 80 – reassigned from "Blank (Spacing)" to "Runway Width" and 5.109 reference added.

4.10.2 Continuation Records

Identical change to that described above for Section 4.2.2.

4.10.3 Simulation Continuation Records

New section added by this Supplement.

4.11.2 Continuation Records

Identical change to that described above for Section 4.2.2.

4.11.3 Simulation Continuation Records

New section added by this Supplement.

4.13 ILS Marker (PM)

New section added by this Supplement.

4.13.1 Primary Records

New section added by this Supplement.

4.14 Airport Communication Records (PV)

New section added by this Supplement.

4.14.1 Primary Records

New section added by this Supplement.

4.14.2 First Simulation Continuation Record

New section added by this Supplement.

4.14.3 Additional Continuation Records

New section added by this Supplement.

4.15.1 Primary Records

New section added by this Supplement.

ARINC STAFF NOTE: The foregoing changes in Chapter 4 necessitated revisions to Figure 4-1 in Specification 424-3.

5.5 Subsection Code (SUB CODE)

Subsection code "M-Airways Markers" added in Section Code "E" part of Table (i). Subsection codes "M-ILS Markers" and "V-Airport Communications" added in Section Code "P" of Table (i).

5.6 Airport Identifier (ARPT IDENT)

Text amended and rearranged for improved clarity.

ORIGINAL TEXT FOLLOWS

5.6 Airport ICAO Identifier (ARPT IDENT)

Definition/Description: The "Airport ICAO Identifier" field contains the ICAO identification code of the airport to which the data contained in the record relates.

Note: ICAO airport identifier codes differ from the perhaps more familiar ATA/IATA airport designators. ATA/IATA codes are not employed in this field.

Source/Content: The content of this field should be derived from ICAO Document No. 7910, "Location Indicators."

Used On:	SID/STAR, Approach Route, ILS, Airport, Gate, Runway and Terminal Waypoint (PC) records
Length:	4 characters
Character Type:	Alpha/numeric
Examples:	KJFK, KMIA, 9V9, CYUL, EDDF

5.26 Outbound Magnetic Course (OB MAG CRS)

Character type changed from "Numeric" to "Alpha/Numeric." Example "194T" added.

5.28 Inbound Magnetic Course (IB MAG CRS)

SID/STAR and Approach Route records deleted from "Used On" list. Character type changed from "Numeric" to "Alpha/Numeric." Example "194T" added.

5.32 Cycle Date (CYCLE)

"Definition/Description" amended to exclude changes in continuation record number and file record numbers from requiring a cycle data change.

ORIGINAL TEXT FOLLOWS

Definition/Description: The "Cycle Date" field identifies the calendar period in which the record was added to the file or last revised. A change in any ARINC 424 field requires a cycle date change. The cycle date will not change if there is no change in the data.

5.34 VOR/NDB Identifier (VOR IDEINT/NDB IDENT)

ILS Marker records added to the "Used On" list.

5.35 NAVAID Class (CLASS)

Table (v) revised to add class of locators at markers and an indicator in column 32 for non-collocated facilities. "Biased DME" facility type changed to "Biased ILS/DME."

5.36 Latitude (LATITUDE)

ILS Marker, Airways Marker and Airport Communications records added to "Used On" list.

5.37 Longitude (LONGITUDE)

ILS Marker, Airways Marker and Airport Communications records added to "Used On" list.

5.39 Magnetic Variation (MAG VAR)

"Used On" list changed from Airport and NDB NAVAID records."

5.46 Runway Identifier (RUNWAY IDENT)

ILS Marker records added to "Used On" list.

5.71 Name Field

Enroute Marker records added to "Used On" list.

5.77 VIA

Note added describing the purpose of Figure 5-6.

5.80 ILS Category (CAT)

"Source/Content" paragraph expanded to describe use of 0 in this field.

5.90 ILS DME Bias

Section title changed from "DME Bias." No other changes.

5.91 Continuation Record Application Type (APPL)

New section added by this Supplement.

5.92 Facility Elevation (FAC ELEV)

New section added by this Supplement.

5.93 Facility Characteristics (FAC CHAR)

New section added by this Supplement.

5.94 True Bearing (TRUE BRG)

New section added by this Supplement.

5.95 Government Source (SOURCE)

New section added by this Supplement.

5.96 Glide Slope Beam Width (GS BEAM WIDTH)

New section added by this Supplement.

5.97 Touchdown Zone Elevation (TDZE)

New section added by this Supplement.

5.98 TDZE Location (LOCATION)

New section added by this Supplement.

5.99 Marker Type (MKR TYPE)

New section added by this Supplement.

5.100 Minor Axis (MINOR AXIS)

New section added by this Supplement.

5.101 Communication Type

New section added by this Supplement.

5.102 Radar (RADAR)

New section added by this Supplement.

5.103 Communications Frequency (COMM FREQ)

New section added by this Supplement.

5.104 Modulation (MODLN)

New section added by this Supplement.

5.105 Call Sign (CALL SIGN)

New section added by this Supplement.

5.106 Additional Service (ADDL SERVICE)

New section added by this Supplement.

5.107 ATA/IATA Designator (ATA/IATA)

New section added by this Supplement.

5.108 IFR Capability (IFR)

New section added by this Supplement.

5.109 Runway Width (WIDTH)

New section added by this Supplement.

5.110 Marker Ident (MARKER IDENT)

New section added by this Supplement.

5.111 Marker Code (MARKER CODE)

New section added by this Supplement.

5.112 Marker Shape (SHAPE)

New section added by this Supplement.

5.113 High/Low (HIGH/LOW)

New section added by this Supplement.

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SUPPLEMENT 5

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: March 18, 1985

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: October 10, 1984

A. PURPOSE OF THIS SUPPLEMENT

This Supplement extends the scope of Specification 424 to cover the navigation data base needs of automated flight planning and introduces a number of corrections for typographical and other minor errors.

B. ORGANIZATION OF THIS SUPPLEMENT

The first part of this document, printed on goldenrod paper, contains descriptions of the changes introduced into the Specification by this Supplement, and where appropriate, extracts from the original text for comparison purposes. The second part consists of replacement white pages for the Specification modified to reflect these changes. The modified and added material on each replacement page is identified with c-5 symbols in the margins.

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C. CHANGES TO ARINC SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This Section presents a complete tabulation of the changes and additions to the Specification introduced by this Supplement. Each change or addition is identified using the section number and title currently employed or the section number and title that will be employed when the Supplement is eventually incorporated. In each case there is included a brief description of the addition or change.

2.2 Special Navigation Terms

The definition of ATC Compulsory Reporting Point is changed, and two new definitions are added.

ORIGINAL TEXT FOLLOWS

ATC Compulsory Reporting Point	A waypoint which normally would be classified as nonessential might be a waypoint at which ATC requires the pilot to make a communications report. Waypoints falling into this category are classified as ATC Compulsory Reporting Points.

3.2 Master Airline User File Content

The paragraphs in this section are renumbered to reflect the hierarchical structure of the master data file.

4.2 VHF NAVAID Record

The wording of the last sentence of the paragraph is changed to clarify the handling of duplicate VOR and TACAN idents.

ORIGINAL TEXT FOLLOWS

For non-frequency paired VOR and TACAN stations having the same identifier, the TACAN is stored unless the VOR is used for enroute airways.

4.2.1 Primary Records VHF NAVAIDS

Columns 6 through 13 reassigned from "Blank (Spacing)." "Reserved" status removed from column 85 assignments and Note 1 deleted. Columns 88 through 93 reassigned from "Reserved (Expansion)."

ORIGINAL TEXT FOLLOWS

Note 1: "NAVAID Figure of Merit No." has no navigation function. Should official approval be forthcoming; however, for the use of NAVAID's beyond ranges specified in the class field, this field could indicate the limitations of such use.

4.2.4 Flight Planning Continuation Records

This section is added.

4.2.5 Flight Planning Continuation Records

This section is added.

4.3.1 Primary Records

Length "(l)" is added to Column 6 Subsection Code.

4.3.4 Flight Planning Continuation Records

This section is added.

4.3.5 Flight Planning Continuation Records

This section is added.

4.4 Waypoint Records (EA OR PC)

A sentence has been added to clarify why waypoints are stored in the EA section.

4.4.1 Primary Records

Column 19 reassigned from "Reserved (6th Char. W.P.)" and columns 52 through 98 reassigned from "Reserved (Expansion)." Length "(2)" is added to columns 30 and 31. Note 3 is deleted.

ORIGINAL TEXT FOLLOWS

Note 3: The standard length for the waypoint identifier field is 5 characters. Some users envisage the need for 6 characters in certain cases. This reserved column will permit this usage.

4.4.3 Flight Planning Continuation Records

This section is added.

4.4.4 Flight Planning Continuation Records

This section is added.

4.5.1 Primary Records

Columns 27 through 29 reassigned from "Blank (Spacing)" and columns 30 through 35 reassigned from "Waypoint Identifier" (30-34) and "Reserved (6th Char. of Waypoint Ident) (35)." Note 1 deleted.

ORIGINAL TEXT FOLLOWS

Note 1: The standard length of the Waypoint Identifier field is five characters. Some users envisage the need for a six-character field in certain cases. This revision column will permit this usage.

4.6 Enroute Airways Records (ER)

The last sentence in this section is deleted.

ORIGINAL TEXT FOLLOWS

4.6 Enroute Airways Records (ER)

The Enroute Airways file will contain the sequential listing of airways by geographic areas. This file may also contain enroute off-airway waypoints by geographic area, without sequence numbers of airway idents.

4.6.1 Primary Records (ER)

Column 35 reassigned from "Reserved" and Note 1 revised. Columns 46 through 49 reassigned from "Blank (Spacing)." Columns 84 through 85 reassigned from "Altitude" and Columns 89 through 123 reassigned from "Reserved (Expansion)."

ORIGINAL TEXT FOLLOWS

Note 1: The standard lengths for the Route Identifier and the Waypoint Identifier fields are each five characters. Some users envisage the need for six-character fields in certain instances. These reserved columns will permit this usage.

4.7.1 Primary Records

Columns 28 through 32 reassigned from "Airport Class" (28-29) "IFR Capability" (30) and "Blank (Spacing)" (31-32). Columns 65 through 70 reassigned from "Reserved (Expansion)."

4.7.3 Flight Planning Continuation Records

This section is added.

4.7.4 Flight Planning Continuation Records

This section is added.

4.9.1 Primary Records

Columns 21 through 25 reassigned from "Transition/RWY No. or Blank." Columns 26 and 35 reassigned from "Blank (Reserved)." Note 1 revised.

ORIGINAL TEXT FOLLOWS

Note 1: The Transition Identifier and the Waypoint Identifier fields are five characters each. Some users envisage the need for six-character fields in certain cases. These reserved columns will permit this usage.

4.9.3 Flight Planning Continuation Records

This section is added.

4.9.4 Flight Planning Continuation Records

This section is added.

4.11 ILS (Localizer and Glide Slope) Records

This section is revised to permit blank fields in the glide slope information fields when the data for them is not available.

4.11.1 Primary Records

Length of "(1)" is added to fields 1 and 6.

4.12.1 Primary Records

Columns 6 through 12 reassigned from "Reserved (Spacing)" (16) and "From Airport/Fix" (7-12). Column 22 reassigned from "To Airport./Fix." Columns 91 through 94 reassigned from "Reserved (Expansion)."

4.13.1 Primary Records

The field name and length for columns 14 through 17 is added (previously "Blank (Spacing))."

4.16 Cruising Tables Records (TC)

This section is added.

4.16.1 Primary Records

This section is added.

4.17 FIR/UIR Records (UF)

This section is added.

4.17.1 Primary Records

This section is added.

4.18 Restrictive Airspace Records (UR)

This section is added.

4.18.1 Primary Records

This section is added.

4.18.2 Continuation Records

This section is added.

4.19 Grid MORA Records (AS)

This section is added.

4.19.1 Primary Records

This section is added.

4.20 MSA (Minimum Safe Altitude Records) (PS)

This section is added.

4.20.1 Primary Records

This section is added.

4.20.1 Primary Records (cont'd)

ARINC STAFF NOTE: The foregoing changes in Chapter 4 necessitated revisions to Figure 4-1 in Specification 424-5.

5.3 Customer/Area Code (CUST/AREA)

"Examples" are changed.

5.5 Subsection Code (SUB CODE)

"Used On" has been revised to read "All records." New sections are added to the chart.

Figure 5-1 Geographical Area Codes

This figure is revised to correct USA coverage to the East and change boundary between SAM and AFR.

5.6 Airport Identifier (ARPT IDENT)

"Used On" section revised to include VHF NAVAID (ILS/DME only) and Airport Communications records.

5.7 Route Type (RT TYPE)

Enroute Airways Records Source/Content Table changed.

ORIGINAL TEXT FOLLOWS

5.7 Route Type (RT TYPE)

Definition/Description: The "Route Type" field defines the type of enroute airway, SID/STAR or approach route of which the record is an element.

Source/Content:

Enroute Airway Records (ER)

Airway Type	Field Content
Airline	A
Both (High/Low)	B
Control	C
Direct	D
high Level	H
Jet	J
Low Level	L
NDB	N
RNAV	R
Victor	V

SID Records (PD)

Route Type	Field Content
SID Runway Transition	1
SID	2
SID Enroute Transition	3
RNAV SID Runway Transition	4
RNAV SID	5
RNAV Enroute Transition	6

STAR Records (PE)

Route Type	Field Content
STAR Enroute Transition	1
STAR	2
STAR Runway Transition	3
RNAV STAR Enroute Transition	4
RNAV STAR	5
RNAV star Runway Transition	6
Profile Descent Enroute Transition	7
Profile Descent	8
Profile Descent Runway	9

Approach Route Records (PF)

Route Type	Field Content
Approach Transition	A
Back Course	B
ILS	I
Localizer	L
RNAV	R
NDB	N
VOR	V
MLS	M

Used On: Enroute Airways, SID/STAR and Approach Route records
Length: 1 character
Character Type: Alpha/Numeric

5.12 Sequence Number (SEQ NR)

Source/Content is revised to include "Enroute airways should be sequenced in order that sequence numbers are not duplicated regardless of area covered."

"Used On" is revised to include Cruise Table, FIR/UIR, and Restrictive Airways records.

"Length" is revised to read "F Characters (Enroute Airways FIR/UIR and Restrictive Airspace, 3 Characters (SID/STAR/Approach and Company Route, 1 Character (Cruise Table)."

5.13 Waypoint Identifier (WAYPOINT IDENT)

"Used On" section is revised to remove "Company Route records." "Examples" section is revised to include RW27L. KGRR.

5.15 File Code (D/E/P)

The source/content table is revised to more accurately define the types of records to which the codes apply.

ORIGINAL TEXT FOLLOWS

5.15 File Code (D/E/P)

Definition/Description The records for a waypoint identifier in a route record may reside in one of the number of major sections of the data base. The "File Code" field permits the section in which a particular waypoint record will be found to be identified.

Source/Content:

Code	Waypoint Record Source
D	NAVAID Section (Waypoint is NAVAID)
E	Enroute Waypoint Section
P	Terminal Section (SID/STAR/APP and Company Route records)

Used On: Enroute Airways, SID/STAR/APP, Holding Pattern and Company Route records
Length: 1 character
Character Type: Alpha

5.16 Continuation Record Number (CONT NR)

"Used On" is revised to add Cruise Table, Grid MORA, and MSA records.

5.19 Level (LEVEL)

Source/Content for "B" is changed to "All Altitudes" (was Both Level Airways (not designated as high level or love level)).

5.23 Recommended VHF NAVAID (REC'D VHF)

This paragraph is reworded to accommodate the addition of the Recommended VHF NAVAID to the airport record.

Two notes are added.

5.27 Route Distance From, Holding Distance/Time (RTE DIST FROM, HOLD DIST/TIME)

A sentence is added to the end of the Source/Content section.

5.30 Altitude (ALTITUDE)

In the examples, the usage of UNKNN and NESTB on enroute records is clarified.

5.31 File Record Number (FRN)

Source/Content is revised to add that if the file reaches 99999, the next record number will start over with 00000.

5.35 NAVAID Class (CLASS)

Table (v) is revised. ("U" in column 30 defined as "Undefined NAVAID. "I," "M" were incorrectly shown in column 29. "M" was used to code Navaid power up to 50 watts.) Note 3 added.

5.37 Longitude (LONGITUDE)

Table (vi) is revised.

ORIGINAL TEXT FOLLOWS

Record File	LAT/LONG Field	Location Defined
VHF NAVAID	VOR (See Note 1)	VOR Antenna
VHF NAVAID	DME (See Note 2)	DME Antenna
Waypoint	Waypoint	Waypoint
ILS Antenna	Localizer	Localizer
ILS	Glide Slope	Glide Slope Antenna
Airport	Airport	Airport Ref. Point
Gate	Gate	Gate
Runway	Runway	Runway Threshold
NDB NAVAID	NDB	NDB Antenna

5.39 Magnetic Variation (MAG VAR)

The Source/Content section's first sentence is revised to clarify the sources for magnetic variations.

5.42 Waypoint Type (TYPE)

The table is expanded to clarify those codes allowed on Enroute Waypoints and Terminal Waypoints, and two new codes are added.

5.44 Localizer Identifier (LOC IDENT)

"Used On" is changed to read "ILS, Runway and ILS marker receives."

5.46 Runway Identifier (RUNWAY IDENT)

Source/Content section is rewritten.

ORIGINAL TEXT FOLLOWS

Source/Content: The two letters RW precede runway identifications derived from official government sources.

5.53 Transition Altitude (TRANS ALTITUDE)

Definition/Description last sentence is replaced.

ORIGINAL TEXT FOLLOWS

If the transition altitude is 18,000 feet, it may be omitted.

5.54 Airport Class (CLASS)

The name of this section is changed to Longest Runway (LONGEST RWY).

In Definition/Description section, "Airport Class" is changed to "Longest Runway." "Character Type" is changed from "Alpha" to "Numeric." "Examples" are added.

5.64 Leg Length (LEG LENGTH_

"See Figure 5-4" is added at the end of the Definition/Description section.

5.66 Station Declination (STN DEC)

ILS records added to the "Used On" list.

5.70 Vertical Angle (VERT ANGLE)

"Character Type" is changed from "Numeric" to "Alpha/Numeric." "Examples" are added.

5.72 Speed List (SPEED LIMIT)

Sentence added to the Source/Content section to cover placing the Speed Limit in the Airport Record.

5.75 From/To – Airport/Fix

This section is revised to delete "airport reference point" and "NAVAID."

ORIGINAL TEXT FOLLOWS

Definition/Description: The "From Airport/Fix" is the airport reference point, the Navaid or Waypoint from that the company route originates. The "To Airport/Fix" is the airport reference point, the Navaid or waypoint that the company route terminates.

5.79 Stopway

"The length of an" is added to the first sentence of the Definition/Description section.

5.81 ATC Indicator (ATC)

The Source/Content section is revised to distinguish between the two different conditions for the use of this field.

ORIGINAL TEXT FOLLOWS

Source/Content: This field will be used when official government source states that the altitude can be modified or assigned by ATC and will contain the alpha character "A."

5.82 Waypoint Usage

"Terminal Use Only" is added to Source/Content section.

5.88 Alternate Distance (ALT DIST)

"…from the destination airport/fix to the alternate airport" added to the last line of the Source/Content section.

5.89 Cost Index

Definition/Description section is revised.

ORIGINAL TEXT FOLLOWS

Definition/Description: The Cost Index field is used to define the cost of different Company Routes between the same city pairs.

The following new Sections added by this Supplement:

5.114 Duplicate Identifier
5.115 Direction Restriction
5.116 FIR/UIR Identifier (FIR/UIR IDENT)
5.117 FIR/UIR Indicator (IND)
5.118 Boundary VIA (BORY VIA)
5.119 Arc Distance (ARC DIST)
5.120 Arc Bearing (ARC BRG)
5.121 Lower/Upper Limit
5.122 FIR/UIR ATC Reporting Units Speed (RUS)
5.123 FIR/UIR ATC Reporting Units Altitude (RUA)
5.124 FIR/UIR Entry Report (ENTRY)
5.125 FIR/UIR Name
5.126 Restrictive Airspace Name
5.127 Maximum Altitude (MAX ALT)
5.128 Restrictive Airspace Type (REST TYPE)
5.129 Restrictive Airspace Designation
5.130 Multiple Code (MULTI CD)
5.131 Time Code (TIME CD)
5.132 NOTAM
5.133 Unit Indicator (UNIT IND)
5.134 Cruise Table Identifier (CRSE TBL IDENT)
5.135 Magnetic Track From/To (MAG TRACK FM/TO)
5.136 Cruise Level From/To
5.137 Vertical Separation
5.138 Time Indicator (TIME IND)
5.139 Time Group
5.140 Controlling Agency
5.141 Starting Latitude
5.142 Starting Longitude
5.143 MORA
5.144 MSA Center
5.145 Radium Limit
5.146 Sector Bearing (SEC BRG)
5.147 Sector Altitude (SEC ALT)
5.148 Enroute Alternate Airport (EAA)
5.149 Figure of Merit (MERIT)
5.150 Frequency Protection Distance (FREQ PRO)
5.151 FIR/UIR Address (ADDRESS)
5.152 Start/End Indicator (S/E IND)
5.153 Start/End Date

6.7.2 Header 1 Label (HDR 1)

In order to identify the cycle during which a tape is created, the Generation Number field is changed to contain the cycle date.

ATTACHMENT 2 – WAYPOINT IDENTIFIERS

CPR29, CPR29A, CPR29, CPR29B are removed from Section E.

Examples are changed to read OM25L, MM09, IM23, RW04, MA18L in Section F.

ATTACHMENT 5 – PATH AND TERMINATOR

Section added.

A family of path types is added based on track legs that are terminated as appropriate.

A new sentence is added in Rule N. Rule W is revised.

Rule AA is deleted and a new rule AA is incorporated.

ORIGINAL TEXT FOLLOWS

When a SID or STAR route is repeated with different TRANSITION ID fields, it shall be defined as a runway or enroute transition route type.

SIDE CODING RULES

SID Rule 1 is revised.

ORIGINAL TEXT FOLLOWS

I. A SID which consists of a runway transition only shall be coded as a route type 2.

SID Rule K is added.

STAR CODING RULES

Rule E is revised.

ORIGINAL TEXT FOLLOWS

E. A STAR which consists of a runway transition only will be coded as a route type 2, 5, or 8.

Rule F is added.

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SUPPLEMENT 6

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: January 31, 1986

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: November 7, 1985

A. PURPOSE OF THIS SUPPLEMENT

This Supplement incorporates airway restriction records into ARINC 424. Changes to ARINC 424 are also included to improve the operational utility of the navigation data base. New rules or clarifications to existing rules are provided by this Supplement. Editorial changes are also made by this Supplement.

B. ORGANIZATION OF THIS SUPPLEMENT

The first part of this document, printed on goldenrod paper, contains descriptions of the changes introduced into the Specification by this Supplement, and where appropriate, extracts from the original text for comparison purposes. The second part consists of replacement white pages for the Specification modified to reflect these changes. The modified and added material on each replacement page is identified with c-6 symbols in the margins.

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C. CHANGES TO ARINC SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This Section presents a complete tabulation of the changes and additions to the Specification introduced by this Supplement. Each change or addition is identified using the section number and title currently employed or the section number and title that will be employed when the Supplement is eventually incorporated. In each case there is included a brief description of the addition or change.

3.2.2.1 VHF NAVAID Section (D) – Subsection (D)

Reference to Section 3.2.5 added.

3.2.3.1 Enroute Waypoints Section (E) – Subsection (EA)

Reference to Section 3.2.3.2 deleted. Reference to Section 3.2.3.4 added.

3.2.3.5 Enroute Airways Restrictions Section (P) Subsection (PV)

New section added.

3.2.4.11 Airport Communications Section (P) Subsection (PV)

New text for airports referenced in Subsection 3.2.4.1 added.

4.3 NDB NAVAID Record (DB)

Clarify text to specify "LF and MF" NDBs and delete obsolete table accidentally included in Supplement 6.

4.4.2 Continuation Records

Add new field.

ORIGINAL TEXT FOLLOWS

4.4.2 Continuation Records

Column	Field Name (Length)	Reference
1 thru 21	Fields as on Primary Records	
22	Continuation Record No. (1)	5.16
23 thru 92	Notes (70)	5.61
93 thru 123	Reserved (Expansion) (34)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.5.1 Primary Records

Added new field.

ORIGINAL TEXT FOLLOWS

4.5.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 10	Region Code (4)	5.41
11 thru 27	Blank (Spacing) (17)	
28 thru 29	Duplicate Identifier (2)	5.114
35	Blank (Spacing) (17)	
36 thru 37	ICAO Code (2)	5.14
38	File Code (1)	5.15
39	Continuation Record No. (1)	5.16
40 thru 43	Inbound Holding Course (4)	5.62
44	Turn Direction (1)	5.63
45 thru 47	Leg Length (3)	5.64
48 thru 49	Leg Time (2)	5.65
50 thru 54	Altitude (5)	5.30
55 thru 98	Reserved (Expansion) (44)	
99 thru 123	Notes (25)	5.60
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.6.3 Flight Planning Continuation Records

Added new section.

4.6.4 Flight Planning Continuation Records

Added new section.

4.14.1 Primary Records

Added new field.

ORIGINAL TEXT FOLLOWS

4.14.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport Identifier/Blank (4)	5.6
11 thru 12	ICAO Code/Blank (2)	5.14
13	Blank (Spacing) (1)	
14 thru 16	Communication Type (3)	5.101
17 thru 19	Service (3)	5.106
20	Radar Service (1)	5.102

Column	Field Name (Length)	Reference
21	Blank (Spacing) (1)	
22	Continuation Record No. (1)	5.16
23 thru 29	Primary Comm Freq (7)	5.103
30	Modulation (1)	5.104
31 thru 32	Blank (Spacing) (2)	
33 thru 41	Facility Latitude (9)	5.36
42 thru 51	Facility Longitude (10)	5.37
52 thru 56	Magnetic Variation (5)	5.39
57 thru 61	Facility Elevation (5)	5.92
62 thru 68	Comm Freq	5.103
69	Modulation (1)	5.104
70 thru 76	Comm Freq (7)	5.103
77	Modulation (1)	5.104
78 thru 84	Comm. Freq (7)	5.103
85	Modulation (1)	5.104
86 thru 92	Comm Freq (7)	5.103
93	Modulation (1)	5.104
94 thru 100	Comm Freq (7)	5.103
101	Modulation (1)	5.104
102 thru 108	Comm Freq (7)	5.103
109	Modulation (1)	5.104
110 thru 116	Comm Freq (7)	5.103
117	Modulation (1)	5.104
118 thru 123	Reserved (Expansion) (6)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.15.1 Primary Records

Added new field.

ORIGINAL TEXT FOLLOWS

4.15.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 13	Blank (Spacing) (7)	
14 thru 17	Marker Ident	5.110
18 thru 19	Blank (Spacing)	
20 thru 21	ICAO Code (2)	5.14
22	Continuation Record No. (1)	5.16
23 thru 26	Marker Code (4)	5.11
27	Blank (Reserved/Spacing) (1)	
28	Marker Shape (1)	5.112
29	Marker Power (1)	5.113
30 thru 33	Blank (Spacing) (4)	
34 thru 41	Marker Latitude (9)	5.36
42 thru 51	Marker Longitude (10)	5.37
52 thru 55	Minor Axis (4)	5.100
56 thru 74	Blank (Reserved) (19)	
75 thru 79	Magnetic Variation (2)	5.39
80 thru 84	Facility Elevation (5)	5.92
85 thru 93	Blank (Reserved) (9)	
94 thru 123	Marker Name (30)	5.71
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.16.1 Primary Records

Added new field.

ORIGINAL TEXT FOLLOWS

4.16.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Blank (Spacing) (3)	
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 8	Cruise Table Identifier (2)	5.134
9	Sequence Number (1)	5.12
10 thru 28	Blank (Spacing) (19)	
29 thru 32	Magnetic Track From (4)	5.135
33 thru 36	Magnetic Track To (4)	5.135
37 thru 39	Blank (Spacing) (3)	
40 thru 44	Cruise Level From (5)	5.136
45 thru 49	Vertical Separation (5)	5.137
50 thru 54	Cruise Level To (5)	5.136
55 thru 59	Cruise Level From (5)	5.136
60 thru 64	Vertical Separation (5)	5.137
65 thru 69	Cruise Level To (5)	5.136
70 thru 74	Cruise Level From (5)	5.136
75 thru 79	Vertical Separation (5)	5.137
80 thru 84	Cruise Level To (5)	5.136
85 thru 89	Cruise Level From (5)	51.36
90 thru 94	Vertical Separation (5)	5.137
95 thru 99	Cruise Level To (5)	5.136
100 thru 123	Reserved (Expansion) (24)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.17.1 Primary Records

Added new field.

ORIGINAL TEXT FOLLOWS

4.17.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type 91)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 10	FIR/UIA Identifier (4)	5.116
11 thru 12	FIR/UIR Address	5.151
13	FIR/UIR Indicator (1)	5.117
14 thru 17	Sequence Number (4)	5.12
18	Continuation Record No. (1)	5.16
19 thru 22	Adjacent FIR Identifier (4)	5.11
23 thru 26	Adjacent UIR Identifier (4)	5.11
27	Reporting Units Speed (1)	5.122
28	Reporting Units Altitude (1)	5.123
29	Entry Report (1)	5.124
30	Blank (Spacing) (3)	
31 thru 32	Boundary Via (2)	5.118
33 thru 41	FIR/UIR Latitude (9)	5.36
42 thru 51	FIR/UIR Longitude (10)	5.37
52 thru 60	Arc Origin Latitude (9)	5.36
61 thru 70	Arc Origin Longitude (10)	5.37
71 thru 74	Arc Distance (4)	5.119
75 thru 78	Arc Bearing (4)	5.120
79 thru 83	FIR Upper Limit (5)	5.121
84 thru 88	UIR Lower Limit (5)	5.121
89 thru 93	UIR Upper Limit (5)	5.121
94 thru 123	FIR/UIR Name (30)	5.125
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.19.1 Primary Records

Added new field.

ORIGINAL TEXT FOLLOWS

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Blank (Spacing) (3)	
5	Section Code (1)	5.4
6	Subsection Code (1)	5.5
7 thru 13	Blank (Spacing) (7)	
14 thru 16	Starting Latitude (3)	5.141
17 thru 20	Starting Longitude (4)	5.142
21 thru 30	Blank (Spacing) (3)	
31 thru 33	MORA (3)	5.143
34 thru 36	MORA (3)	5.143
37 thru 39	MORA (3)	5.143
50 thru 42	MORA (3)	5.143
43 thru 45	MORA (3)	5.143
49 thru 51	MORA (3)	5.143
52 thru 54	MORA (3)	5.143
55 thru 57	MORA (3)	5.143
58 thru 60	MORA (3)	5.143
61 thru 63	MORA (3)	5.143
64 thru 66	MORA (3)	5.143
67 thru 69	MORA (3)	5.143
70 thru 72	MORA (3)	5.143
73 thru 75	MORA (3)	5.143
76 thru 78	MORA (3)	5.143
79 thru 81	MORA (3)	5.143
82 thru 84	MORA (3)	5.143
85 thru 87	MORA (3)	5.143
88 thru 90	MORA (3)	5.143
91 thru 93	MORA (3)	5.143
94 thru 96	MORA (3)	5.143
97 thru 99	MORA (3)	5.143
100 thru 102	MORA (3)	5.143
103 thru 105	MORA (3)	5.143
106 thru 108	MORA (3)	5.143
109 thru 111	MORA (3)	5.143
112 thru 114	MORA (3)	5.143
115 thru 117	MORA (3)	5.143
118 thru 120	MORA (3)	5.143
121 thru 123	Reserved (Expansion) (3)	
124 thru 128	File Record No. (5)	5.31
129 thru 132	Cycle Date (4)	5.32

4.20.1 Primary Records

Added new field.

ORIGINAL TEXT FOLLOWS

4.20.1 Primary Records

Column	Field Name (Length)	Reference
1	Record Type (1)	5.2
2 thru 4	Customer/Area Code (3)	5.3
5	Section Code (1)	5.4
6	Blank (Spacing) (1)	
7 thru 10	Airport (ICAO) Identifier (4)	5.6
11 thru 12	ICAO Code (2)	5.5
14 thru 18	MSA Center (5)	5.144
19 thru 20	ICAO Code (2)	5.14
21	File Code (1)	5.15
22	Subsection Code (1)	5.5
23 thru 24	Radius Limit (2)	5.145
25 thru 27	Sector Bearing (3)	5.146

4.21 Enroute Airways Restriction Records (EU)

New section added.

4.21.1 Primary Records

New record table added.

4.21.2 Continuation Records

New record table added.

4.21.3 Continuation Records

New record table added.

Table 4.1 Record Format

New records and new fields added to reflect changes in Section 4.

ORIGINAL TEXT FOLLOWS

Table 4.1 Record Format (see Pgs. 5-10)

5.3 Customer/Area Code (SUB CODE)

Added new examples.

5.5 Subsection Code (SUB CODE)

Add new subsection code to section code E for airway restrictions (U).

5.17 Waypoint Description Code (DESC CODE)

Expansion of note "++."

ORIGINAL TEXT FOLLOWS

++ Not used in combination with "G" in column 40.

Figure 5-2 7 SUBDIVISIONS FOR UNITED STATES

Figure replaced with a clearer reproduction with the names of states included.

5.23 Recommended VHF NAVAID REC'D VHF)

Expansion of reference for final approach note to indicate referenced NAVAID.

ORIGINAL TEXT FOLLOWS

For Localizer based Approach Transitions ending in an intercept of the localizer or course to Final Approach Course Fix and the Final Approach path, the Recommended VHF NAVAID must be the referenced Localizer. For VOR Approach the final approach path must be the referenced VOR.

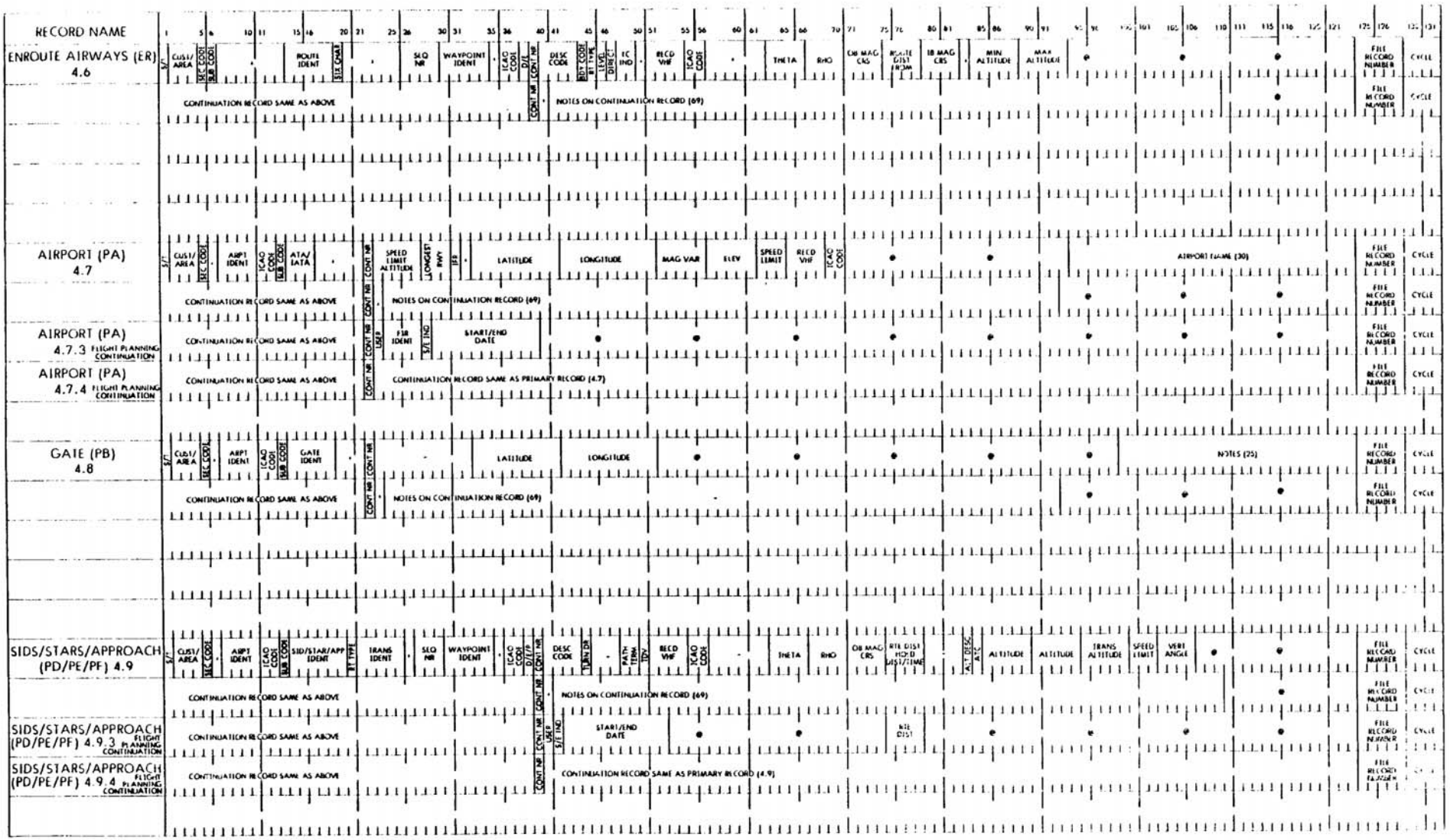
5.30 Altitude (ALTITUDE)

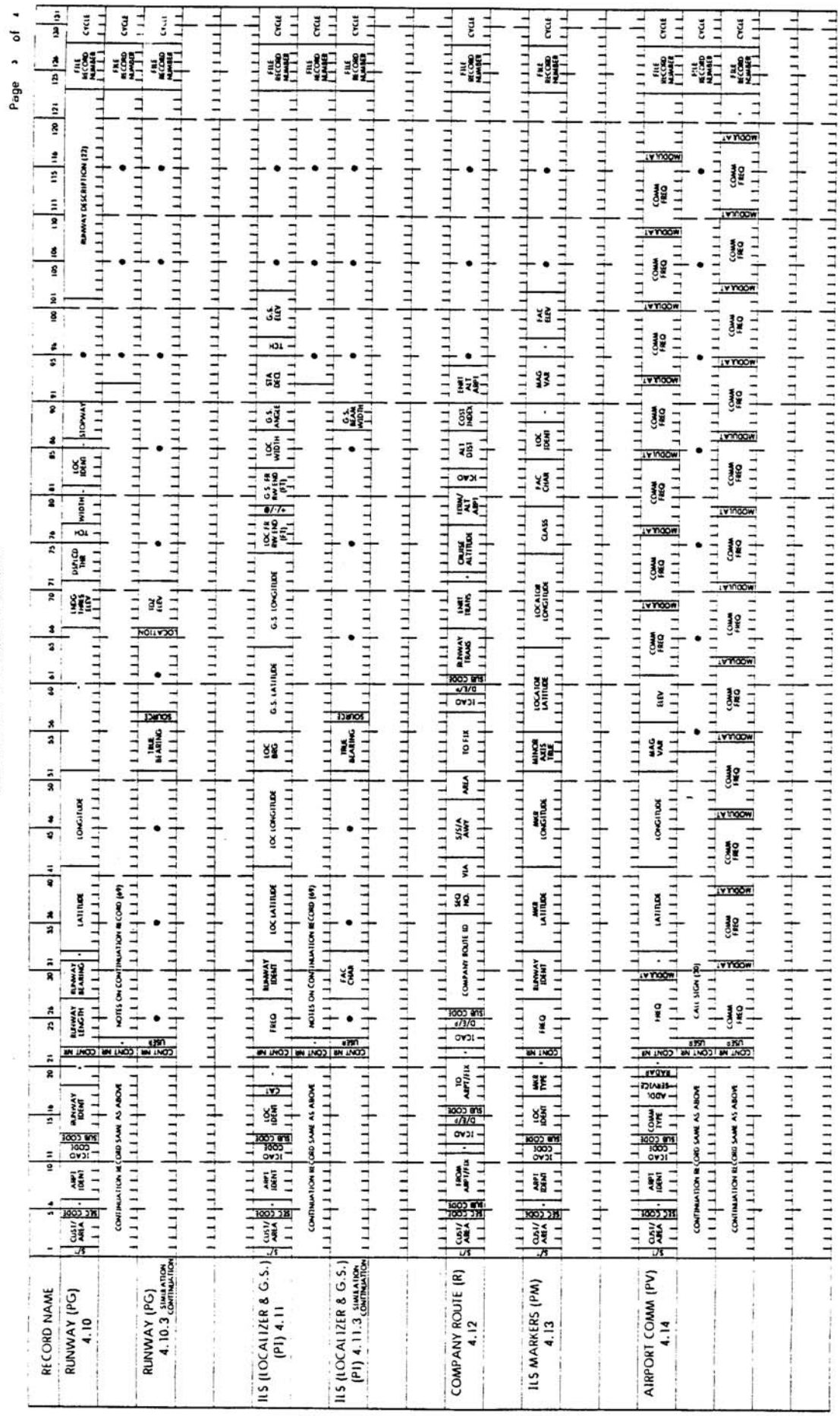
Expand title and text (new paragraph) to include minimum altitude. Delete reference to Attachment 4.

ARINC 424-5 RECORD FORMAT

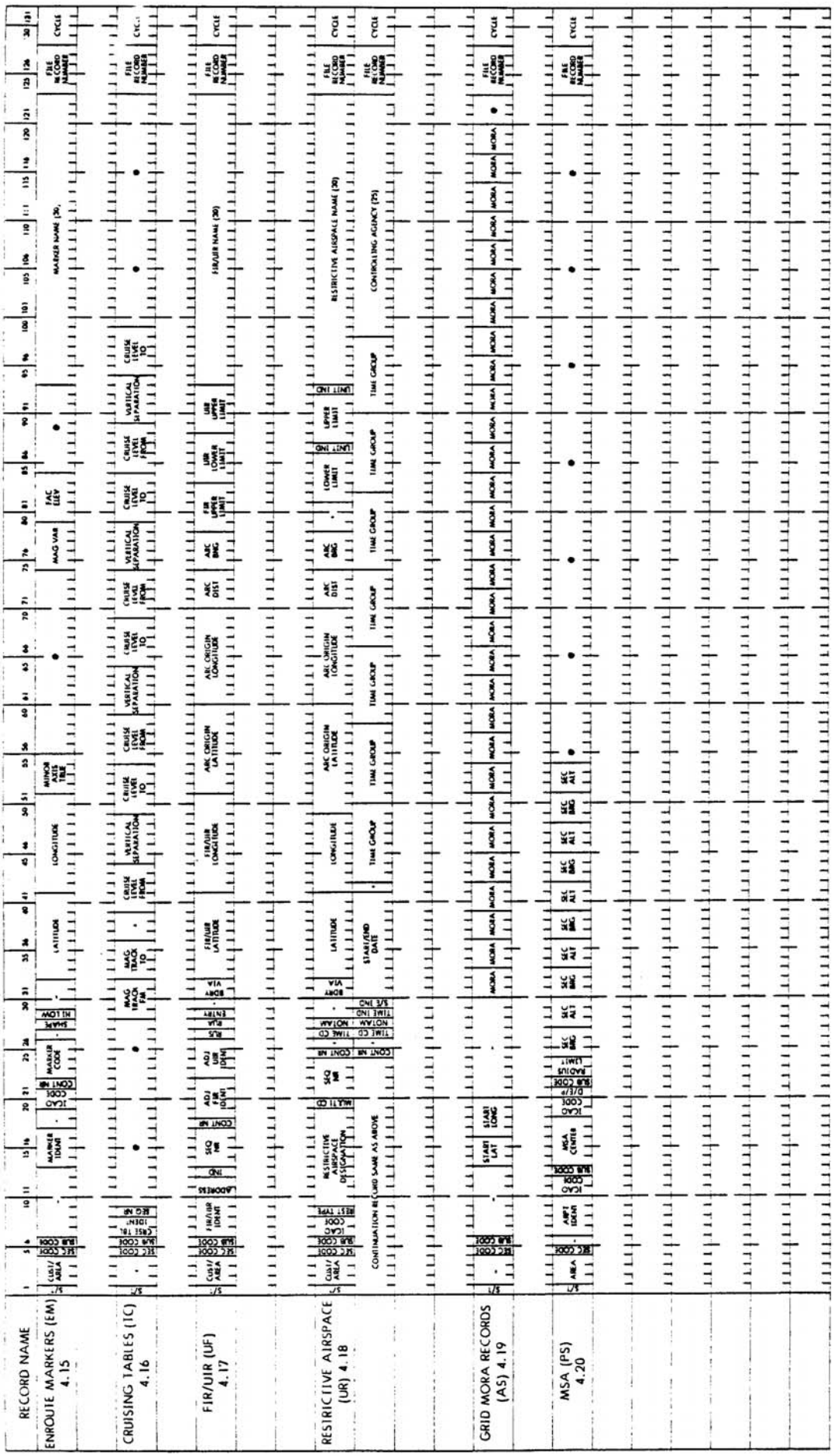
RECORD NAME	1	5	6	10	11	15	16	20	21	25	26	30	31	35	36	40	41	45	46	50	51	55	56	60	61	65	66	70	71	75	76	80	81	85	86	90	91	95	96	100	101	105	106	110	111	115	116	120	121	125	126	130	131
VHF NAVAID (D) 4.2	S/T	CUST/ AREA	SEC CODE	SUB CODE	AIRPT IDENT	ICAO CODE	·	VOR IDENT	·	ICAO CODE	CONT NR	VOR FREQ	VOR CLASS	VOR LATITUDE			VOR LONGITUDE				DME IDENT	DME LATITUDE				DME LONGITUDE				STA DECL	DME ELEV		MERIT	DME BIAS	FREQ PRO	·	VHF NAVAID NAME (30)															FILE RECORD NUMBER	CYCLE
		CONTINUATION RECORD SAME AS ABOVE								CONT NR	·	NOTES ON CONTINUATION RECORD (69)																																															
VHF NAVAID (D) 4.2.3 SIMULATION CONTINUATION		CONTINUATION RECORD SAME AS ABOVE								CONT NR	USER	•	FAC CHAR																	D MAG VAR	FAC ELEV																												
VHF NAVAID (D) 4.2.4 FLIGHT PLANNING CONTINUATION		CONTINUATION RECORD SAME AS ABOVE								CONT NR	USER	FIR IDENT	S/E IND	START/END DATE																																										
VHF NAVAID (D) 4.2.5 FLIGHT PLANNING CONTINUATION		CONTINUATION RECORD SAME AS ABOVE								CONT NR	CONTINUATION RECORD SAME AS PRIMARY RECORD (4.2)																																																	
NDB NAVAID (DB) 4.3	S/T	CUST/ AREA	SEC CODE	SUB CODE	·	NDB IDENT	·	ICAO CODE	CONT NR	NDB FREQ	NDB CLASS	NDB LATITUDE			NDB LONGITUDE															MAG VAR	·						NDB NAVAID NAME (30)															FILE RECORD NUMBER	CYCLE
		CONTINUATION RECORD SAME AS ABOVE								CONT NR	·	NOTES ON CONTINUATION RECORD (69)																																															
NDB NAVAID (DB) 4.3.3 SIMULATION CONTINUATION		CONTINUATION RECORD SAME AS ABOVE								CONT NR	USER	•	FAC CHAR																		FAC ELEV																										
NDB NAVAID (DB) 4.3.4 FLIGHT PLANNING CONTINUATION		CONTINUATION RECORD SAME AS ABOVE								CONT NR	USER	FIR IDENT	S/E IND	START/END DATE																																										
NDB NAVAID (DB) 4.3.5 FLIGHT PLANNING CONTINUATION		CONTINUATION RECORD SAME AS ABOVE								CONT NR	CONTINUATION RECORD SAME AS PRIMARY RECORD (4.3)																																																
WAYPOINT (EA) (PC) 4.4	S/T	CUST/ AREA	SEC CODE	SUB CODE	REGN /ARPT CODE	ICAO CODE	SUB CODE	WAYPOINT IDENT	·	ICAO CODE	CONT NR	·	TYPE	USAGE	·	LATITUDE			LONGITUDE												D MAG VAR							NAME/DESC (25)															FILE RECORD NUMBER	CYCLE
		CONTINUATION RECORD SAME AS ABOVE								CONT NR	·	NOTES ON CONTINUATION RECORD (69)																																															
WAYPOINT (EA) (PC) 4.4.3 FLIGHT PLANNING CONTINUATION		CONTINUATION RECORD SAME AS ABOVE								CONT NR	USER	FIR IDENT	S/E IND	START/END DATE																																										
WAYPOINT (EA) (PC) 4.4.4 FLIGHT PLANNING CONTINUATION		CONTINUATION RECORD SAME AS ABOVE								CONT NR	CONTINUATION RECORD SAME AS PRIMARY RECORD (4.4)																																																
HOLDING PATTERNS (EP) 4.5	S/T	CUST/ AREA	SEC CODE	SUB CODE	REGN CODE	·				DUP IDENT	WAYPOINT IDENT	·	ICAO CODE	D/E	CONT NR	CONT APL	HI HOLD OPS	TURN DR	LEG LENGTH	LEG TIME	MINIMUM ALTITUDE																					NOTES (25)															FILE RECORD NUMBER	CYCLE
		CONTINUATION RECORD SAME AS ABOVE													CONT NR	NOTES ON CONTINUATION RECORD (69)																																																

ARINC 424-5 RECORD FORMAT




ARINC 424 5 RECORD FORMAT
Page 3 of 4
RECORD NAME
RUNWAY (PG) 4.10
RUNWAY (PG) 4.10.3 SIMULATION CONTINUATION
ILS (LOCALIZER & G.S.) (PI) 4.11
ILS (LOCALIZER & G.S.) (PI) 4.11.3 SIMULATION CONTINUATION
COMPANY ROUTE (R) 4.12
ILS MARKERS (PM) 4.13
AIRPORT COMM (PV) 4.14
CONTINUATION RECORD SAME AS ABOVE
NOTES ON CONTINUATION RECORD (4P)
RUNWAY DESCRIPTION (22)
FILE RECORD NUMBER
CYCLE

ARINC 424-5 RECORD FORMAT



5.35 NAVAID Class (CLASS)

VHF NAVAID Table is expanded to include ILS/TACAN installations. Note 4 is added.

ORIGINAL TEXT FOLLOWS

Facility	Record Column				
	28	29	30	31	32
VHF Navaid					
VOR	V				
TACAN Channels 17-50 and 70-117		T			
TACAN Channels 1-16 and 60-69		M			
DME		D			
ILS/MDE		I			
Biased ILS/DME				D	
Terminal			T		See
Low Altitude			L		Note
High Altitude			H		1
Unrestricted (see Note 3)			U		

5.39 Magnetic Variation

Delete "Airport Magnetic Variation" from Terminal Waypoint record.

5.54 Longest Runway (LONGEST RWY)

Table deleted. Runway field entry was changed from alpha characters to numerical length in Supplement 5.

ORIGINAL TEXT FOLLOWS

Available Runway	Field Entry
6000 feet and over	A
5000 to 5999 feet	B
4000 to 4999 feet	C
3000 to 3999 feet	D
Less than 3000 feet	E

Figure 5-3 RUNWAY PLAN AND PROFILE

More comprehensive figure replaced previous figure.

5.115 Direction Restriction

New text encompasses use of direction restriction for airway restriction records.

ORIGINAL TEXT FOLLOWS

Definition/Description: The "Direction Restriction" field will be used to indicate the direction an Enroute Airway is to be flown when such restrictions apply.

Source/Content: One Way direction restrictions should be derived from official government sources.

F = One Way in direction Route is coded.
B = One Way in opposite direction Route is coded.
Blank = No restriction on direction.

Used On: Enroute Airways records
Length: 1 character
Character Type: Alpha

5.118 Boundary Via (BDRY VIA)

Definition/Description: The "Boundary Via" field defines the path of the boundary from the position identified in the record to the next position.

Source/Content: The path of the boundary will be determined from official government sources and the "Boundary Via" selected from the following table:

Column 1	Column 2	Description
A		Arc by edge
C		Circle
	E	End of description, return to original point
G		Great Circle
H		Rhumb Line
L		Counter Clockwise Arc
R		Clockwise Arc

Note: Special Use Airspace designated as following rivers, country boundaries or other political boundaries will be averaged using a straight line so that no path will be greater than two nautical miles from the actual boundary.

Used On: FIR/UIR, Restrictive Airspace records
Length: 2 characters
Character Type: Alpha

5.119 Arc Distance (ARC DIST)

Add "C" to the Arc Distance field content list in Source Content paragraph.

5.121 Lower/Upper Limit

Delete "control airspace" from Definition/Description paragraph.

5.127 Maximum Altitude (MAX ALT)

Definition/Description: The "Maximum Altitude" field is used to indicate the maximum altitude allowed.

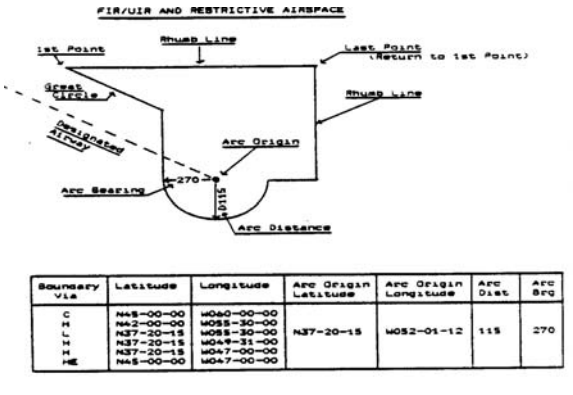
Source/Content: Maximum altitudes should be derived from official government publications describing the upper limit of the airway in feet or flight level.

Used On: Enroute Airway records
Length: 5 characters
Character Type: Alpha/Numeric
Examples: 18000, FL100, FL460

Figure 5-6 FIR/UIR AND RESTRICTIVE AIRSPACE

Revised figure added expanding detail to include shoreline.

ORIGINAL TEXT FOLLOWS



5.131 Time Code (TIME CD)

Clarification added stating that the time code field will be blank wen the active times are only given by NOTAMs.

5.134 Cruise Table Identifier (CRSE TBL IDENT)

Table modified for new field coding. Text modified to include course as either true or magnetic, rather than magnetic only.

ORIGINAL TEXT FOLLOWS

Field Entry	Description
00	ICAO standard cruise table
0A	Exception to ICAO cruise table
10	Modified cruise table
1A	Exception to modified cruise table

5.135 Magnetic TRACK FROM/TO ((MAG TRACK FM/TO)

Text modified to change “magnetic track” to “course.”

ORIGINAL TEXT FOLLOWS

Definition/Description: The “Magnetic Track From” field is used to indicate the lowest track for which a block of cruising levels are prescribed. The “Magnetic Track To” field is used to indicate the highest magnetic track for which a block of cruising levels are prescribed.

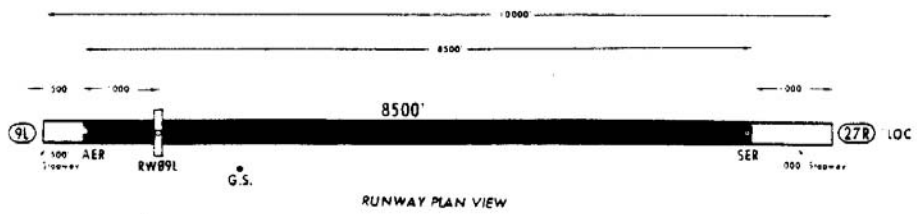
Source/Content: The Magnetic Tracks will be derived from official government sources in degrees and tenths of degree with the decimal point suppressed.

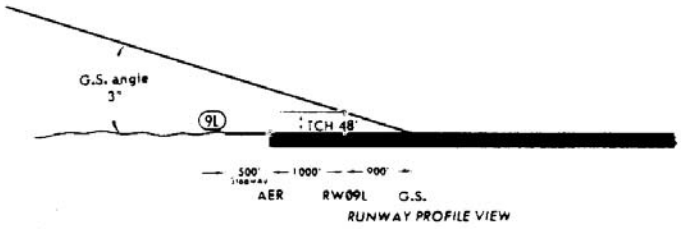
5.136 Cruise Level From/To

Definition/Description revised to complement the use of course in place of magnetic track in Section 5.135.

Figure 5-3

RUNWAY DESCRIPTION





AER-Approach end of runway
N39°45'18.43" W104°53'42.96"

RW09L Landing threshold for runway 9L
N39°45'18.43" W104°53'28.37"

G.S.-Glide Slope
N39°45'15.96" W104°53'14.08"
G.S. Dist from AER- 1900'

TCH- G.S. altitude above landing threshold 48'

SER- Stop end of runway
N39°45'18.43" W104°51'38.72"

LOC-Localizer
N39°45'18.43" W104°51'22.19"
Loc Dist from SER-1300'

Landing distance beyond

	Threshold	Glide Slope
9L	7500'	6600'
27R	8500'	

Landing threshold elevation is elevation at RW09L

ORIGINAL TEXT FOLLOWS

Definition/Description: The "Cruise Level From" field is used to indicate the lowest cruising level prescribed for use within the Magnetic Track From/To fields. The "Cruise Level To" field is used to indicate the highest cruising level prescribed for use within the Magnetic Track From/To fields.

5.139 Time Group

Examples expanded to include references to sunrise and sunset.

5.151 FIR/UIR Address (ADDRESS)

Source/Content revised to define two letter designators to be used when addressing ATS messages.

ORIGINAL TEXT FOLLOWS

Source/Content: Addresses of FIR/UIR will be derived from official government sources and entered in this field.

5.154 Restriction Identifier (REST IDENT)

New field added to define airway restriction record.

5.155 Airway Restriction Note Indicator (NOTE IND)

New field added to define airway restriction record.

5.156 Active Indicator (ACT IND)

New field added to define airway restriction record.

5.157 Airway Restriction Start/End Date (START/END DATE)

New field added to define airway restriction record.

5.158 Airway Restriction Start/End Time (START/END TIME)

New field added to define airway restriction record.

5.159 Weekly Frequency (WEEKLY FREQ)

New field added to define airway restriction record.

5.160 Units of Altitude (UNIT IND)

New field added to define airway restriction record.

5.161 Restriction Altitude (REST ALT)

New field added to define airway restriction record.

5.162 Step Climb Indicator (STEP)

New field added to define airway restriction record.

5.163 Restriction Notes

New field added to define airway restriction record.

5.164 EU Indicator (EU IND)

New field added to define airway restriction record.

5.165 Magnetic/True Indicator (M/T IND)

New field added to define airway restriction record.

ATTACHMENT 2 – WAYPOINT IDENTIFIERS

B. NAMED RNAV Waypoints, Intersections and Reporting Points

The description, addressing only the United States, has been expanded to describe rules covering the naming of points worldwide.

ORIGINAL TEXT FOLLOWS

B. NAMED RNAV Waypoints, Intersections and Reporting Points.

In the United States, these waypoints will be assigned unique 5-character names by the FAA. As the FAA assigns names to new or established waypoints, the identifier will be the same as the name. For all other waypoints, identifiers should be developed using the following rules sequentially until 5-character (or fewer) groups emerge.

B.3 Phonetic Letter Names

The use of the minus (-) sign is deleted from this section.

ORIGINAL TEXT FOLLOWS

3. Phonetic Letter Names

When an ICAO phonetic alpha character is used as a waypoint name (Alpha, Bravo, Charlie, etc.), use the equivalent letter (A, B, C, etc.) followed by a minus (-) sign and the two-letter ICAO code for the country in which the waypoint is located to produce a 4-character identification code. When more than one waypoint in a country has the same phonetic name, obtain uniqueness by adding a numeric character after ICAO country code.

D. Un-Named Turn Points and Intersections

This section is expanded to include bearing/distance waypoints.

F. Terminal Waypoints

F.2 Turn Points

This section deleted. Section F.3 renumbered F.2.

ORIGINAL TEXT FOLLOWS

F. Terminal Waypoints

2. Turn Points

Special turn points required to establish a procedural turn should be assigned identifiers in numerical sequence. The number should be preceded by the characters "TP." Examples: TP1, TP2, TP3.

ATTACHMENT 3 – NAVIGATION CHART/FILE DATA RELATIONSHIP

These figures showing examples are unnecessarily complex. These figures have been replaced with simplified "generic" examples. The old text is not included in this Supplement.

ATTACHMENT 4 – AIRWAY MINIMUM ALTITUDES

Text and figures showing examples are obsolete. These figures have been replaced with simplified "generic" examples which will not require updating. The old text is not included in this Supplement.

ATTACHMENT 5 – PATH AND TERMINATOR

The first paragraph is expanded to indicate the precedence of specific rules over general rules for path terminator assignments.

The first two sentences of the second paragraph are expanded to clarify the selection and use of path terminators.

ORIGINAL TEXT FOLLOWS

PATH AND TERMINATOR

Path Terminators are assigned to all SID/STAR/Approach records in accordance with the rules set forth in this attachment. The general intent of the Path Terminator concept is to permit the coding of standard terminal are procedures without proliferating the number of named waypoints required to support the procedures.

To avoid proliferation of leg types to be implemented within any given system, it is desirable that systems be designed to accept either VX or CX leg types, but not necessarily both. Selection could then be made by system type or area of operation, and the relevant data base coded accordingly. In order to achieve this and to ultimately simplify the path/terminator matrix currently required to define present-day terminal area procedures, it is in the interest of all user airlines to prevail upon their government agencies and ATC authorities to: (1) permit FMS-equipped airplanes to fly tracks instead of procedural headings, and (2) design terminal area procedures to be compatible with the capabilities of increasing number of FMS-equipped airplanes entering service.

5. Intercept Angles

Add new section.

ATTACHMENT 5 – PATH AND TERMINATOR GENERAL RULES

Rule A.

CI coding is deleted. CD and CR coding is added.

ORIGINAL TEXT FOLLOWS

GENERAL RULES

A. The following leg types are NAVAID oriented and the defining parameters will be found in the RECD, VHF, RHO, THETA, OB MAG CRS, and WAYPINT IDENT fields:

AF	CI	FC	FM	VD
CF	FA	FD	PI	VR

Rule N.

The rule is expanded to state that turn direction will be indicated if the turn equals or exceeds 90 degrees.

Rule V.

Delete CI.

ORIGINAL TEXT FOLLOWS

V. When a CI or TI leg is followed by an AF leg, the "course" or "track to" must be to the DME which defines the AF arc.

Rule W.

Add subsection AC, AD and AE.

ATTACHMENT 5 – PATH AND TERMINATORS SID CODING RULES

Rule A.

Revise Rule A to clarify text on CA, VA and FA coding and to state that course legs are preferred over heading legs as a first leg of a SID transition.

ORIGINAL TEXT FOLLOWS

SID CODING RULES

A. If on take-off there is a turn greater than 15° without an altitude specified before the turn, a course from or a heading to an altitude (CA, FA or VA) leg will be coded before the turn, using runway heading for the VA leg terminating at 400 feet above the airport elevation. This altitude may vary with local controlling agency requirements.

ATTACHMENT 5 – PATH AND TERMINATOR APPROACHES AND APPROACH TRANSITIONS

A. General

Add paragraph 10.

B. VOR Approach

Revise Rule 2 to indicate government specified, non-FMS procedures.

ORIGINAL TEXT FOLLOWS

B. VOR Approach Coding

2. The recommended VHF will be the procedure specified NAVAID.

ATTACHMENT 5 – Rule G paragraph 2

Revised to indicate the preference of a course leg over a heading leg as the first leg of a missed approach.

ORIGINAL TEXT FOLLOWS

G. Missed Approaches

2. The first leg of the missed approach will contain an altitude to command a climb. If a turn in excess of 15° from runway heading is required, then a CA, VA or FA leg on the runway heading to 400 feet above the airport elevation will be the first leg of the missed approach followed by the required turn.

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SUPPLEMENT 7

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: January 14, 1987

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: October 8, 1986

A. PURPOSE OF THIS SUPPLEMENT

This Supplement incorporates microwave landing system (MLS) records into ARINC 424. Changes to ARINC 424 are also included to improve the operational utility of the navigation data base. New rules or clarifications to existing rules are provided by this Supplement.

B. ORGANIZATION OF THIS SUPPLEMENT

The first part of this Supplement printed on goldenrod colored paper, contains descriptions of the changes introduced into the Specification by this Supplement. The second part consists of replacement white pages for the Specification modified to reflect these changes. The modified and added material on each replacement page is identified with c-7 symbols in the margins.

Existing copies of Specification 424-6 may be updated by inserting the replacement pages where necessary and destroying the pages they displace. The goldenrod colored Supplement should be inserted inside the rear cover of the Specification following Supplement 6.

C. CHANGES TO ARINC SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This Section presents a complete tabulation of the changes and additions to the Specification introduced by this Supplement. Each change is identified using the Section number and title that will be employed when the Supplement is incorporated. In each case there is included a brief description of the addition or change and, for other than very minor revisions, any text originally contained in the Specification is reproduced for reference.

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SUPPLEMENT 8

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: October 15, 1989

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: February 26, 1989

A. PURPOSE OF THIS SUPPLEMENT

This Supplement incorporates revision of the current Airport Communications record and the addition of Enroute 424. Changes to ARINC Specification 424 are also included to improve the operational utility of the navigation data base. New rules or clarifications to existing rules are provided by this Supplement

B. ORGANIZATION OF THIS SUPPLEMENT

The first part of this Supplement printed on goldenrod colored paper, contains descriptions of the changes introduced into the Specification by this Supplement. The second part consists of replacement white pages for the Specification modified to reflect these changes. The modified and added material on each replacement page is identified with c-8 symbols in the margins.

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C. CHANGES TO ARINC SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This Section presents a complete tabulation of the changes and additions to the Specification introduced by this Supplement. Each change is identified using the Section number and title that will be employed when the Supplement is incorporated. In each case there is included a brief description of the addition or change and, for other than very minor revisions, any text originally contained in the Specification is reproduced for reference.

2.2 Special Navigation Terms

Editorial correction of EA to ER in the Enroute Airway to Restrictive Airspace Link definition.

3.2.2.1 VHF NAVAID

MLS DMEs added to descriptive paragraph.

3.2.3.6 Enroute Communication Section

Add new section.

3.2.4.6 Airport Approach Section (P)

New table added after first paragraph.

3.2.4.11 MSA Section (P)

Statement paragraph revised.

3.2.9 Heliport Section (H)

Add new section.

4.2.4 Flight Planning Continuation Records

Add the UIR Identifier field to the VHF NAVAIDS Flight Planning Continuation record.

4.3.1 Primary Records

Add the Airport Identifier and Airport ICAO fields to the NDB NAVAID Primary record.

4.3.4 Flight Planning Continuation Records

Add the UIR Identifier field to the NDB NAVAID Flight Planning Continuation record.

4.4.3 Flight Planning Continuation Records

Add the UIR identifier field to the Waypoint Flight Planning Continuation record.

4.6.1 Primary Records (ER)

Deleted the reference paragraph number shown for Column 83. Note under box to read Note 1.

4.7.1 Primary Records (PA)

New fields added.

4.7.3 Flight Planning Continuation Records

Add the UIR Identifier field to the Airport Flight Planning Continuation record.

4.9.1 SID/STAR/Approach Primary Records

Column 124 through 128 revised to read File Record Number (5).

4.12.1 Primary Records (R)

Columns 15, 16, 25, 26, 60, 61 and 82 through 83 revised.

4.14 Airport Communications Records (PV)

Replaced Airport Communications Records Section.

4.20.1 Primary Records

Columns 21 and 73 through 123 revised.

4.23 Enroute Communications Records (EV)

New Enroute Communications record added.

4.24 Heliport Records (HA)

New Heliport record added.

Table 4-1 Record Format

New records and new fields added to reflect changes in Section 4.

5.5 Subsection Code

New Subsection Code for Enroute Communications added.

5.6 Airport Identifier (ARPT IDENT)

Title revised to Airport/Heliport Identifier (ARPT/HELI IDENT). Used On revised to contain Airport and Heliport Identifier categories. NDB Navaid Records added to Used

On Airport Identifier category. Heliport Records added to Used On Heliport Identifier category.

5.15 File Code

New file code added for Approach records. Deleted File code for Company Route records.

5.17 Waypoint Description Code

New Waypoint Description for NDB Navaids added.

5.23 Recommended NAVAID

Definition/Description, Used On and Table 5-1 revised.

5.24 Theta

Definition/Description and final note revised. Added new example.

5.25 Rho

Added new example.

5.29 Altitude Descript

First note and Used On revised.

5.30 Altitude/Minimum Altitude

Source/Content revised.

5.36 Latitude

Enroute Communication and Heliport records added to Used On.

5.37 Longitude

Enroute Communication and Heliport records added to Used On. New Heliport entry added to table.

5.39 Magnetic Variation

Enroute Communication and Heliport records added to Used On.

5.42 Waypoint Type

Enroute Waypoint Table revised. Used On, Length and Character Type added.

5.44 Localizer/MLS Identifier (LOC IDENT) (MLS IDENT)

Editorial correction to title.

5.53 Transition Altitude/Level (TRANS ALTITUDE/LEVEL)

Heliport records added to Source/Content and Used On paragraphs.

5.55 Airport Elevation (ELEV)

Title revised to Airport/Heliport Elevation (ELEV). Definition/Description, Source/Content and Used On revised to include Heliports.

5.66 Station Declination

Examples revised.

5.71 Name Field

Heliport records added to Used On.

5.72 Speed Limit

Heliport records added to Used On.

5.73 Speed Limit Altitude

Heliport records added to Used On.

5.92 Facility Elevation

Enroute Communications records added to Used On.

5.94 True Bearing

ILS records revised to read ILS Continuation records in Used On.

5.101 Communications Type

Enroute Communications records added to Source/Content table and Used On.

5.102 Radar

Enroute Communications records added to Source/Content table and Used On.

5.103 Communications Frequency

Section revised to include Enroute Communications records.

5.104 Modulation

Enroute Communications records added to Source/Content table and Used On.

5.105 Call Sign

Section revised to include Enroute Communications records.

5.106 Additional Service (ADDL SERVICE)

Title revised to Service Indicator (SER IND). Entire section revised.

5.108 IFR Capability (IFR)

Heliport records added to Definition/Description and Used On.

5.115 Direction Restriction

Source/Content revised with editorial correction.

5.116 FIR/UIR Identifier

Added text to Source/Content. Heliport records added to Used On.

5.117 FIR/UIR Indicator

Enroute Communications records added to Used On.

5.127 Maximum Altitude

Added new example.

5.130 Multiple Code

Source/Content revised.

5.151 FIR/UIR Address

Enroute Communications records added to Used On.

5.152 Start/End Indicator (S/E IND)

Heliport records added to Used On.

5.153 Start/End Date

Heliport records added to Used On.

The following Sections added by this Supplement:

5.176 Pad Dimensions

5.177 Public/Military Indicator (PUB/MIL)

5.178 Time Zone

5.179 Daylight Time Indicator (DAY TIME)

5.180 PAD Identifier (PAD IDENT)

5.181 H24 Indicator (H24)

5.182 Guard/Transmit (G/T)

5.183 Sectorization (SECTOR)

5.184 Communication Altitude (COMM ALTITUDE)

5.185 Associated Facility (ACCOS FAC)

5.186 Narrative

5.187 Distance Description (DIST DESC)

5.188 Communications Distance (COMM DIST)

5.189 Remote Site Name

5.190 FIR/RDO Identifier (FIR/RDO)

6.7 Labels

Add paragraph clarifying the use of the EOV label type.

6.9 Summary of Tape Data Layout

Add new section.

ATTACHMENT 2 - WAYPOINT IDENTIFIERS

Rule D.1.

Text added to define unnamed turn points, intersections and bearing/distance waypoints of distance equal to or greater than 100 nautical miles.

Rule D.3.

Reporting positions defined by coordinates paragraphs a. and b. revised.

ATTACHMENT 5 - PATH AND TERMINATOR GENERAL RULES

Rule B.

Table under Rule B revised to allow for IF ending leg for STAR Route Type 2, 5 or 8 if there is only one leg.

Rule C.

Table 1 - Leg Sequence

Revised due to the removal of the TI leg.

Rules O. through AB.

Inserted due to editorial omission in Supplement 7.

Table 2 - Leg Type Illustrations

Illustrations revised due to the removal of the TI leg.

Table 3 - Leg Data Fields

Revised due to the removal of the TI leg.

ATTACHMENT 5 - PATH AND TERMINATOR

APPROACHES AND APPROACH TRANSITIONS

Rule H.3.a.

Range for vertical angle revised to -3.00 through -3.77 degrees.

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SUPPLEMENT 9

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: March 2, 1990

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: October 26, 1989

A. PURPOSE OF THIS SUPPLEMENT

This Supplement incorporates the definition of Preferred Routes, LORAN based approach procedures and Non-Precision VOR based approach procedures into ARINC Specification 424. Changes to ARINC Specification 424 are also included to improve the operational utility of the navigation data base. New rules or clarifications to existing rules are provided by this Supplement.

B. ORGANIZATION OF THIS SUPPLEMENT

The first part of this Supplement printed on goldenrod colored paper, contains descriptions of the changes introduced into the Specification by this Supplement. The second part consists of replacement white pages for the Specification modified to reflect these changes. The modified and added material on each replacement page is identified with c-9 symbols in the margins.

Existing copies of Specification 424-8 may be updated by inserting the replacement pages where necessary and destroying the pages they displace. The goldenrod colored Supplement should be inserted inside the rear cover of the Specification following Supplement 8.

C. CHANGES TO ARINC SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This Section presents a complete tabulation of the changes and additions to the Specification introduced by this Supplement. Each change is identified using the Section number and title that will be employed when the Supplement is incorporated. In each case there is included a brief description of the addition or change and, for other than very minor revisions, any text originally contained in the Specification is reproduced for reference.

1.1.2 Coverage of Flight Planning Needs

Add new section.

3.2.2.1 VHF NAVAID Section (D) - Subsection (D)

Statement paragraph revised.

3.2.2.2 Enroute NDB NAVAID Section (D) – Subsection (DB)

Statement paragraph revised.

3.2.4.6 Airport Approaches Section (P) - Subsection (PF)

Statement paragraph revised.

3.2.4.13 Airport Terminal NDB Section (P) - Subsection (PN)

Add new section.

3.2.10 Preferred Routes Section (E) - Subsection (ET)

Add new section.

4.2.1 Primary Record VHF NAVAIDS

Delete editorial text prior to record definition.

4.3 NDB NAVAID Record (DB)

Title revised to NDB NAVAID Record (DB or PN). Statement paragraph revised.

4.4 Waypoint Records (EA or PC)

Statement paragraph revised.

4.4.1 Primary Records

Revise Waypoint Record to add Name Format Identifier to the Primary records.

4.5 Holding Pattern Records (EP)

Statement paragraph revised.

4.5.1 Primary Records (EP)

Revise Holding Pattern Primary records to add Section/Subsection code in place of File Code. Delete File Code.

4.6.1 Primary Records (ER)

Revise Enroute Airway Primary records to add Section/Subsection code in place of File Code. Delete File Code.

4.8.2 Continuation Records

Revise Gate Continuation record. Editorial correction to length field.

4.9.1 Primary Records

Revise SID/STAR/Approach Primary records to add Section/Subsection code in place of File Code. Delete File Code.

4.9.2 SID/STAR/Approach Continuation Records

Add fields for encoding of LORAN data.

4.13 ILS Marker (PM)

Statement paragraph revised.

4.14.3 Additional Continuation Records

Add Airport Communication Additional Continuation record to permit use of Time of Operation field.

4.17.1 Primary Records

Revise the FIR/UIR record to change the FIR/UIR Address from 2 to 4 characters. Data contained after column 12 through 98 is moved 2 columns to accommodate the revision.

4.18.2 Continuation Records

Revise Restrictive Airspace Continuation record to permit use of Time of Operation field.

4.18.3 Flight Planning Continuation Record

Add Restrictive Airspace Flight Planning Continuation record to provide Start/End Indicator and Date for Time of Operation field contained in 4.18.2 Continuation Records.

4.20.1 Primary Records

Revise MSA (Minimum Sector Altitude) Primary records to add Section/Subsection code in place of File Code. Delete File Code.

4.21.1 Primary Records

Revise Enroute Airways Restriction Primary records to add Section/Subsection code in place of File Code. Delete File Code.

4.23 Enroute Communications Records (EV)

Revise FIR/UIR Address from 2 to 4 characters. Revise Remote Site Name length from 30 to 25 characters to accommodate the revision.

4.23.2 Continuation Records

Enroute Communications Continuation record revised for definition of Time of Operation.

4.23.3 Continuation Records

Enroute Communications Continuation record added for definition of Time of Operation.

4.25 Preferred Routes (ET)

Add new section.

4.25.1 Primary Records

Add new record.

4.25.2 Continuation Records

Add new record.

Table 4-1 ARINC 424-9 RECORD FORMAT

Table revised to reflect changes introduced in Supplement 9.

5.5 Subsection Code

Revise Definition/Description. Add Preferred Routes and Airport Terminal NDBs to table (i) Section and Subsection Encoding Scheme.

5.7 Route Type (RT TYPE)

Revise Definition/Description. Add Preferred Route Records (ET) table under Source/Content. Revise Approach Route Records (PF) table.

5.8 Route Identifier (ROUTE IDENT)

Section revised for accommodation of Preferred Routes.

5.9 SID/STAR Route Identifier (SID/STAR IDENT)

Revise Definition/Description to reference new Chapter 7.0 NAMING CONVENTIONS.

5.10 Approach Route Identifier (APPROACH IDENT)

Revise Source/Content and Examples for accommodation of Circling Approaches.

5.12 Sequence Number (SEQ NR)

Revise Length to add Preferred Routes.

5.15 File Code

The File Code is being removed from all records. Revise title to Intentionally left blank and delete remaining text.

5.17 Waypoint Description Code (DESC CODE)

Add Runway Centerline Fix entry to Table (iii) Waypoint Description Codes to add Non-Precision VOR based approach procedures.

5.18 Boundary Code (BDY CODE)

Revise South Pacific Area Boundary Code "I" to "1."

5.19 Level (LEVEL)

Revise Used On to add Preferred Routes and Restrictive Airspace records.

5.23 Recommended Navaid (RECD NAV)

Revise section to add Non-Precision VOR based approach procedures.

5.26 Outbound Magnetic Course (OB MAG CRS)

Revise Definition/Description and Source/Content.

5.28 Inbound Magnetic Course (IB MAG CRS)

Revise Definition/Description and Source/Content.

5.29 Altitude Descript (ALT DESC)

Add "R" descriptor to Source/Content table to add encoding of Non-Precision VOR based procedures.

5.30 Altitude/Minimum Altitude

Revise Definition/Description, Source/Content and Used On to add Preferred Routes.

5.39 Magnetic Variation (MAG VAR)

Revise Source/Content to add records which are oriented to TRUE North.

5.43 Waypoint Name/Description (NAME/DESC)

Revise Source/Content to reference new Chapter 7.0 NAMING CONVENTIONS. Revise Examples.

5.47 Localizer Bearing (LOC BRG)

Revise Source/Content and Examples to add use of localizer courses charted with true courses.

5.58 Runway Magnetic Bearing (RWY BRG)

Revise Source/Content and Examples to add use of runway bearings charted with true bearings.

5.62 Inbound Holding Course (IB HOLD CRS)

Revise Source/Content and Examples to add use of holding courses charted with true bearings.

5.77 VIA

Revise section to add Preferred Routes.

5.78 SID/STAR/App/Awy (S/S/A/AWY)

Revise title to SID/STAR/App/Awy (S/S/A/AWY) SID/STAR/Awy (S/S/AWY)

Revise section to add Preferred Routes.

5.83 To Fix

Revise section to add Preferred Routes.

5.106 Service Indicator (SER IND)

Revise title to Service Indicator (SERV IND). Delete Source/Content Airport Communications Records Terminal Control Area Group I and II coding.

5.114 Duplicate Identifier

Revise Source/Content and Examples to add Holding Pattern Types.

5.127 Maximum Altitude (MAX ALT)

Revise Used On to add Preferred Routes.

5.131 Time Code (TIME CD)

Revise to permit use of Time of Operation field.

5.138 Time Group

Revise Used On to include Restrictive Airspace Continuation and Preferred Route Continuation records only.

5.139 Time Group

The Time Group field is being replaced by Section 5.195 Time of Operation. Revise title to Intentionally left blank and delete remaining text.

5.151 FIR/UIR Address (ADDRESS)

Revise section to incorporate ICAO Document 8585 four character addressing scheme. Revised Used On to include Enroute Communications records.

5.189 Remote Site Name

Revise Length to 25 characters.

The following new Sections added by this Supplement:

5.191 Triad Stations (TRIAD STA)

5.192 Group Repetition Interval (GRI)

5.193 Additional Secondary Phase Factor (ASF)

5.194 Initial/Terminus Airport/Fix

5.195 Time of Operation

5.196 Name Format Indicator (NAME IND)

6.3 Bit Density

Revise text to add 6250 bits per inch bit density capability. Add text requiring external reel label to identify bit density used.

6.4 Coding

Revise text to permit encoding of data using American Standard Code for Information Interchange (ASCII) capability. Add text requiring external reel label to identify encoding standard used.

7.0 NAMING CONVENTIONS

Add new chapter.

ATTACHMENT 2 - WAYPOINT IDENTIFIERS

Revise title to ATTACHMENT 2 - INTENTIONALLY LEFT BLANK. Delete Attachment. Material previously contained in Attachment 2 is now located in Chapter 7.0 Naming Conventions.

ATTACHMENT 5 - PATH AND TERMINATOR GENERAL RULES

Revise the table under Rule B to permit "FM" and "VM" as an ending leg for STAR Runway Transition (Route Type 3, 6 or 9).

ATTACHMENT 5 - PATH AND TERMINATOR APPROACHES AND APPROACH TRANSITIONS

A. GENERAL

Revise Rule 2 for Circling Approaches.

Revise Rule 5 to clarify use of Approach Step-Down Fixes.

Add Rule 11 to permit use of ILS procedures for design of MLS approach.

Add Rule 12 for Non-Precision VOR based approaches.

B. VOR, VORDME, VORTAC AND TACAN Approach Procedure Coding

Add Rule 1.e for Non-Precision VOR based approaches.

Add Rule 1.f for Circling Approaches.

Add Rule 2.e for Non-Precision VOR based approaches.

Add Rule 2.f for Circling Approaches.

Revise Rule 3.a for Non-Precision VOR based approaches.

Add Rule 3.c for Circling Approaches.

C. RNAV and LORAN Approach Procedures Coding

Revise Section C to include LORAN approach procedures.

D. Localizer and Back Course Approach Procedure Coding

Revise Rule 5 to remove vertical limits. Specify only that a vertical angle will be contained in the last point in the final approach fix.

H. NDB Approach Procedure Coding

Add Rule 1.e for Non-Precision VOR based approaches.

Add Rule 1.f for Circling Approaches.

Add Rule 2.e for Non-Precision VOR based approaches.

Add Rule 2.f for Circling Approaches.

Revise Rule 3.a for Non-Precision VOR based approaches.

Add Rule 3.c for Circling Approaches.

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SUPPLEMENT 10

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: August 20, 1993

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: August 16, 1993

A PURPOSE OF THIS DOCUMENT

This Supplement introduces changes to ARINC 424 to improve the operational utility of the navigation data base.

B. ORGANIZATION OF THIS DOCUMENT

The first part of this Supplement printed on goldenrod colored paper, contains descriptions of the changes introduced into the Specification by this Supplement. The second part consists of replacement white pages for the Specification modified to reflect these changes. The modified and added material on each replacement page is identified with c-10 symbols in the margins.

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C. CHANGES TO ARINC SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This section presents a complete tabulation of the changes and additions to the Specification to be introduced by this Supplement. Each change or addition is defined by the section number and the title that will be employed when the Supplement is eventually incorporated. In each case a brief description of the change or addition is included.

2.2 Special Navigation Terms

Definition for Localizer added.

3.2.2.1 VHF NAVAID Section (D) - Subsection (D)

Section revised.

3.2.3.3 Enroute Holding Patterns Section (E) - Subsection (EP)

Section renamed Holding Patterns (E) - Subsection (EP). Section revised.

3.2.4.8 Airport ILS Section (P) - Subsection (PI)

Section renamed Airport Localizer Section (P) - Subsection (PI). Section revised.

3.2.4.10 Airport ILS Markers Section (P) - Subsection (PM)

Section renamed Airport Localizer Markers Section (P) - Subsection (PM). Section revised.

3.2.4.13 Airport Terminal NDB Section (P) - Subsection (PN)

Section revised.

4.2 VHF NAVAID Record (D)

Section revised.

4.2.1 Primary Records

Added Datum Code.

4.2.4 Flight Planning Continuation Records

Section revised.

4.3 NDB NAVAID Record

Section revised.

4.3.1 Primary Records

Added Datum Code.

4.3.4 Flight Planning Continuation Records

Section revised. FIR/UIR Identifier revised to FIR Identifier.

4.4.1 Primary Records

Added references to Note 1 and Note 2. Added Datum Code. Note 2 revised.

4.4.3 Flight Planning Continuation Records

Section revised.

4.5 Holding Pattern Records (EP)

Section revised.

4.5.1 Primary Records

Added ICAO Code following Region Code. Added references to Note 1. Added Note 1.

4.7.1 Primary Records

Added Magnetic/True Indicator and Datum Code.

4.7.3 Flight Planning Continuation Records

Section revised.

4.11 ILS (Localizer and Glide Slope) Records (PI)

Section renamed Localizer and Glide Slope Records (PI).

Section revised.

4.11.2 Continuation Records

Removed Note (i). Renamed Note (ii) to Note (i).

4.13 ILS Marker (PM)

Section renamed Localizer Marker (PM). Section revised.

4.14.1 Primary Records

Added Frequency Units, Signal Emission, Sector Facility, Remote Facility, ICAO Code, Section Code and Subsection Code. Revised Call Sign length.

4.15.1 Primary Records

Added Datum Code.

4.17 FIR/UIR Records (UF)

Section Revised.

4.17.1 Primary Records

Editorial corrections.

4.20.1 Primary Records

Added Magnetic/True Indicator.

4.21 Enroute Airways Restriction Records (EU)

Section revised.

4.21.1 Primary Records

Added provisions for new Enroute Airways Restriction Record types.

4.21.1 AAltitude Exclusion Record (AE)

Added new record.

4.21.2 AAE Continuation Records

Added new record.

4.21.1B Note Restriction Record (NR)

Added new record.

4.21.2B NR Continuation Record

Added new record.

4.21.1C Seasonal Closure Record (SC)

Added new record.

4.21.1D Cruising Table Replacement Record (TC)

Added new record.

4.21.2D TC Continuation Record

Added new record.

4.21.3 Continuation Records

Record removed.

4.23.1 Primary Records

Added Frequency Units and Signal Emission. Renamed Associated Facility to Remote Facility.

4.24.1 Primary Records

Added Datum Code and Magnetic/True Indicator.

4.26 Heliport Communiations Records (HV)

Added new record and associated continuation records.

Table 4-1 ARINC 424-10 RECORD FORMAT

Table revised to reflect changes introduced in Supplement 10.

5.1 General

Section revised.

5.3 Customer/Area Code (CUST/AREA)

Section revised.

5.5 Subsection Code

Section revised.

5.6 Airport/Heliport Identifier (ARPT/HELI IDENT)

Section revised.

5.7 Route Type (RT TYPE)

Section revised.

5.11 Transition Identifier (TRANS IDENT)

Section revised.

5.13 Fix Identifier (FIX IDENT)

Section revised.

5.14 ICAO Code (ICAO CODE)

Section revised.

5.16 Continuation Record Number (CONT NR)

Section revised.

5.20 Turn Direction (TURN DIR)

Section revised.

5.27.1 Route Distance From, Holding Distance/Time (RTE DIST FROM, HOLD DIST/TIME)

Section revised.

5.29 Altitude Descript (ATL DESC)

Section revised.

5.30 Altitude/Minimum Altitude

Section revised.

5.33 VOR/NDB Identifier (VOR IDENT/NDB IDENT)

Section revised.

5.34 VOR/NDB Frequency (VOR/NDB FREQ)

Section revised.

5.35 NAVAID Class (CLASS)

Section revised.

5.37 Longitude (LONGITUDE)

Section revised.

5.38 DME Identifier (DME IDENT)

Section revised.

5.39 Magnetic Variation (MAG VAR)

Section revised.

5.41 Region Code (REGN CODE)

Section revised.

5.42 Waypoint Type (TYPE)

Section revised.

5.46 Runway Identifier (RUNWAY IDENT)

Section revised.

5.69 Threshold Displacement Distance (DSPLCD THR)

Section revised.

5.72 Speed Limit (SPEED LIMIT)

Section revised.

5.77 VIA Code

Section revised.

5.83 To FIX

Section revised.

5.90 ILS/DME Bias

Section revised.

5.99 Marker Type (MKR TYPE)

Section revised.

5.100 Minor Axis Bearing (MINOR AXIS TRUE BRG)

Section revised.

5.101 Communications Type (COMM TYPE)

Section revised.

5.102 Radar (RADAR)

Section revised.

5.103 Communications Frequency (COMM FREQ)

Section revised.

5.104 Modulation (MODLN)

Section renamed Frequency Units (FREQ UNIT). Section revised.

5.105 Call Sign (CALL SIGN)

Section revised.

5.106 Service Indicator (SERV IND)

Section revised.

5.114 Duplicate Indicator (DUP IND)

Section revised.

5.116 FIR/UIR Identifier (FIR/UIR IDENT)

Section revised.

5.131 Time Code (TIME CD)

Section revised.

5.137 Vertical Separation

Section revised.

5.138 Time Indicator (TIME IND)

Section revised.

5.144 MSA Center

Section revised.

5.151 FIR/UIR Address (ADDRESS)

Section revised.

5.155 Airway Restriction Note Indicator (NOTE IND)

Section removed.

5.156 Active Indicator (ACT IND)

Section removed.

5.157 Airway Restriction Start/End Date (START/END DATE)

Section revised.

5.158 Airway Restriction Start/End Time (START/END TIME)

Section removed.

5.159 Weekly Frequency (WEEKLY FREQ)

Section removed.

5.165 Magnetic/True Indicator (M/T IND)

Section revised.

5.167 MLS Azimuth Bearing (MLS AZ BRG)

Section revised.

5.172 Azimuth Coverage Sector (Right/Left) (AZ COV RIGHT/LEFT)

Back Azimuth Coverage Sector (Right/Left) (BAZ COV RIGHT/LEFT)

Editorial correction.

5.178 Time Zone

Section revised.

5.185 Associated Facility (ACCOS FAC)

Section renamed Sector Facility (SEC FAC). Section revised.

5.187 Distance Description (DIST DESC)

Section revised.

5.188 Communications Distance (COMM DIST)

Section revised.

5.191 Triad Stations (TRIAD STA)

Section revised.

5.192 Group Repitition Interval (GRI)

Section revised.

5.193 Additional Secondary Phase Factor (ASF)

Section revised.

5.194 Initial/Terminus Airport/Fix

Section revised.

5.197 Datum Code (DATUM)

Added section.

5.198 Modulation (MODULN)

Added section.

5.199 Signal Emission (SIG EM)

Added section.

5.200 Remote Facility (REM FAC)

Added section.

5.201 Restriction Record Type (REST TYPE)

Added section.

5.202 Exclusion Indicator (EXC IND)

Added section.

5.203 Block Indicator (BLOCK IND)

Added section.

Figure 5-6 Company Route Record (R) Field Content

Renamed Figure 5-7.

6.1 General

Section revised.

6.2 Number of Tape Tracks

Section revised.

6.5 Parity Convention

Section revised.

6.6 Reel-File Relationship

Section revised.

6.7.1 Volume Header Label (VOL)

Section revised.

6.9.1 One File, One Reel

Added section.

6.9.2 One File, Multiple Reels

Added section.

6.9.3 Multiple Files, One Reel

Added section.

7.2.6 Terminal Waypoints

Section revised.

7.4 SID/STAR Procedure Identifiers

Section revised.

7.6 Transition Identifiers

Added section.

ATTACHMENT 2 - LOCAL HORIZONTAL REFERENCE DATUM NAME, DATUM CODE AND ELLIPSOID LIST

Added attachment.

ATTACHMENT 3 - NAVIGATION CHART/FILE DATA RELATIONSHIP

Charts removed. Attachment revised to reflect changes introduced in Supplement 10.

ATTACHMENT 4 - AIRWAY MINIMUM ALTITUDES

Attachment revised.

ATTACHMENT 5 - PATH AND TERMINATOR

Attachment revised.

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SUPPLEMENT 11

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: August 20, 1993

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: August 16, 1993

A. PURPOSE OF THIS DOCUMENT

This Supplement introduces changes necessary to implement curved path approaches.

B. ORGANIZATION OF THIS DOCUMENT

The first part of this Supplement printed on goldenrod colored paper, contains descriptions of the changes introduced into the Specification by this Supplement. The second part consists of replacement white pages for the Specification modified to reflect these changes. The modified and added material on each replacement page is identified with c-11 symbols in the margins.

Existing copies of Specification 424-9 (Supplement 10 changes are being added simultaneously) may be updated by inserting the replacement pages where necessary and destroying the pages they displace. The goldenrod colored Supplement should be inserted inside the rear cover of the Specification following Supplement 9.

C. CHANGES TO SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This section presents a complete tabulation of the changes and additions to the Specification to be introduced by this Supplement. Each change or addition is defined by the section number and the title that will be employed when the Supplement is eventually incorporated. In each case a brief description of the change or addition is included.

2.2 Special Navigation Terms

Definition for Precision ARC added.

4.9 SID/STAR/Approach (PD, PE and PF)

Section revised.

4.9.1 Primary Records

Added ARC Radius. Revised MSA Center to Center Fix.

5.24 Theta (THETA)

Section revised.

5.25 Rho (RHO)

Section revised.

5.26 Outbound Magnetic Course

Section revised.

5.27 Route Distance From, Holding Distance Time (RTE DIST HOLD DIST/TIME)

Section Revised.

5.42 Waypoint Type (TYPE)

Section Revised.

5.144 MSA Center

Section renamed Center Fix (CENTER FIX). Section revised.

5.204 ARC Radius (ARC RAD)

Added section.

7.2.6.d Terminal Waypoints

Added section.

ATTACHMENT 5 PATH AND TERMINATION

Section revised.

1.3 Leg Sequencing

Section revised. Added "RF" leg type.

1.4 Leg Type Description

Added RF Leg Type.

1.5 Leg Data Fields

Section revised. Table revised.

3.0 PATH AND TERMINATOR RELATED RULES VALID FOR ALL PROCEDURE TYPES

Added rules.

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SUPPLEMENT 12

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: September 15, 1994

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: May 17, 1994

A. PURPOSE OF THIS DOCUMENT

This Supplement introduces changes to ARINC Specification 424 to improve the operational utility of the navigation data base.

B. ORGANIZATION OF THIS DOCUMENT

The first part of this Supplement printed on goldenrod colored paper, contains descriptions of the changes introduced into the Specification by this Supplement. The second part consists of replacement white pages for the Specification modified to reflect these changes. The modified and added material on each replacement page is identified with c-12 symbols in the margins.

Existing copies of Specification 424-11 may be updated by inserting the replacement pages where necessary and destroying the pages they displace. The goldenrod colored Supplement should be inserted inside the rear cover of the Specification following Supplement 11.

C. CHANGES TO ARINC SPECIFICATION 424 INTRODUCED BY THIS DRAFT SUPPLEMENT

This section presents a complete tabulation of the changes and additions to the Specification to be introduced by this Supplement. Each change or addition is defined by the section number and the title that will be employed when the Supplement is eventually incorporated. In each case a brief description of the change or addition is included.

2.2 Special Navigation Terms

This section is revised to add definitions for Initial Approach Fix, Intermediate Approach Fix, Gateway Fix and Precision Final Approach Fix. In the addition, the definitions for Final Approach Course Fix and Final Approach Fix have been revised.

4.2.6 Limitation Continuation Record

This record is added to code signal limitations of VHF Navaids.

4.9 SID/STAR/Approach (PD, PE and PF)

This record is revised to add provisions for RNP, and to identity Recommended Navaid Section and Subsection.

4.11.3 Localizer Simulation Continuation Record

This record is revised to add Approach Route Identifier.

5.6 Airport/Heliport Identifier (ARPT IDENT) (HEL IDENT)

This section is revised to provide for coding where no officially published ICAO code is available.

5.7 Route Type

This section is revised to add additional SID, STAR and Approach route types.

5.10 Approach Route Identifier

This section is revised to add the capability for coding of multiple approach procedures.

5.12 Sequence Number (SEQ NR)

This section is revised to support the new VHF Navaid Limitation Continuation record.

5.17 Waypoint Description Code

This section is revised to add additional waypoint types.

5.23 Recommended NAVAID (RECD NAV)

This section is revised based on the Navaid type.

5.24 Theta

This section is revised based on the Recommended Navaid.

5.25 Rho (RHO)

This section is revised based on the Recommended Navaid.

5.26 Outbound Magnetic Course

This section is revised to reflect changes in the coding rules.

5.29 Altitude Description

This section is revised to support the new VHF Navaid Limitation Continuation record.

5.39 Magnetic Variation

This section is revised to support Epoch Year Variation.

5.42 Waypoint Type

This section is revised to add additional enroute and terminal waypoints.

5.70 Vertical Angle (VERT ANGLE)

This section, along with Figure 5-11, is revised to reflect changes in the coding rules.

5.74 Glide Slope Elevation (G.S. ELEV), EL Elevation (EL ELEV)

This section has been renamed Component Elevation (G.S. ELEV, EL ELEV, AZ ELEV, BAZ ELEV). The text has been revised for MLS coding.

5.187 Distance Description

This section is revised to support the new VHF Navaid Limitation Continuation record.

5.205 Navaid Limitation Code

This section is added to support the new VHF Navaid Limitation Continuation record.

5.206 Component Affected Indicator

This section is added to support the new VHF Navaid Limitation Continuation record.

5.207 Sector From/Sector To

This section is added to support the new VHF Navaid Limitation Continuation record.

5.208 Distance Limitation

This section is added to support the new VHF Navaid Limitation Continuation record.

5.209 Altitude Limitation (ALT LIMIT)

This section is added to support the new VHF Navaid Limitation Continuation record.

5.210 Sequence End Indicator (SEQ END)

This section is added to support the new VHF Navaid Limitation Continuation record.

7.2.6.a Terminal Waypoints

This is is revised to add FMS, GPS,. Heliport, NDB CTL and MLS approach types.

ATTACHMENT 2 - LOCAL HORIZONTAL REFERENCE DATUM NAME, DATUM CODE AND ELLIPSOID LIST

This section is revised to add additional datum.

ATTACHMENT 5 - PATH AND TERMINATION

1.2 Beginning and Ending Leg Type

This section is revised to allow FMS and GPS Approach Procedures. In addition, the use of TF legs is expanded to allow coding for final approach segments.

1.3 Leg Sequencing

This section is revised to allow leg DF-DF, IF-HA, IF-HM and IF-PI leg combinations

3.0 Path and Termination Related Rules Valid for Al Procedure Types

This section is revised to allow DF-DF leg types.

4.0 Standard Instrument Departure (SID) Coding Rules

This section is revised to prevent unanticipated turns on departure.

6.0 Approach Procedure Rules Valid for All Procedure Types

This section is revised to add the coding of FMS, GPS, and MLS. In addition, revisions are introduced to add TF legs.

APPENDIX 2 - SUBJECT INDEX

This appendix is added to facilitate easier use of ARINC Specification 424.

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SUPPLEMENT 13
TO
ARINC SPECIFICATION 424
NAVIGATION SYSTEM DATA BASE

Published: December 31, 1995

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: May 23, 1995

A. PURPOSE OF THIS DOCUMENT

This Supplement introduces changes to ARINC Specification 424 to improve the operational utility of the navigation data base.

B. ORGANIZATION OF THIS DOCUMENT

This Supplement introduces a major rework of ARINC Specification 424. The normal practice of publishing a separate supplement to update the existing document has not been followed. The extensiveness of the changes introduced by Supplement 13 has resulted in the impracticality of producing a separate set of replacement pages. Supplement 13 is therefore available only as an integral part of ARINC Specification 424-13. The modified and added material on each page is identified with c-13 symbols in the margins.

C. CHANGES TO ARINC SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This section presents a complete tabulation of the changes and additions to the Specification to be introduced by this Supplement. Each change or addition is defined by the section number and the title. In each case a brief description of the change or addition is included.

2.2 Special Navigation Terms

This section is revised to add the definition for Phantom Waypoint.

3.2.7 Tables Section (T)

This section was previously titled Cruising Table Section (T) - Subsection (TC). Sections 3.7.2.1 and 3.7.2.2 have been added to identify previously include cruising table information and added geographical reference table information.

4.6 Enroute Airway Records (ER)

This section is revised to add provisions for Required Navigation Performance (RNP). Note 1 is revised to describe the use of Column 19 for ATS Route Service.

4.7 Airport Records (PA)

4.7.3 Flight Planning Continuation Records

This section is revised to add an indication if the airport defined in the Primary Record is associated with Controlled Airspace.

4.10 Runway Records (PG)

This section is revised to add runway gradient. In addition, to support MLS, Localizer is revised to Localizer/MLS. The record now permits the coding of a second Localizer/MLS.

4.24 Heliport Records (HA)

4.24.3 Flight Planning Continuation Records

This section is revised to add an indication if the airport defined in the Primary Record is associated with Controlled Airspace.

4.25 Preferred Route Record (ET)

This section is revised to permit expansion of the Route Identifier beyond five characters. In addition, fields are added to support Aircraft Use Group, Direction Restriction and Altitude Description.

4.25.3 Continuation Record (ET)

This section is added to support the Preferred Route Record Continuation Record.

4.27 Controlled Airspace Records (UC)

This section is added to support the encoding of controlled airspace.

4.28 Geographical Reference Tables (TG)

This section is added to support Preferred Routes.

5.3 Customer/Area Code (CUST/AREA)

This section is revised so that Preferred Route Records and Geographical Reference Table Records are indicated as having no area code.

5.5 Subsection Code

This section is revised to include the new Controlled Airspace Record.

5.6 Airport/Heliport Identifier (ARPT/HELI IDENT)

This section is revised to support non-officially published identifiers.

5.7 Route Type (RT TYPE)

This section is revised so that Preferred Route Records have an expanded Route Type capability and SID Route Types are presented in the correct order.

5.8 Route Identifier

This section is revised to expand the Preferred Route Records to 10 characters. Six characters were defined in previous versions of this Specification.

5.10 Approach Route Identifier (APPROACH IDENT)

This section is revised to support Helicopter Approach Procedures.

5.11 Transition Identifier (TRANS IDENT)

This section is revised to support RNAV SID, FMS SID, RNAV STAR and FMS STAR route types.

5.12 Sequence Number (SEQ NR)

This section is revised to support Controlled Airspace and Geographical Reference Table records.

5.17 Waypoint Description Code (WAY DESC)

This section is completely revised.

5.19 Level (LEVEL)

This section is revised to support Controlled Airspace Records.

5.23 Recommended Navaid (RECD NAV)

This section is revised to support Un-Biased ILSDME and Localizer Markers.

5.29 Altitude Description (ALT DESC)

This section is revised to support Preferred Route Records.

5.30 Altitude/Minimum Altitude

This section is revised to describe the application of minimum and maximum altitude as it relates to Preferred Route Records.

5.36 Latitude (LATITUDE)

This section is revised to include Controlled Airspace Records.

5.37 Longitude (LONGITUDE)

This section is revised to include Controlled Airspace Records.

5.42 Waypoint Type (TYPE)

This section is revised to add the arc center fix waypoint to the Terminal Waypoints table.

5.44 Localizer/MLS Identifier (LOC IDENT), (MLS IDENT)

This section is revised to encode multiple localizers associated with a single runway.

5.77 VIA Code

This section is revised to provide Preferred Route Records with two new fix related capabilities.

5.80 Localizer/MLS Category/Classification (CAT/CLASS)

This section was previously titled ILS Category. This section is completely revised.

5.115 Direction Restriction

This section is revised to support Preferred Route Records.

5.118 Boundary VIA

This section is revised to add application rules and to provide support for the Controlled Airspace Record. Figure 5-12 Controlled and Restrictive Airspace and FIR/UIR Boundaries is completely revised.

5.119 Arc Distance (ARC DIST)

This section is revised to support Controlled Airspace Records.

5.120 Arc Bearing (ARC BRG)

This section is revised to support Controlled Airspace Records.

5.121 Lower/Upper Limit

This section is revised to support Controlled Airspace Records.

5.130 Multiple Code (MULTI CD)

This section is revised to support Controlled Airspace Records.

5.131 Time Code (TIME CD)

This section is revised to support Controlled Airspace Records.

5.132 NOTAM

This section is revised to support Controlled Airspace Records.

5.133 Unit Indicator (UNIT IND)

This section is revised to support Controlled Airspace Records.

5.138 Time Indicator (TIME IND)

This section is revised to support Controlled Airspace Records.

5.140 Controlling Agency

This section is revised to support Controlled Airspace Records.

5.149 Figure of Merit (MERIT)

This section is revised to add range values to the descriptions and to designate out of service VHF NAVAIDS.

5.195 Time of Operation (TIME OF OPERATION)

This section is revised to support Controlled Airspace Records.

5.211 Required Navigation Performance (RNP)

This section is added to provide provision for support of RNP.

5.212 Runway Gradient (RWY GRAD)

This section is added to support Runway records.

5.213 Controlled Airspace Type (ARSP TYPE)

This section is added to support Controlled Airspace records.

5.214 Controlled Airspace Center (ARSP CNTR)

This section is added to support Controlled Airspace records.

5.215 Controlled Airspace Classification (ARSP CLASS)

This section is added to support Controlled Airspace records.

5.216 Controlled Airspace Name (ARSP NAME)

This section is added to support Controlled Airspace records.

5.217 Controlled Airspace Indicator (CTLD ARSP IND)

This section is added to support Airport Flight Planning Continuation records.

5.218 Geographical Reference Table Identifier (GEO REF TBL ID)

This section is added to support Geographical Reference Table records.

5.219 Geographical Entity (GEO ENT)

This section is added to support Geographical Reference Table records.

5.220 Preferred Route Use Indicator (ET IND)

This section is added to support Preferred Route and Geographical Reference Table Records.

5.221 Aircraft Use Group (ACFT USE GP)

This section is added to support Preferred Route records.

7.5 Preferred Route Identifiers

This section is completely revised to clearly define and expand the naming conventions.

ATTACHMENT 5 - PATH AND TERMINATION

1.2 Beginning and Ending Leg Type

This section is revised to allow FMS and GPS approach procedures.

1.3 Leg Sequencing

This section is revised to allow the leg IF/RF combination.

3.0 Path and Termination Related Rules Valid for All Procedure Types

This section is revised to add rules for RF legs and Phantom Waypoints.

5.0 Standard Terminal Arrival Route (STAR) Coding Rules

This section is revised to add a rule for Expect Altitudes.

6.0 Approach Procedure Rules Valid for All Procedure Types

This section is revised to includes a number of proposals to modify the Approach Procedure Coding.

7.0 Missed Approach Procedure Rules Valid for All Procedure Types

This section is revised to includes a number of proposals to modify the Missed Approach Procedure Rules.

APPENDIX 2 - STRAIGHT-IN CRITERIA

This appendix is added using material previously contained in Attachment 5, Sections 6.2.7 and 6.2.8.

APPENDIX 3 - SUBJECT INDEX

Previously Appendix 2, this appendix is renumbered Appendix 3.

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SUPPLEMENT 14
TO
ARINC SPECIFICATION 424
NAVIGATION SYSTEM DATA BASE

Published: February 11, 2000

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: August 25, 1999

A. PURPOSE OF THIS SUPPLEMENT

This Supplement introduces revisions and additions to the body and Attachments of ARINC Specification 424. These changes include revisions to Path Point Concept, GLS Record, and DGPS Record and Modification to Paths and Terminators.

B. ORGANIZATION OF THIS DOCUMENT

The material in Supplement 14 is integrated into ARINC Specification 424 to form an updated version of the standard.

The changes introduced by Supplement 14 have been identified using change bars and are labeled in the margin by a "c-14" indicator.

C. CHANGES TO SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This section presents a complete tabulation of the changes and additions to the Specification introduced by this Supplement. Each change or addition is defined by the section number and the title that will be employed. In each case a brief description of the change or addition is included.

1.3.1 Coverage of Helicopter Operation Needs

New section was added.

2.2 Special Navigation

The Final Approach Course Fix definition was revised.

The "Final End Point definition" was added.

2.3 Precision RNAV Terms

This section was added.

ALL TITLES IN SECTIONS 3 WERE MODIFIED TO ONE-LETTER CHARACTERS IN THE SUBSECTIONS

3.1 User File Organization

The second paragraph was added by Supplement 14.

3.2.1 General

Last sentence in first paragraph was modified for clarity.

3.2.2.1 VHF Navaid Section (D), Subsection (blank)

Sections 3.3.5, 3.3.6, 3.3.7, and 3.3.8 were added.

3.2.2.2 NDB Navaid Section (D), Subsection (B)

Sections 3.2.10, 3.3.5, 3.3.6, 3.3.7 and 3.3.8 were added.

3.2.3.1 Enroute Waypoint Section (E), Subsection (A)

Sections 3.2.10, 3.3.5, 3.3.6, and 3.3.7 were added.

3.2.3.5 Enroute Airways Restriction Section (E), Subsection (U)

The Text was revised to be consistent with Section 3.2.3.6.

3.2.4.2 Airport Gates Section (P), Subsection (B)

This section was expanded to include Standard data for airport gates.

3.2.4.6 Airport Approaches Section (P), Subsection (F)

Text was added for clarification.

3.2.4.8 Airport And Heliport Localizer/Glide Slope Section (P), Subsection (I)

This section was revised to support Heliport records.

3.2.4.9 Airport and Heliport MLS Section (P), Subsection (L)

This section was revised to support Heliport records.

3.2.4.10 Airport and Heliport Marker/ Locator Section (P), Subsection (M)

This section was revised to support Heliport records.

3.2.4.13 Airport and Heliport Terminal NBD Section(P), Subsection (N)

Section 3.3.5, 3.3.6, 3.3.7, and 3.3.8 were added to support Heliport records.

3.2.4.14 Airport and Heliport Path Point Section (P), Subsection (P)

This section was added.

3.2.4.15 Flight Planning Arrival / Departure Data Record Section (P), Subsection (R)

This section was added.

3.2.4.16 GNSS Landing System (GLS) Section (P), Subsection (T)

This section was added.

3.2.5 Company Route and Alternation Destination Section (R)

The title was revised to "Company Route and Alternation Destination Section (R)."

3.2.5.1 Company Route Section (R) Subsection (Blank)

The text was revised to support Alternate Destination Record.

3.2.5.2 The Alternate Record Section (R) Subsection (A)

This Section was added to support Alternate Destination Records.

3.2.6.2 FIR/UIR Section (U), Subsection (F)

Existing text was modified for clarity.

3.2.6.3 Controlled Airspace Section (U), Subsection (C)

This section was added.

3.2.7.2 Geographical Reference Table Section (T), Subsection (G)

The reference to Section 3.3.10 was revised to 3.2.9.

3.2.9 Preferred Routes Section (E), Subsection (T)

This section was numbered 3.2.10 prior to 424-14. It was renumbered without change when the former 3.2.9, Heliport Section was moved to the Master Helicopter User File Section 3.3.

3.3 Master Helicopter User File Content

This section was added.

3.3.1 General

This section was added.

3.3.2 Jointly and Specifically used Section / Subsection

This section was added.

3.3.3 Heliport Section (H), Subsection (A)

This section was added.

3.3.4 Heliport Terminal Waypoints Section (H), Subsection (C)

This section was added.

3.3.5 Heliport Standard Instrument Departures (SIDs) Section (H), Subsection (D)

This section was added.

3.3.6 Heliport Standard Terminal Arrival Routes (STARs) Section (H), Subsection (E)

This section was added.

3.3.7 Heliport Approaches Section (H), Subsection (F)

This section was added.

3.3.8 Heliport MSA Section (H), Subsection (S)

This section was added.

3.3.9 Heliport Communications Section (H), Subsection (V)

This section was added.

SECTION 4 HAS BEEN RENUMBERED TO ACCOMMODATE THE INTRODUCTION OF ROTOR-WING RECORDS

4.0.1 General

Was numbered 4.1 prior to 424 -14.

Descriptive paragraph was revised to include Helicopter user file.

4.1 Master Airline User File

This section title added.

4.1.2 VHF NAVAID Record (D)

Was numbered 4.2 prior to 424.14.

4.1.2.1 VHF NAVAID Primary Records

Was numbered 4.2.1 prior to 424-14.

4.1.2.2 VHF NAVAID Continuation Records

Was numbered 4.2.2 prior to 424-14.

4.1.2.3 VHF NAVAID Simulation Continuation Records

Was numbered 4.2.3 prior to 424-14.

4.1.2.4 VHF NAVAID Flight Planning Continuation Records

Was numbered 4.2.4 prior to 424-14.

4.1.2.5 VHF NAVAID Flight Planning Continuation Records

Was numbered 4.2.5 prior to 424-14.

4.1.2.6 VHF NAVAID Limitation Continuation Records

Was numbered 4.2.6 prior to 424-14.

4.1.3 NDB NAVAID Record (DB) or (PN)

Was numbered 4.3 prior to 424-14.

4.1.3.1 NDB NAVAID Primary Records

Was numbered 4.3.1 prior to 424-14.

4.1.3.2 NDB NAVAID Continuation Records

Was numbered 4.3.2 prior to 424-14.

4.1.3.3 NDB NAVAID Simulation Continuation Records

Was numbered 4.3.3 prior to 424-14.

4.1.3.4 NDB NAVAID Flight Planning Continuation Records

Was numbered 4.3.4 prior to 424-14.

4.1.3.5 NDB NAVAID Flight Planning Continuation Records

Was numbered 4.3.5 prior to 424-14.

4.1.4 Waypoint Record (EA) or (PC)

Was numbered 4.4 prior to 424-14.

Text was added to support Helicopter Enroute and Terminal Waypoints

4.1.4.1 Waypoint Primary Records

Was numbered 4.4.1 prior to 424-14.

Text was added to support Helicopter Records.

4.1.4.2 Waypoint Continuation Records

Was numbered 4.4.2 prior to 424-14.

4.1.4.3 Waypoint Flight Planning Continuation Records

Was numbered 4.4.3 prior to 424-14.

4.1.4.4 Waypoint Flight Planning Continuation Records

Was numbered 4.4.4 prior to 424-14.

4.1.5 Holding Pattern Records (EP)

Was numbered 4.5 prior to 424-14.

4.1.5.1 Holding Pattern Primary Records

Was numbered 4.5.1 prior to 424-14.

Note 1 was revised to support Helicopter Records.

4.1.5.2 Holding Pattern Continuation Records

Was numbered 4.5.2 prior to 424-14.

4.1.6 Enroute Airway Records (ER)

Was numbered 4.6 prior to 424-14.

Text was added to support Helicopter Records.

4.1.6.1 Enroute Airways Primary Records

Was numbered 4.6.1 prior to 424-14.

4.1.6.2 Enroute Airways Continuation Records

Was numbered 4.6.2 prior to 424-14.

4.1.6.3 Enroute Airways Flight Planning Continuation Records

Was numbered 4.6.3 prior to 424-14.

4.1.6.4 Enroute Airways Flight Planning Continuation Records

Was numbered 4.6.4 prior to 424-14.

4.1.7 Airport Records (PA)

Was numbered 4.7 prior to 424-14.

4.1.7.1 Airport Primary Records

Was numbered 4.7.1 prior to 424-14 and the title was changed to "Airport Primary Records."

Columns 32 and 69 through 132 were revised.

4.1.7.2 Airport Continuation Records

Was numbered 4.7.2 prior to 424-14 and the title was changed to "Airport Continuation Records."

4.1.7.3 Airport Flight Planning Continuation Records

Was numbered 4.7.3 prior to 424-14 and the title was changed to "Airport Flight Planning Continuation Records."

4.1.7.4 Airport Flight Planning Continuation Records

Was numbered 4.7.4 prior to 424-14 and the title was changed to "Airport Flight Planning Continuation Records."

4.1.8 Airport Gate Records (PB)

Was numbered 4.8 prior to 424-14 and the title was changed to "Airport Gate Records (PB)."

4.1.8.1 Airport Gate Primary Records

Was numbered 4.8.1 prior to 424-14 and the title was changed to "Airport Gate Primary Records."

4.1.8.2 Airport Gate Continuation Records

Was numbered 4.8.2 prior to 424-14 and the title was changed to "Airport Gate Continuation Records."

4.1.9 Airport SID/STAR/APPROACH Records (PD, PE, and PF)

Was numbered 4.9 prior to 424-14 and the title was changed to "Airport SID/STAR/APPROACH Records PD, PE, and PF).

4.1.9.1 SID/STAR/Approach Primary Records

Was numbered 4.9.1 prior to 424-14.

Columns 117 through 123 were revised.

Notes 1 and 2 were added.

4.1.9.2 Airport SID/STAR/APPROACH Continuation Records

Was numbered 4.9.2 prior to 424-14 and the title was changed to "Airport SID/STAR/APPROACH Continuation Records."

Columns 73 through 132 were revised.

Notes 1, 2 and 3 were added.

4.1.9.3 Airport SID/STAR/APPROACH Flight Planning Continuation Records

Was numbered 4.9.3 prior to 424-14 and the title was changed to "Airport SID/STAR/APPROACH Flight Planning Continuation Records."

4.1.9.4 Airport SID/STAR/APPROACH Flight Planning Continuation Records

Was numbered 4.9.4 prior to 424-14 and the title was changed to "Airport SID/STAR/APPROACH Flight Planning Continuation Records."

4.1.10 Runway Records (PG)

Was numbered 4.10 prior to 424-14.

4.1.10.1 Runway Primary Records

Was numbered 4.10.1 prior to 424-14 and the title was changed to "Runway Primary Records."

Section was revised to support GLS records.

4.1.10.2 Runway Continuation Records

Was numbered 4.10.2 prior to 424-14 and the title was changed to "Runway Continuation Records."

4.1.10.3 Runway Simulation Continuation Records

Was numbered 4.10.3 prior to 424-14 and the title was changed to "Runway Simulation Continuation Records."

4.1.11 Airport and Heliport Localizer and Glide Slope Records (PI)

Was numbered 4.11 prior to 424-14 and the title was changed to "Airport and Heliport Localizer and Glide Slope Records (PI)."

4.1.11.1 Airport and Heliport Localizer and Glide Slope Primary Records

Was numbered 4.11.1 prior to 424-14 and the title was changed to "Airport and Heliport Localizer and Glide Slope Primary Records."

4.1.11.2 Airport and Heliport Localizer and Glide Slope Continuation Records

Was numbered 4.11.2 prior to 424-14 and the title was changed to "Airport and Heliport Localizer and Glide Slope Continuation Records."

4.1.11.3 Airport and Heliport Localizer and Glide Slope Simulation Continuation Records

Was numbered 4.11.3 prior to 424-14 and the title was changed to "Airport and Heliport Localizer and Glide Slope Simulation Continuation Records."

4.1.12 Company Route Records (R)

Was numbered 4.12 prior to 424-14.

4.1.13 Airport and Heliport Localizer Marker Records(PM)

Was numbered 4.13 prior to 424-14 and the title was changed to "Airport and Heliport Localizer Marker Records(PM)."

4.1.13.1 Airport and Heliport Localizer Marker Primary Records

Was numbered 4.13.1 prior to 424-14 and the title was changed to "Airport and Heliport Localizer Marker Primary Records."

4.1.14 Airport Communications Records (PV)

Was numbered 4.14 prior to 424-14.

4.1.14.1 Airport Communications Primary Records

Was numbered 4.14.1 prior to 424-14 and the title was changed to "Airport Communications Primary Records."

4.1.14.2 Airport Communications Continuation Records

Was numbered 4.14.2 prior to 424-14 and the title was changed to "Airport Communications Continuation Records."

4.1.14.3 Airport Additional Communications Continuation Records

Was numbered 4.14.3 prior to 424-14 and the title was changed to "Airport Additional Communications Continuation Records."

4.1.15 Airways Marker Records (EM)

Was numbered 4.15 prior to 424-14.

4.1.15.1 Airways Marker Primary Records

Was numbered 4.15.1 prior to 424-14 and the title was changed to "Airways Marker Primary Records."

4.1.16 Cruising Tables Records (TC)

Was numbered 4.16 prior to 424-14.

4.1.16.1 Cruising Tables Primary Records

Was numbered 4.16.1 prior to 424-14 and the title was changed to "Cruising Tables Primary Records."

4.1.17 FIR/UIR Records (UF)

Was numbered 4.17 prior to 424-14.

4.1.17.1 FIR/UIR Primary Records

Was numbered 4.17.1 prior to 424-14 and the title was changed to "FIR/UIR Primary Records."

4.1.18 Restrictive Airspace Records (UR)

Was numbered 4.18 prior to 424-14.

4.1.18.1 Restrictive Airspace Primary Records

Was numbered 4.18.1 prior to 424-14 and the title was changed to "Restrictive Airspace Primary Records."

4.1.18.2 Restrictive Airspace Continuation Records

Was numbered 4.18.2 prior to 424-14 and the title was changed to "Restrictive Airspace Continuation Records."

4.1.18.3 Restrictive Airspace Flight Planning Continuation Records

Was numbered 4.18.3 prior to 424-14 and the title was changed to "Restrictive Airspace Flight Planning Continuation Records."

4.1.19 Grid MORA Records (AS)

Was numbered 4.19 prior to 424-14.

4.1.19.1 Grid MORA Primary Records

Was numbered 4.19.1 prior to 424-14 and the title was changed to "Grid MORA Primary Records."

4.1.20 Airport MSA (Minimum Sector Altitude) Records (PS)

Was numbered 4.20 prior to 424-14 and the title was changed to "Airport MSA (Minimum Sector Altitude) Records (PS)."

4.1.20.1 Airport MSA Primary Records

Was numbered 4.20.1 prior to 424-14 and the title was changed to "Airport MSA Primary Records."

4.1.20.2 Airport MSA Continuation Records

Was numbered 4.20.2 prior to 424-14.

4.1.21 Enroute Airways Restriction Records (EU)

Was numbered 4.21 prior to 424-14.

4.1.21.1 Altitude Exclusion Primary Records

Was numbered 4.21.1 prior to 424-14.

4.1.21.2 Altitude Exclusion Continuation Records

Was numbered 4.21.2 prior to 424-14.

4.1.21A.1 Note Restriction Primary Records

Was numbered 4.21A.1 prior to 424-14.

4.1.21A.2 Note Restriction Continuation Records

Was numbered 4.21A.2 prior to 24-14. 4

4.1.21B.1 Seasonal Closure Primary Records

Was numbered 4.21B.1 prior to 424-14.

4.1.21C.1 Cruising Table Replacement Primary Records

Was numbered 4.21C.1 prior to 424-14.

4.1.21C.2 Cruising Table Replacement Continuation Records

Was numbered 4.21C.2 prior to 424-14.

4.1.22 Airport and Heliport MLS (Azimuth, Elevation and Back Azimuth) Records

Was numbered 4.22 prior to 424-14 and the title was changed to "Airport and Heliport MLS (Azimuth, Elevation and Back Azimuth) Records."

4.1.22.1 Airport and Heliport MLS Primary Records

Was numbered 4.22.1 prior to 424-14 and the title was changed to "Airport and Heliport MLS Primary Records."

4.1.22.2 Airport and Heliport MLS Continuation Records

Was numbered 4.22.2 prior to 424-14 and the title was changed to "Airport and Heliport MLS Continuation Records."

4.1.23 Enroute Communications Record (EV)

Was numbered 4.23 prior to 424-14.

4.1.23.1 Enroute Communications Primary Records

Was numbered 4.23.1 prior to 424-14 and the title was changed to "Enroute Communications Primary Records."

4.1.23.2 Enroute Communications Continuation Records

Was numbered 4.23.2 prior to 424-14 and the title was changed to "Enroute Communications Continuation Records."

4.1.23.3 Enroute Communications Continuation Records

Was numbered 4.23.3 prior to 424-14 and the title was changed to "Enroute Communications Continuation Records."

4.1.24 Preferred Route Records (ET)

Was numbered 4.25 prior to 424-14.

4.1.24.1 Preferred Route Primary Records

Was numbered 4.25.1 prior to 424-14.

4.1.24.2 Preferred Route Continuation Records

Was numbered 4.25.2 prior to 424-14 and the title was changed to "Preferred Route Continuation Records."

4.1.24.3 Preferred Route Continuation Records

Was numbered 4.25.3 prior to 424-14 and the title was changed to "Preferred Route Continuation Records."

4.1.25 Controlled Airspace Records (UC)

Was numbered 4.27 prior to 424-14.

The following text was added to this section:

It includes controlled airspace associated with Airports and Heliports.

4.1.25.1 Controlled Airspace Primary Records

Was numbered 4.27.1 prior to 424-14.

4.1.25.2 Controlled Airspace Continuation Records

Was numbered 4.27.2 prior to 424-14 and the title was changed to "Controlled Airspace Continuation Records

4.1.26 Geographical Reference Table Records (TG)

Was numbered 4.28 prior to 424-14.

4.1.26.1 Geographical Reference Table Primary Records

Was numbered 4.28.1 prior to 424-14 and the title was changed to "Geographical Reference Table Primary Records."

4.1.27 Flight Planning Arrival/Departure Data Record (PR)

This section was added.

4.1.27 Flight Planning Arrival / Departure Data

This section was added.

4.1.27.1 Primary Record

This section was added.

4.1.27.2 Continuation Records

This section was added.

4.1.27.3 Continuation Records

This section was added.

4.1.28 Path Point Records (PP)

This section was added.

4.1.29 GLS Record (PT)

This section was added.

4.1.29.1 Primary Records

This section was added.

4.1.30 Alternate Record (RA)

This section was added.

4.1.30.1 Primary Record

This section was added.

4.2 Master Helicopter User file (HA)

This section was added.

4.2.1 Heliport Records

Was numbered 4.24 prior to 424.14

4.2.1.1 Heliport Primary Records

Was numbered 4.24.1 prior to 424-14 and the title was changed to "Heliport Primary Records."

4.2.1.2 Heliport Continuation Records

Was numbered 4.24.2 prior to 424-14 and the title was changed to "Heliport Continuation Records."

4.2.1.3 Heliport Flight Planning Continuation Records

Was numbered 4.24.3 prior to 424-14 and the title was changed to "Heliport Flight Planning Continuation Records."

4.2.1.4 Heliport Flight Planning Continuation Records

Was numbered 4.24.4 prior to 424-14 and the title was changed to "Heliport Flight Planning Continuation Records."

The Following Sections Have Been Added To Support Rotor-Wing Records:

4.2.2 Heliport Terminal Waypoint Record (HC)

4.2.2.1 Primary Records

4.2.2.2 Continuation Records

4.2.2.3 Flight Planning Continuation Records

4.2.2.4 Flight Planning Continuation Records

4.2.3 Heliport SID/STAR / Approach (HD/HE/HF)

4.2.3.1 Heliport SID/STAR/Approach Primary Records

4.2.3.2 Heliport SID/STAR/Approach Continuation Records

4.2.3.3 Heliport SID/STAR/Approach Flight Planning Continuation Records

4.2.3.4 Heliport SID/STAR/Approach Flight Planning Continuation Records

4.2.4 Heliport MSA (HS)

4.2.4.1 Primary Records

4.2.4.2 Continuation Records

4.2.5 Heliport Communications Records (HV)

Was numbered 4.26 prior to 424-14.

4.2.5.1 Heliport Communications Primary Records

Was numbered 4.26.1 prior to 424-14 and the title was change to "Heliport Communications Primary Records."

4.2.5.2 Heliport Communications Continuation Records

Was numbered 4.26.2 prior to 424-14 and the title was changed to "Heliport Communications Continuation Records."

4.2.5.3 Heliport Communications Continuation Records

Was numbered 4.26.3 prior to 424-14 and the title was changed to "Heliport Communications Continuation Records."

5.5 Subsection Code (SUB CODE)

Section was revised to support Heliport and GLS encoding schemes and the Alternate Record.

5.6 Airport/Heliport Identifier (ARPT/HELI IDENT)

The "Used On" text was expanded to identify airport and heliport identifiers.

5.7 Route Type (RT TYPE):

Section was revised to support Heliport and GLS Route types.

5.8 Route Identifier (ROUTE IDENT)

The existing text was modified to support heliport records.

5.9 SID/STAR Route Identifier (SID/STAR IDENT)

The "Used On" text was revised to support Heliport Records.

5.10 Approach Route Identifier (APPROACH IDENT)

This section was revised to support Heliport Approach Records.

5.11 Transition Identifier (TRANS IDENT)

This section was revised to support Heliport records and the text was modified for clarity.

5.12 Sequence Number (SEQ NR)

The "Used On" text was revised to support Heliport Approach Record.

5.13 Fix Identifier (FIX IDENT)

The "Used On" text was revised to support Heliport Approach Record.

5.17 Waypoint Description Code (DESC CODE)

Text and waypoint description was revised to support Heliport records.

Note 16 was revised to support Final End Point.

Note 19 was added by this Supplement.

5.20 Turn Direction (TURN DIR)

The "Used On" text was revised to support Heliport Approach Record.

5.21 Path and Termination (PATH TERM)

The "Used On" text was revised to support Heliport Approach Record.

5.22 Turn Direction Valid (TDV)

The "Used On" text was revised to support Heliport Approach Record.

5.23 Recommended Navaid (RECD NAV)

Descriptive paragraph was revised to support Heliport.

Item "c" was revised to support GLS navaid.

Item "g" was revised for clarity.

Item "h" and "i" were added were added by Supplement 14.

The "Used On" text was revised to support Heliport Approach Record

5.24 Theta (THETA)

The "Used On" text was revised to support Heliport Approach Records.

5.25 Rho (RHO)

The "Used On" text was revised to support Heliport Approach Records.

5.26 Outbound Magnetic Course (OB MAG CRS)

The "Used On" text was revised to support Heliport Approach Records.

5.27 Route Distance From, Holding Distance/Time (RTE DIST FROM, HOLD DIST/TIME)

The "Used On" text was revised to support Heliport Approach Records.

5.28 Inbound Magnetic Course (IB MAG CRS)

Section was revised to include a paragraph describing the use of "HX" group of Path Terminator codes.

5.29 Altitude Description (ALT DESC)

The last sentence of the Definition/Description paragraph was revised to include "at or above to at or below."

Identifiers "H," "J" and "V" were added to the altitude description table and identifier "R" was remove.

Identifiers B, G, and I of the Waypoint Description Crossing was revised for clarity.

5.30 Altitude/Minimum Altitude

This section revised to support Heliport records.

5.36 Latitude (LATITUDE)

The "Used On" text was revised to support Heliport Records.

5.37 Longitude (LONGITUDE)

The "Used On" text was revised to support Heliport Records.

5.38 DME Identifier (DME IDENT)

The existing text was modified for clarity.

5.39 Magnetic Variation (MAG VAR, D MAG VAR)

The "Used On" text was revised to support GLS Records.

5.42 Waypoint Type

"U" and "V" added to Enroute Waypoints.

"P" added to the Terminal Waypoints.

The "Used On" text was revised to support Heliport Records

5.43 Waypoint Name/Description (NAME/DESC)

The "Used On" text was revised to support Heliport Records.

5.44 Localizer/MLS/GLS Identifier (LOC, MLS, GLS IDENT)

Section revised to support GLS.

5.45 Localizer Frequency (FREQ)

The "Used On" text was revised to included Airport and Heliport.

5.46 Runway Identifier (RUNWAY ID)

The "Used On" text was revised to support Heliport Records.

5.47 Localizer Bearing (LOC BRG)

Character type was changed to Alpha/Numeric.

5.52 Glide Slope Angle (GS ANGLE) Minimum Elevation Angle (MIN ELEV ANGLE)

Section revised to support GLS.

5.53 Transition Altitude/Level (TRANS ALTITUDE/LEVEL.

Section revised to support Heliport records.

5.54 Longest Runway (LONGEST RWY)

The Source/Content paragraph was modified to further define the longest runway.

5.57 Runway Length (RUNWAY LENGTH)

The Source/Content paragraph was modified further to define runway length.

5.58 Runway Magnetic Bearing (RWY BRG)

Character type was changed to Alpha/Numeric.

5.62 Inbound Holding Course (IB HOLD CRS)

Character type was changed to Alpha/Numeric.

5.67 Threshold Crossing Height (TCH)

The "Used On" text was revised to support Heliport Records.

5.70 Vertical Angle (VERT ANGLE)

The "Used On" text was revised to support Heliport Records.

Figure 5-11 and the note on angle constraints were deleted.

5.72 Speed Limit (SPEED LIMIT)

Section revised to support Heliport records.

5.74 Component Elevation (GS ELEV, EL ELEV, AZ ELEV, BAZ ELEV, GLS ELEV)

Section revised to support GLS elevation.

Character type was changed to Alpha/Numeric.

5.75 From/To-Airport/Fix

Section revised to support Alternate Destination Record.

5.77 VIA Code

The note was revised to refer to Figure 5-14.

5.80 ILS Category (CAT)

Section revised to support GLS.

5.81 ATC Indicator (ATC)

The "Used On" text was revised to support Airport and Heliport Records.

5.101 Communication Type (COMM TYPE)

Terminal (TML) was added to the Source/Content.

Airport Comm only column changed to Airport Heliport Comm only.

5.103 Communication Frequency (COMM FREQ)

Source/Content revised to support UHF frequencies.

5.106 Service Indicator (SERV IND)

The column content table header for Enroute Communication Record was changed to 57, 58, and 59.

5.114 Duplicate Indicator (DUP IND)

Note 1, editorial correction were made.

5.115 Directional Restriction

This section is revised to support Preferred Route Records

5.130 Multiple Code (MULTI CD)

The "Used On" text was revised to support Airport and Heliport Records.

5.131 Time Code (TIME CD)

The descriptive paragraph revised for clarity.

Primary and Continuation Records were added when used on Enroute and Airways.

5.134 Cruise Table Identifier (CRSE TBL IDENT)

The "Used On" text was revised reflect Flight Planning and Arrival/Departure Data records.

5.138 Time Indicator (TIME IND)

Greenwich Mean Time was changed to Universal Coordinated Time.

5.144 Center Fix (CENTER FIX)

This section revised to support Heliport records.

5.145 Radius Limit

The "Used On" text was revised to support Airport and Heliport Records.

5.146 Sector Bearing (SEC BRG)

The "Used On" text was revised to support Airport and Heliport Records.

5.147 Sector Altitude (SEC ALT)

The "Used On" text was revised to support Airport and Heliport Records.

5.161 Restriction Altitude (RSTR ALT)

The title of this section was changed.

5.165 Magnetic/True Indicator (M/T IND)

The "Used On" text was revised to support Heliport Records.

5.170 Decision Height (DH)

The "Used On" text was revised to support Airport and Heliport Records.

5.171 Minimum Descent Height (MDH)

The "Used On" text was revised to support Airport and Heliport Records.

5.180 PAD Identifier (PAD IDENT)

Section revised to support Heliport Records.

5.191 Triad Station (TRIAD STA)

Deleted by Supplement 14.

5.192 Group Repetition Interval (GRI)

Deleted by Supplement 14.

5.193 Additional Secondary Phase Factor (ASF)

Deleted by Supplement 14.

5.196 Name Format Indicator (NAME IND)

D, I, and N were added to the table.

Notes 1 and 2 were added.

The "Used On" text was revised to support Airport and Heliport Records.

5.197 Datum Code (DATUM)

The "Used On" text was revised to support Terminal NDB and GLS Transmitted Records.

5.211 Required Navigation Performance (RNP)

Text was rewritten to include ICAO Annex 15 and/or State published rules.

The Following Sections Have Been Added By Supplement 14:

5.222 GPS/FMS Indicator (GPS/FMS IND)

5.223 Operations Type (OPS TYPE)

5.224 Approach Indicator (APP IND)

5.225 Ellipsoidal Height

5.226 Glide Path Angle (GPA)

5.227 Orthometric Height (ORTH HGT)

5.228 Unit of Height (UNIT)

5.229 Path Point Data CRC (CRC)

5.230 Procedure Type (PROC TYPE)

5.231 Along Track Distance (ATD)

5.232 Number of Engines Restriction (NOE)

5.233 Turboprop/Jet Indicator (TURBO)

5.234 RNAV Flag (RNAV)

5.235 ATC Weight Category (ATC WC)

5.236 ATC Identifier (ATC ID)

5.237 Procedure Description (PROC DESC)

5.238 Leg Type Code (LTC)

5.239 Reporting Code (RPT)

5.240 Altitude (ALT)

5.241 Fix Related Transition Code (FRT Code)

5.242 Procedure Category (PROC CAT)

5.243 GLS Station Identifier

5.244 GLS Channel

5.245 Service Volume Radius

5.246 TDMA Slot

5.247 Station Type

5.248 Station Elevation WGS84

5.249 Longest Runway Surface Code (LRSC)

5.250 Alternate Record Type (ART)

5.251 Distance To Alternate (DTA)

5.252 Alternate Type (ALT TYPE)

5.253 Primary and Additional Alternate Identifier (ALT IDENT)

6.10 CRC Calculations

6.10.1 Precision Approach Path Point Cyclic Redundancy Check (CRC) Overview

6.10.2 Generator Polynomials:

6.10.3 32 Bit CRC Calculation

6.11 Application of CRC for Integrity Protection of Straight & Advanced Landing Approach Operations

6.11.2 RNAV GPS/GLS Approach Procedure Path Point Data Field Bits

6.11.3 CRC - Generator Polynomial, G(x)

7.2.2.1 Navaid Waypoint

7.2.2.2 Airport Waypoint

7.2.2.3 Named RNAV Waypoint, Intersections, and Reporting Points

Phonetic Letter Subsection, "County" was changed to "Country."

7.2.5 Reporting Positions Defined by Coordinates

The section was modified to provide a comprehensive definition of reporting positions by coordinates.

7.2.6 Terminal Waypoint

Subsection "A" and "B" was expanded to provide a comprehensive definition of waypoint identifiers.

Subsection "E" and "F" were added.

7.3.4 Navaid Waypoint

This section was added.

7.3.5 Airport Waypoint

This section was added.

ATTACHMENT 5 PATH AND TERMINATOR

Descriptive introductory paragraph revised to clarify the meaning for "must" and "will" and to support rotor-wing procedures.

1.2 Beginning and Ending Leg Types

A Note was added to support rotor-wing procedures.

1.3 Leg Sequencing

Note "&" was expanded to include FC/DF sequences.

1.4 Leg Type Descriptions

Illustrated Leg Types were revised.

1.5 Leg Data fields

Legend "B" was added to leg Data Fields.

2.0 CODING RULES APPLICABLE TO ALL PROCEDURE TYPES

This section was revised to clarify the meaning for "must" and "will" as defined in the introduction of Attachment 5.

Rule 2.3 was expanded to code Termination Altitude to include column 95.

3.0 PATH AND TERMINATION RELATED RULES VALID FOR ALL PROCEDURES TYPES

This section was revised to clarify the meaning for "must" and "will" as defined in the introduction of Attachment 5.

Rule 3.1 the following sentence was added:

For distance terminations, the overfly parameters must be set, otherwise the combination is not permitted.

4.0 STANDARD INSTRUMENT DEPARTURE (SID)CODING RULES

This section was revised to clarify the meaning for "must" and "will" as defined in the introduction of Attachment 5.

Text revised to support rotor-wing procedures.

5.0 STANDARD TERMINAL ARRIVAL ROUTE (STAR) CODING RULES

This section was revised to clarify the meaning for "must" and "will" as defined in the introduction of Attachment 5.

6.0 APPROACH PROCEDURE RULES VALID FOR ALL PROCEDURE TYPES

This section was revised to clarify the meaning for “must” and “will” as defined in the introduction of Attachment 5.

This section was modified to replace “runway” with “runway or helipad” and replace the term “threshold” with “threshold or alighting point.”

Rule 6.9, Loran Procedure Coding and Subsection were deleted by Supplement 14.

This section was revised to add rules for GLS Approach Procedure Coding.

The entire section was modified to support Step-down fixes for approach procedures.

7.0 PRECISION APPROACH PROCEDURES CODING

This section was added.

8.0 NON-PRECISION APPROACH PROCEDURE CODING

This section was added.

9.0 MISSED APPROACH PROCEDURE RULES VALID FOR ALL PROCEDURE

Previously Section 7, prior to Supplement 14.

This section was revised to clarify the meaning for “must” and “will” as defined in the introduction of Attachment 5.

This section was modified to replace “runway” with “runway or helipad” and replace the term “threshold” with “threshold or alighting point.”

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SUPPLEMENT 15
TO
ARINC SPECIFICATION 424
NAVIGATION SYSTEM DATA BASE

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Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: January 12, 2000

A. PURPOSE OF THIS SUPPLEMENT

This Supplement introduces revisions and additions to the body and Attachments of ARINC Specification 424. These changes include revisions to Final Approach Fix record and additions to the Missed Approach Procedures.

B. ORGANIZATION OF THIS DOCUMENT

The material in Supplement 15 is integrated into ARINC Specification 424 to form an updated version of the standard.

The changes introduced by Supplement 15 have been identified using change bars and are labeled in the margin by a "c-15" indicator.

C. CHANGES TO SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This section presents a complete tabulation of the changes and additions to the Specification introduced by this Supplement. Each change or addition is defined by the section number and the title that will be employed. In each case a brief description of the change or addition is included.

1.4 Reference Documentation

The reference to applicable ARINC Standards was updated.

5.7 Route Type (RT TYPE)

The Airport Approach (PF) and Heliport Approach (HF) Records were revised to included Missed Approach, field identifier "Z."

Note 3 were revised to support Qualifiers "P" and "S."

5.11 Transition Identifier (TRANS IDENT)

The Transition Identifier Filed Content was revised to included Missed Approach.

Note 4 was added for Missed Approach Transition.

5.29 Altitude Description (ALT DESC)

Identifiers "G" and "H" were revised for clarity.

The following text was added to this section:

The "V" content will only appear in Approach Route Coding and only for those fixes that are stepdown fixes in the vertical path of the procedure.

5.30 Altitude/Minimum Altitude

The second paragraph of the Source/Content was revised to include the altitude field identifiers.

5.70 Vertical Angle (VERT ANGLE)

The Definition/Description paragraph was revised to clarify the Vertical Angle field.

ATTACHMENT 5 PATH AND TERMINATOR

7.0 PRECISION APPROACH PROCEDURES CODING

Rule 7.1.7 was modified to describe that the vertical angle must be coded in both the Final Approach Fix (FAF) and the fix which carries the Missed Approach Point (MAP) coding.

Rule 7.1.8 was added by this Supplement.

9.0 MISSED APPROACH PROCEDURE RULES VALID FOR ALL PROCEDURE

The following text was added to Rule 9.0:

Identification of multiple missed approach procedures, when coded, will be accomplished through the coding of a specific Transition Identifier which closely aligns with published information.

Rule 9.3.1.4 was added by this Supplement.

Rule 9.3.1.5 was previously 9.3.1.4, additionally 9.3.1.5 was modified to clarify the first leg of the missed approach.

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SUPPLEMENT 16

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: August 30, 2002

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: April 8, 2002

A. PURPOSE OF THIS SUPPLEMENT

This supplement introduces revisions and additions to the body and Attachments of ARINC Specification 424. These changes improve the operational utility of the navigation data base

B. ORGANIZATION OF THIS DOCUMENT

This document, printed on goldenrod paper, contains descriptions of changes introduced into ARINC Specification 424 by this supplement. The material in Supplement 16 is integrated into ARINC Specification 424 to form an updated version of the standard.

Historically, the changes introduced by each supplement are annotated by using a change bar and labeling in the margin with a symbol associated with the supplement number. In preparing this supplement for the latest adoption, Supplement 16, the practicality of identifying all past changes in this manner was determined to be impractical.

Therefore, this supplement will only indicate the latest changes by use of change bars without numeration.

C. CHANGES TO SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This section presents a complete tabulation of the changes and additions to the Specification introduced by this Supplement. Each change or addition is defined by the section number and the title that will be employed. In each case a brief description of the change or addition is included.

1.2 Data Format Standardization Philosophy

The second paragraph was revised for clarity.

2.2 Special Navigation Terms

Editorial corrections were made with no change of intent.

2.3 Precision RNAV Terms

The Landing Threshold Point/Fictitious Threshold Point (LTP/FTP), Flight Path Control Point, Flight Path Alignment Point, Glide Path Angle, and Final Approach Segment (FAS) CRC were revised to support the Path Point concept for precision GNSS Approach Procedures.

3.2.3.1 Enroute Waypoint Section (E), Subsection (A)

The section was revised to include VFR Waypoints.

3.2.4.3 Airport Terminal Waypoints Section (P) Subsection (C)

The section was revised to include VFR Waypoints excluding the landing threshold as a fix (waypoint).

3.2.4.4 Airport Standard Instrument Departures SIDs) Section (P), Subsection (D)

Section title was revised.

3.2.4.5 Airport Standard Terminal Arrival Routes (STARs) Section (P), Subsection (E)

Section title was revised.

3.3.4 Heliport Terminal Waypoints Section (H), Subsection (C)

The section was revised to include VFR Waypoints.

4.1.2.1 VHF NAVAID Primary Records

DME Longitude was revised to columns 65 thru 74.

4.1.4 Waypoint Record (EA) or (PC)

The section was revised to include VFR Waypoints.

4.1.4.1 Waypoint Primary Records

Section title was revised.

4.1.5.1 Holding Pattern Primary Records

The RNP and ARC radius fields were added (column 63 thru 71).

The Notes field (columns 99 thru 123) was changed to Name.

4.1.6.1 Enroute Airways Primary Records

Columns 99 thru 101 were revised to Fixed Radius Transitions Indicator. Section 5.254 was added to support Fixed Radius Transition Indicators in the ER Airway file.

4.1.8.1 Airport Gate Primary Record

The Notes field was changed to Name, Section 5.60.

4.1.9.3 Airport SID/STAR/Approach Flight Planning Continuation Records

Columns 75 thru 78 was revised to Leg Distance Section 5.260.

4.1.12.1 Company Route Primary Records

Section title was revised.

4.1.20.1 Airport MSA

Multiple radius fields were added to this record.

4.1.21.1 Enroute Airways Restriction Altitude Exclusion Primary Records

Column 123 was changed from blank to Block Indicator.

Section title was revised.

4.1.21.2 Altitude Exclusion Continuation Records

Column 123 was changed from blank to Block Indicator.

Section title was revised.

4.1.21A.1 Enroute Airways Restriction Note Restriction Primary Records

Section title was revised.

4.1.21A.2 Enroute Airways Restriction Note Restriction Continuation Records

Section title was revised.

4.1.21B.1 Enroute Airways Restriction Seasonal Closure Primary Records

Section title was revised.

4.1.21C.1 Enroute Airways Restriction Cruising Table Replacement Primary Records

Section title was revised.

4.1.21C.2 Enroute Airways Restriction Cruising Table Replacement Continuation Records

Section title was revised.

4.1.27.1 Flight Planning Primary Record

Section title was revised.

4.1.27.2 Flight Planning Continuation Records

Section title was revised

4.1.27.3 Flight Planning Continuation Records

Section title was revised.

4.1.28.1 Path Point Primary Record

The Path Point Record was revised from columns 27 through 115 to support the Path Point concept for precision GNSS Approach Procedures.

The Note was revised to remove obsolete text and to support the Path Point Record description.

4.1.29.1 GLS Primary Records

Section title was revised.

4.1.30.1 Alternate Primary Records

Section title was revised.

4.2.2 Heliport Terminal Waypoint Records(HC)

The section was revised to include VFR Waypoints.

4.2.2.1 Heliport Terminal Waypoint Primary Records

Section title was revised.

4.2.2.2 Heliport Terminal Waypoint Continuation Records

Section title was revised.

4.2.2.3 Heliport Terminal Waypoint Flight Planning Continuation Records

Section title was revised.

4.2.2.4 Heliport Terminal Waypoint Flight Planning Continuation Records

Section title was revised.

4.2.4.1 Heliport MSA Primary Records

Multiple radius fields were added to this record.

Section title was revised.

4.2.4.2 Heliport MSA Continuation Records

Section title was revised.

5.7 Route Type (RT TYPE)

Deleted the Note 2 from Route Type "J."

Deleted Route Type E.

The Qualifier columns were separated from the Route Type Description into a separate table to aid in clarity.

Added Qualifiers Field 1 codes E, F, G, J, R, and V.

Added Qualifiers Field 2 codes A, B, and E.

Primary and Secondary Missed Approach codes were changed form Qualifier 1 P and S to Qualifier 2 A and B respectively.

Note 1 was revised for clarity.

Note 6 was added.

The Approach examples were revised.

5.10 Approach Route Identifier

Converted the text information into a table to make the information more readable.

5.17 Waypoint Description Code (DESC CODE)

Note 6 was revised for clarification, no change of intent.

5.18 Boundary Code (BDY CODE)

The asterisk Note was added to refer to Figure 5-2.

5.23 Recommended NAVAID (RECD NAV)

The Procedure Use Figure, 5-7 was updated to be consistent with Attachment 5, Leg Data Fields.

5.29 Altitude Description (ALT DESC)

Modified to allow coding of both minimum and VNAV path altitudes from government source or as calculated by the data supplier.

Note was added for Field Content V.

5.30 Altitude/Minimum Altitude

The third paragraph of the Definition/Description was modified to reflect the revised Altitude Description Code V.

5.35 NAVAID Class (CLASS)

The Navaid Class tables were revised for clarity. Notes 6 and 7 were deleted as they clarified in the table.

Code "U" was changed to Undefined from Unrestricted.

5.42 Waypoint Type (TYPE)

VFR Waypoint was added to both the Enroute and Terminal Waypoint tables.

Note 4 was added to support VFR Waypoints.

5.60 Name (NAME)

The section title and text were revised to more accurately reflect the implementation of the record.

5.70 Vertical Angle (VERT ANGLE)

The text was modified to allow indicating published VNAV Angle where appropriate.

5.80 ILS/MLS/GLS Category

Section title was revised to include GLS

5.82 Waypoint Usage

The term Airway was removed from the text and table.

5.101 Communications Type (COMM TYPE)

Enroute Comm Only (EFS) and Ground Comm Outlet (GCO) were added as communication types.

5.103 Communication Frequency (COMM FREQ)

The text was modified to explain that it contains either a frequency or channel to support 8.33 kHz VHF Communication.

5.104 Frequency Units (FREQ UNIT)

Text was added to include a code that indicates "channel" rather than frequency.

5.106 Service Indicator (SERV IND)

The service one indicator of "E" for EFAS was deleted.

5.114 Duplicate Indicator (DUP IND)

New Duplicate Code of 8 was added for "All Altitudes" under multiple holding patterns.

5.118 Boundary Via (BDRY VIA)

The "arc by edge" option was deleted.

5.144 Center Fix (CENTER FIX)

The Source/Content paragraph was revised to support Airport Reference Point for MSA Center Fix.

5.146 Sector Bearing (SEC BRG)

This section was revised to support beginning and ending bearing as well as multiple radii fields.

5.180 Pad Identifier (PAD IDENT)

The character length was changed to 5 characters max.

5.200 Remote Facility (REM FAC)

The text was revised to support that Remote Facility could be a navaid through which the Communication Service is transmitted, using the navaid frequency, or the Remote could be an airport that the Communication Service is transmitted from on unique frequencies.

5.204 ARC Radius (ARC RAD)

The definition/description paragraph and Used On paragraph were revised to support holding patterns for RNP Holding.

5.211 Required Navigation Performance (RNP)

The entire section was revised to support RNP and examples were added to Attachments 4 and 5 to support RNP.

5.223 Operation Types (OPS TYPE)

This section was revised to support the Path Point concept for precision GNSS Approach Procedures.

5.224 Route Indicator (RTE IND)

The title and text was revised to support the Path Point concept for precision GNSS Approach Procedures.

5.225 Ellipsoid Height

The text was revised to remove the reference to feet as the path point concept is only in meters.

5.226 Glide Path Angle (GPA)

Text referring to the illustration on Precision Approach Path Points was added.

Figure 5-15 on Precision Approach Path Points was added by this Supplement.

5.227 Orthometric Height (ORTH HGT)

The text was revised to remove the reference to feet as the path point concept is only in meters.

5.228 Course Width at Threshold (CRS WDTH)

The title and text was revised to support the Path Point concept for precision GNSS Approach Procedures.

5.229 Final Approach Segment data CRC Remainder (FAS CRC)

The title and text was revised to support the Path Point concept for precision GNSS Approach Procedures.

5.254 Fixed Radius Transitions Indicator (FIXED RAD IND)

This Section 5.254 was added to support constant Fixed Radius Transition Indicators in the ER Airway file.

5.255 SBAS Service Provider Identifier (SBAS ID)

This section was added to support the Path Point concept for precision GNSS Approach Procedures.

5.256 Reference Path Data Selector (REF PDS)

This section was added to support the Path Point concept for precision GNSS Approach Procedures.

5.257 Reference Path Identifier (REF ID)

This section was added to support the Path Point concept for precision GNSS Approach Procedures.

5.258 Approach Performance Designator (APD)

This section was added to support the Path Point concept for precision GNSS Approach Procedures.

5.259 Length Offset (OFFSET)

This section was added to support the Path Point concept for precision GNSS Approach Procedures.

5.260 Terminal Procedure Flight Planning Leg Distance (LEG DIST)

This section was added to define the segment distance along the track.

6.7 Labels

The label format was updated to 132 bit character format.

6.7.1 Volume Header Label (VOL)

The label format was updated to 132 bit character format.

6.7.2 Header 1 Label (HDR 1)

The label format was updated to 132 bit character format.

6.7.3 Header 2 Label (HDR 2)

The label format was updated to 132 bit character format.

6.7.4 End-of-File Trailer Label (EOF)

The label format was updated to 132 bit character format.

6.7.5 End-of-Volume Trailer Label (EOV)

The label format was updated to 132 bit character format.

6.10.2 Generator Polynomials

The CRC–32Q polynomial was revised to correlate with Section 6.11.3.

7.3.6 VFR Waypoints

This section was added to support the inclusion of VFR Waypoints.

ATTACHMENT 2 LOCAL HORIZONTAL REFERENCE DATUM NAME, DATUM CODE, AND ELLIPSOID LIST

The Local Horizontal Reference Datum code and Ellipsoid list was updated.

ATTACHMENT 4 AIRWAY MINIMUM ALTITUDE AND REQUIRED NAVIGATION PERFORMANCE (RNP)

Coding Examples for RNP were added.

ATTACHMENT 5 PATH AND TERMINATOR

Introductory Section

A commentary was added for XA legs.

1. The title was changed to Distance to Calculation from Speed.

2. The definition of Bank Angle was revised to state the ground speed of 250 knots.

4. The title was changed to Outbound Leg from Teardrop Procedures.

5. The Intercept Angle, items “A” and “C” were revised to support approach transitions to intercept the localizer approach path.

1.2 Beginning and Ending Leg Types

A “TF” leg was added to the Beginning Leg column for Missed Approach.

A “IF” leg was added to the Beginning Leg column for SID Runway Transition and SID with a note that it be proceeded by the runway as an “IF.”

RF path terminators were added to each Ending Leg column accept Approach Transition.

The Note, “When Approach Transition is for GPS Approach Procedure” was deleted.

1.3 Leg Sequencing

An ampersand was added to the square for “TF/DF” and the corresponding Note was revised to include an “TF/DF” sequence.

The # symbol and the corresponding Note were deleted.

2.0 Coding Rules Applicable to All Procedures

Rule 2.9 was added to support RNP.

5.0 Standard Terminal Arrival Route(STAR) Coding Rules

The following text replaced wording after Approach Transition in Rule 5.7 line 2:

"both the Arrival Route and the Approach Transition will be coded in their entirety in accordance with source documentation."

6.0 Approach Procedure Rules Valid For All Procedure Types

Rule 6.2.1 was revised in correlation with Section 5.29.

Rule 6.2.5 the text was added by this Supplement. Rule 6.2.5.2 was previously Rule 6.2.5.

Rule 6.2.5.1 was added by this Supplement.

Rule 6.2.5.2 was previously Rule 6.2.5 before Rule 6.2.5.1 and 6.2.5.2 were added.

Rule 6.2.5.3 was added to support VNAV coding of approach procedures where the MAP is not at the LTP.

The following text was deleted from Rule 6.2.9.3:

"in cases where the final approach course was designed to cross over the runway threshold."

The following clause was deleted from Rule 6.2.9.4:

"or the published Missed Approach Point is abeam the runway threshold."

Rule 6.2.10.1 was revised in correlation with Section 5.29.

The first sentence of Rule 6.2.10.2.d was revised for clarification.

Rule 6.2.10.3 was revised in correlation with Section 5.29.

Rule 6.3.4.was deleted by Supplement 16.

Rule 6.3.6.4, the following text was added: "angles between 10 and 90 degrees may be coded as required."

Rule 6.6.3.2 was revised to recommend the VORDME RNAV approaches will be the procedure VORDME

7.0 Precision Approach Coding

The following sentence was added to Rule 7.1.7:

"If the altitude 1 and 2 constraints for the FAF are "at" (G) and equal to the FACF altitude with an altitude description of "I", the vertical angel is omitted from the FAF record."

Rule 7.1.8, the Notes were revised for clarity, no change of intent.

8.0 Non-Precision Approach Procedure Coding

Rule 8.1.1 was revised in correlation with Section 5.29.

The first sentence of Rule 8.3.2 was revised for clarification.

Rule 8.3.5.3 the first sentence was revised as follows:

"The recommended navaid is not required unless specified by the state, or if required by leg type. The recommended navaid, must be prior procedure reference VORDME or VORTAC."

The first sentence of rule 8.6 was revised as follows:

"Procedures which are Circle-To-Land may be included in the data base."

Rule 8.6.4 was revised to include the Final End Point.

9.0 Missed Approach Procedures Rules Valid For All Procedures Types

Rule 9.3.1.4 "if not specified by source" was added to end of the first sentence.

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SUPPLEMENT 17

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: August 31, 2004

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: May 5, 2004

A. PURPOSE OF THIS SUPPLEMENT

This Supplement introduces revisions and additions to the body and Attachments of ARINC Specification 424. These changes remove obsolete text, allow using Airport Records as the "TO FIX" in Enroute Airways and improve the operational utility of the navigation data base for RNP Procedures.

B. ORGANIZATION OF THIS DOCUMENT

This document, printed on goldenrod paper, contains descriptions of changes introduced into ARINC Specification 424 by this Supplement. The material in Supplement 17 is integrated into ARINC Specification 424 to form an updated version of the standard.

The changes introduced by Supplement 17 have been identified using change bars.

C. CHANGES TO SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This section presents a complete tabulation of the changes and additions to the Specification introduced by this Supplement. Each change or addition is defined by the section number and the title that will be employed. In each case a brief description of the change or addition is included.

1.1 Purpose of this Document

The obsolete terms Tapes and Cassettes were removed and new text was added.

A statement was included that ARINC 424 is to be an enabling document and enables database suppliers, avionics systems, and other users of the databases to flight plan and fly procedures as prescribed by procedure designers.

1.2 Data Format Standardization Philosophy

The obsolete terms Tapes and Cassettes were removed and revised text was added.

The first Commentary on clarifying the term Cassette was deleted.

1.3 Organization of the Document

The obsolete terms Tapes and Cassettes were removed and revised text was added.

1.3.1 Coverage of Helicopter Operation Needs

Editorial corrections were made.

Figure 1-1 ARINC Specification 424 Information Presentation

The obsolete terms Tapes and Cassettes were removed and revised text was added.

2.1 Data Processing Terms

The term Alpha was revised to describe any letter of the alphabet (A through Z); any punctuation; or any printable character, other than a numeric, including space.

The term Cassette was deleted.

2.2 Special Navigation Terms

The Mandatory Hold definition was added.

3.1 User File Organization

The obsolete terms Tapes and Cassettes were removed and revised text was added.

3.2.4.1 Airport Reference Points Section (P), Subsection (A)

This section was updated to specifically allow using Airport Records as the "TO FIX" in Enroute Airways.

3.2.4.17 Airport Terminal Arrival Altitude Section (P), Subsection (K)

New section added to support Terminal Arrival Altitudes.

3.3.10 Heliport Terminal Arrival Area Section (H), Subsection (K)

New section added to support Terminal Arrival Altitudes.

Figure 3-1 Data Sorting Necessary to Achieve Step 1 of FDSU File Production Process

The Figure title was revised to remove the outdated term tape.

4.1.2.2 VHF NAVAID Continuation Records

Application Type field was added and the Application Note was deleted.

4.1.2.5 VHF NAVAID Flight Planning Continuation Records

A Note was added explaining that Flight Planning continuation records cannot have an Application Type column.

4.1.2.6 VHF NAVAID Limitation Continuation Record

The Application Type field was added.

4.1.3.2 NDB NAVAID Continuation Records

Application Type field was added and the Application Note was deleted.

4.1.3.5 NDB NAVAID Flight Planning Continuation Records

A Note was added explaining that Flight Planning continuation records cannot have an Application Type column.

4.1.4.2 Waypoint Continuation Records

Application Type field was added and the Application Note was deleted.

4.1.4.4 Waypoint Flight Planning Continuation Records

A Note was added explaining that Flight Planning continuation records cannot have an Application Type column.

4.1.5.2 Holding Pattern Continuation Records

Application Type field was added and the Application Note was deleted.

4.1.6.2 Enroute Airways Continuation Records

Application Type field was added and the Application Note was deleted.

4.1.6.4 Enroute Airways Flight Planning Continuation Records

A Note was added explaining that Flight Planning continuation records cannot have an Application Type column.

4.1.7 Airport Records (PA)

This section was updated to specifically allow using Airport Records as the "TO FIX" in Enroute Airways.

4.1.7.2 Airport Continuation Records

Application Type field was added and the Application Note was deleted.

4.1.7.4 Airport Flight Planning Continuation Records

A Note was added explaining that Flight Planning continuation records cannot have an Application Type column.

4.1.8.2 Airport Gate Continuation Records

Application Type field was added and the Application Note was deleted.

4.1.9.1 Airport SID/STAR/Approach Primary Record

Speed Limit Description field was added, column 118.

A TAA Procedure Turn and TAA Sector identifier field was added.

Note 3 was added to support columns 107-116.

4.1.9.2 Airport SID/STAR/Approach Continuation Records

The Application Type field was added and the record was modified to incorporate the procedure TCH and the Altitude three requirements.

Note 1 was deleted on Application Type.

4.1.9.4 Airport SID/STAR Flight Planning Continuation Records

A Note was added explaining that Flight Planning continuation records cannot have an Application Type column.

4.1.10.1 Runway Primary Records

TCH Value Indicated filed was added to support Altitude description in ILS coding.

4.1.10.2 Runway Continuation Records

Application Type field was added and the Application Note was deleted.

4.1.11.1 Airport and Heliport Localizer and Glide Slope

A set of supporting facilities to contain Section, Subsection ID, and ICAO identifier to support MLS DME was added.

4.1.11.2 Airport and Heliport Localizer and Glide Slope Continuation Records

Application Type field was added and the Application Note was deleted.

4.1.13.2 Airport and Heliport Localizer Continuation Records

The Continuation Records was added to be consistent with the philosophy of addressing a new industry standard for providing primary data beyond the 132 character Primary record.

4.1.14.2 Airport Communications Continuation Records

The Application Type field was added.

4.1.14.3 Airport Additional Continuation Records

The Application Type field was added.

4.1.15.2 Airways Marker Continuation Records

The Continuation Records was added to be consistent with the philosophy of addressing a new industry standard for providing primary data beyond the 132 character Primary record.

4.1.17.2 FIR/UIR Continuation Records

The Continuation Records was added to be consistent with the philosophy of addressing a new industry standard for providing primary data beyond the 132 character Primary record.

4.1.18.2 Restrictive Airspace Continuation Record

The Application Type field was added.

4.1.18.3 Restrictive Airspace Flight Planning Continuation Record

The Application Type field was added.

4.1.20.2 Airport MSA Continuation Records

The Application Type field was added.

4.1.22.1 Airport and Heliport MLS Primary Records

A set of supporting facilities to contain Section, Subsection ID, and ICAO identifier to support MLS DME was added.

4.1.21.2 Enroute Airways Restriction Altitude Exclusion Continuation Records

The Application Type field was added.

4.1.21A.2 Enroute Airways Restriction Note Restriction Continuation Records

The Application Type field was added.

4.1.21C.2 Enroute Airways Restriction Cruising Table Replacement Continuation Records

The Application Type field was added.

4.1.22.2 Airport and Heliport MLS Continuation Records

The Application Type field was added.

4.1.23.2 Enroute Communications Continuation Records

The Application Type field was added.

4.1.23.3 Enroute Communications Continuation Records

The Application Type field was added.

4.1.24.2 Preferred Route Continuation Records

The Application Type field was added.

4.1.24.3 Preferred Route Continuation Record (ET)

The Application Type field was added.

4.1.25.2 Controlled Airspace Continuation Records

The Application Type field was added.

4.1.26.2 Geographical Reference Table Continuation Records

The Continuation Records was added to be consistent with the philosophy of addressing a new industry standard for providing primary data beyond the 132 character Primary record.

4.1.27.1 Flight Planning Primary Records

Speed Limit Description field was added.

4.1.27.2 Flight Planning Continuation Records

The Application Type field was added.

4.1.28.1 Path Point Primary Record

The record was updated to be congruent with RTCA DO-229C.

4.1.28.2 Path Point Continuation Records

New section added by this supplement.

4.1.29.2 GLS Continuation Records

The Continuation Records was added to be consistent with the philosophy of addressing a new industry standard for providing primary data beyond the 132 character Primary record.

4.1.31 Airport TAA (PK)

New section added to support Terminal Arrival Altitudes.

4.1.31.1 Airport TAA Primary Records

New section added to support Terminal Arrival Altitudes.

4.1.31.2 Airport Terminal Arrival Area Continuation Records

New Section added to support Terminal Arrival Altitudes.

4.2.1.2 Heliport Continuation Records

The Application Type field was added.

4.2.1.4 Heliport Flight Planning Continuation Records

The Application Type field was added.

4.2.2.2 Heliport Terminal Waypoint Continuation Records

Application Type field was added and the Application Note was deleted.

4.2.3.1 Heliport SID/STAR/Approach Primary Records

Speed Limit Description field was added.

A TAA Procedure Turn and TAA Sector identifier field was added.

Note 2 added to support Columns 119-120.

Note 2 added to support Columns 107-116.

4.2.3.2 Heliport SID/STAR/Approach Continuation Records

Application Type field was added and the record was modified to incorporate the procedure TCH and the Altitude three requirements.

Note 1 was deleted on Application Type.

4.2.3.4 Heliport SID/STAR/Approach Flight Planning Continuation Records

A Note was added explaining that Flight Planning continuation records cannot have an Application Type column.

4.2.4.2 Heliport MSA Continuation Records

Application Type field was added and the Application Note was deleted.

4.2.5.2 Heliport Communications Continuation Records

Application Type field was added.

4.2.5.3 Heliport Communications Continuation Records

The Application Type field was added.

4.2.6.1 Heliport TAA Primary Records

New section added to support Terminal Arrival Altitudes.

4.2.6.2 Heliport Terminal Arrival Area Continuation Records

New section added to support Terminal Arrival Altitudes.

5.5 Subsection Code (SUB CODE)

This section was updated to support TAA.

5.6 Airport/Heliport Identifier (ARPT/HELI IDENT)

This section was updated to specifically allow using Airport Records as the "TO FIX" in Enroute Airways.

Text was added to support TAA.

5.7 Route Type (RT TYPE)

The "H" identifier of the Qualifier 2 Field and the supporting Note 5 were deleted.

The Airport Approach (PF) and Heliport Approach (HF) Records – Route Type table was revised to include the following new or revised Approach Route Types:

VORDME Approach
GNSS Landing System (GLS) Approach
Non-Directional Beacon + DME (NDB+DME) Approach
VOR Approach using VORDME/VORTAC

The Airport Approach (PF) and Heliport (HF) Records – Route Type table was revised to include the following new or revised Route Types:

GBAS Procedure
RNAV, Sensor Not Specified
SBAS Procedure

Note 1 and 3 were revised to support the Qualifier Fields.

Commentary on GLS was added.

The Approach examples were revised for clarity.

5.10 Approach Route Identifier (APPROACH IDENT)

This section was updated to support TAA.

5.12 Sequence Number (SEQ NR)

This section was updated to support TAA.

5.17 Waypoint Description Code (DESC CODE)

This section was updated to specifically allow using Airport Records as the "TO FIX" in Enroute Airways.

5.23 Recommended NAVAID (RECD NAV)

This section was updated to include TACAN.

5.29 Altitude Description (ALT DESC)

Text was added to address Altitude Description in ILS coding for Altitude three.

Field content X and Y was added along with a supporting Note.

5.30 Altitude/Minimum Altitude

Text was added to address Altitude Description in ILS coding for Altitude three.

5.33 VOR/NDB Identifier (VOR IDENT/NDB IDENT)

Airport and Heliport Localizer, and Airport and Heliport MLS records were added to the Used On section.

5.36 Latitude (LATITUDE)

Path Point Record was removed from the Used On text.

5.37 Longitude (LONGITUDE)

Path Point Record was removed from the Used On text.

5.42 Waypoint Type (TYPE)

New codes were added to column 28 to support the FAA High Altitude redesign.

5.67 Threshold Crossing Height (TCH)

The text was revised to clearly identify the data in the Runway record and Commentary was added noting that the single TCH value provided on the Runway Record may be different from the Approach Continuation Record.

5.70 Vertical Angle (VERT ANGLE)

The vertical angle paragraph was revised to address the Localizer Only Vertical Angle.

5.72 Speed Limit (SPEED LIMIT)

The Definition/Description and the Source/Content were revised to designate whether the speed limit coded at the fix in a terminal procedure is mandatory, minimum or maximum.

5.91 Application Type (USER)

User Application A, C, E, L, N, T, U, V, and Q were added.

5.101 Communication Type (COMM Type)

Automatic Surface Observing System (ASO) communication typed was added.

Air to Air communication type was added.

Aerodrome Weather Information Services (AWIS) communication type was added.

Military Frequency communication type was added.

Pilot Activated Lighting (PAL) communication type was added along with a supporting Note.

5.106 Service Indicator (SERV IND)

Air/Air service indicator was added.

5.143 Grid MORA

A Commentary was added that indicates some MORA values are provided as a representation of government source.

5.146 Sector Bearing (SEC BRG)

This section was updated to support TAA.

5.147 Sector Altitude (SEC ALT)

This section was updated to support TAA.

5.149 Figure of Merit (MERIT)

Section was revised to allow a method to identify those TACANs that are part of the National Air Space (NAS) and those that are not.

5.165 Magnetic/True Indicator (M/T IND)

This section was updated to support TAA.

5.204 ARC Radius (ARC RAD)

Text was added stating that the ARC Radius field is also used to specify the turn radius of RNP holding patterns included in SID, STAR, and Approach Records as HA, HF, and HM legs.

5.222 GPS/FMS Indicator (GPS/FMS IND)

Section was modified to update the flagging procedures for WAAS.

5.225 Ellipsoidal Height

The resolution was changed to a tenth of a meter.

5.227 Orthometric Height (ORTH HGT)

The resolution was changed to a tenth of a meter.

5.228 Course Width At Threshold (CRS WDTH)

The width was revised to meters and ends in a resolution of 0.25 meters, and when the procedure is a helicopter alighting point, the value is 38 meters.

5.244 GLS Channel

The section was revised to clearly identify Channel numbers for GBAS and SBAS.

5.259 Length Offset (OFFSET)

The Length Offset value was revised meters.

5.261 Speed Limit Description

New section added to designate whether speed limit coded at a fix in a terminal procedure is mandatory, minimum or maximum speed.

5.262 Approach Type Identifier (ATI)

New section added to support the Path Point Record.

5.263 HAL

New section added to support the Path Point Record.

5.264 VAL

New section added to support the Path Point Record.

5.265 Path Point TCH

New section added to support the Path Point Record.

5.266 TCH Units Indicator

New section added to support the Path Point Record.

5.267 High Precision Latitude (HPLAT)

New section added to support the Path Point Record.

5.268 High Precision Longitude (HPLONG)

New section added to support the Path Point Record.

5.269 Helicopter Procedure Course (HPC)

New section added to support the Path Point Record.

5.270 TCH Value Indicator (TCHVI)

The section was added to address Altitude Description in ILS Coding to address “Altitude Three.

5.271 Procedure Turn (PROC TURN)

This section was added to support TAA.

5.272 TAA Sector Identifier

This section was added to support TAA.

5.273 TAA IAF Waypoint

This section was added to support TAA.

5.274 TAA Sector Radius

This section was added to support TAA.

Figure 5-7 Procedure Use

Figure was updated to support TACAN.

6.1 General

The section was revised to remove the outdated references to magnetic tape formats.

6.2 Header Record

Title and the section were revised to remove the obsolete references to magnetic tape formats.

6.2.1 Header Record 1

New section added by this supplement.

6.2.2 Header Record 2

New section added by this supplement.

6.3 Bit Density

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.4 Coding

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.5 Parity Convention

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.6 Reel-File Relationship

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.7 Labels

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.7.1 Volume Header Label (VOL)

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.7.2 Header 1 Label (HDR 1)

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.7.3 Header 2 Label (HDR 2)

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.7.4 End-of-File Trailer Label (EOF)

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.7.5 End-of-Volume Trailer Label (EOV)

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.8 Tape Marks

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.9 Summary of Tape Data Layout

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.9.1 One File, One Reel

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.9.2 One File, Multiple Reels

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.9.3 Multiple Files, One Reel

This section was deleted and replaced by Section 6.2.1 and 6.2.2.

6.11 Application of CRC for Integrity Protection of Straight & Advanced Landing Approach Operations

Text was added to refer RTCA DO-229C for Final Approach Segment Data Block and CRC standards.

6.11.1 Data Block Structure,M(x)

This section was deleted when Section 6.11 was revised to refer to RTCA DO-229C.

6.11.2 RNAV-GPS/GLS Approach Procedure Path Point Data Fields Bits

This section was deleted when Section 6.11 was revised to refer to RTCA DO-229C.

6.11.3 CRC – Generator Polynomial G(x)

This section was deleted when Section 6.11 was revised to refer to RTCA DO-229C.

ATTACHMENT 1 FLOW DIAGRAM

The obsolete terms Tapes and Cassettes were removed and revised text was added.

The Notes were revised to remove obsolete text.

ATTACHMENT 5 PATH AND TERMINATOR

1.2 Beginning and Ending Leg Types

SID Runway Transition, removed flag Note 1 from FM and VM.

SID, Beginning Leg List, removed flag note 1 from FM and VM, and flag note 3 from VM. Ending Leg, added FM and VM.

1.3 Leg Sequencing

FA/IF leg combination was changed to a valid leg combination.

1.4 Leg Type Description

Revised the text for HA, HF, and HM in the description block.

1.5 Leg Data Field

HA, HF, and HM leg in the "HLD" column was changed to invalid. AF, CF, DF, IF, RF, and TF changed to optional.

HA, HF, and HM legs in the TD field changed to required from optional.

HA, HF, and HM in the Overfly column changed to optional, and TF leg was changed to "B."

RF in the MAP field was changed to Optional.

AF, IF, RF, and TF field was changed to Optional.

RF in the THETA field changed to I and Legend I was added for inbound tangential track.

VD in the TM/DST field changed to DME Distance.

Legend T and I were added.

3.0 Path and Termination Related Rules Valid For All Procedure Types

Rule 3.5 and 3.6 were revised to include RF.

Rule 3.14 was revised to exclude leg combination IF/RF, RF/RF or Hx.

4.0 STANDARD INSTRUMENT DEPARTURE (SID) CODING R.ULES

Rule 4.1 was revised so that RNAV Departure Procedure requires an initial turn is excess of 15 degrees unless certain conditions are met.

Rule 4.3 and Rule 4.4 was revised in correlation with Section 1.2, Beginning and Ending Leg Types to allow a single VM leg for a SID.

6.0 APPROACH PROCEDURE RULES VALID FOR ALL PROCEDURE TYPES

The following text was added to Rule 6.2.3:

"When a VORDME or VORTAC is coded as the recommended Navaid in approach procedure coding, the Navaid will be within 40 NM of the fix in which it is coded. GPS approach procedures do not include a recommended Navaid. GLS approach procedures will reference the GLS facility. For specific rules on recommended Navaid for NDB + DME Approach Procedures, see Rule 6.8.1.4.b of this Attachment."

Rule 6.2.5 was revised to specify what approach procedure coding will include a Final Approach Course Fix.

Rules 6.2.5.1, 6.2.5.2, 6.2.5.3, and 6.2.5.4 were revised to support the FACF coding.

Rule 6.2.10.2.d was modified to require an altitude in the leg from FEP to MAP.

Rule 6.5.3.2 was revised to remove the reference to the overfly bit.

Rule 6.3.6.1 was revised to support FACF coding.

Rule 6.3.6.2 was revised to include TACAN

Rule 6.3.6.3 was deleted.

Rule 6.3.6.3.4 was revised to state if the angles between 10 and 90 degrees may be coded as required by source documentation, provide the resulting intercept is within the reception area of the localizer.

Rule 6.3.6.5 and 6.4.2.1 were revised to include FACF.

Rule 6.5.3.2 was revised to remove obsolete text.

The following text was added to Rule 6.6.1.2.b:

"The final approach will include a FACF when one is required by Rule 6.2.5."

Rule 6.6.3.1 was revised to include missed approach point.

The following text was added to Rule 6.6.3.3:

"RF legs may be used in the final approach segment of RNAV approaches."

Rule 6.7, 6.7.1, 6.7.1.1, and 6.7.1.2 were deleted and Rules 6.7.2, 6.7.2.1, and 6.7.2.2 were renumbered to 6.7, 6.7.1 and 6.7.2.

The following text was added to Rule 6.7.2.2:

"The final approach will include a FACF when one is required by Rule 6.2.5."

Rule 6.8.1.1.was deleted as it is redundant with Rule 6.2.5.2.

The following text was added to Rule 6.8.1.2:

"The final approach will include a FACF when one is required by Rule 6.2.5."

Rule 6.8.1.3 was revised to include "or FACF when coded."

Rule 6.8.1.4 was revised to include FACF when coded.

7.0 Precision Approach Procedure Coding

Rule 7.4.3.2 was added for RF legs.

8.0 NON-PRECISION APPROACH PROCEDURE CODING

Rule 8.1.1 was revised to refer to Rule 6.2.5.2.

Rule 8.1.2 was added to indicated that all must be coded in the Missed Approach Point (MAP).

Rule 8.3.2 was revised to refer to Rule 6.2.5.

The following text was added to Rule 8.3.4.2:

"The final approach will include a FACF when one is required by Rule 6.2.5."

Rules 8.3.6, 8.3.6.1, 8.3.6.2, and 8.3.6.3 were deleted.

Rule 8.4.2 was deleted as 6.2.5.2 was revised and the rule is now redundant.

The following text was added to Rule 8.4.3:

"The final approach will include a FACF when one is required by Rule 6.2.5."

Rule 8.7.1 was revised to support approach transition can end in a leg to the FAF and that the FACF fix is required only on Localizer based procedures and optional when provided by source as FACF or IF.

Rule 8.9.1.1 was added to support TCH.

Rule 8.9.9 was added to support step-down fix coding.

Rule 8.10, 8.10.1, 8.10.2, 8.10.3, and the associated VNAV Approach Coding examples 14, 15, A, B, C, and C were added by this supplement.

9.0 Missed Approach Procedure Rules Valid For All Procedure Types

Rules 9.3.1.2, 9.3.1.3, 9.3.1.4, and 9.3.1.5 were revised in their entirety to clarify the coding of first leg of Missed Approach coding.

Ruled 9.4, 9.4.1, 94.1.1, 9.4.1.2, 9.4.1.3, 9.4.1.4, and 9.4.1.5 were deleted.

Rule 9.5.2 was revised in its entirety that the reference Navaid must be used as the Recommended Navaid when required.

Rule 9.5.2.2 was deleted.

Rule 9.5.2.4 was added to so that when the first leg if a Missed Approach Procedure is coded as HA or HM leg the coding will also include a government source provided altitude.

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SUPPLEMENT 18

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: November 23, 2005

Prepared by the Airlines Electronic Engineering Committee

Adopted by the Airlines Electronic Engineering Committee: October 5, 2005

A. PURPOSE OF THIS SUPPLEMENT

This supplement introduces revisions and additions to the body and attachments of ARINC Specification 424. The majority of the changes were added to support level of service based on published procedure operating minimums information for sensor independent RNAV Approach procedures.

B. ORGANIZATION OF THIS DOCUMENT

This document, printed on goldenrod paper, contains descriptions of changes introduced into ARINC Specification 424 by this supplement. The material in Supplement 18 is integrated into ARINC Specification 424 to form an updated version of the standard.

The changes introduced by Supplement 18 have been identified using change bars.

C. CHANGES TO SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This section presents a complete tabulation of the changes and additions to the Specification introduced by this supplement. Each change or addition is defined by the section number and the title that will be employed. In each case a brief description of the change or addition is included.

2.3 Precision RNAV Terms

Final Approach Segment (FAS) Data Block and Level of Service (LPV, LNAV/VNAV, and LNAV) were added.

4.1.9 Airport SID/STAR/Approach (PD, PE and PF)

Text was added to clarify that Approach file contains the sequential listing of published Airport Standard instrument Approach Procedures

4.1.9.1 Airport SID/STAR/Approach Primary Records

Note 2 was revised for clarity. Note 4 was added to support RNP criteria.

4.1.9.2 Airport SID/STAR/Approach Primary Extension Continuation Records

Column 86 thru 89 was revised to Blank from Procedure Category. Note 2 was revised to support RNP Procedures with multiple RNP values.

4.1.9.5 Airport Procedure Data Continuation Record

New section added to contain information for airport Approach Procedures for which SBAS is authorized.

4.1.10.1 Runway Primary Records

LTP Ellipsoid Height was added to columns 61-66.

4.2.3 Heliport SID/STAR/Approach (HD/HE/HF)

Text was added to indicate the STAR File contains the sequential listing of published Heliport Standard Terminal Arrival Routes. The Approach File contains the sequential listing of published Heliport Standard Instrument Approach Procedures.

4.2.3.1 Heliport SID/STAR/Approach Primary Records

Note 2 was revised for clarity.

4.2.3.2 Heliport SID/STAR/Approach Primary Extension Continuation Records

Column 86 thru 89 was revised to Blank from Procedure Category. Note 2 was revised to support RNP Procedures with multiple RNP values.

4.2.3.5 Heliport Procedure Continuation Records

New section added to contain information for airport Approach Procedures for which SBAS is authorized.

5.5 Subsection Code (SUB CODE)

The section and subsection encoding scheme table were revised to support RNAV Name Table.

5.7 Route Type (RT TYPE)

The Airport Approach (PF) and Heliport (HF) Record table were revised. Notes 1, 2, 3, 4, and 5 were revised and Note 6 was added.

5.91 Continuation Record Application Type (APPL)

Field Content “W” was added for SBAS authorized.

5.222 GNSS/FMS Indicator (GNSS/FMS IND)

The title and text was revised to GNSS from GPS as RNAV is part of the new ICAO procedure naming convention.

Note 1 was added for GNSS overlay.

5.225 Ellipsoidal Height

Source/Content text was revised for clarity and the Runway Record was added to the Used on text.

5.275 Level of Service Name (LSN)

New section added to support level of service based on published procedure operating minimums information for sensor independent RNAV Approach Procedures.

5.276 Level of Service Authorized

New section added to support the level of operating criteria for RNAV approach procedures identified by FAA TSO-C146a.

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SUPPLEMENT 19

TO

ARINC SPECIFICATION 424

NAVIGATION SYSTEM DATA BASE

Published: December 19, 2008

Prepared by the AEEC

Adopted by the AEEC Executive Committee: October 22, 2008

A. PURPOSE OF THIS SUPPLEMENT

This supplement introduces revisions and additions to the body and attachments of ARINC Specification 424. The majority of the changes were added to support level of service based on published procedure operating minimums information for sensor independent RNAV Approach procedures.

B. ORGANIZATION OF THIS DOCUMENT

This document, printed on goldenrod paper, contains descriptions of changes introduced into ARINC Specification 424 by this supplement. The material in Supplement 19 is integrated into ARINC Specification 424 to form an updated version of the standard.

The changes introduced by Supplement 19 have been identified using change bars.

C. CHANGES TO SPECIFICATION 424 INTRODUCED BY THIS SUPPLEMENT

This section presents a complete tabulation of the changes and additions to the Specification introduced by this Supplement. Each change or addition is defined by the section number and the title that will be employed. In each case a brief description of the change or addition is included.

1.3.1 Coverage of Helicopter Operation Needs

The following text was added:

Supplement 19 of this document added material required to provide for Helicopter Company Routes.

2.2 Special Navigation Terms

The Navigation term section was divided into four sections: Point Data Terms, Procedure and Route Terms, Support Terms, and Precision RNAV Terms.

2.2.1 Special Data Terms

Uncharted Airway Intersection definition was added.

2.2.2 Procedure and Route Terms

The following definitions were added to the Procedure and Route Terms:

Approach Transitions
Final Approach Coding
Final Approach Course
Missed Approach Point
Missed Approach Procedure
Precision Final Approach Fix
Terminal Procedure
VNAV Path

The following definitions were revised to the Procedure and Route Terms:

Final Approach Course Fix
Final Approach Fix
Final End Point
Initial Approach Coding
Intermediate Fix

2.2.3 Support Terms

The following definitions were revised to the Support Terms:

Enroute Airway to Restrictive Airspace Link
Localizer

2.2.4 Precision RNAV Terms

Final Approach Flight Path definition was added.

The following terms were revised to the Precision RNAV terms:

Final Approach Segment CRC
Flight Path Alignment Point
Flight Path Control Point
Glide Path Angle
Landing Threshold Point/Fictitious Threshold Point

3.2.5.1 Company Route Section (R), Subsection (Blank)

Section was revised to support helicopter Company Routes.

3.2.5.2 The Alternate Record Section (R), Subsection (A)

Section was revised to support helicopter Company Routes.

3.3.2 Jointly and Specifically Used Sections/Subsections

Section was revised to support helicopter Company Routes.

3.3.11 Helicopter Operations Company Route Section (R) Subsection (H)

New section added for helicopter operation specific company routes.

4.1.2.1 VHF NAVAID Primary Records

DME Operational Service Volume was added, column 123.

4.1.2.4 VHF NAVAID Flight Planning Continuation Records

The ability for Off-Cycle Changes was deleted.

4.1.2.5 VHF NAVAID Flight Planning Continuation Records

Section deleted.

4.1.3.4 NDB NAVAID Flight Planning Continuation Records

The ability for Off-Cycle Changes was deleted.

4.1.3.5 NDB NAVAID Flight Planning Continuation Records

Section deleted.

4.1.4.3 Waypoint Flight Planning Continuation Records

The ability for Off-Cycle Changes was deleted.

4.1.4.4 Waypoint Flight Planning Continuation Records

Section deleted.

4.1.5.1 Holding Pattern Primary Records

Vertical Scale Factor, RVSM Minimum, and RVSM Maximum were added to support Radius Valid for the MDA.

4.1.6.1 Enroute Airways Primary Records

Vertical Scale Factor, RVSM Minimum, and RVSM Maximum were added to support Radius Valid for the MDA.

4.1.6.3 Enroute Airways Flight Planning Continuation Records

The ability for Off-Cycle Changes was deleted.

4.1.6.4 Enroute Airways Flight Planning Continuation Records

Section deleted.

4.1.7.3 Airport Flight Planning Continuation Records

The ability for Off-Cycle Changes was deleted.

4.1.7.4 Airport Flight Planning Continuation Records

Section deleted.

4.1.9.1 Airport SID/STAR/Approach Primary Records

Vertical Scale factor was added to column 121 thru 123.

4.1.9.2 Airport SID/STAR/ Primary Extension Approach Continuation Records

The Airport SID/STAR primary record was revised to support providing government source procedures design magnetic variation in the SID/STAR Approach Record.

4.1.9.3 Airport SID/STAR/Approach Flight Planning Continuation Records

The ability for Off-Cycle Changes was deleted.

4.1.9.4 Airport SID/STAR Flight Planning Continuation Records

Section deleted.

4.1.9.5 Airport Procedure Data Continuation Record

Columns 74 -118 were revised to add RNP Authorized and RNP Level of Service value.

4.14.1 Airport Communications Primary Records

Communication records were restructured to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.1.14.2 Airport Communications Continuation Records

Communication records were restructured to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.1.14.3 Airport Additional Continuation Records

Communication records were restructured to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.14.4 Airport Additional Continuation Records

New record added to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.14.5 Airport Additional Continuation Records

New record added to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.1.20.2 Airport MSA Primary Record Extension

A new primary record added to support MSA.

4.1.20.3 Airport MSA Continuation Records

Section renumbered.

4.1.23.1 Enroute Communications Primary Records

Communication records were restructured to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.1.23.2 Enroute Communications Continuation Records

Communication records were restructured to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.1.23.3 Enroute Communications Continuation Record

Communication records were restructured to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.1.23.4 Enroute Communication Continuation Record

New record added to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.1.24.1 Preferred Route Primary Records

Columns 106-123 were revised to support SID?STAT/Awy Route Type.

4.1.28.1 Path Point Primary Records

Note 2 was revised clarify the at the CRC data wrap refer to RTCA/DO-229.

4.1.31.1 Airport TAA Primary Records

Section revised to remove the classification of NoPT for a group of sectors, and instead provide means of indication for each sector, based on a particular fix whether a procedure turn is required or not.

4.1.31.2 Airport Terminal Arrival Altitude Continuation Records

Section revised to remove the classification of NoPT for a group of sectors, and instead provide means of indication for each sector, based on a particular fix whether a procedure turn is required or not.

4.1.33 TACAN-Only NAVAID Record (DT)

New section added to support VOR/DME and TACAN within the same ICAO region.

4.1.33.1 TACAN-Only NAVAID Primary Records

New section added to support VOR/DME and TACAN within the same ICAO region.

4.1.34 Special Activity AREA (ES)

New Section added to support Special Activity Area (e.g., parachute jumping, hang gliding, etc.).

4.1.34.1 Special Activity Area Primary Record

New Section added to support Special Activity Area (e.g., parachute jumping, hang gliding, etc.).

4.1.35 Communication Type Translation (TV)

New record added to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.2.1.3 Heliport Flight Planning Continuation Records

The ability for Off-Cycle Changes was deleted.

4.2.1.4 Heliport Flight Planning Continuation Records

Section deleted.

2.2.3 Heliport Terminal Waypoint Flight Planning Continuation Records

The ability for Off-Cycle Changes was deleted.

4.2.2.4 Heliport Terminal Waypoint Flight Planning Continuation Records

Section deleted.

4.2.3.1 Heliport SID/STAR/Approach Primary Records

Vertical Scale factor was added to columns 121 thru 123.

4.2.3.2 Heliport SID/STAR/Approach Primary Extension Continuation Records

The Airport SID/STAR primary record was revised to support providing government source procedures design magnetic variation in the SID/STAR Approach Record.

4.2.3.3 Heliport SID/STAR/Approach Flight Planning Continuation Records

The ability for Off-Cycle Changes was deleted.

4.2.3.4 Heliport SID/STAR/Approach Flight Planning Continuation Records

Section deleted.

4.2.4.2 Heliport MSA Primary Record Extension

A new primary record added to support MSA.

4.2.4.3 Heliport MSA Continuation Records

Section renumbered.

4.2.5.1 Heliport Communications Primary Records

Communication records were restructured to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.2.5.2 Heliport Communications Primary Records

Communication records were restructured to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.2.5.3 Heliport Communication Continuation Records

Communication records were restructured to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.2.5.4 Heliport Communication Continuation Records

Communication records were restructured to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.2.5.5 Heliport Communication Continuation Records

New record added to assist in determining which communications types are recognized through the industry versus those that are unique to a given country.

4.2.6.1 Heliport TAA Primary Records

Section revised to remove the classification of NoPT for a group of sectors, and instead provide means of indication for each sector, based on a particular fix whether a procedure turn is required or not.

4.2.6.2 Heliport Terminal Arrival Altitude Continuation Records

Section revised to remove the classification of NoPT for a group of sectors, and instead provide means of indication for each sector, based on a particular fix whether a procedure turn is required or not.

4.2.7 Helicopter Operations Company Route Records (RH)

New section added.

4.2.7.1 Helicopter Operations Company Route Primary Records

New record to support Helicopter company Routes.

5.5 Subsection Code (SUB CODE)

Added new subsection code "V" to section code "T" to support the new communication structure.

Added new subsection code H to support Helicopter operation.

5.7 Route Type (RT TYPE)

Definitions were revised to support the new RNAV Coding Rules.

Helicopter straight-in and circle to land minimums were added.

Note 7 was added.

TACAN Airway Type T was added.

H, F and A Route Type Field Content were added.

Note 8 was added for Route H.

Used on portion was updated to include Helicopter Operations.

5.10 Approach Route Identifier (APPROACH IDENT)

Section was revised to support Circle-to-Land procedures.

5.11 Transition Identifier (TRANS IDENT)

Definitions were revised to support the new RNAV Coding Rules.

5.12 Sequence Number (SEQ NR)

Paragraph added to support the new communication structure.

Used on portion was updated to include Helicopter Operations.

5.17 Waypoint Description Code (DESC CODE)

Definitions were revised to support the new RNAV Coding Rules.

Engine Out SID Missed Approach Disarm Point and Note 5 were added to the Waypoint Description Table.

5.23 Recommended NAVAID (RECD NAV)

The following text was added between the Source/Content lead paragraph and detail sections:

"Procedures that use coding which require leg types referenced to specific navaids are covered by the procedure coding rules in Attachment Five to this Specification".

The general rules for field content were updated.

Note j and the Figure 5-7 were added to support the use on non-collocated facilities.

5.29 Altitude Description (ALT DESC)

Item D was added to the Field Content. The second note was revised to support the Field Content of D.

Added references to the two notes associated with altitude descriptors G, H, I, and J.

The Note for the B entry was revised.

5.30 Altitude/Minimum Altitude

Section was revised to correlate to Section 5.29.

5.35 NAVAID Class (CLASS)

The section was revise to bring into alignment with ICAO.

5.36 Latitude (LATITUDE)

Airport Heliport and helipad were added to the Used on paragraph.

5.37 Longitude (LONGITUDE)

Amended the Airport Comm, Enroute Comm, Heliport Comm location definitions table, and Notes 7 and 8 to support the new communication record structure.

Airport Heliport and helipad were added to the Used on paragraph and Figure 5-8 was revised to support Heliport Reverence Point.

5.39 Magnetic Variation (MAG VAR, D MAG VAR)

New Paragraph added to the Source/Content to support the revised communication record structure.

5.42 Waypoint Type (TYPE)

Definitions were revised to support the new RNAV Coding Rules.

5.58 Runway Magnetic Bearing (RWY BRG)

Section was revised to support the growing number of new heliports being created from retired runways.

5.67 Threshold Crossing Height (TCH)

The Source/Content paragraph was revised to support a default 40 foot TCH.

5.70 Vertical Angle (VERT ANGLE)

The entire section was rewritten to support the implementation of VNAV.

5.75 From/To – Airport/Heliport/Fix

Section revised to support Helicopter operation.

5.76 Company Route Ident

Used on portion was updated to include Helicopter Operations.

5.77 VIA Code

APT (Approach Transition) was added to the VIA Field and to Figure 5.11. The next VIA Table was added for clarity.

Used on portion was updated to include Helicopter Operations.

5.78 SID/STAR/App/AWY (S/S/A/AWY)
SID/STAR/Awy (S/S/AWY)

The Definition/Description paragraph was revised for clarity.

Used on portion was updated to include Helicopter Operations.

5.83 To FIX

Section revised to support Helicopter operation.

5.84 RUNWAY TRANS

Section revised to support Helicopter operation.

5.85 ENRT TRANS

Section revised to support Helicopter operation.

5.86 Cruise Altitude

Section revised to support Helicopter operation.

5.87 Terminal/Alternate Airport (TERM/ALT ARPT)

Section revised to support Helicopter operation.

5.88 Alternate Distance (ALT DIST)

Section revised to support Helicopter operation.

5.91 Continuation Record Application Type (APPL)

The descriptions of the field content were updated to more precisely identity the type of data contained in the continuation record.

5.92 Facility Elevation (FAC ELEV)

New Paragraph added to the Source/Content to support the revised communication record structure.

5.101 Communications Type (COMM TYPE)

The following communications types were added:

CBA – Class B Airspace
CCA – Class C Airspace
CFT – Common Traffic Advisory Frequencies
MBZ – Mandatory Broadcast Zone

The Definition/Description and the Source/Content text was updated to correlate with the new communication record structure.

5.102 Radar (RADAR)

The Definition/Description and the Source/Content text was updated to correlate with the new communication record structure.

5.103 Communications Frequency (COMM FREQ)

The Definition/Description text was updated to correlate with the new communication record structure.

5.104 Frequency Units (FREQ UNIT)

Field Content codes were revised to support the new communication record structure.

5.105 Call Sign (CALL SIGN)

The Definition/Description text was updated to correlate with the new communication record structure.

5.106 Service Indicator (SERV IND)

Remove the following from the Source/Content tables in support of the new communication record structure.

Airport/Heliport Comm
Column 27, remove the P capability
Column 28, remove the R capability
Column 29, remove the A, G and R capabilities

Enroute Comm,
Column 57, no change
Column 58, remove the R capability
Column 59, remove the G and R capabilities

5.114 Duplicate Indicator (DUP IND)

The Source/Content paragraph was revised for clarity.

5.117 FIR/UIR Indicator (IND)

The Definition/Description text was updated to correlate with the new communication record structure.

5.128 Restrictive Airspace Type (REST TYPE)

National Security Area was added to Source/Content field.

5.131 Time Code (TIME CD)

The Definition/Description text was updated to correlate with the new communication record structure.

The following identifiers were added to the Field content Primary and Primary Extension Continuation Records:

P Active Times announced by NOTAM
U Active Times are not specified in source documentation

5.144 Center Fix (CENTER FIX)

Section revised in its entirety to support new RNAV RNP approaches designed with RF path to the FAF.

5.146 Sector Bearing (SEC BRG)

The section was revived to better address information on coding MSA.

5.147 Sector Altitude (SEC ALT)

The section was revived to better address information on coding MSA.

5.151 FIR/UIR Address (ADDRESS)

The Source/Content text was updated to correlate with the new communication record structure.

5.148 Enroute Alternate Airport/Heliport (EAA)

The Section revised to support Helicopter operation.

5.153 Start/End Date

The definition was changed to denote that the start and end dates are in effect when there are restriction.

5.167 MLS Azimuth Bearing (MLS AZ BRG)
MLS Back Azimuth Bearing (MLS BAZ BRG)

The Source/Content paragraph was rewritten to support the implementation of VNAV.

5.168 Azimuth Proportional Angle Right/Left (AZ PRO RIGHT/LEFT)

Back Azimuth Proportional Angle Right/Left (BAZ PRO RIGHT/LEFT)

The section was rewritten to support the implementation of VNAV.

5.172 Azimuth Coverage Sector Right/Left (AZ COV RIGHT/LEFT)

Back Azimuth Coverage Sector Right/Left (BAZ COV RIGHT/LEFT)

The section was rewritten to support the implementation of VNAV.

5.176 Pad Dimensions

Section was revised to support the growing number of new heliports being created from retired runways.

5.177 Public/Military Indicator (PUB/MIL)

The content was revised to support a fourth category, joint use civil and military, for airports.

5.178 Time Zone

The Time zone section was updated to support differences greater 12 hours.

5.180 Pad Identifier (PAD IDENT)

The used on portion was updated to Helicopter operation Company Routes.

5.181 H24 Indicator (H24)

The Definition/Description and the Source/Content text was updated to correlate with the new communication record structure.

5.182 Guard/Transmit (G/T)

This Section was deleted as transmit only, receive only or both for a given frequency is provided by transmit and receive frequency columns of the communication record.

5.183 Sectorization (SECTOR)

The Source/Content text was updated to correlate with the new communication record structure.

5.184 Communication Altitude (COMM ALTITUDE)

The Definition/Description and the Source/Content text was updated to correlate with the new communication record structure. The character length was change to 3 characters from 5.

5.186 Sectorization Narrative

Section title was renamed to Sectorization Narrative from Narrative.

The Definition/Description and the Source/Content text was updated to correlate with the new communication record structure.

5.189 Position Narrative

Section title was renamed to Position Narrative from Remote Site Name.

The Definition/Description and the Source/Content text was updated to correlate with the new communication record structure.

5.190 FIR/RDO Identifier (FIR/RDO)

The Definition/Description and the Source/Content text was updated to correlate with the new communication record structure.

5.198 Modulation (MODULN)

The Source/Content text was revised to indicate that the field will be set to “A” unless the source documentation specifies otherwise.

5.199 Signal Emission (SIG EM)

The Source/Content text was revised to indicate that field content will be set to “3” unless the source documentation specifies otherwise.

5.200 Remote Facility (REM FAC)

The Definition/Description and the Source/Content text was updated to correlate with the new communication record structure.

5.211 Required Navigation Performance (RNP)

Used on text was revised to include SID/STAR/Approach continuation records.

5.222 GNSS/FMS Indicator (GNSS/FMS IND)

Note 1 was added to Field Content B.

Note 2 was added to support GNSS/FMS IND.

5.223 Operation Type (OPS TYPE)

The Definition/Description was revised to indicate whether the operation is an approach procedure, an advanced operation or other operational to be defined later.

Commentary was added describing Advance Operations.

5.224 Route Indicator (RTE IND)

Text was added stating that the final approach segments are contained in the Final Approach Coding.

5.229 Final Approach Segment Data CRC Remainder (FAS CRC)

The text was amended to refer to RTCA/DO-229.

5.258 Approach Performance Designator (APD)

The character type was listed as Alpha, it should have been numeric.

5.270 TCH Value Indicator (TCHVI)

The TCH was modified to add the two default values.

5.271 Procedure Turn (PROC TURN)

Section revised to correspond to the revision made in Section 4.1.31.1, 4.1.31.2, 4.2.6.1, and 4.2.6.2.

5.272 TAA Sector Identifier

Section revised to correspond to the revision made in Section 4.1.31.1, 4.1.31.2, 4.2.6.1, and 4.2.6.2.

5.275 Level of Service Name (LSN)

Note 4 was added on RNO Level of Service.

5.277 DME Operational Service Volume (D-OVS)

New section added to support VOR/DME and TACAN within the same ICAO region.

5.278 Activity Type

New Section added to support Special Activity Area (e.g., parachute jumping, hang gliding, etc.).

5.279 Activity Identifier

New Section added to support Special Activity Area (e.g., parachute jumping, hang gliding, etc.).

5.280 Special Activity Area Size

New Section added to support Special Activity Area (e.g., parachute jumping, hang gliding, etc.).

5.281 Special Activity Area Volume

New Section added to support Special Activity Area (e.g., parachute jumping, hang gliding, etc.).

5.282 Special Activity Area Operating Times

New Section added to support Special Activity Area (e.g., parachute jumping, hang gliding, etc.).

5.283 Communications Class (Comm Class)

New section added to assist in determining communications types.

5.284 Assigned Sector Name (ASN)

New section added to assist in determining communications types.

5.285 Time Narrative

New section added to assist in determining communications types.

5.286 Multi-Sector Indicator (MSEC IND)

New section added to assist in determining communications types.

5.287 Type Recognized By (TRB)

New section added to assist in determining communications types.

5.288 Translation

New section added to assist in determining communications types.

5.289 Used On

New section added to assist in determining communications types.

5.290 Procedure Design Mag Var. (PDMV)

The section was added to support Section 4.1.9.2.

5.291 Procedure Design Mag Var Indicator (PDMVI)

The section was added to support Section 4.1.9.2.

5.292 Category Distance

Section was added to support the growing number of new heliports being created from retired runways.

5.293 Vertical Scale Factor (VSF)

Section was added to support the new column in 4.1.61.

5.294 RVSM Minimum Level

Section was added to support the new column in 4.1.61.

5.295 RVSM Maximum Level

Section was added to support the new column in 4.1.61.

5.296 RNP Level of Service (LSN)

Section added to support minimums information for Approach Procedures for RNP.

5.297 Route Inappropriate Navaid Indicator

Section added to support Inappropriate Navaid Indicator.

6.2.1 Header Record 1

Column 125-132 was revised to refer to ARINC Report 665.

6.10 CRC Calculations

Section 6.10 and sub-section were deleted.

6.11 Application of CRC for Integrity Protection of Straight & Advanced Landing Approach Operations

Section was deleted as all references to the CRC were reference to RTCA DO-229.

6.11.1 Data Block Structure, M(x)

Section was deleted as all references to the CRC were reference to RTCA DO-229.

6.11.2 RNAV - GPS/GLS Approach Procedure Path Point Data Field Bits

Section was deleted as all references to the CRC were reference to RTCA DO-229.

6.11.2 CRC - Generator Polynomial, G(x)

Section was deleted as all references to the CRC were reference to RTCA DO-229.

7.2.6 Terminal Waypoints

The Naming Convention Table was updated.

7.3.2 Unnamed Waypoints

More specific conventions were added to provide the same level of industry standardization as is applied to waypoint and procedure identifiers.

7.6 Transition Identifiers

Section was revised in its entirety for clarification.

ATTACHMENT 2 LOCAL HORIZONTAL REFERENCE DATUM NAME, DATUM CODE, AND ELLIPSOID LIST

The following reference points were added:

- Austria NS
- Nouvelle Triangulation de France
- Portugese Datum 1973

ATTACHMENT 5 PATH AND TERMINATOR

1.2 Beginning and Ending Leg Types

SID Runway Transition, added flag Note 9 to IF.

Procedure Approach Transition text Route type A deleted.

Procedure Approach was revised to Final Approach Coding.

Procedure Missed Approach, Route type Z was deleted.

1.3 Leg Sequencing

The @ symbol and the supporting Note were added to the Leg Sequencing table.

1.3.1 Conditional Leg Sequencing

New section added to support the @ leg combinations in the Leg Sequencing Table in 1.3.

1.5 Leg Data Fields

MAP column was deleted.

Legend Notes I and T were deleted.

2.0 Coding Rules Applicable To All Procedures

Rule 2.9, the explanation under Instrument Approach Procedure, was revised for clarity.

3.0 Path and Termination Related Rules Valid For All Procedures Types

Rule 3.5 was revised to include IF leg.

Rule 3.14 was deleted.

Rules 3.17 through 3.19 were added by this supplement to support Path and Termination related rules valid for all RNAV procedures that do not reference ground based navaids.

4.0 Standard Instrument Departure (SID) Coding Rules

Rule 4.8 was revised to state all enroute waypoints used in SID common or Vector SID.

Rule 4.9 was revised to remove coded Route Types of 5 or M.

Rule 4.10 was revised to remove coded Route Types of 5 or M.

Rule 4.11 was revised to code the Enroute Transition(s), Route Type as V. Additionally, the text was added that

the DF leg must carry a distance value in the continuation record.

Rule 4.12 was revised to remove coded Runway Transition Route Type of 4, F or V. Further, Rule 4.12 was revised to state when a SID Route is repeated with different fix identifiers in the Transition Identifier, it must be coded as an Enroute Transition, Route Type of 3 or V.

The SID and STAR coding example provided by Jeppesen were removed and new generic examples were added.

5.0 Standard Terminal Arrival Route (STAR) Coding Rules

Rule 5.2 was deleted.

Rule 5.5 was revised to state a STAR or Profile Descent which consists of a single path from an origination fix to a termination fix will be coded as a Route Type 2.

Rule 5.6 was revised to indicate that Transition Identifier must be coded as a Runway Transition Route Type 3 and when a STAR Route/Profile Descent Route or portion of a STAR Route/Profile Descent Route is repeated with different Fix Identifiers in the Transition Identifier, it must be coded as an Enroute Transition Route Type 1.

Rule 5.8 was added to support a STAR which consist of Enroute Transitions.

6.0 Approach Procedure Rules Valid For All Procedure Types

Rule 6.1.1 was revised to support GLS operations.

Rule 6.1.3 was revised for clarity to support the implementation of VNAV.

Rule 6.2.3 was revised for clarity to support the implementation of VNAV.

Rule 6.2.5 was revised for clarity to support the implementation of VNAV.

Rule 6.2.5.4 was revised to clarify FEP.

Rules 6.2.6 and 6.2.7 were deleted by this supplement.

Rule 6.2.9.4 was revised in its entirety to clarify FEP.

Rule 6.2.10.2.c and Rule 6.2.10.2.d was revised in its entirety to clarify FEP.

Rules 6.2.10.2.a, 6.2.10.2.b, 6.2.10.2.c, 6.2.10.2.d, and 6.2.10.2.e were revised to indicate if no procedure TCH is specfieid by the source then use 40 or 50 ft.

Rule 6.2.10.4 was added to support RF leg fixes not at a procedure fix locations.

Rule 6.3.6.5 was revised to support GLS operations.

Rule 6.4.1 was revised for clarity to support the implementation of VNAV.

Rule 6.4.1.1 the following text was deleted “A Runway Centerline Intercept (RCI) point may be coded in some non precision, Localizer based procedures”.

Rule 6.4.1.7 was added in support of FEP.

Rule 6.4.2.1 was revised to support GLS operations.

Rule 6.4.2.2 was deleted.

Rule 6.5 was revised for clarity to support the implementation of VNAV.

Rule 6.5.3 the following text was deleted “Every leg preceding or following an “RF” leg will be tangent to the “RF” leg at that point”.

Rules 6.6, 6.6.1, 6.6.1.1.a, 6.6.1.1.b, 6.6.1.1.c, 6.6.1.2.a, 6.6.1.2.b, and 6.6.1.2.c were revised for clarity to support the implementation of VNAV.

Rule 6.6.3 was revised to state that RNAV-Area Navigation procedures in the context of this Attachment are the so called RNAV Procedures that require referencing ground-based navigational aids. These procedures are also referred to as VOR DME RNAV Procedures.

Rule 6.6.3.1 was revised to indicate all RNAV-Area Navigation approach procedures will be coded to a runway threshold as the last leg in the Final Approach Coding sequence. The runway threshold may be a source defined name waypoint.

Rule 6.6.3.2 was revised for clarity to support the implementation of VNAV.

Rule was 6.6.3.3 was revised to indicate that Final Approach Coding will be accomplished using IF and CF or TF legs only. RF legs may be used in the final approach coding, FACF to FAF and FAF to MAP sequence.

Rule 6.6.3.4 was revised for clarity to support the implementation of VNAV.

Rules 6.7, 6.7.1, 6.7.2, 6.7.2.1, and 6.7.3 were revised for clarity to support the implementation of VNAV.

Rule 6.7.2.2 was deleted.

Rules 6.8, 6.8.1, 6.8.1.1, 6.8.1.2, 6.8.1.3, and 6.8.1.4.b were revised for clarity to support the implementation of VNAV.

Rules 6.10.1, 6.10.2, 6.10.2.1, 6.10.2.2, 6.10.3, 6.10.3.1, 6.10.3.2, 6.10.3.3, 610.3.4, and 610.3.5 have been revised to support coding of approach procedures with circle-to-land minimums.

Rule 6.10.2.3 was revised if a runway is in alignment, the first altitude field in the missed approach point fix record will be coded as the LTP elevation + published TCH (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification]).

Rule 6.10.3.1 was revised for clarity to support the implementation of VNAV.

Rules 6.11, 6.11.1 and 6.11.2 were deleted.

Rules 6.12 and 6.12.1 were revised for clarity to support the implementation of VNAV.

7.0 Precision Approach Procedure Coding

Rule 7.1 was revised to indicate that it will be applied to the final approach coding of LDA and SDF procedures when those procedures include reference to an electronic glide slope, to all types of MLS Approach Procedure.

Rule 7.1.7 was revised to support the altitude angle for the FAF.

Rules 7.3 and 7.3.1 were removed from Attachment 5 and added to the new Attachment 6.

Rule 7.4.3 the following text was deleted "Every leg preceding or following an "RF" leg will be tangent to the "RF" leg at that point".

Rule 7.5, Transition Identifiers, and sub-Rules (7.5.1, 7.5.2, 7.5.3, and 7.5.4) were added.

Rule 7.6 and the associated sub-rules were added to support Transition Identifier.

8.0 Non-precision Approach Procedure Coding

Rule 8.1.2 was revised to indicate that the government source Vertical Angle will also be repeated on fixes associated with an RF Leg as the start or end of the arc, when these fixes are not at procedure fix locations and the RF Leg is in the FAF to MAP portion of the final approach coding.

Rule 8.1.3, was revised in its entirety to support FEP.

Rules 8.2, 8.2.5, 8.3, 8.3.1, 8.3.2, 8.3.4.1, 8.3.4.2, 8.3.4.3, 8.3.5.1, and 8.3.5.2 were revised for clarity to support the implementation of VNAV.

Rule 8.3.5.3 was revised to state the recommended Navaid, Theta and Rho are never provided on fixes associated with the start and end of a precision arc (RF) leg used in the final approach coding sequences.

Rules 8.4, 8.4.1, 8.4.3, 8.6, 8.6.1, and 8.7 were revised for clarity to support the implementation of VNAV.

Rules 8.6.2 and 8.6.3 were revised to clarify the published missed approach point.

Rule 8.6.4 was to support a Vertical Angle must be provided on the missed approach point fix

The following text was added to Rule 8.7.1:

> The rules on RF leg coding to not exclude the use of a RF leg in between a FACF and a TF leg to the FAF or between a FAF and the final TF leg to the missed approach point. Such RF legs must be coded with 3rd character of the Waypoint Description set to "R" and the 4th character of the Waypoint Description field blank.

Rule 8.9 was revised to indicate that Rules 8.9.1 through 8.9.8 apply to the FAF to MAP segment only. Rule 8.9.9 applies to the FACF to FAF segment only.

Rule 8.9.1 was revised for clarity.

Rule 8.9.1 revised to indicate that the published TCH for a procedure used in these calculations is the procedure TCH (if no procedure TCH is specified by source use 40 or 50 feet). TCH is further defined in Chapter Five, Sections 5.67 and 5.265 of this document.

Rule 8.9.3 was revised to include Final Approach Coding segment to FAF and MAP.

Rule 8.9.5 was revised in it entirety to clarify FEP and 40 50 feet for TCH.

Rules 8.9.7, 8.9.8, 8.9.9 and 8.9.10 were revised for clarity to support the implementation of VNAV and TCH.

Examples 1 though 15 and examples A, B and C were revised to clarify the text to support the implementation of VNAV.

Examples of Non-Precision Approach coding were revised for consistent so that VNAV path angle is calculated for the LTP + TCH elevation (if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification].

Commentary was added for FEP coding and delivery.

9.0 Missed Approach Procedure Rules Valid For All Procedure Types

Rule 9.3.1.1 revised to support if no procedure TCH is specified by source use 40 or 50 feet [see Section 5.67 of this Specification.

Rules 9.3.1.3 and 9.3.1.4 were revised for clarity.

Rule 9.3.1.5 was revised to include that (CA, FA, VA) leg must be coded as the first leg of the missed approach and must included a command to climb before the turning leg.

FEP delivery format examples added (1-4).

APPENDIX 2 STRAIGHT-IN CRITERIA

Appendix deleted.

ARINC Standard – Errata Report

1. Document Title

ARINC Specification 474-19: *Navigation System Data Base*
Published December 19, 20080

2. Reference

Page Number:_____ Section Number: _____ Date of Submission: _____

3. Error

(Reproduce the material in error, as it appears in the standard.)

4. Recommended Correction

(Reproduce the correction as it would appear in the corrected version of the material.)

5. Reason for Correction

(State why the correction is necessary.)

6. Submitter (Optional)

(Name, organization, contact information, e.g., phone, email address.)

Note: Items 2-5 may be repeated for additional errata. All recommendations will be evaluated by the staff. Any substantive changes will require submission to the relevant subcommittee for incorporation into a subsequent supplement.

Please return comments to fax +1 410-266-2047 or standards@arinc.com

ARINC IA Project Initiation/Modification (APIM)

1.0 Name of Proposed Project **APIM #: _____**

(Insert name of proposed project.)

2.0 Subcommittee Assignment and Project Support

2.1 Identify AEEC Group

(Identify an existing or new AEEC group.)

2.2 Support for the activity

Airlines: *(Identify each company by name.)*
Airframe Manufacturers:
Suppliers:
Others:

2.3 Commitment for resources (Identify each company by name.)

Airlines:
Airframe Manufacturers:
Suppliers:
Others:

2.4 Chairman: (Recommended name of Chairman.)

2.5 Recommended Coordination with other groups

(List other AEEC subcommittees or other groups.)

3.0 Project Scope (why and when standard is needed)

3.1 Description

(Insert scope. Use the following symbol to check yes or no below. ☒)

3.2 Planned usage of the envisioned specification

New aircraft developments planned to use this specification yes ❑ no ❑
- Airbus: *(aircraft & date)*
- Boeing: *(aircraft & date)*
- Other: *(manufacturer, aircraft & date)*

Modification/retrofit requirement yes ❑ no ❑
- Specify: *(aircraft & date)*

Needed for airframe manufacturer or airline project yes ❑ no ❑
- Specify: *(aircraft & date)*

Mandate/regulatory requirement yes ❑ no ❑
- Program and date: *(program & date)*

Is the activity defining/changing an infrastructure standard? yes ❑ no ❑
- Specify *(e.g., ARINC 429)*

When is the ARINC standard required?
_____*(month/year)*_____

What is driving this date? _____*(state reason)*_____

Are 18 months (min) available for standardization work? yes ❑ no ❑

If NO please specify solution: _____

Are Patent(s) involved? yes ❑

If YES please describe, identify patent holder: _____

3.3 Issues to be worked

(Describe any major issues to be addressed.)

4.0 Benefits

4.1 Basic benefits

Operational enhancements yes ❑ no ❑

For equipment standards:

a. Is this a hardware characteristic? yes ❑ no ❑

b. Is this a softwareware characteristic? yes ❑ no ❑

c. Interchangeable interface definition? yes ❑ no ❑

d. Interchangeable function definition? yes ❑ no ❑

If not fully interchangeable, please explain: _____

Is this a software interface and protocol standard*?* yes ❑ no ❑

Specify: _____

Product offered by more than one supplier yes ❑ no ❑

Identify: *(company name)*

4.2 Specific project benefits

(Describe overall project benefits.)

4.3 Benefits for Airlines

(Describe any benefits unique to the airline point of view.)

4.4 Benefits for Airframe Manufacturers

(Describe any benefits unique to the airframe manufacturer's point of view.)

4.5 Benefits for Avionics Equipment Suppliers

(Describe any benefit unique to the equipment supplier's point of view.)

5.0 Documents to be Produced and Date of Expected Result

5.1 Meetings and Expected Document Completion

The following table identifies the number of meetings and proposed meeting days needed to produce the documents described above.

Activity	Mtgs	Mtg-Days (Total)	Expected Start Date	Expected Completion Date
Document a	*# of mtgs*	*# of mtg days*	*mm/yyyy*	*mm/yyyy*
Document b	*# of mtgs*	*# of mtg days*	*mm/yyyy*	*mm/yyyy*

6.0 Comments

(Insert any other information.)